

49.2 Epithelial tissue forms membranes and glands.

Characteristics of Epithelial Tissue

An epithelial membrane, or **epithelium**, covers every surface of the vertebrate body. Epithelial membranes are derived from all three germ layers. The epidermis, derived from ectoderm, constitutes the outer portion of the skin. The inner surface of the digestive tract is lined by an epithelium derived from endoderm, and the inner surfaces of the body cavities are lined with an epithelium derived from mesoderm.

Because all body surfaces are covered by epithelial membranes, a substance must pass through an epithelium in order to enter or leave the body. Epithelial membranes thus provide a barrier that can impede the passage of some substances while facilitating the passage of others. For land-dwelling vertebrates, the relative impermeability of the surface epithelium (the epidermis) to water offers essential protection from dehydration and from airborne pathogens (disease-causing organisms). On the other hand, the epithelial lining of the digestive tract must allow selective entry of the products of digestion while providing a barrier to toxic substances, and the epithelium of the lungs must allow for the rapid diffusion of gases.

Some epithelia become modified in the course of embryonic development into glands, which are specialized for secretion. A characteristic of all epithelia is that the cells are tightly bound together, with very little space between them. As a consequence, blood vessels cannot be interposed between adjacent epithelial cells. Therefore, nutrients and oxygen must diffuse to the epithelial cells from blood vessels in nearby tissues. This places a limit on the thickness of epithelial membranes; most are only one or a few cell layers thick.

Epithelium possesses remarkable regenerative powers, constantly replacing its cells throughout the life of the animal. For example, the liver, a gland formed from epithelial tissue, can readily regenerate after substantial portions of it have been surgically removed. The epidermis is renewed every two weeks, and the epithelium inside the stomach is replaced every two to three days.

There are two general classes of epithelial membranes: simple and stratified. These classes are further subdivided into squamous, cuboidal, and columnar, based upon the shape of the cells (table 49.2). Squamous cells are flat, cuboidal cells are about as thick as they are tall, and columnar cells are taller than they are wide.

Types of Epithelial Tissues

Simple epithelial membranes are one cell layer thick. A *simple, squamous epithelium* is composed of squamous epithelial cells that have an irregular, flattened shape with tapered edges. Such membranes line the lungs and blood

capillaries, for example, where the thin, delicate nature of these membranes permits the rapid movement of molecules (such as the diffusion of gases). A *simple cuboidal epithelium* lines the small ducts of some glands, and a *simple columnar epithelium* is found in the airways of the respiratory tract and in the gastrointestinal tract, among other locations. Interspersed among the columnar epithelial cells are numerous *goblet cells*, specialized to secrete mucus. The columnar epithelial cells of the respiratory airways contain cilia on their apical surface (the surface facing the lumen, or cavity), which move mucus toward the throat. In the small intestine, the apical surface of the columnar epithelial cells form fingerlike projections called *microvilli*, that increase the surface area for the absorption of food.

Stratified epithelial membranes are several cell layers thick and are named according to the features of their uppermost layers. For example, the epidermis is a *stratified squamous epithelium*. In terrestrial vertebrates it is further characterized as a *keratinized epithelium*, because its upper layer consists of dead squamous cells and filled with a water-resistant protein called *keratin*. The deposition of keratin in the skin can be increased in response to abrasion, producing calluses. The water-resistant property of keratin is evident when the skin is compared with the red portion of the lips, which can easily become dried and chapped because it is covered by a nonkeratinized, stratified squamous epithelium.

The glands of vertebrates are derived from invaginated epithelium. In **exocrine glands**, the connection between the gland and the epithelial membrane is maintained as a duct. The duct channels the product of the gland to the surface of the epithelial membrane and thus to the external environment (or to an interior compartment that opens to the exterior, such as the digestive tract). Examples of exocrine glands include sweat and sebaceous (oil) glands, which secrete to the external surface of the skin, and accessory digestive glands such as the salivary glands, liver, and pancreas, which secrete to the surface of the epithelium lining the digestive tract.

Endocrine glands are ductless glands; their connections with the epithelium from which they were derived are lost during development. Therefore, their secretions, called *hormones*, are not channeled onto an epithelial membrane. Instead, hormones enter blood capillaries and thus stay within the body. Endocrine glands are discussed in more detail in chapter 56.

Epithelial tissues include membranes that cover all body surfaces and glands. The epidermis of the skin is an epithelial membrane specialized for protection, whereas membranes that cover the surfaces of hollow organs are often specialized for transport.

Table 49.2 Epithelial Tissue

Simple Epithelium



SQUAMOUS

Typical Location

Lining of lungs, capillary walls, and blood vessels

Function

Cells very thin; provides thin layer across which diffusion can readily occur

Characteristic Cell Types

Epithelial cells



CUBOIDAL

Typical Location

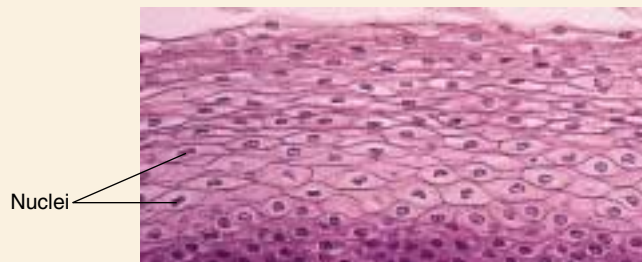
Lining of some glands and kidney tubules; covering of ovaries

Function

Cells rich in specific transport channels; functions in secretion and absorption

Characteristic Cell Types

Gland cells



COLUMNAR

Typical Location

Surface lining of stomach, intestines, and parts of respiratory tract

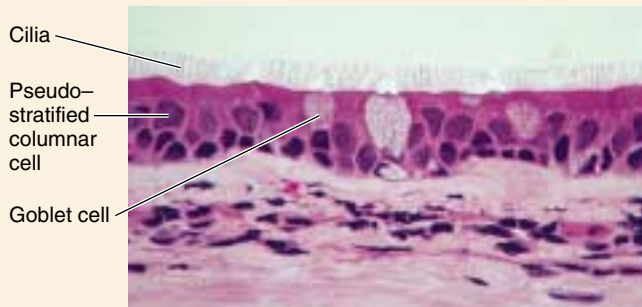
Function

Thicker cell layer; provides protection and functions in secretion and absorption

Characteristic Cell Types

Epithelial cells

Stratified Epithelium



SQUAMOUS

Typical Location

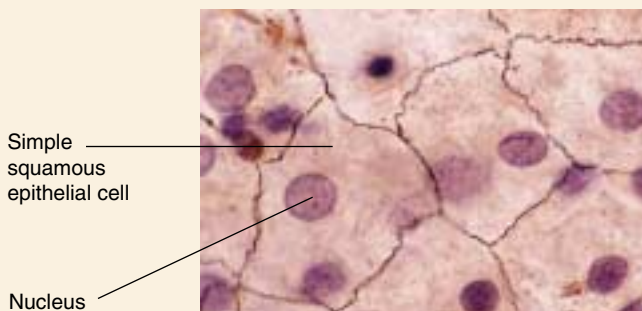
Outer layer of skin; lining of mouth

Function

Tough layer of cells; provides protection

Characteristic Cell Types

Epithelial cells



PSEUDOSTRATIFIED COLUMNAR

Typical Location

Lining parts of the respiratory tract

Function

Secretes mucus; dense with cilia that aid in movement of mucus; provides protection

Characteristic Cell Types

Gland cells; ciliated epithelial cells

49.3 Connective tissues contain abundant extracellular material.

Connective Tissue Proper

Connective tissues are derived from embryonic mesoderm and occur in many different forms (table 49.3). These various forms are divided into two major classes: **connective tissue proper**, which is further divided into loose and dense connective tissues; and **special connective tissues** that include cartilage, bone, and blood. At first glance, it may seem odd that such diverse tissues are placed in the same category. Yet all connective tissues do share a common structural feature: they all have abundant extracellular material because their cells are spaced widely apart. This extracellular material is generically known as the **matrix** of the tissue. In bone, the extracellular matrix contains crystals that make the bones hard; in blood, the extracellular matrix is plasma, the fluid portion of the blood.

Loose connective tissue consists of cells scattered within an amorphous mass of proteins that form a **ground substance**. This gelatinous material is strengthened by a loose scattering of protein fibers such as *collagen* (figure 49.6), *elastin*, which makes the tissue elastic, and *reticulin*, which supports the tissue by forming a collagenous meshwork. The flavored gelatin we eat for dessert consists of the extracellular material from loose connective tissues. The cells that secrete collagen and other fibrous proteins are known as *fibroblasts*.

Loose connective tissue contains other cells as well, including *mast cells* that produce histamine (a blood vessel dilator) and heparin (an anticoagulant) and *macrophages*, the immune system's first defense against invading organisms, as will be described in detail in chapter 57.

Adipose cells are found in loose connective tissue, usually in large groups that form what is referred to as *adipose tissue* (figure 49.7). Each adipose cell contains a droplet of fat (triglycerides) within a storage vesicle. When that fat is needed for energy, the adipose cell hydrolyzes its stored triglyceride and secretes fatty acids into the blood for oxidation by the cells of the muscles, liver, and other organs. The number of adipose cells in an adult is generally fixed. When a person gains weight, the cells become larger, and when weight is lost, the cells shrink.

Dense connective tissue contains tightly packed collagen fibers, making it stronger than loose connective tissue. It consists of two types: regular and irregular. The collagen fibers of **dense regular connective tissue** are lined up in parallel, like the strands of a rope. This is the structure of *tendons*, which bind muscle to bone, and *ligaments*, which bind bone to bone. In contrast, the collagen fibers of **dense irregular connective tissue** have many different orientations. This type of connective tissue produces the tough coverings that package organs, such as

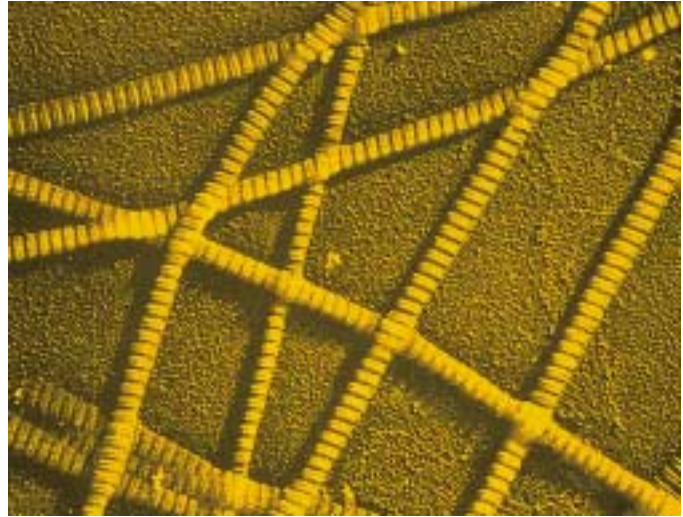


FIGURE 49.6
Collagen fibers. Each fiber is composed of many individual collagen strands and can be very strong under tension.

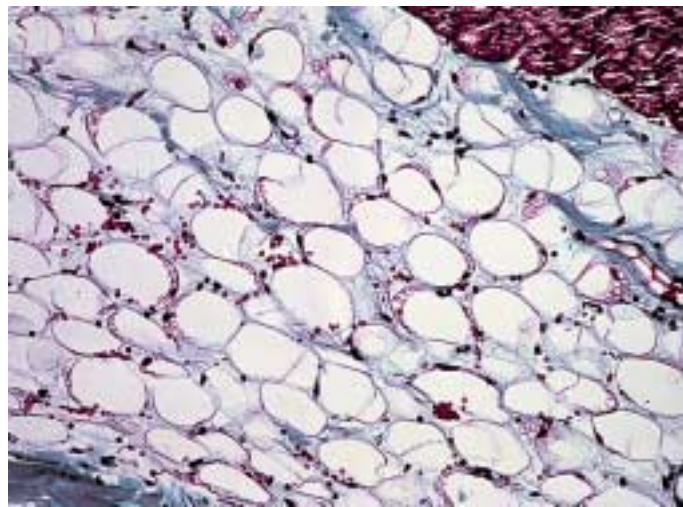
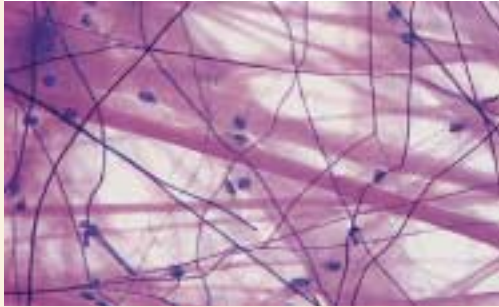


FIGURE 49.7
Adipose tissue. Fat is stored in globules of adipose tissue, a type of loose connective tissue. As a person gains or loses weight, the size of the fat globules increases or decreases. A person cannot decrease the number of fat cells by losing weight.

the *capsules* of the kidneys and adrenal glands. It also covers muscle as *epimysium*, nerves as *perineurium*, and bones as *periosteum*.

Connective tissues are characterized by abundant extracellular materials in the matrix between cells.
Connective tissue proper may be either loose or dense.

Table 49.3 Connective Tissue



LOOSE CONNECTIVE TISSUE

Typical Location

Beneath skin; between organs

Function

Provides support, insulation, food storage, and nourishment for epithelium

Characteristic Cell Types

Fibroblasts, macrophages, mast cells, fat cells



DENSE CONNECTIVE TISSUE

Typical Location

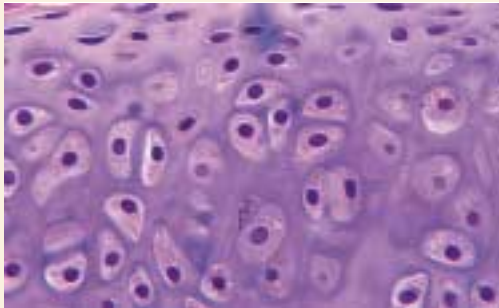
Tendons; sheath around muscles; kidney; liver; dermis of skin

Function

Provides flexible, strong connections

Characteristic Cell Types

Fibroblasts



CARTILAGE

Typical Location

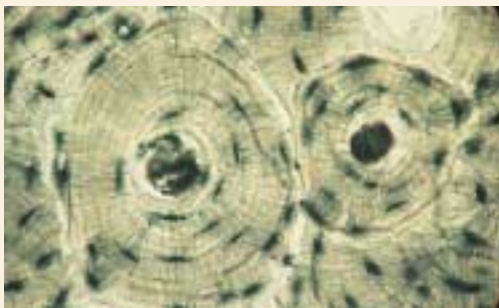
Spinal discs; knees and other joints; ear; nose; tracheal rings

Function

Provides flexible support, shock absorption, and reduction of friction on load-bearing surfaces

Characteristic Cell Types

Chondrocytes



BONE

Typical Location

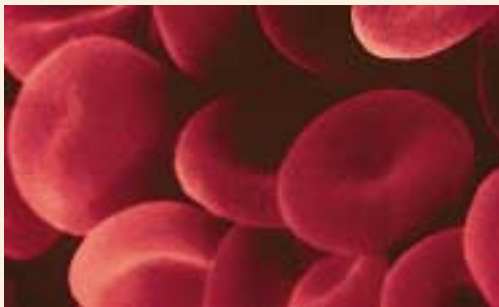
Most of skeleton

Function

Protects internal organs; provides rigid support for muscle attachment

Characteristic Cell Types

Osteocytes



BLOOD

Typical Location

Circulatory system

Function

Functions as highway of immune system and primary means of communication between organs

Characteristic Cell Types

Erythrocytes, leukocytes

Special Connective Tissues

The special connective tissues—cartilage, bone, and blood—each have unique cells and extracellular matrices that allow them to perform their specialized functions.

Cartilage

Cartilage (figure 49.8) is a specialized connective tissue in which the ground substance is formed from a characteristic type of glycoprotein, and the collagen fibers are laid down along the lines of stress in long, parallel arrays. The result is a firm and flexible tissue that does not stretch, is far tougher than loose or dense connective tissue, and has great tensile strength. Cartilage makes up the entire skeletal system of the modern agnathans and cartilaginous fishes (see chapter 48), replacing the bony skeletons that were characteristic of the ancestors of these vertebrate groups. In most adult vertebrates, however, cartilage is restricted to the articular (joint) surfaces of bones that form freely movable joints and to other specific locations. In humans, for example, the tip of the nose, the pinna (outer ear flap), the intervertebral discs of the backbone, the larynx (voice box) and a few other structures are composed of cartilage.

Chondrocytes, the cells of the cartilage, live within spaces called *lacunae* within the cartilage ground substance. These cells remain alive, even though there are no blood vessels within the cartilage matrix, because they receive oxygen and nutrients by diffusion through the cartilage ground substance from surrounding blood vessels. This diffusion can only occur because the cartilage matrix is not calcified, as is bone.

Bone

In the course of fetal development, the bones of vertebrate fins, arms, and legs, among others, are first “modeled” in cartilage. The cartilage matrix then calcifies at particular locations, so that the chondrocytes are no longer able to obtain oxygen and nutrients by diffusion through the matrix. The dying and degenerating cartilage is then replaced by living bone. Bone cells, or osteocytes, can remain alive even though the extracellular matrix becomes hardened with crystals of calcium phosphate. This is because blood vessels travel through central canals into the bone. Osteo-

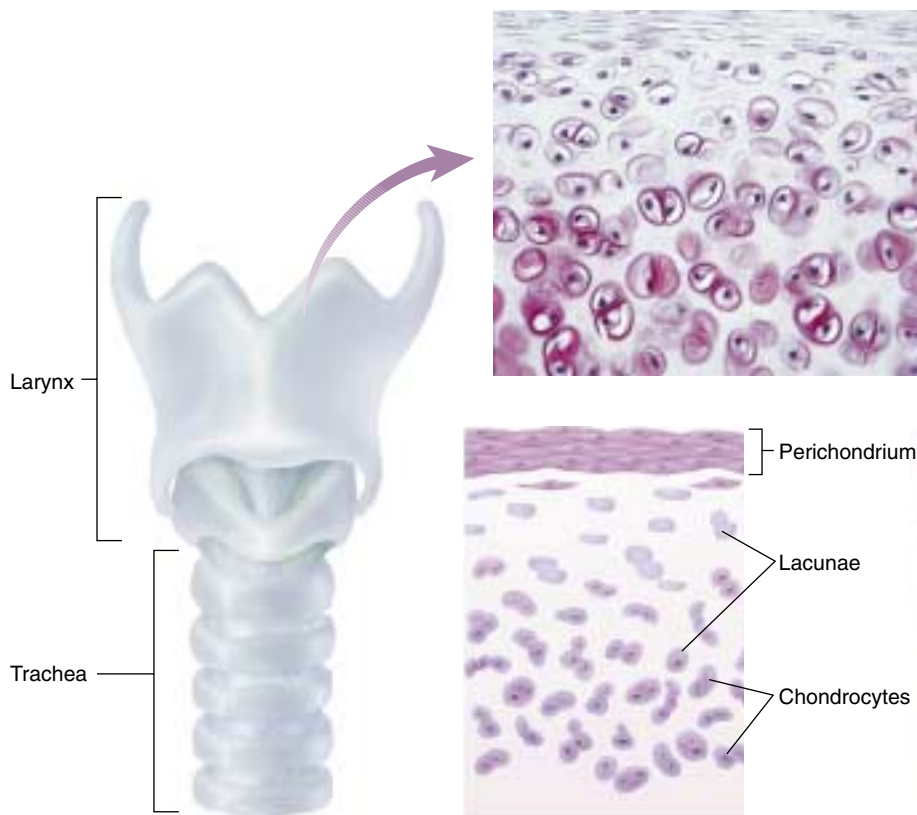


FIGURE 49.8

Cartilage is a strong, flexible tissue that makes up the larynx (voice box) and several other structures in the human body. The larynx (a) is seen under the light microscope in (b), where the cartilage cells, or chondrocytes, are visible within cavities, or lacunae, in the matrix (extracellular material) of the cartilage. This is diagrammed in (c).

cytes extend cytoplasmic processes toward neighboring osteocytes through tiny canals, or *canaliculi* (figure 49.9). Osteocytes communicate with the blood vessels in the central canal through this cytoplasmic network.

It should be noted here that some bones, such as those of the cranium, are not formed first as cartilage models. These bones instead develop within a membrane of dense, irregular connective tissue. The structure and formation of bone are discussed in chapter 50.

Blood

Blood is classified as a connective tissue because it contains abundant extracellular material, the fluid plasma. The cells of blood are erythrocytes, or red blood cells, and leukocytes, or white blood cells (figure 49.10). Blood also contains platelets, or *thrombocytes*, which are fragments of a type of bone marrow cell.

Erythrocytes are the most common blood cells; there are about 5 billion in every milliliter of blood. During their maturation in mammals, they lose their nucleus, mitochondria, and endoplasmic reticulum. As a result, mammalian erythrocytes are relatively inactive metabolically. Each erythrocyte

contains about 300 million molecules of the iron-containing protein *hemoglobin*, the principal carrier of oxygen in vertebrates and in many other groups of animals.

There are several types of leukocytes, but together they are only one-thousandth as numerous as erythrocytes. Unlike mammalian erythrocytes, leukocytes have nuclei and mitochondria but lack the red pigment hemoglobin. These cells are therefore hard to see under a microscope without special staining. The names *neutrophils*, *eosinophils*, and *basophils* distinguish three types of leukocytes on the basis of their staining properties; other leukocytes include *lymphocytes* and *monocytes*. These different types of leukocytes play critical roles in immunity, as will be described in chapter 57.

The blood plasma is the “commons” of the body; it (or a derivative of it) travels to and from every cell in the body. As the plasma circulates, it carries nourishment, waste products, heat, and regulatory molecules. Practically every substance used by cells, including sugars, lipids, and amino acids, is delivered by the plasma to the body cells. Waste products from the cells are carried by the plasma to the kidneys, liver, and lungs or gills for disposal, and regulatory molecules (hormones) that endocrine gland cells secrete are carried by the plasma to regulate the activities of most organs of the body. The plasma also contains sodium, calcium, and other inorganic ions that all cells need, as well as numerous proteins. Plasma proteins include *fibrinogen*, produced by the liver, which helps blood to clot; *albumin*, also produced by the liver, which exerts an osmotic force needed for fluid balance; and *antibodies* produced by lymphocytes and needed for immunity.

Special connective tissues each have a unique extracellular matrix between cells. The matrix of cartilage is composed of organic material, whereas that of bone is impregnated with calcium phosphate crystals. The matrix of blood is fluid, the plasma.

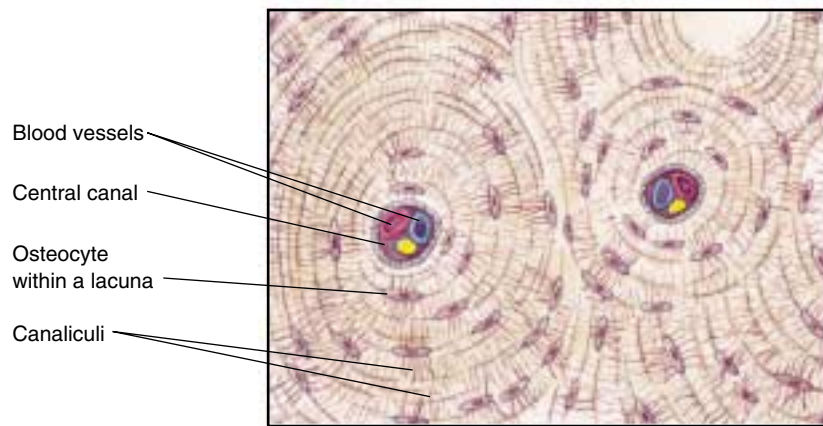
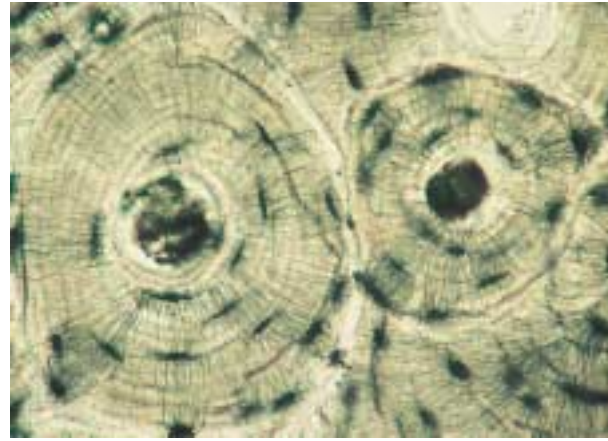


FIGURE 49.9

The structure of bone. A photomicrograph (a) and diagram (b) of the structure of bone, showing the bone cells, or osteocytes, within their lacunae (cavities) in the bone matrix. Though the bone matrix is calcified, the osteocytes remain alive because they can be nourished by blood vessels in the central cavity. Nourishment is carried between the osteocytes through a network of cytoplasmic processes extending through tiny canals, or canaliculi.

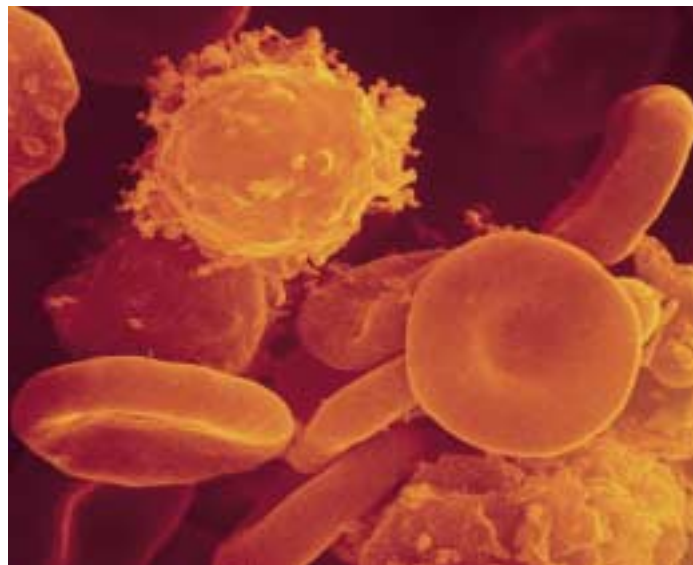


FIGURE 49.10
White and red blood cells

(500×). White blood cells, or leukocytes, are roughly spherical and have irregular surfaces with numerous extending pili. Red blood cells, or erythrocytes, are flattened spheres, typically with a depressed center, forming biconcave discs.

49.4 Muscle tissue provides for movement, and nerve tissue provides for control.

Muscle Tissue

Muscle cells are the motors of the vertebrate body. The characteristic that makes them unique is the relative abundance and organization of actin and myosin filaments within them. Although these filaments form a fine network in all eukaryotic cells, where they contribute to cellular movements, they are far more common in muscle cells, which are specialized for contraction. Vertebrates possess three kinds of muscle: smooth, skeletal, and cardiac (table 49.4). Skeletal and cardiac muscles are also known as **striated muscles** because their cells have transverse stripes when viewed in longitudinal section under the microscope. The contraction of each skeletal muscle is under voluntary control, whereas the contraction of cardiac and smooth muscles is generally involuntary. Muscles are described in more detail in chapter 50.

Smooth Muscle

Smooth muscle was the earliest form of muscle to evolve, and it is found throughout the animal kingdom. In vertebrates, smooth muscle is found in the organs of the internal environment, or *viscera*, and is sometimes known as visceral muscle. Smooth muscle tissue is organized into sheets of long, spindle-shaped cells, each cell containing a single nucleus. In some tissues, the cells contract only when they are

stimulated by a nerve, and then all of the cells in the sheet contract as a unit. In vertebrates, muscles of this type line the walls of many blood vessels and make up the iris of the eye. In other smooth muscle tissues, such as those in the wall of the gut, the muscle cells themselves may spontaneously initiate electric impulses and contract, leading to a slow, steady contraction of the tissue. Nerves regulate, rather than cause, this activity.

Skeletal Muscle

Skeletal muscles are usually attached by tendons to bones, so that, when the muscles contract, they cause the bones to move at their joints. A skeletal muscle is made up of numerous, very long muscle cells, called *muscle fibers*, which lie parallel to each other within the muscle and insert into the tendons on the ends of the muscle. Each skeletal muscle fiber is stimulated to contract by a nerve fiber; therefore, a stronger muscle contraction will result when more of the muscle fibers are stimulated by nerve fibers to contract. In this way, the nervous system can vary the strength of skeletal muscle contraction. Each muscle fiber contracts by means of substructures called *myofibrils* (figure 49.11) that contain highly ordered arrays of actin and myosin myofilaments, that, when aligned, give the muscle fiber its striated appearance. Skeletal muscle fibers are produced during development by the fusion of several cells, end to end. This

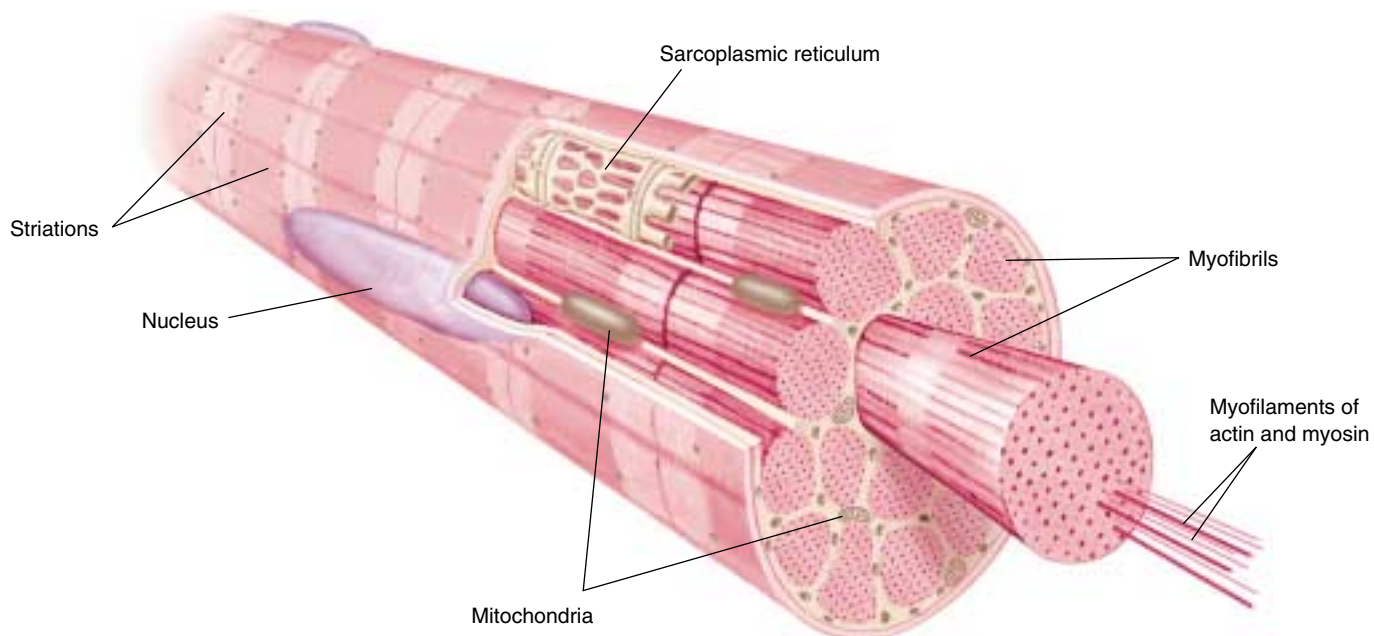
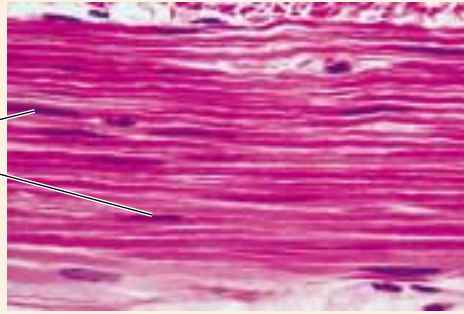


FIGURE 49.11

A muscle fiber, or muscle cell. Each muscle fiber is composed of numerous myofibrils, which, in turn, are composed of actin and myosin filaments. Each muscle fiber is multinucleate as a result of its embryological development from the fusion of smaller cells. Muscle cells have a modified endoplasmic reticulum called the sarcoplasmic reticulum.

Table 49.4 Muscle Tissue

Nuclei



SMOOTH MUSCLE

Typical Location

Walls of blood vessels, stomach, and intestines

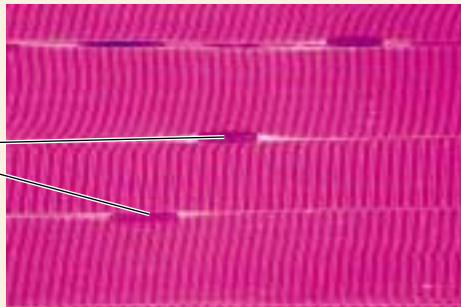
Function

Powers rhythmic, involuntary contractions commanded by the central nervous system

Characteristic Cell Types

Smooth muscle cells

Nuclei



SKELETAL MUSCLE

Typical Location

Voluntary muscles

Function

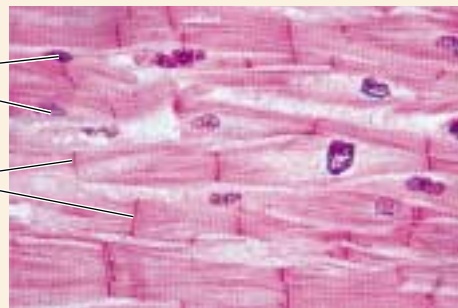
Powers walking, lifting, talking, and all other voluntary movement

Characteristic Cell Types

Skeletal muscle cells

Nuclei

Intercalated discs



CARDIAC

Typical Location

Walls of heart

Function

Highly interconnected cells; promotes rapid spread of signal initiating contraction

Characteristic Cell Types

Cardiac muscle cells

embryological development explains why a mature muscle fiber contains many nuclei. The structure and function of skeletal muscle is explained in more detail in chapter 50.

Cardiac Muscle

The hearts of vertebrates are composed of striated muscle cells arranged very differently from the fibers of skeletal muscle. Instead of having very long, multinucleate cells running the length of the muscle, cardiac muscle is composed of smaller, interconnected cells, each with a single nucleus. The interconnections between adjacent cells appear under the microscope as dark lines called *intercalated discs*. In reality, these lines are regions where adjacent cells are linked by *gap junctions*. As we noted in chapter 7, gap

junctions have openings that permit the movement of small substances and electric charges from one cell to another. These interconnections enable the cardiac muscle cells to form a single, functioning unit known as a myocardium. Certain cardiac muscle cells generate electric impulses spontaneously, and these impulses spread across the gap junctions from cell to cell, causing all of the cells in the myocardium to contract. We will describe this process more fully in chapter 52.

Skeletal muscles enable the vertebrate body to move. Cardiac muscle powers the heartbeat, while smooth muscles provide a variety of visceral functions.

Nerve Tissue

The fourth major class of vertebrate tissue is nerve tissue (table 49.5). Its cells include neurons and neuroglia, or supporting cells. Neurons are specialized to produce and conduct electrochemical events, or “impulses.” Each neuron consists of three parts: cell body, dendrites, and axon (figure 49.12). The cell body of a neuron contains the nucleus. Dendrites are thin, highly branched extensions that receive incoming stimulation and conduct electric events to the cell body. As a result of this stimulation and the electric events produced in the cell body, outgoing impulses may be produced at the origin of the axon. The axon is a single extension of cytoplasm that conducts impulses away from the cell body. Some axons can be quite long. The cell bodies of neurons that control the muscles in your feet, for example, lie in the spinal cord, and their axons may extend over a meter to your feet.

Neuroglia do not conduct electrical impulses but instead support and insulate neurons and eliminate foreign materials in and around neurons. In many neurons, neuroglia cells associate with the axons and form an insulating covering, a *myelin sheath*, produced by successive wrapping of the membrane around the axon (figure 49.13). Adjacent neuroglia cells are separated by interruptions known as *nodes of Ranvier*, which serve as sites for accelerating an impulse (see chapter 54).

The nervous system is divided into the central nervous system (CNS), which includes the brain and spinal cord, and the peripheral nervous system (PNS), which includes *nerves* and *ganglia*. Nerves consist of axons in the PNS that are bundled together in much the same way as wires are bundled together in a cable. Ganglia are collections of neuron cell bodies.

There are different types of neurons, but all are specialized to receive, produce, and conduct electrical signals. Neuroglia do not conduct electrical impulses but have various functions, including insulating axons to accelerate an electrical impulse. Both neurons and neuroglia are present in the CNS and the PNS.

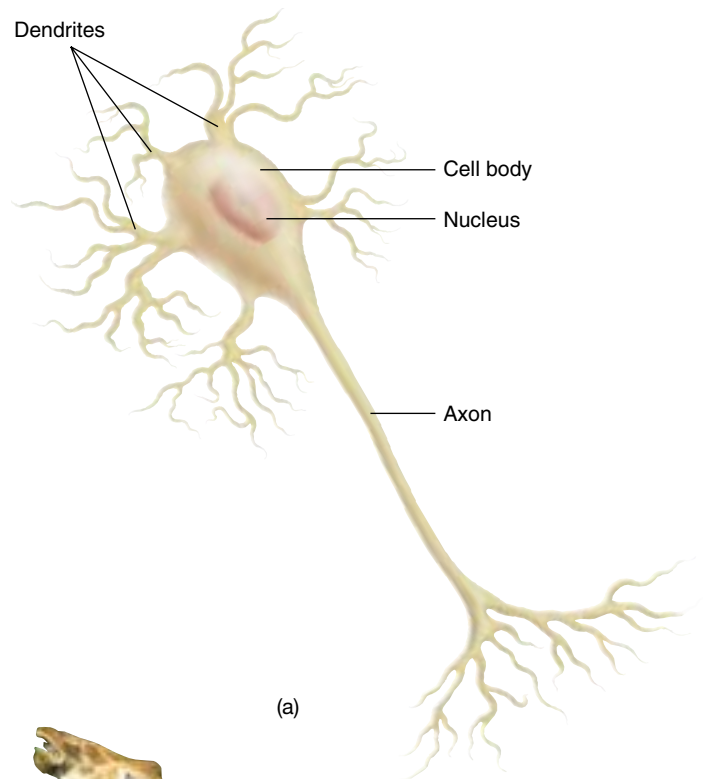
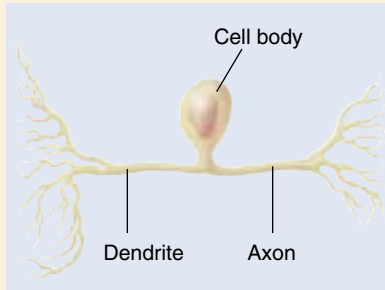


FIGURE 49.12

A neuron has a very long projection called an axon. (a) A nerve impulse is received by the dendrites and then passed to the cell body and out through the axon. **(b)** Axons can be very long; single axons extend from the skull down several meters through a giraffe's neck to its pelvis.

Table 49.5 Nerve Tissue



SENSORY NEURONS

Typical Location

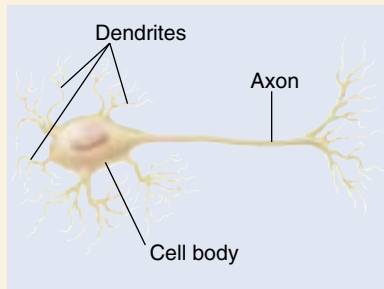
Eyes; ears; surface of skin

Function

Receive information about body's condition and external environment; send impulses from sensory receptors to CNS

Characteristic Cell Types

Rods and cones; muscle stretch receptors



MOTOR NEURONS

Typical Location

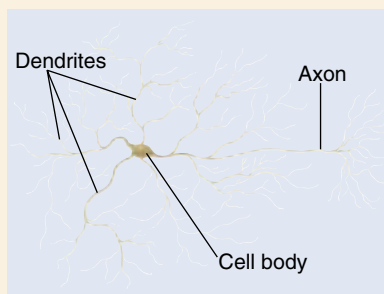
Brain and spinal cord

Function

Stimulate muscles and glands; conduct impulses out of CNS toward muscles and glands

Characteristic Cell Types

Motor neurons



ASSOCIATION NEURONS

Typical Location

Brain and spinal cord

Function

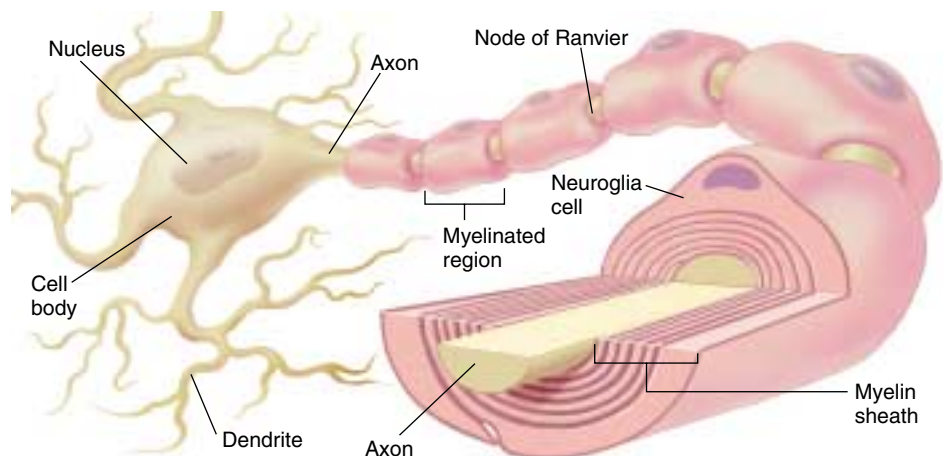
Integrate information; conduct impulses between neurons within CNS

Characteristic Cell Types

Association neurons

FIGURE 49.13

A myelinated neuron. Many dendrites arise from the cell body, as does a single long axon. In some neurons specialized for rapid signal conduction, the axon is encased in a myelin sheath that is interrupted at intervals. At its far end, the axon may branch to terminate on more than one cell.





Chapter 49

Summary

Questions

Media Resources

49.1 The bodies of vertebrates are organized into functional systems.

- The vertebrate body is organized into cells, tissues, organs, and organ systems, which are specialized for different functions.
- The four primary tissues of the vertebrate adult body—epithelial, connective, muscle, and nerve—are derived from three embryonic germ layers.

1. What is a tissue? What is an organ? What is an organ system?



- Art Activity: Mammalian body cavities

49.2 Epithelial tissue forms membranes and glands.

- Epithelial membranes cover all body surfaces.
- Stratified membranes, particularly the keratinized epithelium of the epidermis, provides protection, whereas simple membranes are more adapted for secretion and transport.
- Exocrine glands secrete into ducts that conduct the secretion to the surface of an epithelial membrane; endocrine glands secrete hormones into the blood.

2. What are the different types of epithelial membranes, and how do they differ in structure and function?

3. What are the two types of glands, and how do they differ in structure and function?



- Epithelial tissue
- Epithelial glands

49.3 Connective tissues contain abundant extracellular material.

- Connective tissues are characterized by abundant extracellular matrix, which is composed of fibrous proteins and a gel-like ground substance in connective tissue proper.
- Loose connective tissues contain many cell types such as adipose cells and mast cells; dense regular connective tissues form tendons and ligaments.
- Special connective tissues include cartilage, bone, and blood. Nutrients can diffuse through the cartilage matrix but not through the calcified matrix of bone, which contains canaliculi for that purpose.

4. What feature do all connective tissues share? What are the different categories of connective tissue? Give an example of each.

5. What is the structure of a ligament? How do cartilage and bone differ? Why is blood considered to be a connective tissue?



- Connective tissue

49.4 Muscle tissue provides for movement, and nerve tissue provides for control.

- Smooth muscles are composed of spindle-shaped cells and are found in the organs of the internal environment and in the walls of blood vessels.
- Skeletal and cardiac muscles are striated; skeletal muscles, however, are under voluntary control whereas cardiac muscle is involuntary.
- Neurons consist of a cell body with one or more dendrites and one axon. Neuron cell bodies form ganglia, and their axons form nerves in the peripheral nervous system.
- Neuroglia are supporting cells with various functions including insulating axons to accelerate an electrical impulse.

6. From what embryonic tissue is muscle derived? What two contractile proteins are abundant in muscle? What are the three categories of muscle tissue? Which two are striated?

7. Why are skeletal muscle fibers multinucleated? What is the functional significance of intercalated discs in heart muscle?



- Tissues
- Nerve tissue
- Nervous tissue
- Muscle tissue



50

Locomotion

Concept Outline

50.1 A skeletal system supports movement in animals.

Types of Skeletons. There are three types of skeletal systems found in animals: hydrostatic skeletons, exoskeletons, and endoskeletons. Hydrostatic skeletons function by the movement of fluid in a body cavity. Exoskeletons are made of tough exterior coverings on which muscles attach to move the body. Endoskeletons are rigid internal bones or cartilage which move the body by the contraction of muscles attached to the skeleton.

The Structure of Bone. The human skeleton, an example of an endoskeleton, is made of bone that contains cells called osteocytes within a calcified matrix.

50.2 Skeletal muscles contract to produce movements at joints.

Types of Joints. The joints where bones meet may be immovable, slightly movable, or freely movable.

Actions of Skeletal Muscles. Synergistic and antagonistic muscles act on the skeleton to move the body.

50.3 Muscle contraction powers animal locomotion.

The Sliding Filament Mechanism of Contraction. Thick and thin myofilaments slide past one another to cause muscle shortening.

The Control of Muscle Contraction. During contraction Ca^{++} moves aside a regulatory protein which had been preventing cross-bridges from attaching to the thin filaments. Nerves stimulate the release of Ca^{++} from its storage depot so that contraction can occur.

Types of Muscle Fibers. Muscle fibers can be categorized as slow-twitch (slow to fatigue) or fast-twitch (fatigue quickly but can provide a fast source of power).

Comparing Cardiac and Smooth Muscles. Cardiac muscle cells are interconnected to form a single functioning unit. Smooth muscles lack the myofilament organization found in striated muscle but they still contract via the sliding filament mechanism.

Modes of Animal Locomotion. Animals rarely move in straight lines. Their movements are adjusted both by mechanical feedback and by neural control. Muscles generate power for movement, and also act as springs, brakes, struts, and shock absorbers.



FIGURE 50.1

On the move. The movements made by this sidewinder rattlesnake are the result of strong muscle contractions acting on the bones of the skeleton. Without muscles and some type of skeletal system, complex locomotion as shown here would not be possible.

Plants and fungi move only by growing, or as the passive passengers of wind and water. Of the three multicellular kingdoms, only animals explore their environment in an active way, through locomotion. In this chapter we examine how vertebrates use muscles connected to bones to achieve movement. The rattlesnake in figure 50.1 slithers across the sand by a rhythmic contraction of the muscles sheathing its body. Humans walk by contracting muscles in their legs. Although our focus in this chapter will be on vertebrates, it is important to realize that essentially all animals employ muscles. When a mosquito flies, its wings are moved rapidly through the air by quickly contracting flight muscles. When an earthworm burrows through the soil, its movement is driven by strong muscles pushing its body past the surrounding dirt.

50.1 A skeletal system supports movement in animals.

Types of Skeletons

Animal locomotion is accomplished through the force of muscles acting on a rigid skeletal system. There are three types of skeletal systems in the animal kingdom: hydraulic skeletons, exoskeletons, and endoskeletons.

Hydrostatic skeletons are primarily found in soft-bodied invertebrates such as earthworms and jellyfish. In this case, a fluid-filled cavity is encircled by muscle fibers. As the muscles contract, the fluid in the cavity moves and changes the shape of the cavity. In an earthworm, for example, a wave of contractions of circular muscles begins anteriorly and compresses each segment of the body, so that the fluid pressure pushes it forward. Contractions of longitudinal muscles then pull the rear of the body forward (figure 50.2).

Exoskeletons surround the body as a rigid hard case in most animals. Arthropods, such as crustaceans and insects, have exoskeletons made of the polysaccharide *chitin* (figure 50.3*a*). An exoskeleton offers great protection to internal organs and resists bending. However, in order to grow, the animal must periodically molt. During molting, the animal is particularly vulnerable to predation because its old exoskeleton has been shed. Having an exoskeleton also limits the size of the animal. An animal with an exoskeleton cannot get too large because its exoskeleton would have to become thicker and heavier, in order to prevent collapse, as the animal grew larger. If an insect were the size of a human being, its exoskeleton would have to be so thick and heavy it would be unable to move.

Endoskeletons, found in vertebrates and echinoderms, are rigid internal skeletons to which muscles are attached. Vertebrates have a flexible exterior that accommodates the movements of their skeleton. The endoskeleton of vertebrates is composed of cartilage or bone. Unlike chitin, bone is a cellular, living tissue capable of growth, self-repair, and remodeling in response to physical stresses.

The Vertebrate Skeleton

A vertebrate endoskeleton (figure 50.3*b*) is divided into an axial and an appendicular skeleton. The axial skeleton's bones form the axis of the body and support and protect the organs of the head, neck, and chest. The appendicular skeleton's bones include the bones of the limbs, and the pectoral and pelvic girdles that attach them to the axial skeleton.

The bones of the skeletal system support and protect the body, and serve as levers for the forces produced by contraction of skeletal muscles. Blood cells form within the bone marrow, and the calcified matrix of bones acts as a reservoir for calcium and phosphate ions.

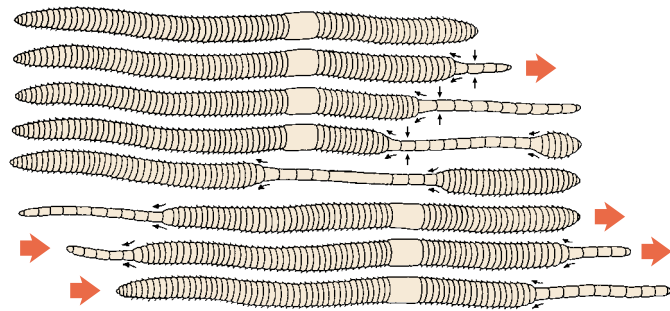


FIGURE 50.2

Locomotion in earthworms. The hydrostatic skeleton of the earthworm uses muscles to move fluid within the segmented body cavity changing the shape of the animal. When an earthworm's circular muscles contract, the internal fluid presses on the longitudinal muscles, which then stretch to elongate segments of the earthworms. A wave of contractions down the body of the earthworm produces forward movement.

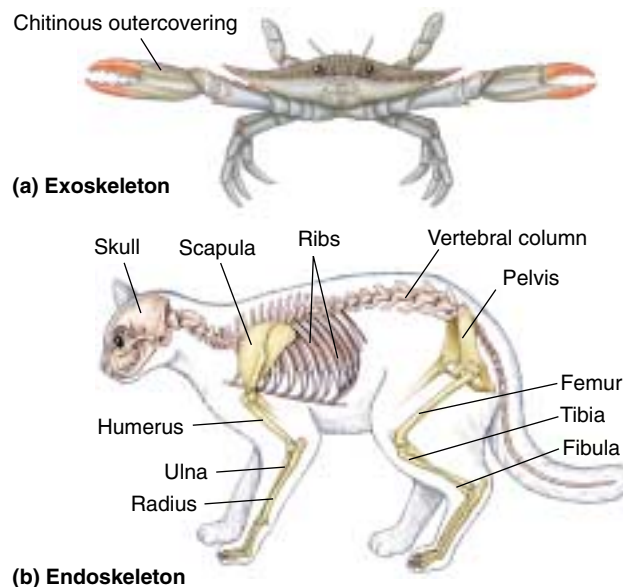


FIGURE 50.3

Exoskeleton and endoskeleton. (a) The hard, tough outcovering of an arthropod, such as this crab, is its exoskeleton. (b) Vertebrates, such as this cat, have endoskeletons. The axial skeleton is shown in the peach shade, the appendicular skeleton in the yellow shade. Some of the major bones are labeled.

There are three types of animal skeletons: hydrostatic skeleton, exoskeleton, and endoskeleton. The endoskeletons found in vertebrates are composed of bone or cartilage and are organized into axial and appendicular portions.

The Structure of Bone

Bone, the building material of the vertebrate skeleton, is a special form of connective tissue (see chapter 49). In bone, an organic extracellular matrix containing collagen fibers is impregnated with small, needle-shaped crystals of calcium phosphate in the form of hydroxyapatite crystals. Hydroxyapatite is brittle but rigid, giving bone great strength. Collagen, on the other hand, is flexible but weak. As a result, bone is both strong and flexible. The collagen acts to spread the stress over many crystals, making bone more resistant to fracture than hydroxyapatite is by itself.

Bone is a dynamic, living tissue that is constantly reconstructed throughout the life of an individual. New bone is formed by *osteoblasts*, which secrete the collagen-containing organic matrix in which calcium phosphate is later deposited. After the calcium phosphate is deposited, the cells, now known as *osteocytes*, are encased within spaces called *lacunae* in the calcified matrix. Yet another type of bone cells, called *osteoclasts*, act to dissolve bone and thereby aid in the remodeling of bone in response to physical stress.

Bone is constructed in thin, concentric layers, or *lamellae*, which are laid down around narrow channels called *Haversian canals* that run parallel to the length of the bone. Haversian canals contain nerve fibers and blood vessels, which keep the osteocytes alive even though they are entombed in a calcified matrix. The concentric lamellae of bone, with their entrapped osteocytes, that surround a Haversian canal form the basic unit of bone structure, called a Haversian system.

Bone formation occurs in two ways. In flat bones, such as those of the skull, osteoblasts located in a web of dense connective tissue produce bone within that tissue. In long bones, the bone is first “modeled” in cartilage. Calcification then occurs, and bone is formed as the cartilage degenerates. At the end of this process, cartilage remains only at the articular (joint) surfaces of the bones and at the growth plates located in the necks of the long bones. A child grows taller as the cartilage thickens in the growth plates and then is partly replaced with bone. A person stops growing (usually by the late teenage years) when the entire cartilage growth plate becomes replaced

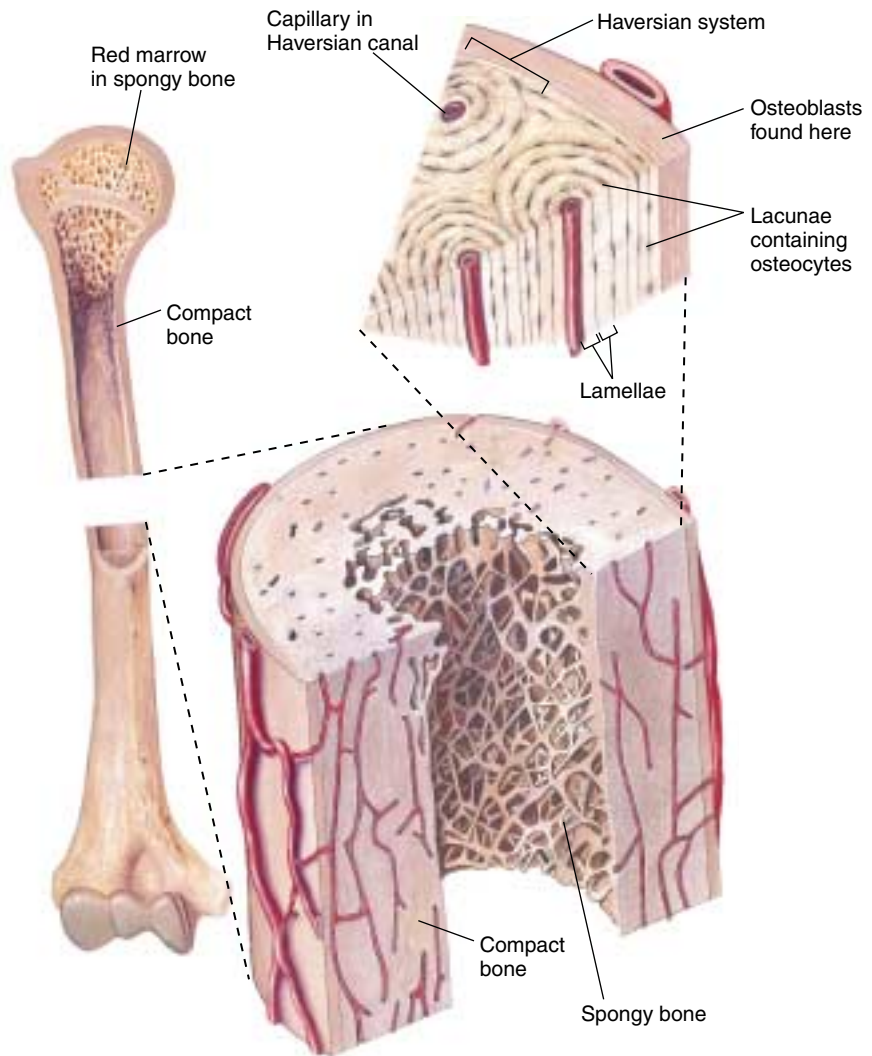


FIGURE 50.4

The organization of bone, shown at three levels of detail. Some parts of bone are dense and compact, giving the bone strength. Other parts are spongy, with a more open lattice; it is there that most blood cells are formed.

by bone. At this point, only the articular cartilage at the ends of the bone remains.

The ends and interiors of long bones are composed of an open lattice of bone called *spongy bone*. The spaces within contain marrow, where most blood cells are formed (figure 50.4). Surrounding the spongy bone tissue are concentric layers of *compact bone*, where the bone is much denser. Compact bone tissue gives bone the strength to withstand mechanical stress.

Bone consists of cells and an extracellular matrix that contains collagen fibers, which provide flexibility, and calcium phosphate, which provides strength. Bone contains blood vessels and nerves and is capable of growth and remodeling.

50.2 Skeletal muscles contract to produce movements at joints.

Types of Joints

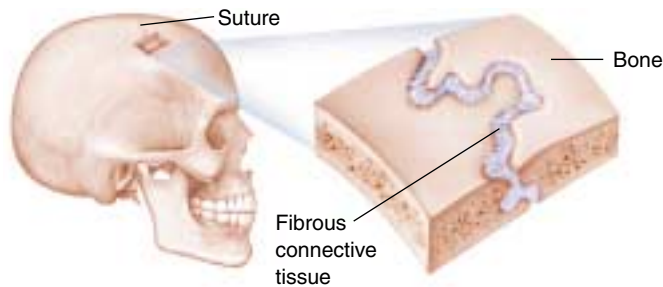
The skeletal movements of the body are produced by contraction and shortening of muscles. Skeletal muscles are generally attached by tendons to bones, so when the muscles shorten, the attached bones move. These movements

of the skeleton occur at joints, or articulations, where one bone meets another. There are three main classes of joints:

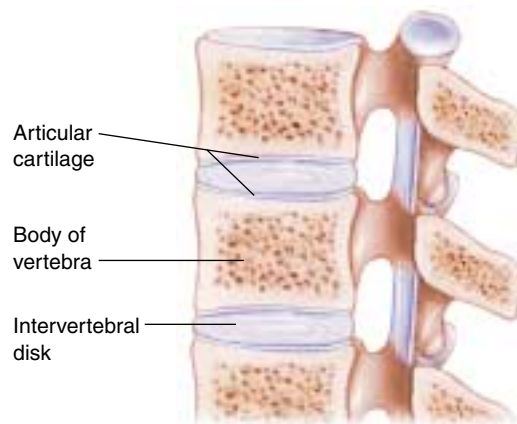
1. **Immovable joints** include the *sutures* that join the bones of the skull (figure 50.5a). In a fetus, the skull bones are not fully formed, and there are open areas of dense connective tissue (“soft spots,” or *fontanelles*) between the bones. These areas allow the bones to shift slightly as the fetus moves through the birth canal during childbirth. Later, bone replaces most of this connective tissue.
2. **Slightly movable joints** include those in which the bones are bridged by cartilage. The vertebral bones of the spine are separated by pads of cartilage called *intervertebral discs* (figure 50.5b). These *cartilaginous joints* allow some movement while acting as efficient shock absorbers.
3. **Freely movable joints** include many types of joints and are also called synovial joints, because the articulating ends of the bones are located within a *synovial capsule* filled with a lubricating fluid. The ends of the bones are capped with cartilage, and the synovial capsule is strengthened by ligaments that hold the articulating bones in place.

Synovial joints allow the bones to move in directions dictated by the structure of the joint. For example, a joint in the finger allows only a hingelike movement, while the joint between the thigh bone (femur) and pelvis has a ball-and-socket structure that permits a variety of different movements (figure 50.5c).

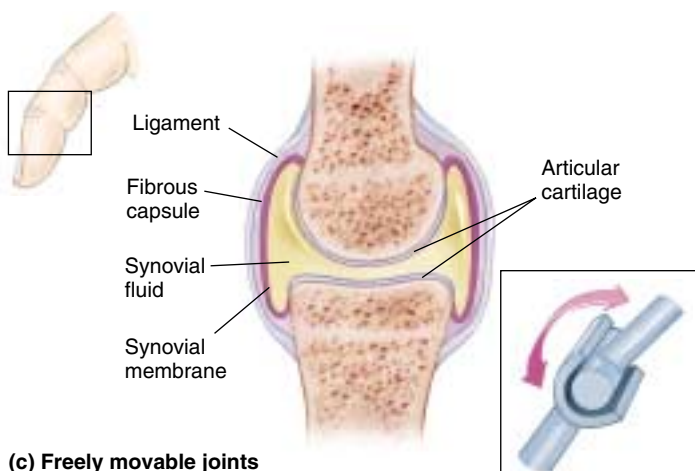
Joints confer flexibility to a rigid skeleton, allowing a range of motions determined by the type of joint.



(a) Immovable joint



(b) Slightly movable joints



(c) Freely movable joints

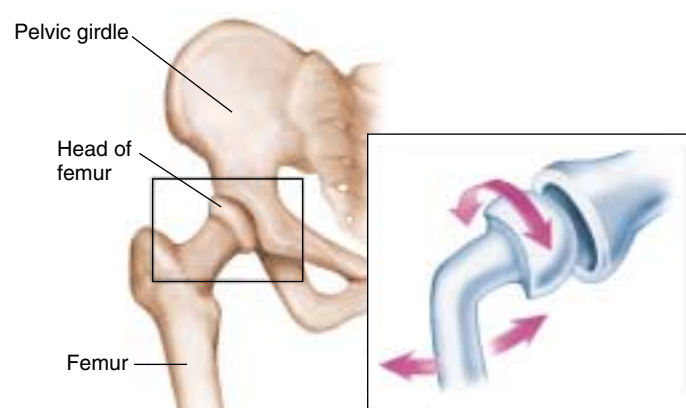


FIGURE 50.5

Three types of joints. (a) Immovable joints include the sutures of the skull; (b) slightly movable joints include the cartilaginous joints between the vertebrae; and (c) freely movable joints are the synovial joints, such as a finger joint and or a hip joint.

Actions of Skeletal Muscles

Skeletal muscles produce movement of the skeleton when they contract. Usually, the two ends of a skeletal muscle are attached to different bones (although in some cases, one or both ends may be connected to some other kind of structure, such as skin). The attachments to bone are made by means of dense connective tissue straps called *tendons*. Tendons have elastic properties that allow “give-and-take” during muscle contraction. One attachment of the muscle, the **origin**, remains relatively stationary during a contraction. The other end of the muscle, the **insertion**, is attached to the bone that moves when the muscle contracts. For example, contraction of the biceps muscle in the upper arm causes the forearm (the insertion of the muscle) to move toward the shoulder (the origin of the muscle).

Muscles that cause the same action at a joint are **synergists**. For example, the various muscles of the quadriceps group in humans are synergists: they all act to extend the knee joint. Muscles that produce opposing actions are **antagonists**. For example, muscles that flex a joint are antagonist to muscles that extend that joint (figure 50.6a). In humans, when the hamstring muscles contract, they cause flexion of the knee joint (figure 50.6b). Therefore, the quadriceps and hamstrings are antagonists to each other. In general, the muscles that antagonize a given movement are relaxed when that movement is performed. Thus, when the hamstrings flex the knee joint, the quadriceps muscles relax.

Isotonic and Isometric Contractions

In order for muscle fibers to shorten when they contract, they must generate a force that is greater than the opposing forces that act to prevent movement of the muscle’s insertion. When you lift a weight by contracting muscles in your biceps, for example, the force produced by the muscle is greater than the force of gravity on the object you are lifting. In this case, the muscle and all of its fibers shorten in length. This type of contraction is referred to as **isotonic contraction**, because the force of contraction remains relatively constant throughout the shortening process (*iso* = same; *tonic* = strength).

Preceding an isotonic contraction, the muscle begins to contract but the tension is absorbed by the tendons and other elastic tissue associated with the muscle. The muscle does not change in length and so this is called **isometric** (literally, “same length”) **contraction**. Isometric contractions occur as a phase of normal muscle contraction but also exist to provide tautness and stability to the body.

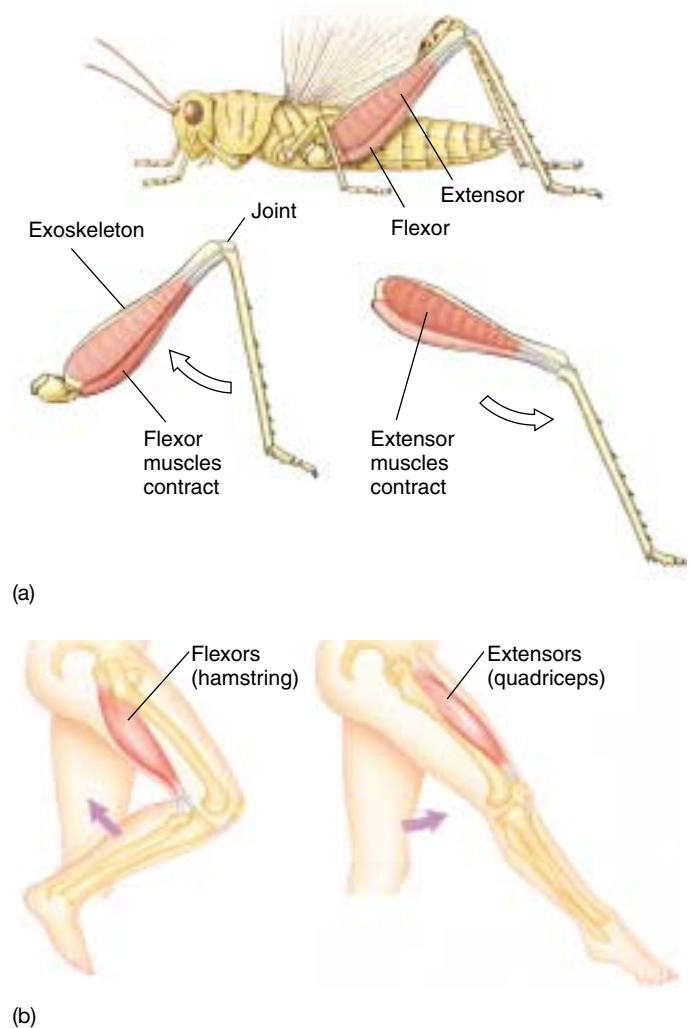


FIGURE 50.6

Flexor and extensor muscles of the leg. (a) Antagonistic muscles control the movement of an animal with an exoskeleton, such as the jumping of a grasshopper. When the smaller flexor tibia muscle contracts it pulls the lower leg in toward the upper leg. Contraction of the extensor tibia muscles straightens out the leg and sends the insect into the air. (b) Similarly, antagonistic muscles can act on an endoskeleton. In humans, the hamstrings, a group of three muscles, produce flexion of the knee joint, whereas the quadriceps, a group of four muscles, produce extension.

Synergistic muscles have the same action, whereas antagonistic muscles have opposite actions. Both muscle groups are involved in locomotion. Isotonic contractions involve the shortening of muscle, while isometric contractions do not alter the length of the muscle.

50.3 Muscle contraction powers animal locomotion.

The Sliding Filament Mechanism of Contraction

Each skeletal muscle contains numerous **muscle fibers**, as described in chapter 49. Each muscle fiber encloses a bundle of 4 to 20 elongated structures called **myofibrils**. Each myofibril, in turn, is composed of **thick** and **thin myofilaments** (figure 50.7). The muscle fiber is striated (has cross-striping) because its myofibrils are striated, with dark and light bands. The banding pattern results from the organization of the myofilaments within the myofibril. The thick myofilaments are stacked together to produce the dark

bands, called *A bands*; the thin filaments alone are found in the light bands, or *I bands*.

Each I band in a myofibril is divided in half by a disc of protein, called a *Z line* because of its appearance in electron micrographs. The thin filaments are anchored to these discs of proteins that form the Z lines. If you look at an electron micrograph of a myofibril (figure 50.8), you will see that the structure of the myofibril repeats from Z line to Z line. This repeating structure, called a **sarcomere**, is the smallest subunit of muscle contraction.

The thin filaments stick partway into the stack of thick filaments on each side of an A band, but, in a resting

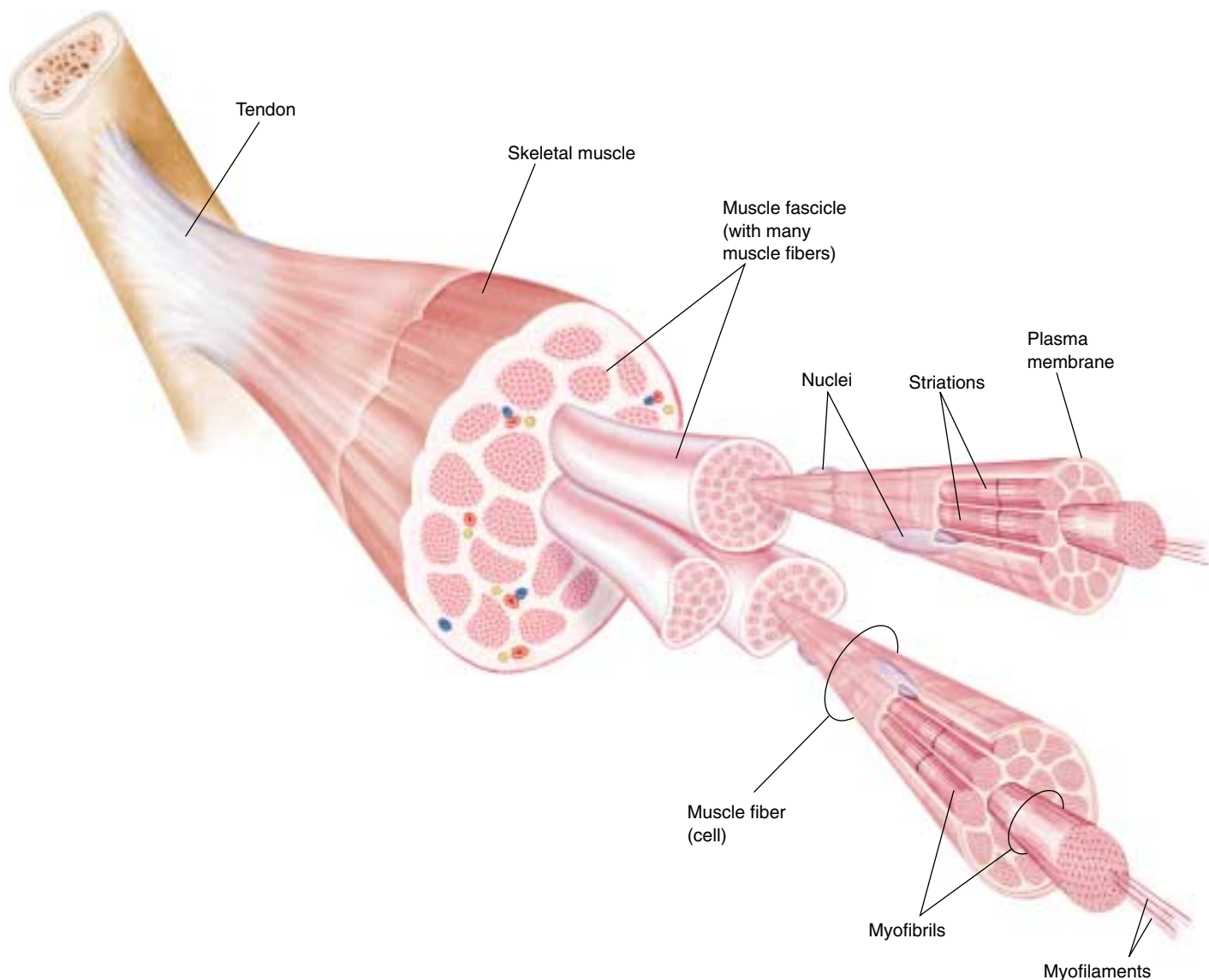


FIGURE 50.7

The organization of skeletal muscle. Each muscle is composed of many fascicles, which are bundles of muscle cells, or fibers. Each fiber is composed of many myofibrils, which are each, in turn, composed of myofilaments.

muscle, do not project all the way to the center of the A band. As a result, the center of an A band (called an *H band*) is lighter than each side, with its interdigitating thick and thin filaments. This appearance of the sarcomeres changes when the muscle contracts.

A muscle contracts and shortens because its myofibrils contract and shorten. When this occurs, the myofilaments do *not* shorten; instead, the thin filaments slide deeper into the A bands (figure 50.9). This makes the H bands narrower until, at maximal shortening, they disappear entirely. It also makes the I bands narrower, because the dark A bands are brought closer together. This is the **sliding filament mechanism** of contraction.

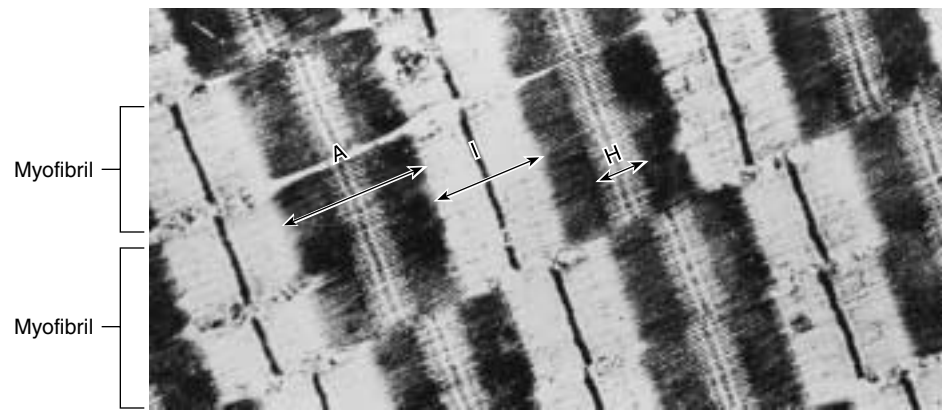
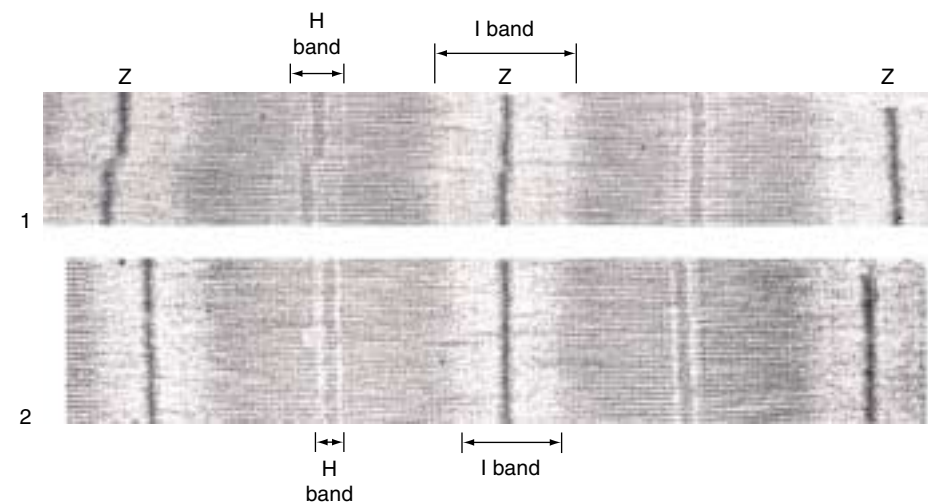
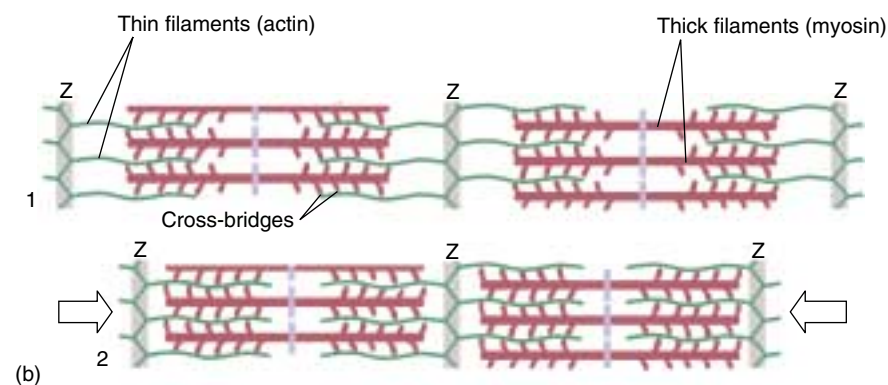


FIGURE 50.8

An electron micrograph of a skeletal muscle fiber. The Z lines that serve as the borders of the sarcomeres are clearly seen within each myofibril. The thick filaments comprise the A bands; the thin filaments are within the I bands and stick partway into the A bands, overlapping with the thick filaments. There is no overlap of thick and thin filaments at the central region of an A band, which is therefore lighter in appearance. This is the H band.



(a)



(b)

FIGURE 50.9

Electron micrograph (a) and diagram (b) of the sliding filament mechanism of contraction. As the thin filaments slide deeper into the centers of the sarcomeres, the Z lines are brought closer together. (1) Relaxed muscle; (2) partially contracted muscle.

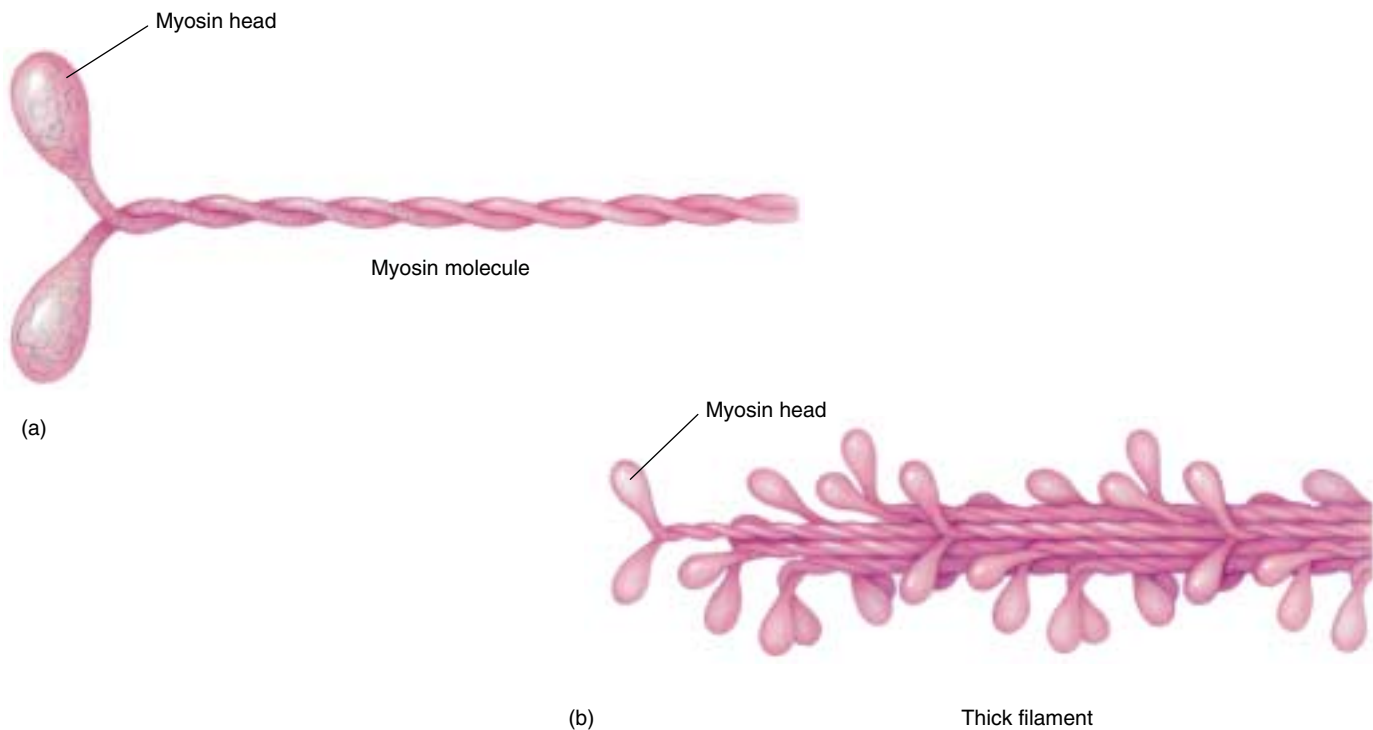


FIGURE 50.10

Thick filaments are composed of myosin. (a) Each myosin molecule consists of two polypeptide chains wrapped around each other; at the end of each chain is a globular region referred to as the “head.” (b) Thick filaments consist of myosin molecules combined into bundles from which the heads protrude at regular intervals.

Electron micrographs reveal **cross-bridges** that extend from the thick to the thin filaments, suggesting a mechanism that might cause the filaments to slide. To understand how this is accomplished, we have to examine the thick and thin filaments at a molecular level. Biochemical studies show that each thick filament is composed of many **myosin** proteins packed together, and every myosin molecule has a “head” region that protrudes from the thick filaments (figure 50.10). These myosin heads form the cross-bridges seen in electron micrographs. Biochemical studies also show that each thin filament consists primarily of many globular **actin** proteins twisted into a double helix (figure 50.11). Therefore, if we were able to see a sarcomere at a molecular level, it would have the structure depicted in figure 50.12a.

Before the myosin heads bind to the actin of the thin filaments, they act as ATPase enzymes, splitting ATP into ADP and P_i . This activates the heads, “cocking” them so that they can bind to actin and form cross-bridges. Once a myosin head binds to actin, it undergoes a conformational (shape) change, pulling the thin filament toward the cen-

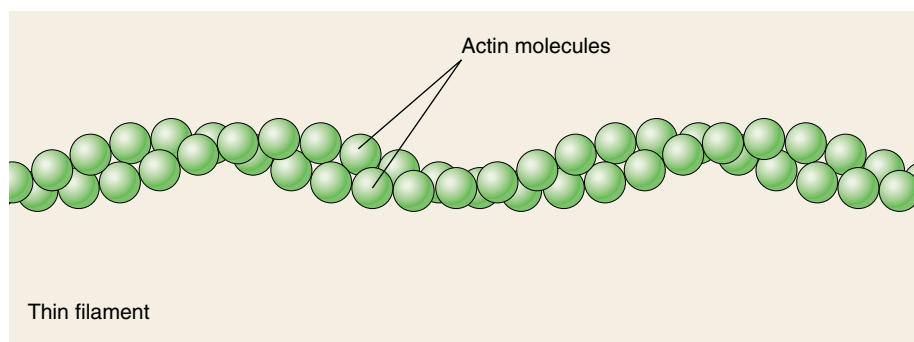


FIGURE 50.11

Thin filaments are composed of globular actin proteins. Two rows of actin proteins are twisted together in a helix to produce the thin filaments.

ter of the sarcomere (figure 50.12b) in a *power stroke*. At the end of the power stroke, the myosin head binds to a new molecule of ATP. This allows the head to detach from actin and continue the **cross-bridge cycle** (figure 50.13), which repeats as long as the muscle is stimulated to contract.

In death, the cell can no longer produce ATP and therefore the cross-bridges cannot be broken—this causes the muscle stiffness of death, or *rigor mortis*. A living cell, however, always has enough ATP to allow the

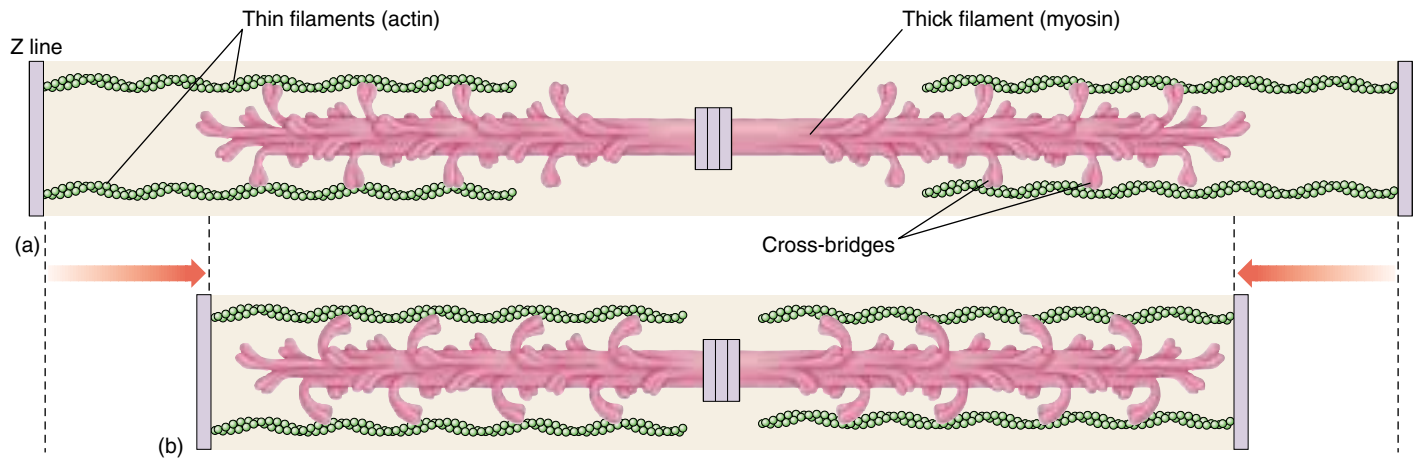


FIGURE 50.12

The interaction of thick and thin filaments in striated muscle sarcomeres. The heads on the two ends of the thick filaments are oriented in opposite directions (a), so that the cross-bridges pull the thin filaments and the Z lines on each side of the sarcomere toward the center. (b) This sliding of the filaments produces muscle contraction.

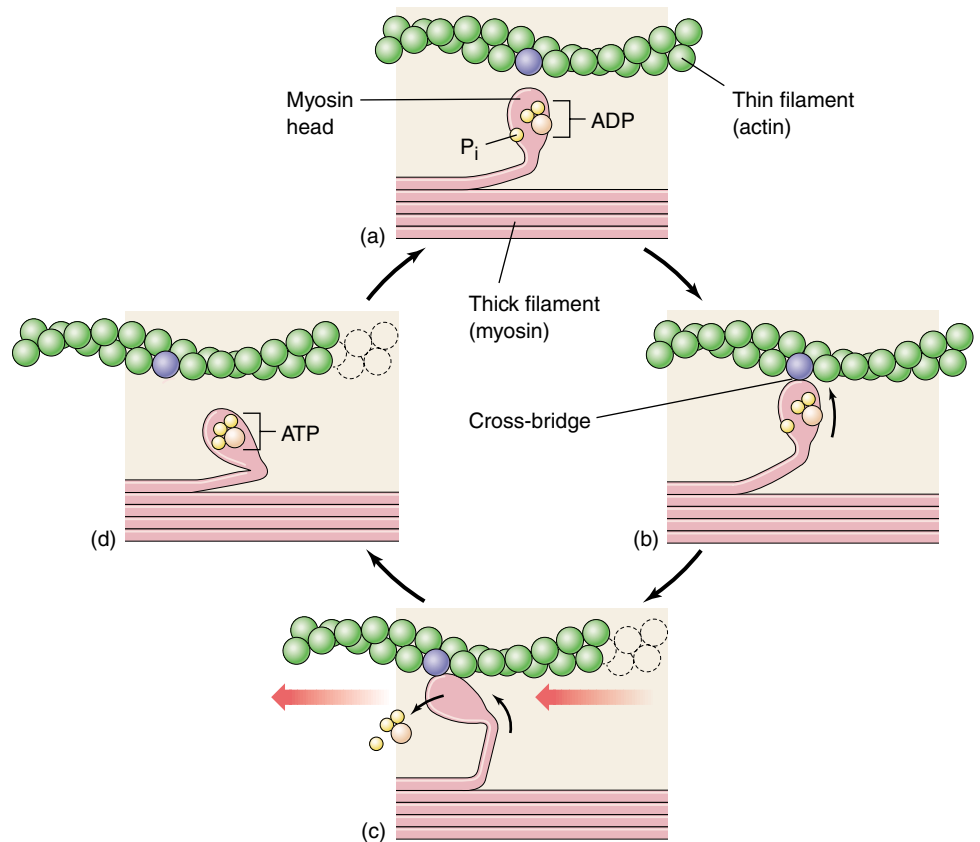


FIGURE 50.13

The cross-bridge cycle in muscle contraction. (a) With ADP and P_i attached to the myosin head, (b) the head is in a conformation that can bind to actin and form a cross-bridge. (c) Binding causes the myosin head to assume a more bent conformation, moving the thin filament along the thick filament (to the left in this diagram) and releasing ADP and P_i . (d) Binding of ATP to the head detaches the cross-bridge; cleavage of ATP into ADP and P_i puts the head into its original conformation, allowing the cycle to begin again.

myosin heads to detach from actin. How, then, is the cross-bridge cycle arrested so that the muscle can relax? The regulation of muscle contraction and relaxation requires additional factors that we will discuss in the next section.

Thick and thin filaments are arranged to form sarcomeres within the myofibrils. Myosin proteins comprise the thick filaments, and the heads of the myosin form cross-bridges with the actin proteins of the thin filaments. ATP provides the energy for the cross-bridge cycle and muscle contraction.

The Control of Muscle Contraction

The Role of Ca^{++} in Contraction

When a muscle is relaxed, its myosin heads are “cocked” and ready, through the splitting of ATP, but are unable to bind to actin. This is because the attachment sites for the myosin heads on the actin are physically blocked by another protein, known as **tropomyosin**, in the thin filaments. Cross-bridges therefore cannot form in the relaxed muscle, and the filaments cannot slide.

In order to contract a muscle, the tropomyosin must be moved out of the way so that the myosin heads can bind to actin. This requires the function of **troponin**, a regulatory protein that binds to the tropomyosin. The troponin and tropomyosin form a complex that is regulated by the calcium ion (Ca^{++}) concentration of the muscle cell cytoplasm.

When the Ca^{++} concentration of the muscle cell cytoplasm is low, tropomyosin inhibits cross-bridge formation and the muscle is relaxed (figure 50.14). When the Ca^{++} concentration is raised, Ca^{++} binds to troponin. This causes the troponin-tropomyosin complex to be shifted away from the attachment sites for the myosin heads on the actin. Cross-bridges can thus form, undergo power strokes, and produce muscle contraction.

Where does the Ca^{++} come from? Muscle fibers store Ca^{++} in a modified endoplasmic reticulum called a sarcoplasmic reticulum, or SR (figure 50.15). When a muscle fiber is stimulated to contract, an electrical impulse travels into the muscle fiber down invaginations called the **transverse tubules (T tubules)**. This triggers the release of Ca^{++} from the SR. Ca^{++} then diffuses into the myofibrils, where it binds to troponin and causes contraction. The contraction of muscles is regulated by nerve activity, and so nerves must influence the distribution of Ca^{++} in the muscle fiber.

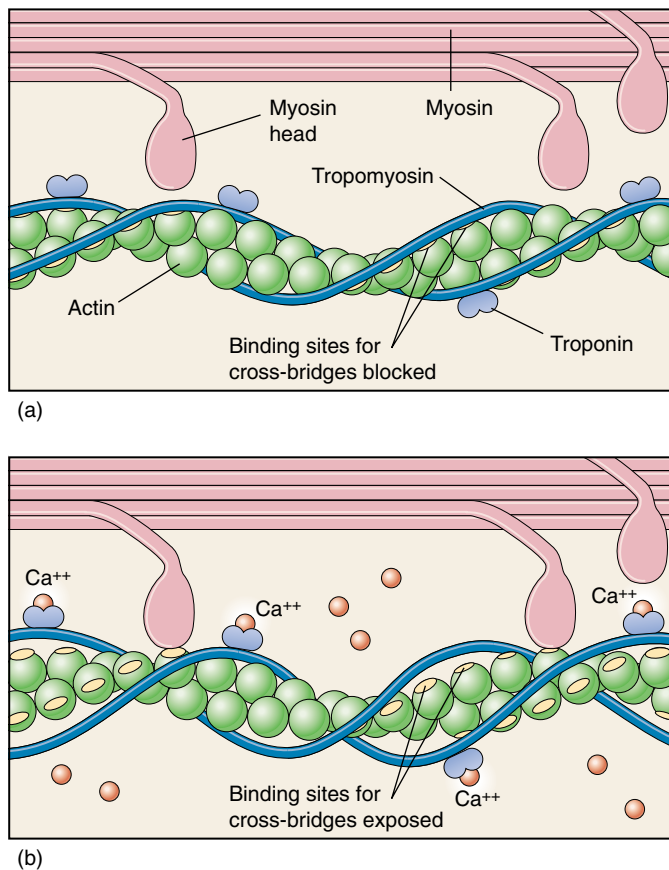


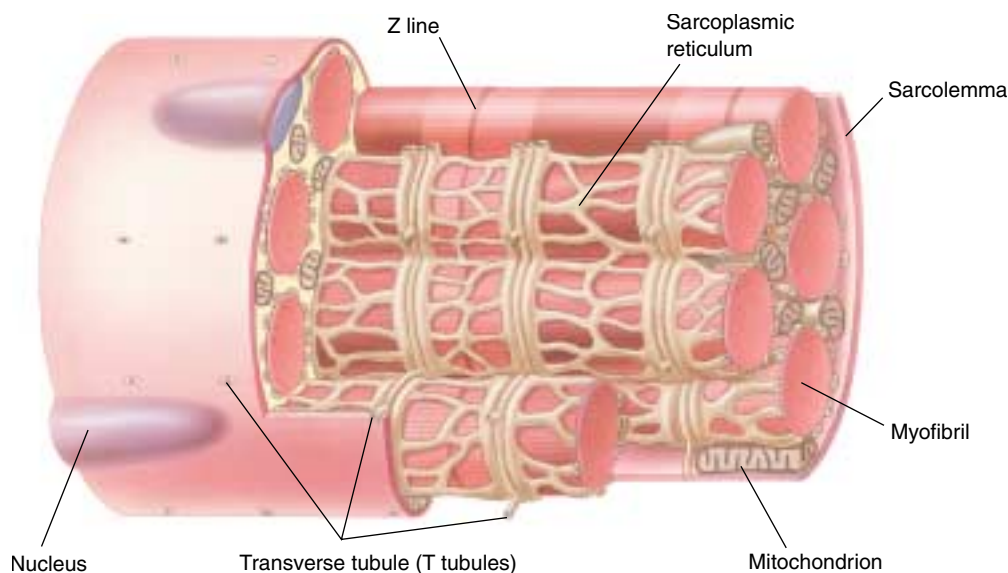
FIGURE 50.14

How calcium controls striated muscle contraction. (a) When the muscle is at rest, a long filament of the protein tropomyosin blocks the myosin-binding sites on the actin molecule. Because myosin is unable to form cross-bridges with actin at these sites, muscle contraction cannot occur. (b) When Ca^{++} binds to another protein, troponin, the Ca^{++} -troponin complex displaces tropomyosin and exposes the myosin-binding sites on actin, permitting cross-bridges to form and contraction to occur.

FIGURE 50.15

The relationship between the myofibrils, transverse tubules, and sarcoplasmic reticulum.

Impulses travel down the axon of a motor neuron that synapses with a muscle fiber. The impulses are conducted along the transverse tubules and stimulate the release of Ca^{++} from the sarcoplasmic reticulum into the cytoplasm. Ca^{++} diffuses toward the myofibrils and causes contraction.



Nerves Stimulate Contraction

Muscles are stimulated to contract by motor neurons. The particular motor neurons that stimulate skeletal muscles, as opposed to cardiac and smooth muscles, are called *somatic motor neurons*. The axon (see figure 49.12) of a somatic motor neuron extends from the neuron cell body and branches to make functional connections, or *synapses*, with a number of muscle fibers. (Synapses are discussed in more detail in chapter 54.) One axon can stimulate many muscle fibers, and in some animals a muscle fiber may be innervated by more than one motor neuron. However, in humans each muscle fiber only has a single synapse with a branch of one axon.

When a somatic motor neuron produces electrochemical impulses, it stimulates contraction of the muscle fibers it innervates (makes synapses with) through the following events:

1. The motor neuron, at its synapse with the muscle fibers, releases a chemical known as a *neurotransmitter*. The specific neurotransmitter released by somatic motor neurons is **acetylcholine (ACh)**. ACh acts on the muscle fiber membrane to stimulate the muscle fiber to produce its own electrochemical impulses.
2. The impulses spread along the membrane of the muscle fiber and are carried into the muscle fibers through the T tubules.
3. The T tubules conduct the impulses toward the sarcoplasmic reticulum, which then release Ca^{++} . As described earlier, the Ca^{++} binds to troponin, which exposes the cross-bridge binding sites on the actin myofilaments, stimulating muscle contraction.

When impulses from the nerve stop, the nerve stops releasing ACh. This stops the production of impulses in the muscle fiber. When the T tubules no longer produce impulses, Ca^{++} is brought back into the SR by active transport. Troponin is no longer bound to Ca^{++} , so tropomyosin returns to its inhibitory position, allowing the muscle to relax.

The involvement of Ca^{++} in muscle contraction is called, **excitation-contraction coupling** because it is the release of Ca^{++} that links the excitation of the muscle fiber by the motor neuron to the contraction of the muscle.

Motor Units and Recruitment

A single muscle fiber responds in an all-or-none fashion to stimulation. The response of an entire muscle depends upon the number of individual fibers involved. The set of muscle fibers innervated by all axonal branches of a given motor neuron is defined as a **motor unit** (figure 50.16). Every time the motor neuron produces impulses, all muscle fibers in that motor unit contract together. The division of the muscle into motor units allows the muscle's strength of contraction to be finely graded, a requirement

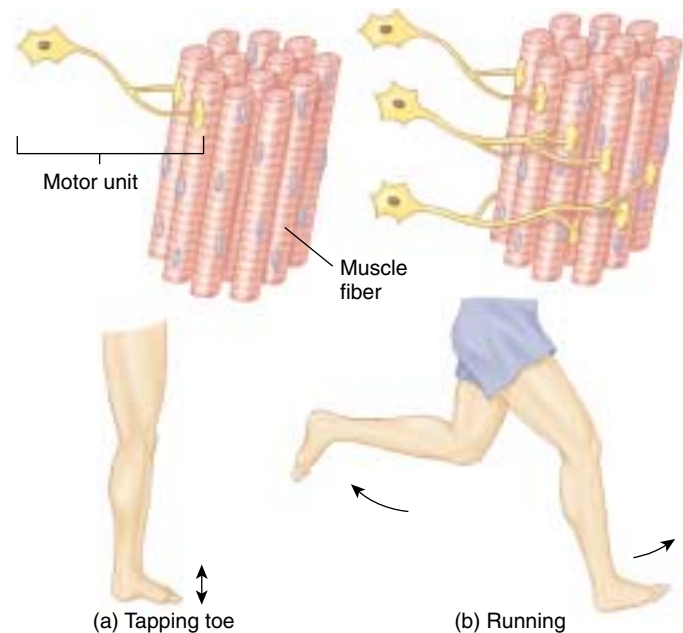


FIGURE 50.16

The number and size of motor units. (a) Weak, precise muscle contractions use smaller and fewer motor units. (b) Larger and stronger movements require additional motor units that are larger.

for coordinated movements of the skeleton. Muscles that require a finer degree of control have smaller motor units (fewer muscle fibers per neuron) than muscles that require less precise control but must exert more force. For example, there are only a few muscle fibers per motor neuron in the muscles that move the eyes, while there are several hundred per motor neuron in the large muscles of the legs.

Most muscles contain motor units in a variety of sizes, which can be selectively activated by the nervous system. The weakest contractions of a muscle involve the activation of a few small motor units. If a slightly stronger contraction is necessary, additional small motor units are also activated. The initial increments to the total force generated by the muscle are therefore relatively small. As ever greater forces are required, more and larger motor units are brought into action, and the force increments become larger. The nervous system's use of increased numbers and sizes of motor units to produce a stronger contraction is termed **recruitment**.

The cross-bridges are prevented from binding to actin by tropomyosin in a relaxed muscle. In order for a muscle to contract, Ca^{++} must be released from the sarcoplasmic reticulum, where it is stored, so that it can bind to troponin and cause the tropomyosin to shift its position in the thin filaments. Muscle contraction is stimulated by neurons. Varying sizes and numbers of motor units are used to produce different types of muscle contractions.

Types of Muscle Fibers

Muscle Fiber Twitches

An isolated skeletal muscle can be studied by stimulating it artificially with electric shocks. If a muscle is stimulated with a single electric shock, it will quickly contract and relax in a response called a **twitch**. Increasing the stimulus voltage increases the strength of the twitch up to a maximum. If a second electric shock is delivered immediately after the first, it will produce a second twitch that may partially “ride piggyback” on the first. This cumulative response is called **summation** (figure 50.17).

If the stimulator is set to deliver an increasing frequency of electric shocks automatically, the relaxation time between successive twitches will get shorter and shorter, as the strength of contraction increases. Finally, at a particular frequency of stimulation, there is no visible relaxation between successive twitches. Contraction is smooth and sustained, as it is during normal muscle contraction in the body. This smooth, sustained contraction is called **tetanus**. (The term *tetanus* should not be confused with the disease of the same name, which is accompanied by a painful state of muscle contracture, or *tetany*.)

Skeletal muscle fibers can be divided on the basis of their contraction speed into **slow-twitch**, or **type I, fibers**, and **fast-twitch**, or **type II, fibers**. The muscles that move the eyes, for example, have a high proportion of fast-twitch fibers and reach maximum tension in about 7.3 milliseconds; the soleus muscle in the leg, by contrast, has a high proportion of slow-twitch fibers and requires about 100 milliseconds to reach maximum tension (figure 50.18).

Muscles like the soleus must be able to sustain a contraction for a long period of time without fatigue. The resistance to fatigue demonstrated by these muscles is aided by other characteristics of slow-twitch (type I) fibers that endow them with a high capacity for aerobic respiration. Slow-twitch fibers have a rich capillary supply, numerous mitochondria and aerobic respiratory enzymes, and a high concentration of **myoglobin** pigment. Myoglobin is a red pigment, similar to the hemoglobin in red blood cells, but its higher affinity for oxygen improves the delivery of oxygen to the slow-twitch fibers. Because of their high myoglobin content, slow-twitch fibers are also called *red fibers*.

The thicker, fast-twitch (type II) fibers have fewer capillaries and mitochondria than slow-twitch fibers and not as much myoglobin; hence, these fibers are also called *white fibers*. Fast-twitch fibers are adapted to respire anaerobically by using a large store of glycogen and high concentrations of glycolytic enzymes. Fast-twitch fibers are adapted for the rapid generation of power and can grow thicker and stronger in response to weight training. The “dark meat” and “white meat” found in meat such as chicken and turkey consists of muscles with primarily red and white fibers, respectively.

In addition to the type I (slow-twitch) and type II (fast-twitch) fibers, human muscles also have an intermediate form of fibers that are fast-twitch but also have a high oxidative capacity, and so are more resistant to fatigue. Endurance training increases the proportion of these fibers in muscles.

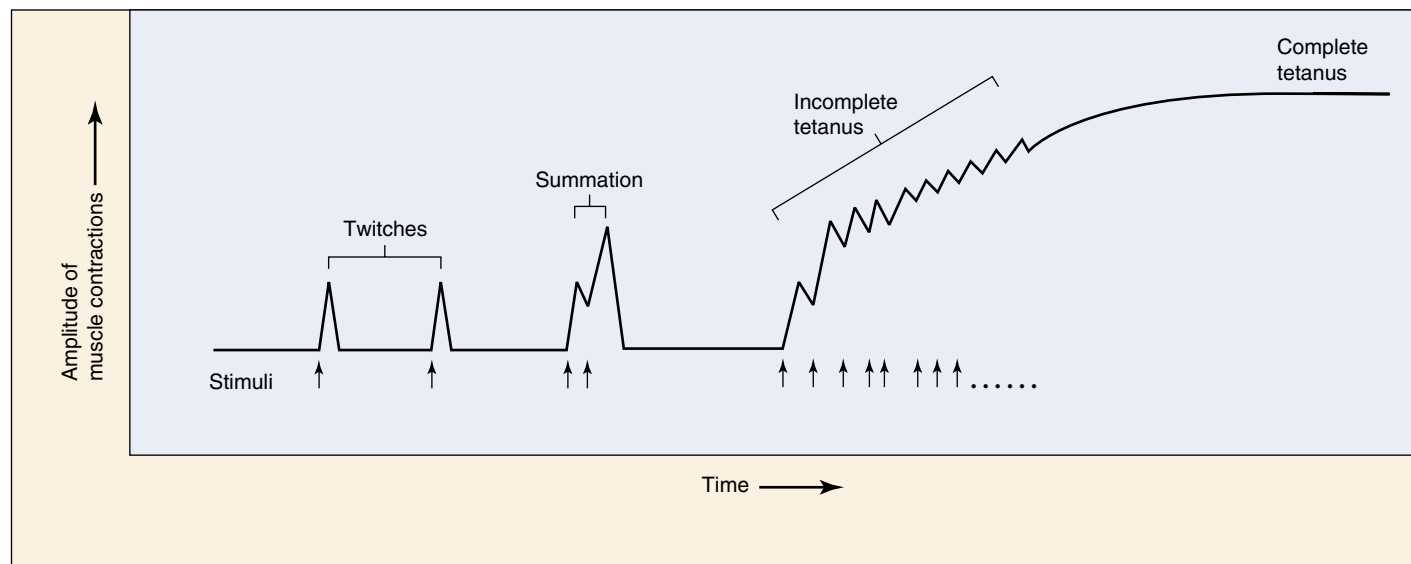
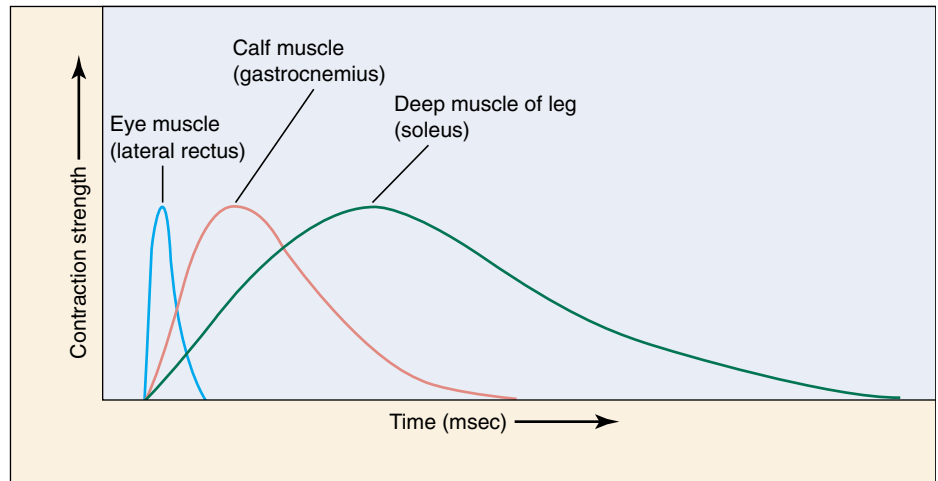


FIGURE 50.17

Muscle twitches summate to produce a sustained, tetanized contraction. This pattern is produced when the muscle is stimulated electrically or naturally by neurons. Tetanus, a smooth, sustained contraction, is the normal type of muscle contraction in the body.

FIGURE 50.18

Skeletal muscles have different proportions of fast-twitch and slow-twitch fibers. The muscles that move the eye contain mostly fast-twitch fibers, whereas the deep muscle of the leg (the soleus) contains mostly slow-twitch fibers. The calf muscle (gastrocnemius) is intermediate in its composition.



Muscle Metabolism during Rest and Exercise

Skeletal muscles at rest obtain most of their energy from the aerobic respiration of fatty acids. During exercise, muscle glycogen and blood glucose are also used as energy sources. The energy obtained by cell respiration is used to make ATP, which is needed for (1) the movement of the cross-bridges during muscle contraction and (2) the pumping of Ca^{++} into the sarcoplasmic reticulum for muscle relaxation. ATP can be obtained by skeletal muscles quickly by combining ADP with phosphate derived from creatine phosphate. This compound was produced previously in the resting muscle by combining creatine with phosphate derived from the ATP generated in cell respiration.

Skeletal muscles respire anaerobically for the first 45 to 90 seconds of moderate-to-heavy exercise, because the cardiopulmonary system requires this amount of time to sufficiently increase the oxygen supply to the exercising muscles. If exercise is moderate, aerobic respiration contributes the major portion of the skeletal muscle energy requirements following the first 2 minutes of exercise.

Whether exercise is light, moderate, or intense for a given person depends upon that person's maximal capacity for aerobic exercise. The maximum rate of oxygen consumption in the body (by aerobic respiration) is called the maximal oxygen uptake, or the aerobic capacity. The intensity of exercise can also be defined by the lactate threshold. This is the percentage of the maximal oxygen uptake at which a significant rise in blood lactate levels occurs as a result of anaerobic respiration. For average, healthy people, for example, a significant amount of blood lactate appears when exercise is performed at about 50 to 70% of the maximal oxygen uptake.

Muscle Fatigue and Physical Training

Muscle fatigue refers to the use-dependant decrease in the ability of a muscle to generate force. The reasons for fa-

tigue are not entirely understood. In most cases, however, muscle fatigue is correlated with the production of lactic acid by the exercising muscles. Lactic acid is produced by the anaerobic respiration of glucose, and glucose is obtained from muscle glycogen and from the blood. Lactate production and muscle fatigue are therefore also related to the depletion of muscle glycogen.

Because the depletion of muscle glycogen places a limit on exercise, any adaptation that spares muscle glycogen will improve physical endurance. Trained athletes have an increased proportion of energy derived from the aerobic respiration of fatty acids, resulting in a slower depletion of their muscle glycogen reserve. The greater the level of physical training, the higher the proportion of energy derived from the aerobic respiration of fatty acids. Because the aerobic capacity of endurance-trained athletes is higher than that of untrained people, athletes can perform more exercise before lactic acid production and glycogen depletion cause muscle fatigue.

Endurance training does not increase muscle size. Muscle enlargement is produced only by frequent periods of high-intensity exercise in which muscles work against high resistance, as in weight lifting. As a result of resistance training, type II (fast-twitch) muscle fibers become thicker as a result of the increased size and number of their myofibrils. Weight training, therefore, causes skeletal muscles to grow by **hypertrophy** (increased cell size) rather than by cell division and an increased number of cells.

Muscles contract through summation of the contractions of their fibers, producing tension that may result in shortening of the muscle. Slow-twitch skeletal muscle fibers are adapted for aerobic respiration and are slower to fatigue than fast-twitch fibers, which are more adapted for the rapid generation of power.

Comparing Cardiac and Smooth Muscles

Cardiac and smooth muscle are similar in that both are found within internal organs and both are generally not under conscious control. Cardiac muscle, however, is like skeletal muscle in that it is striated and contracts by means of a sliding filament mechanism. Smooth muscle (as its name implies) is not striated. Smooth muscle does contain actin and myosin filaments, but they are arranged less regularly within the cell.

Cardiac Muscle

Cardiac muscle in the vertebrate heart is composed of striated muscle cells that are arranged differently from the fibers in a skeletal muscle. Instead of the long, multinucleate cells that form skeletal muscle, cardiac muscle is composed of shorter, branched cells, each with its own nucleus, that interconnect with one another at intercalated discs (figure 50.19). Intercalated discs are regions where the membranes of two cells fuse together, and the fused membranes are pierced by **gap junctions** (chapter 7). The gap junctions permit the diffusion of ions, and thus the spread of electric excitation, from one cell to the next. The mass of interconnected cardiac muscle cells forms a single, functioning unit called a *myocardium*. Electric impulses begin spontaneously in a specific region of the myocardium known as the *pacemaker*. These impulses are *not* initiated by impulses in motor neurons, as they are in skeletal muscle, but rather are produced by the cardiac muscle cells themselves. From the pacemaker, the impulses spread throughout the myocardium via gap junctions, causing contraction.

The heart has two myocardia, one that receives blood from the body and one that ejects blood into the body. Because all of the cells in a myocardium are stimulated as a unit, cardiac muscle cannot produce summated contractions or tetanus. This would interfere with the alternation between contraction and relaxation that is necessary for pumping.

Smooth Muscle

Smooth muscle surrounds hollow internal organs, including the stomach, intestines, bladder, and uterus, as well as all blood vessels except capillaries. Smooth muscle cells are long and spindle-shaped, and each contains a single nucleus. They also contain actin and myosin, but these contractile proteins are not organized into sarcomeres. Parallel arrangements of thick and thin filaments cross diagonally

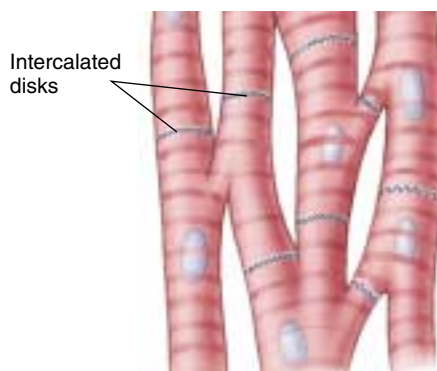


FIGURE 50.19

Cardiac muscle. Cells are organized into long branching chains that interconnect, forming a lattice; neighboring cells are linked by structures called intercalated discs.

from one side of the cell to the other. The thick filaments are attached either to structures called dense bodies, the functional equivalents of Z lines, or to the plasma membrane. Most smooth muscle cells have 10 to 15 thin filaments per thick filament, compared to 3 per thick filament in striated muscle fibers.

Smooth muscle cells do not have a sarcoplasmic reticulum; during a contraction, Ca^{++} enters from the extracellular fluid. In the cytoplasm, Ca^{++} binds to calmodulin, a protein that is structurally similar to troponin. The Ca^{++} -calmodulin complex activates an enzyme that phosphorylates (adds a phosphate group to) the myosin heads. Unlike the case with striated muscles, this phosphorylation is required for the

myosin heads to form cross-bridges with actin.

This mechanism allows gradations in the strength of contraction in a smooth muscle cell, increasing contraction strength as more Ca^{++} enters the cytoplasm. Heart patients sometimes take drugs that block Ca^{++} entry into smooth muscle cells, reducing the cells' ability to contract. This treatment causes vascular smooth muscle to relax, dilating the blood vessels and reducing the amount of work the heart must do to pump blood through them.

In some smooth muscle tissues, the cells contract only when they are stimulated by the nervous system. These muscles line the walls of many blood vessels and make up the iris of the eye. Other smooth muscle tissues, like those in the wall of the gut, contain cells that produce electric impulses spontaneously. These impulses spread to adjoining cells through gap junctions, leading to a slow, steady contraction of the tissue.

Neither skeletal nor cardiac muscle can be greatly stretched because if the thick and thin filaments no longer overlap in the sarcomere, cross-bridges cannot form. Unlike these striated muscles, smooth muscle can contract even when it is greatly stretched. If one considers the degree to which some internal organs may be stretched—a uterus during pregnancy, for example—it is no wonder that these organs contain smooth muscle instead of striated muscle.

Cardiac muscle cells interconnect physically and electrically to form a single, functioning unit called a **myocardium**, which produces its own impulses at a **pacemaker region**. Smooth muscles lack the organization of myofilaments into sarcomeres and lack sarcoplasmic reticulum but contraction still occurs as myofilaments slide past one another by use of cross-bridges.

Modes of Animal Locomotion

Animals are unique among multicellular organisms in their ability to actively move from one place to another. Locomotion requires both a propulsive mechanism and a control mechanism. Animals employ a wide variety of propulsive mechanisms, most involving contracting muscles to generate the necessary force. The quantity, quality, and position of contractions are initiated and coordinated by the nervous system. In large animals, active locomotion is almost always produced by appendages that oscillate—*appendicular locomotion*—or by bodies that undulate, pulse, or undergo peristaltic waves—*axial locomotion*.

While animal locomotion occurs in many different forms, the general principles remain much the same in all groups. The physical restraints to movement—gravity and frictional drag—are the same in every environment, differing only in degree. You can conveniently divide the environments through which animals move into three types, each involving its own forms of locomotion: water, land, and air.

Locomotion in Water

Many aquatic and marine invertebrates move along the bottom using the same form of locomotion employed by terrestrial animals moving over the land surface. Flatworms employ ciliary activity to brush themselves along, roundworms a peristaltic slither, leeches a contract-anchor-extend creeping. Crabs walk using limbs to pull themselves along; mollusks use a muscular foot, while starfish use unique tube feet to do the same thing.

Moving directly through the water, or swimming, presents quite a different challenge. Water's buoyancy reduces the influence of gravity. The primary force retarding forward movement is frictional drag, so body shape is important in reducing the friction and turbulence produced by swimming through the water.

Some marine invertebrates swim using hydraulic propulsion. Scallops clap their shells together forcefully, while squids and octopuses squirt water like a marine jet. All aquatic and marine vertebrates, however, swim.

Swimming involves using the body or its appendages to push against the water. An eel swims by sinuous undulations of its whole body (figure 50.20a). The undulating body waves of eel-like swimming are created by waves of muscle contraction alternating between the left and right axial musculature. As each body segment in turn pushes against the water, the moving wave forces the eel forward.

Fish, reptiles, and aquatic amphibians swim in a way similar to eels, but only undulate the posterior (back) portion of the body (figure 50.20b) and sometimes only the caudal (rear) fin. This allows considerable specialization of the front end of the body, while sacrificing little propulsive force.

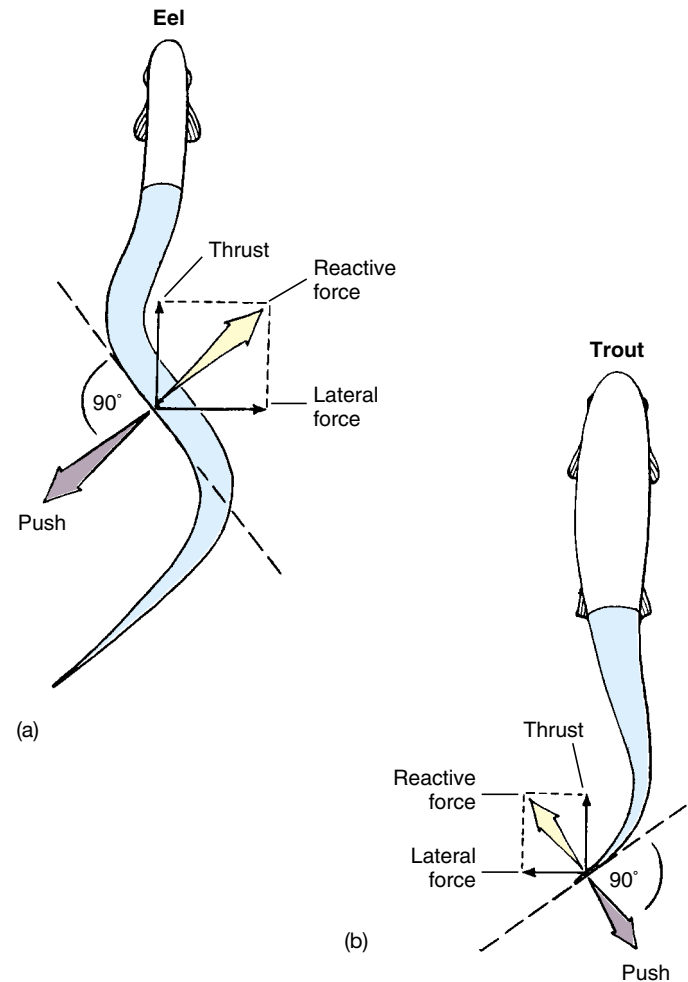


FIGURE 50.20

Movements of swimming fishes. (a) An eel pushes against the water with its whole body, (b) a trout only with its posterior half.

Whales also swim using undulating body waves, but unlike any of the fishes, the waves pass from top to bottom and not from side to side. The body musculature of eels and fish is highly segmental; that is, a muscle segment alternates with each vertebra. This arrangement permits the smooth passage of undulatory waves along the body. Whales are unable to produce lateral undulations because mammals do not have this arrangement.

Many tetrapod vertebrates swim, usually with appendicular locomotion. Most birds that swim, like ducks and geese, propel themselves through the water by pushing against it with their hind legs, which typically have webbed feet. Frogs, turtles, and most marine mammals also swim with their hind legs and have webbed feet. Tetrapod vertebrates that swim with their forelegs usually have these limbs modified as flippers, and pull themselves through the water. These include sea turtles, penguins, and fur seals. A few principally terrestrial tetrapod vertebrates, like polar bears and platypuses, swim with walking forelimbs not modified for swimming.



FIGURE 50.21

Animals that hop or leap use their rear legs to propel themselves through the air. The powerful leg muscles of this frog allow it to explode from a crouched position to a takeoff in about 100 milliseconds.

Locomotion on Land

The three great groups of terrestrial animals—mollusks, arthropods, and vertebrates—each move over land in different ways.

Mollusk locomotion is far less efficient than that of the other groups. Snails, slugs, and other terrestrial mollusks secrete a path of mucus that they glide along, pushing with a muscular foot.

Only vertebrates and arthropods (insects, spiders, and crustaceans) have developed a means of rapid surface locomotion. In both groups, the body is raised above the ground and moved forward by pushing against the ground with a series of jointed appendages, the legs.

Because legs must provide support as well as propulsion, it is important that the sequence of their movements not shove the body's center of gravity outside of the legs' zone of support. If they do, the animal loses its balance and falls. It is the necessity to maintain stability that determines the sequence of leg movements, which are similar in vertebrates and arthropods.

The apparent differences in the walking gaits of these two groups reflects the differences in leg number. Vertebrates are tetrapods (four limbs), while all arthropods have six or more limbs. Although having many legs increases stability during locomotion, they also appear to reduce the maximum speed that can be attained.

The basic walking pattern of all tetrapod vertebrates is left hind leg (LH), left foreleg (LF), right hindleg (RH), right foreleg (RF), and then the same sequence again and again. Unlike insects, vertebrates can begin to walk with any of the four legs, and not just the posterior pair. Both

arthropods and vertebrates achieve faster gaits by overlapping the leg movements of the left and right sides. For example, a horse can convert a walk to a trot, by moving diagonally opposite legs simultaneously.

The highest running speeds of tetrapod vertebrates, such as the gallop of a horse, are obtained with asymmetric gaits. When galloping, a horse is never supported by more than two legs, and occasionally is supported by none. This reduces friction against the ground to an absolute minimum, increasing speed. With their larger number of legs, arthropods cannot have these speedy asymmetric gaits, because the movements of the legs would interfere with each other.

Not all animals walk or run on land. Many insects, like grasshoppers, leap using strong rear legs to propel themselves through the air. Vertebrates such as kangaroos, rabbits, and frogs are also effective leapers (figure 50.21).

Many invertebrates use peristaltic motion to slide over the surface. Among vertebrates, this form of locomotion is exhibited by snakes and caecilians (legless amphibians). Most snakes employ serpentine locomotion, in which the body is thrown into a series of sinuous curves. The movements superficially resemble those of eel-like swimming, but the similarity is more apparent than real. Propulsion is not by a wave of contraction undulating the body, but by a simultaneous lateral thrust in all segments of the body in contact with the ground. To go forward, it is necessary that the strongest muscular thrust push against the ground opposite the direction of movement. Because of this, thrust tends to occur at the anterior (outside) end of the inward-curving side of the loop of the snake's body.

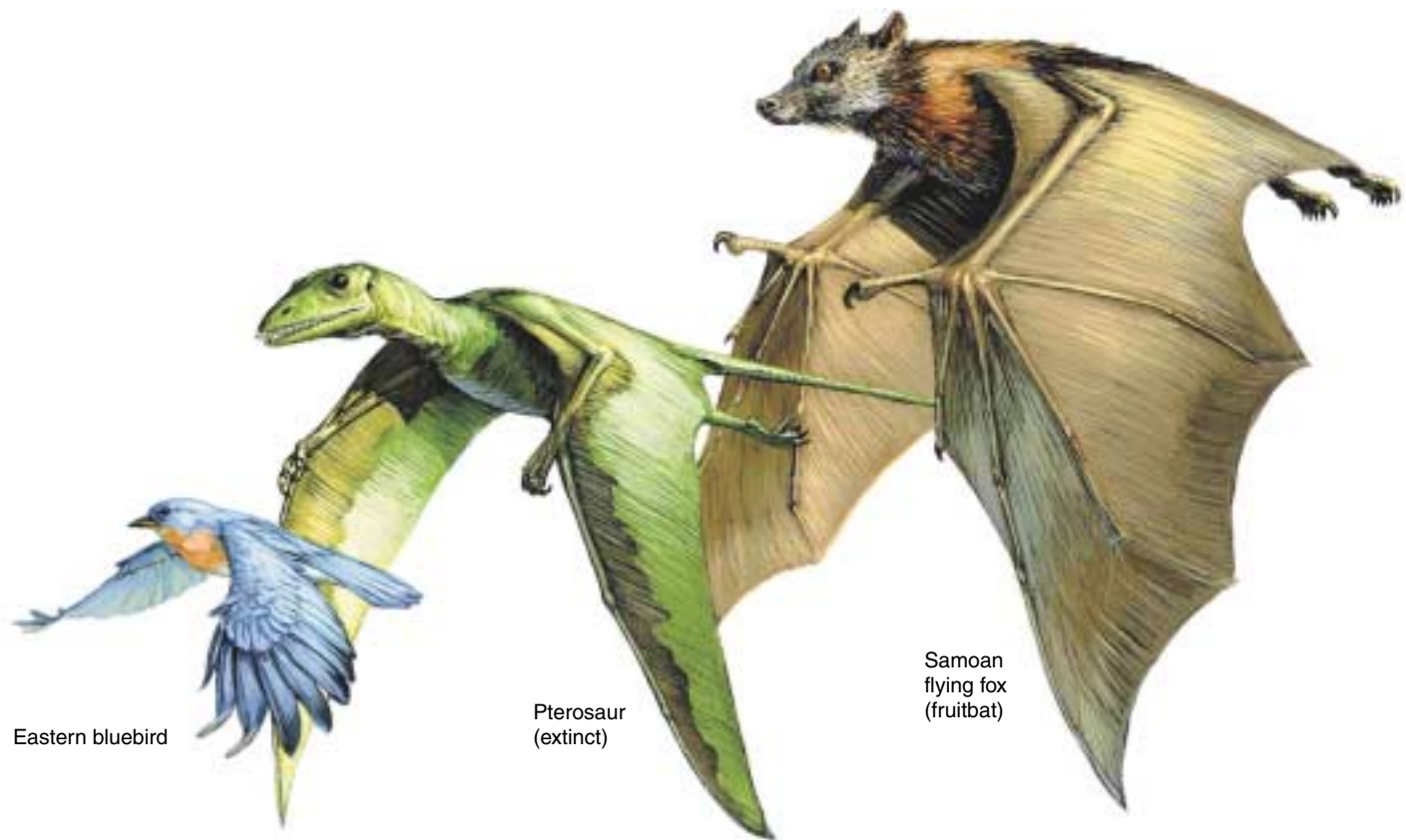


FIGURE 50.22

Flight has evolved three times among the vertebrates. These three very different vertebrates all have lightened bones and forelimbs transformed into wings.

Locomotion in Air

Flight has evolved among the animals four times: insects, pterosaurs (extinct flying reptiles), birds, and bats. In all four groups, active flying takes place in much the same way. Propulsion is achieved by pushing down against the air with wings. This provides enough lift to keep insects in the air. Vertebrates, being larger, need greater lift, obtaining it with wings that are convex in cross section. Because air must travel farther over the top surface, it moves faster, creating lift over the wing.

In birds and most insects, the raising and lowering of the wings is achieved by the alternate contraction of extensor muscles (elevators) and flexor muscles (depressors). Four insect orders (containing flies, mosquitoes, wasps, bees, and beetles), however, beat their wings at frequencies from 100 to more than 1000 times per second, faster than nerves can carry successive impulses! In these insects, the flight muscles are not attached to the wings at all but rather to the stiff wall of the thorax, which is distorted in and out by their contraction. The reason that these muscles can beat so fast is that the contraction of one set stretches the other, triggering its contraction in turn without waiting for the arrival of a nerve impulse.

Among vertebrates (figure 50.22), flight first evolved some 200 million years ago among flying reptiles called pterosaurs. A very successful and diverse group, pterosaurs ranged in size from individuals no bigger than sparrows to pterodons the size of a fighter plane. For much of this time, they shared the skies with birds, which most paleontologists believe evolved from feathered dinosaurs about 150 million years ago. How did they share their ecological world for 100 million years without competition driving one or the other from the skies? No one knows for sure. Perhaps these early birds were night fliers, while pterosaurs flew by day.

Such an arrangement for sharing resources is not as unlikely as it might at first appear. Bats, flying mammals which evolved after the pterosaurs disappeared with the dinosaurs, are night fliers. By flying at night bats are able to shop in a store with few other customers and a wealth of food: night-flying insects. It has proven to be a very successful approach. One-quarter of all mammal species are bats.

Locomotion in larger animals is almost always produced by appendages that push against the surroundings in some fashion, or by shoving the entire body forward by an undulation.



Summary

Questions

Media Resources

50.1 A skeletal system supports movement in animals.

- There are three types of skeleton: hydrostatic skeletons, exoskeletons, and endoskeletons.
- Bone is formed by the secretion of an organic matrix by osteoblasts; this organic matrix becomes calcified.

1. What are the two major components of the extracellular matrix in bone? What structural properties does each component have? How do the two components combine to make bone resistant to fracture?



• On Science Article: Running improperly



• Walking

50.2 Skeletal muscles contract to produce movements at joints.

- Freely movable joints surround the articulating bones with a synovial capsule filled with a lubricating fluid.
- Skeletal muscles can work together as synergists, or oppose each other as antagonists.

2. What are the three types of joints in a vertebrate skeleton? Give an example of where each type is found in the body.



• Bioethics case study: Sports and fitness

3. What is the difference between a skeletal muscle's origin and its insertion?

50.3 Muscle contraction powers animal locomotion.

- A muscle fiber contains numerous myofibrils, which consist of thick filaments composed of myosin and thin filaments of actin.
- There are small cross-bridges of myosin that extend out toward the actin; the cross-bridges are activated by the hydrolysis of ATP so that it can bind to actin and undergo a power stroke that causes the sliding of the myofilaments.
- When Ca^{++} binds to troponin, the tropomyosin shifts position in the thin filament, allowing the cross-bridges to bind to actin and undergo a power stroke.
- The release of Ca^{++} from the sarcoplasmic reticulum is stimulated by impulses in the muscle fiber produced by neural stimulation.
- Slow-twitch fibers are adapted for aerobic respiration and are resistant to fatigue; fast-twitch fibers can provide power quickly but produce lactic acid and fatigue quickly.
- Cardiac muscle cells have gap junctions that permit the spread of electric impulses from one cell to the next.
- Cardiac and smooth muscles are involuntary and regulated by autonomic nerves; the contractions are automatically produced in cardiac muscle and some smooth muscles.
- Animals have adapted modes of locomotion to three different environments: water, land, and air.

4. Of what proteins are thick and thin filaments composed?



• On Science Article: Climbing the walls

5. Describe the steps involved in the cross-bridge cycle. What functions does ATP perform in the cycle?



- Straited muscle contraction
- Muscle contraction action potential
- Detailed straited muscle
- Actin-myosin crossbridges

6. Describe the steps involved in excitation-contraction coupling. What functions do acetylcholine and Ca^{++} perform in this process?

7. How does a somatic motor neuron stimulate a muscle fiber to contract?



- Activity: Muscle contraction
- Muscle cell function
- Body musculature
- Head and neck muscles
- Trunk muscles
- Upper limb muscles
- Lower limb muscles
- Muscle characteristics

8. What is the difference between a muscle twitch and tetanus?

9. Why can't a myocardium produce a sustained contraction?

10. How does smooth muscle differ from skeletal muscle in terms of thick and thin filament organization, the role of Ca^{++} in contraction, and the effect of stretching on the muscle's ability to contract?

11. What do all modes of locomotion have in common?

51

Fueling Body Activities: Digestion

Concept Outline

51.1 Animals employ a digestive system to prepare food for assimilation by cells.

Types of Digestive Systems. Some invertebrates have a gastrovascular cavity, but vertebrates have a digestive tract that chemically digests and absorbs the food.

Vertebrate Digestive Systems. The different regions of the gastrointestinal tract are adapted for different functions.

51.2 Food is ingested, swallowed, and transported to the stomach.

The Mouth and Teeth. Carnivores, herbivores, and omnivores display differences in the structure of their teeth.

Esophagus and Stomach. The esophagus delivers food to the stomach, which secretes hydrochloric acid and pepsin.

51.3 The small and large intestines have very different functions.

The Small Intestine. The small intestine has mucosal folds called villi and smaller folds called microvilli that absorb glucose, amino acids, and fatty acids into the blood.

The Large Intestine. The large intestine absorbs water, ions, and vitamin K, and excretes what remains as feces.

Variations in Vertebrate Digestive Systems. Digestive systems are adapted to particular diets.

51.4 Accessory organs, neural stimulation, and endocrine secretions assist in digestion.

Accessory Organs. The pancreas secretes digestive enzymes and the hormones insulin and glucagon. The liver produces bile, which emulsifies fat; the gallbladder stores the bile.

Neural and Hormonal Regulation of Digestion. Nerves and hormones help regulate digestive functions.

51.5 All animals require food energy and essential nutrients.

Food Energy and Energy Expenditure. The intake of food energy must balance the energy expended by the body in order to maintain a stable weight.

Essential Nutrients. Food must contain vitamins, minerals, and specific amino acids and fatty acids for health.



FIGURE 51.1

Animals are heterotrophs. All animals must consume plant material or other animals in order to live. The nuts in this chipmunk's cheeks will be consumed and converted to body tissue, energy, and refuse.

Plants and other photosynthetic organisms can produce the organic molecules they need from inorganic components. Therefore, they are autotrophs, or self-sustaining. Animals are heterotrophs: they must consume organic molecules present in other organisms (figure 51.1). The molecules heterotrophs eat must be digested into smaller molecules in order to be absorbed into the animal's body. Once these products of digestion enter the body, the animal can use them for energy in cell respiration or for the construction of the larger molecules that make up its tissues. The process of animal digestion is the focus of this chapter.

51.1 Animals employ a digestive system to prepare food for assimilation by cells.

Types of Digestive Systems

Heterotrophs are divided into three groups on the basis of their food sources. Animals that eat plants exclusively are classified as **herbivores**; common examples include cows, horses, rabbits and sparrows. Animals that are meat-eaters, such as cats, eagles, trout, and frogs, are **carnivores**. **Omnivores** are animals that eat both plants and other animals. Humans are omnivores, as are pigs, bears, and crows.

Single-celled organisms (as well as sponges) digest their food intracellularly. Other animals digest their food extracellularly, within a digestive cavity. In this case, the digestive enzymes are released into a cavity that is continuous with the animal's external environment. In coelenterates and flatworms (such as *Planaria*), the digestive cavity has only one opening that serves as both mouth and anus. There can be no specialization within this type of digestive system, called a *gastrovascular cavity*, because every cell is exposed to all stages of food digestion (figure 51.2).

Specialization occurs when the digestive tract, or alimentary canal, has a separate mouth and anus, so that transport of food is one-way. The most primitive digestive tract is seen in nematodes (phylum Nematoda), where it is simply a tubular *gut* lined by an epithelial membrane. Earthworms (phylum Annelida) have a digestive tract spe-

cialized in different regions for the ingestion, storage, fragmentation, digestion, and absorption of food. All higher animal groups, including all vertebrates, show similar specializations (figure 51.3).

The ingested food may be stored in a specialized region of the digestive tract or may first be subjected to physical fragmentation. This fragmentation may occur through the chewing action of teeth (in the mouth of many vertebrates), or the grinding action of pebbles (in the gizzard of earthworms and birds). Chemical digestion then occurs, breaking down the larger food molecules of polysaccharides and disaccharides, fats, and proteins into their smallest subunits. Chemical digestion involves hydrolysis reactions that liberate the subunit molecules—primarily monosaccharides, amino acids, and fatty acids—from the food. These products of chemical digestion pass through the epithelial lining of the gut into the blood, in a process known as absorption. Any molecules in the food that are not absorbed cannot be used by the animal. These waste products are excreted, or defecated, from the anus.

Most animals digest their food extracellularly. The digestive tract, with a one-way transport of food and specialization of regions for different functions, allows food to be ingested, physically fragmented, chemically digested, and absorbed.

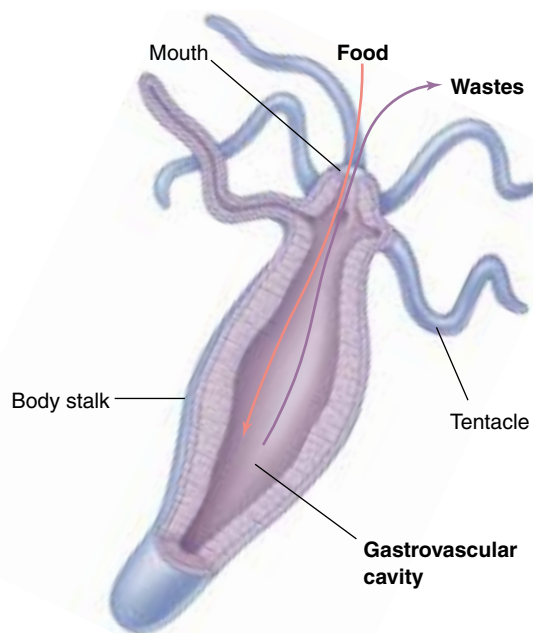


FIGURE 51.2

The gastrovascular cavity of *Hydra*, a coelenterate. Because there is only one opening, the mouth is also the anus, and no specialization is possible in the different regions that participate in extracellular digestion.

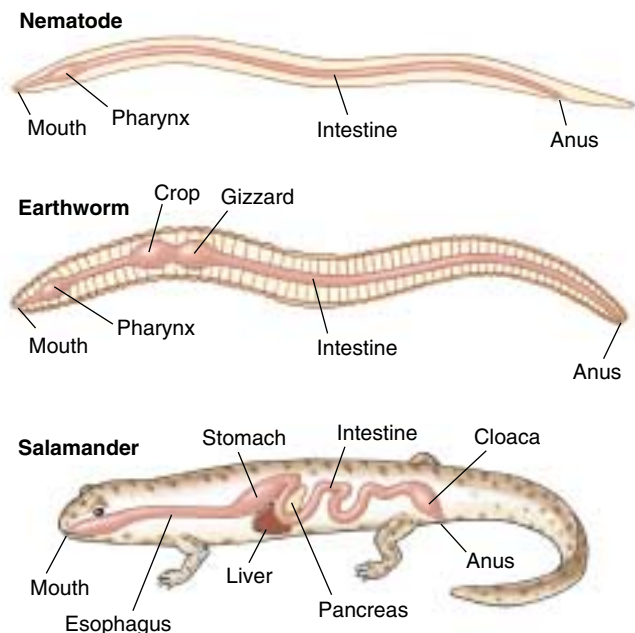


FIGURE 51.3

The one-way digestive tract of nematodes, earthworms, and vertebrates. One-way movement through the digestive tract allows different regions of the digestive system to become specialized for different functions.

Vertebrate Digestive Systems

In humans and other vertebrates, the digestive system consists of a tubular gastrointestinal tract and accessory digestive organs (figure 51.4). The initial components of the gastrointestinal tract are the mouth and the pharynx, which is the common passage of the oral and nasal cavities. The pharynx leads to the esophagus, a muscular tube that delivers food to the stomach, where some preliminary digestion occurs. From the stomach, food passes to the first part of the small intestine, where a battery of digestive enzymes continues the digestive process. The products of digestion then pass across the wall of the small intestine into the bloodstream. The small intestine empties what remains into the large intestine, where water and minerals are absorbed. In most vertebrates other than mammals, the waste products emerge from the large intestine into a cavity called the cloaca (see figure 51.3), which also receives the products of the urinary and reproductive systems. In mammals, the urogenital products are separated from the fecal material in the large intestine; the fecal material enters the rectum and is expelled through the anus.

In general, carnivores have shorter intestines for their size than do herbivores. A short intestine is adequate for a carnivore, but herbivores ingest a large amount of plant cellulose, which resists digestion. These animals have a long, convoluted small intestine. In addition, mammals called *ruminants* (such as cows) that consume grass and other vegetation have stomachs with multiple chambers, where bacteria aid in the digestion of cellulose. Other herbivores, including rabbits and horses, digest cellulose (with the aid of bacteria) in a blind pouch called the **cecum** located at the beginning of the large intestine.

The accessory digestive organs (described in detail later in the chapter) include the liver, which produces *bile* (a green solution that emulsifies fat), the gallbladder, which stores and concentrates the bile, and the pancreas. The pancreas produces *pancreatic juice*, which contains digestive enzymes and bicarbonate. Both bile and pancreatic juice are secreted into the first region of the small intestine and aid digestion.

The tubular gastrointestinal tract of a vertebrate has a characteristic layered structure (figure 51.5). The innermost layer is the mucosa, an epithelium that lines the interior of the tract (the lumen). The next major tissue layer, made of connective tissue, is called the submucosa. Just outside the submucosa is the muscularis, which consists of a double layer of smooth muscles. The muscles in the inner layer have a circular orientation, and those in the outer layer are arranged longitudinally. Another connective tissue layer, the serosa, covers the external surface of the tract. Nerves, intertwined in regions called *plexuses*, are located in the submucosa and help regulate the gastrointestinal activities.

The vertebrate digestive system consists of a tubular gastrointestinal tract, which is modified in different animals, composed of a series of tissue layers.

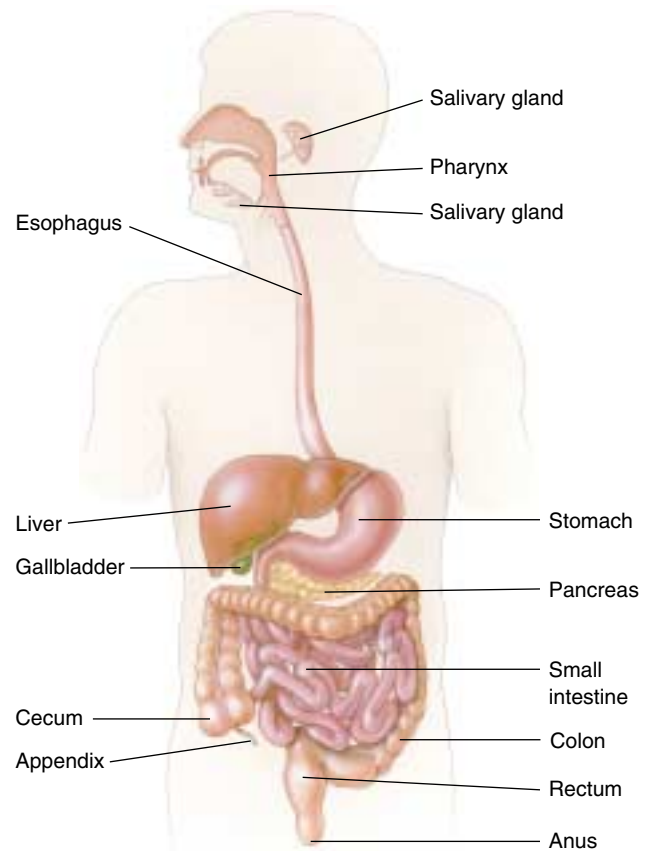


FIGURE 51.4

The human digestive system. Humans, like all placental mammals, lack a cloaca and have a separate exit from the digestive tract through the rectum and anus.

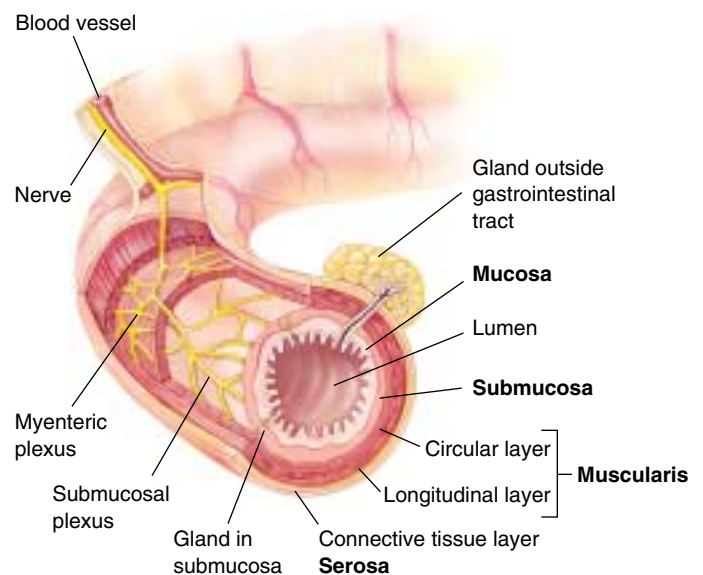


FIGURE 51.5

The layers of the gastrointestinal tract. The mucosa contains a lining epithelium; the submucosa is composed of connective tissue (as is the serosa), and the muscularis consists of smooth muscles.

51.2 Food is ingested, swallowed, and transported to the stomach.

The Mouth and Teeth

Specializations of the digestive systems in different kinds of vertebrates reflect differences in the way these animals live. Fishes have a large pharynx with gill slits, while air-breathing vertebrates have a greatly reduced pharynx. Many vertebrates have teeth (figure 51.6), and chewing (*mastication*) breaks up food into small particles and mixes it with fluid secretions. Birds, which lack teeth, break up food in their two-chambered stomachs (figure 51.7). In one of these chambers, the gizzard, small pebbles ingested by the bird are churned together with the food by muscular action. This churning grinds up the seeds and other hard plant material into smaller chunks that can be digested more easily.

Vertebrate Teeth

Carnivorous mammals have pointed teeth that lack flat grinding surfaces. Such teeth are adapted for cutting and shearing. Carnivores often tear off pieces of their prey but have little need to chew them, because digestive enzymes can act directly on animal cells. (Recall how a cat or dog gulps down its food.) By contrast, grass-eating herbivores, such as cows and horses, must pulverize the cellulose cell walls of plant tissue before digesting it. These animals have large, flat teeth with complex ridges well-suited to grinding.

Human teeth are specialized for eating both plant and animal food. Viewed simply, humans are carnivores in the front of the mouth and herbivores in the back (figure 51.8). The four front teeth in the upper and lower jaws are sharp, chisel-shaped incisors used for biting. On each side of the incisors are sharp, pointed teeth called cuspids (sometimes referred to as “canine” teeth), which are used for tearing food. Behind the canines are two premolars and three molars, all with flattened, ridged surfaces for grinding and crushing food. Children have only 20 teeth, but these deciduous teeth are lost during childhood and are replaced by 32 adult teeth.

The Mouth

Inside the mouth, the tongue mixes food with a mucous solution, saliva. In humans, three pairs of salivary glands secrete saliva into the mouth through ducts in the mouth’s mucosal lining. Saliva moistens and lubricates the food so that it is easier to swallow and does not abrade the tissue it passes on its way through the esophagus. Saliva also contains the hydrolytic enzyme salivary amylase, which initiates the breakdown of the polysaccharide starch into the disaccharide maltose. This digestion is usually minimal in humans, however, because most people don’t chew their food very long.

The secretions of the salivary glands are controlled by the nervous system, which in humans maintains a constant

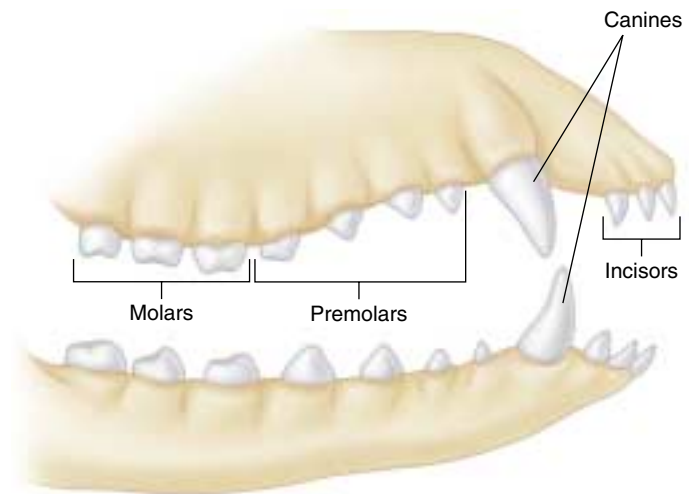


FIGURE 51.6

Diagram of generalized vertebrate dentition. Different vertebrates will have specific variations from this generalized pattern, depending on whether the vertebrate is an herbivore, carnivore, or omnivore.

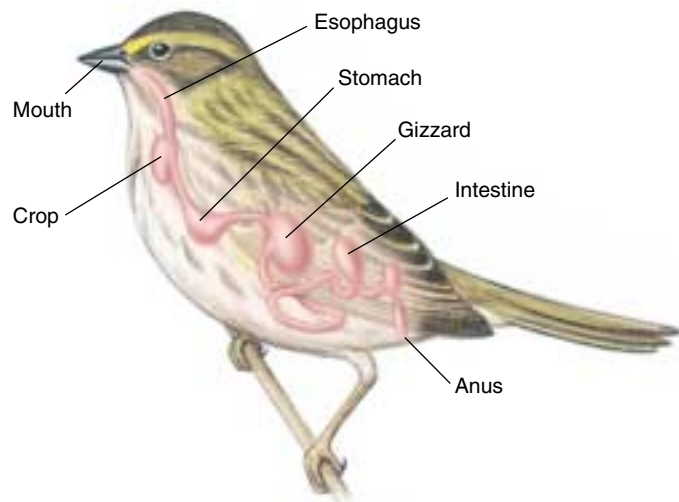


FIGURE 51.7

Birds store food in the crop and grind it up in the gizzard. Birds lack teeth but have a muscular chamber called the gizzard that works to break down food. Birds swallow gritty objects or pebbles that lodge in the gizzard and pulverize food before it passes into the small intestine.

flow of about half a milliliter per minute when the mouth is empty of food. This continuous secretion keeps the mouth moist. The presence of food in the mouth triggers an increased rate of secretion, as taste-sensitive neurons in the mouth send impulses to the brain, which responds by stimulating the salivary glands. The most potent stimuli are

acidic solutions; lemon juice, for example, can increase the rate of salivation eightfold. The sight, sound, or smell of food can stimulate salivation markedly in dogs, but in humans, these stimuli are much less effective than thinking or talking about food.

When food is ready to be swallowed, the tongue moves it to the back of the mouth. In mammals, the process of swallowing begins when the soft palate elevates, pushing against the back wall of the pharynx (figure 51.9). Elevation of the soft palate seals off the nasal cavity and prevents food from entering it. Pressure against the pharynx triggers an automatic, involuntary response called a reflex. In this reflex, pressure on the pharynx stimulates neurons within its walls, which send impulses to the swallowing center in the brain. In response, electrical impulses in motor neurons stimulate muscles to contract and raise the *larynx* (voice box). This pushes the *glottis*, the opening from the larynx into the trachea (windpipe), against a flap of tissue called the *epiglottis*. These actions keep food out of the respiratory tract, directing it instead into the esophagus.

In many vertebrates ingested food is fragmented through the tearing or grinding action of specialized teeth. In birds, this is accomplished through the grinding action of pebbles in the gizzard. Food mixed with saliva is swallowed and enters the esophagus.

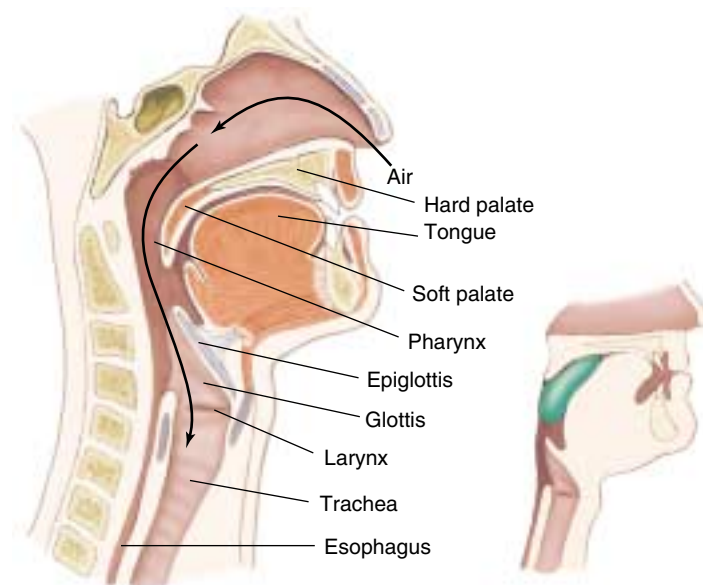


FIGURE 51.9

The human pharynx, palate, and larynx. Food that enters the pharynx is prevented from entering the nasal cavity by elevation of the soft palate, and is prevented from entering the larynx and trachea (the airways of the respiratory system) by elevation of the larynx against the epiglottis.

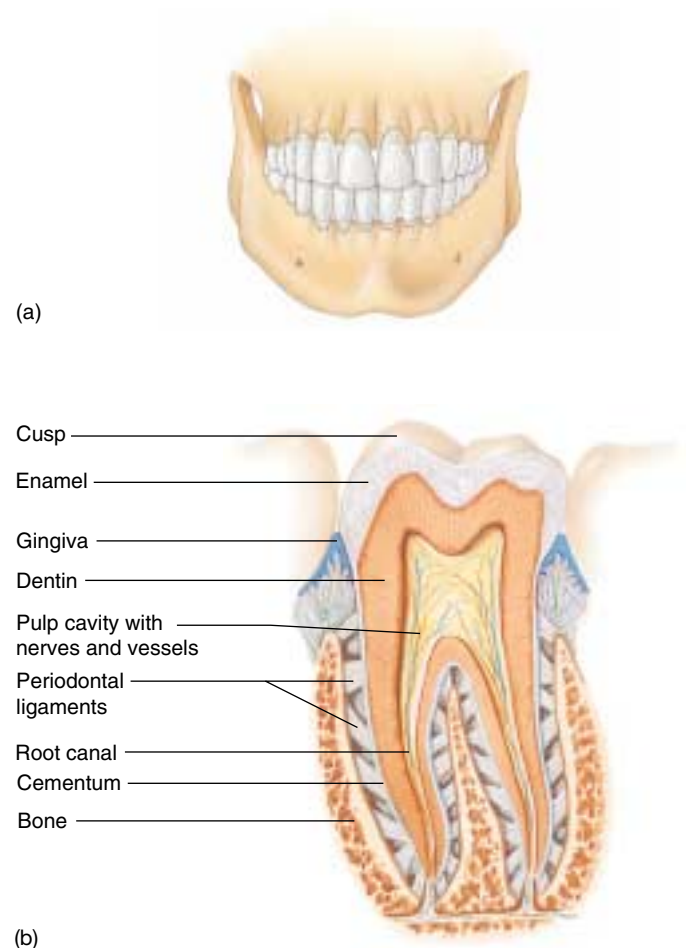


FIGURE 51.8

Human teeth. (a) The front six teeth on the upper and lower jaws are cuspids and incisors. The remaining teeth, running along the sides of the mouth, are grinders called premolars and molars. Hence, humans have carnivore-like teeth in the front of their mouth and herbivore-like teeth in the back. (b) Each tooth is alive, with a central pulp containing nerves and blood vessels. The actual chewing surface is a hard enamel layered over the softer dentin, which forms the body of the tooth.



Esophagus and Stomach

Structure and Function of the Esophagus

Swallowed food enters a muscular tube called the esophagus, which connects the pharynx to the stomach. In adult humans, the esophagus is about 25 centimeters long; the upper third is enveloped in skeletal muscle, for voluntary control of swallowing, while the lower two-thirds is surrounded by involuntary smooth muscle. The swallowing center stimulates successive waves of contraction in these muscles that move food along the esophagus to the stomach. These rhythmic waves of muscular contraction are called peristalsis (figure 51.10); they enable humans and other vertebrates to swallow even if they are upside down.

In many vertebrates, the movement of food from the esophagus into the stomach is controlled by a ring of circular smooth muscle, or a *sphincter*, that opens in response to the pressure exerted by the food. Contraction of this sphincter prevents food in the stomach from moving back into the esophagus. Rodents and horses have a true sphincter at this site and thus cannot regurgitate, while humans lack a true sphincter and so are able to regurgitate. Normally, however, the human esophagus is closed off except during swallowing.

Structure and Function of the Stomach

The stomach (figure 51.11) is a saclike portion of the digestive tract. Its inner surface is highly convoluted, enabling it to fold up when empty and open out like an expanding balloon as it fills with food. Thus, while the human stomach has a volume of only about 50 milliliters when empty, it may expand to contain 2 to 4 liters of food when full. Carnivores that engage in sporadic gorging as an important survival strategy possess stomachs that are able to distend much more than that.

Secretory Systems

The stomach contains an extra layer of smooth muscle for churning food and mixing it with *gastric juice*, an acidic secretion of the tubular gastric glands of the mucosa (figure 51.11). These exocrine glands contain two kinds of secretory cells: parietal cells, which secrete hydrochloric acid (HCl); and chief cells, which secrete pepsinogen, a weak protease (protein-digesting enzyme) that requires a very low pH to be active. This low pH is provided by the HCl. Activated pepsinogen molecules then cleave one another at specific sites, producing a much more active protease, pepsin. This process of secreting a relatively inactive enzyme that is then converted into a more active enzyme outside the cell prevents the chief cells from digesting themselves. It should be noted that only proteins are partially digested in the stomach—there is no significant digestion of carbohydrates or fats.

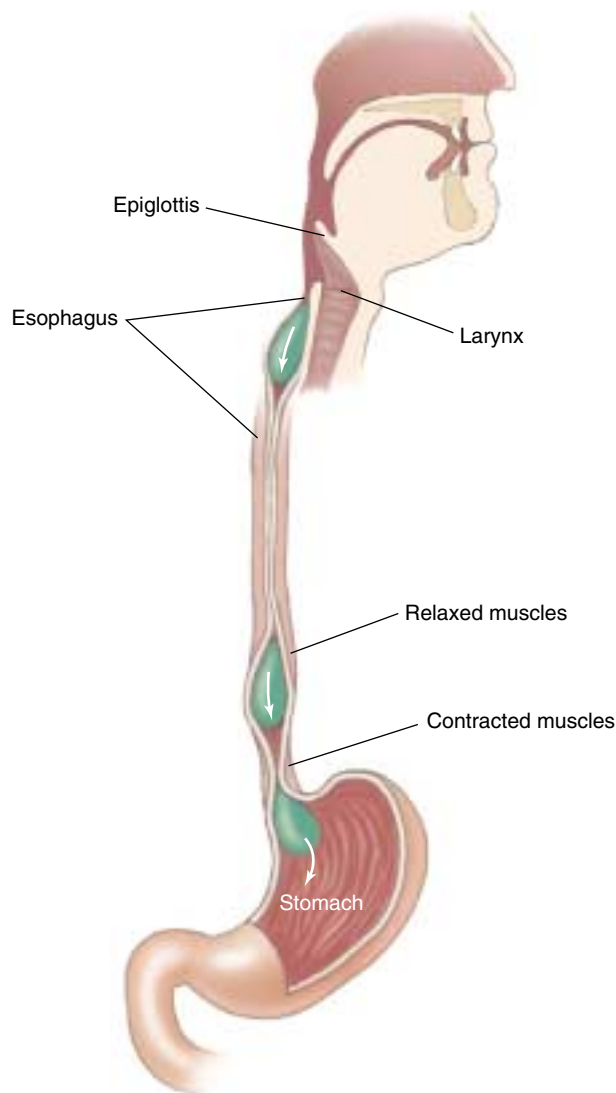


FIGURE 51.10
The esophagus and peristalsis. After food has entered the esophagus, rhythmic waves of muscular contraction, called peristalsis, move the food down to the stomach.

Action of Acid

The human stomach produces about 2 liters of HCl and other gastric secretions every day, creating a very acidic solution inside the stomach. The concentration of HCl in this solution is about 10 millimolar, corresponding to a pH of 2. Thus, gastric juice is about 250,000 times more acidic than blood, whose normal pH is 7.4. The low pH in the stomach helps denature food proteins, making them easier to digest, and keeps pepsin maximally active. Active pepsin hydrolyzes food proteins into shorter chains of polypeptides that are not fully digested until the mixture enters the small intestine. The mixture of partially digested food and gastric juice is called chyme.

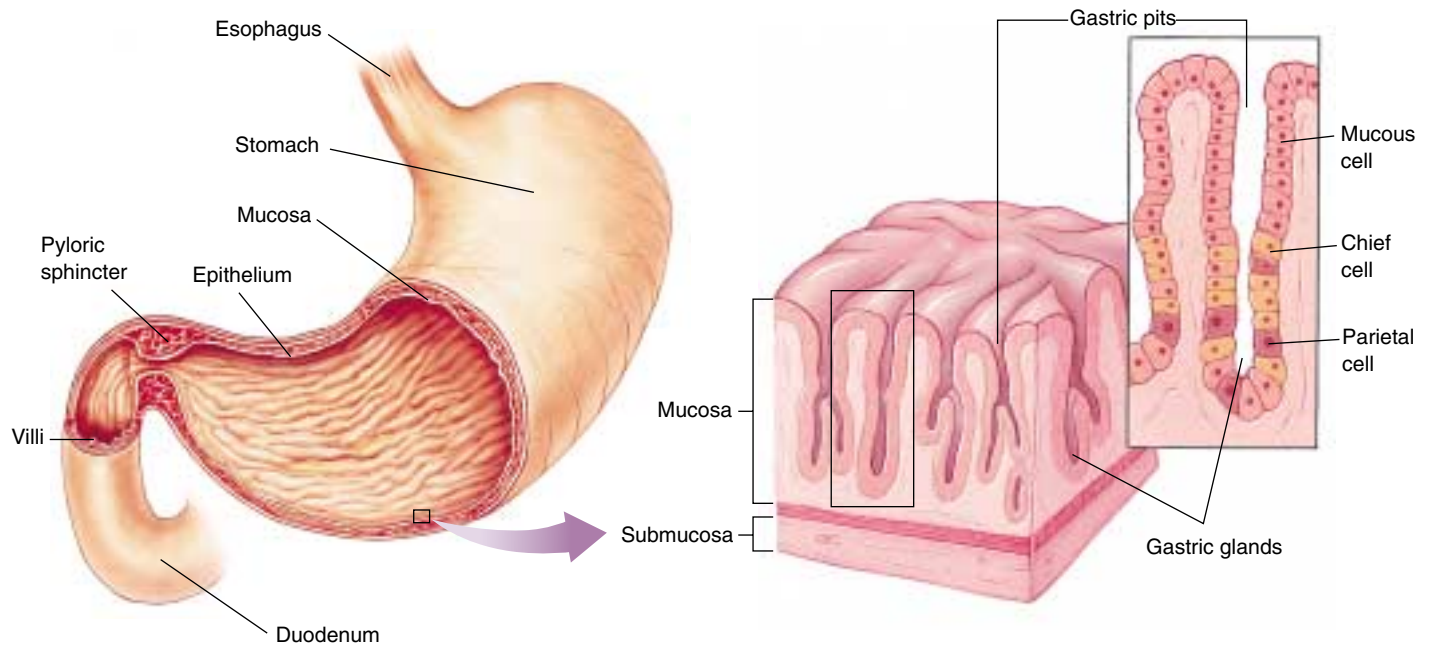


FIGURE 51.11

The stomach and duodenum. Food enters the stomach from the esophagus. A band of smooth muscle called the pyloric sphincter controls the entrance to the duodenum, the upper part of the small intestine. The epithelial walls of the stomach are dotted with gastric pits, which contain gastric glands that secrete hydrochloric acid and the enzyme pepsinogen. The gastric glands consist of mucous cells, chief cells that secrete pepsinogen, and parietal cells that secrete HCl. Gastric pits are the openings of the gastric glands.

The acidic solution within the stomach also kills most of the bacteria that are ingested with the food. The few bacteria that survive the stomach and enter the intestine intact are able to grow and multiply there, particularly in the large intestine. In fact, most vertebrates harbor thriving colonies of bacteria within their intestines, and bacteria are a major component of feces. As we will discuss later, bacteria that live within the digestive tract of cows and other ruminants play a key role in the ability of these mammals to digest cellulose.

Ulcers

Overproduction of gastric acid can occasionally eat a hole through the wall of the stomach. Such *gastric ulcers* are rare, however, because epithelial cells in the mucosa of the stomach are protected somewhat by a layer of alkaline mucus, and because those cells are rapidly replaced by cell division if they become damaged (gastric epithelial cells are replaced every 2 to 3 days). Over 90% of gastrointestinal ulcers are *duodenal ulcers*. These may be produced when excessive amounts of acidic chyme are delivered into the duodenum, so that the acid cannot be properly neutralized through the action of alkaline pancreatic juice (described later). Susceptibility to ulcers is increased when the mucosal barriers to self-digestion are weakened by an infection of the bacterium *Helicobacter pylori*. Indeed, modern

antibiotic treatments of this infection can reduce symptoms and often even cure the ulcer.

In addition to producing HCl, the parietal cells of the stomach also secrete intrinsic factor, a polypeptide needed for the intestinal absorption of vitamin B₁₂. Because this vitamin is required for the production of red blood cells, persons who lack sufficient intrinsic factor develop a type of anemia (low red blood cell count) called *pernicious anemia*.

Leaving the Stomach

Chyme leaves the stomach through the *pyloric sphincter* (see figure 51.11) to enter the small intestine. This is where all terminal digestion of carbohydrates, lipids, and proteins occurs, and where the products of digestions—amino acids, glucose, and so on—are absorbed into the blood. Only some of the water in chyme and a few substances such as aspirin and alcohol are absorbed through the wall of the stomach.

Peristaltic waves of contraction propel food along the esophagus to the stomach. Gastric juice contains strong hydrochloric acid and the protein-digesting enzyme pepsin, which begins the digestion of proteins into shorter polypeptides. The acidic chyme is then transferred through the pyloric sphincter to the small intestine.

51.3 The small and large intestines have very different functions.

The Small Intestine

Digestion in the Small Intestine

The capacity of the small intestine is limited, and its digestive processes take time. Consequently, efficient digestion requires that only relatively small amounts of chyme be introduced from the stomach into the small intestine at any one time. Coordination between gastric and intestinal activities is regulated by neural and hormonal signals, which we will describe in a later section.

The small intestine is approximately 4.5 meters long in a living person, but is 6 meters long at autopsy when the muscles relax. The first 25 centimeters is the **duodenum**; the remainder of the small intestine is divided into the **jejunum** and the **ileum**. The duodenum receives acidic chyme from the stomach, digestive enzymes and bicarbonate from the pancreas, and bile from the liver and gallbladder. The pancreatic juice enzymes digest larger food molecules into smaller fragments. This occurs primarily in the duodenum and jejunum.

The epithelial wall of the small intestine is covered with tiny, fingerlike projections called villi (singular, villus; figure

51.12). In turn, each of the epithelial cells lining the villi is covered on its apical surface (the side facing the lumen) by many foldings of the plasma membrane that form cytoplasmic extensions called *microvilli*. These are quite tiny and can be seen clearly only with an electron microscope (figure 51.13). In a light micrograph, the microvilli resemble the bristles of a brush, and for that reason the epithelial wall of the small intestine is also called a brush border.

The villi and microvilli greatly increase the surface area of the small intestine; in humans, this surface area is 300 square meters! It is over this vast surface that the products of digestion are absorbed. The microvilli also participate in digestion because a number of digestive enzymes are embedded within the epithelial cells' plasma membranes, with their active sites exposed to the chyme (figure 51.14). These brush border enzymes include those that hydrolyze the disaccharides lactose and sucrose, among others (table 51.1). Many adult humans lose the ability to produce the brush border enzyme *lactase* and therefore cannot digest lactose (milk sugar), a rather common condition called *lactose intolerance*. The brush border enzymes complete the digestive process that started with the action of the pancreatic enzymes released into the duodenum.

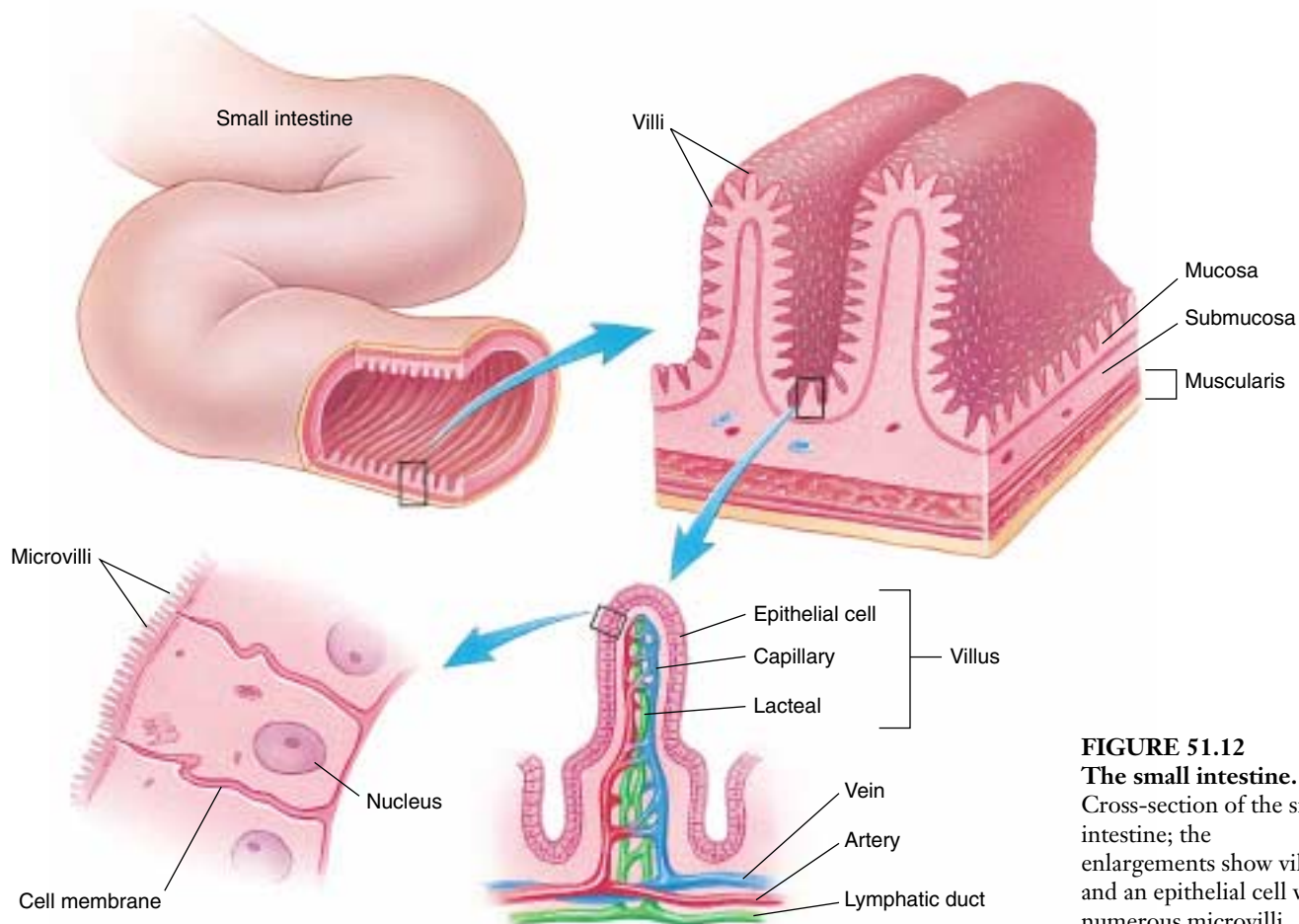


FIGURE 51.12

The small intestine.

Cross-section of the small intestine; the enlargements show villi and an epithelial cell with numerous microvilli.

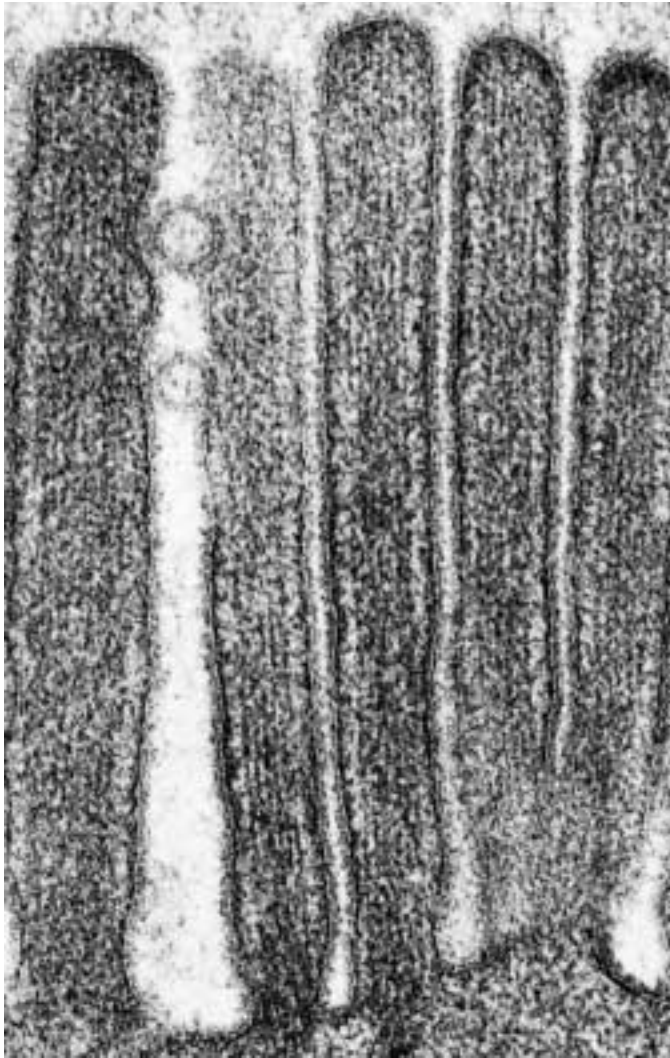


FIGURE 51.13
Intestinal microvilli. Microvilli, shown in an electron micrograph, are very densely clustered, giving the small intestine an enormous surface area important in efficient absorption of the digestion products.

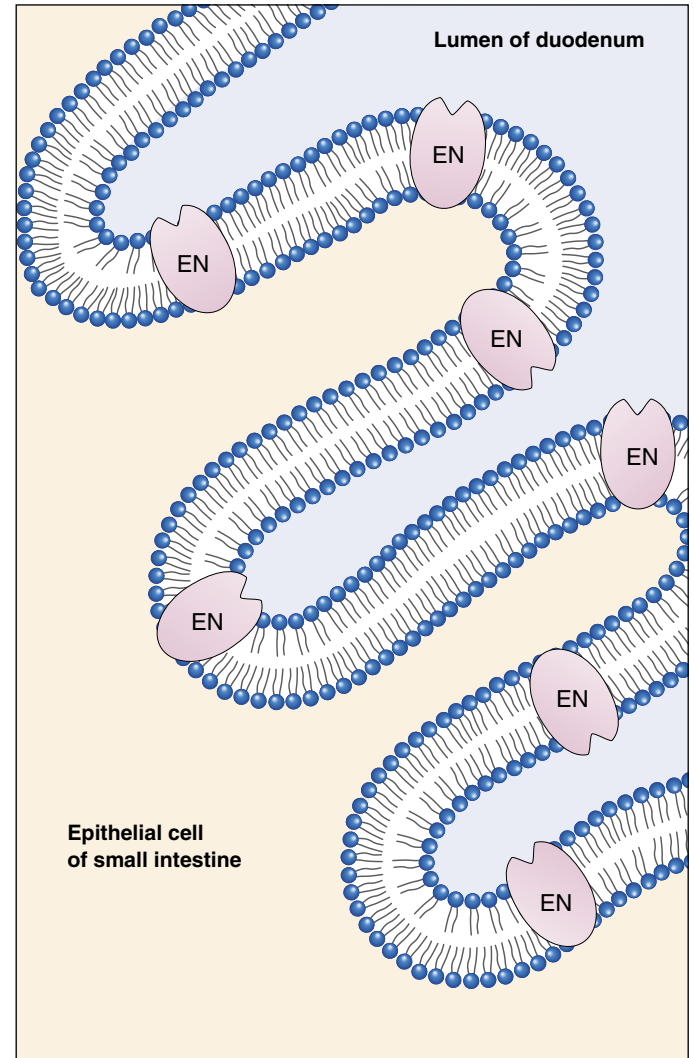


FIGURE 51.14
Brush border enzymes. These enzymes, which are labeled “EN” in this diagram, are part of the plasma membrane of the microvilli in the small intestine. They catalyze many of the terminal steps in digestion.

Table 51.1 Digestive Enzymes			
Location	Enzymes	Substrates	Digestion Products
Salivary glands	Amylase	Starch, glycogen	Disaccharides
Stomach	Pepsin	Proteins	Short peptides
Small intestine (brush border)	Peptidases	Short peptides	Amino acids
	Nucleases	DNA, RNA	Sugars, nucleic acid bases
	Lactase, maltase, sucrase	Disaccharides	Monosaccharides
Pancreas	Lipase	Triglycerides	Fatty acids, glycerol
	Trypsin, chymotrypsin	Proteins	Peptides
	DNase	DNA	Nucleotides
	RNase	RNA	Nucleotides

Absorption in the Small Intestine

The amino acids and monosaccharides resulting from the digestion of proteins and carbohydrates, respectively, are transported across the brush border into the epithelial cells that line the intestine (figure 51.15*a*). They then move to the other side of the epithelial cells, and from there are transported across the membrane and into the blood capillaries within the villi. The blood carries these products of digestion from the intestine to the liver via the hepatic portal vein. The term *portal* here refers to a special arrangement of vessels, seen only in a couple of instances, where one organ (the liver, in this case) is located “downstream” from another organ (the intestine). As a result, the second organ receives blood-borne molecules from the first. Because of the hepatic portal vein, the liver is the first organ to receive most of the products of digestion. This arrangement is important for the functions of the liver, as will be described in a later section.

The products of fat digestion are absorbed by a different mechanism (figure 51.15*b*). Fats (triglycerides) are hydrolyzed into fatty acids and monoglycerides, which are absorbed into the intestinal epithelial cells and reassembled into triglycerides. The triglycerides then combine with proteins to form small particles called chylomicrons. In-

stead of entering the hepatic portal circulation, the chylomicrons are absorbed into lymphatic capillaries (see chapter 52), which empty their contents into the blood in veins near the neck. Chylomicrons can make the blood plasma appear cloudy if a sample of blood is drawn after a fatty meal.

The amount of fluid passing through the small intestine in a day is startlingly large: approximately 9 liters. However, almost all of this fluid is absorbed into the body rather than eliminated in the feces. About 8.5 liters are absorbed in the small intestine and an additional 350 milliliters in the large intestine. Only about 50 grams of solid and 100 milliliters of liquid leave the body as feces. The normal fluid absorption efficiency of the human digestive tract thus approaches 99%, which is very high indeed.

Digestion occurs primarily in the duodenum, which receives the pancreatic juice enzymes. The small intestine provides a large surface area for absorption. Glucose and amino acids from food are absorbed through the small intestine and enter the blood via the hepatic portal vein, going to the liver. Fat from food enters the lymphatic system.

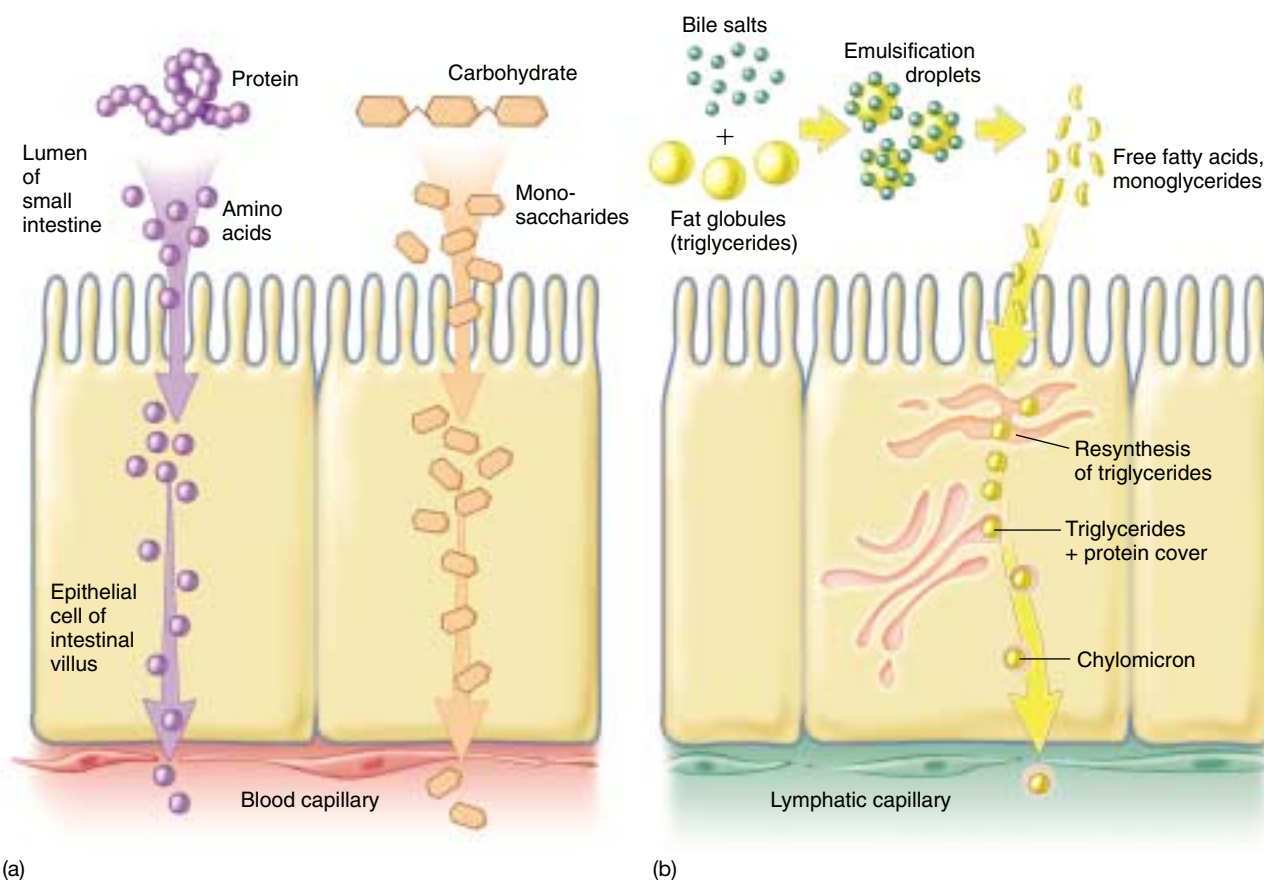


FIGURE 51.15

Absorption of the products of digestion. (a) Monosaccharides and amino acids are transported into blood capillaries. (b) Fatty acids and monoglycerides within the intestinal lumen are absorbed and converted within the intestinal epithelial cells into triglycerides. These are then coated with proteins to form tiny structures called chylomicrons, which enter lymphatic capillaries.

The Large Intestine

The large intestine, or colon, is much shorter than the small intestine, occupying approximately the last meter of the digestive tract; it is called “large” only because of its larger diameter. The small intestine empties directly into the large intestine at a junction where two vestigial structures, the cecum and the appendix, remain (figure 51.16). No digestion takes place within the large intestine, and only about 4% of the absorption of fluids by the intestine occurs there. The large intestine is not as convoluted as the small intestine, and its inner surface has no villi. Consequently, the large intestine has less than one-thirtieth the absorptive surface area of the small intestine. Although sodium, vitamin K, and some products of bacterial metabolism are absorbed across its wall, the primary function of the large intestine is to concentrate waste material. Within it, undigested material, primarily bacterial fragments and cellulose, is compacted and stored. Many bacteria live and reproduce within the large intestine, and the excess bacteria are incorporated into the refuse material, called *feces*. Bacterial fermentation produces gas within the colon at a rate of about 500 milliliters per day. This rate increases greatly after the consumption of beans or other vegetable matter because the passage of undigested plant material (fiber) into the large intestine provides substrates for fermentation.

The human colon has evolved to process food with a relatively high fiber content. Diets that are low in fiber, which are common in the United States, result in a slower passage of food through the colon. Low dietary fiber content is thought to be associated with the level of colon cancer in the United States, which is among the highest in the world.

Compacted feces, driven by peristaltic contractions of the large intestine, pass from the large intestine into a short tube called the rectum. From the rectum, the feces exit the body through the anus. Two sphincters control passage through the anus. The first is composed of smooth muscle and opens involuntarily in response to pressure inside the rectum. The second, composed of striated muscle, can be controlled voluntarily by the brain, thus permitting a conscious decision to delay defecation.

In all vertebrates except most mammals, the reproductive and urinary tracts empty together with the digestive tract into a common cavity, the cloaca. In some reptiles and birds, additional water from either the feces or urine may be absorbed in the cloaca before the products are expelled from the body.

The large intestine concentrates wastes for excretion by absorbing water. Some ions and vitamin K are also absorbed by the large intestine.

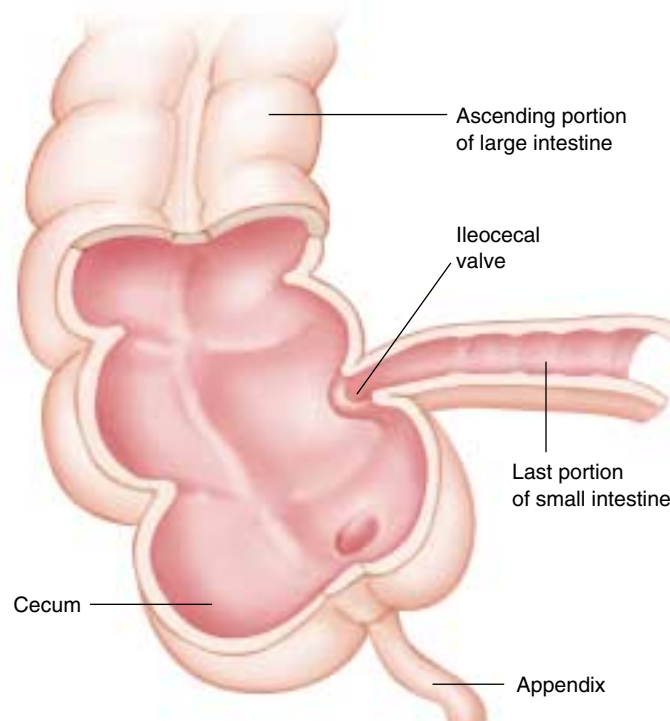


FIGURE 51.16

The junction of the small and large intestines in humans. The large intestine, or colon, starts with the cecum, which is relatively small in humans compared with that in other mammals. A vestigial structure called the appendix extends from the cecum.

Variations in Vertebrate Digestive Systems

Most animals lack the enzymes necessary to digest cellulose, the carbohydrate that functions as the chief structural component of plants. The digestive tracts of some animals, however, contain bacteria and protists that convert cellulose into substances the host can digest. Although digestion by gastrointestinal microorganisms plays a relatively small role in human nutrition, it is an essential element in the nutrition of many other kinds of animals, including insects like termites and cockroaches, and a few groups of herbivorous mammals. The relationships between these microorganisms and their animal hosts are mutually beneficial and provide an excellent example of symbiosis.

Cows, deer, and other ruminants have large, divided stomachs (figure 51.17). The first portion consists of the rumen and a smaller chamber, the reticulum; the second portion consists of two additional chambers: the omasum and abomasum. The rumen which may hold up to 50 gallons, serves as a fermentation vat in which bacteria and protozoa convert cellulose and other molecules into a variety of simpler compounds. The location of the rumen at the front of the four chambers is important because it allows the animal to regurgitate and rechew the contents of the rumen, an activity called *rumination*, or “chewing the cud.” The cud is then swallowed and enters the reticulum, from which it passes to the omasum and then the abomasum, where it is finally mixed with gastric juice. Hence, only the abomasum is equivalent to the human stomach in its function. This process leads to a far more efficient digestion of cellulose in ruminants than in mammals that lack a rumen, such as horses.

In horses, rodents, and lagomorphs (rabbits and hares), the digestion of cellulose by microorganisms takes place in the cecum, which is greatly enlarged (figure 51.18). Because the cecum is located beyond the stomach, regurgitation of its contents is impossible. However, rodents and lagomorphs have evolved another way to digest cellulose that achieves a degree of efficiency similar to that of ruminant digestion. They do this by eating their feces, thus passing the food through their digestive tract a second time. The second passage makes it possible for the animal to absorb the nutrients produced by the microorganisms in its cecum. Animals that engage in this practice of **coprophagy** (from the Greek words *copros*, “excrement,” and *phagein*, “eat”) cannot remain healthy if they are prevented from eating their feces.

Cellulose is not the only plant product that vertebrates can use as a food source because of the digestive activities

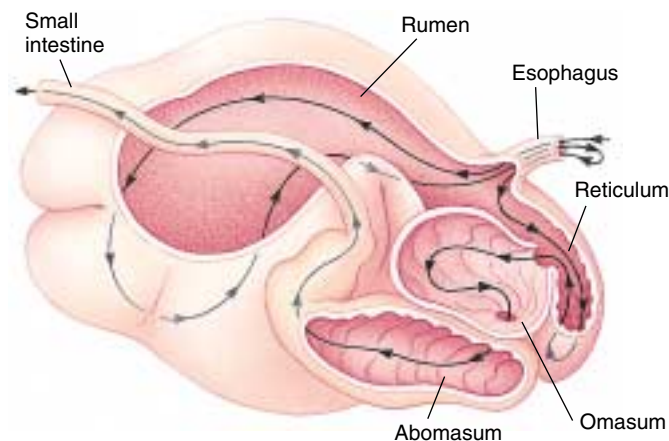


FIGURE 51.17

Four-chambered stomach of a ruminant. The grass and other plants that a ruminant, such as a cow, eats enter the rumen, where they are partially digested. Before moving into a second chamber, the reticulum, the food may be regurgitated and rechewed. The food is then transferred to the rear two chambers: the omasum and abomasum. Only the abomasum is equivalent to the human stomach in its function of secreting gastric juice.

of intestinal microorganisms. Wax, a substance indigestible by most terrestrial animals, is digested by symbiotic bacteria living in the gut of honey guides, African birds that eat the wax in bee nests. In the marine food chain, wax is a major constituent of copepods (crustaceans in the plankton), and many marine fish and birds appear to be able to digest wax with the aid of symbiotic microorganisms.

Another example of the way intestinal microorganisms function in the metabolism of their animal hosts is provided by the synthesis of vitamin K. All mammals rely on intestinal bacteria to synthesize this vitamin, which is necessary for the clotting of blood. Birds, which lack these bacteria, must consume the required quantities of vitamin K in their food. In humans, prolonged treatment with antibiotics greatly reduces the populations of bacteria in the intestine; under such circumstances, it may be necessary to provide supplementary vitamin K.

Much of the food value of plants is tied up in cellulose, and the digestive tract of many animals harbors colonies of cellulose-digesting microorganisms. Intestinal microorganisms also produce molecules such as vitamin K that are important to the well-being of their vertebrate hosts.

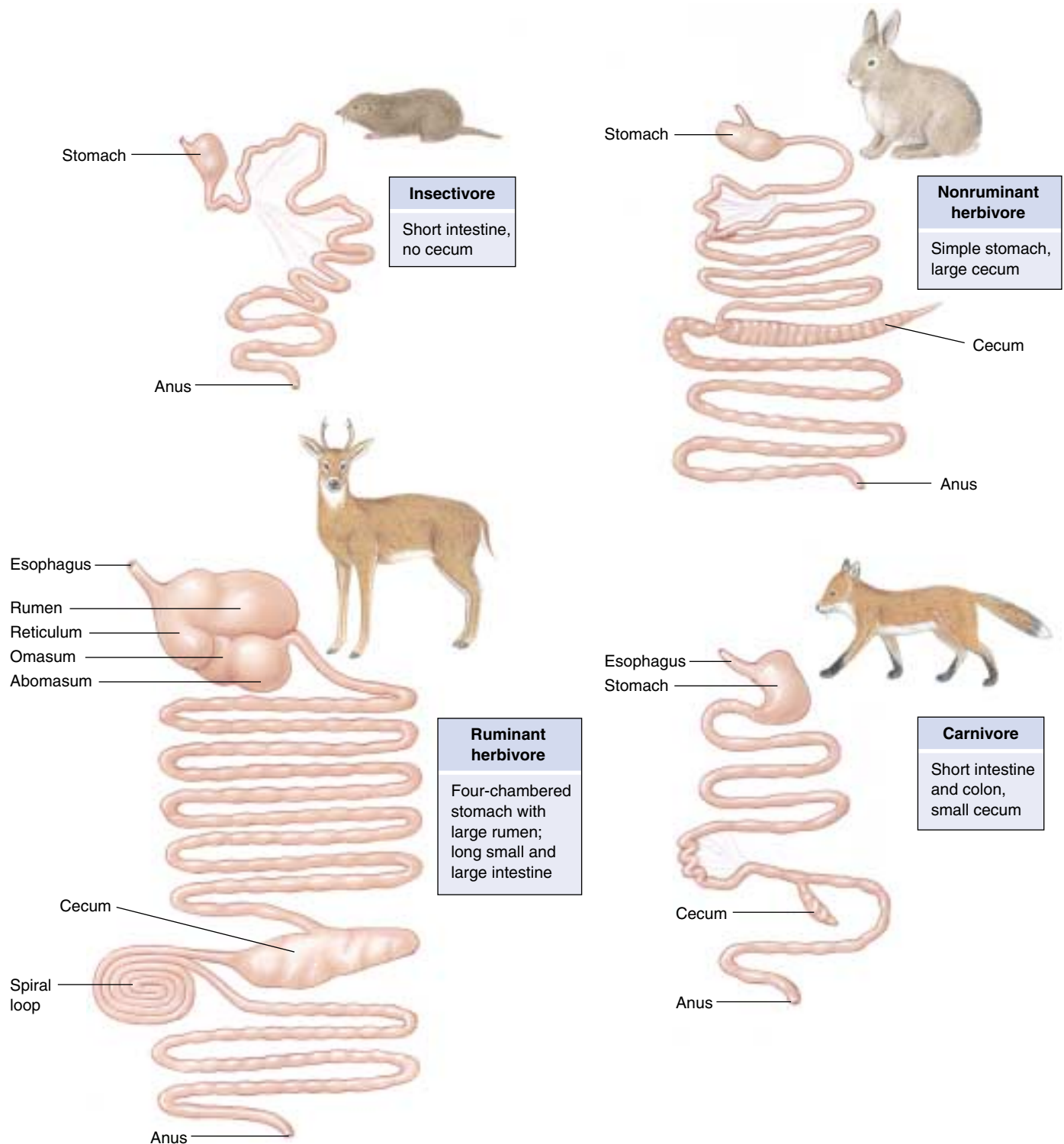


FIGURE 51.18

The digestive systems of different mammals reflect their diets. Herbivores require long digestive tracts with specialized compartments for the breakdown of plant matter. Protein diets are more easily digested; thus, insectivorous and carnivorous mammals have short digestive tracts with few specialized pouches.

51.4 Accessory organs, neural stimulation, and endocrine secretions assist in digestion.

Accessory Organs

Secretions of the Pancreas

The pancreas (figure 51.19), a large gland situated near the junction of the stomach and the small intestine, is one of the accessory organs that contribute secretions to the digestive tract. Pancreatic fluid is secreted into the duodenum through the *pancreatic duct*; thus, the pancreas functions as an exocrine organ. This fluid contains a host of enzymes, including trypsin and chymotrypsin, which digest proteins; pancreatic amylase, which digests starch; and lipase, which digests fat. These enzymes are released into the duodenum primarily as inactive zymogens and are then activated by the brush border enzymes of the intestine. Pancreatic enzymes digest proteins into smaller polypeptides, polysaccharides into shorter chains of sugars, and fat into free fatty acids and other products. The digestion of these molecules is then completed by the brush border enzymes.

Pancreatic fluid also contains bicarbonate, which neutralizes the HCl from the stomach and gives the chyme in the duodenum a slightly alkaline pH. The digestive enzymes and bicarbonate are produced by clusters of secretory cells known as *acini*.

In addition to its exocrine role in digestion, the pancreas also functions as an endocrine gland, secreting several hormones into the blood that control the blood levels of glucose and other nutrients. These hormones are produced in the **islets of Langerhans**, clusters of endocrine cells scattered throughout the pancreas. The two most important pancreatic hormones, insulin and glucagon, are discussed later in this chapter.

The Liver and Gallbladder

The liver is the largest internal organ of the body (see figure 51.4). In an adult human, the liver weighs about 1.5 kilograms and is the size of a football. The main exocrine secretion of the liver is bile, a fluid mixture consisting of *bile pigments* and *bile salts* that is delivered into the duodenum during the digestion of a meal. The bile pigments do not participate in digestion; they are waste products resulting from the liver's destruction of old red blood cells and

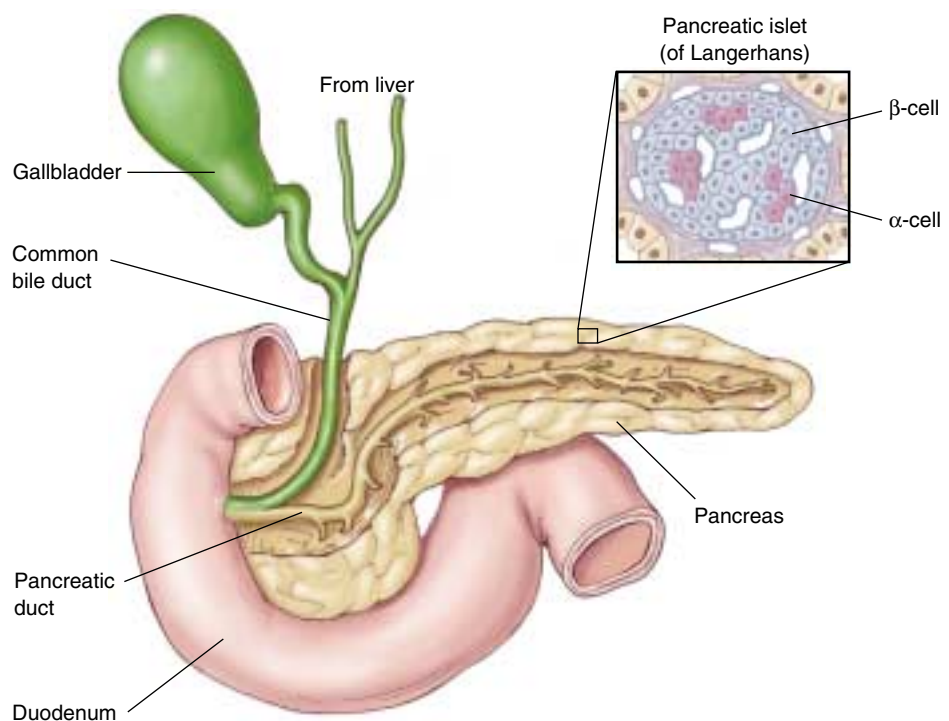


FIGURE 51.19

The pancreas and bile duct empty into the duodenum. The pancreas secretes pancreatic juice into the pancreatic duct. The pancreatic islets of Langerhans secrete hormones into the blood; α -cells secrete glucagon and β -cells secrete insulin.

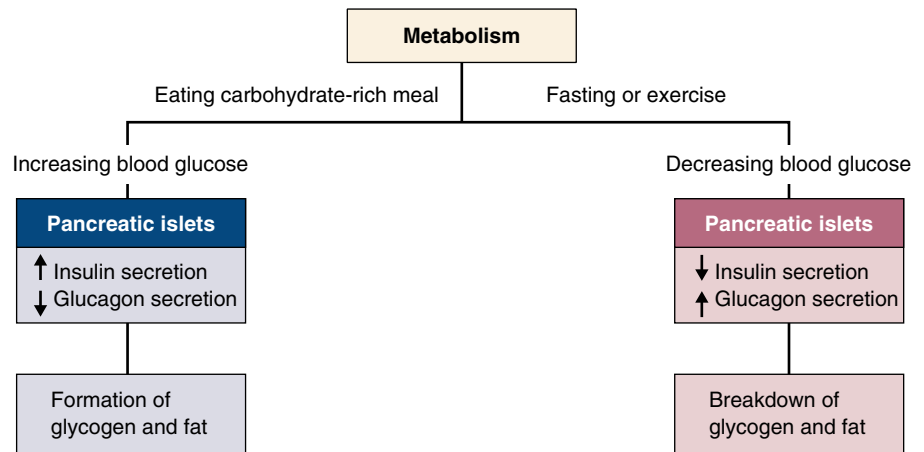
ultimately are eliminated with the feces. If the excretion of bile pigments by the liver is blocked, the pigments can accumulate in the blood and cause a yellow staining of the tissues known as *jaundice*.

In contrast, the bile salts play a very important role in the digestion of fats. Because fats are insoluble in water, they enter the intestine as drops within the watery chyme. The bile salts, which are partly lipid-soluble and partly water-soluble, work like detergents, dispersing the large drops of fat into a fine suspension of smaller droplets. This emulsification process produces a greater surface area of fat upon which the lipase enzymes can act, and thus allows the digestion of fat to proceed more rapidly.

After it is produced in the liver, bile is stored and concentrated in the gallbladder. The arrival of fatty food in the duodenum triggers a neural and endocrine reflex (discussed later) that stimulates the gallbladder to contract, causing bile to be transported through the common bile duct and injected into the duodenum. If the bile duct is blocked by a *gallstone* (formed from a hardened precipitate of cholesterol), contraction of the gallbladder will cause pain generally felt under the right scapula (shoulder blade).

FIGURE 51.20**The actions of insulin and glucagon.**

After a meal, an increased secretion of insulin by the β cells of the pancreatic islets promotes the deposition of glycogen and fat. During fasting or exercising, an increased glucagon secretion by the α cells of the pancreatic islets and a decreased insulin secretion promote the breakdown (through hydrolysis reactions) of glycogen and fat.

**Regulatory Functions of the Liver**

Because the hepatic portal vein carries blood from the stomach and intestine directly to the liver, the liver is in a position to chemically modify the substances absorbed in the gastrointestinal tract before they reach the rest of the body. For example, ingested alcohol and other drugs are taken into liver cells and metabolized; this is why the liver is often damaged as a result of alcohol and drug abuse. The liver also removes toxins, pesticides, carcinogens, and other poisons, converting them into less toxic forms. An important example of this is the liver's conversion of the toxic ammonia produced by intestinal bacteria into urea, a compound that can be contained safely and carried by the blood at higher concentrations.

Similarly, the liver regulates the levels of many compounds produced within the body. Steroid hormones, for instance, are converted into less active and more water-soluble forms by the liver. These molecules are then included in the bile and eliminated from the body in the feces, or carried by the blood to the kidneys and excreted in the urine.

The liver also produces most of the proteins found in blood plasma. The total concentration of plasma proteins is significant because it must be kept within normal limits in order to maintain osmotic balance between blood and interstitial (tissue) fluid. If the concentration of plasma proteins drops too low, as can happen as a result of liver disease such as cirrhosis, fluid accumulates in the tissues, a condition called *edema*.

Regulation of Blood Glucose Concentration

The neurons in the brain obtain their energy primarily from the aerobic respiration of glucose obtained from the blood plasma. It is therefore extremely important that the

blood glucose concentration not fall too low, as might happen during fasting or prolonged exercise. It is also important that the blood glucose concentration not stay at too high a level, as it does in people with uncorrected *diabetes mellitus*, because this can lead to tissue damage.

After a carbohydrate-rich meal, the liver and skeletal muscles remove excess glucose from the blood and store it as the polysaccharide glycogen. This process is stimulated by the hormone insulin, secreted by the β (*beta*) cells in the islets of Langerhans of the pancreas. When blood glucose levels decrease, as they do between meals, during periods of fasting, and during exercise, the liver secretes glucose into the blood. This glucose is obtained in part from the breakdown of liver glycogen to glucose-6-phosphate, a process called glycogenolysis. The phosphate group is then removed, and free glucose is secreted into the blood. Skeletal muscles lack the enzyme needed to remove the phosphate group, and so, even though they have glycogen stores, they cannot secrete glucose into the blood. The breakdown of liver glycogen is stimulated by another hormone, glucagon, which is secreted by the α (*alpha*) cells of the islets of Langerhans in the pancreas (figure 51.20).

If fasting or exercise continues, the liver begins to convert other molecules, such as amino acids and lactic acid, into glucose. This process is called **gluconeogenesis** ("new formation of glucose"). The amino acids used for gluconeogenesis are obtained from muscle protein, which explains the severe muscle wasting that occurs during prolonged fasting.

The pancreas secretes digestive enzymes and bicarbonate into the pancreatic duct. The liver produces bile, which is stored and concentrated in the gallbladder. The liver and the pancreatic hormones regulate blood glucose concentration.

Neural and Hormonal Regulation of Digestion

The activities of the gastrointestinal tract are coordinated by the nervous system and the endocrine system. The nervous system, for example, stimulates salivary and gastric secretions in response to the sight and smell of food. When food arrives in the stomach, proteins in the food stimulate the secretion of a stomach hormone called gastrin (table 51.2), which in turn stimulates the secretion of pepsinogen and HCl from the gastric glands (figure 51.21). The secreted HCl then lowers the pH of the gastric juice, which acts to inhibit further secretion of gastrin. Because inhibition of gastrin secretion will reduce the amount of HCl released into the gastric juice, a negative feedback loop is completed. In this way, the secretion of gastric acid is kept under tight control.

The passage of chyme from the stomach into the duodenum inhibits the contractions of the stomach, so that no additional chyme can enter the duodenum until the previous amount has been processed. This inhibition is mediated by a neural reflex and by a hormone secreted by the small intestine that inhibits gastric emptying. The hormone is known generically as an enterogastrone (*entero* refers to the intestine; *gastro* to the stomach). The chemical identity of the enterogastrone is currently controver-

sial. A hormone known as gastric inhibitory peptide (GIP), released by the duodenum, was named for this function but may not be the only, or even the major, enterogastrone. The secretion of enterogastrone is stimulated most strongly by the presence of fat in the chyme. Fatty meals therefore remain in the stomach longer than meals low in fat.

The duodenum secretes two additional hormones. Cholecystokinin (CCK), like enterogastrone, is secreted in response to the presence of fat in the chyme. CCK stimulates the contractions of the gallbladder, injecting bile into the duodenum so that fat can be emulsified and more efficiently digested. The other duodenal hormone is secretin. Released in response to the acidity of the chyme that arrives in the duodenum, secretin stimulates the pancreas to release bicarbonate, which then neutralizes some of the acidity. Secretin has the distinction of being the first hormone ever discovered.

Neural and hormonal reflexes regulate the activity of the digestive system. The stomach's secretions are regulated by food and by the hormone gastrin. Other hormones, secreted by the duodenum, inhibit stomach emptying and promote the release of bile from the gallbladder and the secretion of bicarbonate in pancreatic juice.

Table 51.2 Hormones of Digestion					
Hormone	Class	Source	Stimulus	Action	Note
Gastrin	Polypeptide	Pyloric portion of stomach	Entry of food into stomach	Stimulates secretion of HCl and pepsinogen by stomach	Unusual in that it acts on same organ that secretes it
Cholecystokinin	Polypeptide	Duodenum	Fatty chyme in duodenum	Stimulates gallbladder contraction and secretion of digestive enzymes by pancreas	Structurally similar to gastrin
Gastric inhibitory peptide	Polypeptide	Duodenum	Fatty chyme in duodenum	Inhibits stomach emptying	Also stimulates insulin secretion
Secretin	Polypeptide	Duodenum	Acidic chyme in duodenum	Stimulates secretion of bicarbonate by pancreas	The first hormone to be discovered (1902)

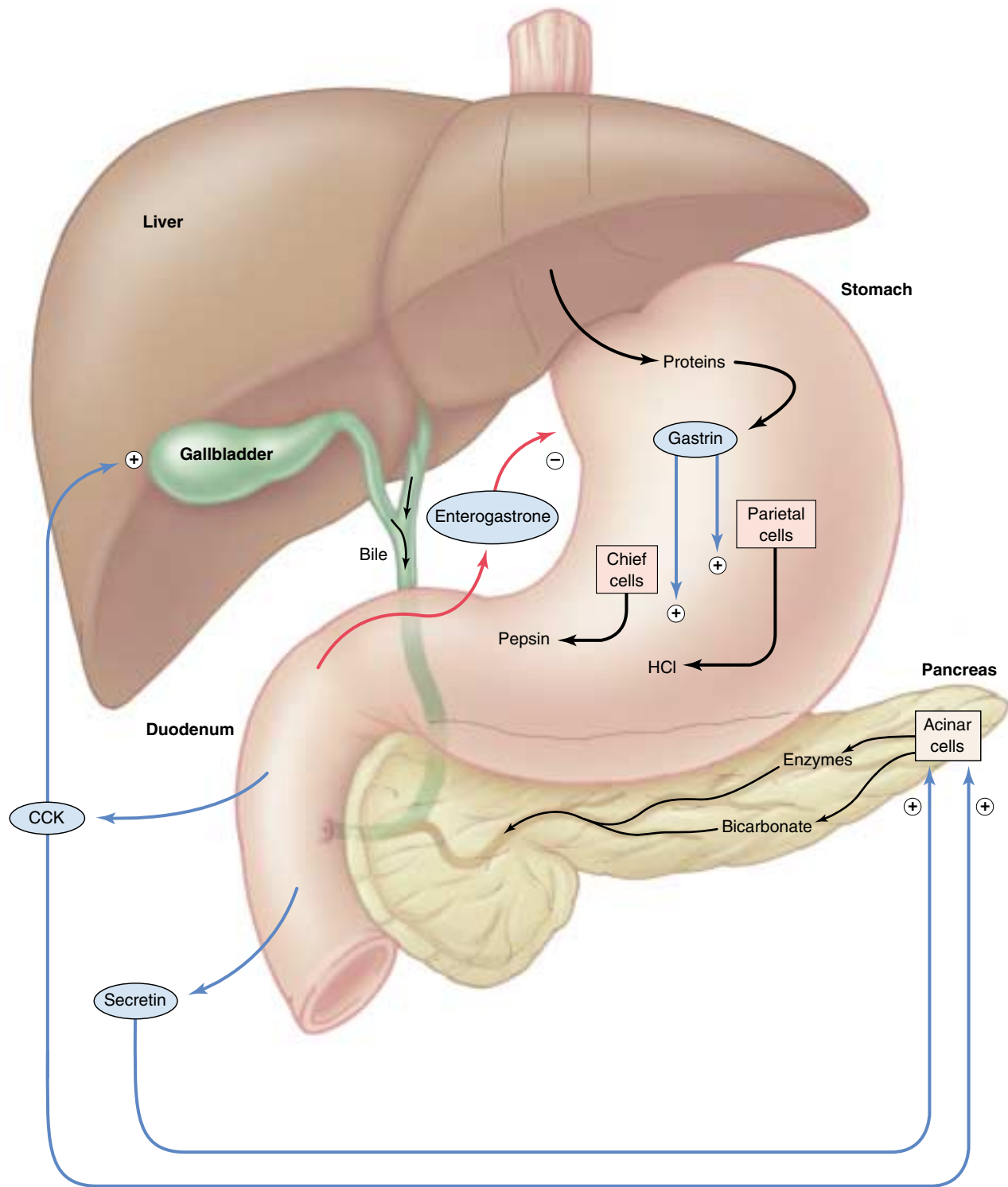


FIGURE 51.21

Hormonal control of the gastrointestinal tract. Gastrin is secreted by the mucosa of the stomach and stimulates the secretion of pepsinogen (which is converted into pepsin) and HCl. The duodenum secretes three hormones: cholecystokinin (CCK), which stimulates contraction of the gallbladder and secretion of pancreatic enzymes; secretin, which stimulates secretion of pancreatic bicarbonate; and an enterogastrone, which inhibits stomach emptying.

51.5 All animals require food energy and essential nutrients.

Food Energy and Energy Expenditure

The ingestion of food serves two primary functions: it provides a source of energy, and it provides raw materials the animal is unable to manufacture for itself. Even an animal that is completely at rest requires energy to support its metabolism; the minimum rate of energy consumption under defined resting conditions is called the basal metabolic rate (BMR). The BMR is relatively constant for a given individual, depending primarily on the person's age, sex, and body size.

Exercise raises the metabolic rate above the basal levels, so the amount of energy that the body consumes per day is determined not only by the BMR but also by the level of physical activity. If food energy taken in is greater than the energy consumed per day, the excess energy will be stored in glycogen and fat. Because glycogen reserves are limited, however, continued ingestion of excess food energy results primarily in the accumulation of fat. The intake of food energy is measured in kilocalories (1 kilocalorie = 1000 calories; nutritionists use Calorie with a capital C instead of kilocalorie). The measurement of kilocalories in food is determined by the amount of heat generated when the food is "burned," either literally, or when the caloric content of food is measured using a calorimeter, or in the body when the food is digested. Caloric intake can be altered by the choice of diet, and the amount of energy expended in exercise can be changed by the choice of lifestyle. The daily energy expenditures (metabolic rates) of people vary between 1300 and 5000 kilocalories per day, depending on the person's BMR and level of physical activity. If the food kilocalories ingested exceed the metabolic rate for a sustained period, the person will accumulate an amount of fat that is deleterious to health, a condition called **obesity**. In the United States, about 30% of middle-aged women and 15% of middle-aged men are classified as obese, which means they weigh at least 20% more than the average weight for their height.

Regulation of Food Intake

Scientists have for years suspected that adipose tissue secretes a hormonal *satiety factor* (a circulating chemical that decreases appetite), because genetically obese mice lose weight when their circulatory systems are surgically joined with those of normal mice. Apparently, some weight-loss hormone was passing into the obese mice! The satiety factor secreted by adipose tissue has recently been identified. It is the product of a gene first observed in a strain of mice known as *ob/ob* (*ob* stands for "obese"; the double symbols indicate that the mice are homozygous for this gene—they inherit it from both parents). The *ob* gene has been cloned in mice, and more recently



FIGURE 51.22

Injection of the hormone leptin causes genetically obese mice to lose weight. These two mice are identical twins, both members of a mutant strain of obese mice. The mouse on the right has been injected with the hormone leptin. It lost 30% of its body weight in just two weeks, with no apparent side effects.

in humans, and has been found to be expressed (that is, to produce mRNA) only in adipocytes. The protein product of this gene, the presumed satiety factor, is called leptin. The *ob* mice produce a mutated and ineffective form of leptin, and it is this defect that causes their obesity. When injected with normal leptin, they stop eating and lose weight (figure 51.22).

More recent studies in humans show that the activity of the *ob* gene and the blood concentrations of leptin are actually higher in obese than in lean people, and that the leptin produced by obese people appears to be normal. It has therefore been suggested that most cases of human obesity may result from a reduced sensitivity to the actions of leptin in the brain rather than from reduced leptin production by adipose cells. Aggressive research is ongoing, as might be expected from the possible medical and commercial applications of these findings.

In the United States, serious eating disorders have become much more common since the mid-1970s. The most common of these disorders are *anorexia nervosa*, a condition in which the afflicted individuals literally starve themselves, and *bulimia*, in which individuals gorge themselves and then vomit, so that their weight stays constant. Ninety to 95% of those suffering from these disorders are female; researchers estimate that 2 to 5% of the adolescent girls and young women in the United States have eating disorders.

The amount of caloric energy expended by the body depends on the basal metabolic rate and the additional calories consumed by exercise. Obesity results if the ingested food energy exceeds the energy expenditure by the body over a prolonged period.

Essential Nutrients

Over the course of their evolution, many animals have lost the ability to synthesize specific substances that nevertheless continue to play critical roles in their metabolism. Substances that an animal cannot manufacture for itself but which are necessary for its health must be obtained in the diet and are referred to as **essential nutrients**.

Included among the essential nutrients are *vitamins*, certain organic substances required in trace amounts. Humans, apes, monkeys, and guinea pigs, for example, have lost the ability to synthesize ascorbic acid (vitamin C). If vitamin C is not supplied in sufficient quantities in their diets, they will develop scurvy, a potentially fatal disease. Humans require at least 13 different vitamins (table 51.3).

Some essential nutrients are required in more than trace amounts. Many vertebrates, for example, are unable to synthesize 1 or more of the 20 amino acids used in making proteins. These *essential amino acids* must be obtained from proteins in the food they eat. There are nine essential amino acids for humans. People who are vegetarians must

choose their foods so that the essential amino acids in one food complement those in another.

In addition, all vertebrates have lost the ability to synthesize certain unsaturated fatty acids and therefore must obtain them in food. On the other hand, some essential nutrients that vertebrates can synthesize cannot be manufactured by the members of other animal groups. For example, vertebrates can synthesize cholesterol, a key component of steroid hormones, but some carnivorous insects cannot.

Food also supplies **essential minerals** such as calcium, phosphorus, and other inorganic substances, including a wide variety of *trace elements* such as zinc and molybdenum which are required in very small amounts (see table 2.1). Animals obtain trace elements either directly from plants or from animals that have eaten plants.

The body requires vitamins and minerals obtained in food. Also, food must provide particular essential amino acids and fatty acids that the body cannot manufacture by itself.

Table 51.3 Major Vitamins

Vitamin	Function	Source	Deficiency Symptoms
Vitamin A (retinol)	Used in making visual pigments, maintenance of epithelial tissues	Green vegetables, milk products, liver	Night blindness, flaky skin
B-complex vitamins			
B ₁	Coenzyme in CO ₂ removal during cellular respiration	Meat, grains, legumes	Beriberi, weakening of heart, edema
B ₂ (riboflavin)	Part of coenzymes FAD and FMN, which play metabolic roles	Many different kinds of foods	Inflammation and breakdown of skin, eye irritation
B ₃ (niacin)	Part of coenzymes NAD ⁺ and NADP ⁺	Liver, lean meats, grains	Pellagra, inflammation of nerves, mental disorders
B ₅ (pantothenic acid)	Part of coenzyme-A, a key connection between carbohydrate and fat metabolism	Many different kinds of foods	Rare: fatigue, loss of coordination
B ₆ (pyridoxine)	Coenzyme in many phases of amino acid metabolism	Cereals, vegetables, meats	Anemia, convulsions, irritability
B ₁₂ (cyanocobalamin)	Coenzyme in the production of nucleic acids	Red meats, dairy products	Pernicious anemia
Biotin	Coenzyme in fat synthesis and amino acid metabolism	Meat, vegetables	Rare: depression, nausea
Folic acid	Coenzyme in amino acid and nucleic acid metabolism	Green vegetables	Anemia, diarrhea
Vitamin C	Important in forming collagen, cement of bone, teeth, connective tissue of blood vessels; may help maintain resistance to infection	Fruit, green leafy vegetables	Scurvy, breakdown of skin, blood vessels
Vitamin D (calciferol)	Increases absorption of calcium and promotes bone formation	Dairy products, cod liver oil	Rickets, bone deformities
Vitamin E (tocopherol)	Protects fatty acids and cell membranes from oxidation	Margarine, seeds, green leafy vegetables	Rare
Vitamin K	Essential to blood clotting	Green leafy vegetables	Severe bleeding

**Summary****Questions****Media Resources****51.1 Animals employ a digestive system to prepare food for assimilation by cells.**

- The digestive system of vertebrates consists of a gastrointestinal tract and accessory digestive organs.
- Different regions of the digestive tract display specializations of structure and function.

1. What are the layers that make up the wall of the vertebrate gastrointestinal tract? What type of tissue is found in each layer?



- Art Activity: Digestive tract wall



- Introduction to digestion
- Human digestion
- Digestion overview

51.2 Food is ingested, swallowed, and transported to the stomach.

- The teeth of carnivores are different from those of herbivores
- The esophagus contracts in peristaltic waves to drive the swallowed food to the stomach.
- Cells of the gastric mucosa secrete hydrochloric acid, which activates pepsin, an enzyme that promotes the partial hydrolysis of ingested proteins.

2. How does tooth structure vary among carnivores, herbivores, and omnivores?
3. What normally prevents regurgitation in humans, and why can't horses regurgitate?
4. What inorganic substance is secreted by parietal cells?



- Art Activities: Digestive system
Mouth
Teeth
Swallowing
Glottis function

51.3 The small and large intestines have very different functions.

- The duodenum receives pancreatic juice and bile, which help digest the chyme that arrives from the stomach through the pyloric valve.
- Digestive enzymes in the small intestine finish the breakdown of food into molecules that can be absorbed by the small intestine.
- The large intestine absorbs water and ions, as well as certain organic molecules such as vitamin K; the remaining material passes out of the anus.

5. How are the products of protein and carbohydrate digestion absorbed across the intestinal wall, and where do they go after they are absorbed?
6. What anatomical and behavioral specializations do ruminants have for making use of microorganisms?



- Art Activities: Small intestine anatomy
Hepatic lobules



- Stomach to small intestine
- Small intestine digestion

51.4 Accessory organs, neural stimulation, and endocrine secretions assist in digestion.

- Pancreatic juice contains bicarbonate to neutralize the acid chyme from the stomach. Bile contains bile pigment and bile salts, which emulsify fat. The liver metabolizes toxins and hormones that are delivered to it in the hepatic portal vein; the liver also helps to regulate the blood glucose concentration.
- The stomach secretes the hormone gastrin, and the small intestine secretes various hormones that help to regulate the digestive system.

7. What are the main exocrine secretions of the pancreas, and what are their functions?
8. What is the function of bile salts in digestion?
9. Describe the role of gastrin and secretin in digestion.



- Art Activity: Digestive system



- formation of gallstones

51.5 All animals require food energy and essential nutrients.

- The basal metabolic rate (BMR) is the lowest level of energy consumption of the body.
- Vitamins, minerals, and the essential amino acids and fatty acids must be supplied in the diet.

10. What is a vitamin? What is the difference between an essential amino acid and any other amino acid?



- Nutrition

52

Circulation

Concept Outline

52.1 The circulatory systems of animals may be open or closed.

Open and Closed Circulatory Systems. All vertebrates have a closed circulation, while many invertebrate animals have open circulatory systems.

52.2 A network of vessels transports blood through the body.

The Blood Plasma. The blood plasma transports a variety of solutes, including ions, metabolites, proteins, and hormones.

The Blood Cells. The blood cells include erythrocytes, which transport oxygen, leukocytes, which provide defenses for the body, and platelets, which function in blood clotting.

Characteristics of Blood Vessels. Blood leaves the heart in arteries and returns in veins; in between, the blood passes through capillaries, where all exchanges with tissues occur.

The Lymphatic System. The lymphatic system returns interstitial fluid to the bloodstream.

52.3 The vertebrate heart has undergone progressive evolutionary change.

The Fish Heart. The fish heart consists of a row of four chambers that receives blood in the posterior end from the body and pumps blood from the anterior end to the gills.

Amphibian and Reptile Circulation. Land vertebrates have a double circulation, where blood from the lungs returns to the heart to be pumped to the rest of the body.

Mammalian and Bird Hearts. Mammals and birds have a complete separation between the two sides of the heart.

52.4 The cardiac cycle drives the cardiovascular system.

The Cardiac Cycle. The right and left sides of the heart rest and receive blood at the same time, then pump the blood into arteries at the same time.

Electrical Excitation and Contraction of the Heart. The impulse begins in one area of the heart and is conducted to the rest of the heart.

Blood Flow and Blood Pressure. Blood flow and blood pressure depend on the diameter of the arterial vessels and on the amount of blood pumped by the heart.



FIGURE 52.1

Red blood cells. This ruptured blood vessel, seen in a scanning electron micrograph, is full of red blood cells, which move through vessels transporting oxygen from one place to another in the body.

Every cell in the animal body must acquire the energy it needs for living from other molecules in food. Like residents of a city whose food is imported from farms in the countryside, cells in the body need trucks to carry the food, highways for the trucks to travel on, and a way to cook the food when it arrives. In animals, the circulatory system provides blood and blood vessels (the trucks and highways), and is discussed in this chapter (figure 52.1). The respiratory system provides the glucose (fuel) and oxygen (fuel to cook the food), and will be discussed in the following chapter.

52.1 The circulatory systems of animals may be open or closed.

Open and Closed Circulatory Systems

Among the unicellular protists, oxygen and nutrients are obtained directly from the aqueous external environment by simple diffusion. The body wall is only two cell layers thick in cnidarians, such as *Hydra*, and flatworms, such as *Planaria*. Each cell layer is in direct contact with either the external environment or the gastrovascular cavity (figure 52.2a). The gastrovascular cavity of *Hydra* (see chapter 51) extends from the body cavity into the tentacles, and that of *Planaria* branches extensively to supply every cell with oxygen and nutrients. Larger animals, however, have tissues that are several cell layers thick, so that many cells are too far away from the body surface or digestive cavity to exchange materials directly with the environment. Instead, oxygen and nutrients are transported from the environment and digestive cavity to the body cells by an internal fluid within a *circulatory system*.

There are two main types of circulatory systems: *open* or *closed*. In an **open circulatory system**, such as that found in mollusks and arthropods (figure 52.2b), there is no distinction between the circulating fluid (blood) and the extracellular fluid of the body tissues (interstitial fluid or lymph). This fluid is thus called **hemolymph**. In insects, the heart is a muscular tube that pumps hemolymph through a network of channels and cavities in the body. The fluid then drains back into the central cavity.

In a **closed circulatory system**, the circulating fluid, or blood, is always enclosed within blood vessels that transport blood away from and back to a pump, the **heart**. Annelids (see chapter 45) and all vertebrates have a closed circulatory system. In annelids such as an earthworm, a dorsal vessel contracts rhythmically to function as a pump. Blood is pumped through five small connecting arteries which also function as pumps, to a ventral vessel, which transports

the blood posteriorly until it eventually reenters the dorsal vessel. Smaller vessels branch from each artery to supply the tissues of the earthworm with oxygen and nutrients and to transport waste products (figure 52.2c).

The Functions of Vertebrate Circulatory Systems

The vertebrate circulatory system is more elaborate than the invertebrate circulatory system. It functions in transporting oxygen and nutrients to tissues by the cardiovascular system. Blood vessels form a tubular network that permits blood to flow from the heart to all the cells of the body and then back to the heart. *Arteries* carry blood away from the heart, whereas *veins* return blood to the heart. Blood passes from the arterial to the venous system in *capillaries*, which are the thinnest and most numerous of the blood vessels.

As blood plasma passes through capillaries, the pressure of the blood forces some of this fluid out of the capillary walls. Fluid derived from plasma that passes out of capillary walls into the surrounding tissues is called **interstitial fluid**. Some of this fluid returns directly to capillaries, and some enters into **lymph vessels**, located in the connective tissues around the blood vessels. This fluid, now called *lymph*, is returned to the venous blood at specific sites. The lymphatic system is considered a part of the circulatory system and is discussed later in this chapter.

The vertebrate circulatory system has three principal functions: transportation, regulation, and protection.

1. Transportation. All of the substances essential for cellular metabolism are transported by the circulatory system. These substances can be categorized as follows:

a. Respiratory. Red blood cells, or *erythrocytes*, transport oxygen to the tissue cells. In the capillaries of

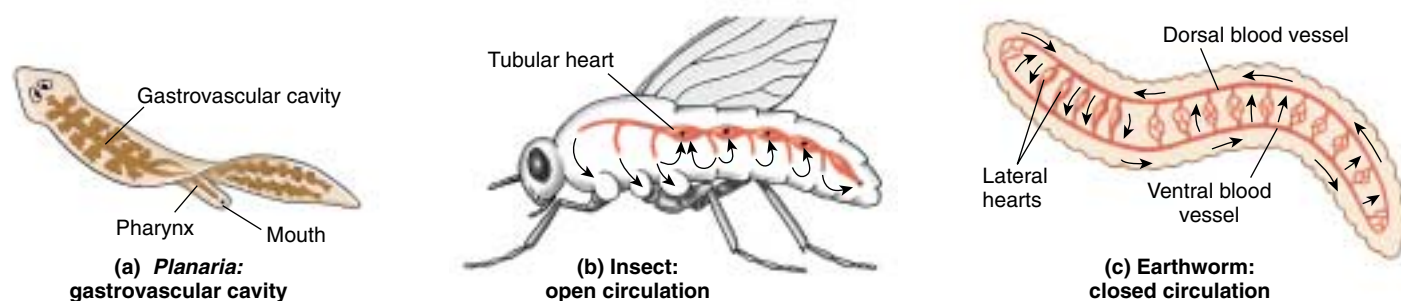


FIGURE 52.2

Circulatory systems of the animal kingdom. (a) The gastrovascular cavity of *Planaria* serves as both a digestive and circulatory system, delivering nutrients directly to the tissue cells by diffusion from the digestive cavity. (b) In the open circulation of an insect, hemolymph is pumped from a tubular heart into cavities in the insect's body; the hemolymph then returns to the blood vessels so that it can be recirculated. (c) In the closed circulation of the earthworm, blood pumped from the hearts remains within a system of vessels that returns it to the hearts. All vertebrates also have closed circulatory systems.

the lungs or gills, oxygen attaches to hemoglobin molecules within the erythrocytes and is transported to the cells for aerobic respiration. Carbon dioxide, a by-product of cell respiration, is carried by the blood to the lungs or gills for elimination.

b. Nutritive. The digestive system is responsible for the breakdown of food into molecules so that nutrients can be absorbed through the intestinal wall and into the blood vessels of the circulatory system. The blood then carries these absorbed products of digestion through the liver and to the cells of the body.

c. Excretory. Metabolic wastes, excessive water and ions, and other molecules in the fluid portion of blood are filtered through the capillaries of the kidneys and excreted in urine.

2. Regulation. The cardiovascular system transports regulatory hormones and participates in temperature regulation.

a. Hormone transport. The blood carries hormones from the endocrine glands, where they are secreted, to the distant target organs they regulate.

b. Temperature regulation. In warm-blooded vertebrates, or **endotherms**, a constant body temperature is maintained regardless of the ambient temperature. This is accomplished in part by blood vessels located just under the epidermis. When the ambient temperature is cold, the superficial vessels constrict to divert the warm blood to deeper vessels. When the ambient temperature is warm, the superficial vessels dilate so that the warmth of the blood can be lost by radiation (figure 52.3).

Some vertebrates also retain heat in a cold environment by using a **countercurrent heat exchange** (also see chapter 53). In this process, a vessel carrying warm blood from deep within the body passes next to a vessel carrying cold blood from the surface of the body (figure 52.4). The warm blood going out heats the cold blood returning from the body surface, so that this blood is no longer cold when it reaches the interior of the body.

3. Protection. The circulatory system protects against injury and foreign microbes or toxins introduced into the body.

a. Blood clotting. The clotting mechanism protects against blood loss when vessels are damaged. This clotting mechanism involves both proteins from the blood plasma and cell fragments called platelets (discussed in the next section).

b. Immune defense. The blood contains white blood cells, or leukocytes, that provide immunity against many disease-causing agents. Some white blood cells are phagocytic, some produce antibodies, and some act by other mechanisms to protect the body.

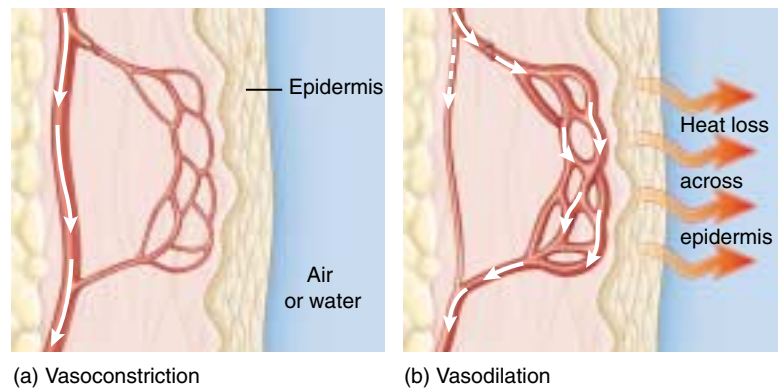


FIGURE 52.3

Regulation of heat loss. The amount of heat lost at the body's surface can be regulated by controlling the flow of blood to the surface. (a) Constriction of surface blood vessels limits flow and heat loss; (b) dilation of these vessels increases flow and heat loss.

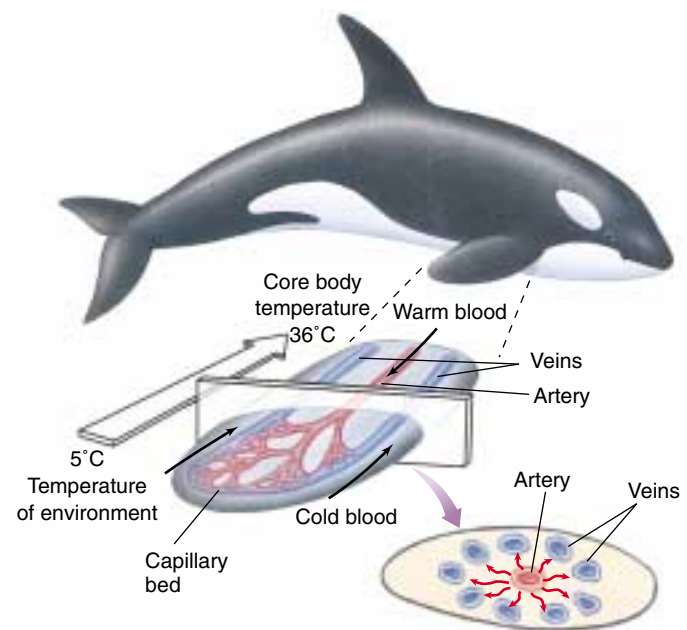


FIGURE 52.4

Countercurrent heat exchange. Many marine animals, such as this killer whale, limit heat loss in cold water using countercurrent heat exchange. The warm blood pumped from within the body in arteries loses heat to the cold blood returning from the skin in veins. This warms the venous blood so that the core body temperature can remain constant in cold water and cools the arterial blood so that less heat is lost when the arterial blood reaches the tip of the extremity.

Circulatory systems may be open or closed. All vertebrates have a closed circulatory system, in which blood circulates away from the heart in arteries and back to the heart in veins. The circulatory system serves a variety of functions, including transportation, regulation, and protection.

52.2 A network of vessels transports blood through the body.

The Blood Plasma

Blood is composed of a fluid **plasma** and several different kinds of cells that circulate within that fluid (figure 52.5). Blood platelets, although included in figure 52.5, are not complete cells; rather, they are fragments of cells that reside in the bone marrow. Blood plasma is the matrix in which blood cells and platelets are suspended. Interstitial (extracellular) fluids originate from the fluid present in plasma.

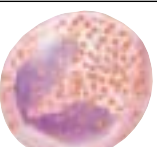
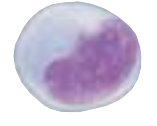
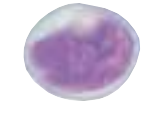
Plasma contains the following solutes:

- 1. Metabolites, wastes, and hormones.** Dissolved within the plasma are all of the metabolites used by cells, including glucose, amino acids, and vitamins. Also dissolved in the plasma are hormones that regulate cellular activities, wastes such as nitrogen compounds, and CO₂ produced by metabolizing cells. CO₂ is carried in the blood as bicarbonate because free carbon dioxide would decrease blood pH.
- 2. Ions.** Like the water of the seas in which life arose, blood plasma is a dilute salt solution. The predominant plasma ions are sodium, chloride, and bicarbonate ions. In addition, there are trace amounts of other ions such as calcium, magnesium, copper, potassium, and zinc. The composition of the plasma, therefore, is similar to seawater, but plasma has a lower total ion concentration than that of present-day seawater.
- 3. Proteins.** The liver produces most of the plasma proteins, including **albumin**, which comprises most of the plasma protein; the alpha (α) and beta (β) **globulins**, which serve as carriers of lipids and steroid hormones; and **fibrinogen**, which is required for blood clotting. Following an injury of a blood vessel, platelets release clotting factors (proteins) into the blood. In the presence of these clotting factors, fibrinogen is converted into insoluble threads of *fibrin*. Fibrin then aggregates to form the clot. Blood plasma which has had fibrinogen removed is called **serum**.

Plasma, the liquid portion of the blood, contains different types of proteins, ions, metabolites, wastes, and hormones. This liquid, and fluids derived from it, provide the extracellular environment of most the cells of the body.

FIGURE 52.5

Types of blood cells. Erythrocytes are red blood cells, platelets are fragments of a bone marrow cell, and all the other cells are different types of leukocytes, or white blood cells.

Blood cell	Life span in blood	Function
 Erythrocyte	120 days	O ₂ and CO ₂ transport
 Neutrophil	7 hours	Immune defenses
 Eosinophil	Unknown	Defense against parasites
 Basophil	Unknown	Inflammatory response
 Monocyte	3 days	Immune surveillance (precursor of tissue macrophage)
 B - lymphocyte	Unknown	Antibody production (precursor of plasma cells)
 T - lymphocyte	Unknown	Cellular immune response
 Platelets	7 - 8 days	Blood clotting

The Blood Cells

Red blood cells function in oxygen transport, white blood cells in immunological defenses, and platelets in blood clotting (see figure 52.5).

Erythrocytes and Oxygen Transport

Each cubic millimeter of blood contains about 5 million **red blood cells**, or **erythrocytes**. The fraction of the total blood volume that is occupied by erythrocytes is called the blood's *hematocrit*; in humans, it is typically around 45%. A disc with a central depression, each erythrocyte resembles a doughnut with a hole that does not go all the way through. As we've already seen, the erythrocytes of vertebrates contain hemoglobin, a pigment which binds and transports oxygen. In vertebrates, hemoglobin is found only in erythrocytes. In invertebrates, the oxygen binding pigment (not always hemoglobin) is also present in plasma.

Erythrocytes develop from unspecialized cells, called *stem cells*. When plasma oxygen levels decrease, the kidney converts a plasma protein into the hormone, *erythropoietin*. Erythropoietin then stimulates the production of erythrocytes in bone marrow. In mammals, maturing erythrocytes lose their nuclei through a process called *erythropoiesis*. This is different from the mature erythrocytes of all other vertebrates, which remain nucleated. As mammalian erythrocytes age, they are removed from the blood by phagocytic cells of the spleen, bone marrow, and liver. Balancing this loss, new erythrocytes are constantly formed in the bone marrow.

Leukocytes Defend the Body

Less than 1% of the cells in human blood are **leukocytes**, or **white blood cells**; there are only 1 or 2 leukocytes for every 1000 erythrocytes. Leukocytes are larger than erythrocytes and have nuclei. Furthermore, leukocytes are not confined to the blood as erythrocytes are, but can migrate out of capillaries into the interstitial (tissue) fluid.

There are several kinds of leukocytes, each of which plays a specific role in defending the body against invading microorganisms and other foreign substances, as described in Chapter 57. **Granular leukocytes** include **neutrophils**, **eosinophils**, and **basophils**, which are named according to the staining properties of granules in their cytoplasm. **Nongranular leukocytes** include **monocytes** and **lymphocytes**. Neutrophils are the most numerous of the leukocytes, followed in order by lymphocytes, monocytes, eosinophils, and basophils.

Platelets Help Blood to Clot

Megakaryocytes are large cells present in bone marrow. Pieces of cytoplasm are pinched off of the megakaryocytes and become **platelets**. Platelets play an important role in

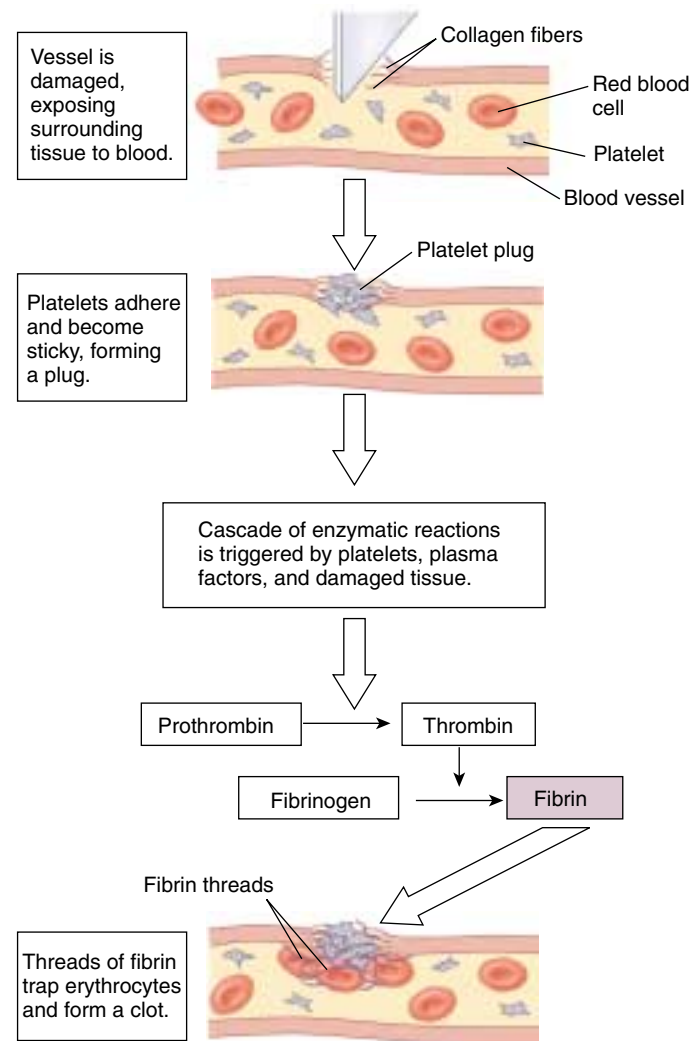


FIGURE 52.6

Blood clotting. Fibrin is formed from a soluble protein, fibrinogen, in the plasma. This reaction is catalyzed by the enzyme thrombin, which is formed from an inactive enzyme called prothrombin. The activation of thrombin is the last step in a cascade of enzymatic reactions that produces a blood clot when a blood vessel is damaged.

blood clotting. When a blood vessel is broken, smooth muscle in the vessel walls contracts, causing the vessel to constrict. Platelets then accumulate at the injured site and form a plug by sticking to each other and to the surrounding tissues. This plug is reinforced by threads of the protein **fibrin** (figure 52.6), which contract to form a tighter mass. The tightening plug of platelets, fibrin, and often trapped erythrocytes constitutes a blood clot.

Erythrocytes contain hemoglobin and serve in oxygen transport. The different types of leukocytes have specialized functions that serve to protect the body from invading pathogens, and the platelets participate in blood clotting.

Characteristics of Blood Vessels

Blood leaves the heart through vessels known as **arteries**. These continually branch, forming a hollow “tree” that enters each of the organs of the body. The finest, microscopically-sized branches of the arterial trees are the **arterioles**. Blood from the arterioles enters the **capillaries** (from the Latin *capillus*, “a hair”), an elaborate latticework of very narrow, thin-walled tubes. After traversing the capillaries, the blood is collected into **venules**; the venules lead to larger vessels called **veins**, which carry blood back to the heart.

Arteries, arterioles, veins, and venules all have the same basic structure (figure 52.7). The innermost layer is an epithelial sheet called the *endothelium*. Covering the endothelium is a thin layer of elastic fibers, a smooth muscle layer, and a connective tissue layer. The walls of these vessels are thus too thick to permit any exchange of materials between the blood and the tissues outside the vessels. The walls of capillaries, however, are made up of only the endothelium, so molecules and ions can leave the blood plasma by diffusion, by filtration through pores in the capillary walls, and by transport through the endothelial cells. Therefore, it is while blood is in the capillaries that gases and metabolites are exchanged with the cells of the body.

Arteries and Arterioles

Arteries function in transporting blood away from the heart. The larger arteries contain extra elastic fibers in their walls, allowing them to recoil each time they receive a volume of blood pumped by the heart. Smaller arteries and arterioles are less elastic, but their disproportionately thick smooth muscle layer enables them to resist bursting.

The vast tree of arteries presents a frictional resistance to blood flow. The narrower the vessel, the greater the frictional resistance to flow. In fact, a vessel that is half the diameter of another has 16 times the frictional resistance! This is because the resistance to blood flow is inversely proportional to the radius of the vessel. Therefore, within the arterial tree, it is the small arteries and arterioles that provide the greatest resistance to blood flow. Contraction of the smooth muscle layer of the arterioles results in **vasoconstriction**, which greatly increases resistance and decreases flow. Relaxation of the smooth muscle layer results in **vasodilation**, decreasing resistance and increasing blood flow to an organ (see figure 52.3).

In addition, blood flow through some organs is regulated by rings of smooth muscle around arterioles near the region where they empty into capillaries. These **precapillary sphincters** (figure 52.8) can close off specific capillary beds completely. For example, the closure of precapillary sphincters in the skin contributes to the vasoconstriction that limits heat loss in cold environments.

Exchange in the Capillaries

Each time the heart contracts, it must produce sufficient pressure to pump blood against the resistance of the arterial tree and into the capillaries. The vast number and extensive branching of the capillaries ensure that *every cell in the body is within 100 μm of a capillary*. On the average, capillaries are about 1 mm long and 8 μm in diameter, only slightly larger than a red blood cell (5 to 7 μm in diameter). Despite the close fit, red blood cells can squeeze through capillaries without difficulty.

Although each capillary is very narrow, there are so many of them that the capillaries have the greatest *total* cross-sectional area of any other type of vessel. Conse-

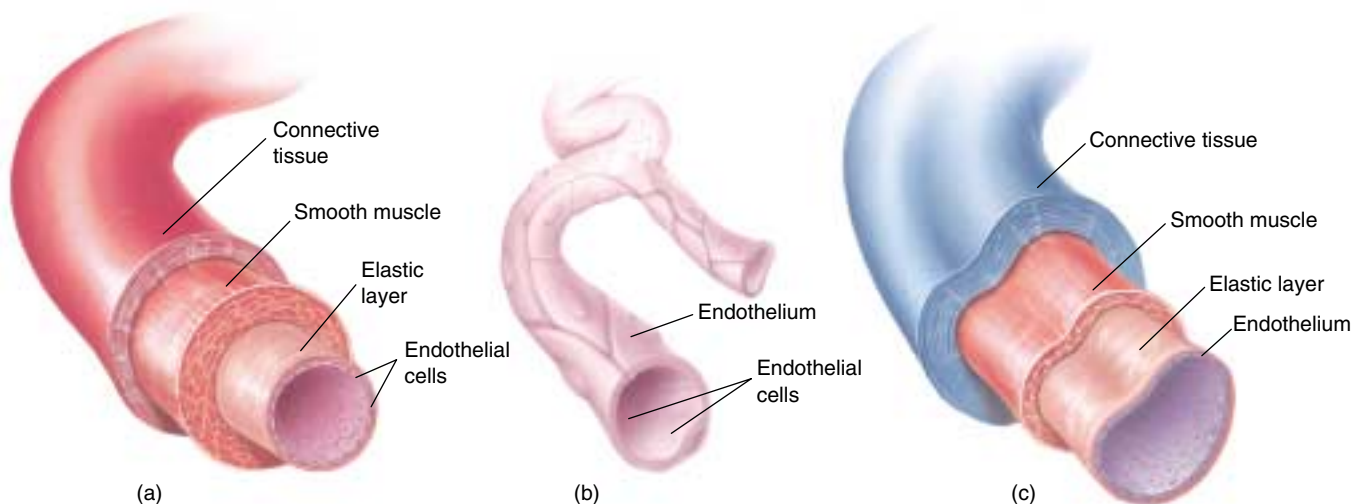
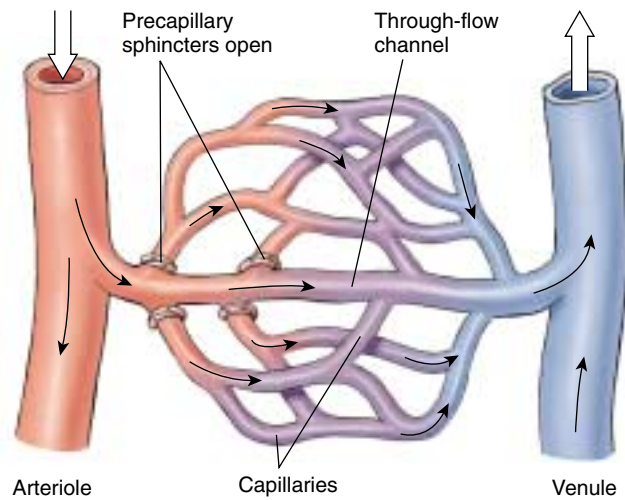
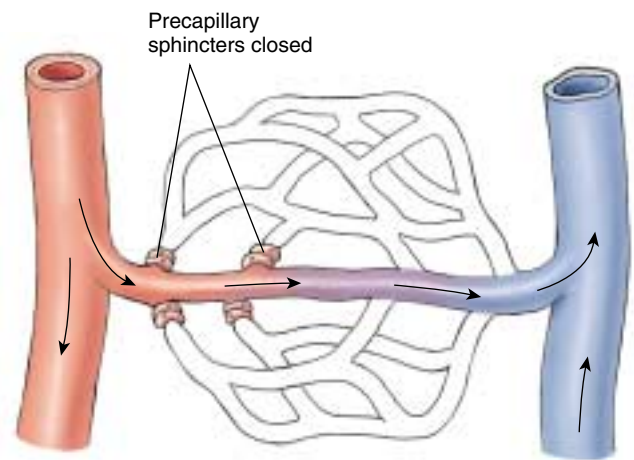


FIGURE 52.7

The structure of blood vessels. (a) Arteries and (c) veins have the same tissue layers. (b) Capillaries are composed of only a single layer of endothelial cells. (not to scale)



(a) Blood flows through capillary network



(b) Blood flow in capillary network is limited

FIGURE 52.8

The capillary network connects arteries with veins. (a) Most of the exchange between the blood and the extracellular fluid occurs while the blood is in the capillaries. Entrance to the capillaries is controlled by bands of muscle called precapillary sphincters at the entrance to each capillary. (b) When a sphincter contracts, it closes off the capillary. By contracting these sphincters, the body can limit the amount of blood in the capillary network of a particular tissue, and thus control the rate of exchange in that tissue.

quently, the blood decreases in velocity as it passes through the capillary beds, allowing more time for it to exchange materials with the surrounding extracellular fluid. By the time the blood reaches the end of a capillary, it has released some of its oxygen and nutrients and picked up carbon dioxide and other waste products. Blood also loses most of its pressure in passing through the vast capillary networks, and so is under very low pressure when it enters the veins.

Venules and Veins

Blood flows from the venules to ever larger veins, and ultimately back to the heart. Venules and veins have the same tissue layers as arteries, but they have a thinner layer of smooth muscle. Less muscle is needed because the pressure in the veins is only about one-tenth that in the arteries. Most of the blood in the cardiovascular system is contained within veins, which can expand when needed to hold additional amounts of blood. You can see the expanded veins in your feet when you stand for a long time.

When the blood pressure in the veins is so low, how does the blood return to the heart from the feet and legs? The venous pressure alone is not sufficient, but several sources provide help. Most significantly, skeletal muscles surrounding the veins can contract to move blood by squeezing the veins. Blood moves in one direction through the veins back to the heart with the help of **venous valves** (figure 52.9). When a person's veins expand too much with blood, the venous valves may no longer work and the blood may pool in the veins. Veins in this condition are known as varicose veins.

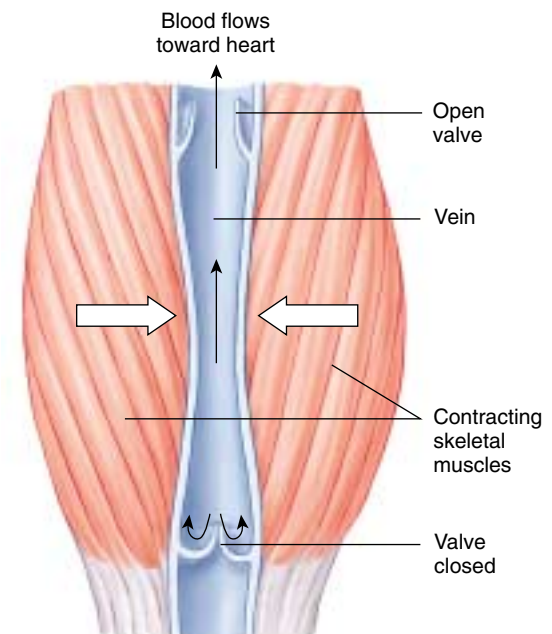


FIGURE 52.9

One-way flow of blood through veins. Venous valves ensure that blood moves through the veins in only one direction, back to the heart.

Blood is pumped from the heart into the arterial system, which branches into fine arterioles. This blood is delivered into the thinnest and most numerous of vessels, the capillaries, where exchanges with the tissues occur. Blood returns to the heart through veins.

The Lymphatic System

The cardiovascular system is considered to be a closed system because all of its vessels are connected with one another—none are simply open-ended. However, some water and solutes in the blood plasma do filter through the walls of the capillaries to form the interstitial (tissue) fluid. This filtration is driven by the pressure of the blood, and it helps supply the tissue cells with oxygen and nutrients. Most of the fluid is filtered from the capillaries near their arteriolar ends, where the blood pressure is higher, and returned to the capillaries near their venular ends. This return of fluid occurs by osmosis, which is driven by a higher solute concentration within the capillaries. Most of the plasma proteins cannot escape through the capillary pores because of their large size and so the concentration of proteins in the plasma is greater than the protein concentration in the interstitial fluid. The difference in protein concentration produces an osmotic pressure, called the *oncotic pressure*, that causes osmosis of water into the capillaries (figure 52.10).

Because interstitial fluid is produced because of the blood pressure, high capillary blood pressure could cause too much interstitial fluid to be produced. A common example of this occurs in pregnant women, when the fetus compresses veins and thereby increases the capillary blood pressure in the mother's lower limbs. The increased interstitial fluid can cause swelling of the tissues, or *edema*, of the feet. Edema may also result if the plasma protein concentration (and thus the oncotic pressure) is too low. Fluids will not return to the capillaries but will remain as interstitial fluid. This may be caused either by liver disease, because the liver produces most of the plasma proteins, or by protein malnutrition (*kwashiorkor*).

Even under normal conditions, the amount of fluid filtered out of the capillaries is greater than the amount that returns to the capillaries by osmosis. The remainder does eventually return to the cardiovascular system, however, by way of an *open* circulatory system called the **lymphatic system**. The lymphatic system consists of lymphatic capillaries, lymphatic vessels, lymph nodes, and lymphatic organs, including the spleen and thymus. Excess fluid in the tissues drains into blind-ended lymph capillaries with highly permeable walls. This fluid, now called **lymph**, passes into progressively larger lymphatic vessels, which resemble veins and have one-way valves (figure 52.11). The lymph eventually enters two major lymphatic vessels, which drain into veins on each side of the neck.

Movement of lymph in mammals is accomplished by skeletal muscles squeezing against the lymphatic vessels, a mechanism similar to the one that moves blood through veins. In some cases, the lymphatic vessels also contract rhythmically. In many fishes, all amphibians and reptiles, bird embryos, and some adult birds, movement of lymph is propelled by **lymph hearts**.

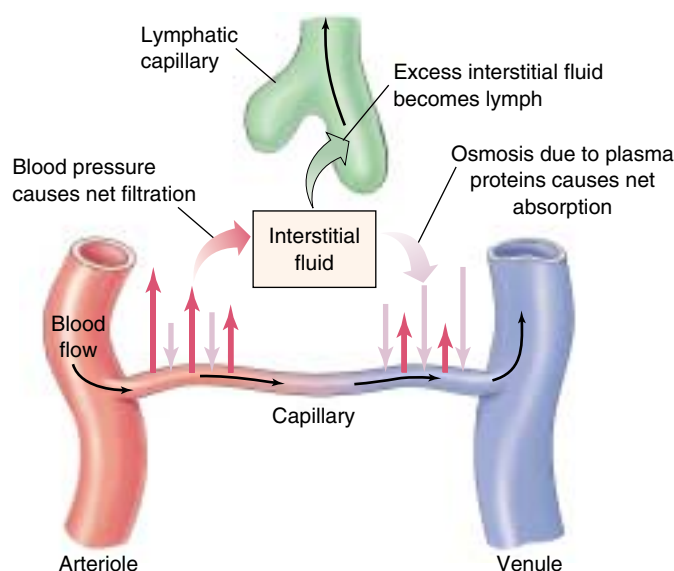


FIGURE 52.10
Plasma fluid, minus proteins, is filtered out of capillaries. This forms interstitial fluid, which bathes the tissues. Much of the interstitial fluid is returned to the capillaries by the osmotic pressure generated by the higher protein concentration in plasma. The excess interstitial fluid is drained into open-ended lymphatic capillaries, which ultimately return the fluid to the cardiovascular system.

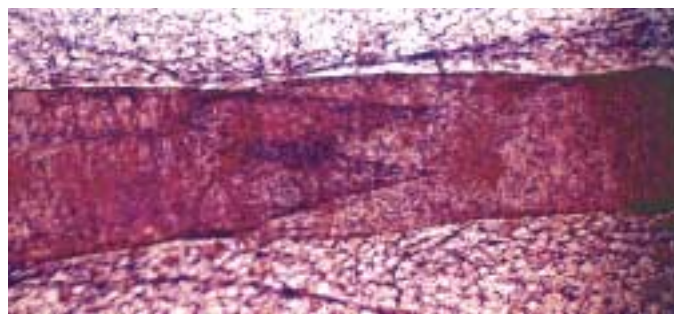


FIGURE 52.11
A lymphatic vessel valve (25 $\times$). Valves allow lymph to flow in one direction (from left to right in this figure) but not in the reverse direction.

As the lymph moves through lymph nodes and lymphatic organs, it is modified by phagocytic cells that line the channels of those organs. In addition, the lymph nodes and lymphatic organs contain *germinal centers* for the production of lymphocytes, a type of white blood cell critically important in immunity.

Lymphatic vessels carry excess interstitial fluid back to the vascular system. This fluid, called lymph, travels through lymph nodes and lymphatic organs where it encounters the immune cells called lymphocytes that are produced in these organs.

52.3 The vertebrate heart has undergone progressive evolutionary change.

The Fish Heart

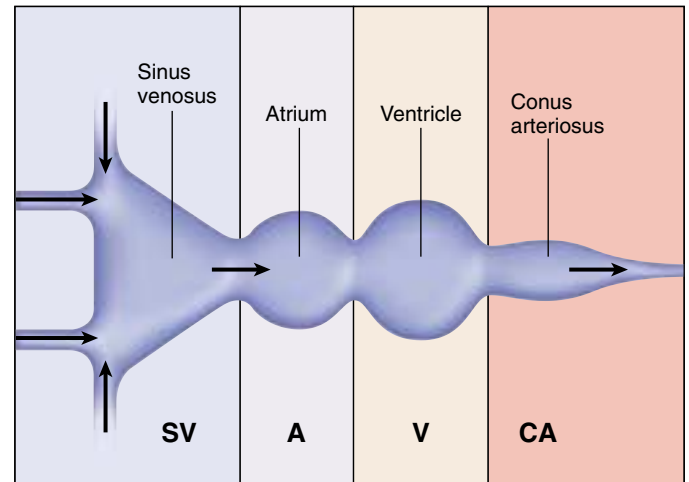
The chordates that were ancestral to the vertebrates are thought to have had simple tubular hearts, similar to those now seen in lancelets (see chapter 48). The heart was little more than a specialized zone of the ventral artery, more heavily muscled than the rest of the arteries, which contracted in simple peristaltic waves. A pumping action results because the uncontracted portions of the vessel have a larger diameter than the contracted portion, and thus present less resistance to blood flow.

The development of gills by fishes required a more efficient pump, and in fishes we see the evolution of a true chamber-pump heart. The fish heart is, in essence, a tube with four chambers arrayed one after the other (figure 52.12*a*). The first two chambers—the **sinus venosus** and **atrium**—are collection chambers, while the second two, the **ventricle** and **conus arteriosus**, are pumping chambers.

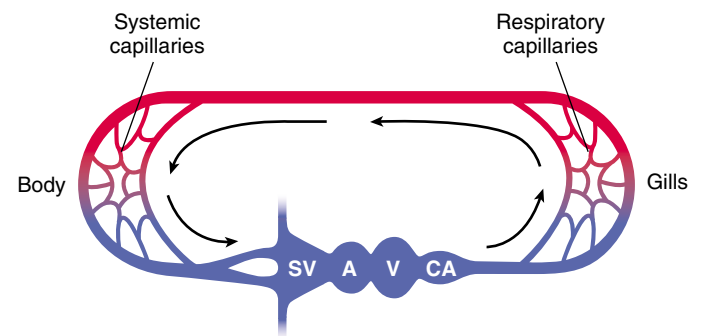
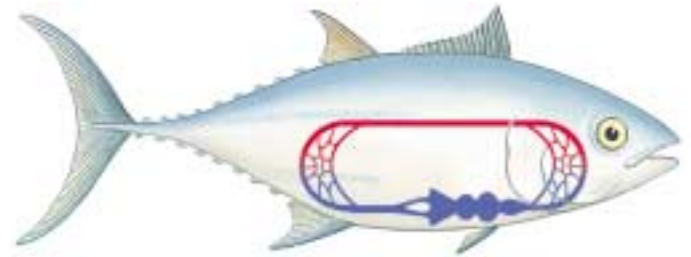
As might be expected, the sequence of the heartbeat in fishes is a peristaltic sequence, starting at the rear and moving to the front, similar to the early chordate heart. The first of the four chambers to contract is the sinus venosus, followed by the atrium, the ventricle, and finally the conus arteriosus. Despite shifts in the relative positions of the chambers in the vertebrates that evolved later, this heartbeat sequence is maintained in all vertebrates. In fish, the electrical impulse that produces the contraction is initiated in the sinus venosus; in other vertebrates, the electrical impulse is initiated by their equivalent of the sinus venosus.

The fish heart is remarkably well suited to the gill respiratory apparatus and represents one of the major evolutionary innovations in the vertebrates. Perhaps its greatest advantage is that the blood that moves through the gills is fully oxygenated when it moves into the tissues. After blood leaves the conus arteriosus, it moves through the gills, where it becomes oxygenated; from the gills, it flows through a network of arteries to the rest of the body; then it returns to the heart through the veins (figure 52.12*b*). This arrangement has one great limitation, however. In passing through the capillaries in the gills, the blood loses much of the pressure developed by the contraction of the heart, so the circulation from the gills through the rest of the body is sluggish. This feature limits the rate of oxygen delivery to the rest of the body.

The fish heart is a modified tube, consisting of a series of four chambers. Blood first enters the heart at the sinus venosus, where the wavelike contraction of the heart begins.



(a)



(b)

FIGURE 52.12

The heart and circulation of a fish. (a) Diagram of a fish heart, showing the chambers in series with each other. (b) Diagram of fish circulation, showing that blood is pumped by the ventricle through the gills and then to the body. Blood rich in oxygen (oxygenated) is shown in red; blood low in oxygen (deoxygenated) is shown in blue.

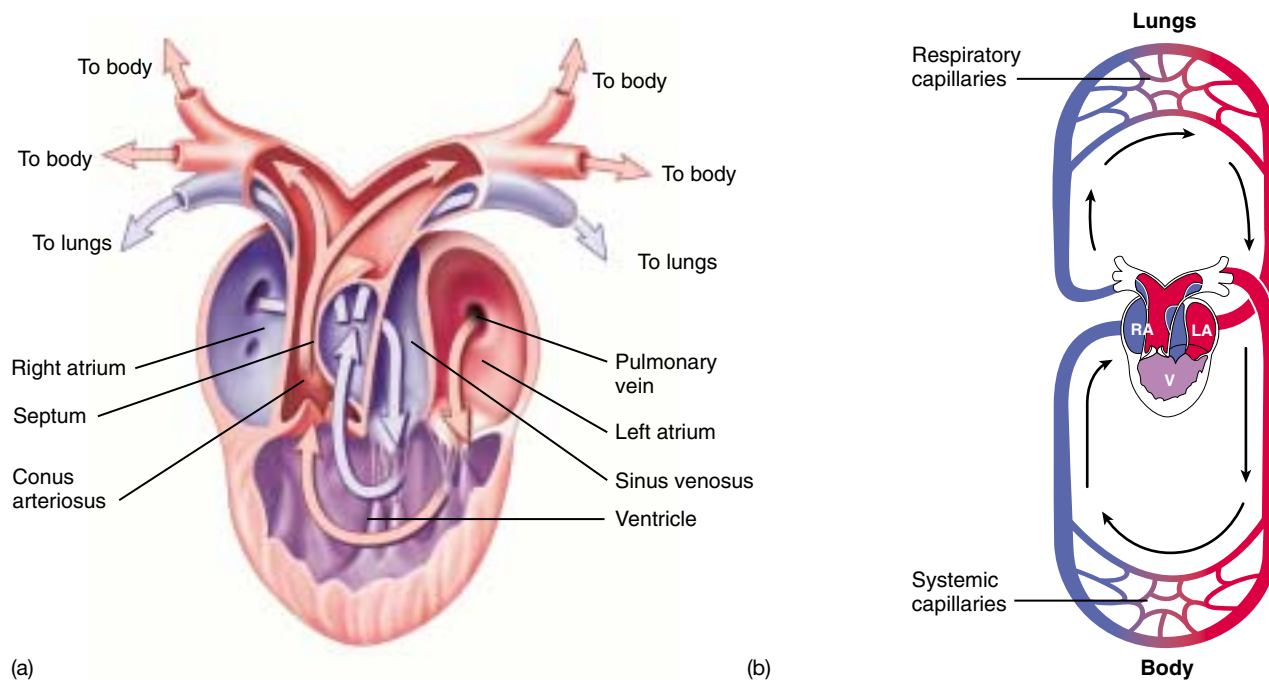


FIGURE 52.13

The heart and circulation of an amphibian. (a) The frog heart has two atria but only one ventricle, which pumps blood both to the lungs and to the body. (b) Despite the potential for mixing, the oxygenated and deoxygenated bloods (red and blue, respectively) mix very little as they are pumped to the body and lungs. The slight mixing is shown in purple. RA = right atrium; LA = left atrium; V = ventricle.

Amphibian and Reptile Circulation

The advent of lungs involved a major change in the pattern of circulation. After blood is pumped by the heart through the *pulmonary arteries* to the lungs, it does not go directly to the tissues of the body but is instead returned via the *pulmonary veins* to the heart. This results in two circulations: one between heart and lungs, called the **pulmonary circulation**, and one between the heart and the rest of the body, called the **systemic circulation**.

If no changes had occurred in the structure of the heart, the oxygenated blood from the lungs would be mixed in the heart with the deoxygenated blood returning from the rest of the body. Consequently, the heart would pump a mixture of oxygenated and deoxygenated blood rather than fully oxygenated blood. The amphibian heart has two structural features that help reduce this mixing (figure 52.13). First, the atrium is divided into two chambers: the right atrium receives deoxygenated blood from the systemic circulation, and the left atrium receives oxygenated blood from the lungs. These two stores of blood therefore do not mix in the atria, and little mixing occurs when the contents of each atrium enter the single, common ventricle, due to internal channels created by recesses in the ventricular wall. The conus arteriosus is partially separated by a dividing wall which directs deoxygenated blood into the pulmonary arteries to the lungs and oxygenated blood into the *aorta*, the major artery of the systemic circulation to the body.

Because there is only one ventricle in an amphibian heart, the separation of the pulmonary and systemic circulations is incomplete. Amphibians in water, however, can obtain additional oxygen by diffusion through their skin. This process, called **cutaneous respiration**, helps to supplement the oxygenation of the blood in these vertebrates.

Among reptiles, additional modifications have reduced the mixing of blood in the heart still further. In addition to having two separate atria, reptiles have a septum that partially subdivides the ventricle. This results in an even greater separation of oxygenated and deoxygenated blood within the heart. The separation is complete in one order of reptiles, the crocodiles, which have two separate ventricles divided by a complete septum. Crocodiles therefore have a completely divided pulmonary and systemic circulation. Another change in the circulation of reptiles is that the conus arteriosus has become incorporated into the trunks of the large arteries leaving the heart.

Amphibians and reptiles have two circulations, pulmonary and systemic, that deliver blood to the lungs and rest of the body, respectively. The oxygenated blood from the lungs is kept relatively separate from the deoxygenated blood from the rest of the body by incomplete divisions within the heart.

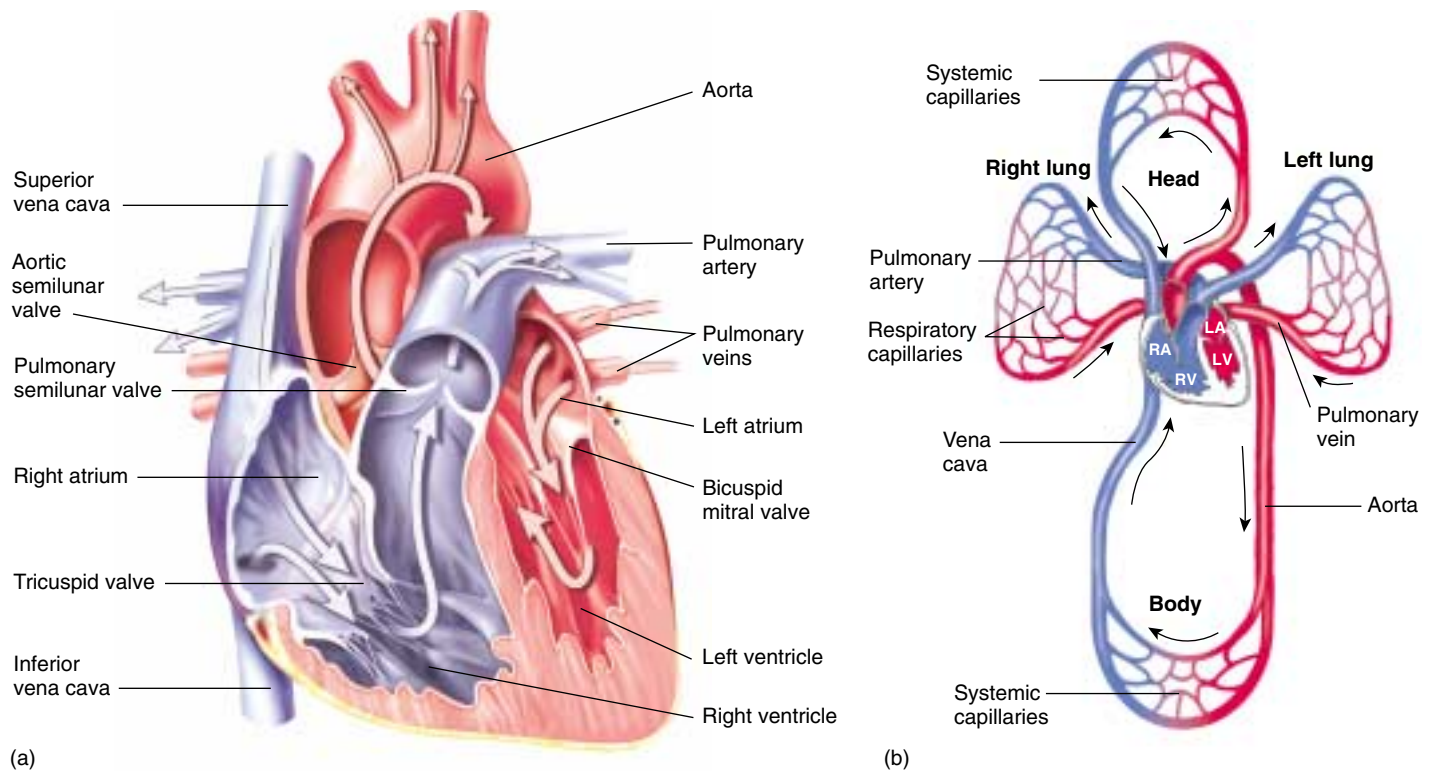


FIGURE 52.14

The heart and circulation of mammals and birds. (a) The path of blood through the four-chambered heart. (b) The right side of the heart receives deoxygenated blood and pumps it to the lungs; the left side of the heart receives oxygenated blood and pumps it to the body. In this way, the pulmonary and systemic circulations are kept completely separate. RA = right atrium; LA = left atrium; RV = right ventricle; LV = left ventricle.

Mammalian and Bird Hearts

Mammals, birds, and crocodiles have a four-chambered heart with two separate atria and two separate ventricles (figure 52.14). The right atrium receives deoxygenated blood from the body and delivers it to the right ventricle, which pumps the blood to the lungs. The left atrium receives oxygenated blood from the lungs and delivers it to the left ventricle, which pumps the oxygenated blood to the rest of the body. This completely double circulation is powered by a two-cycle pump. Both atria fill with blood and simultaneously contract, emptying their blood into the ventricles. Both ventricles contract at the same time, pushing blood simultaneously into the pulmonary and systemic circulations. The increased efficiency of the double circulatory system in mammals and birds is thought to have been important in the evolution of endothermy (warm-bloodedness), because a more efficient circulation is necessary to support the high metabolic rate required.

Because the overall circulatory system is closed, the same volume of blood must move through the pulmonary circulation as through the much larger systemic circulation with each heartbeat. Therefore, the right and left ventricles must pump the same amount of blood each time they contract. If the output of one ventricle did not match that of

the other, fluid would accumulate and pressure would increase in one of the circuits. The result would be increased filtration out of the capillaries and edema (as occurs in congestive heart failure, for example). Although the volume of blood pumped by the two ventricles is the same, the pressure they generate is not. The left ventricle, which pumps blood through the higher-resistance systemic pathway, is more muscular and generates more pressure than does the right ventricle.

Throughout the evolutionary history of the vertebrate heart, the sinus venosus has served as a pacemaker, the site where the impulses that initiate the heartbeat originate. Although it constitutes a major chamber in the fish heart, it is reduced in size in amphibians and further reduced in reptiles. In mammals and birds, the sinus venosus is no longer evident as a separate chamber, but its disappearance is not really complete. Some of its tissue remains in the wall of the right atrium, near the point where the systemic veins empty into the atrium. This tissue, which is called the *sinoatrial (SA) node*, is still the site where each heartbeat originates.

The oxygenated blood from the lungs returns to the left atrium and is pumped out the left ventricle. The deoxygenated blood from the body returns to the right atrium and out the right ventricle to the lungs.

52.4 The cardiac cycle drives the cardiovascular system.

The Cardiac Cycle

The human heart, like that of all mammals and birds, is really two separate pumping systems operating within a single organ. The right pump sends blood to the lungs, and the left pump sends blood to the rest of the body.

The heart has two pairs of valves. One pair, the **atrioventricular (AV) valves**, guards the opening between the atria and ventricles. The AV valve on the right side is the **tricuspid valve**, and the AV valve on the left is the **bicuspid**, or **mitral, valve**. Another pair of valves, together called the **semilunar valves**, guard the exits from the ventricles to the arterial system; the **pulmonary valve** is located at the exit of the right ventricle, and the **aortic valve** is located at the exit of the left ventricle. These valves open and close as the heart goes through its **cardiac cycle** of rest (*diastole*) and contraction (*systole*). The sound of these valves closing produces the “lub-dub” sounds heard with a stethoscope.

Blood returns to the resting heart through veins that empty into the right and left atria. As the atria fill and the pressure in them rises, the AV valves open to admit the blood into the ventricles. The ventricles become about 80% filled during this time. Contraction of the atria wrings out the final 20% of the 80 milliliters of blood the ventricles will receive, on average, in a resting person. These events occur while the ventricles are relaxing, a period called ventricular **diastole**.

After a slight delay, the ventricles contract; this period of contraction is known as ventricular **systole**. Contraction of each ventricle increases the pressure within each chamber, causing the AV valves to forcefully close (the “lub” sound), thereby preventing blood from backing up into the atria. Immediately after the AV valves close, the pressure in the ventricles forces the semilunar valves open so that blood can be pushed out into the arterial system. As the ventricles relax, closing of the semilunar valves prevents back flow (the “dub” sound).

The right and left **pulmonary arteries** deliver oxygen-depleted blood to the right and left lungs. As previously mentioned, these return blood to the left atrium of the heart via the **pulmonary veins**. The **aorta** and all its branches are systemic arteries (figure 52.15), carrying oxygen-rich blood from the left ventricle to all parts of the body. The **coronary arteries** are the first branches off the aorta; these supply the heart muscle itself. Other systemic arteries branch from the aorta as it makes an arch above the heart, and as it descends and traverses the thoracic and abdominal cavities. These branches provide all body organs with oxygenated blood. The blood from the body organs, now lower in oxygen, returns to the heart in the systemic veins. These eventually empty into two major veins: the **superior vena cava**, which drains the upper body, and the **inferior vena cava**, which drains the lower body. These veins empty into the right atrium and thereby complete the systemic circulation.

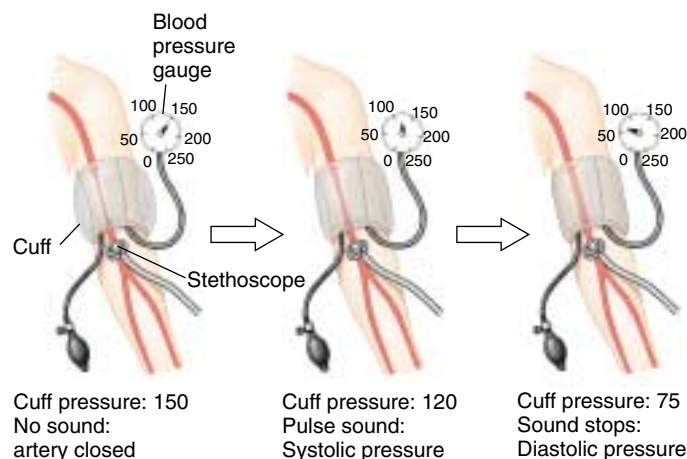


FIGURE 52.15
Measurement of blood pressure.

Measuring Arterial Blood Pressure

As the ventricles contract, great pressure is generated in the arteries throughout the body. You can tell this by feeling your pulse, either on the inside of your wrist, below the thumb or on the sides of your neck below your ear and jawbone. The contraction of the ventricles has to be strong enough to force blood through capillary beds but not too strong as to cause damage to smaller arteries and arterioles. Doctors measure your blood pressure to determine how hard your heart is working.

The measuring device used is called a sphygmomanometer and measures the blood pressure of the brachial artery found on the inside part of the arm, at the elbow (figure 52.15). A cuff wrapped around the upper part of the arm is tightened enough to stop the flow of blood to the lower part of the arm. As the cuff is loosened, blood will begin pulsating through the artery and can be detected using a stethoscope. Two measurements are recorded: the systolic and the diastolic pressure. The systolic pressure is the peak pressure during ventricular systole (contraction of the ventricle). The diastolic pressure is the minimum pressure between heartbeats (repolarization of the ventricles). The blood pressure is written as a ratio of systolic over diastolic pressure, and for a healthy person in his or her twenties, a typical blood pressure is 120/75 (measurement in mm of mercury). A condition called *hypertension* (high blood pressure) occurs when the ventricles experience very strong contractions, and the blood pressure is elevated, either systolic pressure greater than 150 or diastolic pressures greater than 90.

The cardiac cycle consists of systole and diastole; the ventricles contract at systole and relax at diastole.

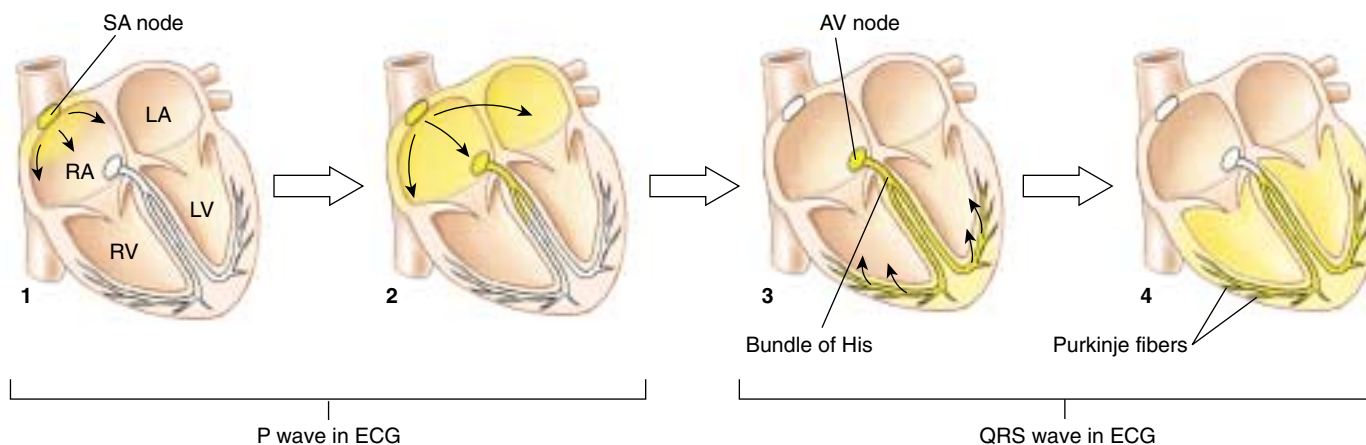
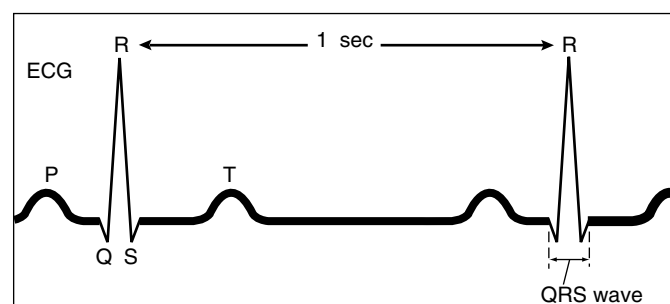


FIGURE 52.16

The path of electrical excitation in the heart. A wave of depolarization begins at the sinoatrial (SA) node. After passing over the atria and causing them to contract (forming the P wave on the ECG), the depolarization reaches the atrioventricular (AV) node, from which it passes to the ventricles along the septum by the bundle of His. Finer Purkinje fibers carry the depolarization into the right and left ventricular muscles (forming the QRS wave on the ECG). The T wave on the ECG corresponds to the repolarization of the ventricles.



Electrical Excitation and Contraction of the Heart

As in other types of muscle, contraction of heart muscle is stimulated by membrane **depolarization**, a reversal of the electrical polarity that normally exists across the plasma membrane (see chapter 50). In skeletal muscles, the nervous system initiates depolarization. However, in the heart, the depolarization is triggered by the **sinoatrial (SA) node** (figure 52.16), the small cluster of cardiac muscle cells derived from the sinus venosus. The SA node acts as a **pacemaker** for the rest of the heart by producing depolarization impulses spontaneously at a particular rate. Each depolarization initiated within this pacemaker region passes quickly from one cardiac muscle cell to another in a wave that envelops the right and left atria nearly simultaneously. The spread of depolarization is possible because the cardiac muscle cells are electrically coupled by gap junctions.

After a delay of almost 0.1 second, the wave of depolarization spreads to the ventricles. The reason for this delay is that connective tissue separates the atria from the ventricles, and connective tissue cannot transmit depolarization. The depolarization would not pass to the ventricles at all, were it not for a group of specialized cardiac muscle cells known as the **atrioventricular (AV) node**. The cells in the AV node transmit the depolarization slowly, causing the delay. This delay permits the atria to finish contracting and emptying their blood into the ventricles before the ventricles contract.

From the AV node, the wave of depolarization is conducted rapidly over both ventricles by a network of fibers called the **atrioventricular bundle** or **bundle of His**. It is then transmitted by **Purkinje fibers**, which directly stimulate the myocardial cells of the ventricles. The rapid conduction of the depolarization along the bundle of His and the Purkinje fibers causes the almost simultaneous contraction of the left and right ventricles. The rate can be increased or decreased by neural regulation or increased by the hormone epinephrine.

The spread of electrical activity through the heart creates currents that can be recorded from the surface of the body with electrodes placed on the limbs and chest. The recording, called an **electrocardiogram (ECG or EKG)**, shows how the cells of the heart depolarize and repolarize during the cardiac cycle (see figure 52.16). As was explained in chapter 50, depolarization causes contraction of a muscle (including the heart), while repolarization causes relaxation. The first peak in the recording, P, is produced by the depolarization of the atria, and thus is associated with atrial systole. The second, larger peak, QRS, is produced by ventricular depolarization; during this time, the ventricles contract (ventricular systole) and eject blood into the arteries. The last peak, T, is produced by ventricular repolarization; at this time the ventricles begin diastole.

The SA node in the right atrium initiates waves of depolarization that stimulate first the atria and then the ventricles to contract.

Blood Flow and Blood Pressure

Cardiac Output

Cardiac output is the volume of blood pumped by each ventricle per minute. Because humans (like all vertebrates) have a closed circulation, the cardiac output is the same as the volume of blood that traverses the systemic or pulmonary circulations per minute. It is calculated by multiplying the heart rate by the *stroke volume*, which is the volume of blood ejected by each ventricle per beat. For example, if the heart rate is 72 beats per minute and the stroke volume is 70 milliliters, the cardiac output is 5 liters per minute, which is about average in a resting person.

Cardiac output increases during exercise because of an increase in heart rate and stroke volume. When exercise begins, the heart rate increases up to about 100 beats per minute. As exercise becomes more intense, skeletal muscles squeeze on veins more vigorously, returning blood to the heart more rapidly. In addition, the ventricles contract more strongly, so they empty more completely with each beat.

During exercise, the cardiac output increases to a maximum of about 25 liters per minute in an average young adult. Although the cardiac output has increased five times, not all organs receive five times the blood flow; some receive more, others less. This is because the arterioles in some organs, such as in the digestive system, constrict, while the arterioles in the exercising muscles and heart dilate. As previously mentioned, the resistance to flow decreases as the radius of the vessel increases. As a consequence, vasodilation greatly increases and vasoconstriction greatly decreases blood flow.

Blood Pressure and the Baroreceptor Reflex

The arterial blood pressure depends on two factors: how much blood the ventricles pump (the cardiac output) and how great a resistance to flow the blood encounters in the entire arterial system. An increased blood pressure, therefore, could be produced by an increased heart rate or an increased blood volume (because both increase the cardiac output) or by vasoconstriction, which increases the resistance to blood flow. Conversely, blood pressure will fall if the heart rate slows or if the blood volume is reduced, for example by dehydration or excessive bleeding (hemorrhage).

Changes in the arterial blood pressure are detected by **baroreceptors** located in the arch of the aorta and in the carotid arteries. These receptors activate sensory neurons that relay information to *cardiovascular control centers* in the medulla oblongata, a region of the brain stem. When the baroreceptors detect a fall in blood pressure, they stimulate neurons that go to blood vessels in the skin and viscera, causing arterioles in these organs to constrict and raise the blood pressure. This baroreceptor reflex therefore completes a negative feedback loop that acts to correct the fall in blood pressure and restore homeostasis.

Blood Volume Reflexes

Blood pressure depends in part on the total blood volume. A decrease in blood volume, therefore, will decrease blood pressure, if all else remains equal. Blood volume regulation involves the effects of four hormones: (1) antidiuretic hormone; (2) aldosterone; (3) atrial natriuretic hormone; and (4) nitric oxide.

Antidiuretic Hormone. *Antidiuretic hormone* (ADH), also called *vasopressin*, is secreted by the posterior pituitary gland in response to an increase in the osmotic concentration of the blood plasma. Dehydration, for example, causes the blood volume to decrease while the remaining plasma becomes more concentrated. This stimulates *osmoreceptors* in the hypothalamus of the brain, a region located immediately above the pituitary. The osmoreceptors promote thirst and stimulate ADH secretion from the posterior pituitary gland. ADH, in turn, stimulates the kidneys to retain more water in the blood, excreting less in the urine (urine is derived from blood plasma—see chapter 58). A dehydrated person thus drinks more and urinates less, helping to raise the blood volume and restore homeostasis.

Aldosterone. If a person's blood volume is lowered (by dehydration, for example), the flow of blood through the organs will be reduced if no compensation occurs. Whenever the kidneys experience a decreased blood flow, a group of kidney cells initiate the release of a short polypeptide known as *angiotensin II*. This is a very powerful molecule: it stimulates vasoconstriction throughout the body while it also stimulates the adrenal cortex (the outer region of the adrenal glands) to secrete the hormone *aldosterone*. This important steroid hormone is necessary for life; it acts on the kidneys to promote the retention of Na^+ and water in the blood. An animal that lacks aldosterone will die if untreated, because so much of the blood volume is lost in urine that the blood pressure falls too low to sustain life.

Atrial Natriuretic Hormone. When the body needs to eliminate excessive Na^+ , less aldosterone is secreted by the adrenals, so that less Na^+ is retained by the kidneys. In recent years, scientists have learned that Na^+ excretion in the urine is promoted by another hormone. Surprisingly, this hormone is secreted by the right atrium of the heart—the heart is an endocrine gland! The right atrium secretes *atrial natriuretic hormone* in response to stretching of the atrium by an increased blood volume. The action of atrial natriuretic hormone completes a negative feedback loop, because it promotes the elimination of Na^+ and water, which will lower the blood volume and pressure.

Nitric Oxide. *Nitric oxide* (NO) is a gas that acts as a hormone in vertebrates, regulating blood pressure and blood flow. As described in chapter 7, nitric oxide gas is a paracrine hormone, is produced by one cell, penetrates through membranes, and alters the activities of other

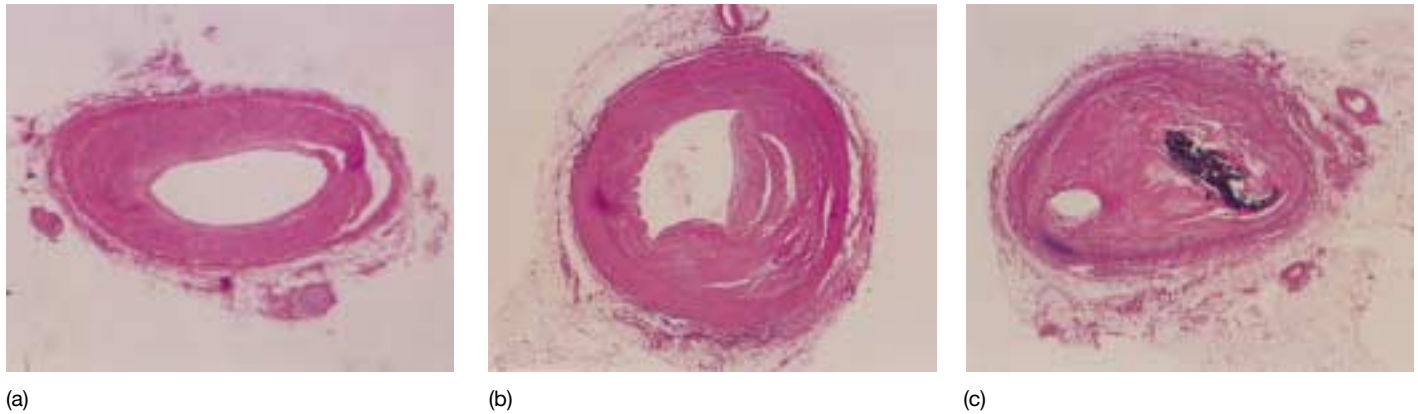


FIGURE 52.17

Atherosclerosis. (a) The coronary artery shows only minor blockage. (b) The artery exhibits severe atherosclerosis—much of the passage is blocked by build-up on the interior walls of the artery. (c) The coronary artery is essentially completely blocked.

neighboring cells. In 1998 the Nobel Prize for Medicine was awarded for the discovery of this signal transmission activity. How does NO regulate blood pressure? Nitric oxide gas produced by the surface endothelial cells of blood vessels passes inward through the cell layers of the vessel, causing the smooth muscles that encase it to relax and the blood vessel to dilate (become wider). For over a century, heart patients have been prescribed nitroglycerin to relieve chest pain, but only now has it become clear that nitroglycerin acts by releasing nitric oxide gas.

Cardiovascular Diseases

Cardiovascular diseases are the leading cause of death in the United States; more than 42 million people have some form of cardiovascular disease. Heart attacks are the main cause of cardiovascular deaths in the United States, accounting for about a fifth of all deaths. They result from an insufficient supply of blood reaching one or more parts of the heart muscle, which causes myocardial cells in those parts to die. Heart attacks may be caused by a blood clot forming somewhere in the coronary arteries (the arteries that supply the heart muscle with blood) and blocking the passage of blood through those vessels. They may also result if an artery is blocked by atherosclerosis (see below). Recovery from a heart attack is possible if the portion of the heart that was damaged is small enough that the other blood vessels in the heart can enlarge their capacity and re-supply the damaged tissues. **Angina pectoris**, which literally means “chest pain,” occurs for reasons similar to those that cause heart attacks, but it is not as severe. The pain may occur in the heart and often also in the left arm and shoulder. Angina pectoris is a warning sign that the blood supply to the heart is inadequate but still sufficient to avoid myocardial cell death.

Strokes are caused by an interference with the blood supply to the brain. They may occur when a blood vessel

bursts in the brain, or when blood flow in a cerebral artery is blocked by a thrombus (blood clot) or by atherosclerosis. The effects of a stroke depend on how severe the damage is and where in the brain the stroke occurs.

Atherosclerosis is an accumulation within the arteries of fatty materials, abnormal amounts of smooth muscle, deposits of cholesterol or fibrin, or various kinds of cellular debris. These accumulations cause blood flow to be reduced (figure 52.17). The lumen (interior) of the artery may be further reduced in size by a clot that forms as a result of the atherosclerosis. In the severest cases, the artery may be blocked completely. Atherosclerosis is promoted by genetic factors, smoking, hypertension (high blood pressure), and high blood cholesterol levels. Diets low in cholesterol and saturated fats (from which cholesterol can be made) can help lower the level of blood cholesterol, and therapy for hypertension can reduce that risk factor. Stopping smoking, however, is the single most effective action a smoker can take to reduce the risk of atherosclerosis.

Arteriosclerosis, or hardening of the arteries, occurs when calcium is deposited in arterial walls. It tends to occur when atherosclerosis is severe. Not only do such arteries have restricted blood flow, but they also lack the ability to expand as normal arteries do to accommodate the volume of blood pumped out by the heart. This inflexibility forces the heart to work harder.

Cardiac output depends on the rate of the heart and how much blood is ejected per beat. Blood flow is regulated by the degree of constriction of the arteries, which affects the resistance to flow. Blood pressure is influenced by blood volume. The volume of water retained in the vascular system is regulated by hormones that act on the kidneys and blood vessels. Many cardiovascular diseases are associated with the accumulation of fatty materials on the inner surfaces of arteries.



Summary

Questions

Media Resources

52.1 The circulatory systems of animals may be open or closed.

- Vertebrates have a closed circulation, where the blood stays within vessels as it travels away from and back to the heart.
- The circulatory system serves a variety of functions, including transport, regulation, and protection.

1. What is the difference between a closed circulatory system and an open circulatory system? In what types of animals would you find each?



- Bioethics case study: Heart transplant



- Types of systems

52.2 A network of vessels transports blood through the body.

- Plasma is the liquid portion of the blood. A variety of plasma proteins, ions, metabolites, wastes, and hormones are dissolved in the plasma.
- Erythrocytes, or red blood cells, contain hemoglobin and function to transport oxygen; the leukocytes, or white blood cells, function in immunological defenses.
- The heart pumps blood into arteries, which branch into smaller arterioles.
- Blood from the arterial system empties into capillaries with thin walls; all exchanges between the blood and tissues pass across the walls of capillaries.
- Blood returns to the heart in veins, which have one-way valves to ensure that blood travels toward the heart only.
- Lymphatic vessels return interstitial fluid to the venous system.

2. What are the major components of blood plasma?

3. Describe the structure of arteries and veins, explaining their similarities and differences. Why do arteries differ in structure from veins?

4. What is the relationship between vessel diameter and the resistance to blood flow? How do the arterial trees adjust their resistance to flow?

5. What drives the flow of fluid within the lymphatic system, and in what direction does the fluid flow?



- Art Activities:
Blood vessels
Capillary bed anatomy
Human circulatory system
Lymphatic system
Lymphoid organs



- Lymphatic system



- Vessels and pressure
- Blood
- Lymph system
- Plasma

52.3 The vertebrate heart has undergone progressive evolutionary change.

- The fish heart consists of four chambers in a row; the beat originates in the sinus venosus and spreads through the atrium, ventricle, and conus arteriosus.
- In the circulation of fishes, blood from the heart goes to the gills and then to the rest of the body before returning to the heart; in terrestrial vertebrates, blood returns from the lungs to the heart before it is pumped to the body.

6. Describe the pattern of circulation through a fish and an amphibian, and compare the structure of their hearts. What new circulatory pattern accompanies the evolution of lungs?



- On Science Article: Dinosaur hearts



- Art Activities:
External heart anatomy
Internal view of heart



- Cardiac cycle blood flow

52.4 The cardiac cycle drives the cardiovascular system.

- Electrical excitation of the heart is initiated by the SA (sinoatrial) node, spreads through gap junctions between myocardial cells in the atria, and then is conducted into the ventricles by specialized conducting tissue.
- The cardiac output is regulated by nerves that influence the cardiac rate and by factors that influence the stroke volume.

7. How does the baroreceptor reflex help to maintain blood pressure? How do ADH and aldosterone maintain blood volume and pressure? What causes their secretion?



- Art Activity: Plaque



- Blood flow
- Cardiac cycle
- Blood pressure

53

Respiration

Concept Outline

53.1 Respiration involves the diffusion of gases.

Fick's Law of Diffusion. The rate of diffusion across a membrane depends on the surface area of the membrane, the concentration gradients, and the distance across the membrane.

How Animals Maximize the Rate of Diffusion. The diffusion rate increases when surface area or concentration gradient increases.

53.2 Gills are used for respiration by aquatic vertebrates.

The Gill as a Respiratory Structure. Water is forced past the gill surface, and blood flows through the gills.

53.3 Lungs are used for respiration by terrestrial vertebrates.

Respiration in Air-Breathing Animals. In insects, oxygen diffuses directly from the air into body cells; in vertebrates, oxygen diffuses into blood and then into body cells.

Respiration in Amphibians and Reptiles. Amphibians force air into their lungs, whereas reptiles, birds, and mammals draw air in by expanding their rib cage.

Respiration in Mammals. In mammals, gas exchange occurs across millions of tiny air sacs called alveoli.

Respiration in Birds. In birds, air flows through the lung unidirectionally.

53.4 Mammalian breathing is a dynamic process.

Structures and Mechanisms of Breathing. The rib cage and lung volumes are expanded during inspiration by the contraction of the diaphragm and other muscles.

Mechanisms That Regulate Breathing. The respiratory control center in the brain is influenced by reflexes triggered by the blood levels of carbon dioxide and blood pH.

53.5 Blood transports oxygen and carbon dioxide.

Hemoglobin and Oxygen Transport. Hemoglobin, a molecule within the red blood cells, loads with oxygen in the lungs and unloads its oxygen in the tissue capillaries.

Carbon Dioxide and Nitric Oxide Transport. Carbon dioxide is converted into carbonic acid in erythrocytes and is transported as bicarbonate.



FIGURE 53.1

Elephant seals are respiratory champions. Diving to depths greater than those of all other marine animals, including sperm whales and sea turtles, elephant seals can hold their breath for over two hours, descend and ascend rapidly in the water, and endure repeated dives without suffering any apparent respiratory distress.

Animals pry energy out of food molecules using the biochemical process called cellular respiration. While the term cellular respiration pertains to the use of oxygen and production of carbon dioxide at the cellular level, the general term respiration describes the uptake of oxygen from the environment and the disposal of carbon dioxide into the environment at the body system level. Respiration at the body system level involves a host of processes not found at the cellular level, like the mechanics of breathing and the exchange of oxygen and carbon dioxide in the capillaries. These processes, one of the principal physiological challenges facing all animals (figure 53.1), are the subject of this chapter.

53.1 Respiration involves the diffusion of gases.

Fick's Law of Diffusion

Respiration involves the diffusion of gases across plasma membranes. Because plasma membranes must be surrounded by water to be stable, the external environment in gas exchange is always aqueous. This is true even in terrestrial animals; in these cases, oxygen from air dissolves in a thin layer of fluid that covers the respiratory surfaces, such as the alveoli in lungs.

In vertebrates, the gases diffuse into the aqueous layer covering the epithelial cells that line the respiratory organs. The diffusion process is passive, driven only by the difference in O_2 and CO_2 concentrations on the two sides of the membranes. In general, the rate of diffusion between two regions is governed by a relationship known as **Fick's Law of Diffusion**:

$$R = D \times A \frac{\Delta p}{d}$$

In this equation,

R = the rate of diffusion; the amount of oxygen or carbon dioxide diffusing per unit of time;

D = the diffusion constant;

A = the area over which diffusion takes place;

Δp = the difference in concentration (for gases, the difference in their partial pressures) between the interior of the organism and the external environment; and

d = the distance across which diffusion takes place.

Major changes in the mechanism of respiration have occurred during the evolution of animals (figure 53.2) that have tended to optimize the rate of diffusion R . By inspecting Fick's Law, you can see that natural selection can optimize R by favoring changes that (1) increase the surface area A ; (2) decrease the distance d ; or (3) increase the concentration difference, as indicated by Δp . The evolution of respiratory systems has involved changes in all of these factors.

Fick's Law of Diffusion states that the rate of diffusion across a membrane depends on surface area, concentration (partial pressure) difference, and distance.

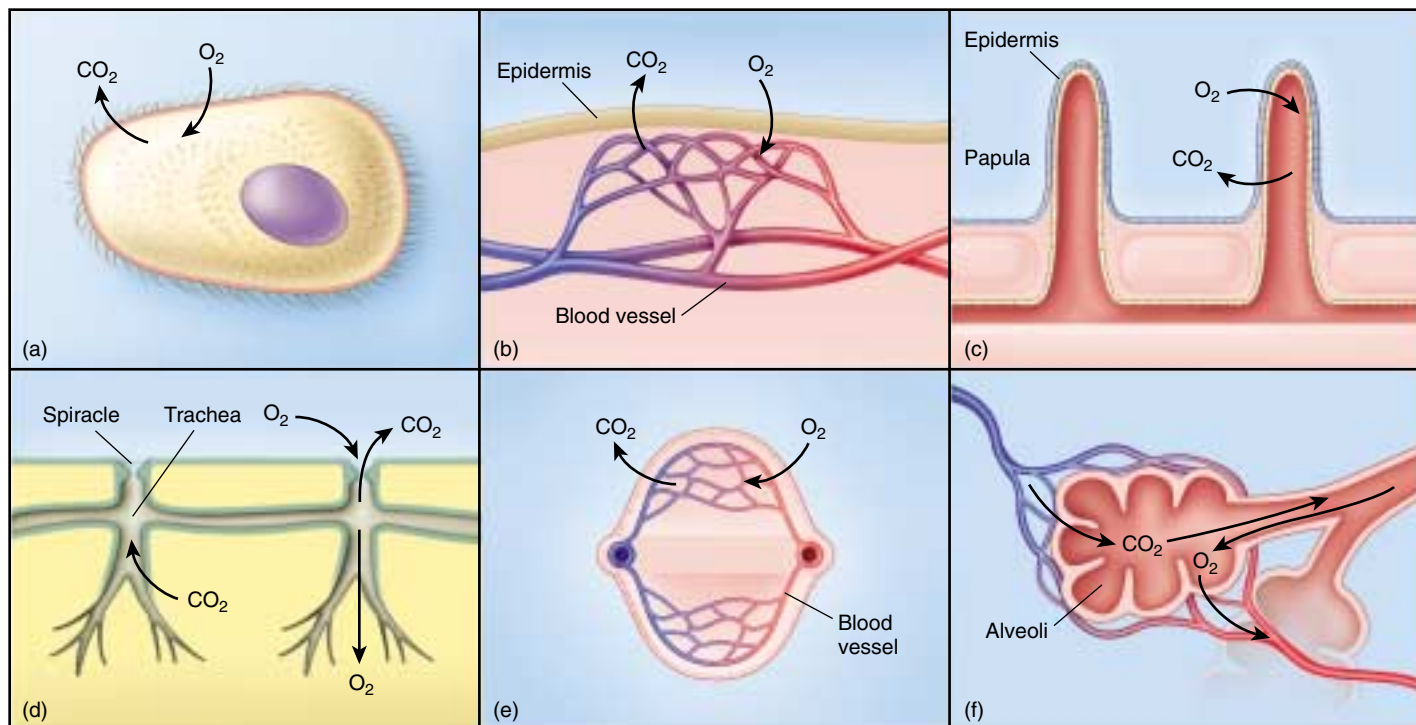


FIGURE 53.2

Gas exchange may take place in a variety of ways. (a) Gases diffuse directly into single-celled organisms. (b) Amphibians and many other animals respire across their skin. (c) Echinoderms have protruding papulae, which provide an increased respiratory surface area. (d) Insects respire through an extensive tracheal system. (e) The gills of fishes provide a very large respiratory surface area and countercurrent exchange. (f) The alveoli in mammalian lungs provide a large respiratory surface area but do not permit countercurrent exchange.

How Animals Maximize the Rate of Diffusion

The levels of oxygen required by oxidative metabolism cannot be obtained by diffusion alone over distances greater than about 0.5 millimeter. This restriction severely limits the size of organisms that obtain their oxygen entirely by diffusion directly from the environment. Protists are small enough that such diffusion can be adequate (see figure 53.2a), but most multicellular animals are much too large.

Most of the more primitive phyla of invertebrates lack special respiratory organs, but they have developed means of improving the movement of water over respiratory structures. In a number of different ways, many of which involve beating cilia, these organisms create a *water current* that continuously replaces the water over the respiratory surfaces. Because of this continuous replenishment with water containing fresh oxygen, *the external oxygen concentration does not decrease along the diffusion pathway*. Although each oxygen molecule that passes into the organism has been removed from the surrounding water, new water continuously replaces the oxygen-depleted water. This increases the rate of diffusion by maximizing the concentration difference—the Δp of the Fick equation.

All of the more advanced invertebrates (mollusks, arthropods, echinoderms), as well as vertebrates, possess respiratory organs that increase the surface area available for diffusion and bring the external environment (either water or air) close to the internal fluid, which is usually circulated throughout the body. The respiratory organs thus increase the rate of diffusion by maximizing surface area and decreasing the distance the diffusing gases must travel (the A and d factors, respectively, in the Fick equation).

Atmospheric Pressure and Partial Pressures

Dry air contains 78.09% nitrogen (N_2), 20.95% oxygen, 0.93% argon and other inert gases, and 0.03% carbon dioxide. Convection currents cause air to maintain a constant composition to altitudes of at least 100 kilometers, although the *amount* (number of molecules) of air that is present decreases with altitude (figure 53.3).

Imagine a column of air extending from the ground to the limits of the atmosphere. All of the gas molecules in this column experience the force of gravity, so they have weight and can exert pressure. If this column were on top of one end of a U-shaped tube of mercury at sea level, it would exert enough pressure to raise the other end of the tube 760 millimeters under a set of specified, standard conditions (see figure 53.3). An apparatus that measures air pressure is called a barometer, and 760 mm Hg (millimeters of mercury) is the barometric pressure of the air at sea level. A pressure of 760 mm Hg is also defined as **one atmosphere** of pressure.

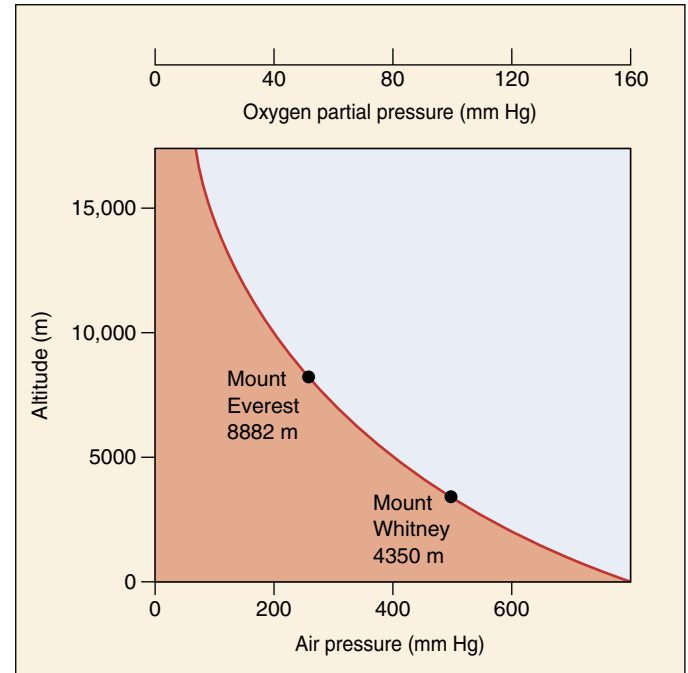


FIGURE 53.3

The relationship between air pressure and altitude above sea level. At the high altitudes characteristic of mountaintops, air pressure is much less than at sea level. At the top of Mount Everest, the world's highest mountain, the air pressure is only one-third that at sea level.

Each type of gas contributes to the total atmospheric pressure according to its fraction of the total molecules present. That fraction contributed by a gas is called its **partial pressure** and is indicated by P_{N_2} , P_{O_2} , P_{CO_2} , and so on. The total pressure is the sum of the partial pressures of all gases present. For dry air, the partial pressures are calculated simply by multiplying the fractional composition of each gas in the air by the atmospheric pressure. Thus, at sea level, the partial pressures of N_2 , inert gases, O_2 , and CO_2 are:

$$\begin{aligned}P_{N_2} &= 760 \times 7902\% = 6006 \text{ mm Hg,} \\P_{O_2} &= 760 \times 2095\% = 1592 \text{ mm Hg, and} \\P_{CO_2} &= 760 \times 0.03\% = 0.2 \text{ mm Hg.}\end{aligned}$$

Humans do not survive long at altitudes above 6000 meters. Although the air at these altitudes still contains 20.95% oxygen, the atmospheric pressure is only about 380 mm Hg, so its P_{O_2} is only 80 mm Hg ($380 \times 20.95\%$), only half the amount of oxygen available at sea level.

The exchange of oxygen and carbon dioxide between an organism and its environment occurs by diffusion of dissolved gases across plasma membranes and is maximized by increasing the concentration gradient and the surface area and by decreasing the distance that the diffusing gases must travel.

53.2 Gills are used for respiration by aquatic vertebrates.

The Gill as a Respiratory Structure

Aquatic respiratory organs increase the diffusion surface area by extensions of tissue, called *gills*, that project out into the water. Gills can be simple, as in the papulae of echinoderms (see figure 53.2c), or complex, as in the highly convoluted gills of fish (see figure 53.2e). The great increase in diffusion surface area provided by gills enables aquatic organisms to extract far more oxygen from water than would be possible from their body surface alone.

External gills (gills that are not enclosed within body structures) provide a greatly increased surface area for gas exchange. Examples of vertebrates with external gills are the larvae of many fish and amphibians, as well as developmentally arrested (*neotenic*) amphibian larvae that remain permanently aquatic, such as the axolotl. One of the disadvantages of external gills is that they must constantly be moved or the surrounding water becomes depleted in oxygen as the oxygen diffuses from the water to the blood of the gills. The highly branched gills, however, offer significant resistance to movement, making this form of respiration ineffective except in smaller animals. Another disadvantage is that external gills are easily damaged. The thin epithelium required for gas exchange is not consistent with a protective external layer of skin.

Other types of aquatic animals evolved specialized *branchial chambers*, which provide a means of pumping water past stationary gills. Mollusks, for example, have an internal *mantle cavity* that opens to the outside and contains the gills. Contraction of the muscular walls of the mantle cavity draws water in and then expels it. In crustaceans, the branchial chamber lies between the bulk of the body and the hard exoskeleton of the animal. This chamber contains gills and opens to the surface beneath a limb. Movement of the limb draws water through the branchial chamber, thus creating currents over the gills.

The Gills of Bony Fishes

The gills of bony fishes are located between the *buccal* (mouth) *cavity* and the *opercular cavities* (figure 53.4). The buccal cavity can be opened and closed by opening and closing the mouth, and the opercular cavity can be opened and closed by movements of the **operculum**, or gill cover. The two sets of cavities function as pumps that expand alternately to move water into the mouth, through the gills, and out of the fish through the open operculum. Water is brought into the buccal cavity by lowering the jaw and floor of the mouth, and then is moved through the gills into the opercular cavity by the opening of the operculum. The lower pressure in the opercular cavity causes water to move in the correct direction across the gills, and tissue that acts as valves ensures that the movement is one-way.

Some fishes that swim continuously, such as tuna, have practically immobile opercula. These fishes swim with their mouths partly open, constantly forcing water over the gills in a form of **ram ventilation**. Most bony fishes, however, have flexible gill covers that permit a pumping action. For example, the remora, a fish that rides “piggyback” on sharks, uses ram ventilation while the shark swims and the pumping action of its opercula when the shark stops swimming.

There are four **gill arches** on each side of the fish head. Each gill arch is composed of two rows of *gill filaments*, and each gill filament contains thin membranous plates, or *lamellae*, that project out into the flow of water (figure 53.5). Water flows past the lamellae in one direction only. Within each lamella, blood flows in a direction that is *opposite* the direction of water movement. This arrangement is called **countercurrent flow**, and it acts to maximize the oxygenation of the blood by increasing the concentration gradient of oxygen along the pathway for diffusion, increasing Δp in Fick’s Law of Diffusion.

The advantages of a countercurrent flow system were discussed in chapter 52 in relation to temperature regulation and are again shown here in figure 53.6a. Blood low in oxygen en-

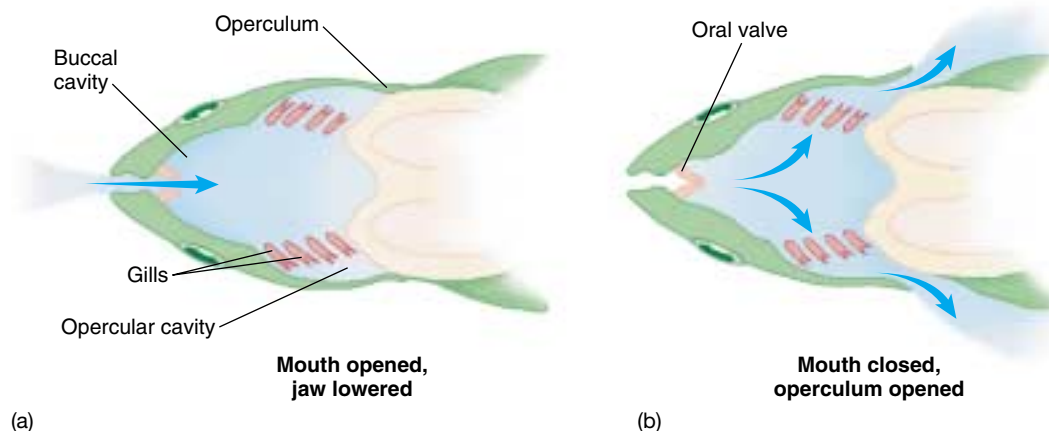


FIGURE 53.4

How most bony fishes respire.

The gills are suspended between the buccal (mouth) cavity and the opercular cavity. Respiration occurs in two stages. (a) The oral valve in the mouth is opened and the jaw is depressed, drawing water into the buccal cavity while the opercular cavity is closed. (b) The oral valve is closed and the operculum is opened, drawing water through the gills to the outside.

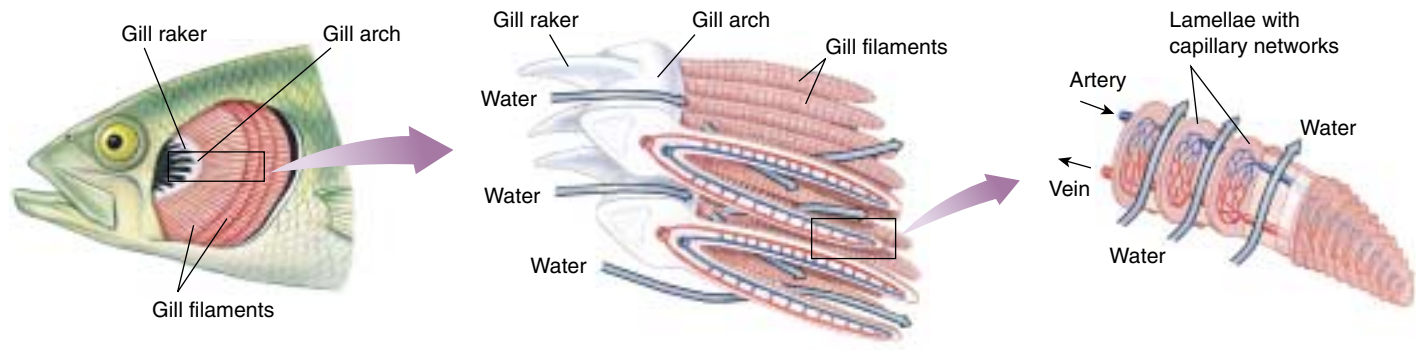


FIGURE 53.5

Structure of a fish gill. Water passes from the gill arch over the filaments (from left to right in the diagram). Water always passes the lamellae in a direction that is opposite to the direction of blood flow through the lamellae. The success of the gill's operation critically depends on this countercurrent flow of water and blood.

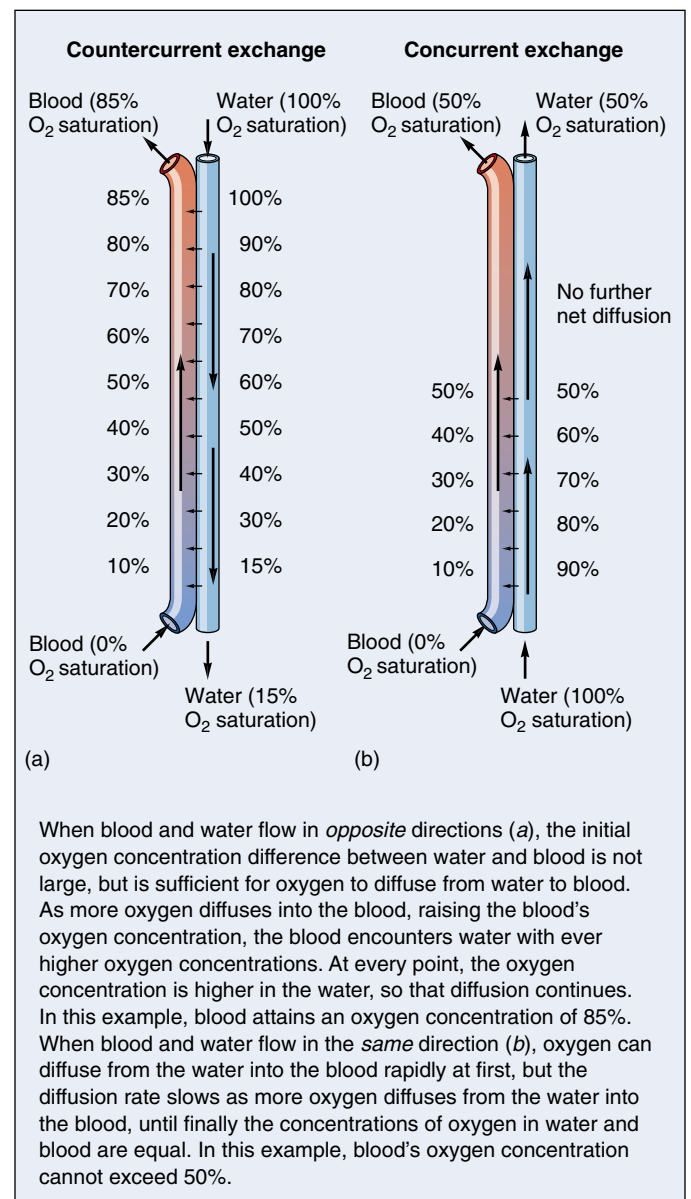
ters the back of the lamella, where it comes in close proximity to water that has already had most of its oxygen removed as it flowed through the lamella in the opposite direction. The water still has a higher oxygen concentration than the blood at this point, however, so oxygen diffuses from the water to the blood. As the blood flows toward the front of the lamella, it runs next to water that has a still higher oxygen content, so oxygen continuously diffuses from the water to the blood. Thus, countercurrent flow ensures that a concentration gradient remains between blood and water throughout the flow. This permits oxygen to continue to diffuse all along the lamellae, so that the blood leaving the gills has nearly as high an oxygen concentration as the water entering the gills.

This concept is easier to understand if we look at what would happen if blood and water flowed in the same direction, that is, had a *concurrent flow*. The difference in oxygen concentration would be very high at the front of each lamella, where oxygen-depleted blood would meet oxygen-rich water entering the gill (figure 53.6b). The concentration difference would fall rapidly, however, as the water lost oxygen to the blood. Net diffusion of oxygen would cease when the oxygen concentration of blood matched that of the water. At this point, much less oxygen would have been transferred to the blood than is the case with countercurrent flow. The flow of blood and water in a fish gill is in fact countercurrent, and because of the countercurrent exchange of gases, fish gills are the most efficient of all respiratory organs.

In bony fishes, water is forced past gills by the pumping action of the buccal and opercular cavities, or by active swimming in ram ventilation. In the gills, blood flows in an opposite direction to the flow of water. This countercurrent flow maximizes gas exchange, making the fish's gill an efficient respiratory organ.

FIGURE 53.6

Countercurrent exchange. This process allows for the most efficient blood oxygenation known in nature.



53.3 Lungs are used for respiration by terrestrial vertebrates.

Respiration in Air-Breathing Animals

Despite the high efficiency of gills as respiratory organs in aquatic environments, gills were replaced in terrestrial animals for two principal reasons:

1. **Air is less buoyant than water.** The fine membranous lamellae of gills lack structural strength and rely on water for their support. A fish out of water, although awash in oxygen (water contains only 5 to 10 mL O₂/L, compared with air with 210 mL O₂/L), soon suffocates because its gills collapse into a mass of tissue. This collapse greatly reduces the diffusion surface area of the gills. Unlike gills, internal air passages can remain open, because the body itself provides the necessary structural support.
2. **Water diffuses into air through evaporation.** Atmospheric air is rarely saturated with water vapor, except immediately after a rainstorm. Consequently, terrestrial organisms that are surrounded by air constantly lose water to the atmosphere. Gills would provide an enormous surface area for water loss.

Two main types of respiratory organs are used by terrestrial animals, and both sacrifice respiratory efficiency to some extent in exchange for reduced evaporation. The first are the **tracheae** of insects (see chapter 46 and figure 53.2*d*). Tracheae comprise an extensive series of air-filled passages connecting the surface of an insect to all portions of its body. Oxygen diffuses from these passages directly into cells, without the intervention of a circulatory system. Piping air directly from the external environment to the cells works very well in insects because their small bodies give them a high surface area-to-volume ratio. Insects prevent excessive water loss by closing the external openings of the tracheae whenever their internal CO₂ levels fall below a certain point.

The other main type of terrestrial respiratory organ is the **lung** (figure 53.7). A lung minimizes evaporation by moving air through a branched tubular passage; the air becomes saturated with water vapor before reaching the portion of the lung where a thin, wet membrane permits gas exchange. The lungs of all terrestrial vertebrates ex-



FIGURE 53.7

Human lungs. This chest X ray (dorsal view) was color-enhanced to show the lungs clearly. The heart is the pear-shaped object behind the vertical white column that is the esophagus.

cept birds use a **uniform pool** of air that is in contact with the gas exchange surface. Unlike the one-way flow of water that is so effective in the respiratory function of gills, air moves in and out by way of the same airway passages, a two-way flow system. Let us now examine the structure and function of lungs in the four classes of terrestrial vertebrates.

Air is piped directly to the body cells of insects, but the cells of terrestrial vertebrates obtain oxygen from the blood. The blood obtains its oxygen from a uniform pool of air by diffusion across the wet membranes of the lungs, which are filled with air in the process of ventilation.

Respiration in Amphibians and Reptiles

The lungs of amphibians are formed as saclike outpouchings of the gut (figure 53.8). Although the internal surface area of these sacs is increased by folds, much less surface area is available for gas exchange in amphibian lungs than in the lungs of other terrestrial vertebrates. Each amphibian lung is connected to the rear of the oral cavity, or pharynx, and the opening to each lung is controlled by a valve, the glottis.

Amphibians do not breathe the same way other terrestrial vertebrates do. Amphibians force air into their lungs by creating a greater-than-atmospheric pressure (positive pressure) in the air outside their lungs. They do this by filling their *buccal cavity* with air, closing their mouth and nostrils, and then elevating the floor of their oral cavity. This pushes air into their lungs in the same way that a pressurized tank of air is used to fill balloons. This is called positive pressure breathing; in humans, it would be analogous to forcing air into a victim's lungs by performing mouth-to-mouth resuscitation.

All other terrestrial vertebrates breathe by expanding their lungs and thereby creating a lower-than-atmospheric pressure (a negative pressure) within the lungs. This is called negative pressure breathing and is analogous to taking air into an accordion by pulling the accordion out to a greater volume. In reptiles, birds, and mammals, this is ac-

complished by expanding the thoracic (chest) cavity through muscular contractions, as will be described in a later section.

The oxygenation of amphibian blood by the lungs is supplemented by cutaneous respiration—the exchange of gases across the skin, which is wet and well vascularized in amphibians. Cutaneous respiration is actually more significant than pulmonary (lung) ventilation in frogs during winter, when their metabolisms are slow. Lung function becomes more important during the summer as the frog's metabolism increases. Although not common, some terrestrial amphibians, such as plethodontid salamanders, rely on cutaneous respiration exclusively.

Reptiles expand their rib cages by muscular contraction, and thereby take air into their lungs through negative pressure breathing. Their lungs have somewhat more surface area than the lungs of amphibians and so are more efficient at gas exchange. Terrestrial reptiles have dry, tough, scaly skins that prevent desiccation, and so cannot have cutaneous respiration. Cutaneous respiration, however, has been demonstrated in marine sea snakes.

Amphibians force air into their lungs by positive pressure breathing, whereas reptiles and all other terrestrial vertebrates take air into their lungs by expanding their lungs when they increase rib cage volume through muscular contractions. This creates a subatmospheric pressure in the lungs.

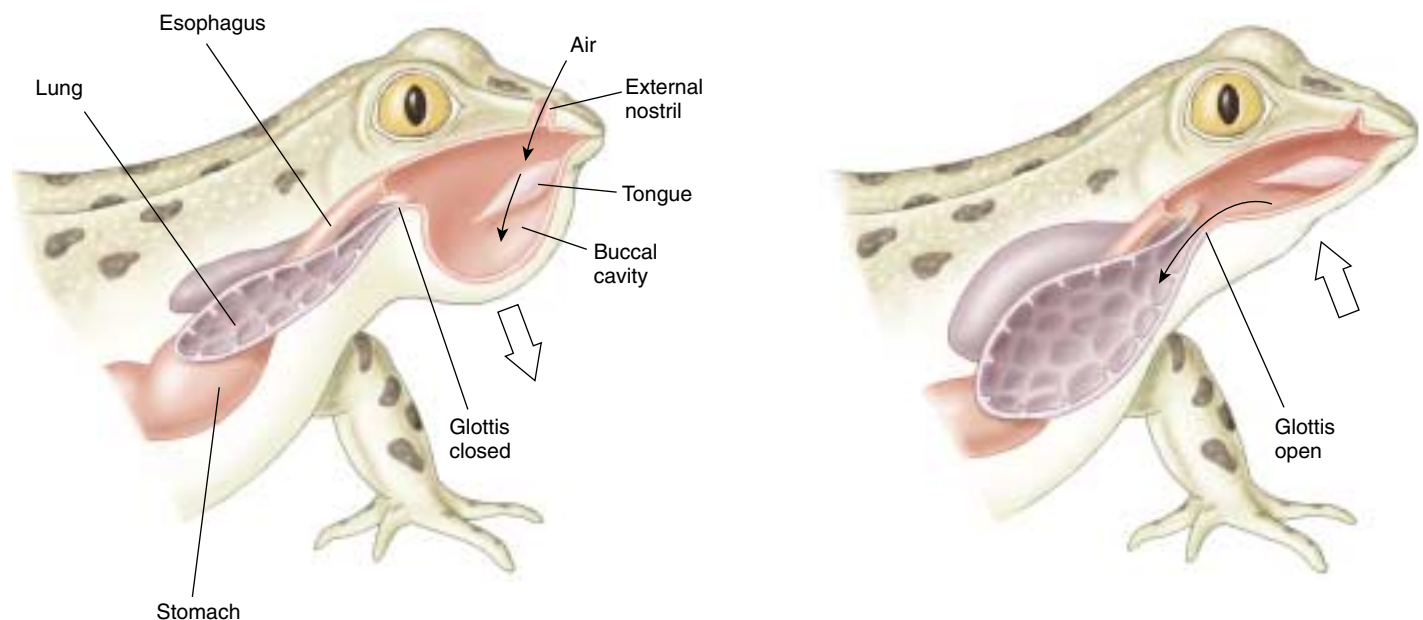


FIGURE 53.8

Amphibian lungs. Each lung of this frog is an outpouching of the gut and is filled with air by the creation of a positive pressure in the buccal cavity. The amphibian lung lacks the structures present in the lungs of other terrestrial vertebrates that provide an enormous surface area for gas exchange, and so are not as efficient as the lungs of other vertebrates.

Respiration in Mammals

The metabolic rate, and therefore the demand for oxygen, is much greater in birds and mammals, which are endothermic and thus require a more efficient respiratory system.

The lungs of mammals are packed with millions of *alveoli*, tiny sacs clustered like grapes (figure 53.9). This provides each lung with an enormous surface area for gas exchange. Air is brought to the alveoli through a system of air passages. Inhaled air is taken in through the mouth and nose past the pharynx to the **larynx** (voice box), where it passes through an opening in the vocal cords, the *glottis*, into a tube supported by C-shaped rings of cartilage, the **trachea** (windpipe). The trachea bifurcates into right and left **bronchi** (singular, *bronchus*), which enter each lung and further subdivide into **bronchioles** that deliver the air into blind-ended sacs called **alveoli**. The alveoli are surrounded

by an extremely extensive capillary network. All gas exchange between the air and blood takes place across the walls of the alveoli.

The branching of bronchioles and the vast number of alveoli combine to increase the respiratory surface area (A in Fick's Law) far above that of amphibians or reptiles. In humans, there are about 300 million alveoli in each of the two lungs, and the total surface area available for diffusion can be as much as 80 square meters, or about 42 times the surface area of the body. Respiration in mammals will be considered in more detail in a separate section later.

Mammalian lungs are composed of millions of alveoli that provide a huge surface area for gas exchange. Air enters and leaves these alveoli through the same system of airways.

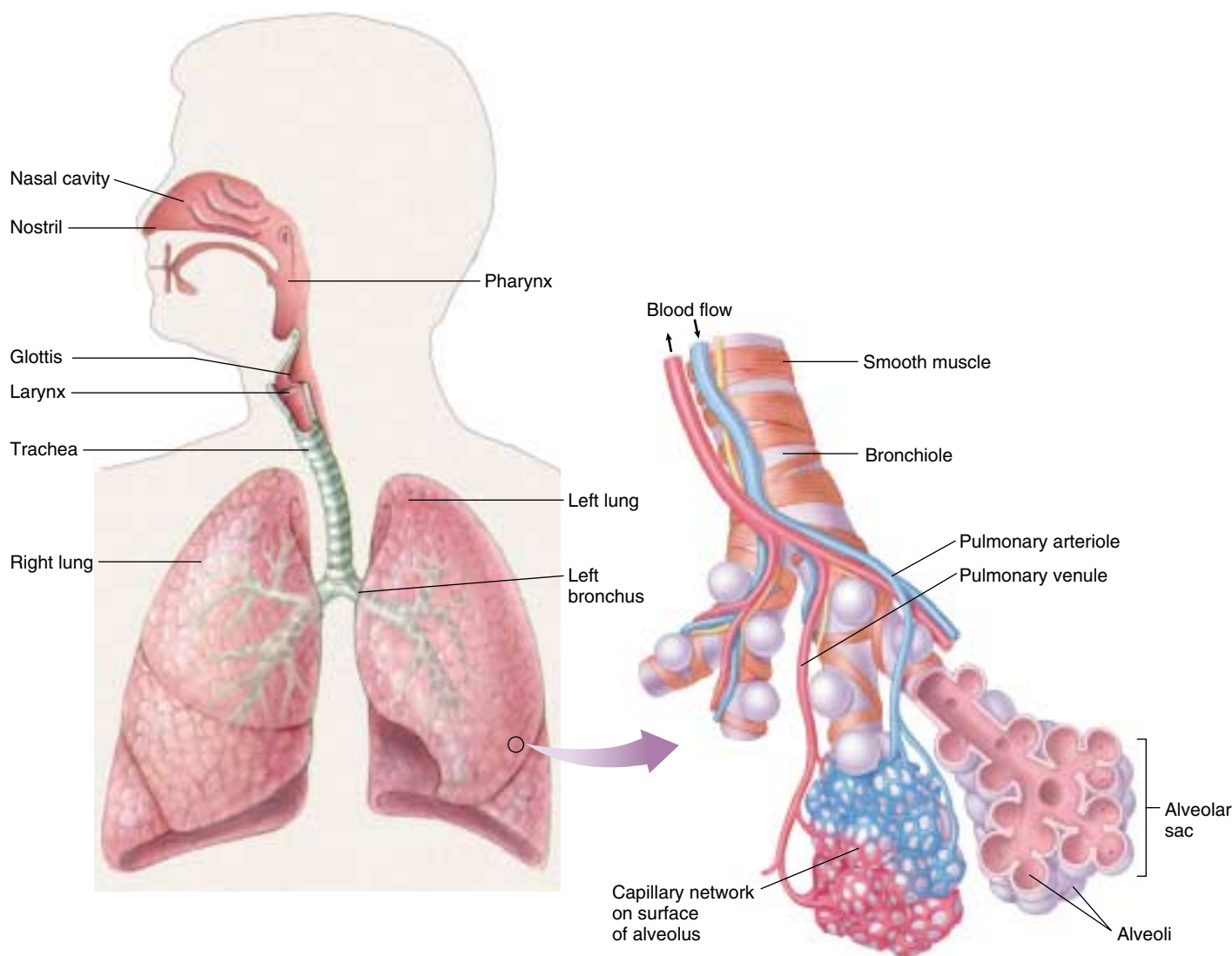


FIGURE 53.9

The human respiratory system and the structure of the mammalian lung. The lungs of mammals have an enormous surface area because of the millions of alveoli that cluster at the ends of the bronchioles. This provides for efficient gas exchange with the blood.

Respiration in Birds

The avian respiratory system has a unique structure that affords birds the most efficient respiration of all terrestrial vertebrates. Unlike the blind-ended alveoli in the lungs of mammals, the bird lung channels air through tiny air vessels called parabronchi, where gas exchange occurs (figure 53.10*a*). Air flows through the parabronchi in one direction only; this is similar to the unidirectional flow of water through a fish gill, but markedly different from the two-way flow of air through the airways of other terrestrial vertebrates. In other terrestrial vertebrates, the inhaled fresh air is mixed with “old” oxygen-depleted air that was not exhaled from the previous breathing cycle. In birds, only fresh air enters the parabronchi of the lung, and the old air exits the lung by a different route.

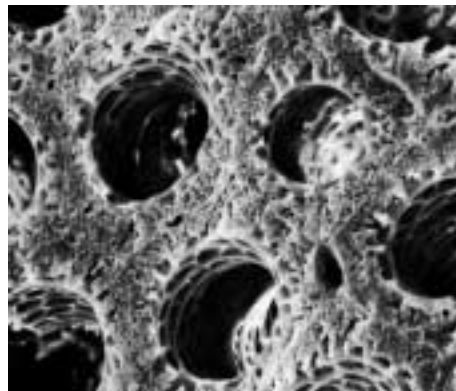
The unidirectional flow of air through the parabronchi of an avian lung is achieved through the action of air sacs, which are unique to birds (figure 53.10*b*). There are two groups of air sacs, anterior and posterior. When they are expanded during *inspiration* they take in air, and when they are compressed during *expiration* they push air into and through the lungs.

If we follow the path of air through the avian respiratory system, we will see that respiration occurs in *two cycles*. Each cycle has an inspiration and expiration phase—but the

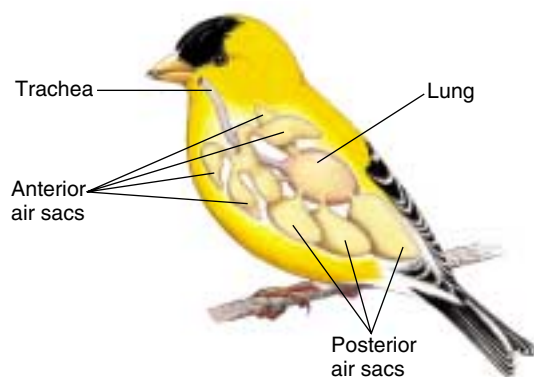
air inhaled in one cycle is not exhaled until the second cycle. Upon inspiration, both anterior and posterior air sacs expand and take in air. The inhaled air, however, only enters the posterior air sacs; the anterior air sacs fill with air from the lungs (figure 53.10*c*). Upon expiration, the air forced out of the anterior air sacs is exhaled, but the air forced out of the posterior air sacs enters the lungs. This process is repeated in the second cycle, so that air flows through the lungs in one direction and is exhaled at the end of the second cycle.

The unidirectional flow of air also permits a second respiratory efficiency: the flow of blood through the avian lung runs at a 90° angle to the air flow. This **cross-current flow** is not as efficient as the 180° countercurrent flow in fish gills, but it has the capacity to extract more oxygen from the air than a mammalian lung can. Because of the unidirectional air flow in the parabronchi and cross-current blood flow, a sparrow can be active at an altitude of 6000 meters while a mouse, which has a similar body mass and metabolic rate, cannot respire successfully at that elevation.

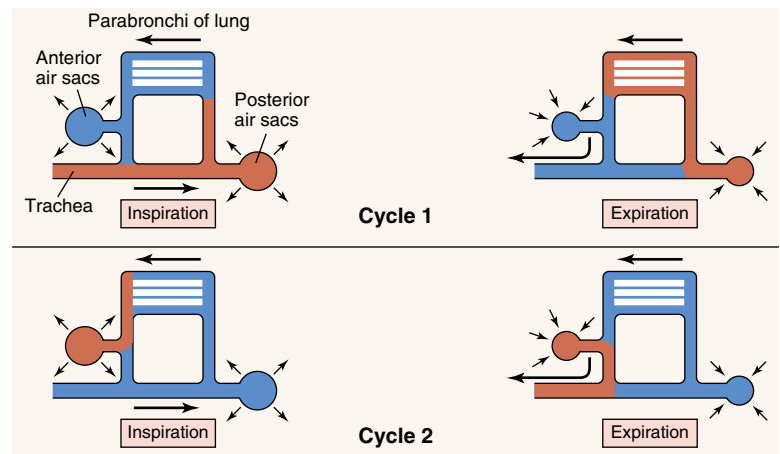
The avian respiration system is the most efficient among terrestrial vertebrates because it has unidirectional air flow and cross-current blood flow through the lungs.



(a)



(b)



(c)

FIGURE 53.10

How a bird breathes. (a) Cross section of lung of a domestic chicken (75×). Air travels through tiny tunnels in the lungs, called parabronchi, while blood circulates within the fine lattice at right angles to the air flow. This cross-current flow makes the bird lung very efficient at extracting oxygen. (b) Birds have a system of air sacs, divided into an anterior group and posterior group, that extend between the internal organs and into the bones. (c) Breathing occurs in two cycles. *Cycle 1*: Inhaled air (shown in red) is drawn from the trachea into the posterior air sacs and then is exhaled into the lungs. *Cycle 2*: Air is drawn from the lungs into the anterior air sacs and then is exhaled through the trachea. Passage of air through the lungs is always in the same direction, from posterior to anterior (right to left in this diagram).

53.4 Mammalian breathing is a dynamic process.

Structures and Mechanisms of Breathing

In mammals, inspired air travels through the trachea, bronchi, and bronchioles to reach the alveoli, where gas exchange occurs. Each alveolus is composed of an epithelium only one cell thick, and is surrounded by blood capillaries with walls that are also only one cell layer thick. There are about 30 billion capillaries in both lungs, or about 100 capillaries per alveolus. Thus, an alveolus can be visualized as a microscopic air bubble whose entire surface is bathed by blood. Because the alveolar air and the capillary blood are separated by only two cell layers, the distance between the air and blood is only 0.5 to 1.5 micrometers, allowing for the rapid exchange of gases by diffusion by decreasing d in Fick's Law.

The blood leaving the lungs, as a result of this gas exchange, normally contains a partial oxygen pressure (P_{O_2}) of about 100 millimeters of mercury. As previously discussed, the P_{O_2} is a measure of the concentration of dissolved oxygen—you can think of it as indicating the plasma oxygen. Because the P_{O_2} of the blood leaving the lungs is close to the P_{O_2} of the air in the alveoli (about 105 mm Hg), the lungs do a very effective, but not perfect, job of oxygenating the blood. After gas exchange in the systemic capillaries, the blood that returns to the right side of the heart is depleted in oxygen, with a P_{O_2} of about 40 millimeters of mercury. These changes in the P_{O_2} of the blood, as well as the changes in plasma carbon dioxide (indicated as the P_{CO_2}), are shown in figure 53.11.

The outside of each lung is covered by a thin membrane called the **visceral pleural membrane**. A second, **parietal pleural membrane** lines the inner wall of the thoracic cavity. The space between these two membranes, the **pleural cavity**, is normally very small and filled with fluid. This fluid links the two membranes in the same way a thin film of water can hold two plates of glass together, effectively coupling the lungs to the thoracic cavity. The pleural membranes package each lung separately—if one collapses due to a perforation of the membranes, the other lung can still function.

Mechanics of Breathing

As in all other terrestrial vertebrates except amphibians, air is drawn into the lungs by the creation of a negative, or subatmospheric, pressure. In accordance with *Boyle's Law*, when the volume of a given quantity of gas increases its pressure decreases. This occurs because the volume of the thorax is increased during inspiration (inhalation), and the lungs likewise expand because of the adherence of the visceral and parietal pleural membranes. When the pressure within the lungs is lower than the atmospheric pressure, air enters the lungs.

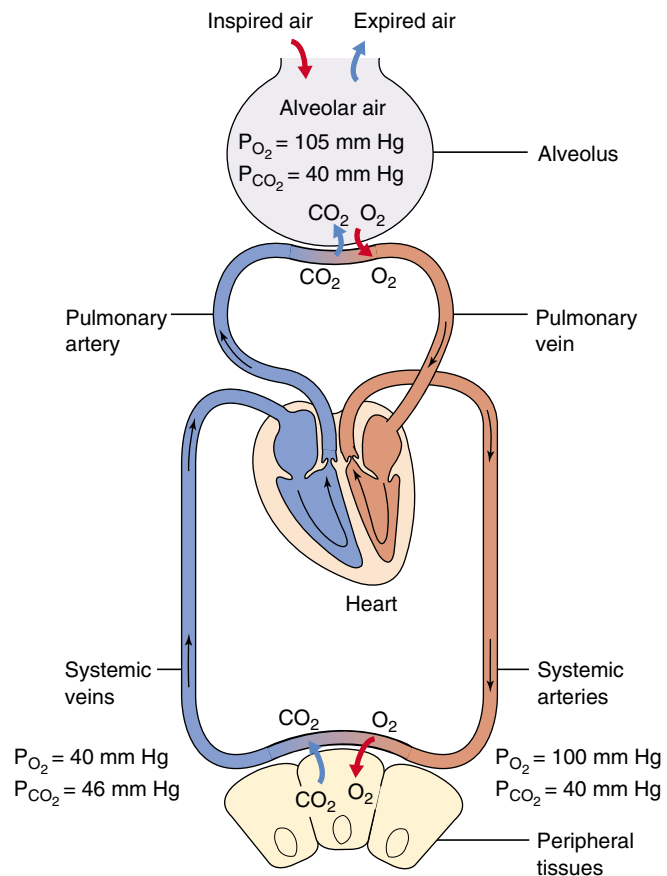


FIGURE 53.11

Gas exchange in the blood capillaries of the lungs and systemic circulation. As a result of gas exchange in the lungs, the systemic arteries carry oxygenated blood with a relatively low carbon dioxide concentration. After the oxygen is unloaded to the tissues, the blood in the systemic veins has a lowered oxygen content and an increased carbon dioxide concentration.

The thoracic volume is increased through contraction of two sets of muscles: the *external intercostals* and the *diaphragm*. During inspiration, contraction of the external intercostal muscles between the ribs raises the ribs and expands the rib cage. Contraction of the diaphragm, a convex sheet of striated muscle separating the thoracic cavity from the abdominal cavity, causes the diaphragm to lower and assume a more flattened shape. This expands the volume of the thorax and lungs while it increases the pressure on the abdomen (causing the belly to protrude). You can force a deeper inspiration by contracting other muscles that insert on the sternum or rib cage and expand the thoracic cavity and lungs to a greater extent (figure 53.12a).

The thorax and lungs have a degree of *elasticity*—they tend to resist distension and they recoil when the distend-

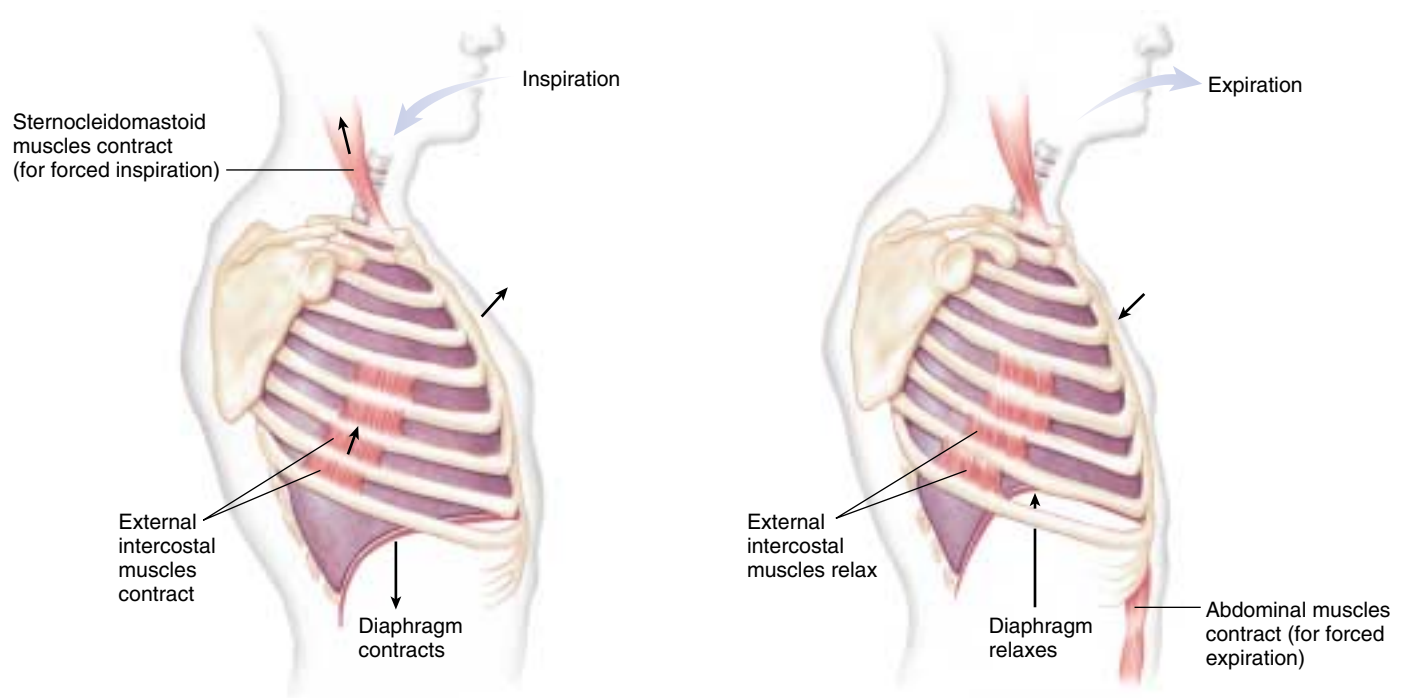


FIGURE 53.12

How a human breathes. (a) Inspiration. The diaphragm contracts and the walls of the chest cavity expand, increasing the volume of the chest cavity and lungs. As a result of the larger volume, air is drawn into the lungs. (b) Expiration. The diaphragm and chest walls return to their normal positions as a result of elastic recoil, reducing the volume of the chest cavity and forcing air out of the lungs through the trachea. Note that inspiration can be forced by contracting accessory respiratory muscles (such as the sternocleidomastoid), and expiration can be forced by contracting abdominal muscles.

ing force subsides. Expansion of the thorax and lungs during inspiration places these structures under elastic tension. It is the relaxation of the external intercostal muscles and diaphragm that produces unforced expiration, because it relieves that elastic tension and allows the thorax and lungs to recoil. You can force a greater expiration by contracting your abdominal muscles and thereby pressing the abdominal organs up against the diaphragm (figure 53.12b).

Breathing Measurements

A variety of terms are used to describe the volume changes of the lung during breathing. At rest, each breath moves a **tidal volume** of about 500 milliliters of air into and out of the lungs. About 150 milliliters of the tidal volume is contained in the tubular passages (trachea, bronchi, and bronchioles), where no gas exchange occurs. The air in this *anatomical dead space* mixes with fresh air during inspiration. This is one of the reasons why respiration in mammals is not as efficient as in birds, where air flow through the lungs is one-way.

The maximum amount of air that can be expired after a forceful, maximum inspiration is called the **vital capacity**. This measurement, which averages 4.6 liters in young men and 3.1 liters in young women, can be clinically important, because an abnormally low vital capacity may indicate dam-

age to the alveoli in various pulmonary disorders. For example, in **emphysema**, a potentially fatal condition usually caused by cigarette smoking, vital capacity is reduced as the alveoli are progressively destroyed.

A person normally breathes at a rate and depth that properly oxygenate the blood and remove carbon dioxide, keeping the blood P_{O_2} and P_{CO_2} within a normal range. If breathing is insufficient to maintain normal blood gas measurements (a rise in the blood P_{CO_2} is the best indicator), the person is **hypoventilating**. If breathing is excessive for a particular metabolic rate, so that the blood P_{CO_2} is abnormally lowered, the person is said to be **hyperventilating**. Perhaps surprisingly, the increased breathing that occurs during moderate exercise is not necessarily hyperventilation, because the faster breathing is matched to the faster metabolic rate, and blood gas measurements remain normal. The next section describes how breathing is regulated to keep pace with metabolism.

Humans inspire by contracting muscles that insert on the rib cage and by contracting the diaphragm. Expiration is produced primarily by muscle relaxation and elastic recoil. As a result, the blood oxygen and carbon dioxide levels are maintained in a normal range through adjustments in the depth and rate of breathing.

Mechanisms That Regulate Breathing

Each breath is initiated by neurons in a *respiratory control center* located in the medulla oblongata, a part of the brain stem (see chapter 54). These neurons send impulses to the diaphragm and external intercostal muscles, stimulating them to contract, and contractions of these muscles expand the chest cavity, causing inspiration. When these neurons stop producing impulses, the inspiratory muscles relax and expiration occurs. Although the muscles of breathing are skeletal muscles, they are usually controlled automatically. This control can be voluntarily overridden, however, as in hypoventilation (breath holding) or hyperventilation.

A proper rate and depth of breathing is required to maintain the blood oxygen and carbon dioxide levels in the normal range. Thus, although the automatic breathing cycle is driven by neurons in the brain stem, these neurons must be responsive to changes in blood P_{O_2} and P_{CO_2} in order to maintain homeostasis. You can demonstrate this mechanism by simply holding your breath. Your blood carbon dioxide immediately rises and your blood oxygen falls. After a short time, the urge to breathe induced by the changes in blood gases becomes overpowering. This is due primarily to the rise in blood carbon dioxide, as indicated by a rise in P_{CO_2} , rather than to the fall in oxygen levels.

A rise in P_{CO_2} causes an increased production of carbonic acid (H_2CO_3), which is formed from carbon dioxide and water and acts to lower the blood pH (carbonic acid dissociates into HCO_3^- and H^+ , thereby increasing blood H^+ concentration). A fall in blood pH stimulates neurons in the **aortic** and **carotid bodies**, which are sensory structures known as *peripheral chemoreceptors* in the aorta and the carotid artery. These receptors send impulses to the respiratory control center in the medulla oblongata, which then stimulates increased breathing. The brain also contains chemoreceptors, but they cannot be stimulated by blood H^+ because the blood is unable to enter the brain. After a brief delay, however, the increased blood P_{CO_2} also causes a decrease in the pH of the cerebrospinal fluid (CSF) bathing the brain. This stimulates the **central chemoreceptors** in the brain (figure 53.13).

The peripheral chemoreceptors are responsible for the immediate stimulation of breathing when the blood P_{CO_2} rises, but this immediate stimulation only accounts for about 30% of increased ventilation. The central chemore-

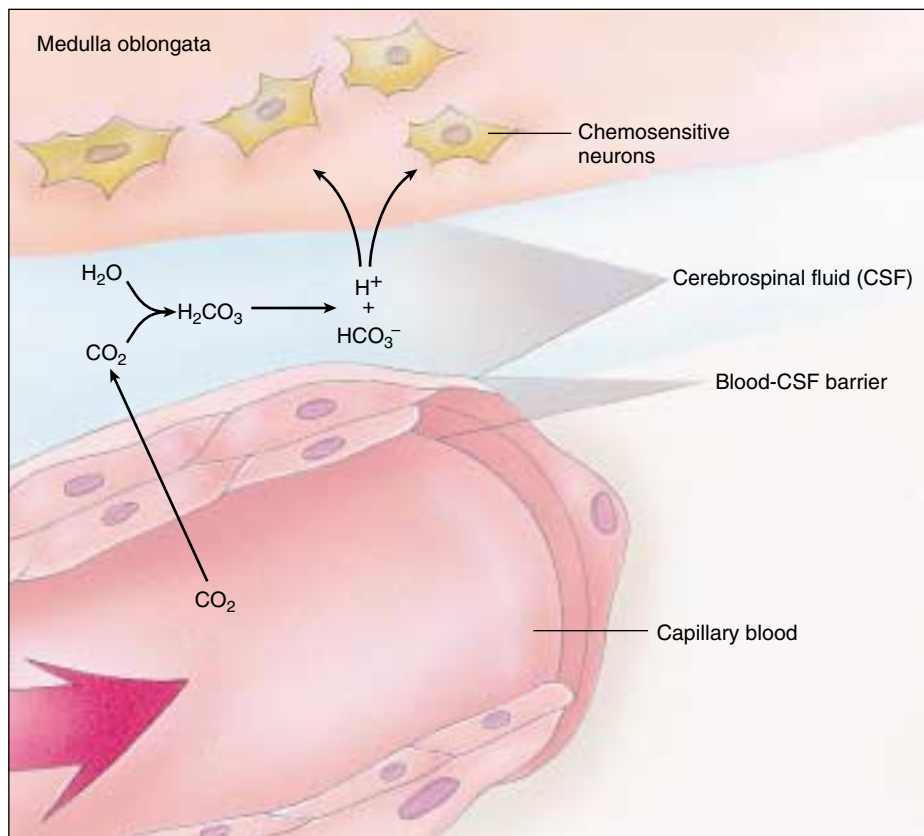


FIGURE 53.13

The effect of blood CO_2 on cerebrospinal fluid (CSF). Changes in the pH of the CSF are detected by chemosensitive neurons in the brain that help regulate breathing.

ceptors are responsible for the sustained increase in ventilation if P_{CO_2} remains elevated. The increased respiratory rate then acts to eliminate the extra CO_2 , bringing the blood pH back to normal (figure 53.14).

A person cannot voluntarily hyperventilate for too long. The decrease in plasma P_{CO_2} and increase in pH of plasma and CSF caused by hyperventilation extinguish the reflex drive to breathe. They also lead to constriction of cerebral blood vessels, causing dizziness. People can hold their breath longer if they hyperventilate first, because it takes longer for the CO_2 levels to build back up, not because hyperventilation increases the P_{O_2} of the blood. Actually, in people with normal lungs, P_{O_2} becomes a significant stimulus for breathing only at high altitudes, where the P_{O_2} is low. Low P_{O_2} can also stimulate breathing in patients with emphysema, where the lungs are so damaged that blood CO_2 can never be adequately eliminated.

Breathing serves to keep the blood gases and pH in the normal range and is under the reflex control of peripheral and central chemoreceptors. These chemoreceptors sense the pH of the blood and cerebrospinal fluid, and they regulate the respiratory control center in the medulla oblongata of the brain.

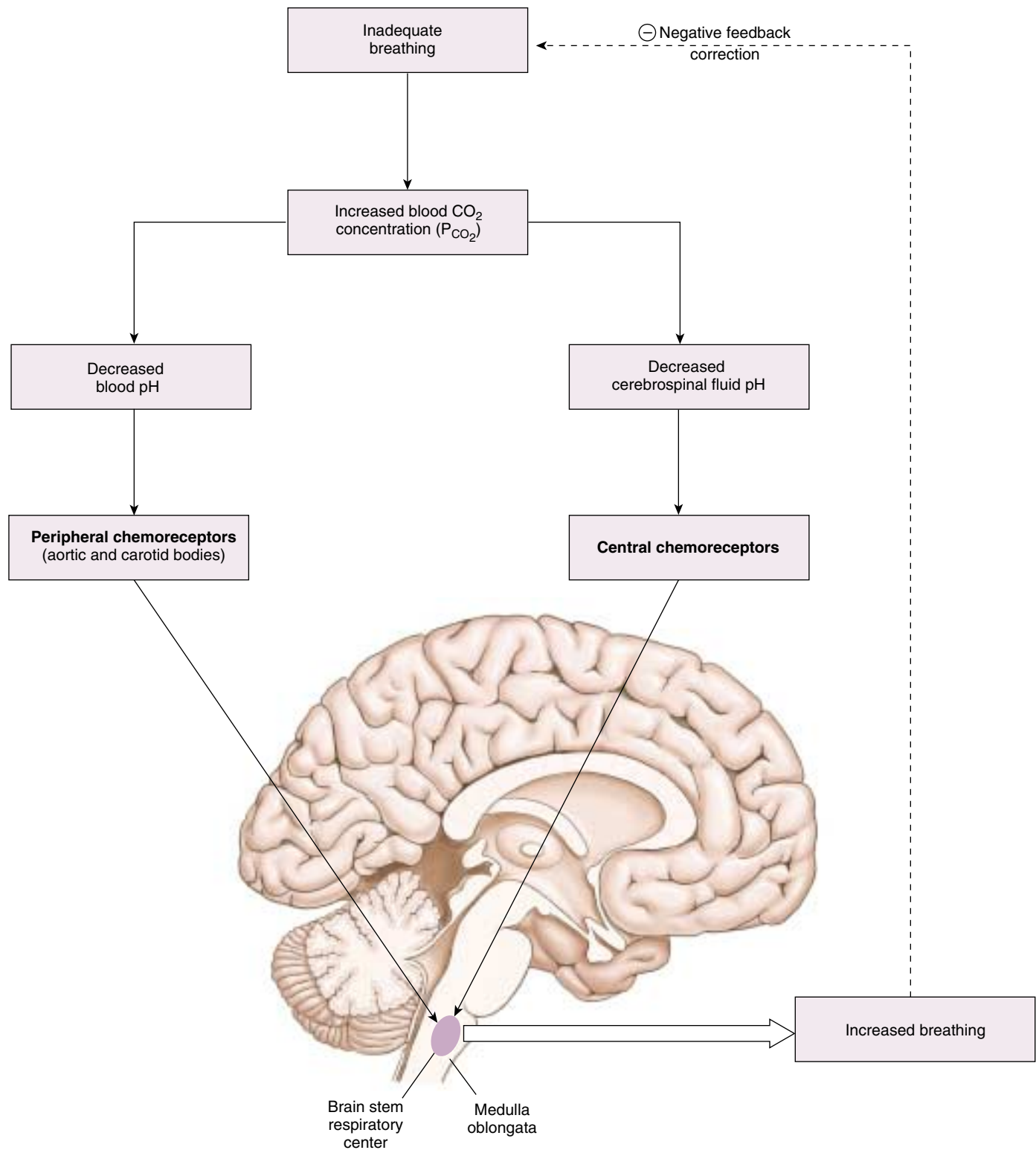


FIGURE 53.14

The regulation of breathing by chemoreceptors. Peripheral and central chemoreceptors sense a fall in the pH of blood and cerebrospinal fluid, respectively, when the blood carbon dioxide levels rise as a result of inadequate breathing. In response, they stimulate the respiratory control center in the medulla oblongata, which directs an increase in breathing. As a result, the blood carbon dioxide concentration is returned to normal, completing the negative feedback loop.

53.5 Blood transports oxygen and carbon dioxide.

Hemoglobin and Oxygen Transport

When oxygen diffuses from the alveoli into the blood, its journey is just beginning. The circulatory system delivers oxygen to tissues for respiration and carries away carbon dioxide. The transport of oxygen and carbon dioxide by the blood is itself an interesting and physiologically important process.

The amount of oxygen that can be dissolved in the blood plasma depends directly on the P_{O_2} of the air in the alveoli, as we explained earlier. When the lungs are functioning normally, the blood plasma leaving the lungs has almost as much dissolved oxygen as is theoretically possible, given the P_{O_2} of the air. Because of oxygen's low solubility in water, however, blood plasma can contain a maximum of only about 3 milliliters O_2 per liter. Yet whole blood carries almost 200 milliliters O_2 per liter! Most of the oxygen is bound to molecules of hemoglobin inside the red blood cells.

Hemoglobin is a protein composed of four polypeptide chains and four organic compounds called *heme groups*. At the center of each heme group is an atom of iron, which can bind to a molecule of oxygen (figure 53.15). Thus, each hemoglobin molecule can carry up to four molecules of oxygen. Hemoglobin loads up with oxygen in the lungs, forming **oxyhemoglobin**. This molecule has a bright red, tomato juice color. As blood passes through capillaries in the rest of the body, some of the oxyhemoglobin releases oxygen and becomes **deoxyhemoglobin**. Deoxyhemoglobin has a dark red color (the color of blood that is collected from the veins of blood donors), but it imparts a bluish tinge to tissues. Because of these color changes, vessels that carry oxygenated blood are always shown in artwork with a red color, and vessels that carry oxygen-depleted blood are indicated with a blue color.

Hemoglobin is an ancient protein that is not only the oxygen-carrying molecule in all vertebrates, but is also used as an oxygen carrier by many invertebrates, including annelids, mollusks, echinoderms, flatworms, and even some protists. Many other invertebrates, however, employ different oxygen carriers, such as *hemocyanin*. In hemocyanin, the oxygen-binding atom is copper instead of iron. Hemocyanin is not found in blood cells, but is instead dissolved in the circulating fluid (hemolymph) of invertebrates.

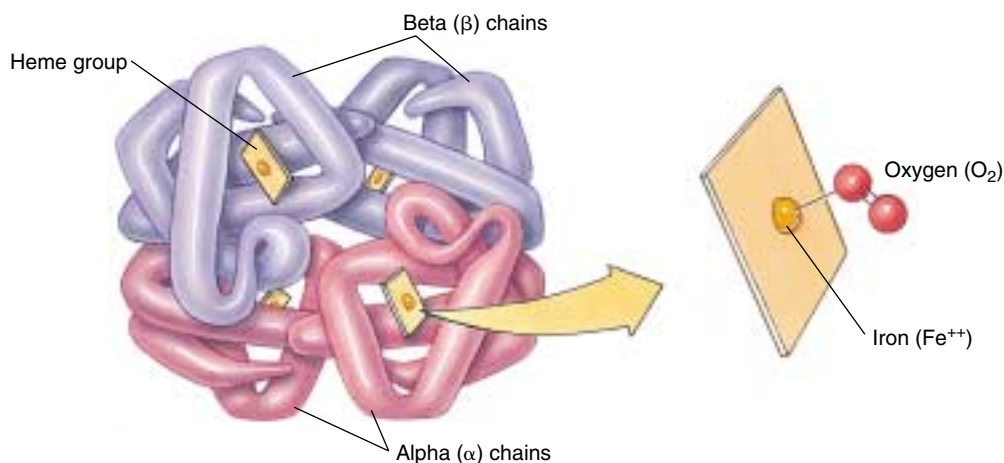


FIGURE 53.15

Hemoglobin consists of four polypeptide chains—two alpha (α) chains and two beta (β) chains. Each chain is associated with a heme group, and each heme group has a central iron atom, which can bind to a molecule of O_2 .

Oxygen Transport

The P_{O_2} of the air within alveoli at sea level is approximately 105 millimeters of mercury (mm Hg), which is less than the P_{O_2} of the atmosphere because of the mixing of freshly inspired air with “old” air in the anatomical dead space of the respiratory system. The P_{O_2} of the blood leaving the alveoli is slightly less than this, about 100 mm Hg, because the blood plasma is not completely saturated with oxygen due to slight inefficiencies in lung function. At a blood P_{O_2} of 100 mm Hg, approximately 97% of the hemoglobin within red blood cells is in the form of oxyhemoglobin—indicated as a percent oxyhemoglobin saturation of 97%.

As the blood travels through the systemic blood capillaries, oxygen leaves the blood and diffuses into the tissues. Consequently, the blood that leaves the tissue in the veins has a P_{O_2} that is decreased (in a resting person) to about 40 mm Hg. At this lower P_{O_2} , the percent saturation of hemoglobin is only 75%. A graphic representation of these changes is called an oxyhemoglobin dissociation curve (figure 53.16). In a person at rest, therefore, 22% (97% minus 75%) of the oxyhemoglobin has released its oxygen to the tissues. Put another way, roughly one-fifth of the oxygen is unloaded in the tissues, leaving four-fifths of the oxygen in the blood as a reserve.

This large reserve of oxygen serves an important function. It enables the blood to supply the body's oxygen needs during exercise as well as at rest. During exercise, the muscles' accelerated metabolism uses more oxygen from the capillary blood and thus decreases the venous blood P_{O_2} . For example, the P_{O_2} of the venous blood could drop to 20 mm Hg; in this case, the percent saturation of hemoglobin will be only 35% (see figure 53.16).

Because arterial blood still contains 97% oxyhemoglobin (ventilation increases proportionately with exercise), the amount of oxygen unloaded is now 62% (97% minus 35%), instead of the 22% at rest. In addition to this function, the oxygen reserve also ensures that the blood contains enough oxygen to maintain life for four to five minutes if breathing is interrupted or if the heart stops pumping.

Oxygen transport in the blood is affected by other conditions. The CO_2 produced by metabolizing tissues as a product of aerobic respiration combines with H_2O to ultimately form bicarbonate and H^+ , lowering the pH of the blood. This reaction occurs primarily inside red blood cells, where the lowered pH reduces hemoglobin's affinity for oxygen and thus causes it to release oxygen more readily. The effect of pH on hemoglobin's affinity for oxygen is known as the Bohr effect and is shown graphically by a shift of the oxyhemoglobin dissociation curve to the right (figure 53.17a). Increasing temperature has a similar effect on hemoglobin's affinity for oxygen (figure 53.17b). Because skeletal muscles produce carbon dioxide more rapidly during exercise and active muscles produce heat, the blood unloads a higher percentage of the oxygen it carries during exercise.

Deoxyhemoglobin combines with oxygen in the lungs to form oxyhemoglobin, which dissociates in the tissue capillaries to release its oxygen. The degree to which the loading reaction occurs depends on ventilation; the degree of unloading is influenced by such factors as pH and temperature.

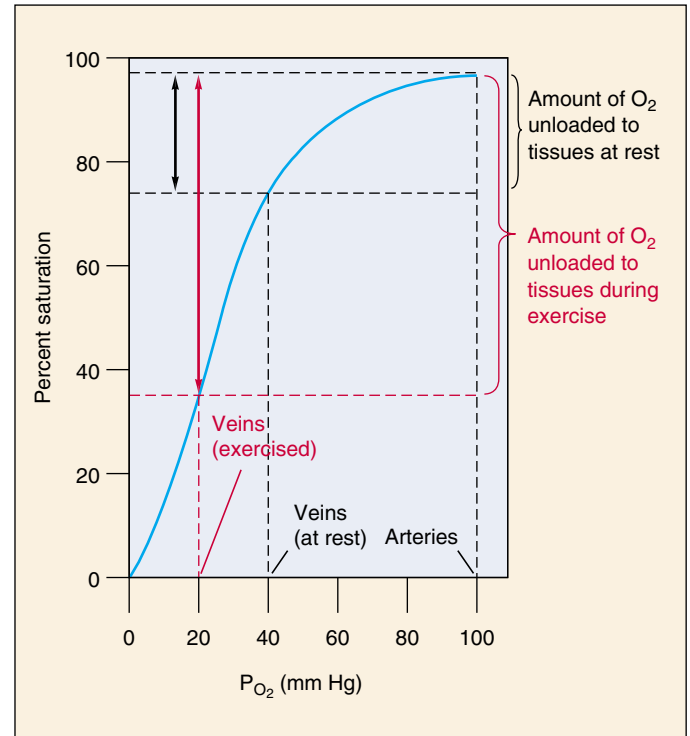


FIGURE 53.16

The oxyhemoglobin dissociation curve. Hemoglobin combines with O_2 in the lungs, and this oxygenated blood is carried by arteries to the body cells. After oxygen is removed from the blood to support cell respiration, the blood entering the veins contains less oxygen. The difference in O_2 content between arteries and veins during rest and exercise shows how much O_2 was unloaded to the tissues.

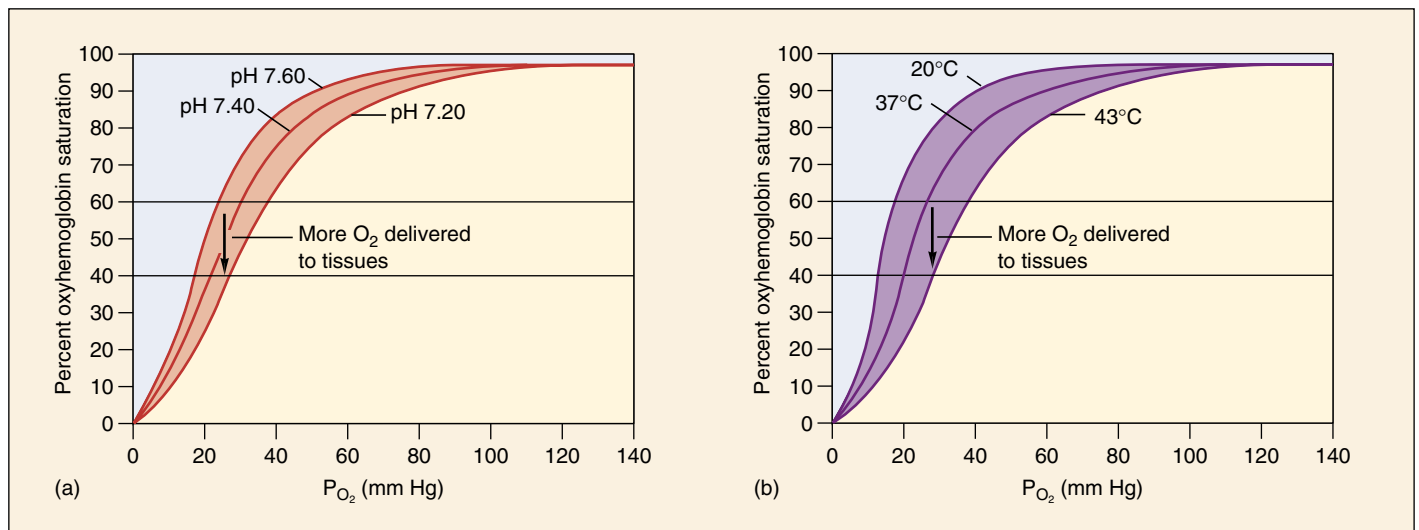


FIGURE 53.17

The effect of pH and temperature on the oxyhemoglobin dissociation curve. Lower blood pH (a) and higher blood temperatures (b) shift the oxyhemoglobin dissociation curve to the right, facilitating oxygen unloading. This can be seen as a lowering of the oxyhemoglobin percent saturation from 60 to 40% in the example shown, indicating that the difference of 20% more oxygen is unloaded to the tissues.

Carbon Dioxide and Nitric Oxide Transport

The systemic capillaries deliver oxygen to the tissues and remove carbon dioxide. About 8% of the CO_2 in blood is simply dissolved in plasma; another 20% is bound to hemoglobin. (Because CO_2 binds to the protein portion of hemoglobin, however, and not to the heme irons, it does not compete with oxygen.) The remaining 72% of the CO_2 diffuses into the red blood cells, where the enzyme carbonic anhydrase catalyzes the combination of CO_2 with water to form carbonic acid (H_2CO_3). Carbonic acid dissociates into bicarbonate (HCO_3^-) and hydrogen (H^+) ions. The H^+ binds to deoxyhemoglobin, and the bicarbonate moves out of the erythrocyte into the plasma via a transporter that exchanges one chloride ion for a bicarbonate (this is called the “chloride shift”). This reaction removes large amounts of CO_2 from the plasma, facilitating the diffusion of additional CO_2 into the plasma from the surrounding tissues (figure 53.18). The formation of carbonic acid is also important in maintaining the acid-base balance of the blood, because bicarbonate serves as the major buffer of the blood plasma.

The blood carries CO_2 in these forms to the lungs. The lower P_{CO_2} of the air inside the alveoli causes the carbonic anhydrase reaction to proceed in the reverse direction,

converting H_2CO_3 into H_2O and CO_2 (see figure 53.18). The CO_2 diffuses out of the red blood cells and into the alveoli, so that it can leave the body in the next exhalation (figure 53.19).

Nitric Oxide Transport

Hemoglobin also has the ability to hold and release nitric oxide gas (NO). Although a noxious gas in the atmosphere, nitric oxide has an important physiological role in the body and acts on many kinds of cells to change their shapes and functions. For example, in blood vessels the presence of NO causes the blood vessels to expand because it relaxes the surrounding muscle cells (see chapters 7 and 52). Thus, blood flow and blood pressure are regulated by the amount of NO released into the bloodstream.

A current hypothesis proposes that hemoglobin carries NO in a special form called super nitric oxide. In this form, NO has acquired an extra electron and is able to bind to the amino acid cysteine in hemoglobin. In the lungs, hemoglobin that is dumping CO_2 and picking up O_2 also picks up NO as super nitric oxide. In blood vessels at the tissues, hemoglobin that is releasing its O_2 and picking up CO_2 can do one of two things with nitric oxide. To increase blood flow, hemoglobin can release the

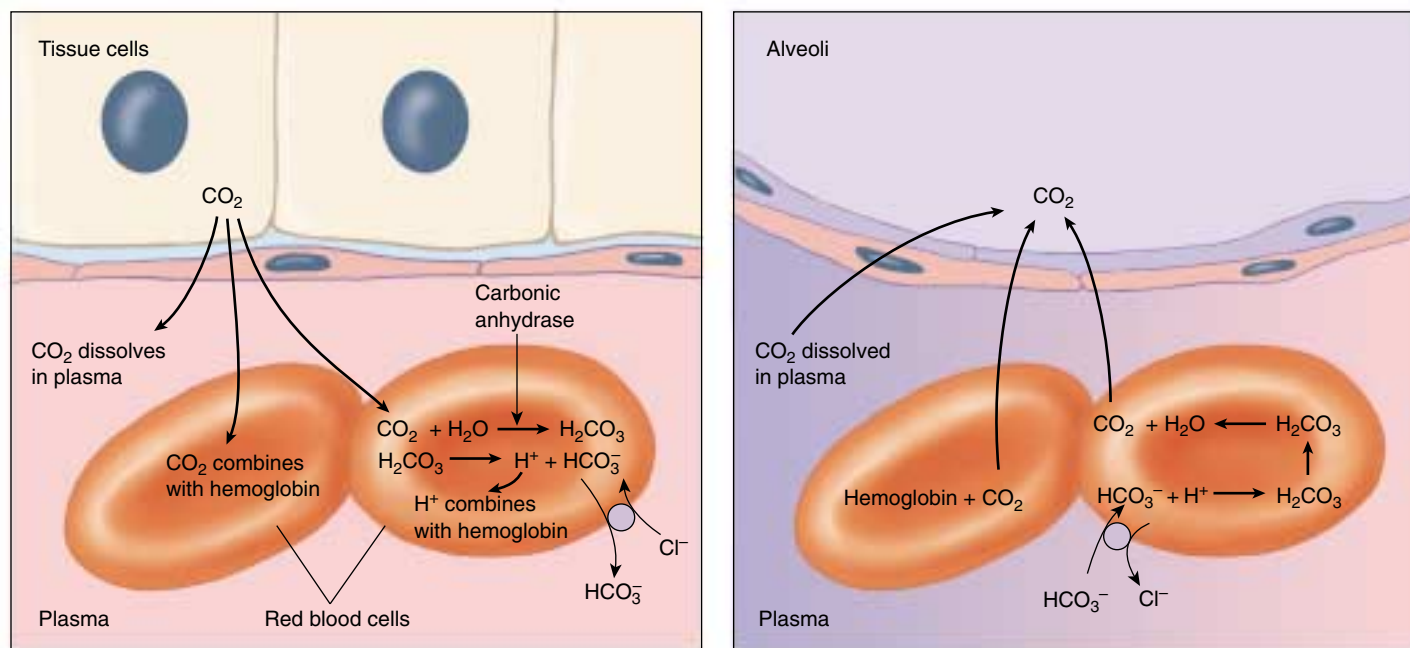


FIGURE 53.18

The transport of carbon dioxide by the blood. CO_2 is transported in three ways: dissolved in plasma, bound to the protein portion of hemoglobin, and as carbonic acid and bicarbonate, which form in the red blood cells. When the blood passes through the pulmonary capillaries, these reactions are reversed so that CO_2 gas is formed, which is exhaled.

GAS EXCHANGE DURING RESPIRATION

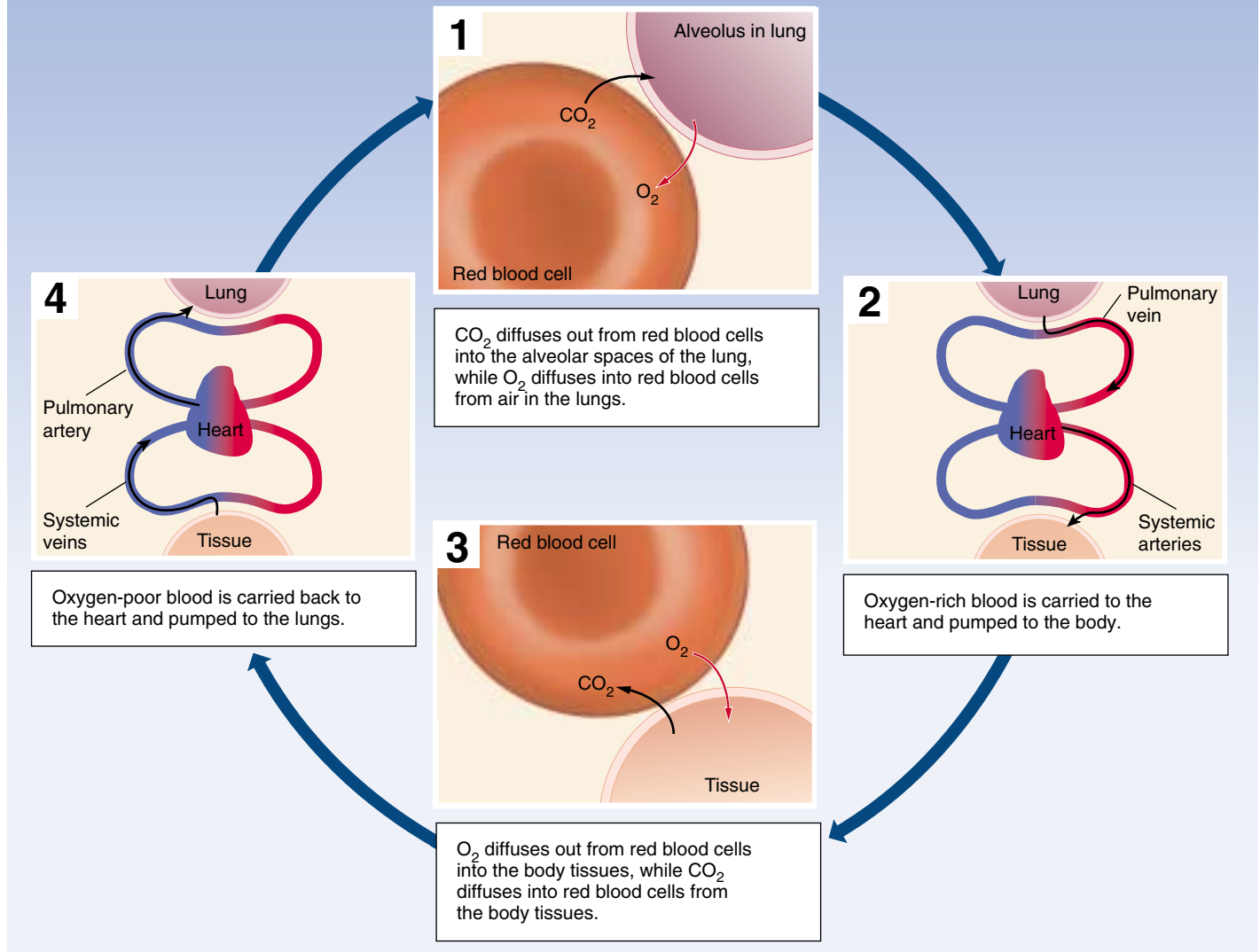


FIGURE 53.19
Summary of respiratory gas exchange.

super nitric oxide as NO into the blood, making blood vessels expand because NO acts as a relaxing agent. Or, hemoglobin can trap any excess of NO on its iron atoms left vacant by the release of oxygen, causing blood vessels to constrict. When the red blood cells return to the lungs, hemoglobin dumps its CO₂ and the regular form of NO bound to the iron atoms. It is then ready to pick up O₂ and super nitric oxide and continue the cycle.

Carbon dioxide is transported in the blood in three ways: dissolved in the plasma, bound to hemoglobin, and the majority as bicarbonate in the plasma following an enzyme-catalyzed reaction in the red blood cells. Nitric oxide is also transported in the blood providing yet another explanation of NO actions on blood vessels.



Summary

Questions

Media Resources

53.1 Respiration involves the diffusion of gases.

- The factors that influence the rate of diffusion, surface area, concentration gradient, and diffusion distance, are described by Fick's Law.
- Animals have evolved to maximize the diffusion rate across respiratory membranes by increasing the respiratory surface area, increasing the concentration gradient across the membrane, or decreasing the diffusion distance.

1. Approximately what percentage of dry air is oxygen, and what percentage is carbon dioxide?
2. Why is it that only very small organisms can satisfy their respiratory requirements by direct diffusion to all cells from the body surface?



- Respiration



- Gas exchange systems
- Respiratory overview
- Gas exchange

53.2 Gills are used for respiration by aquatic vertebrates.

- As water flows past a gill's lamellae, it comes close to blood flowing in an opposite, or countercurrent, direction; this maximizes the concentration difference between the two fluids, thereby maximizing the diffusion of gases.

3. What is countercurrent flow, and how does it help make the fish gill the most efficient respiratory organ?

53.3 Lungs are used for respiration by terrestrial vertebrates.

- Reptiles, birds, and mammals use negative pressure breathing; air is taken into the lungs when the lung volume is expanded to create a partial vacuum.
- Mammals have lungs composed of millions of alveoli, where gas exchange occurs; this is very efficient, but because inspiration and expiration occur through the same airways, new air going into the lungs is mixed with some old air.

4. How do amphibians get air into their lungs? How do other terrestrial vertebrates get air into their lungs?
5. What two features in birds make theirs the most efficient of all terrestrial respiratory systems?



- Art Activities:
Respiratory tract
Upper respiratory tract
Section of larynx



- Gas exchange

53.4 Mammalian breathing is a dynamic process.

- The lungs are covered with a wet membrane that sticks to the wet membrane lining the thoracic cavity, so the lungs expand as the chest expands through muscular contractions.
- Breathing is controlled by centers in the medulla oblongata of the brain; breathing is stimulated by a rise in blood CO₂, and consequent fall in blood pH, as sensed by chemoreceptors located in the aorta and carotid artery.

6. How are the lungs connected to and supported within the thoracic cavity?
7. How does the brain control inspiration and expiration? How do peripheral and central chemoreceptors influence the brain's control of breathing?



- Boyle's Law
- Breathing



- Breathing
- Mechanics of ventilation
- Control of respiration

53.5 Blood transports oxygen and carbon dioxide.

- Hemoglobin loads with oxygen in the lungs; this oxyhemoglobin then unloads oxygen as the blood goes through the systemic capillaries.
- Carbon dioxide combines with water as the carbon dioxide is transported to the lungs for exhalation.

8. In what form does most of the carbon dioxide travel in the blood? How and where is this molecule produced?



- Art Activity:
Hemoglobin module



- Hemoglobin

Regulating the Animal Body

Are Pollutants Affecting the Sexual Development of Florida's Alligators?

Alligators are among the most interesting of animals for a biologist to study. Their ecology is closely tied to the environment, and their reptilian biology offers an interesting contrast to that of mammals like ourselves. Studies of alligator development offer powerful general lessons well worthy of our attention.

In no area of biology is this more true than in investigation of alligator sexual development. This importance is not because sexual development in alligators is unusual. It is not. As with all vertebrates, sexual development in alligator males—particularly development of their external sexual organs—is largely dependent on the androgen sex hormone testosterone and its derivatives. In the alligator embryo, these steroid hormones are responsible for the differentiation of the male internal duct system, as well as the formation of the external genitalia. After the alligator's birth, androgen hormones are essential for normal maturation and growth of the juvenile male reproductive system, particularly during puberty.

The strong dependence of a male alligator's sexual development on androgens is not unusual—mammals show the same strong dependence. So why should researchers be interested in alligators? In a nutshell, we humans don't spend our lives sloshing around in an aquatic environment, and alligators do. Florida alligators live in the many lakes that pepper the state, and, living in these lakes, they are exposed all their lives to whatever chemicals happen to be added to the lakewater by chemical spills, industrial wastes, and agricultural runoff.

The androgen-dependent sexual development of alligators provides a sensitive barometer to environmental pollution, because the androgen response can be blocked by a class of pollutant chemicals called endocrine disrupters. When endocrine-disrupting pollutants contaminate Florida lakes, their presence can be detected by its impact on the sexual development of the lakes' resident alligators. Just as the death of coal miners' canaries warn of the buildup of dangerous gas within the shaft of coal mines, so disruption of the sexual development of alligators can warn us of dangerous chemicals in the environment around us.

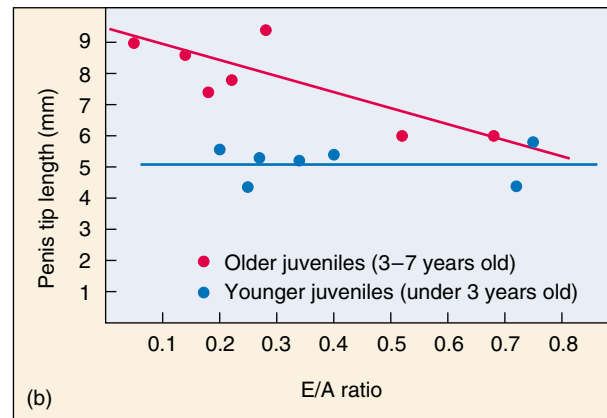
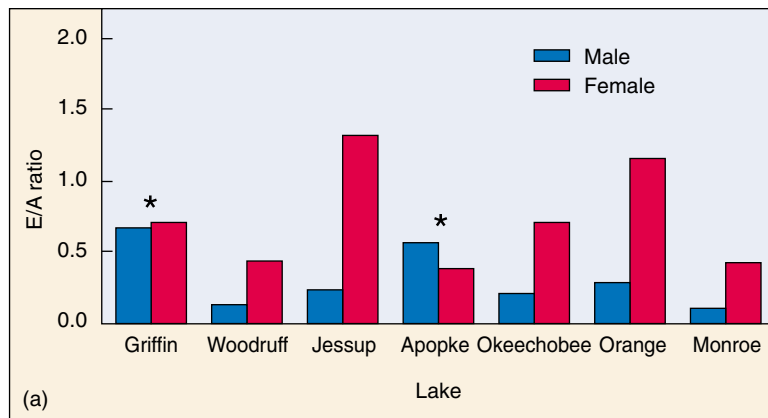


Catching alligators is a job best done at night. Alligators in Florida lakes, like the one shown here in the hands of Professor Guillette, seem to be experiencing developmental abnormalities, perhaps due to pollution of many of Florida's lakes by endocrine-disrupting chemicals.

One of the great joys of biological research is being able to choose research that is fun to do. Few research projects offer the particular joys of studying alligators. With State Game Commission permits, researchers go to lakes in central Florida, wait till after dark, then spend the night on the lake in small boats hand-capturing the animals. As you might guess, researchers mostly choose juvenile individuals. The captured animals are confined in cloth bags until sex can be determined, body measurements made, and blood samples collected, and then released.

For over six years, Louis Guillette of the University of Florida, Gainesville, and his students have been carrying out just this sort of research. Their goal has been to assess the degree to which agricultural and other chemicals have polluted the lakes of central Florida, using as their gauge the disruption of normal sexual development in alligators.

To assess hormonal changes that might be expected to inhibit male sexual development, Guillette's team looked at the relative ratio of androgens (which promote male development) to estrogens (which promote female development) in each captured alligator. Some male endocrine disrupters act like estrogens, while others decrease native androgen levels. In either case, the ratio of estrogen to androgen (the E/A ratio) increases, producing a more estrogenic environment and so retarding male sexual development. Particularly after puberty, the growth of male alligators' external sexual organs is very dependent upon a high-androgen environment. Any pollutant that raises the E/A ratio would be expected to markedly inhibit this development.



Alligator sexual development inhibited by contamination. (a) Ratio of estrogen/androgen (E/A) plasma concentrations in large juvenile alligators 3–7 years old. A relatively larger ratio in males is atypical and indicates an estrogenic hormonal environment, as opposed to the expected androgenic hormonal environment. (b) Sexual development in male alligators, measured by penis length as a function of E/A ratio. In small juveniles under 3 years old, there is no apparent influence. In older juveniles 3–7 years old, there is a pronounced effect, higher E/A ratios retarding sexual development.

The Experiment

Guillette's team first looked at animals in two lakes and then expanded the research to look at animals from several other lakes. Alligators were initially collected from Lake Woodruff National Wildlife Refuge and from Lake Apopka. Lake Woodruff is a relatively pristine lake with no agricultural or industrial runoff. Lake Apopka, on the other hand, is a large eutrophic lake exposed to various agricultural and municipal contaminants. In 1980, the lake experienced a sulfuric acid spill from a chemical company, and has a history of pesticide contamination by DDT.

Clear comparisons of alligators collected from different lakes required that animals be captured as nearly as possible at the same time, to minimize possible variation due to photoperiod, temperature, and nutrition. This experimental requirement led to truly prodigious feats of alligator catching by the research team. On a single night in 1994, 40 male alligators were hand-captured from Lake Woodruff. The following night, 54 males were captured from Lake Apopka. In a broader study of seven lakes carried out the following year, 528 animals were captured during a 17-day period.

The external genitalia and total body length were measured on captured animals. Body-length is a good indicator of the age of the alligator. Alligators reach puberty at about 3 years of age, and this must be taken into account when making comparisons.

Blood samples were taken from each animal in order to determine the plasma levels of estrogen and testosterone. Investigators measured plasma concentrations of estradiol- 17β and testosterone. By comparing the ratio of the two values, the researchers estimated the E/A ratio, and thus if the internal environment was androgenic or estrogenic.

The Results

In most of the seven lakes studied, female alligators showed

a much higher E/A ratio than males (graph *a* above), a normal result. The exceptions are Lake Griffin and Lake Apopka, the most polluted of the lakes. The larger E/A ratio observed in male alligators caught from these two lakes indicates an estrogenic hormonal environment in these animals rather than the normal androgenic one.

Does this estrogenic environment have an impact on juvenile sexual development? Yes. Researchers observed that postpuberty juvenile males from Lake Apopka and Lake Griffin (where E/A ratios were elevated) exhibited stunted reproductive organs compared to those found in Lake Woodruff and other lakes (graph *b* above).

Prepuberty males did not show this effect, exhibiting the same size external reproductive organs whatever the E/A ratio. This is as you would expect, as organ growth occurs primarily after puberty, in response to androgen hormones released from the testes.

A primary contaminant found in alligators' eggs in Lake Apopka is *p,p'*-DDE, a major metabolite of DDT. *p,p'*-DDE has been shown to bind to androgen receptors, and functions as an antiandrogen. The presence of *p,p'*-DDE reduces the androgen effect in cells, creating a more estrogenic environment.

The researchers also measured levels of plasma testosterone. The plasma levels of testosterone were significantly reduced in alligators from Lake Apopka compared to the control animals removed from Lake Woodruff. These reduced levels of plasma testosterone from the Lake Apopka alligators also act to reduce the E/A ratio, and so to produce the observed abnormalities in reproductive structures.



To explore this experiment further, go to the Virtual Lab at www.mhhe.com/raven6/vlab14.mhtml

54

The Nervous System

Concept Outline

54.1 The nervous system consists of neurons and supporting cells.

Neuron Organization. Neurons and neuroglia are organized into the central nervous system (the brain and spinal cord) and the peripheral nervous system (sensory and motor neurons).

54.2 Nerve impulses are produced on the axon membrane.

The Resting Membrane Potential. The inside of the membrane is electrically negative in comparison with the outside.

Action Potentials. In response to a stimulus that depolarizes the membrane, voltage-gated channels open, producing a nerve impulse. One action potential stimulates the production of the next along the axon.

54.3 Neurons form junctions called synapses with other cells.

Structure of Synapses. Neurotransmitters diffuse across to the postsynaptic cell and combine with receptor proteins.

Neurotransmitters and Their Functions. Some neurotransmitters cause a depolarization in the postsynaptic membrane; others produce inhibition by hyperpolarization.

54.4 The central nervous system consists of the brain and spinal cord.

The Evolution of the Vertebrate Brain. Vertebrate brains include a forebrain, midbrain, and hindbrain.

The Human Forebrain. The cerebral cortex contains areas specialized for different functions.

The Spinal Cord. Reflex responses and messages to and from the brain are coordinated by the spinal cord.

54.5 The peripheral nervous system consists of sensory and motor neurons.

Components of the Peripheral Nervous System. A spinal nerve contains sensory and motor neurons.

The Autonomic Nervous System. Sympathetic motor neurons arouse the body for fight or flight; parasympathetic motor neurons have antagonistic actions.

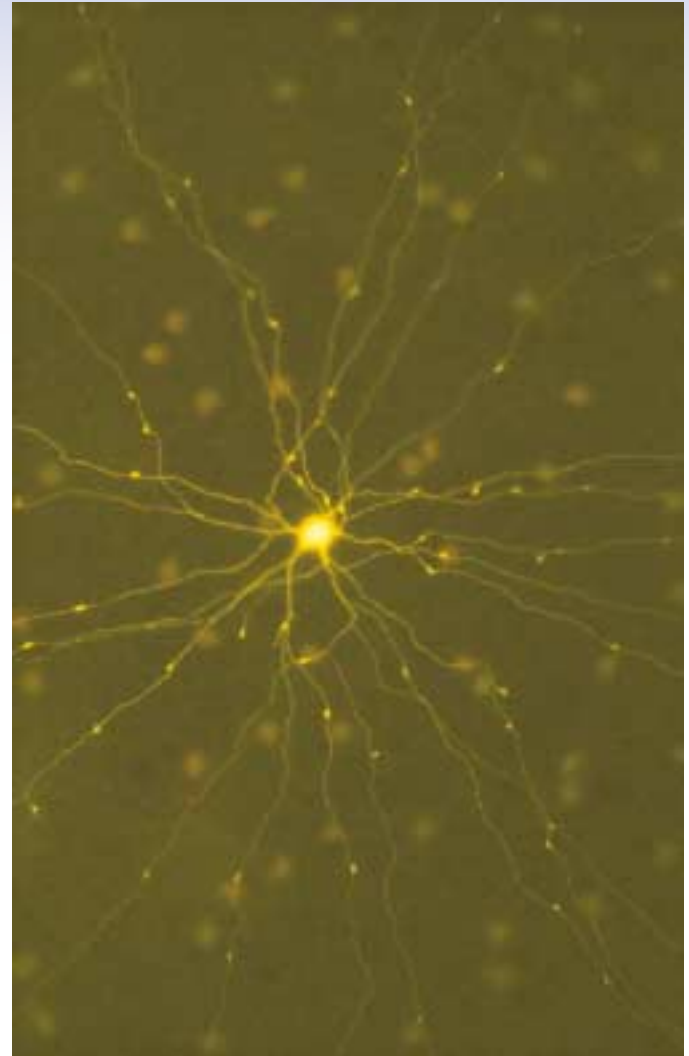


FIGURE 54.1

A neuron in the retina of the eye (500×). This neuron has been injected with a fluorescent dye, making its cell body and long dendrites readily apparent.

All animals except sponges use a network of nerve cells to gather information about the body's condition and the external environment, to process and integrate that information, and to issue commands to the body's muscles and glands. Just as telephone cables run from every compartment of a submarine to the conning tower, where the captain controls the ship, so bundles of nerve cells called neurons connect every part of an animal's body to its command and control center, the brain and spinal cord (figure 54.1). The animal body is run just like a submarine, with status information about what is happening in organs and outside the body flowing into the command center, which analyzes the data and issues commands to glands and muscles.

54.1 The nervous system consists of neurons and supporting cells.

Neuron Organization

An animal must be able to respond to environmental stimuli. A fly escapes a swat; the antennae of a crayfish detect food and the crayfish moves toward it. To do this, it must have sensory receptors that can detect the stimulus and motor *effectors* that can respond to it. In most invertebrate phyla and in all vertebrate classes, sensory receptors and motor effectors are linked by way of the nervous system. As described in chapter 49, the nervous system consists of neurons and supporting cells. **Sensory** (or **afferent**) **neurons** carry impulses from sensory receptors to the central nervous system (CNS); **motor** (or **efferent**) **neurons** carry impulses from the CNS to effectors—muscles and glands (figure 54.2).

In addition to sensory and motor neurons, a third type of neuron is present in the nervous systems of most invertebrates and all vertebrates: **association neurons** (or **interneurons**). These neurons are located in the brain and spinal cord of vertebrates, together called the **central nervous system (CNS)**, where they help provide more complex reflexes and higher associative functions, including learning and memory. Sensory neurons carry impulses into the CNS, and motor neurons carry impulses away from the CNS. Together, sensory and motor neurons constitute the **peripheral nervous system (PNS)** of vertebrates. Motor neurons that stimulate skeletal muscles to contract are **somatic motor neurons**, and those that regulate the activity of the smooth muscles, cardiac muscle, and glands are **autonomic motor neurons**. The autonomic motor neurons are further subdivided into the **sympathetic** and **parasympa-**

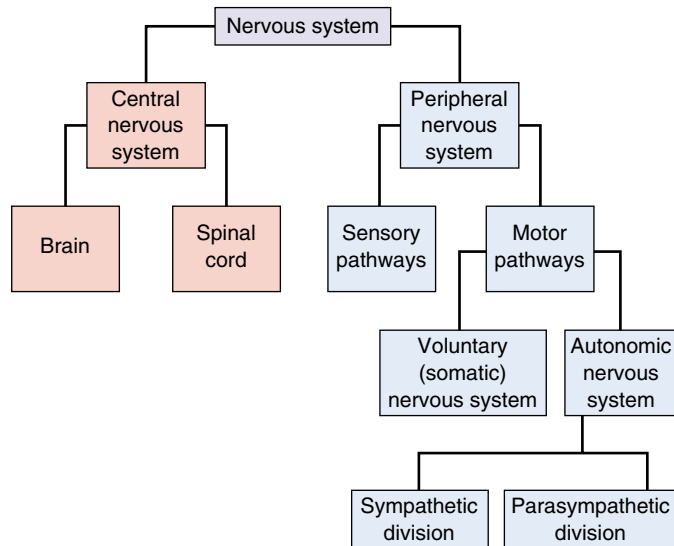


FIGURE 54.3

The divisions of the vertebrate nervous system. The major divisions are the central and peripheral nervous systems.

thetic systems, which act to counterbalance each other (figure 54.3).

Despite their varied appearances, most neurons have the same functional architecture (figure 54.4). The cell body is an enlarged region containing the nucleus. Extending from the cell body are one or more cytoplasmic extensions called **dendrites**. Motor and association neurons possess a profusion of highly branched dendrites, enabling those cells to

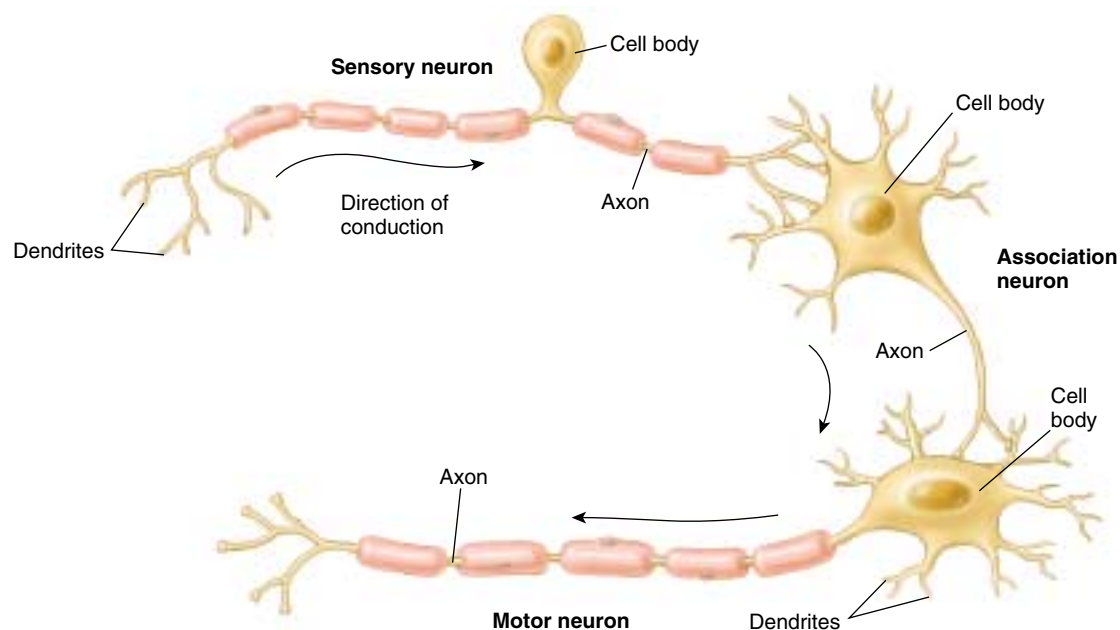


FIGURE 54.2

Three types of neurons

Sensory neurons carry information about the environment to the brain and spinal cord. **Association neurons** are found in the brain and spinal cord and often provide links between sensory and motor neurons. **Motor neurons** carry impulses or “commands” to muscles and glands (effectors).

receive information from many different sources simultaneously. Some neurons have extensions from the dendrites called *dendritic spines* that increase the surface area available to receive stimuli. The surface of the cell body integrates the information arriving at its dendrites. If the resulting membrane excitation is sufficient, it triggers impulses that are conducted away from the cell body along an **axon**. Each neuron has a single axon leaving its cell body, although an axon may produce small terminal branches to stimulate a number of cells. An axon can be quite long: the axons controlling the muscles in your feet are more than a meter long, and the axons that extend from the skull to the pelvis in a giraffe are about three meters long!

Neurons are supported both structurally and functionally by **supporting cells**, which are called **neuroglia**. These cells are ten times more numerous than neurons and serve a variety of functions, including supplying the neurons with nutrients, removing wastes from neurons, guiding axon migration, and providing immune functions. Two of the most important kinds of neuroglia in vertebrates are **Schwann cells** and **oligodendrocytes**, which produce **myelin sheaths** that surround the axons of many neurons. Schwann cells produce myelin in the PNS, while oligodendrocytes produce myelin in the CNS. During development, these cells wrap themselves around each axon several times to form the myelin sheath, an insulating covering consisting of multiple layers of membrane (figure 54.5). Axons that have myelin sheaths are said to be myelinated, and those that don't are unmyelinated. In the CNS, myelinated axons form the **white matter**, and the unmyelinated dendrites and cell bodies form the **gray matter**. In the PNS, both myelinated and unmyelinated axons are bundled together, much like wires in a cable, to form **nerves**.

The myelin sheath is interrupted at intervals of 1 to 2 mm by small gaps known as **nodes of Ranvier** (see figure 54.4). The role of the myelin sheath in impulse conduction will be discussed later in this chapter.

Neurons and neuroglia make up the central and peripheral nervous systems in vertebrates. Sensory, motor, and association neurons play different roles in the nervous system, and the neuroglia aid their function, in part by producing myelin sheaths.

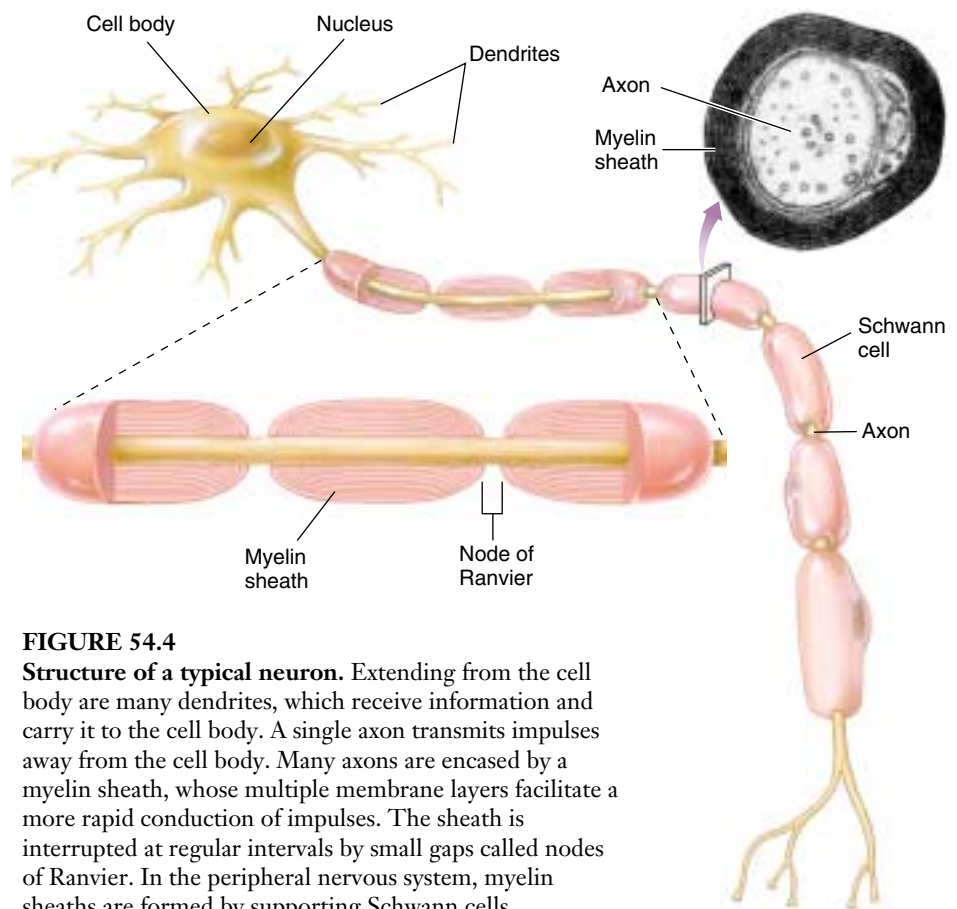


FIGURE 54.4

Structure of a typical neuron. Extending from the cell body are many dendrites, which receive information and carry it to the cell body. A single axon transmits impulses away from the cell body. Many axons are encased by a myelin sheath, whose multiple membrane layers facilitate a more rapid conduction of impulses. The sheath is interrupted at regular intervals by small gaps called nodes of Ranvier. In the peripheral nervous system, myelin sheaths are formed by supporting Schwann cells.

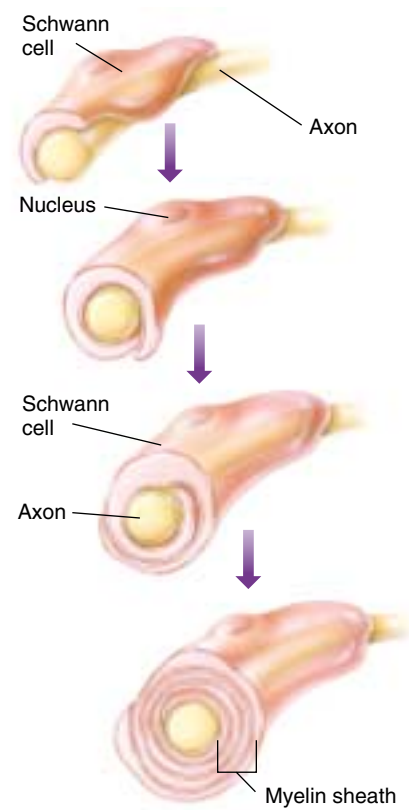


FIGURE 54.5

The formation of the myelin sheath around a peripheral axon. The myelin sheath is formed by successive wrappings of Schwann cell membranes, leaving most of the Schwann cell cytoplasm outside the myelin.

54.2 Nerve impulses are produced on the axon membrane.

The Resting Membrane Potential

Neurons communicate through changes in electrical properties of the plasma membrane that travel from one cell to another. The architecture of the neuron aids the spread of these electrical signals called nerve impulses. To understand how these signals are generated and transmitted within the nervous system, we must first examine some of the electrical properties of plasma membranes.

The battery in a car or a flashlight separates electrical charges between its two poles. There is said to be a *potential difference*, or voltage, between the poles, with one pole being positive and the other negative. Similarly, a potential difference exists across every cell's plasma membrane. The side of the membrane exposed to the cytoplasm is the negative pole, and the side exposed to the extracellular fluid is the positive pole. This potential difference is called the membrane potential.

The inside of the cell is more negatively charged in relation to the outside because of three factors: (1) Large molecules like proteins and nucleic acids that are negatively charged are more abundant inside the cell and cannot diffuse out. These molecules are called *fixed anions*. (2) The sodium-potassium pump brings only two potassium ions (K^+) into the cell for every three sodium ions (Na^+) it pumps out (figure 54.6). In addition to contributing to the electrical potential, this also establishes concentration gradients for Na^+ and K^+ . (3) Ion channels allow more K^+ to diffuse out of the cell than Na^+ to diffuse into the cell. Na^+ and K^+ channels in the plasma membrane have *gates*, portions of the channel protein that open or close the channel's pore. In the axons of neurons and in muscle fibers, the

gates are closed or open depending on the membrane potential. Such channels are therefore known as *voltage-gated ion channels* (figure 54.7).

In most cells, the permeability of ions through the membrane is constant, and the net negativity on the inside of the cell remains constant. The plasma membranes of muscle and neurons, however, are excitable because the permeability of their ion channels can be altered by various stimuli. When a neuron is not being stimulated, it maintains a *resting membrane potential*. A cell is very small, and so its membrane potential is very small. The electrical potential of a car battery is typically 12 volts, but a cell's resting membrane potential is about -70 millivolts (-70 mV or 0.07 volts). The negative sign indicates that the inside of the cell is negative with respect to the outside.

We know that the resting membrane potential is -70 mV because of an unequal distribution of electrical charges across the membrane. But why -70 mV rather than -50 mV or -10 mV? To understand this, we need to remember that there are two forces acting on the ions involved in establishing the resting membrane potential: (1) ions are attracted to ions or molecules of opposite charge; and (2) ions respond to concentration gradients by moving from an area of high concentration to an area of lower concentration.

The positively charged ions, called *cations*, outside the cell are attracted to the negatively charged fixed anions inside the cell. However, the resting plasma membrane is more permeable to K^+ than to other cations, so K^+ enters the cell. Other cations enter the cell, but the leakage of K^+ into the cell has the dominant effect on the resting membrane potential. In addition to the electrical gradient dri-

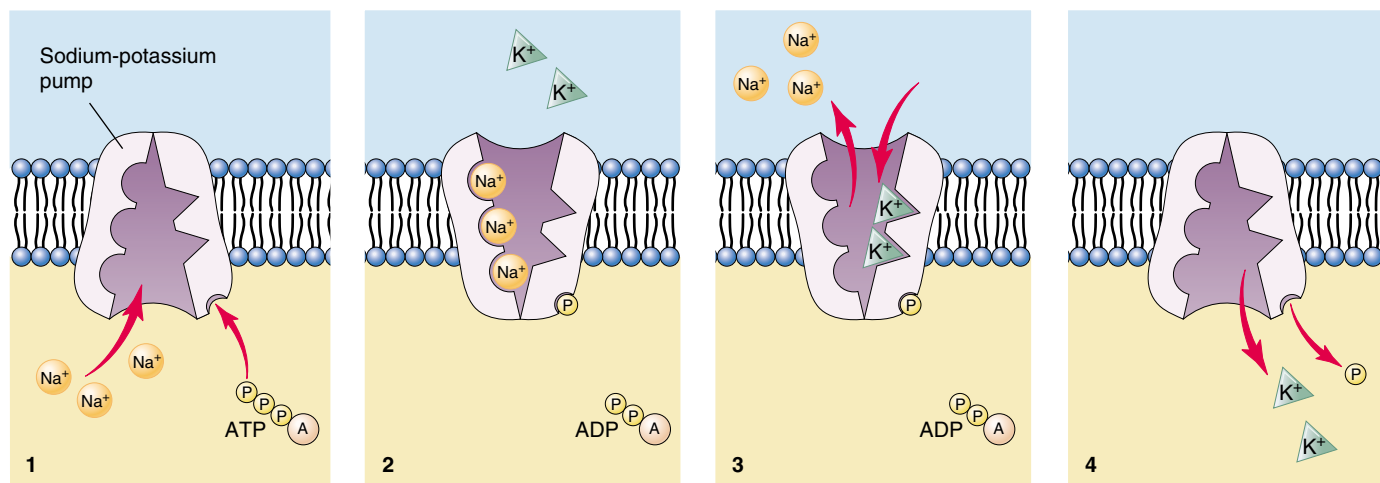


FIGURE 54.6

The sodium-potassium pump. This pump transports three Na^+ to the outside of the cell and simultaneously transports two K^+ to the inside of the cell. This is an active transport carrier requiring the breakdown of ATP.

Table 54.1 The Ionic Composition of Cytoplasm and Extracellular Fluid (ECF)

Ion	Concentration in Cytoplasm (mM)	Concentration in ECF (mM)	Ratio	Equilibrium Potential (mV)
Na ⁺	15	150	10:1	+60
K ⁺	150	5	1:30	−90
Cl [−]	7	110	15:1	−70

ving K⁺ into the cell, there is also a concentration gradient established by the sodium-potassium pump that is driving K⁺ out of the cell. At a point, these two forces balance each other, and the voltage at which the influx of K⁺ equals the efflux of K⁺ is called the **equilibrium potential** (table 54.1). For potassium, the equilibrium potential is −90 mV. At −80 mV, K⁺ will diffuse out of the cell and at −100 mV K⁺ will diffuse into the cell.

If K⁺ were the only cation involved, the resting membrane potential of the cell would be −90 mV. However, the membrane is also slightly permeable to Na⁺, and its equilibrium potential is +60 mV. The effects of Na⁺ leaking into the cell make the resting membrane potential less negative. With a membrane potential less negative than −90 mV, K⁺ diffuses into the cell, and the combined effect bring the equilibrium potential for the resting cell to −70 mV. The resting membrane potential of a neuron can be seen using a voltmeter and a pair of electrodes, one outside and one inside the cell (figure 54.8)

When a nerve or muscle cell is stimulated, sodium channels become more permeable, and Na⁺ rushes into the cell, down its concentration gradient. This sudden influx of positive charges reduces the negativity on the inside of the cell and causes the cell to *depolarize* (move toward a polarity above that of the resting potential). After a slight delay, potassium channels also become more permeable, and K⁺ flows out of the cell down its concentration gradient. Similarly, the membrane becomes more permeable to Cl[−], and Cl[−] flows into the cell. But the effects of Cl[−] on the membrane potential are far less than those of K⁺. The inside of the cell again becomes more negative and causes the cell to *hyperpolarize* (move its polarity below that of the resting potential).

The resting plasma membrane maintains a potential difference as a result of the uneven distribution of charges, where the inside of the membrane is negatively charged in comparison with the outside (−70 mV). The magnitude, measured in millivolts, of this potential difference primarily reflects the difference in K⁺ concentration.

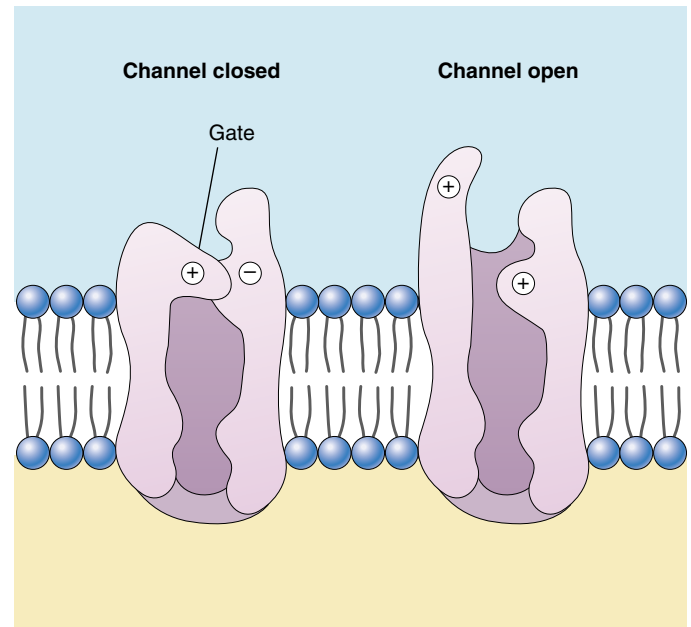


FIGURE 54.7

Voltage-gated ion channels. In neurons and muscle cells, the channels for Na⁺ and K⁺ have gates that are closed at the resting membrane potential but open when a threshold level of depolarization is attained.

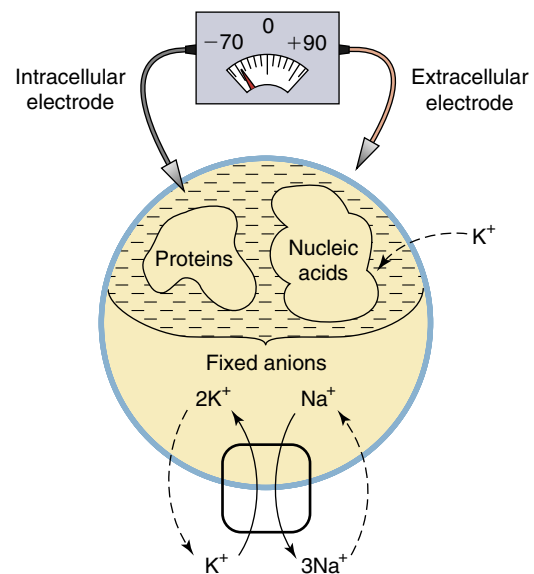


FIGURE 54.8

The establishment of the resting membrane potential. The fixed anions (primarily proteins and nucleic acids) attract cations from the extracellular fluid. If the membrane were only permeable to K⁺, an equilibrium would be established and the membrane potential would be −90 mV. A true resting membrane potential is about −70 mV, because the membrane does allow a low rate of Na⁺ diffusion into the cell. This is not quite sufficiently negative to prevent the outward diffusion of K⁺, so the cell is not at equilibrium and the action of the sodium-potassium pumps is required to maintain stability.

Action Potentials

Generation of Action Potentials

If the plasma membrane is depolarized slightly, an oscilloscope will show a small upward deflection of the line that soon decays back to the resting membrane potential. These small changes in membrane potential are called *graded potentials* because their amplitudes depend on the strength of the stimulus. Graded potentials can be either depolarizing or hyperpolarizing and can add together to amplify or reduce their effects, just as two waves add to make one bigger one when they meet in synchrony or cancel each other out when a trough meets with a crest. The ability of graded potentials to combine is called summation (figure 54.9). Once a particular level of depolarization is reached (about -55 mV in mammalian axons), however, a nerve impulse, or **action potential**, is produced. The level of depolarization needed to produce an action potential is called the *threshold*.

A depolarization that reaches or exceeds the threshold opens both the Na^+ and the K^+ channels, but the Na^+ channels open first. The rapid diffusion of Na^+ into the cell shifts the membrane potential toward the equilibrium potential for Na^+ ($+60$ mV—recall that the positive sign indicates that the membrane reverses polarity as Na^+ rushes in).

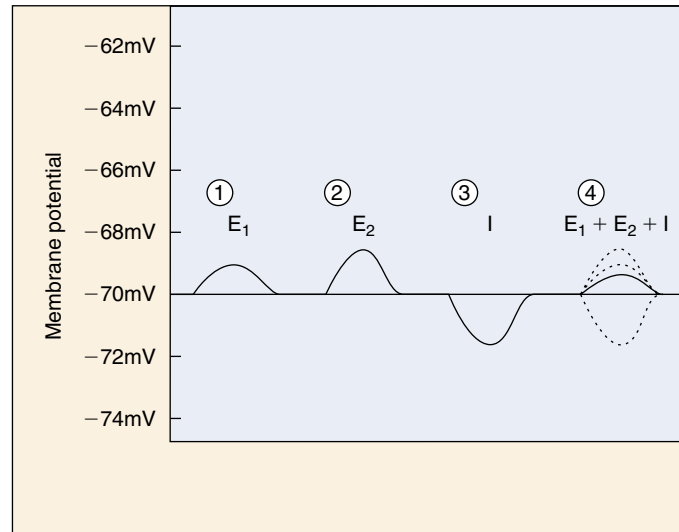


FIGURE 54.9

Graded potentials. (1) A weak excitatory stimulus, E_1 , elicits a smaller depolarization than (2) a stronger stimulus, E_2 . (3) An inhibitory stimulus, I , produces a hyperpolarization. (4) Because graded potentials can summate, if all three stimuli occur very close together, the resulting polarity change will be the algebraic sum of the three changes individually.

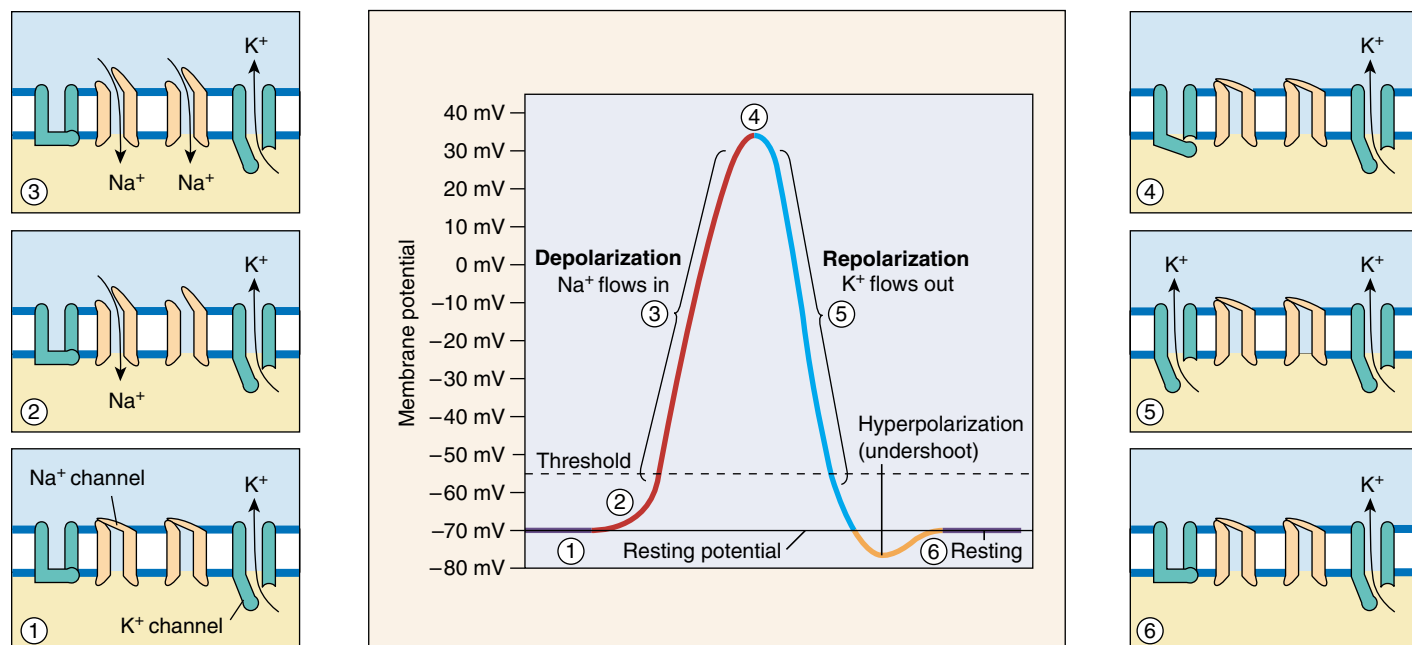


FIGURE 54.10

The action potential. (1) At resting membrane potential, some K^+ channels are open. (2) In response to a stimulus, the cell begins to depolarize, and once the threshold level is reached, an action potential is produced. (3) Rapid depolarization occurs (the rising portion of the spike) because sodium channels open, allowing Na^+ to diffuse into the axon. (4) At the top of the spike, Na^+ channels close, and K^+ channels that were previously closed begin to open. (5) With the K^+ channels open, repolarization occurs because of the diffusion of K^+ out of the axon. (6) An undershoot occurs before the membrane returns to its original resting potential.

When the action potential is recorded on an oscilloscope, this part of the action potential appears as the *rising phase* of a spike (figure 54.10). The membrane potential never quite reaches +60 mV because the Na^+ channels close and, at about the same time, the K^+ channels that were previously closed begin to open. The action potential thus peaks at about +30 mV. Opening the K^+ channels allows K^+ to diffuse out of the cell, repolarizing the plasma membrane. On an oscilloscope, this repolarization of the membrane appears as the *falling phase* of the action potential. In many cases, the repolarization carries the membrane potential to a value slightly more negative than the resting potential for a brief period because K^+ channels remain open, resulting in an *undershoot*. The entire sequence of events in an action potential is over in a few milliseconds.

Action potentials have two distinguishing characteristics. First, they follow an all-or-none law: each depolarization produces either a full action potential, because the voltage-gated Na^+ channels open completely at threshold, or none at all. Secondly, action potentials are always separate events; they cannot add together or interfere with one another as graded potentials can because the membrane enters a brief refractory period after it generates an action potential during which time voltage-gated Na^+ channels cannot reopen.

The production of an action potential results entirely from the passive diffusion of ions. However, at the end of each action potential, the cytoplasm has a little more Na^+ and a little less K^+ than it did at rest. The constant activity of the sodium-potassium pumps compensates for these changes. Thus, although active transport is not required to produce action potentials, it is needed to maintain the ion gradients.

Propagation of Action Potentials

Although we often speak of axons as conducting action potentials (impulses), action potentials do not really travel along an axon—they are events that are reproduced at different points along the axon membrane. This can occur for two reasons: action potentials are stimulated by depolarization, and an action potential can serve as a depolarization stimulus for the next region of the axon. Each action potential, during its rising phase, reflects a reversal in membrane polarity (from -70 mV to +30 mV) as Na^+ diffuses rapidly into the axon. The positive charges can depolarize the next region of membrane to threshold, so that the next region produces its own action potential (figure 54.11). Meanwhile, the previous region of membrane repolarizes back to the resting membrane potential. This is analogous to people in a stadium performing the “wave”: individuals stay in place as they stand up (depolarize), raise their hands (peak of the action potential), and sit down again (repolarize).

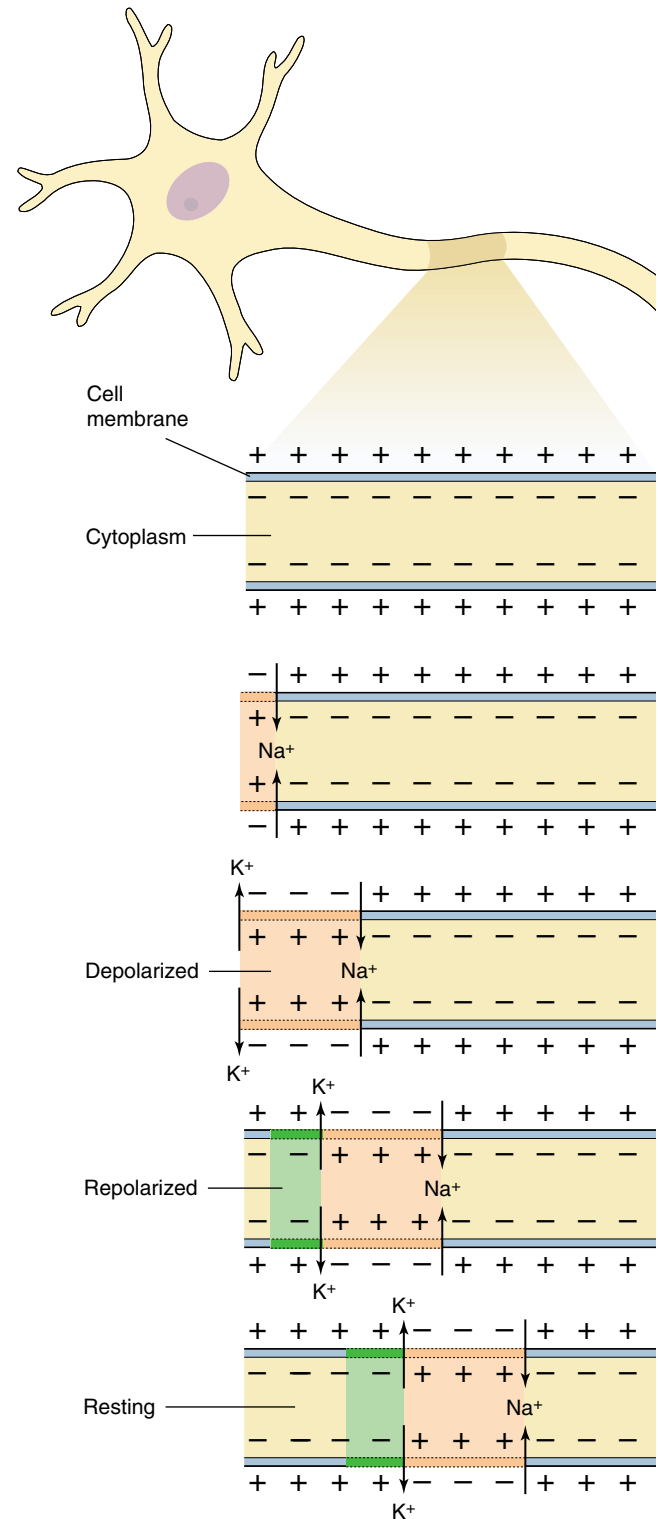


FIGURE 54.11

Propagation of an action potential in an unmyelinated axon. When one region produces an action potential and undergoes a reversal of polarity, it serves as a depolarization stimulus for the next region of the axon. In this way, action potentials are regenerated along each small region of the unmyelinated axon membrane.

Saltatory Conduction

Action potentials are conducted without decrement (without decreasing in amplitude); thus, the last action potential at the end of the axon is just as large as the first action potential. The velocity of conduction is greater if the diameter of the axon is large or if the axon is myelinated (table 54.2). Myelinated axons conduct impulses more rapidly than nonmyelinated axons because the action potentials in myelinated axons are only produced at the nodes of Ranvier. One action potential still serves as the depolarization stimulus for the next, but the depolarization at one node must spread to the next before the voltage-gated channels can be opened. The impulses therefore seem to jump from node to node (figure 54.12) in a process called **saltatory conduction** (Latin *saltare*, “to jump”).

To see how saltatory conduction speeds nervous transmission, return for a moment to the “wave” analogy used on the previous page to describe propagation of an action potential. The “wave” moves across the seats of a crowded stadium as fans standing up in one section trigger the next section to stand up in turn. Because the “wave” will skip sections of empty bleachers, it actually progresses around the stadium even faster with more empty sections. The wave doesn’t have to wait for the missing people to stand, simply “jumping” the gaps—just as saltatory conduction jumps the nonconduction “gaps” of myelin between exposed nodes.

The rapid inward diffusion of Na⁺ followed by the outward diffusion of K⁺ produces a rapid change in the membrane potential called an action potential. Action potentials are all-or-none events and cannot summate. Action potentials are regenerated along an axon as one action potential serves as the depolarization stimulus for the next action potential.

Table 54.2 Conduction Velocities of Some Axons			
	Axon Diameter (mm)	Myelin	Conduction Velocity (m/s)
Squid giant axon	500	No	25
Large motor axon to human leg muscle	20	Yes	120
Axon from human skin pressure receptor	10	Yes	50
Axon from human skin temperature receptor	5	Yes	20
Motor axon to human internal organ	1	No	2

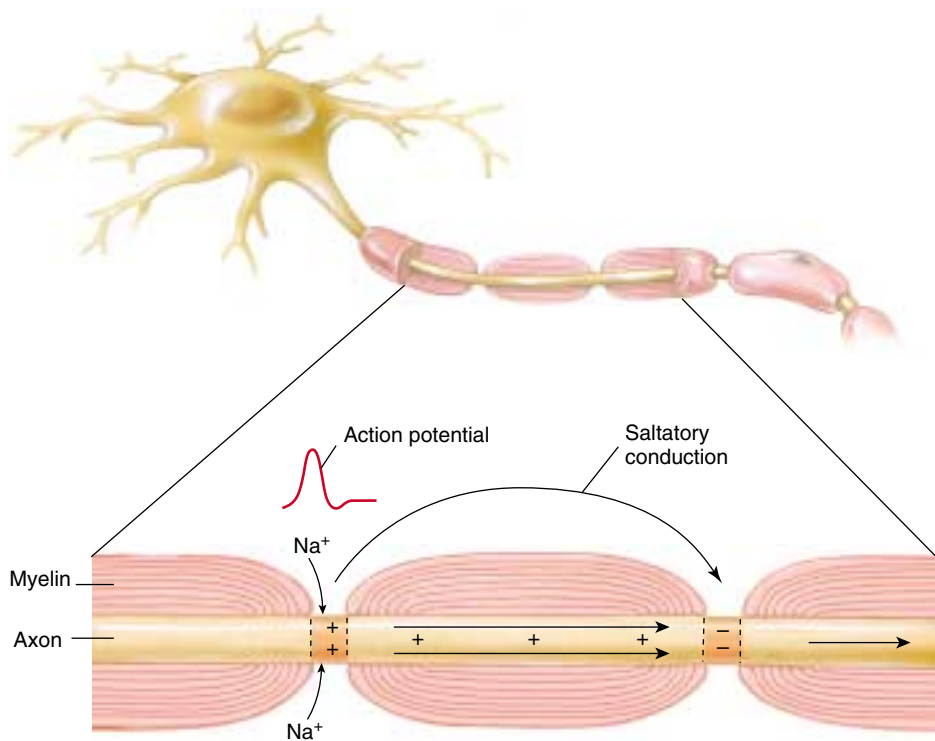


FIGURE 54.12
Saltatory conduction in a myelinated axon. Action potentials are only produced at the nodes of Ranvier in a myelinated axon. One node depolarizes the next node so that the action potentials can skip between nodes. As a result, saltatory (“leaping”) conduction in a myelinated axon is more rapid than conduction in an unmyelinated axon.

54.3 Neurons form junctions called synapses with other cells.

Structure of Synapses

An action potential passing down an axon eventually reaches the end of the axon and all of its branches. These branches may form junctions with the dendrites of other neurons, with muscle cells, or with gland cells. Such inter-cellular junctions are called **synapses**. The neuron whose axon transmits action potentials to the synapse is the *presynaptic cell*, while the cell on the other side of the synapse is the *postsynaptic cell*. Although the presynaptic and postsynaptic cells may appear to touch when the synapse is seen under a light microscope, examination with an electron microscope reveals that most synapses have a **synaptic cleft**, a narrow space that separates these two cells (figure 54.13).

The end of the presynaptic axon is swollen and contains numerous **synaptic vesicles**, which are each packed with chemicals called **neurotransmitters**. When action potentials arrive at the end of the axon, they stimulate the opening of voltage-gated Ca^{++} channels, causing a rapid inward diffusion of Ca^{++} . This serves as the stimulus for the fusion of the synaptic vesicles membrane with the plasma membrane of the axon, so that the contents of the vesicles can be released by exocytosis (figure 54.14). The higher the frequency of action potentials in the presynaptic axon, the more vesicles will release their contents of neurotransmitters. The neurotransmitters diffuse rapidly to the other side of the cleft and bind to **receptor proteins** in the membrane of the postsynaptic cell. There are different types of neurotransmitters, and different ones act in different ways. We will next consider the action of a few of the important neurotransmitter chemicals.

The presynaptic axon is separated from the postsynaptic cell by a narrow synaptic cleft. Neurotransmitters diffuse across it to transmit a nerve impulse.

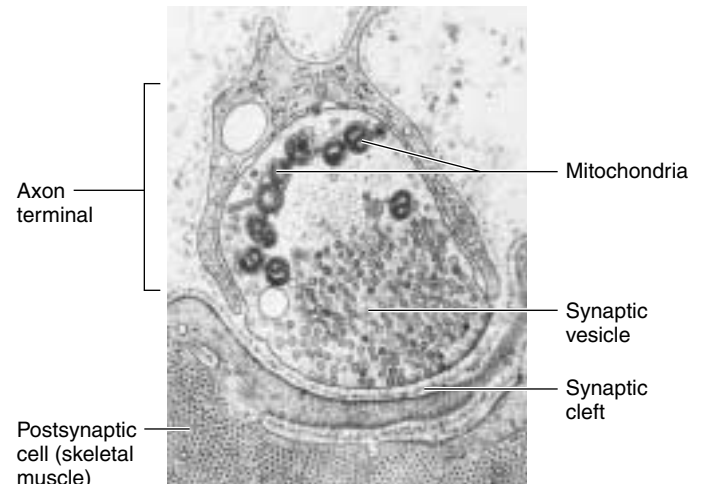


FIGURE 54.13
A synaptic cleft. An electron micrograph showing a neuromuscular synapse.

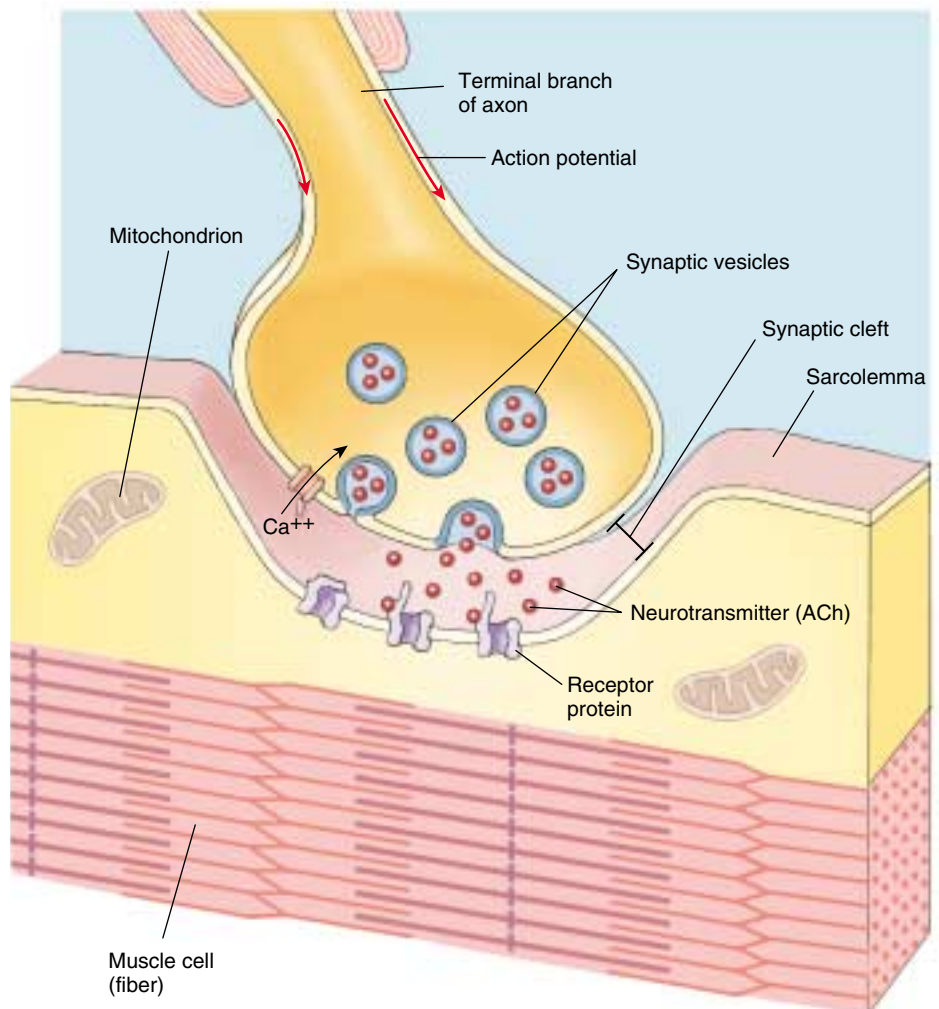


FIGURE 54.14
The release of neurotransmitter. Action potentials arriving at the end of an axon trigger the uptake of Ca^{++} , which causes synaptic vesicles to fuse with the plasma membrane and release their neurotransmitters (acetylcholine [ACh] in this case), which diffuse across the synaptic gap and bind to receptors in the postsynaptic membrane.

Neurotransmitters and Their Functions

Acetylcholine was the first neurotransmitter chemical to be discovered and is widely used in the nervous system. Many other neurotransmitter chemicals have been shown to play important roles, however, and ongoing research continues to produce new information about neurotransmitter function.

Acetylcholine

Acetylcholine (ACh) is the neurotransmitter that crosses the synapse between a motor neuron and a muscle fiber. This synapse is called a *neuromuscular junction* (figure 54.15). Acetylcholine binds to its receptor proteins in the postsynaptic membrane and thereby causes ion channels within these proteins to open (figure 54.16). The gates to these ion channels are said to be *chemically gated* because they open in response to ACh, rather than in response to depolarization. The opening of the chemically regulated channels permits Na^+ to diffuse into the postsynaptic cell and K^+ to diffuse out. Although both ions move at the same time, the inward diffusion of Na^+ occurs at a faster rate and has the predominant effect. As a result, that site on the

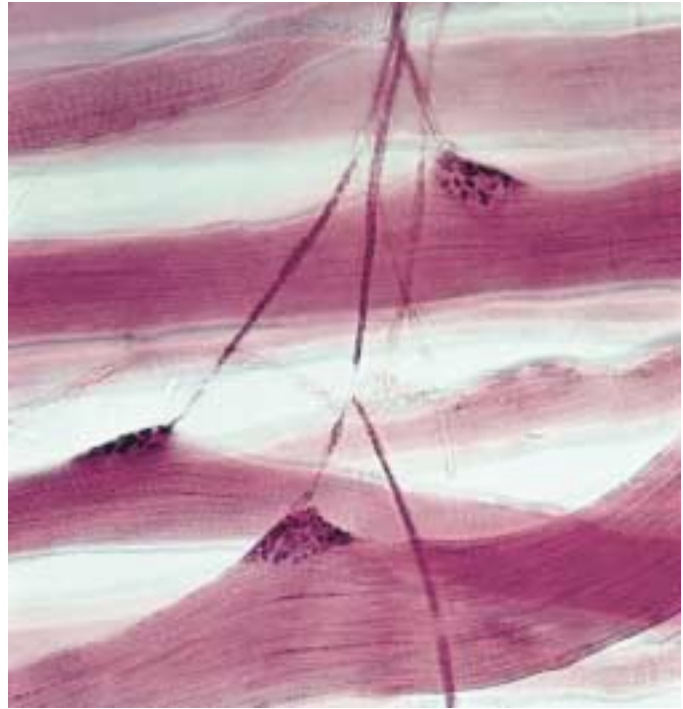


FIGURE 54.15
Neuromuscular junctions. A light micrograph shows axons branching to make contact with several individual muscle fibers.

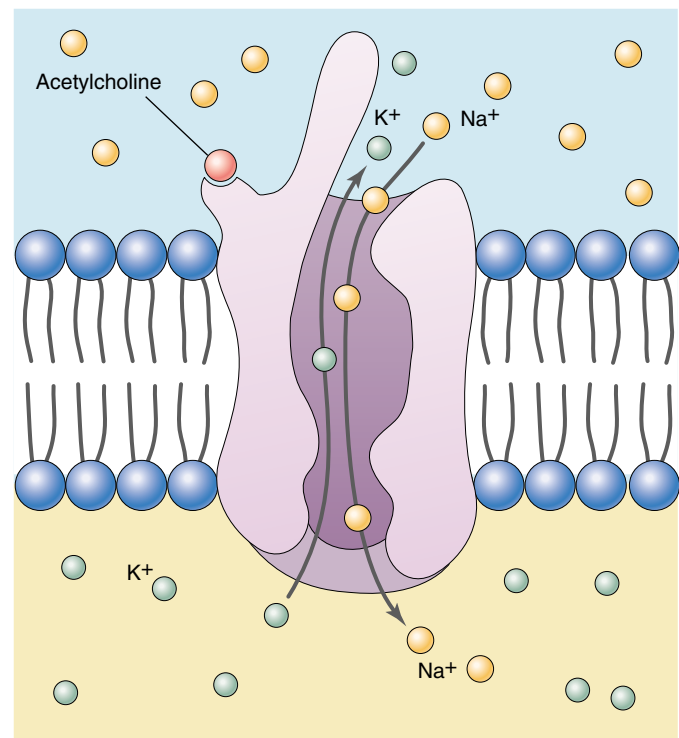
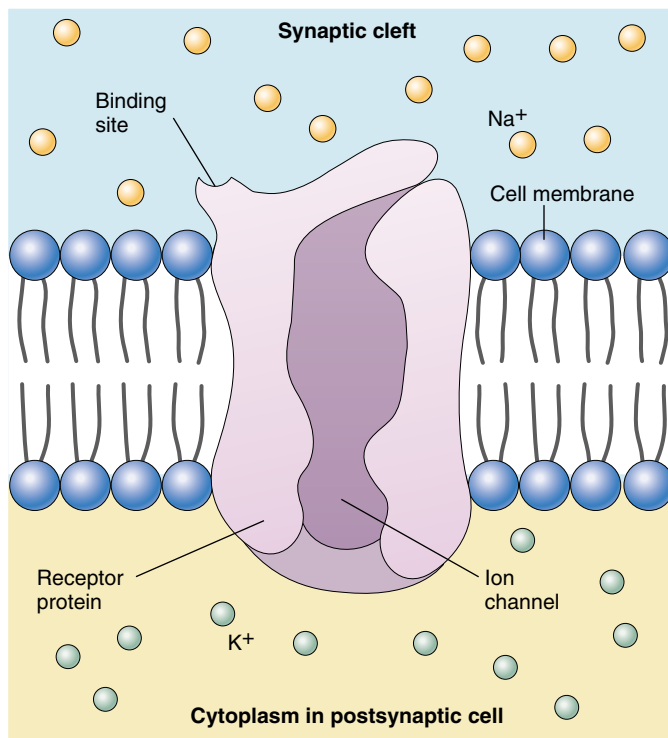


FIGURE 54.16
The binding of ACh to its receptor opens ion channels. The chemically regulated gates to these channels open when the neurotransmitter ACh binds to the receptor.

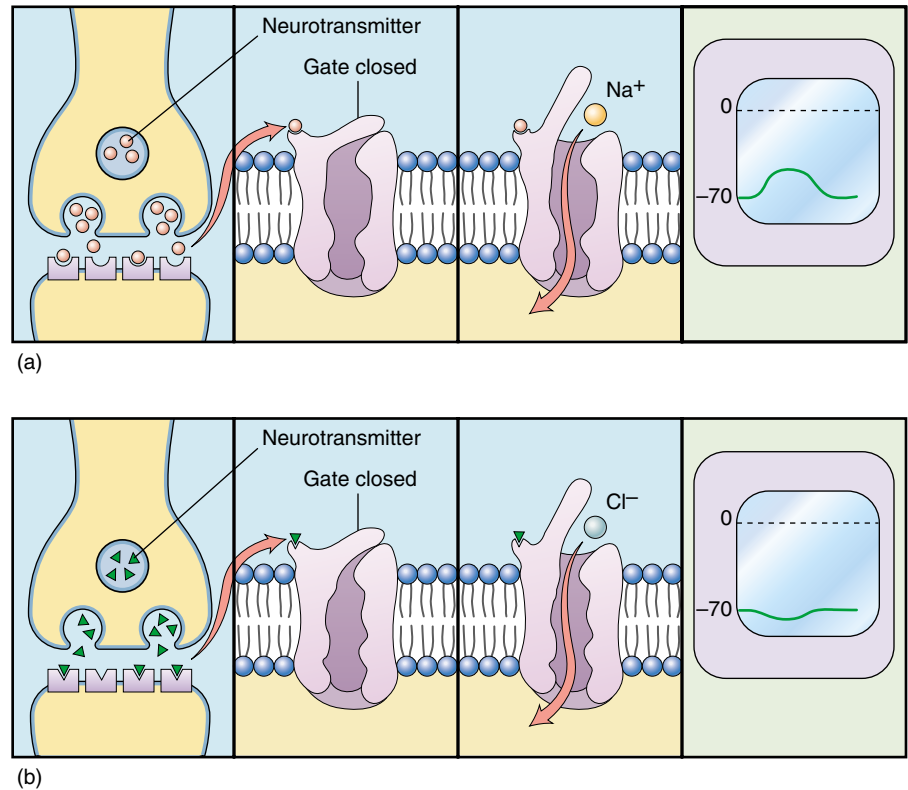


FIGURE 54.17
Different neurotransmitters can have different effects. (a) An excitatory neurotransmitter promotes a depolarization, or excitatory postsynaptic potential (EPSP). (b) An inhibitory neurotransmitter promotes a hyperpolarization, or inhibitory postsynaptic potential (IPSP).

postsynaptic membrane produces a depolarization (figure 54.17a) called an **excitatory postsynaptic potential (EPSP)**. The EPSP can now open the voltage-gated channels for Na^+ and K^+ that are responsible for action potentials. Because the postsynaptic cell we are discussing is a skeletal muscle cell, the action potentials it produces stimulate muscle contraction through the mechanisms discussed in chapter 50.

If ACh stimulates muscle contraction, we must be able to eliminate ACh from the synaptic cleft in order to relax our muscles. This illustrates a general principle: molecules such as neurotransmitters and certain hormones must be quickly eliminated after secretion if they are to be effective regulators. In the case of ACh, the elimination is achieved by an enzyme in the postsynaptic membrane called **acetylcholinesterase (AChE)**. This enzyme is one of the fastest known, cleaving ACh into inactive fragments. Nerve gas and the agricultural insecticide parathion are potent inhibitors of AChE and in humans can produce severe spastic paralysis and even death if the respiratory muscles become paralyzed.

Although ACh acts as a neurotransmitter between motor neurons and skeletal muscle cells, many neurons also use ACh as a neurotransmitter at their synapses with other neurons; in these cases, the postsynaptic membrane is generally on the dendrites or cell body of the second neuron. The EPSPs produced must then travel through the dendrites and cell body to the initial segment of the axon, where the first voltage-regulated channels needed for action potentials are located. This is where the first action

potentials will be produced, providing that the EPSP depolarization is above the threshold needed to trigger action potentials.

Glutamate, Glycine, and GABA

Glutamate is the major excitatory neurotransmitter in the vertebrate CNS, producing EPSPs and stimulating action potentials in the postsynaptic neurons. Although normal amounts produce physiological stimulation, excessive stimulation by glutamate has been shown to cause neurodegeneration, as in Huntington's chorea.

Glycine and **GABA** (an acronym for gamma-aminobutyric acid) are inhibitory neurotransmitters. If you remember that action potentials are triggered by a threshold level of depolarization, you will understand why hyperpolarization of the membrane would cause inhibition. These neurotransmitters cause the opening of chemically regulated gated channels for Cl^- , which has a concentration gradient favoring its diffusion into the neuron. Because Cl^- is negatively charged, it makes the inside of the membrane even more negative than it is at rest—from -70 mV to -85 mV, for example (figure 54.17b). This hyperpolarization is called an **inhibitory postsynaptic potential (IPSP)**, and is very important for neural control of body movements and other brain functions. Interestingly, the drug diazepam (Valium) causes its sedative and other effects by enhancing the binding of GABA to its receptors and thereby increasing the effectiveness of GABA at the synapse.

Biogenic Amines

The **biogenic amines** include the hormone epinephrine (adrenaline), together with the neurotransmitters dopamine, norepinephrine, and serotonin. Epinephrine, norepinephrine, and dopamine are derived from the amino acid tyrosine and are included in the subcategory of *catecholamines*. Serotonin is a biogenic amine derived from a different amino acid, tryptophan.

Dopamine is a very important neurotransmitter used in the brain to control body movements and other functions. Degeneration of particular dopamine-releasing neurons produces the resting muscle tremors of Parkinson's disease, and people with this condition are treated with L-dopa (an acronym for dihydroxyphenylalanine), a precursor of dopamine. Additionally, studies suggest that excessive activity of dopamine-releasing neurons in other areas of the brain is associated with schizophrenia. As a result, patients with schizophrenia are sometimes helped by drugs that block the production of dopamine.

Norepinephrine is used by neurons in the brain and also by particular autonomic neurons, where its action as a neurotransmitter complements the action of the hormone epinephrine, secreted by the adrenal gland. The autonomic nervous system will be discussed in a later section of this chapter.

Serotonin is a neurotransmitter involved in the regulation of sleep and is also implicated in various emotional states. Insufficient activity of neurons that release serotonin may be one cause of clinical depression; this is suggested by the fact that antidepressant drugs, particularly fluoxetine (Prozac) and related compounds, specifically block the elimination of serotonin from the synaptic cleft (figure 54.18). The drug lysergic acid diethylamide (LSD) specifically blocks serotonin receptors in a region of the brain stem known as the raphe nuclei.

Other Neurotransmitters

Axons also release various polypeptides, called **neuropeptides**, at synapses. These neuropeptides may have a neurotransmitter function or they may have more subtle, long-term action on the postsynaptic neurons. In the latter case, they are often referred to as **neuromodulators**. A given axon generally releases only one kind of neurotransmitter, but many can release both a neurotransmitter and a neuromodulator.

One important neuropeptide is **substance P**, which is released at synapses in the CNS by sensory neurons activated by painful stimuli. The perception of pain, however, can vary depending on circumstances; an injured football

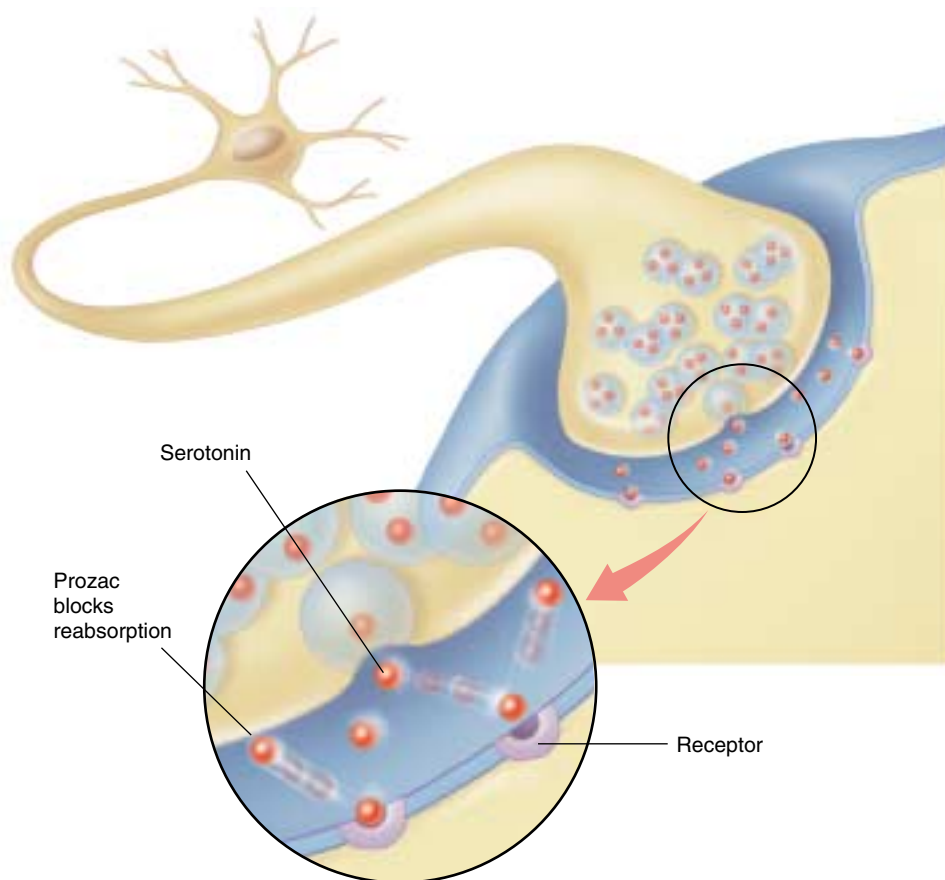
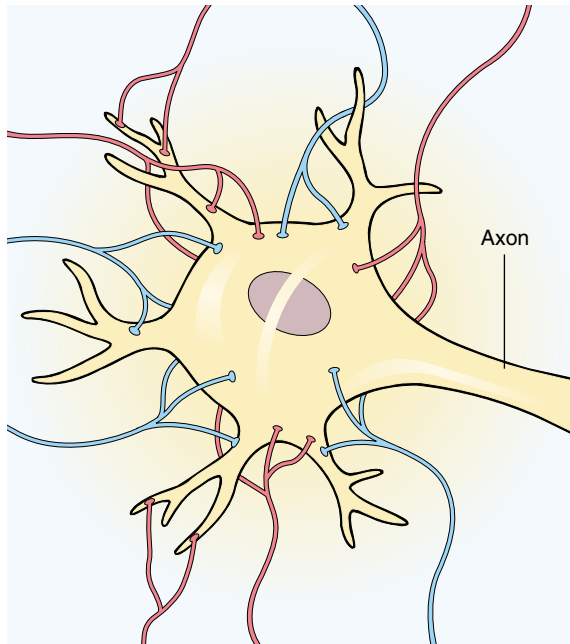
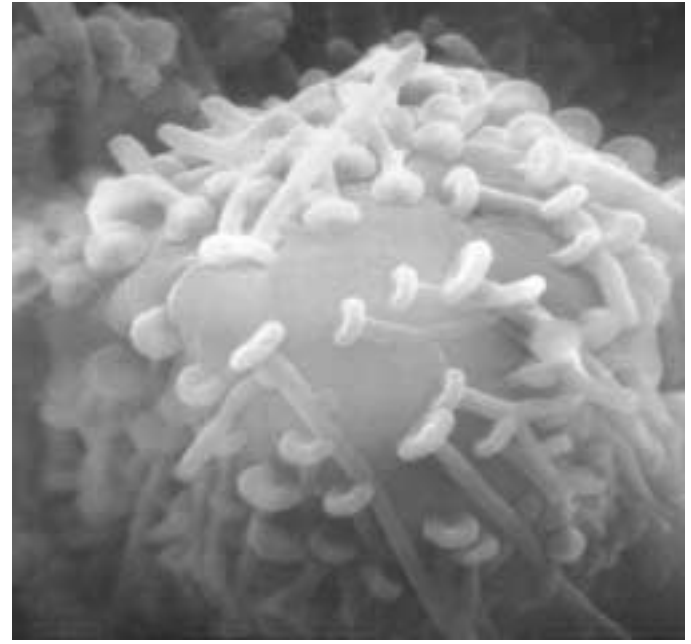


FIGURE 54.18

Serotonin and depression. Depression can result from a shortage of the neurotransmitter serotonin. The antidepressant drug Prozac works by blocking reabsorption of serotonin in the synapse, making up for the shortage.



(a)



(b)

FIGURE 54.19

Integration of EPSPs and IPSPs takes place on the neuronal cell body. (a) The synapses made by some axons are excitatory (blue); the synapses made by other axons are inhibitory (red). The summed influence of all of these inputs determines whether the axonal membrane of the postsynaptic cell will be sufficiently depolarized to produce an action potential. (b) Micrograph of a neuronal cell body with numerous synapses (15,000 $\times$).

player may not feel the full extent of his trauma, for example, until he's taken out of the game. The intensity with which pain is perceived partly depends on the effects of neuropeptides called **enkephalins** and **endorphins**. Enkephalins are released by axons descending from the brain and inhibit the passage of pain information to the brain. Endorphins are released by neurons in the brain stem and also block the perception of pain. Opium and its derivatives, morphine and heroin, have an analgesic (pain-reducing) effect because they are similar enough in chemical structure to bind to the receptors normally utilized by enkephalins and endorphins. For this reason, the enkephalins and the endorphins are referred to as *endogenous opiates*.

Nitric oxide (NO) is the first gas known to act as a regulatory molecule in the body. Because NO is a gas, it diffuses through membranes so it cannot be stored in vesicles. It is produced as needed from the amino acid arginine. Nitric oxide's actions are very different from those of the more familiar nitrous oxide (N_2O), or laughing gas, sometimes used by dentists. Nitric oxide diffuses out of the presynaptic axon and into neighboring cells by simply passing through the lipid portions of the cell membranes. In the PNS, nitric oxide is released by some neurons that innervate the gastrointestinal tract, penis, respiratory passages, and cerebral blood vessels. These are

autonomic neurons that cause smooth muscle relaxation in their target organs. This can produce, for example, the engorgement of the spongy tissue of the penis with blood, causing erection. The drug sildenafil (Viagra) increases the release of nitric oxide in the penis, prolonging erection. Nitric oxide is also released as a neurotransmitter in the brain, and has been implicated in the processes of learning and memory.

Synaptic Integration

The activity of a postsynaptic neuron in the brain and spinal cord of vertebrates is influenced by different types of input from a number of presynaptic neurons. For example, a single motor neuron in the spinal cord can receive as many as 50,000 synapses from presynaptic axons! Each postsynaptic neuron may receive both excitatory and inhibitory synapses. The EPSPs (depolarizations) and IPSPs (hyperpolarizations) from these synapses interact with each other when they reach the cell body of the neuron. Small EPSPs add together to bring the membrane potential closer to the threshold, while IPSPs subtract from the depolarizing effect of the EPSPs, keeping the membrane potential below the threshold (figure 54.19). This process is called **synaptic integration**.

Neurotransmitters and Drug Addiction

When a cell of your body is exposed to a stimulus that produces a chemically mediated signal for a prolonged period, it tends to lose its ability to respond to the stimulus with its original intensity. (You are familiar with this loss of sensitivity—when you sit in a chair, how long are you aware of the chair?) Nerve cells are particularly prone to this loss of sensitivity. If receptor proteins within synapses are exposed to high levels of neurotransmitter molecules for prolonged periods, that nerve cell often responds by inserting fewer receptor proteins into the membrane. This feedback is a normal function in all neurons, one of several mechanisms that have evolved to make the cell more efficient, in this case, adjusting the number of “tools” (receptor proteins) in the membrane “workshop” to suit the workload.

Cocaine. The drug cocaine causes abnormally large amounts of neurotransmitter to remain in the synapses for long periods of time. Cocaine affects nerve cells in the brain’s pleasure pathways (the so-called limbic system). These cells transmit pleasure messages using the neurotransmitter dopamine. Using radioactively labeled cocaine molecules, investigators found that cocaine binds tightly to the transporter proteins in synaptic clefts. These proteins normally remove the neurotransmitter dopamine after it has acted. Like a game of musical chairs in which all the chairs become occupied, there are no unoccupied carrier proteins available to the dopamine molecules, so the dopamine stays in the cleft, firing the receptors again and again. As new signals arrive, more and more dopamine is added, firing the pleasure pathway more and more often (figure 54.20).

When receptor proteins on limbic system nerve cells are exposed to high levels of dopamine neurotransmitter molecules for prolonged periods of time, the nerve cells “turn down the volume” of the signal by lowering the number of receptor proteins on their surfaces. They respond to the greater number of neurotransmitter molecules by simply reducing the number of targets available for these molecules to hit. The cocaine user is now addicted (figure 54.21). With so few receptors, the user needs the drug to maintain even normal levels of limbic activity.

Is Nicotine an Addictive Drug? Investigators attempting to explore the habit-forming nature of nicotine used what had been learned about cocaine to carry out what seemed a reasonable experiment—they introduced radioactively labeled nicotine into the brain and looked to see what sort of carrier protein it attached itself to. To their great surprise, the nicotine ignored proteins in the synaptic clefts and instead bound directly to a specific receptor on the postsynaptic cell! This was totally unexpected, as nicotine does not normally occur in the brain—why should it have a receptor there?

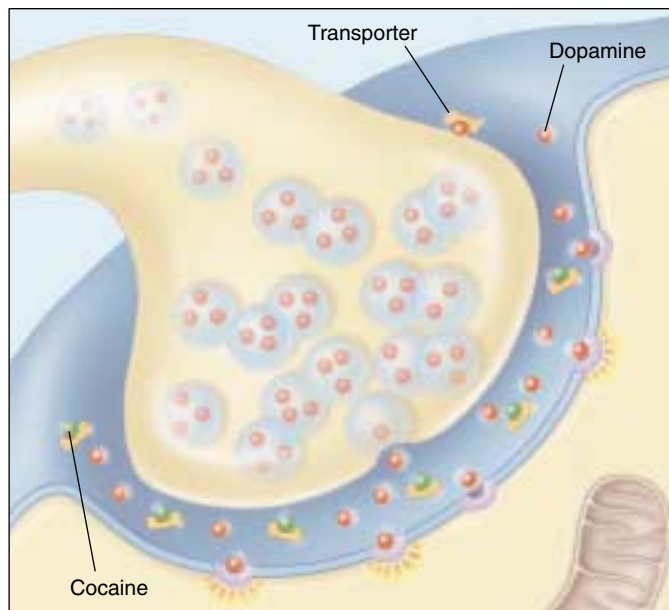


FIGURE 54.20

How cocaine alters events at the synapse. When cocaine binds to the dopamine transporters, the neurotransmitter survives longer in the synapse and continues to stimulate the postsynaptic cell. Cocaine thus acts to intensify pleasurable sensations.

Intensive research followed, and researchers soon learned that the “nicotine receptors” were a class of receptors that normally served to bind the neurotransmitter acetylcholine. There are other types of ACh receptors that don’t respond to nicotine. It was just an accident of nature that nicotine, an obscure chemical from a tobacco plant, was also able to bind to them. What, then, is the normal function of these receptors? The target of considerable research, these receptors turned out to be one of the brain’s most important tools. The brain uses them to coordinate the activities of many other kinds of receptors, acting to “fine tune” the sensitivity of a wide variety of behaviors.

When neurobiologists compare the nerve cells in the brains of smokers to those of nonsmokers, they find changes in both the number of nicotine receptors and in the levels of RNA used to make the receptors. They have found that the brain adjusts to prolonged exposure to nicotine by “turning down the volume” in two ways: (1) by making fewer receptor proteins to which nicotine can bind; and (2) by altering the pattern of *activation* of the nicotine receptors (that is, their sensitivity to neurotransmitter).

It is this second adjustment that is responsible for the profound effect smoking has on the brain’s activities. By overriding the normal system used by the brain to coordinate its many activities, nicotine alters the pattern of release into synaptic clefts of many neurotransmitters, including acetylcholine, dopamine, serotonin, and many others.

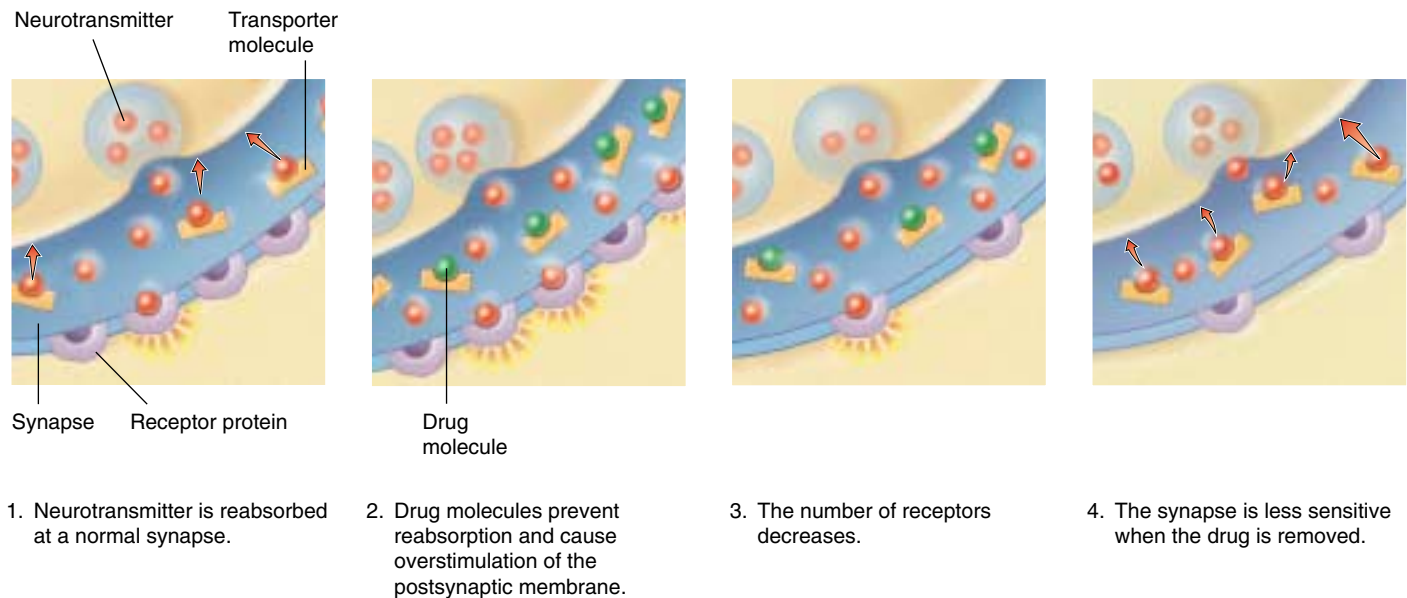


FIGURE 54.21

Drug addiction. (1) In a normal synapse, the neurotransmitter binds to a transporter molecule and is rapidly reabsorbed after it has acted. (2) When a drug molecule binds to the transporters, reabsorption of the neurotransmitter is blocked, and the postsynaptic cell is overstimulated by the increased amount of neurotransmitter left in the synapse. (3) The central nervous system adjusts to the increased firing by producing fewer receptors in the postsynaptic membrane. The result is addiction. (4) When the drug is removed, normal absorption of the neurotransmitter resumes, and the decreased number of receptors creates a less-sensitive nerve pathway. Physiologically, the only way a person can then maintain normal functioning is to continue to take the drug. Only if the drug is removed permanently will the nervous system eventually adjust again and restore the original amount of receptors.

As a result, changes in level of activity occur in a wide variety of nerve pathways within the brain.

Addiction occurs when chronic exposure to nicotine induces the nervous system to adapt physiologically. The brain compensates for the many changes nicotine induces by making other changes. Adjustments are made to the numbers and sensitivities of many kinds of receptors within the brain, restoring an appropriate balance of activity.

Now what happens if you stop smoking? Everything is out of whack! The newly coordinated system *requires* nicotine to achieve an appropriate balance of nerve pathway activities. This is addiction in any sensible use of the term. The body's physiological response is profound and unavoidable. There is no way to prevent addiction to nicotine with willpower, any more than willpower can stop a bullet when playing Russian roulette with a loaded gun. If you smoke cigarettes for a prolonged period, you will become addicted.

What do you do if you are addicted to smoking cigarettes and you want to stop? When use of an addictive drug like nicotine is stopped, the level of signaling will change to levels far from normal. If the drug is not reintroduced, the altered level of signaling will eventually induce the nerve cells to once again make compensatory changes that restore an appropriate balance of activities within the brain. Over time, receptor numbers, their sensitivity, and patterns of release of neurotransmitters all revert to normal, once

again producing normal levels of signaling along the pathways. There is no way to avoid the down side of addiction. The pleasure pathways will not function at normal levels until the number of receptors on the affected nerve cells have time to readjust.

Many people attempt to quit smoking by using patches containing nicotine; the idea is that providing gradually lesser doses of nicotine allows the smoker to be weaned of his or her craving for cigarettes. The patches do reduce the craving for cigarettes—so long as you keep using the patches! Actually, using such patches simply substitutes one (admittedly less dangerous) nicotine source for another. If you are going to quit smoking, there is no way to avoid the necessity of eliminating the drug to which you are addicted. Hard as it is to hear the bad news, there is no easy way out. The only way to quit is to quit.

Acetylcholine stimulates the opening of chemically regulated ion channels, causing a depolarization called an excitatory postsynaptic potential (EPSP). Glycine and GABA are inhibitory neurotransmitters that produce hyperpolarization of the postsynaptic membrane. There are also many other neurotransmitters, including the biogenic amines: dopamine, norepinephrine, and serotonin. The effects of different neurotransmitters are integrated through summation of depolarizations and hyperpolarizations.

54.4 The central nervous system consists of the brain and spinal cord.

The Evolution of the Vertebrate Brain

Sponges are the only major phylum of multicellular animals that lack nerves. The simplest nervous systems occur among cnidarians (figure 54.22): all neurons are similar and are linked to one another in a web, or *nerve net*. There is no associative activity, no control of complex actions, and little coordination. The simplest animals with associative activity in the nervous system are the free-living flatworms, phylum Platyhelminthes. Running down the bodies of these flatworms are two nerve cords; peripheral nerves extend outward to the muscles of the body. The two nerve cords converge at the front end of the body, forming an enlarged mass of nervous tissue that also contains associative neurons with synapses connecting neurons to one another. This primitive “brain” is a rudimentary central nervous system and permits a far more complex control of muscular responses than is possible in cnidarians.

All of the subsequent evolutionary changes in nervous systems can be viewed as a series of elaborations on the characteristics already present in flatworms. For example, earthworms exhibit a central nervous system that is connected to all other parts of the body by peripheral nerves. And, in arthropods, the central coordination of complex response is increasingly localized in the front end of the nerve cord. As this region evolved, it came to contain a progressively larger number of associative interneurons, and to develop tracts, which are highways within the brain that connect associative elements.

Casts of the interior braincases of fossil agnathans, fishes that swam 500 million years ago, have revealed much about the early evolutionary stages of the vertebrate brain. Although small, these brains already had the three divisions that characterize the brains of all contemporary vertebrates: (1) the hindbrain, or rhombencephalon; (2) the midbrain, or mesencephalon; and (3) the forebrain, or prosencephalon (figure 54.23).

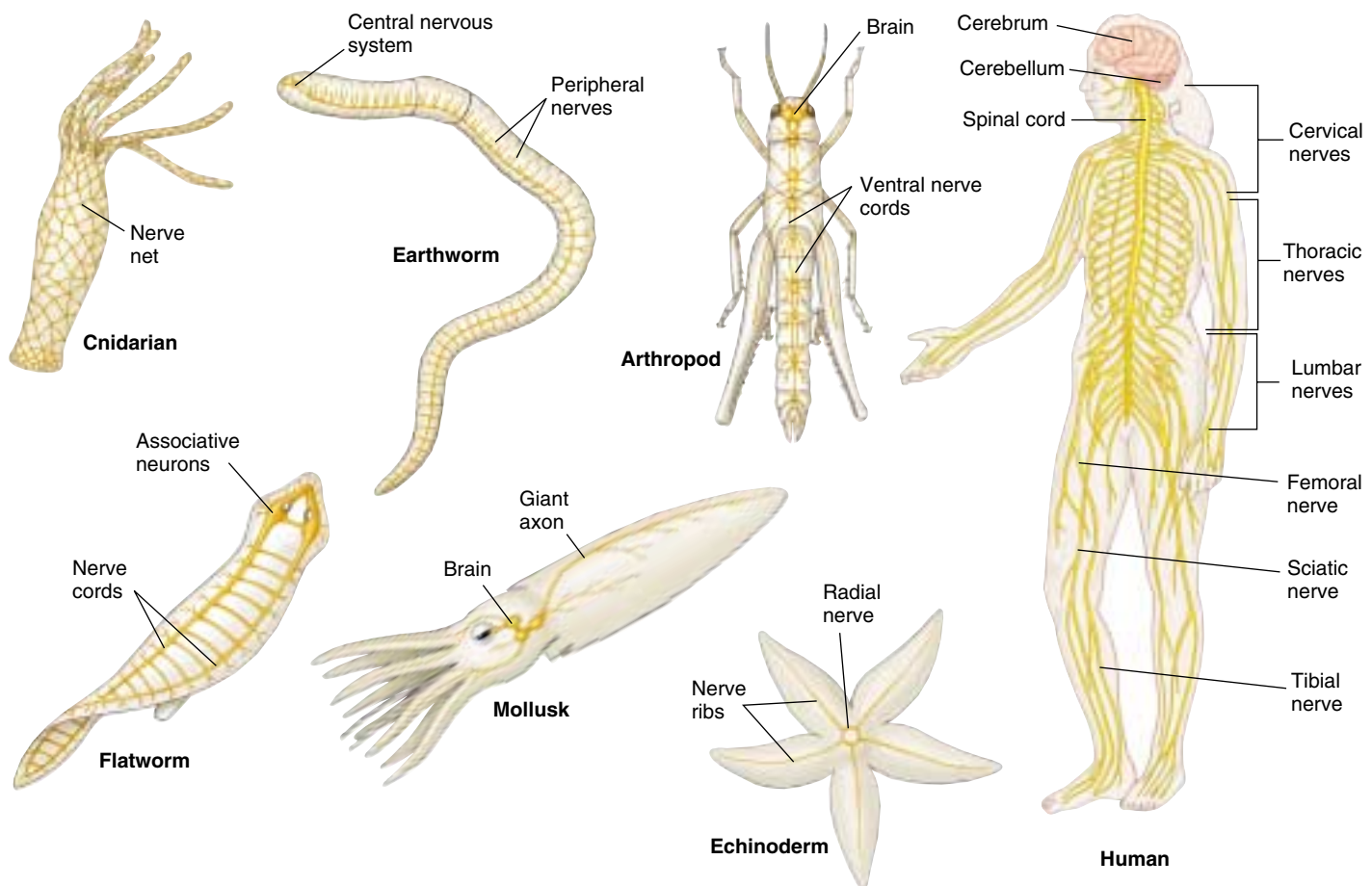


FIGURE 54.22

Evolution of the nervous system. Animals exhibit a progressive elaboration of organized nerve cords and the centralization of complex responses in the front end of the nerve cord.

The hindbrain was the major component of these early brains, as it still is in fishes today. Composed of the *cerebellum*, *pons*, and *medulla oblongata*, the hindbrain may be considered an extension of the spinal cord devoted primarily to coordinating motor reflexes. Tracts containing large numbers of axons run like cables up and down the spinal cord to the hindbrain. The hindbrain, in turn, integrates the many sensory signals coming from the muscles and coordinates the pattern of motor responses.

Much of this coordination is carried on within a small extension of the hindbrain called the cerebellum (“little cerebrum”). In more advanced vertebrates, the cerebellum plays an increasingly important role as a coordinating center for movement and is correspondingly larger than it is in the fishes. In all vertebrates, the cerebellum processes data on the current position and movement of each limb, the state of relaxation or contraction of the muscles involved, and the general position of the body and its relation to the outside world. These data are gathered in the cerebellum, synthesized, and the resulting commands issued to efferent pathways.

In fishes, the remainder of the brain is devoted to the reception and processing of sensory information. The midbrain is composed primarily of the **optic lobes**, which receive and process visual information, while the forebrain is devoted to the processing of *olfactory* (smell) information. The brains of fishes continue growing throughout their lives. This continued growth is in marked contrast to the brains of other classes of vertebrates, which generally complete their development by infancy (figure 54.24). The

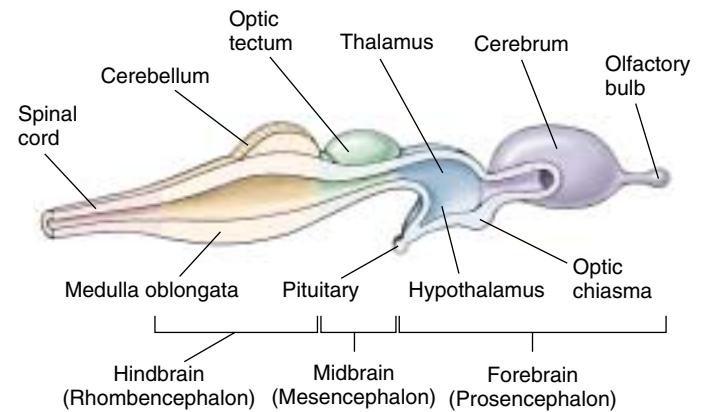


FIGURE 54.23

The basic organization of the vertebrate brain can be seen in the brains of primitive fishes. The brain is divided into three regions that are found in differing proportions in all vertebrates: the hindbrain, which is the largest portion of the brain in fishes; the midbrain, which in fishes is devoted primarily to processing visual information; and the forebrain, which is concerned mainly with olfaction (the sense of smell) in fishes. In terrestrial vertebrates, the forebrain plays a far more dominant role in neural processing than it does in fishes.

human brain continues to develop through early childhood, but no new neurons are produced once development has ceased, except in the tiny hippocampus, which controls which experiences are filed away into long-term memory and which are forgotten.

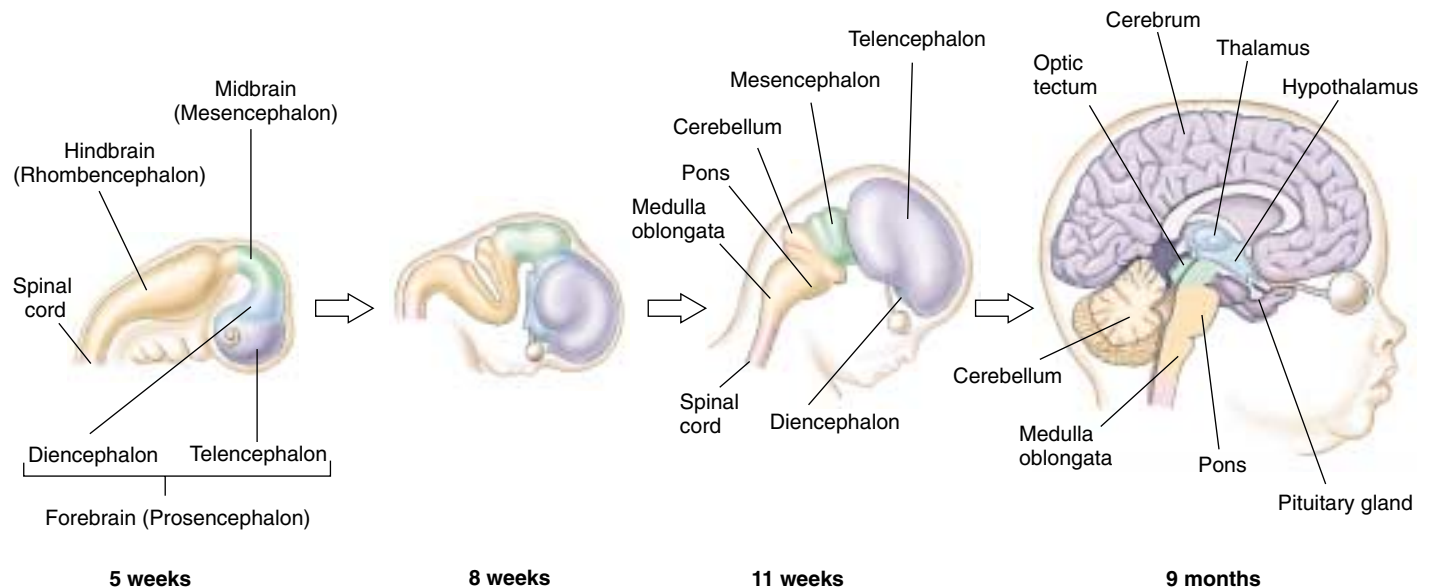


FIGURE 54.24

Development of the brain in humans. The main regions of the brain form during fetal development.

The Dominant Forebrain

Starting with the amphibians and continuing more prominently in the reptiles, processing of sensory information is increasingly centered in the forebrain. This pattern was the dominant evolutionary trend in the further development of the vertebrate brain (figure 54.25).

The forebrain in reptiles, amphibians, birds, and mammals is composed of two elements that have distinct functions. The *diencephalon* (Greek *dia*, “between”) consists of the thalamus and hypothalamus. The **thalamus** is an integrating and relay center between incoming sensory infor-

mation and the cerebrum. The **hypothalamus** participates in basic drives and emotions and controls the secretions of the pituitary gland. The *telencephalon*, or “end brain” (Greek *telos*, “end”), is located at the front of the forebrain and is devoted largely to associative activity. In mammals, the telencephalon is called the **cerebrum**.

The Expansion of the Cerebrum

In examining the relationship between brain mass and body mass among the vertebrates (figure 54.26), you can

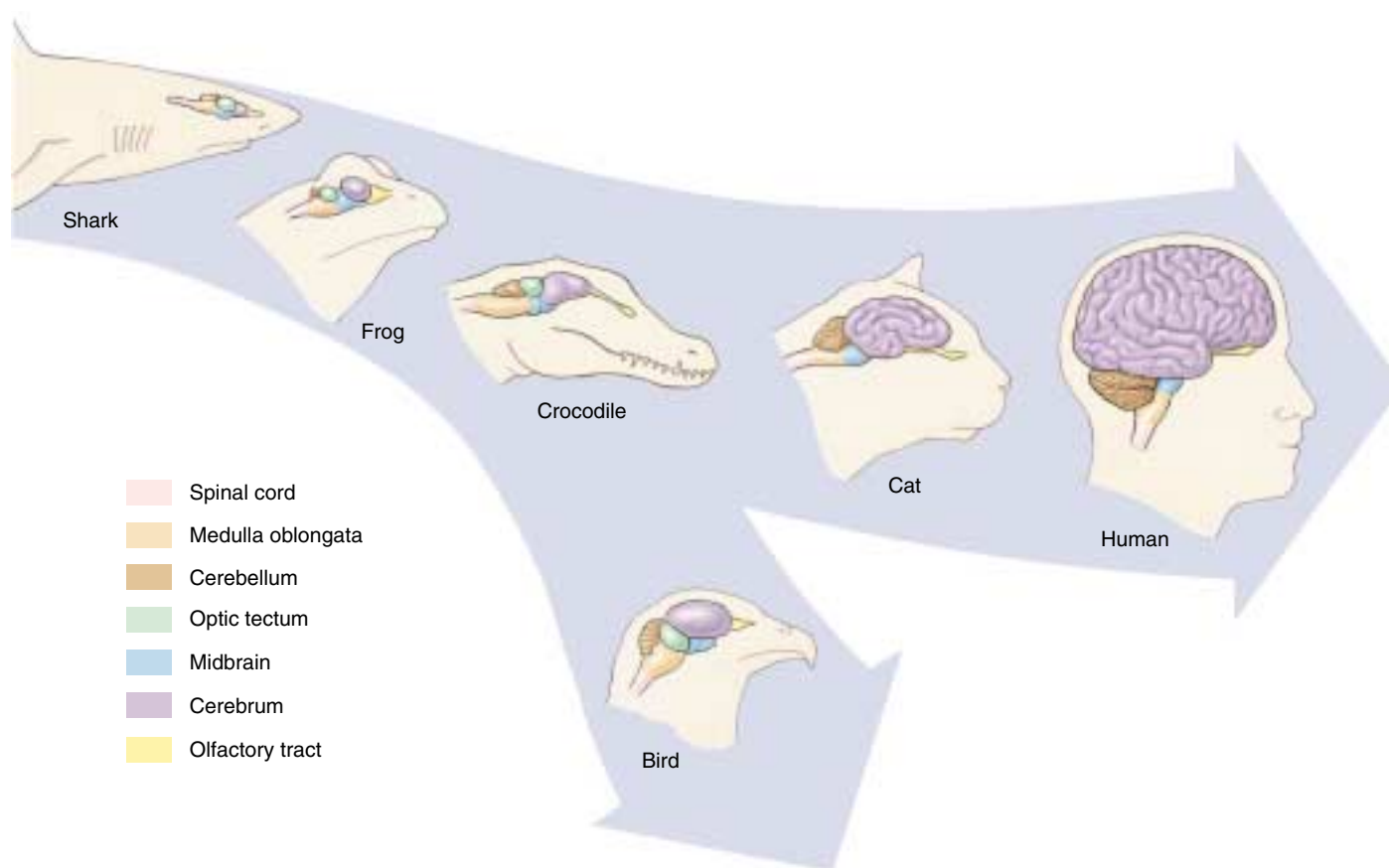


FIGURE 54.25

The evolution of the vertebrate brain involved changes in the relative sizes of different brain regions. In sharks and other fishes, the hindbrain is predominant, and the rest of the brain serves primarily to process sensory information. In amphibians and reptiles, the forebrain is far larger, and it contains a larger cerebrum devoted to associative activity. In birds, which evolved from reptiles, the cerebrum is even more pronounced. In mammals, the cerebrum covers the optic tectum and is the largest portion of the brain. The dominance of the cerebrum is greatest in humans, in whom it envelops much of the rest of the brain.

see a remarkable difference between fishes and reptiles, on the one hand, and birds and mammals, on the other. Mammals have brains that are particularly large relative to their body mass. This is especially true of porpoises and humans; the human brain weighs about 1.4 kilograms. The increase in brain size in the mammals largely reflects the great enlargement of the cerebrum, the dominant part of the mammalian brain. The cerebrum is the center for correlation, association, and learning in the mammalian brain. It receives sensory data from the thalamus and issues motor commands to the spinal cord via descending tracts of axons.

In vertebrates, the central nervous system is composed of the brain and the spinal cord (table 54.3). These two structures are responsible for most of the information processing within the nervous system and consist primarily of interneurons and neuroglia. Ascending tracts carry sensory information to the brain. Descending tracts carry impulses from the brain to the motor neurons and interneurons in the spinal cord that control the muscles of the body.

The vertebrate brain consists of three primary regions: the forebrain, midbrain, and hindbrain. The hindbrain was the principal component of the brain of early vertebrates; it was devoted to the control of motor activity. In vertebrates more advanced than fishes, the processing of information is increasingly centered in the forebrain.

Table 54.3 Subdivisions of the Central Nervous System

Major Subdivision	Function
SPINAL CORD	Spinal reflexes; relays sensory information
HINDBRAIN (rhombencephalon)	
Medulla oblongata	Sensory nuclei; reticular activating system; visceral control
Pons	Reticular activating system; visceral control
Cerebellum	Coordination of movements; balance
MIDBRAIN (Mesencephalon)	Reflexes involving eyes and ears
FOREBRAIN (Prosencephalon)	
Thalamus	Relay station for ascending sensory and descending tracts; visceral control
Hypothalamus	Visceral control; neuroendocrine control
Telencephalon (cerebrum)	
Basal ganglia	Motor control
Corpus callosum	Connects the two hemispheres
Hippocampus (limbic system)	Memory; emotion
Cerebral cortex	Higher functions

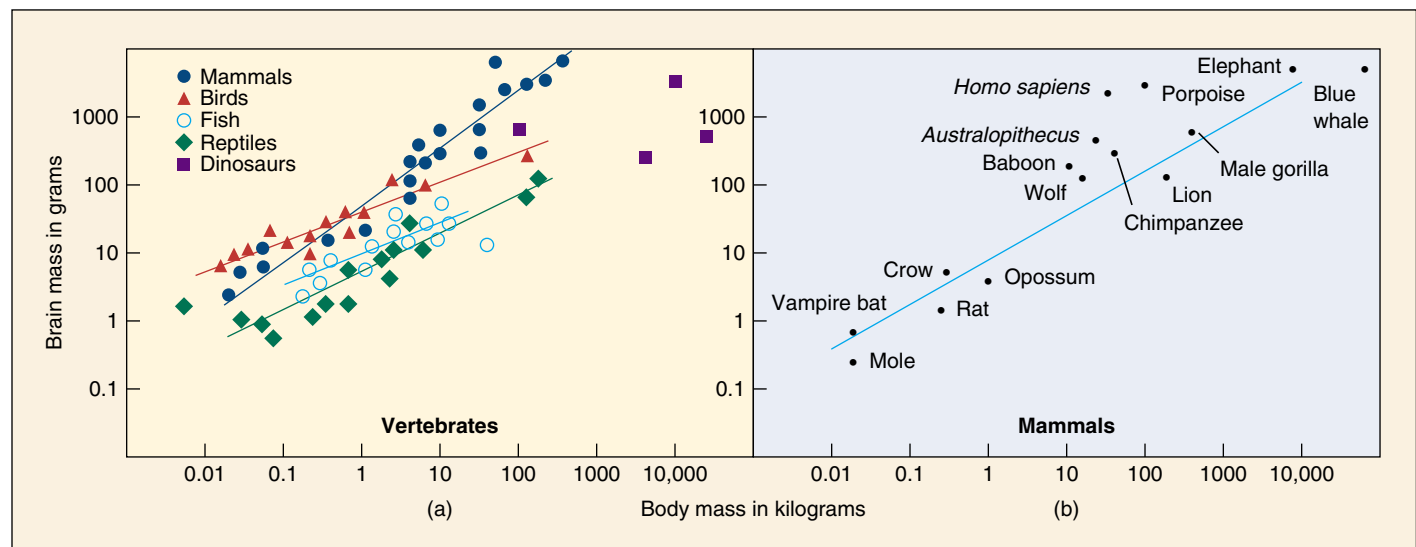


FIGURE 54.26

Brain mass versus body mass. Among most vertebrates, brain mass is a relatively constant proportion of body mass, so that a plot of brain mass versus body mass gives a straight line. (a) However, the proportion of brain mass to body mass is much greater in birds than in reptiles, and it is greater still in mammals. (b) Among mammals, humans have the greatest brain mass per unit of body mass (that is, the farthest perpendicular distance from the plotted line). In second place are the porpoises.

The Human Forebrain

The human cerebrum is so large that it appears to envelop the rest of the brain (figure 54.27). It is split into right and left cerebral hemispheres, which are connected by a tract called the *corpus callosum*. The hemispheres are further divided into the *frontal*, *parietal*, *temporal*, and *occipital lobes*.

Each hemisphere receives sensory input from the opposite, or contralateral, side of the body and exerts motor control primarily over that side. Therefore, a touch on the right hand, for example, is relayed primarily to the left hemisphere, which may then initiate movement of that hand in response to the touch. Damage to one hemisphere due to a stroke often results in a loss of sensation and paralysis on the contralateral side of the body.

Cerebral Cortex

Much of the neural activity of the cerebrum occurs within a layer of gray matter only a few millimeters thick on its outer surface. This layer, called the **cerebral cortex**, is densely packed with nerve cells. In humans, it contains over 10 billion nerve cells, amounting to roughly 10% of all the neurons in the brain. The surface of the cerebral cortex is highly convoluted; this is particularly true in the human brain, where the convolutions increase the surface area of the cortex threefold.

The activities of the cerebral cortex fall into one of three general categories: motor, sensory, and associative. The primary motor cortex lies along the *gyrus* (convolution) on the posterior border of the frontal lobe, just in front of the central *sulcus* (crease) (figure 54.28). Each point on its surface is associated with the movement of a different part of the body (figure 54.29). Just behind the central sulcus, on the anterior edge of the parietal lobe, lies the primary somatosensory cortex. Each point in this area receives input from sensory neurons serving cutaneous and muscle senses in a particular part of the body. Large areas of the motor cortex and primary somatosensory cortex are devoted to the fingers, lips, and tongue because of the need for manual dexterity and speech. The auditory cortex lies within the temporal lobe, and different regions of this cortex deal with different sound frequencies. The visual cortex lies on the occipital lobe, with different sites processing information from different positions on the retina, equivalent to particular points in the visual fields of the eyes.

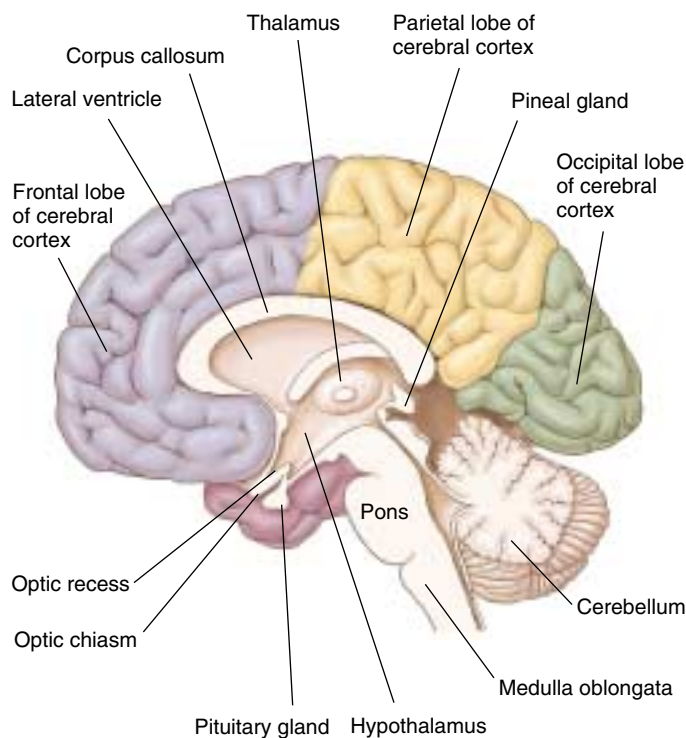


FIGURE 54.27

A section through the human brain. In this sagittal section showing one cerebral hemisphere, the corpus callosum, a fiber tract connecting the two cerebral hemispheres, can be clearly seen.

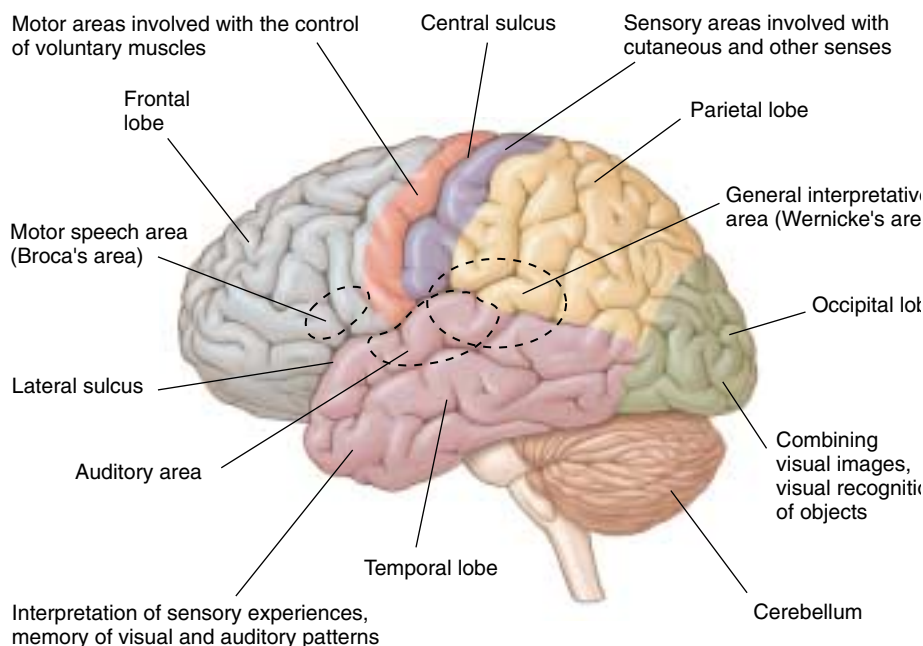


FIGURE 54.28

The lobes of the cerebrum. Some of the known regions of specialization are indicated in this diagram.

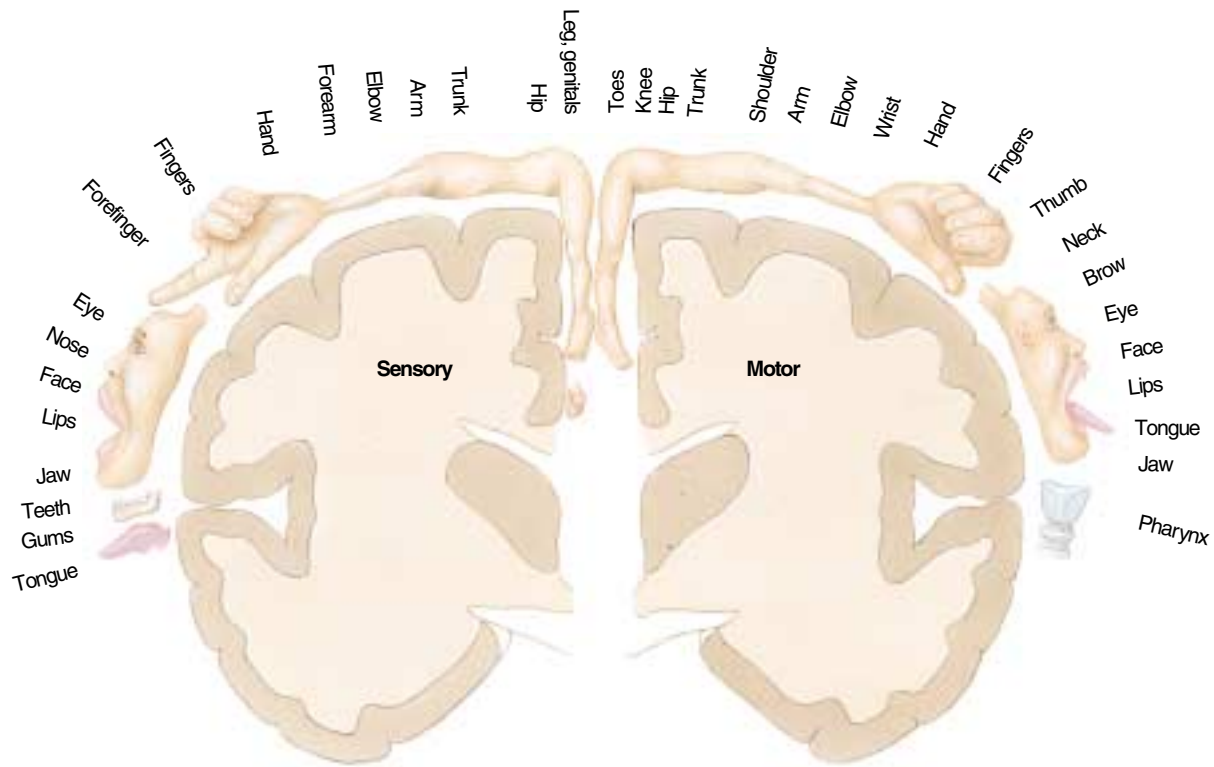


FIGURE 54.29

The primary somatosensory cortex (left) and the primary motor cortex (right). Each of these regions of the cerebral cortex is associated with a different region of the body, as indicated in this stylized map. The areas of the body are drawn in proportion to the amount of cortex dedicated to their sensation or control. For example, the hands have large areas of sensory and motor control, while the pharynx has a considerable area of motor control but little area devoted to the sensations of the pharynx.

The portion of the cerebral cortex that is not occupied by these motor and sensory cortices is referred to as *association cortex*. The site of higher mental activities, the association cortex reaches its greatest extent in primates, especially humans, where it makes up 95% of the surface of the cerebral cortex.

Basal Ganglia

Buried deep within the white matter of the cerebrum are several collections of cell bodies and dendrites that produce islands of gray matter. These aggregates of neuron cell bodies, which are collectively termed the basal ganglia, receive sensory information from ascending tracts and motor commands from the cerebral cortex and cerebellum. Outputs from the basal ganglia are sent down the spinal cord, where they participate in the control of body movements. Damage to specific regions of the basal ganglia can produce the resting tremor of muscles that is characteristic of people with Parkinson's disease.

Thalamus and Hypothalamus

The thalamus is a primary site of sensory integration in the brain. Visual, auditory, and somatosensory information is

sent to the thalamus, where the sensory tracts synapse with association neurons. The sensory information is then relayed via the thalamus to the occipital, temporal, and parietal lobes of the cerebral cortex, respectively. The transfer of each of these types of sensory information is handled by specific aggregations of neuron cell bodies within the thalamus.

The hypothalamus integrates the visceral activities. It helps regulate body temperature, hunger and satiety, thirst, and—along with the limbic system—various emotional states. The hypothalamus also controls the pituitary gland, which in turn regulates many of the other endocrine glands of the body. By means of its interconnections with the cerebral cortex and with control centers in the brain stem (a term used to refer collectively to the midbrain, pons, and medulla oblongata), the hypothalamus helps coordinate the neural and hormonal responses to many internal stimuli and emotions.

The *hippocampus* and *amygdala* are, together with the hypothalamus, the major components of the **limbic system**. This is an evolutionarily ancient group of linked structures deep within the cerebrum that are responsible for emotional responses. The hippocampus is also believed to be important in the formation and recall of memories, a topic we will discuss later.

Language and Other Functions

Arousal and Sleep. The brain stem contains a diffuse collection of neurons referred to as the *reticular formation*. One part of this formation, the reticular activating system, controls consciousness and alertness. All of the sensory pathways feed into this system, which monitors the information coming into the brain and identifies important stimuli. When the reticular activating system has been stimulated to arousal, it increases the level of activity in many parts of the brain. Neural pathways from the reticular formation to the cortex and other brain regions are depressed by anesthetics and barbiturates.

The reticular activating system controls both sleep and the waking state. It is easier to sleep in a dark room than in a lighted one because there are fewer visual stimuli to stimulate the reticular activating system. In addition, activity in this system is reduced by serotonin, a neurotransmitter we previously discussed. Serotonin causes the level of brain activity to fall, bringing on sleep.

Sleep is not the loss of consciousness. Rather, it is an active process whose multiple states can be revealed by recording the electrical activity of the brain in an electroencephalogram (EEG). In a relaxed but awake individual whose eyes are shut, the EEG consists primarily of large, slow waves that occur at a frequency of 8 to 13 hertz (cycles per second). These waves are referred to as *alpha waves*. In an alert subject whose eyes are open, the EEG waves are more rapid (*beta waves* are seen at frequencies of 13 to 30 hertz) and is more desynchronized because multiple sensory inputs are being received, processed, and translated into motor activities.

Theta waves (4 to 7 hertz) and *delta waves* (0.5 to 4 hertz) are seen in various stages of sleep. The first change seen in the EEG with the onset of drowsiness is a slowing and reduction in the overall amplitude of the waves. This slow-wave sleep has several stages but is generally characterized by decreases in arousability, skeletal muscle tone, heart rate, blood pressure, and respiratory rate. During REM sleep (named for the rapid eye movements that occur during this stage), the EEG resembles that of a relaxed, awake individual, and the heart rate, blood pressure, and respiratory rate are all increased. Paradoxically, individuals in REM sleep are difficult to arouse and are more likely to



FIGURE 54.30

Different brain regions control various activities. This illustration shows how the brain reacts in human subjects asked to listen to a spoken word, to read that same word silently, to repeat the word out loud, and then to speak a word related to the first. Regions of white, red, and yellow show the greatest activity. Compare this with figure 54.28 to see how regions of the brain are mapped.

awaken spontaneously. Dreaming occurs during REM sleep, and the rapid eye movements resemble the tracking movements made by the eyes when awake, suggesting that dreamers “watch” their dreams.

Language and Spatial Recognition. Although the two cerebral hemispheres seem structurally similar, they are responsible for different activities. The most thoroughly investigated example of this lateralization of function is language. The left hemisphere is the “dominant” hemisphere for language—the hemisphere in which most neural processing related to language is performed—in 90% of right-handed people and nearly two-thirds of left-handed people. There are two language areas in the dominant hemisphere. Wernicke’s area (see figure 54.28), located in the parietal lobe between the primary auditory and visual areas, is important for language comprehension and the formulation of thoughts into speech (figure 54.30). Broca’s area, found near the part of the motor cortex controlling the face, is responsible for the generation of motor output needed for language communication. Damage to these brain areas can

cause language disorders known as *aphasias*. For example, if Wernicke's area is damaged, the person's speech is rapid and fluid but lacks meaning; words are tossed together as in a "word salad."

While the dominant hemisphere for language is adept at sequential reasoning, like that needed to formulate a sentence, the nondominant hemisphere (the right hemisphere in most people) is adept at spatial reasoning, the type of reasoning needed to assemble a puzzle or draw a picture. It is also the hemisphere primarily involved in musical ability—a person with damage to Broca's speech area in the left hemisphere may not be able to speak but may retain the ability to sing! Damage to the nondominant hemisphere may lead to an inability to appreciate spatial relationships and may impair musical activities such as singing. Even more specifically, damage to the inferior temporal cortex in that hemisphere eliminates the capacity to recall faces. Reading, writing, and oral comprehension remain normal, and patients with this disability can still recognize acquaintances by their voices. The nondominant hemisphere is also important for the consolidation of memories of non-verbal experiences.

Memory and Learning. One of the great mysteries of the brain is the basis of memory and learning. There is no one part of the brain in which all aspects of a memory appear to reside. Specific cortical sites cannot be identified for particular memories because relatively extensive cortical damage does not selectively remove memories. Although memory is impaired if portions of the brain, particularly the temporal lobes, are removed, it is not lost entirely. Many memories persist in spite of the damage, and the ability to access them gradually recovers with time. Therefore, investigators who have tried to probe the physical mechanisms underlying memory often have felt that they were grasping at a shadow. Although we still do not have a complete understanding of these mechanisms, we have learned a good deal about the basic processes in which memories are formed.

There appear to be fundamental differences between short-term and long-term memory. Short-term memory is transient, lasting only a few moments. Such memories can readily be erased by the application of an electrical shock, leaving previously stored long-term memories intact. This result suggests that short-term memories are stored electrically in the form of a transient neural excitation. Long-term memory, in contrast, appears to involve structural changes in certain neural connections within the brain. Two parts of the temporal lobes, the hippocampus and the amygdala, are involved in both short-term memory and its consolidation into long-term memory. Damage to these structures impairs the ability to process recent events into long-term memories.

Synapses that are used intensively for a short period of time display more effective synaptic transmission upon subsequent use. This phenomenon is called long-term potentiation (LTP). During LTP, the presynaptic neuron may release increased amounts of neurotransmitter with each action potential, and the postsynaptic neuron may become increasingly sensitive to the neurotransmitter. It is believed that these changes in synaptic transmission may be responsible for some aspects of memory storage.

Mechanism of Alzheimer's Disease Still a Mystery

In the past, little was known about *Alzheimer's disease*, a condition in which the memory and thought processes of the brain become dysfunctional. Drug companies are eager to develop new products for the treatment of Alzheimer's, but they have little concrete evidence to go on. Scientists disagree about the biological nature of the disease and its cause. Two hypotheses have been proposed: one that nerve cells in the brain are killed from the outside in, and the other that the cells are killed from the inside out.

In the first hypothesis, external proteins called β -amyloid peptides kill nerve cells. A mistake in protein processing produces an abnormal form of the peptide, which then forms aggregates, or plaques. The plaques begin to fill in the brain and then damage and kill nerve cells. However, these amyloid plaques have been found in autopsies of people that did not have Alzheimer's disease.

The second hypothesis maintains that the nerve cells are killed by an abnormal form of an internal protein. This protein, called tau (τ), normally functions to maintain protein transport microtubules. Abnormal forms of τ assemble into helical segments that form tangles, which interfere with the normal functioning of the nerve cells. Researchers continue to study whether tangles and plaques are causes or effects of Alzheimer's disease.

Progress has been made in identifying genes that increase the likelihood of developing Alzheimer's and genes that, when mutated, can cause Alzheimer's disease. However, the genes may not reveal much about Alzheimer's as they do not show up in most Alzheimer's patients, and they cause symptoms that start much earlier than when most Alzheimer's patients show symptoms.

The cerebrum is composed of two cerebral hemispheres. Each hemisphere consists of the gray matter of the cerebral cortex overlying white matter and islands of gray matter (nuclei) called the basal ganglia. These areas are involved in the integration of sensory information, control of body movements, and such associative functions as learning and memory.

The Spinal Cord

The spinal cord is a cable of neurons extending from the brain down through the backbone (figure 54.31). It is enclosed and protected by the vertebral column and layers of membranes called *meninges*, which also cover the brain. Inside the spinal cord there are two zones. The inner zone, called gray matter, consists of interneurons and the cell bodies of motor neurons. The outer zone, called white matter, contains the axons and dendrites of nerve cells. Messages from the body and the brain run up and down the spinal cord, an “information highway.”

In addition to relaying messages, the spinal cord also functions in reflexes, the sudden, involuntary movement of muscles. A reflex produces a rapid motor response to a stimulus because the sensory neuron passes its information to a motor neuron in the spinal cord, without higher level processing. One of the most frequently used reflexes in your body is blinking, a reflex that protects your eyes. If anything, such as an insect or a cloud of dust, approaches your eye, the eyelid blinks before you realize what has happened. The reflex occurs before the cerebrum is aware the eye is in danger.



FIGURE 54.31

A view down the human spinal cord. Pairs of spinal nerves can be seen extending from the spinal cord. It is along these nerves, as well as the cranial nerves that arise from the brain, that the central nervous system communicates with the rest of the body.

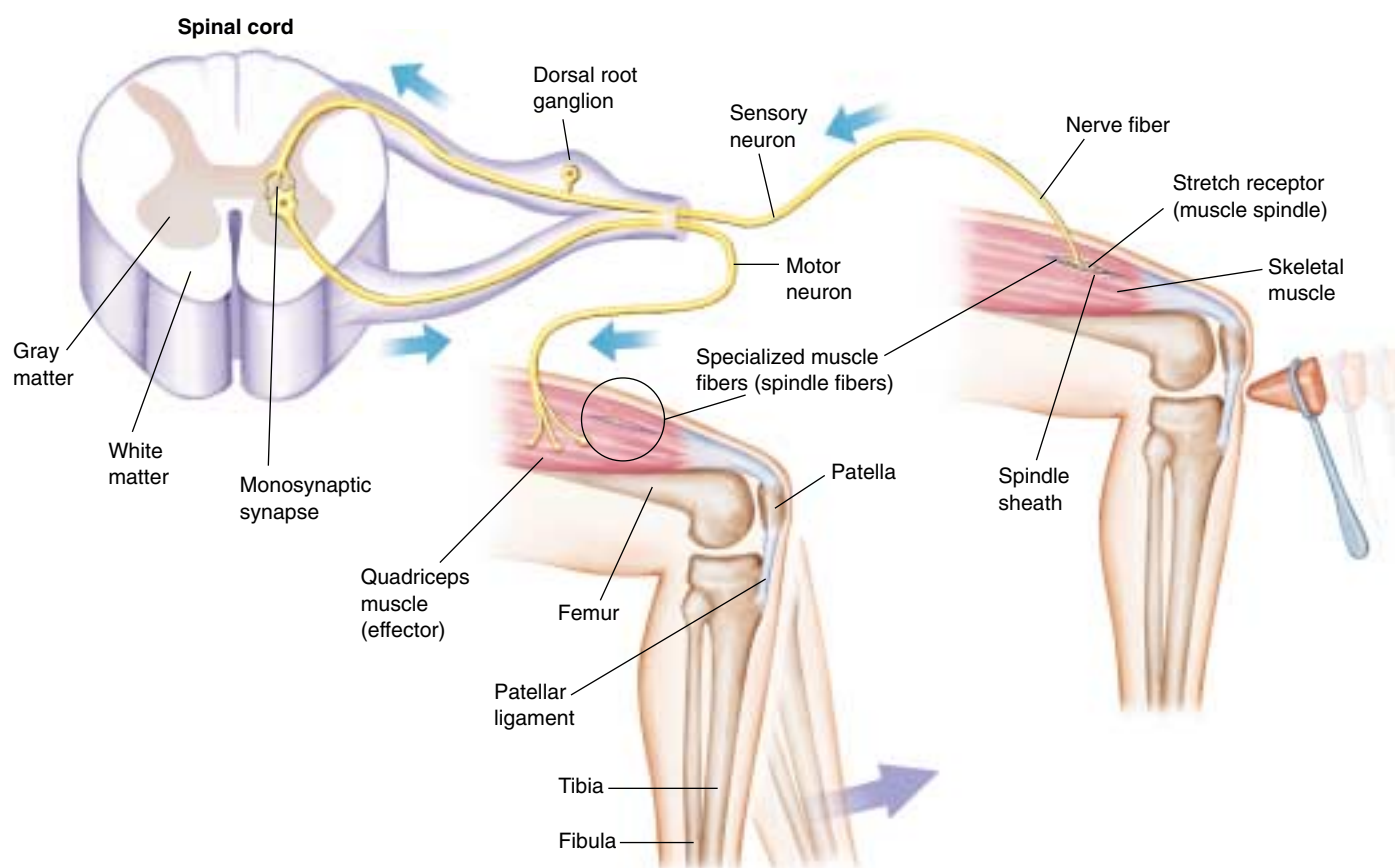


FIGURE 54.32

The knee-jerk reflex. This is the simplest reflex, involving only sensory and motor neurons.

Because they pass information along only a few neurons, reflexes are very fast. Many reflexes never reach the brain. The nerve impulse travels only as far as the spinal cord and then comes right back as a motor response. A few reflexes, like the knee-jerk reflex (figure 54.32), are monosynaptic reflex arcs. In these, the sensory nerve cell makes synaptic contact directly with a motor neuron in the spinal cord whose axon travels directly back to the muscle. The knee-jerk reflex is also an example of a *muscle stretch reflex*. When the muscle is briefly stretched by tapping the patellar ligament with a rubber mallet, the *muscle spindle apparatus* is also stretched. The spindle apparatus is embedded within the muscle, and, like the muscle fibers outside the spindle, is stretched along with the muscle. Stretching of the spindle activates sensory neurons that synapse directly with somatic motor neurons within the spinal cord. As a result, the somatic motor neurons conduct action potentials to the skeletal muscle fibers and stimulate the muscle to contract. This reflex is the simplest in the vertebrate body because only one synapse is crossed in the reflex arc.

Most reflexes in vertebrates, however, involve a single connecting interneuron between the sensory and the motor neuron. The withdrawal of a hand from a hot stove or the blinking of an eye in response to a puff of air involve a relay of information from a sensory neuron through one or more interneurons to a motor neuron. The motor neuron then stimulates the appropriate muscle to contract (figure 54.33).

Spinal Cord Regeneration

In the past, scientists have tried to repair severed spinal cords by installing nerves from another part of the body to bridge the gap and act as guides for the spinal cord to regenerate. But most of these experiments have failed because although axons may regenerate through the implanted nerves, they cannot penetrate the spinal cord tissue once they leave the implant. Also, there is a factor that inhibits nerve growth in the spinal cord. After discovering that fibroblast growth factor stimulates nerve growth, neurobiologists tried gluing on the nerves, from the implant to the spinal cord, with fibrin that had been mixed with the fibroblast growth factor. Three months later, rats with the nerve bridges began to show movement in their lower bodies. In further analyses of the experimental animals, dye tests indicated that the spinal cord nerves had regrown from both sides of the gap. Many scientists are encouraged by the potential to use a similar treatment in human medicine. However, most spinal cord injuries in humans do not involve a completely severed spinal cord; often, nerves are crushed, which results in different tissue damage. Also, while the rats with nerve bridges did regain some locomotory ability, tests indicated that they were barely able to walk or stand.

The spinal cord relays messages to and from the brain and processes some sensory information directly.

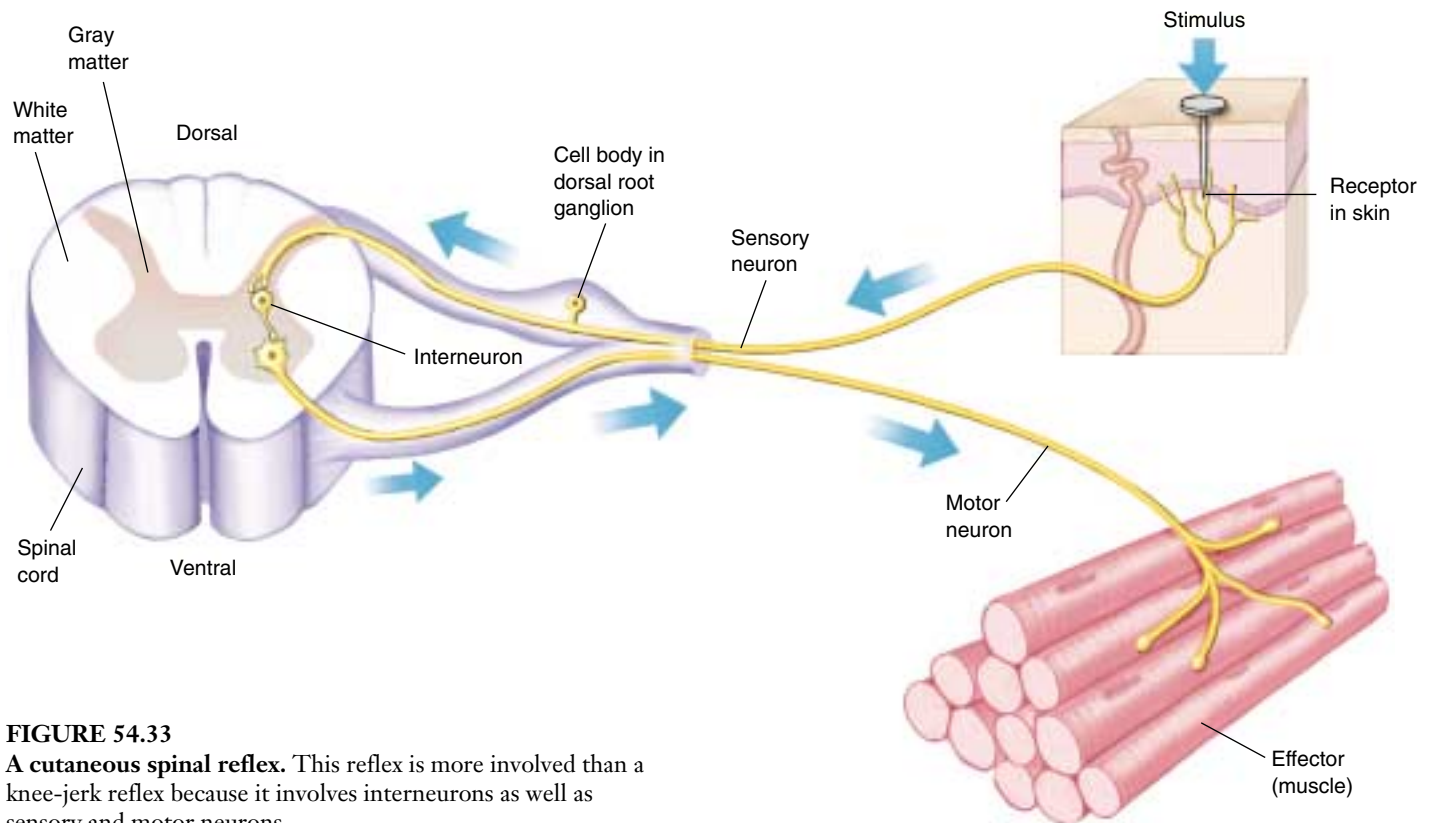


FIGURE 54.33

A cutaneous spinal reflex. This reflex is more involved than a knee-jerk reflex because it involves interneurons as well as sensory and motor neurons.

54.5 The peripheral nervous system consists of sensory and motor neurons.

Components of the Peripheral Nervous System

The peripheral nervous system consists of nerves and ganglia. Nerves are cablelike collections of axons (figure 54.34), usually containing both sensory and motor neurons. Ganglia are aggregations of neuron cell bodies located outside the central nervous system.

At its origin, a spinal nerve separates into sensory and motor components. The axons of sensory neurons enter the dorsal surface of the spinal cord and form the **dorsal root** of the spinal nerve, whereas motor axons leave from the ventral surface of the spinal nerve and form the **ventral root** of the spinal nerve. The cell bodies of sensory neurons are grouped together outside each level of the spinal cord in the **dorsal root ganglia**. The cell bodies of somatic motor neurons, on the other hand, are located within the spinal cord and so are not located in ganglia.

Somatic motor neurons stimulate skeletal muscles to contract, and autonomic motor neurons innervate involuntary effectors—smooth muscles, cardiac muscle, and glands. A comparison of the somatic and autonomic nervous systems is provided in table 54.4 and each will be discussed in turn. Somatic motor neurons stimulate the skeletal muscles of the body to contract in response to conscious commands and as part of reflexes that do not require conscious control. Conscious control of skeletal muscles is achieved by activation of tracts of axons that descend from the cerebrum to the appropriate level of the spinal cord. Some of these descending axons will stimulate spinal cord motor neurons directly, while others will activate interneurons that in turn stimulate the spinal motor neurons. When a particular muscle is stimulated to contract, however, its antagonist must be inhibited. In order to flex the arm, for example, the flexor muscles must be stimulated while the antagonistic extensor muscle is inhibited (see figure 50.6). Descending motor axons produce this necessary inhibition by causing hyperpolarizations (IPSPs) of the spinal motor neurons that innervate the antagonistic muscles.

A spinal nerve contains sensory neurons that enter the dorsal root and motor neurons that enter the ventral root of the nerve. Somatic motor neurons innervate skeletal muscles and stimulate the muscles to contract.

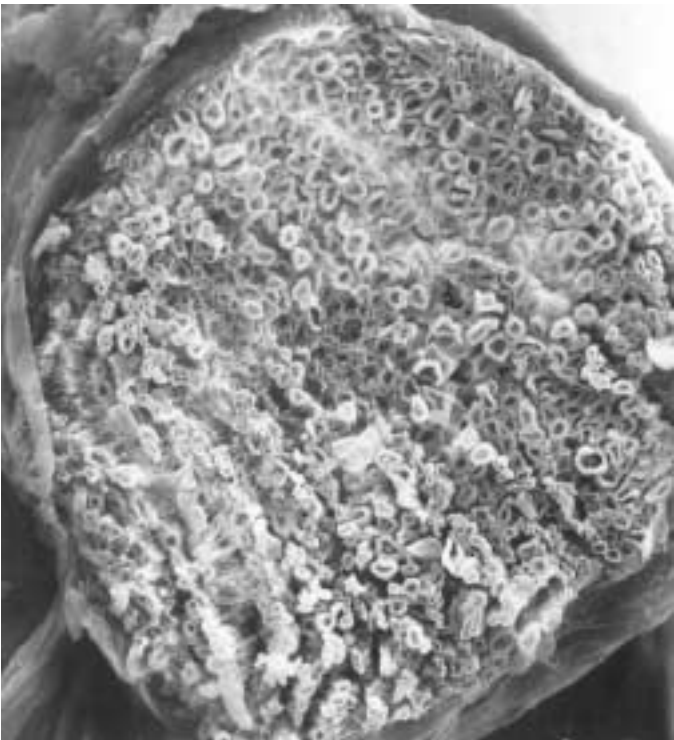


FIGURE 54.34
Nerves in the peripheral nervous system. Photomicrograph (1600×) showing a cross section of a bullfrog nerve. The nerve is a bundle of axons bound together by connective tissue. Many myelinated axons are visible, each looking somewhat like a doughnut.

Table 54.4 Comparison of the Somatic and Autonomic Nervous Systems		
Characteristic	Somatic	Autonomic
Effectors	Skeletal muscle	Cardiac muscle Smooth muscle Gastrointestinal tract Blood vessels Airways Exocrine glands
Effect on motor nerves	Excitation	Excitation or inhibition
Innervation of effector cells	Always single	Typically dual
Number of neurons in path to effector	One	Two
Neurotransmitter	Acetylcholine	Acetylcholine Norepinephrine

The Autonomic Nervous System

The autonomic nervous system is composed of the sympathetic and parasympathetic divisions and the medulla oblongata of the hindbrain, which coordinates this system. Though they differ, the sympathetic and parasympathetic divisions share several features. In both, the efferent motor pathway involves two neurons: the first has its cell body in the CNS and sends an axon to an autonomic ganglion, while the second has its cell body in the autonomic ganglion and sends its axon to synapse with a smooth muscle, cardiac muscle, or gland cell (figure 54.35). The first neuron is called a *preganglionic neuron*, and it always releases ACh at its synapse. The second neuron is a *postganglionic neuron*; those in the parasympathetic division release ACh, while those in the sympathetic division release norepinephrine.

In the sympathetic division, the preganglionic neurons originate in the thoracic and lumbar regions of the spinal cord (figure 54.36). Most of the axons from these neurons synapse in two parallel chains of ganglia immediately outside the spinal cord. These structures are usually called the *sympathetic chain* of ganglia. The sympathetic chain contains the cell bodies of postganglionic neurons, and it is the axons from these neurons that innervate the different visceral organs. There are some exceptions to this general pattern, however. Most importantly, the axons of some preganglionic sympathetic neurons pass through the

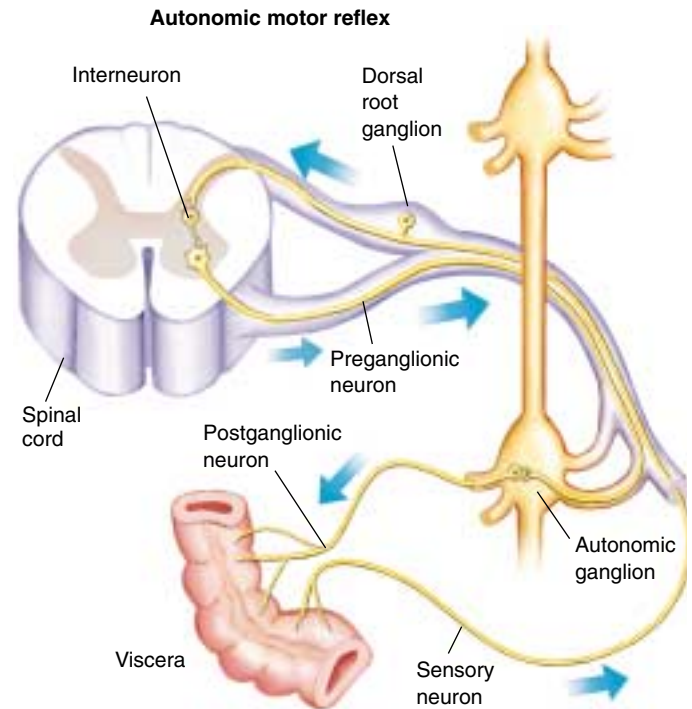


FIGURE 54.35

An autonomic reflex. There are two motor neurons in the efferent pathway. The first, or preganglionic neuron, exits the CNS and synapses at an autonomic ganglion. The second, or postganglionic neuron, exits the ganglion and regulates the visceral effectors (smooth muscle, cardiac muscle, or glands).

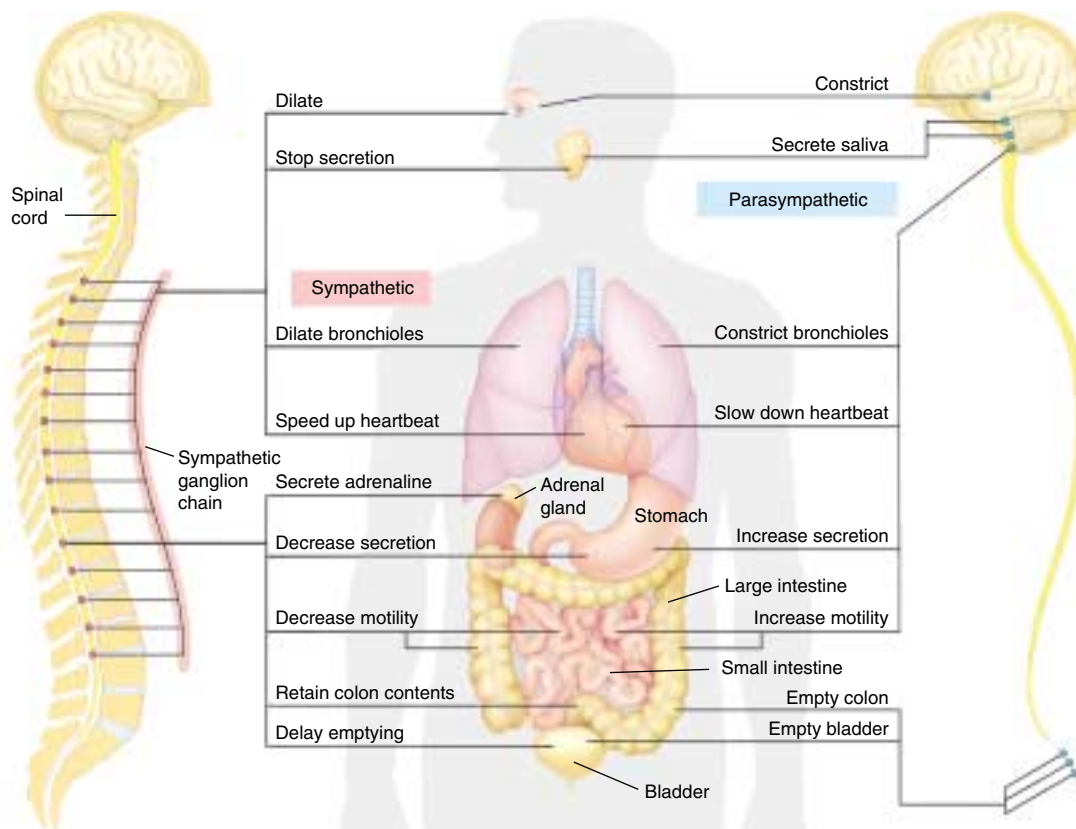


FIGURE 54.36

The sympathetic and parasympathetic divisions of the autonomic nervous system. The preganglionic neurons of the sympathetic division exit the thoracic and lumbar regions of the spinal cord, while those of the parasympathetic division exit the brain and sacral region of the spinal cord. The ganglia of the sympathetic division are located near the spinal cord, while those of the parasympathetic division are located near the organs they innervate. Most of the internal organs are innervated by both divisions.

Table 54.5 Autonomic Innervation of Target Tissues

Target Tissue	Sympathetic Stimulation	Parasympathetic Stimulation
Pupil of eye	Dilation	Constriction
Glands		
Salivary	Vasoconstriction; slight secretion	Vasodilation; copious secretion
Gastric	Inhibition of secretion	Stimulation of gastric activity
Liver	Stimulation of glucose secretion	Inhibition of glucose secretion
Sweat	Sweating	None
Gastrointestinal tract		
Sphincters	Increased tone	Decreased tone
Wall	Decreased tone	Increased motility
Gallbladder	Relaxation	Contraction
Urinary bladder		
Muscle	Relaxation	Contraction
Sphincter	Contraction	Relaxation
Heart muscle	Increased rate and strength	Decreased rate
Lungs	Dilation of bronchioles	Constriction of bronchioles
Blood vessels		
In muscles	Dilation	None
In skin	Constriction	None
In viscera	Constriction	Dilation

sympathetic chain without synapsing and, instead, terminate within the adrenal gland. The adrenal gland consists of an outer part, or cortex, and an inner part, or medulla. The adrenal medulla receives sympathetic nerve innervation and secretes the hormone epinephrine (adrenaline) in response.

When the sympathetic division becomes activated, epinephrine is released into the blood as a hormonal secretion, and norepinephrine is released at the synapses of the postganglionic neurons. Epinephrine and norepinephrine act to prepare the body for fight or flight (figure 54.37). The heart beats faster and stronger, blood glucose concentration increases, blood flow is diverted to the muscles and heart, and the bronchioles dilate (table 54.5).

These responses are antagonized by the parasympathetic division. Preganglionic parasympathetic neurons originate in the brain and sacral regions of the spinal cord. Because of this origin, there cannot be a chain of parasympathetic ganglia analogous to the sympathetic chain. Instead, the preganglionic axons, many of which travel in the vagus (the tenth cranial) nerve, terminate

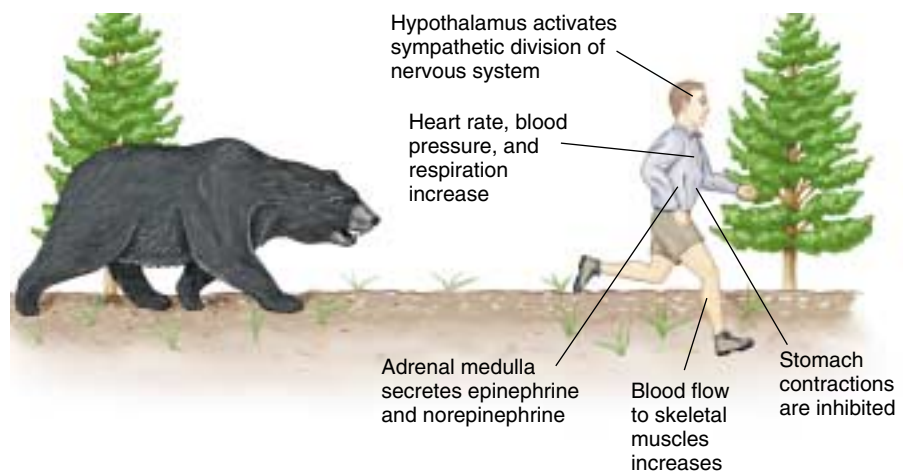


FIGURE 54.37

The sympathetic division of the nervous system in action. To prepare the body for fight or flight, the sympathetic division is activated and causes changes in many organs, glands, and body processes.

in ganglia located near or even within the internal organs. The postganglionic neurons then regulate the internal organs by releasing ACh at their synapses. Parasympathetic nerve effects include a slowing of the heart, increased secretions and activities of digestive organs, and so on.

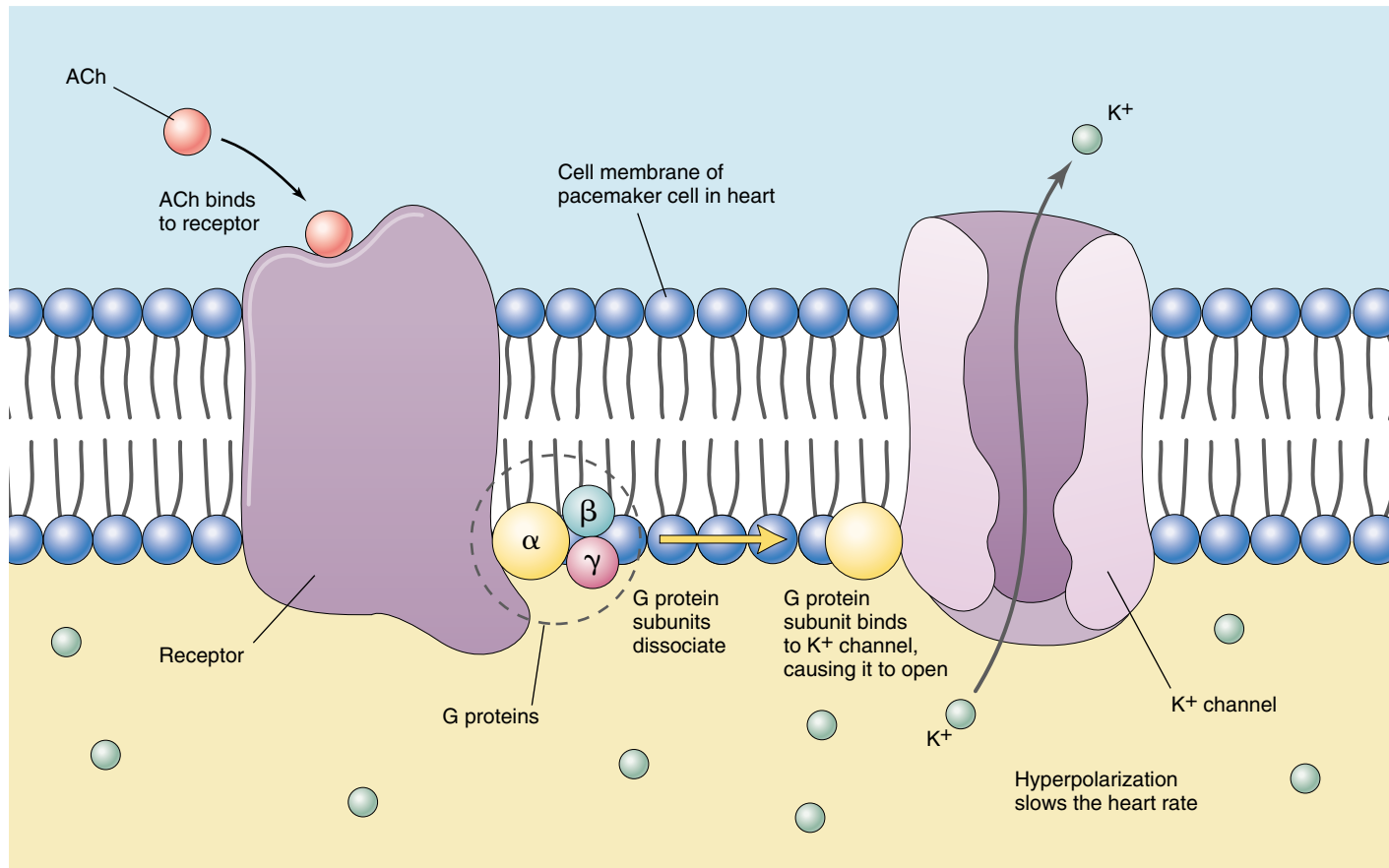


FIGURE 54.38

The parasympathetic effects of ACh require the action of G proteins. The binding of ACh to its receptor causes dissociation of a G protein complex, releasing some components of this complex to move within the membrane and bind to other proteins that form ion channels. The example shown here is the effects of ACh on the heart, where the G protein components cause the opening of potassium channels. This leads to outward diffusion of potassium and hyperpolarization, slowing the heart rate.

G Proteins Mediate Cell Responses to Autonomic Nerves

You might wonder how ACh can slow the heart rate—an inhibitory effect—when it has excitatory effects elsewhere. This inhibitory effect in the pacemaker cells of the heart is produced because ACh causes the opening of potassium channels, leading to the outward diffusion of potassium and thus to hyperpolarization. This and other parasympathetic effects of ACh are produced indirectly, using a group of membrane proteins called **G proteins** (so-called because they are regulated by guanosine diphosphate and guanosine triphosphate [GDP and GTP]). Because the ion channels are located some distance away from the receptor proteins for ACh, the G proteins are needed to serve as connecting links between them.

There are three G protein subunits, designated α, β, and γ, bound together and attached to the receptor protein for ACh. When ACh, released by parasympathetic endings, binds to its receptor, the G protein subunits dissociate (figure 54.38). Specific G protein components move within the

membrane to the potassium channel and cause it to open, producing hyperpolarization and a slowing of the heart. In other organs, the G proteins have different effects that lead to excitation. In this way, for example, the parasympathetic nerves that innervate the stomach can cause increased gastric secretions and contractions.

The sympathetic nerve effects also involve the action of G proteins. Stimulation by norepinephrine from sympathetic nerve endings and epinephrine from the adrenal medulla requires G proteins to activate the target cells. We will describe this in more detail, together with hormone action, in chapter 56.

The sympathetic division of the autonomic system, together with the adrenal medulla, activates the body for fight-or-flight responses, whereas the parasympathetic division generally has antagonistic effects. The actions of parasympathetic nerves are produced by ACh, whereas the actions of sympathetic nerves are produced by norepinephrine.



Summary

Questions

Media Resources

54.1 The nervous system consists of neurons and supporting cells.

- The nervous system is subdivided into the central nervous system (CNS) and peripheral nervous system (PNS).

1. What are the differences and similarities among the three types of neurons?



- Nervous system divisions
- Nervous system cells I
- Nervous system cells II

54.2 Nerve impulses are produced on the axon membrane.

- The resting axon has a membrane potential of -70 mV; the magnitude of this voltage is produced primarily by the distribution of K^+ .
- A depolarization stimulus opens voltage-regulated Na^+ channels and then K^+ channels, producing first the upward phase and then the repolarization phase of the action potential.
- Action potentials are all or none and are conducted without decrease in amplitude because each action potential serves as the stimulus for the production of the next action potential along the axon.

2. Which cation is most concentrated in the cytoplasm of a cell, and which is most concentrated in the extracellular fluid? How are these concentration differences maintained?



- Membrane potential
- Action potential

3. What is a voltage-gated ion channel?



- Activities: Action potential 1 Sodium-potassium pump Action potential 2
- Action potential 1
- Membrane potential
- Local potential

4. What happens to the size of an action potential as it is propagated?

54.3 Neurons form junctions called synapses with other cells.

- The presynaptic axon releases neurotransmitter chemicals that diffuse across the synapse and stimulate the production of either a depolarization or a hyperpolarization in the postsynaptic membrane.
- Depolarizations and hyperpolarization can summate in the dendrites and cell bodies of the postsynaptic neuron, allowing integration of information.

5. If a nerve impulse can jump from node to node along a myelinated axon, why can't it jump from the presynaptic cell to the postsynaptic cell across a synaptic cleft?



- Bioethics case study: Smoking bar
- Student Research: Neural development in Moths
- On *Science* Articles: Is smoking addictive? Nobel prize 2000

54.4 The central nervous system consists of the brain and spinal cord.

- The vertebrate brain is divided into a forebrain, midbrain, and hindbrain, and these are further subdivided into other brain regions. The cerebral cortex has a primary motor area and a primary somatosensory area, as well as areas devoted to the analysis of vision and hearing and the integration and association of information.
- The spinal cord carries information to and from the brain and coordinates many reflex movements.

6. Where are the basal ganglia located, and what is their function?



- Art activities: Central nervous system Spinal cord anatomy Human brain

7. How are short-term and long-term memory thought to differ in terms of their basic underlying mechanisms?



- Reflex arc

54.5 The peripheral nervous system consists of sensory and motor neurons.

- The sympathetic division is activated during fight-or-flight responses; the parasympathetic division opposes the action of the sympathetic division in most activities.

8. How do the sympathetic and parasympathetic divisions differ in the locations of the ganglionic neurons?



- Art activity: Peripheral nervous system

55

Sensory Systems

Concept Outline

55.1 Animals employ a wide variety of sensory receptors.

Categories of Sensory Receptors and Their Actions.

Sensory receptors can be classified according to the type of stimuli to which they can respond.

55.2 Mechanical and chemical receptors sense the body's condition.

Detecting Temperature and Pressure. Receptors within the skin respond to touch, pressure, pain, heat and cold.

Sensing Muscle Contraction and Blood Pressure. A muscle spindle responds to stretching of the muscle; receptors in arteries monitor changes in blood pressure.

Sensing Taste, Smell, and Body Position. Receptors that respond to chemicals produce sensations of taste and smell. Hair cells send nerve impulses when they are bent.

55.3 Auditory receptors detect pressure waves in the air.

The Ears and Hearing. Sound causes vibrations in the ear that bend hair cell processes, initiating a nerve impulse.

Sonar. Bats orient themselves in space by emitting sounds and detecting the time required for the sounds to bounce off objects and return to their ears.

55.4 Optic receptors detect light over a broad range of wavelengths.

Evolution of the Eye. True image-forming eyes evolved independently in several phyla.

Vertebrate Photoreceptors. Light causes a pigment molecule in a rod or cone cell to dissociate; this “bleaching” reaction activates the photoreceptor.

Visual Processing in the Vertebrate Retina. Action potentials travel from the retina of the eyes to the brain for visual perception.

55.5 Some vertebrates use heat, electricity, or magnetism for orientation.

Diversity of Sensory Experiences. Special receptors can detect heat, electrical currents, and magnetic fields.



FIGURE 55.1

Photoreceptors in the vertebrate eye. Rods, the broad, tubular cells, allow black-and-white vision, while cones, the short, tapered cells, are responsible for color vision. Not all vertebrates have both types of receptors.

All input from sensory neurons to the central nervous system arrives in the same form, as action potentials propagated by afferent (inward-conducting) sensory neurons. Different sensory neurons project to different brain regions, and so are associated with different sensory modalities (figure 55.1). The intensity of the sensation depends on the frequency of action potentials conducted by the sensory neuron. A sunset, a symphony, and a searing pain are distinguished by the brain only in terms of the identity of the sensory neuron carrying the action potentials and the frequency of these impulses. Thus, if the auditory nerve is artificially stimulated, the brain perceives the stimulation as sound. But if the optic nerve is artificially stimulated in exactly the same manner and degree, the brain perceives a flash of light.

55.1 Animals employ a wide variety of sensory receptors.

Categories of Sensory Receptors and Their Actions

Sensory information is conveyed to the CNS and perceived in a four-step process (figure 55.2): (1) *stimulation*—a physical stimulus impinges on a sensory neuron or an accessory structure; (2) *transduction*—the stimulus energy is used to produce electrochemical nerve impulses in the dendrites of the sensory neuron; (3) *transmission*—the axon of the sensory neuron conducts action potentials along an afferent pathway to the CNS; and (4) *interpretation*—the brain creates a sensory perception from the electrochemical events produced by afferent stimulation. We actually see (as well as hear, touch, taste, and smell) with our brains, not with our sense organs.

Sensory receptors differ with respect to the nature of the environmental stimulus that best activates their sensory dendrites. Broadly speaking, we can recognize three classes of environmental stimuli: (1) mechanical forces, which stimulate **mechanoreceptors**; (2) chemicals, which stimulate **chemoreceptors**; and (3) electromagnetic and thermal energy, which stimulate a variety of receptors, including the **photoreceptors** of the eyes (table 55.1).

The simplest sensory receptors are *free nerve endings* that respond to bending or stretching of the sensory neuron membrane, to changes in temperature, or to chemicals like oxygen in the extracellular fluid. Other sensory receptors are more complex, involving the association of the sensory neurons with specialized epithelial cells.

Sensing the External and Internal Environments

Exteroceptors are receptors that sense stimuli that arise in the external environment. Almost all of a vertebrate's exterior senses evolved in water before vertebrates invaded the land. Consequently, many senses of terrestrial vertebrates emphasize stimuli that travel well in water, using receptors that have been retained in the transition from the sea to the land. Mammalian hearing, for example, converts an airborne stimulus into a waterborne one, using receptors similar to those that originally evolved in the water. A few vertebrate sensory systems that function well in the water, such as the electrical organs of fish, cannot function in the air and are not found among terrestrial vertebrates. On the other hand, some land-dwellers have sensory systems, such as infrared receptors, that could not function in the sea.

Sensory systems can provide several levels of information about the external environment. Some sensory sys-

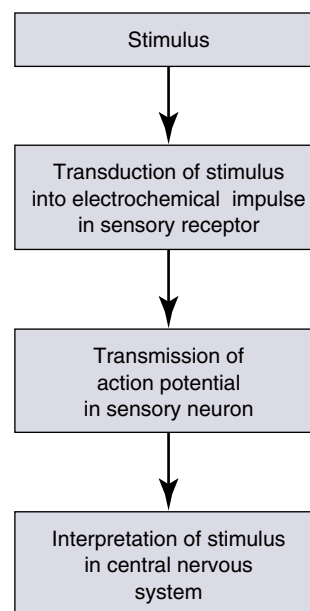


FIGURE 55.2

The path of sensory information. Sensory stimuli must be transduced into electrochemical nerve impulses that are conducted to the brain for interpretation.

Table 55.1 Classes of Environmental Stimuli

Mechanical Forces	Chemicals	Electromagnetic Energy
Pressure	Taste	Light
Gravity	Smell	Heat
Inertia	Humidity	Electricity
Sound		Magnetism
Touch		
Vibration		

tems provide only enough information to determine that an object is present; they call the animal's attention to the object but give little or no indication of where it is located. Other sensory systems provide information about the location of an object, permitting the animal to move toward it. Still other sensory systems enable the brain to construct a three-dimensional image of an object and its surroundings.

Interoceptors sense stimuli that arise from within the body. These internal receptors detect stimuli related to

Table 55.2 Sensory Transduction Among the Vertebrates

Stimulus	Receptor	Location	Structure	Transduction Process
INTEROCEPTION				
Temperature	Heat receptors and cold receptors	Skin, hypothalamus	Free nerve ending	Temperature change opens/closes ion channels in membrane
Touch	Meissner's corpuscles, Merkel cells	Surface of skin	Nerve ending within elastic capsule	Rapid or extended change in pressure deforms membrane
Vibration	Pacinian corpuscles	Deep within skin	Nerve ending within elastic capsule	Severe change in pressure deforms membrane
Pain	Nociceptors	Throughout body	Free nerve ending	Chemicals or changes in pressure or temperature open/close ion channels in membrane
Muscle stretch	Stretch receptors	Within muscles	Spiral nerve endings wrapped around muscle spindle	Stretch of spindle deforms membrane
Blood pressure	Baroreceptors	Arterial branches	Nerve endings over thin part of arterial wall	Stretch of arterial wall deforms membrane
EXTEROCEPTION				
Gravity	Statocysts	Outer chambers of inner ear	Otoliths and hair cells	Otoliths deform hair cells
Motion	Cupula	Semicircular canals of inner ear	Collection of hair cells	Fluid movement deforms hair cells
	Lateral line organ	Within grooves on body surface of fish	Collection of hair cells	Fluid movement deforms hair cells
Taste	Taste bud cells	Mouth; skin of fish	Chemoreceptors: epithelial cells with microvilli	Chemicals bind to membrane receptors
Smell	Olfactory neurons	Nasal passages	Chemoreceptors: ciliated neurons	Chemicals bind to membrane receptors
Hearing	Organ of Corti	Cochlea of inner ear	Hair cells between basilar and tectorial membranes	Sound waves in fluid deform membranes
Vision	Rod and cone cells	Retina of eye	Array of photosensitive pigments	Light initiates process that closes ion channels
Heat	Pit organ	Face of snake	Temperature receptors in two chambers	Receptors compare temperatures of surface and interior chambers
Electricity	Ampullae of Lorenzini	Within skin of fishes	Closed vesicles with asymmetrical ion channel distribution	Electrical field alters ion distribution on membranes
Magnetism	Unknown	Unknown	Unknown	Deflection at magnetic field initiates nerve impulses?

muscle length and tension, limb position, pain, blood chemistry, blood volume and pressure, and body temperature. Many of these receptors are simpler than those that monitor the external environment and are believed to bear

a closer resemblance to primitive sensory receptors. In the rest of this chapter, we will consider the different types of interoceptors and exteroceptors according to the kind of stimulus each is specialized to detect (table 55.2).

Sensory Transduction

Sensory cells respond to stimuli because they possess *stimulus-gated ion channels* in their membranes. The sensory stimulus causes these ion channels to open or close, depending on the sensory system involved. In doing so, a sensory stimulus produces a change in the membrane potential of the receptor cell. In most cases, the sensory stimulus produces a depolarization of the receptor cell, analogous to the excitatory postsynaptic potential (EPSP, described in chapter 54) produced in a postsynaptic cell in response to neurotransmitter. A depolarization that occurs in a sensory receptor upon stimulation is referred to as a **receptor potential** (figure 55.3*a*).

Like an EPSP, a receptor potential is graded: the larger the sensory stimulus, the greater the degree of depolarization. Receptor potentials also decrease in size (*decrement*) with distance from their source. This prevents small, irrelevant stimuli from reaching the cell body of the sensory

neuron. Once a threshold level of depolarization is reached, the receptor potential stimulates the production of action potentials that are conducted by a sensory axon into the CNS (figure 55.3*b*). The greater the sensory stimulus, the greater the depolarization of the receptor potential and the higher the frequency of action potentials. There is generally a logarithmic relationship between stimulus intensity and action potential frequency—a sensory stimulus that is ten times greater than another stimulus will produce action potentials at twice the frequency of the other stimulus. This allows the brain to interpret the incoming signals as indicating a sensory stimulus of a particular strength.

Sensory receptors transduce stimuli in the internal or external environment into graded depolarizations, which stimulates the production of action potentials. Sensory receptors may be classified on the basis of the type of stimulus energy to which they respond.

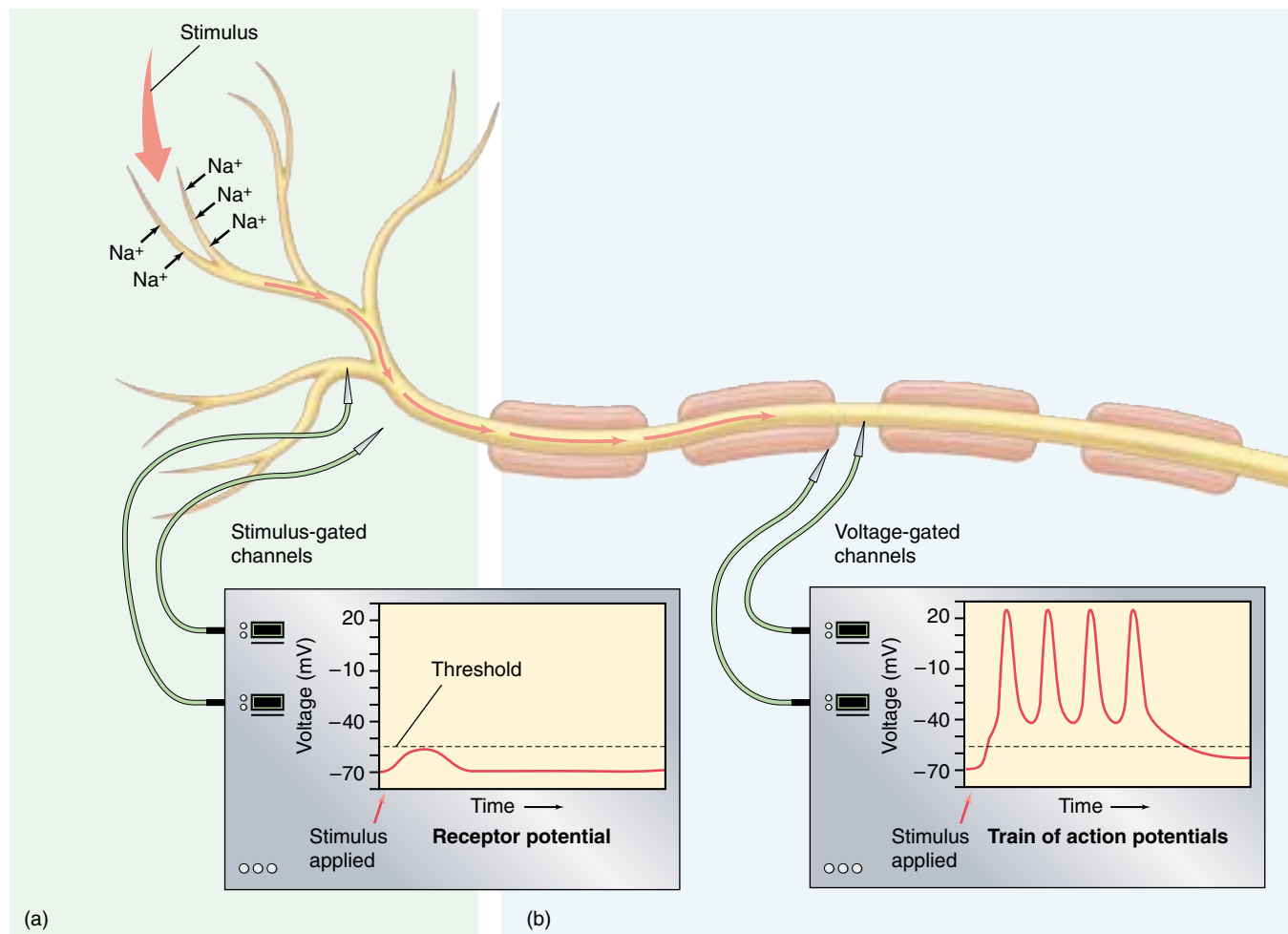


FIGURE 55.3

Events in sensory transduction. (a) Depolarization of a free nerve ending leads to a receptor potential that spreads by local current flow to the axon. (b) Action potentials are produced in the axon in response to a sufficiently large receptor potential.

55.2 Mechanical and chemical receptors sense the body's condition.

Detecting Temperature and Pressure

While the receptors of the skin, called the **cutaneous receptors**, are classified as interoceptors, they in fact respond to stimuli at the border between the external and internal environments. These receptors serve as good examples of the specialization of receptor structure and function, responding to heat, cold, pain, touch, and pressure.

The skin contains two populations of **thermoreceptors**, which are naked dendritic endings of sensory neurons that are sensitive to changes in temperature. *Cold receptors* are stimulated by a fall in temperature and inhibited by warming, while *warm receptors* are stimulated by a rise in temperature and inhibited by cooling. Cold receptors are located immediately below the epidermis, while warm receptors are located slightly deeper, in the dermis. Thermoreceptors are also found within the hypothalamus of the brain, where they monitor the temperature of the circulating blood and thus provide the CNS with information on the body's internal (core) temperature.

A stimulus that causes or is about to cause tissue damage is perceived as pain. The receptors that transmit impulses that are perceived by the brain as pain are called **nociceptors**. They consist of free nerve endings located throughout the body, especially near surfaces where damage is most likely to occur. Different nociceptors may respond to extremes in temperature, very intense mechanical stimulation, or specific chemicals in the extracellular fluid, including some that are released by injured cells. The thresholds of these sensory cells vary; some nociceptors are sensitive only to actual tissue damage, while others respond before damage has occurred.

Several types of mechanoreceptors are present in the skin, some in the dermis and others in the underlying subcutaneous tissue (figure 55.4). Morphologically specialized receptors that respond to fine touch are most concentrated on areas such as the fingertips and face. They are used to localize cutaneous stimuli very precisely and can be either phasic (intermittently activated) or tonic (continuously activated). The phasic receptors include *hair follicle receptors* and *Meissner's corpuscles*, which are present on body surfaces that do not contain hair, such as the fingers, palms, and nipples. The tonic receptors consist of *Ruffini endings* in the dermis and *touch dome endings* (*Merkel cells*) located near the surface of the skin. These receptors monitor the duration of a touch and the extent to which it is applied.

Deep below the skin in the subcutaneous tissue lie phasic, pressure-sensitive receptors called **Pacinian corpuscles**. Each of these receptors consists of the end of an afferent axon, surrounded by a capsule of alternating layers of connective tissue cells and extracellular fluid. When sustained pressure is applied to the corpuscle, the elastic capsule absorbs much of the pressure and the axon ceases to produce impulses. Pacinian corpuscles thus monitor only the onset and removal of pressure, as may occur repeatedly when something that vibrates is placed against the skin.

Different cutaneous receptors respond to touch, pressure, heat, cold, and pain. Some of these receptors are naked dendrites of sensory neurons, while others have supporting cells that modify the activities of their sensory dendrites.

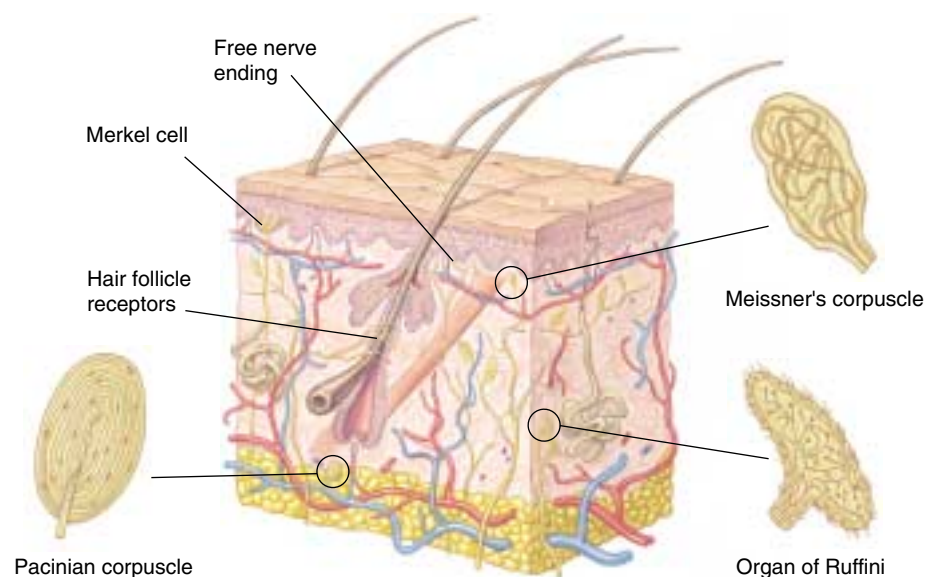


FIGURE 55.4

Sensory receptors in human skin.

Cutaneous receptors may be free nerve endings or sensory dendrites in association with other supporting structures.

Sensing Muscle Contraction and Blood Pressure

Mechanoreceptors contain sensory cells with ion channels that are sensitive to a mechanical force applied to the membrane. These channels open in response to mechanical distortion of the membrane, initiating a depolarization (receptor potential) that causes the sensory neuron to generate action potentials.

Muscle Length and Tension

Buried within the skeletal muscles of all vertebrates except the bony fishes are muscle spindles, sensory stretch receptors that lie in parallel with the rest of the fibers in the muscle (figure 55.5). Each spindle consists of several thin muscle fibers wrapped together and innervated by a sensory neuron, which becomes activated when the muscle, and therefore the spindle, is stretched. Muscle spindles, together with other receptors in tendons and joints, are known as **proprioceptors**, which are sensory receptors that provide information about the relative position or movement of the animal's body parts. The sensory neurons conduct action potentials into the spinal cord, where they synapse with somatic motor neurons that innervate the muscle. This pathway constitutes the muscle stretch reflex, including the knee-jerk reflex, previously discussed in chapter 54.

When a muscle contracts, it exerts tension on the tendons attached to it. The Golgi tendon organs, another type

of proprioceptor, monitor this tension; if it becomes too high, they elicit a reflex that inhibits the motor neurons innervating the muscle. This reflex helps to ensure that muscles do not contract so strongly that they damage the tendons to which they are attached.

Blood Pressure

Blood pressure is monitored at two main sites in the body. One is the *carotid sinus*, an enlargement of the left and right internal carotid arteries, which supply blood to the brain. The other is the *aortic arch*, the portion of the aorta very close to its emergence from the heart. The walls of the blood vessels at both sites contain a highly branched network of afferent neurons called **baroreceptors**, which detect tension in the walls. When the blood pressure decreases, the frequency of impulses produced by the baroreceptors decreases. The CNS responds to this reduced input by stimulating the sympathetic division of the autonomic nervous system, causing an increase in heart rate and vasoconstriction. Both effects help to raise the blood pressure, thus maintaining homeostasis. A rise in blood pressure, conversely, reduces sympathetic activity and stimulates the parasympathetic division, slowing the heart and lowering the blood pressure.

Mechanical distortion of the plasma membrane of mechanoreceptors produces nerve impulses that serve to monitor muscle length from skeletal muscle spindles and to monitor blood pressure from baroreceptors within arteries.

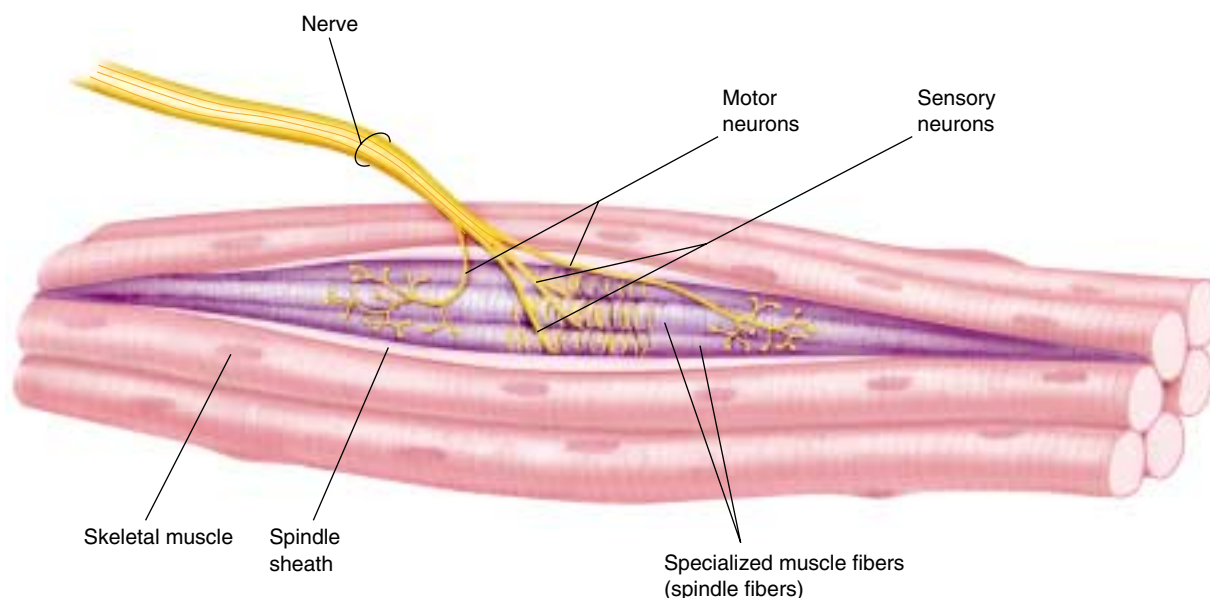


FIGURE 55.5

A muscle spindle is a stretch receptor embedded within skeletal muscle. Stretching of the muscle elongates the spindle fibers and stimulates the sensory dendritic endings wrapped around them. This causes the sensory neurons to send impulses to the CNS, where they synapse with motor neurons.

Sensing Taste, Smell, and Body Position

Some sensory cells, called **chemoreceptors**, contain membrane proteins that can bind to particular chemicals in the extracellular fluid. In response to this chemical interaction, the membrane of the sensory neuron becomes depolarized, leading to the production of action potentials. Chemoreceptors are used in the senses of taste and smell and are also important in monitoring the chemical composition of the blood and cerebrospinal fluid.

Taste

Taste buds—collections of chemosensitive epithelial cells associated with afferent neurons—mediate the sense of taste in vertebrates. In a fish, the taste buds are scattered over the surface of the body. These are the most sensitive vertebrate chemoreceptors known. They are particularly sensitive to amino acids; a catfish, for example, can distinguish between two different amino acids at a concentration of less than 100 parts per billion (1 g in 10,000 L of water)! The ability to taste the surrounding water is very important to bottom-feeding fish, enabling them to sense the presence of food in an often murky environment.

The taste buds of all terrestrial vertebrates are located in the epithelium of the tongue and oral cavity, within raised areas called *papillae* (figure 55.6). Humans have four kinds of taste buds—salty, sweet, sour, and bitter. The salty taste is produced by the effects of sodium (Na^+) and the sour taste by the effects of hydrogen (H^+). Organic molecules that produce the sweet and bitter tastes, such as sugars and quinine, respectively, are varied in structure. Taste buds that respond best to specific tastes are concentrated in specific regions of the tongue: sweet at the tip, sour at the sides, bitter at the back, and salty over most of the tongue's surface. Our complex perception of taste is the result of different combinations of impulses in the sensory neurons from these four kinds of taste buds, together with information related to smell. The effect of smell on the sense of taste can easily be demonstrated by eating an onion with the nose open and then eating it with the nose plugged.

Like vertebrates, many arthropods also have taste chemoreceptors. For example, flies, because of their mode of searching for food, have taste receptors in sensory hairs located on their feet. The sensory hairs contain different chemoreceptors that are able to detect sugars, salts, and

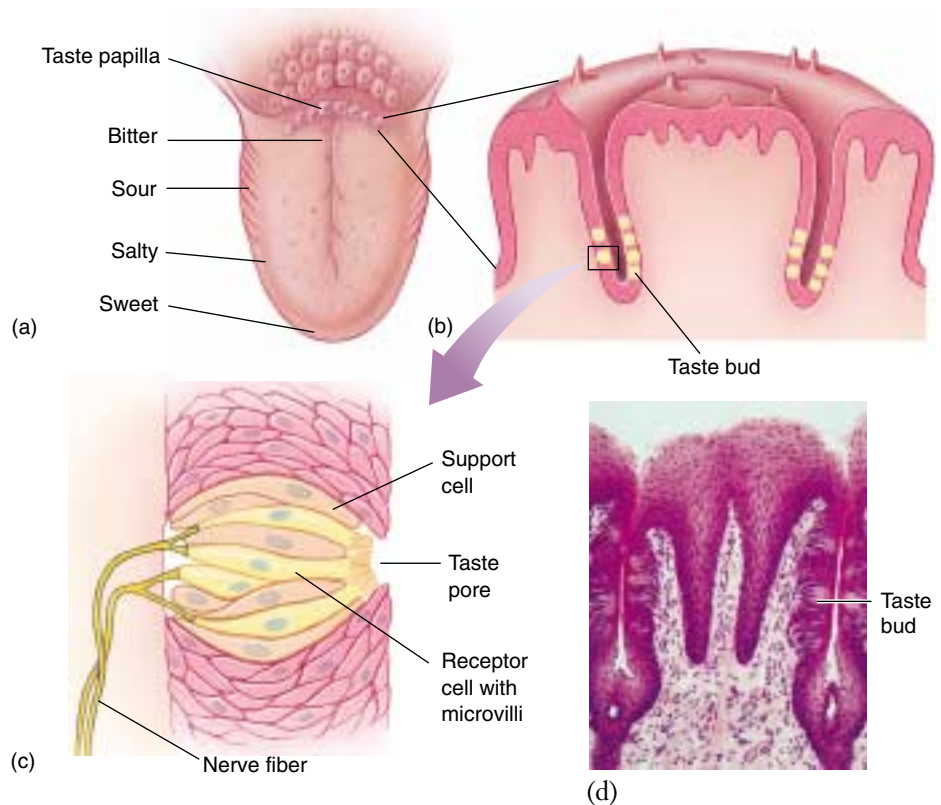


FIGURE 55.6

Taste. (a) Human beings have four kinds of taste buds (bitter, sour, salty, and sweet), located on different regions of the tongue. (b) Groups of taste buds are typically organized in sensory projections called papillae. (c) Individual taste buds are bulb-shaped collections of chemosensitive receptors that open out into the mouth through a pore. (d) Photomicrograph of taste buds in papillae.

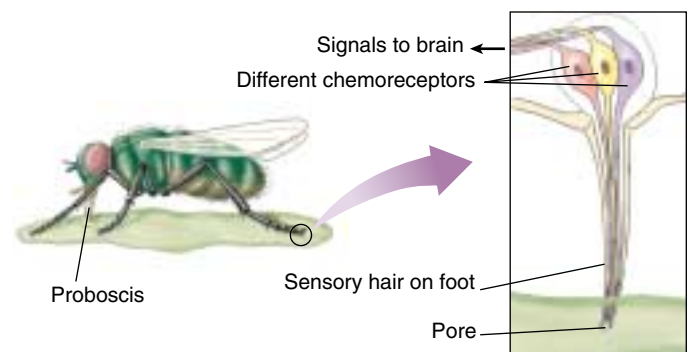


FIGURE 55.7

Many insects taste with their feet. In the blowfly shown here, chemoreceptors extend into the sensory hairs on the foot. Each different chemoreceptor detects a different type of food molecule. When the fly steps in a food substance, it can taste the different food molecules and extend its proboscis for feeding.

other molecules (figure 55.7). They can detect a wide variety of tastes by the integration of stimuli from these chemoreceptors. If they step on potential food, the proboscis (the tubular feeding apparatus) extends to feed.

Smell

In terrestrial vertebrates, the sense of smell, or olfaction, involves chemoreceptors located in the upper portion of the nasal passages (figure 55.8). These receptors are bipolar neurons whose dendrites end in tassels of cilia that project into the nasal mucosa, and whose axon projects directly into the cerebral cortex. A terrestrial vertebrate uses its sense of smell in much the same way that a fish uses its sense of taste—to sample the chemical environment around it. Because terrestrial vertebrates are surrounded by air rather than water, their sense of smell has become specialized to detect airborne particles (but these particles must first dissolve in extracellular fluid before they can activate the olfactory receptors). The sense of smell can be extremely acute in many mammals, so much so that a single odorant molecule may be all that is needed to excite a given receptor.

Although humans can detect only four modalities of taste, they can discern thousands of different smells. New research suggests that there may be as many as a thousand different genes coding for different receptor proteins for smell. The particular set of olfactory neurons that respond to a given odor might serve as a “fingerprint” the brain can use to identify the odor.

Internal Chemoreceptors

Sensory receptors within the body detect a variety of chemical characteristics of the blood or fluids derived from the blood, including cerebrospinal fluid. Included among these receptors are the *peripheral chemoreceptors* of the aortic and carotid bodies, which are sensitive primarily to plasma pH, and the *central chemoreceptors* in the medulla oblongata of the brain, which are sensitive to the pH of cerebrospinal fluid. These receptors were discussed together with the regulation of breathing in chapter 53. When the breathing rate is too low, the concentration of plasma CO_2 increases, producing more carbonic acid and causing a fall in the blood pH. The carbon dioxide can also enter the cerebrospinal fluid and cause a lowering of the pH, thereby stimulating the central chemoreceptors. This chemoreceptor stimulation indirectly affects the respiratory control center of the brain stem, which increases the breathing rate. The aortic bodies can also respond to a lowering of blood oxygen concentrations, but this effect is normally not significant unless a person goes to a high altitude.

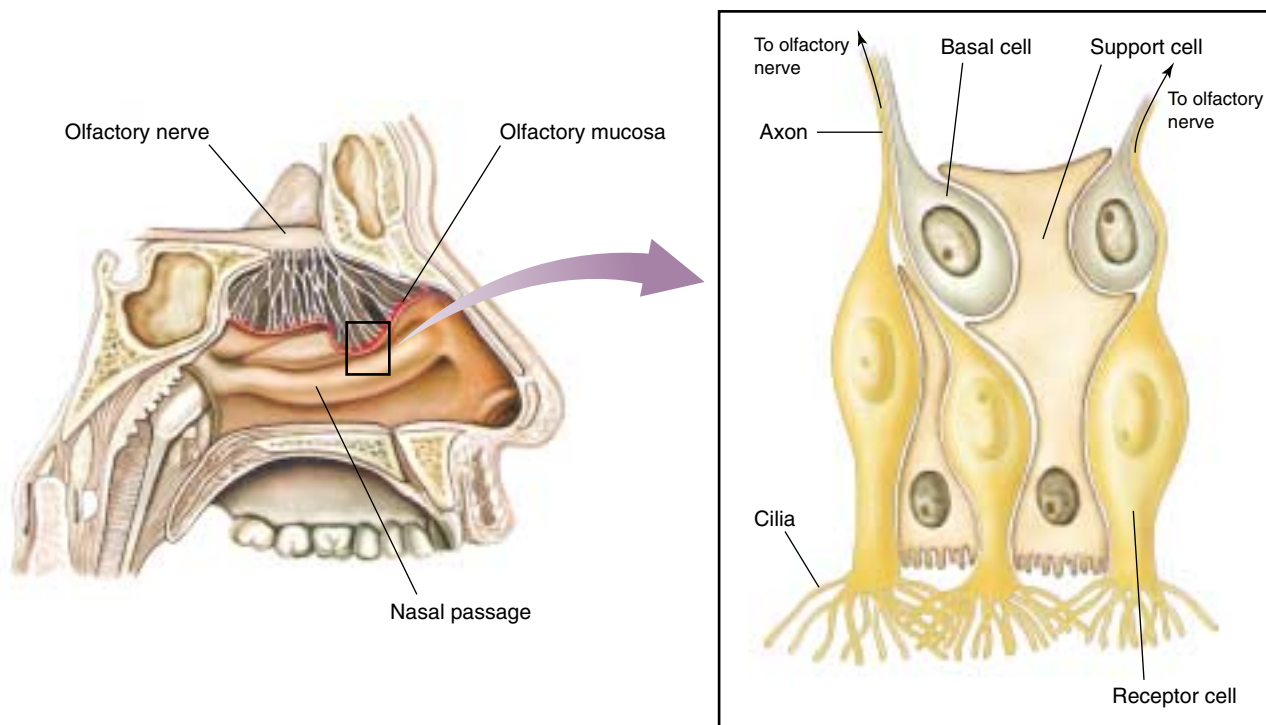


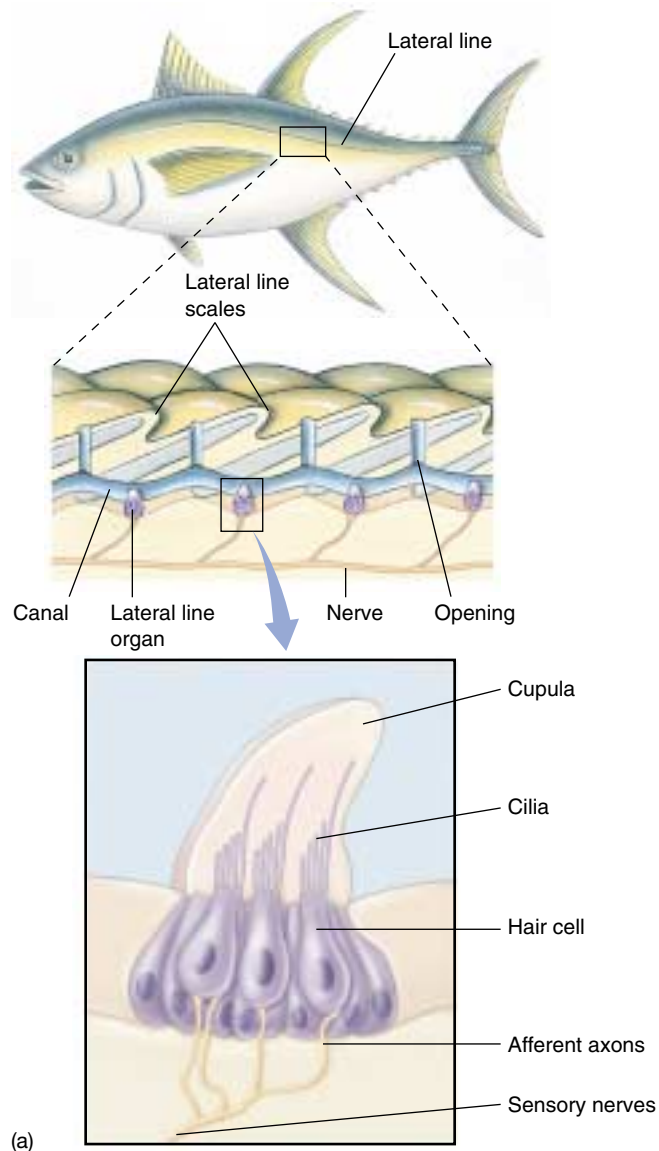
FIGURE 55.8

Smell. Humans detect smells by means of olfactory neurons located in the lining of the nasal passages. The axons of these neurons transmit impulses directly to the brain via the olfactory nerve. Basal cells regenerate new olfactory neurons to replace dead or damaged cells. Olfactory neurons typically live about one month.

The Lateral Line System

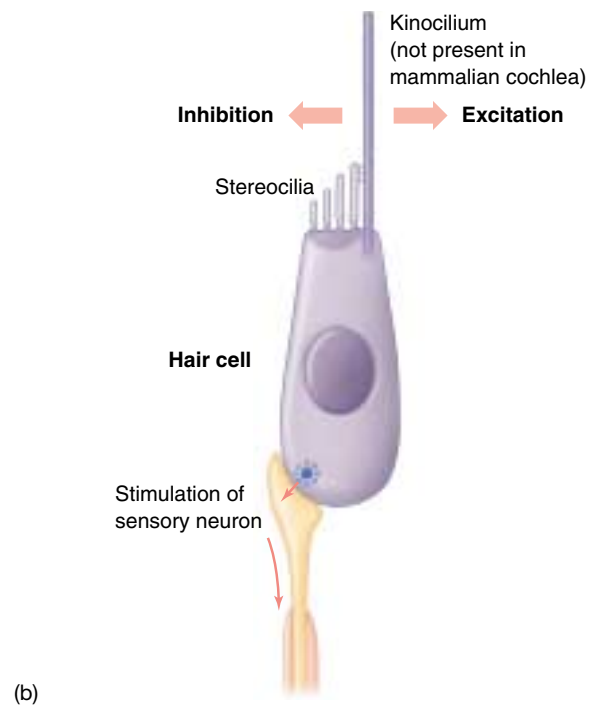
The lateral line system provides fish with a sense of “distant touch,” enabling them to sense objects that reflect pressure waves and low-frequency vibrations. This enables a fish to detect prey, for example, and to swim in synchrony with the rest of its school. It also enables a blind cave fish to sense its environment by monitoring changes in the patterns of water flow past the lateral line receptors. The lateral line system is found in amphibian larvae, but is lost at metamorphosis and is not present in any terrestrial vertebrate. The sense provided by the lateral line system supplements the fish’s sense of hearing, which is performed by a different sensory structure. The structures and mechanisms involved in hearing will be described in a later section.

The lateral line system consists of sensory structures within a longitudinal canal in the fish’s skin that extends



along each side of the body and within several canals in the head (figure 55.9a). The sensory structures are known as hair cells because they have hairlike processes at their surface that project into a gelatinous membrane called a *cupula* (Latin, “little cup”). The hair cells are innervated by sensory neurons that transmit impulses to the brain.

Hair cells have several hairlike processes of approximately the same length, called *stereocilia*, and one longer process called a *kinocilium* (figure 55.9b). Vibrations carried through the fish’s environment produce movements of the cupula, which cause the hairs to bend. When the stereocilia bend in the direction of the kinocilium, the associated sensory neurons are stimulated and generate a receptor potential. As a result, the frequency of action potentials produced by the sensory neuron is increased. If the stereocilia are bent in the opposite direction, on the other hand, the activity of the sensory neuron is inhibited.



(b)

FIGURE 55.9

The lateral line system. (a) This system consists of canals running the length of the fish’s body beneath the surface of the skin. Within these canals are sensory structures containing hair cells with cilia that project into a gelatinous cupula. Pressure waves traveling through the water in the canals deflect the cilia and depolarize the sensory neurons associated with the hair cells. (b) Hair cells are mechanoreceptors with hairlike cilia that project into a gelatinous membrane. The hair cells of the lateral line system (and the membranous labyrinth of the vertebrate inner ear) have a number of smaller cilia called stereocilia and one larger kinocilium. When the cilia bend in the direction of the kinocilium, the hair cell releases a chemical transmitter that depolarizes the associated sensory neuron. Bending of the cilia in the opposite direction has an inhibitory effect.

Gravity and Angular Acceleration

Most invertebrates can orient themselves with respect to gravity due to a sensory structure called a *statocyst*. Statocysts generally consist of ciliated hair cells with the cilia embedded in a gelatinous membrane containing crystals of calcium carbonate. These “stones,” or *statoliths*, increase the mass of the gelatinous membrane so that it can bend the cilia when the animal’s position changes. If the animal tilts to the right, for example, the statolith membrane will bend the cilia on the right side and activate associated sensory neurons.

A similar structure is found in the membranous labyrinth of the inner ear of vertebrates. The labyrinth is a system of fluid-filled membranous chambers and tubes that constitute the organs of equilibrium and hearing in vertebrates. This membranous labyrinth is surrounded by bone and perilymph, which is similar in ionic content to interstitial fluid. Inside, the chambers and tubes are filled with endolymph fluid, which is similar in ionic content to intracellular fluid. Though intricate, the entire structure is very small; in a human, it is about the size of a pea.

The receptors for gravity in vertebrates consist of two chambers of the membranous labyrinth called the utricle and saccule (figure 55.10). Within these structures are hair cells with stereocilia and a kinocilium, similar to those in the lateral line system of fish. The hairlike processes are embedded within a gelatinous membrane containing calcium carbonate crystals; this is known as an *otolith membrane*, because of its location in the inner ear (*oto* is derived from the Greek word for ear). Because the otolith organ is oriented differently in the utricle and saccule, the utricle is more sensi-

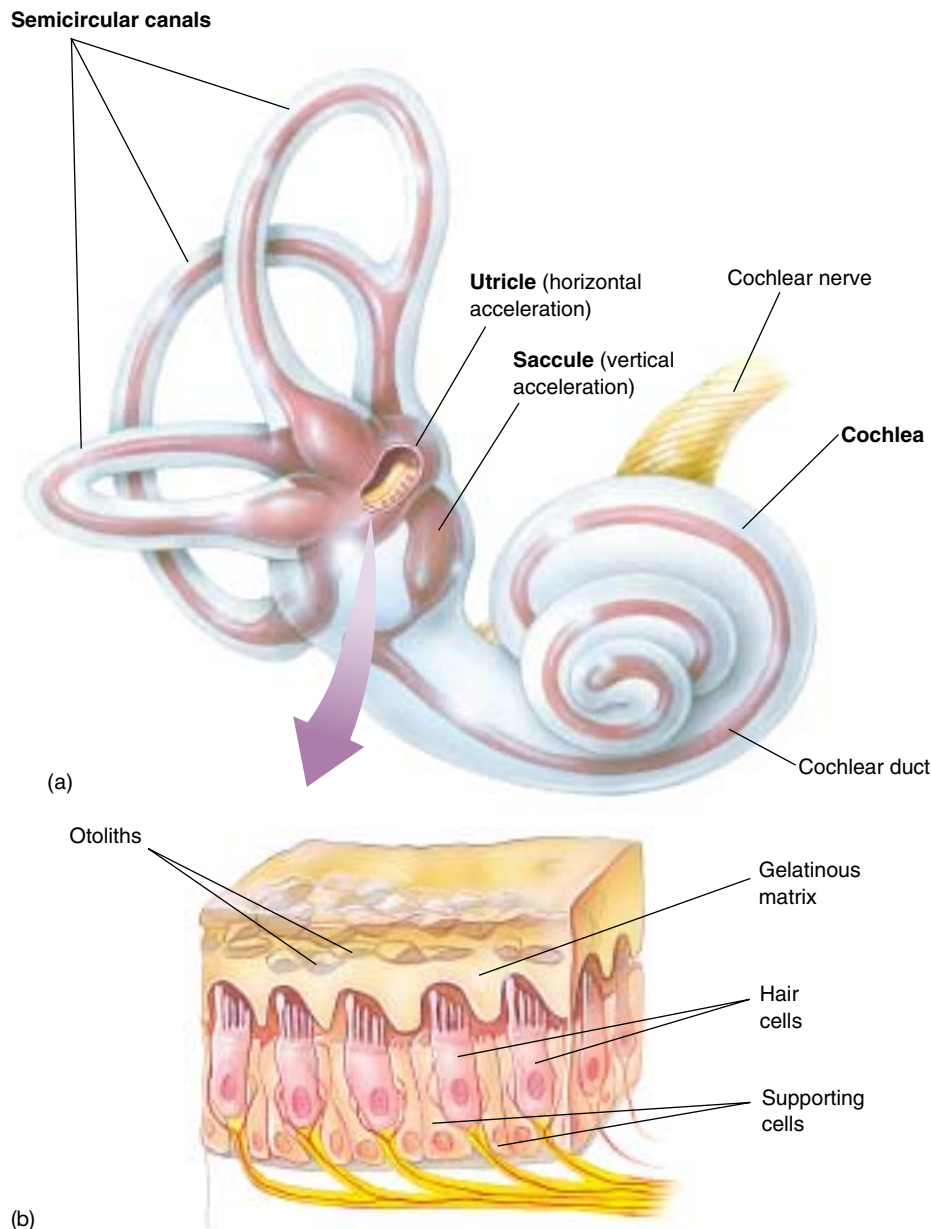


FIGURE 55.10

The structure of the utricle and saccule. (a) The relative positions of the utricle and saccule within the membranous labyrinth of the human inner ear. (b) Enlargement of a section of the utricle or saccule showing the otoliths embedded in the gelatinous matrix that covers the hair cells.

tive to horizontal acceleration (as in a moving car) and the saccule to vertical acceleration (as in an elevator). In both cases, the acceleration causes the stereocilia to bend and consequently produces action potentials in an associated sensory neuron.

The membranous labyrinth of the utricle and saccule is continuous with three semicircular canals, oriented in different planes so that angular acceleration in any direction can be detected (figure 55.11). At the ends of the canals are swollen chambers called *ampullae*, into which protrude the cilia of another group of hair cells. The tips of the cilia are embedded within a sail-like wedge of gelatinous material called a *cupula* (similar to the cupula of the fish lateral line system) that protrudes into the endolymph fluid of each semicircular canal.

When the head rotates, the fluid inside the semicircular canals pushes against the cupula and causes the cilia to bend. This bending either depolarizes or hyperpolarizes the hair cells, depending on the direction in which the cilia are bent. This is similar to the way the lateral line system

works in a fish: if the stereocilia are bent in the direction of the kinocilium, a depolarization (receptor potential) is produced, which stimulates the production of action potentials in associated sensory neurons.

The saccule, utricle, and semicircular canals are collectively referred to as the *vestibular apparatus*. While the saccule and utricle provide a sense of linear acceleration, the semicircular canals provide a sense of angular acceleration. The brain uses information that comes from the vestibular apparatus about the body's position in space to maintain balance and equilibrium.

Receptors that sense chemicals originating outside the body are responsible for the senses of odor, smell, and taste. Internal chemoreceptors help to monitor chemicals produced within the body and are needed for the regulation of breathing. Hair cells in the lateral line organ of fishes detect water movements, and hair cells in the vestibular apparatus of terrestrial vertebrates provide a sense of acceleration.

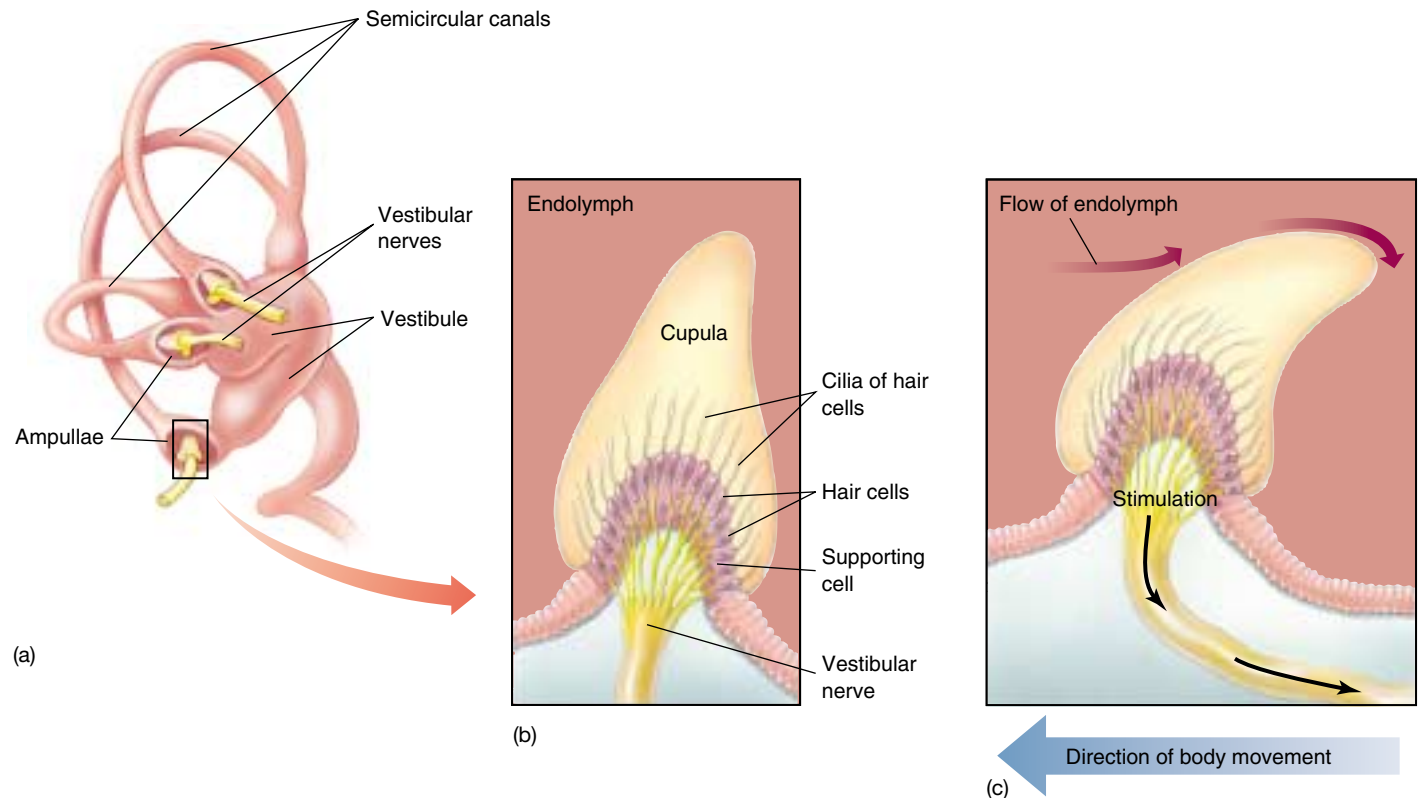


FIGURE 55.11

The structure of the semicircular canals. (a) The position of the semicircular canals in relation to the rest of the inner ear.

(b) Enlargement of a section of one ampulla, showing how hair cell cilia insert into the cupula. (c) Angular acceleration in the plane of the semicircular canal causes bending of the cupula, thereby stimulating the hair cells.

55.3 Auditory receptors detect pressure waves in the air.

The Ears and Hearing

Fish detect vibrational pressure waves in water by means of their lateral line system. Terrestrial vertebrates detect similar vibrational pressure waves in air by means of similar hair cell mechanoreceptors in the inner ear. Hearing actually works better in water than in air because water transmits pressure waves more efficiently. Despite this limitation, hearing is widely used by terrestrial vertebrates to monitor their environments, communicate with other members of the same species, and to detect possible sources of danger (figure 55.12). Auditory stimuli travel farther and more quickly than chemical ones, and auditory receptors provide better directional information than do chemoreceptors. Auditory stimuli alone, however, provide little information about distance.

Structure of the Ear

Fish use their lateral line system to detect water movements and vibrations emanating from relatively nearby objects, and their hearing system to detect vibrations that originate from a greater distance. The hearing system of fish consists of the otolith organs in the membranous labyrinth (utricle and saccule) previously described, together with a very small outpouching of the membranous labyrinth called the lagena. Sound waves travel through the body of the fish as easily as through the surrounding water, as the body is composed primarily of water. Therefore, an object of different density is needed in order for the sound to be detected. This function is served by the otolith (calcium carbonate crystals) in many fish. In catfish, minnows, and suckers, however, this function is served by an air-filled swim bladder that vibrates with the sound. A chain of small bones, Weberian ossicles, then transmits the vibrations to the saccule in some of these fish.

In the ears of terrestrial vertebrates, vibrations in air may be channeled through an ear canal to the eardrum, or tympanic membrane. These structures are part of the *outer ear*. Vibrations of the tympanic membrane cause movement of three small bones (ossicles)—the *malleus* (hammer), *incus* (anvil), and *stapes* (stirrup)—that are located in a bony cavity known as the *middle ear* (figure 55.13). These middle ear ossicles are analogous to the Weberian ossicles in fish. The



FIGURE 55.12

Kangaroo rats have specialized ears. Kangaroo rats (*Dipodomys*) are unique in having an enlarged tympanic membrane (eardrum), a lengthened and freely rotating malleus (ear bone), and an increased volume of air-filled chambers in the middle ear. These and other specializations result in increased sensitivity to sound, especially to low-frequency sounds. Experiments have shown that the kangaroo rat's ears are adapted to nocturnal life and allow them to hear the low-frequency sounds of their predators, such as an owl's wingbeats or a sidewinder rattlesnake's scales rubbing against the ground. Also, the ears seem to be adapted to the poor sound-carrying quality of dry, desert air.

middle ear is connected to the throat by the *Eustachian tube*, which equalizes the air pressure between the middle ear and the external environment. The “ear popping” you may have experienced when flying in an airplane or driving on a mountain is caused by pressure equalization between the two sides of the eardrum.

The stapes vibrates against a flexible membrane, the *oval window*, which leads into the *inner ear*. Because the oval window is smaller in diameter than the tympanic membrane, vibrations against it produce more force per unit area, transmitted into the inner ear. The inner ear consists of the cochlea (Latin for “snail”), a bony structure containing part of the membranous labyrinth called the cochlear duct. The cochlear duct is located in the center of the cochlea; the area above the cochlear duct is the *vestibular canal*, and the area below is the *tympanic canal*. All three chambers are filled with fluid, as previously described. The oval window opens to the upper vestibular canal, so that when the stapes causes it to vibrate, it produces pressure waves of fluid. These pressure waves travel down to the tympanic canal, pushing another flexible membrane, the *round window*, that transmits the pressure back into the middle ear cavity (see figure 55.13).

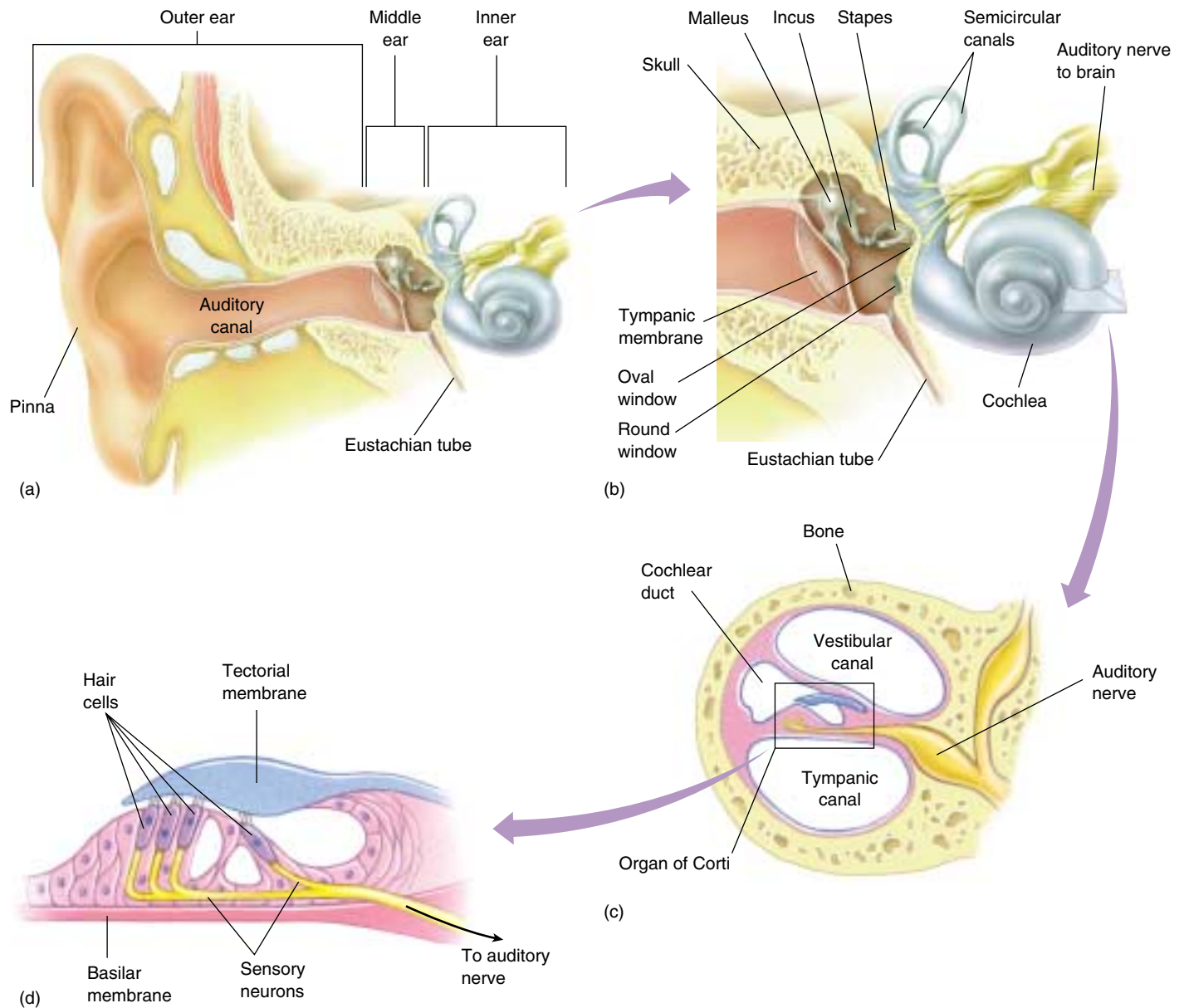


FIGURE 55.13

Structure of the human ear. (a) Sound waves passing through the ear canal produce vibrations of the tympanic membrane, which cause movement of the (b) middle ear ossicles (the malleus, incus, and stapes) against an inner membrane called the oval window. Vibration of the oval window sets up pressure waves that (c and d) travel through the fluid in the vestibular and tympanic canals of the cochlea.

Transduction in the Cochlea

As the pressure waves produced by vibrations of the oval window are transmitted through the cochlea to the round window, they cause the cochlear duct to vibrate. The bottom of the cochlear duct, called the *basilar membrane*, is quite flexible and vibrates in response to these pressure waves. The surface of the basilar membrane contains sensory hair cells, similar to those of the vestibular apparatus and lateral line system but lacking a kinocilium. The cilia from the hair cells project into an overhanging gelatinous membrane, the *tectorial membrane*. This sensory apparatus,

consisting of the basilar membrane, hair cells with associated sensory neurons, and tectorial membrane, is known as the **organ of Corti**.

As the basilar membrane vibrates, the cilia of the hair cells bend in response to the movement of the basilar membrane relative to the tectorial membrane. As in the lateral line organs and the vestibular apparatus, the bending of these cilia depolarizes the hair cells. The hair cells, in turn, stimulate the production of action potentials in sensory neurons that project to the brain, where they are interpreted as sound.

Frequency Localization in the Cochlea

The basilar membrane of the cochlea consists of elastic fibers of varying length and stiffness, like the strings of a musical instrument, embedded in a gelatinous material. At the base of the cochlea (near the oval window), the fibers of the basilar membrane are short and stiff. At the far end of the cochlea (the apex), the fibers are 5 times longer and 100 times more flexible. Therefore, the resonant frequency of the basilar membrane is higher at the base than the apex; the base responds to higher pitches, the apex to lower.

When a wave of sound energy enters the cochlea from the oval window, it initiates a traveling up-and-down motion of the basilar membrane. However, this wave imparts most of its energy to that part of the basilar membrane with a resonant frequency near the frequency of the sound wave, resulting in a maximum deflection of the basilar membrane at that point (figure 55.14). As a result, the hair cell depolarization is greatest in that region, and the afferent axons from that region are stimulated to produce action potentials more than those from other regions. When these action potentials arrive in the brain, they are interpreted as representing a sound of that particular frequency, or pitch.

The flexibility of the basilar membrane limits the frequency range of human hearing to between approximately 20 and 20,000 cycles per second (hertz) in children. Our ability to hear high-pitched sounds decays progressively throughout middle age. Other vertebrates can detect sounds at frequencies lower than 20 hertz and much higher than 20,000 hertz. Dogs, for example, can detect sounds at 40,000 hertz, enabling them to hear high-pitched dog whistles that seem silent to a human listener.

Hair cells are also innervated by efferent axons from the brain, and impulses in those axons can make hair cells less sensitive. This central control of receptor sensitivity can increase an individual's ability to concentrate on a particular auditory signal (for example, a single voice) in the midst of background noise, which is effectively "tuned out" by the efferent axons.

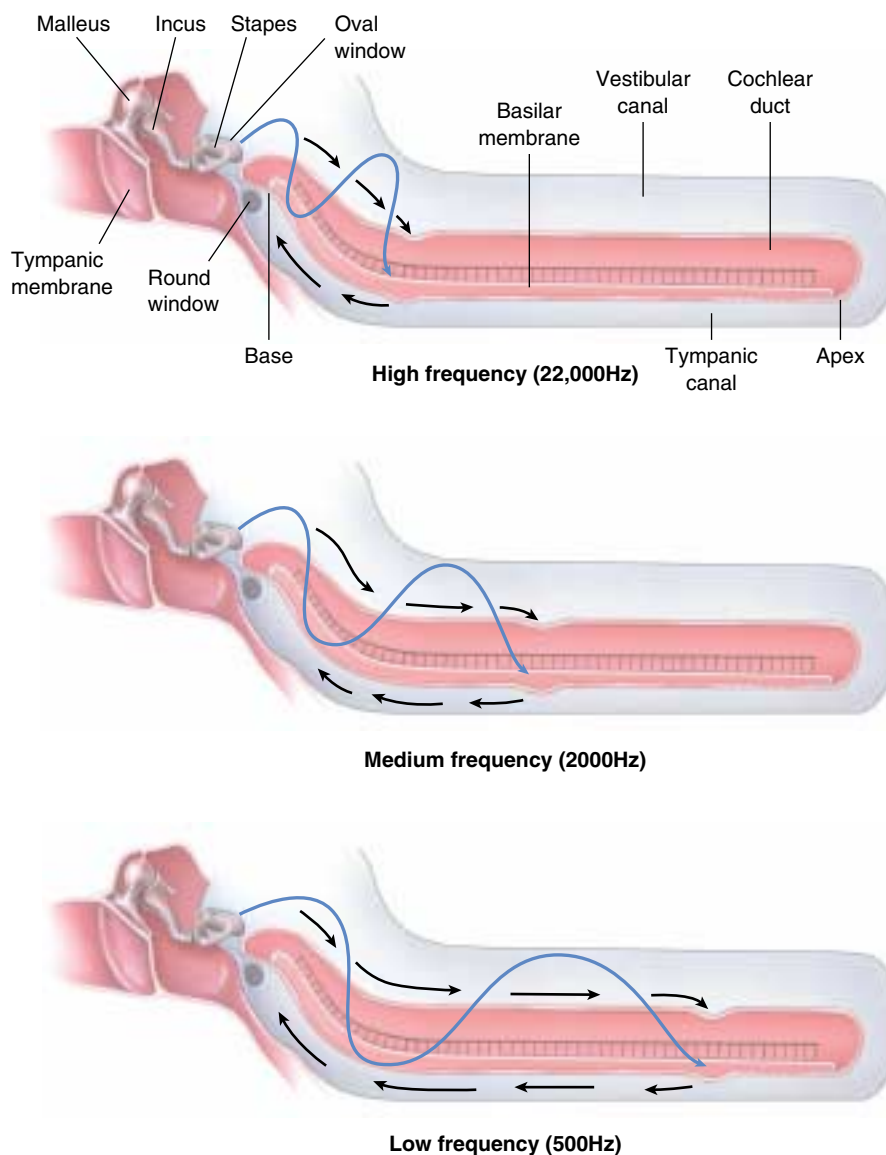


FIGURE 55.14

Frequency localization in the cochlea. The cochlea is shown unwound, so that the length of the basilar membrane can be seen. The fibers within the basilar membrane vibrate in response to different frequencies of sound, related to the pitch of the sound. Thus, regions of the basilar membrane show maximum vibrations in response to different sound frequencies. Notice that low-frequency (pitch) sounds vibrate the basilar membrane more toward the apex, while high frequencies cause vibrations more toward the base.

The middle ear ossicles vibrate in response to sound waves, creating fluid vibrations within the inner ear. This causes the hair cells to bend, transducing the sound into action potentials. The pitch of a sound is determined by which hair cells (and thus which sensory neurons) are activated by the vibration of the basilar membrane.

Sonar

Because terrestrial vertebrates have two ears located on opposite sides of the head, the information provided by hearing can be used by the CNS to determine *direction* of a sound source with some precision. Sound sources vary in strength, however, and sounds are attenuated (weakened) to varying degrees by the presence of objects in the environment. For these reasons, auditory sensors do not provide a reliable measure of *distance*.

A few groups of mammals that live and obtain their food in dark environments have circumvented the limitations of darkness. A bat flying in a completely dark room easily avoids objects that are placed in its path—even a wire less than a millimeter in diameter (figure 55.15). Shrews use a similar form of “lightless vision” beneath the ground, as do whales and dolphins beneath the sea. All of these mammals perceive distance by means of sonar. They emit sounds and then determine the time it takes these sounds to reach an

object and return to the animal. This process is called echolocation. A bat, for example, produces clicks that last 2 to 3 milliseconds and are repeated several hundred times per second. The three-dimensional imaging achieved with such an auditory sonar system is quite sophisticated.

Being able to “see in the dark” has opened a new ecological niche to bats, one largely closed to birds because birds must rely on vision. There are no truly nocturnal birds; even owls rely on vision to hunt, and do not fly on dark nights. Because bats are able to be active and efficient in total darkness, they are one of the most numerous and widespread of all orders of mammals.

Some mammals emit sounds and then determine the time it takes for the sound to return, using the method of sonar to locate themselves and other objects in a totally dark environment by the characteristics of the echo. Bats are the most adept at this echolocation.



FIGURE 55.15

Sonar. As it flies, a bat emits high-frequency “chirps” and listens for the return of the chirps after they are reflected by objects such as moths. By timing how long it takes for a chirp to return, the bat can locate its prey and catch it even in total darkness.

55.4 Optic receptors detect light over a broad range of wavelengths.

Evolution of the Eye

Vision begins with the capture of light energy by photoreceptors. Because light travels in a straight line and arrives virtually instantaneously, visual information can be used to determine both the direction and the distance of an object. No other stimulus provides as much detailed information.

Many invertebrates have simple visual systems with photoreceptors clustered in an eyespot. Simple eyespots can be made sensitive to the direction of a light source by the addition of a pigment layer which shades one side of the eye. Flatworms have a screening pigmented layer on the inner and back sides of both eyespots allowing stimulation of the photoreceptor cells only by light from the front of the animal (figure 55.16). The flatworm will turn and swim in the direction in which the photoreceptor cells are the least stimulated. Although an eyespot can perceive the direction of light, it cannot be used to construct a visual image. The members of four phyla—annelids, mollusks, arthropods, and chordates—have evolved well-developed, image-forming eyes. True image-forming eyes in these phyla, though strikingly similar in structure, are believed to have evolved independently (figure 55.17). Interestingly, the photoreceptors in all of them use the same light-capturing molecule, suggesting that not many alternative molecules are able to play this role.

Structure of the Vertebrate Eye

The eye of a human is typical of the vertebrate eye (figure 55.18). The “white of the eye” is the sclera, formed of tough connective tissue. Light enters the eye through a transparent cornea, which begins to focus the light. This

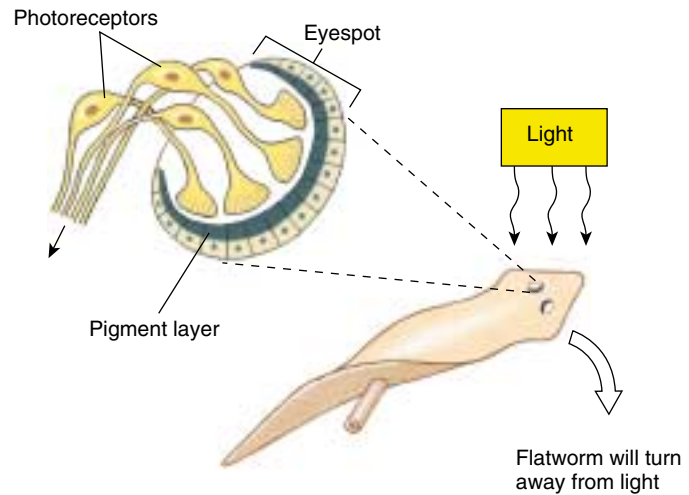


FIGURE 55.16
Simple eyespots in the flatworm. Eyespots will detect the direction of light because a pigmented layer on one side of the eyespot screens out light coming from the back of the animal. Light is thus the strongest coming from the front of the animal; flatworms will respond by turning away from the light.

occurs because light is refracted (bent) when it travels into a medium of different density. The colored portion of the eye is the iris; contraction of the iris muscles in bright light decreases the size of its opening, the pupil. Light passes through the pupil to the lens, a transparent structure that completes the focusing of the light onto the retina at the back of the eye. The lens is attached by the *suspensory ligament* to the ciliary muscles.

The shape of the lens is influenced by the amount of tension in the suspensory ligament, which surrounds the

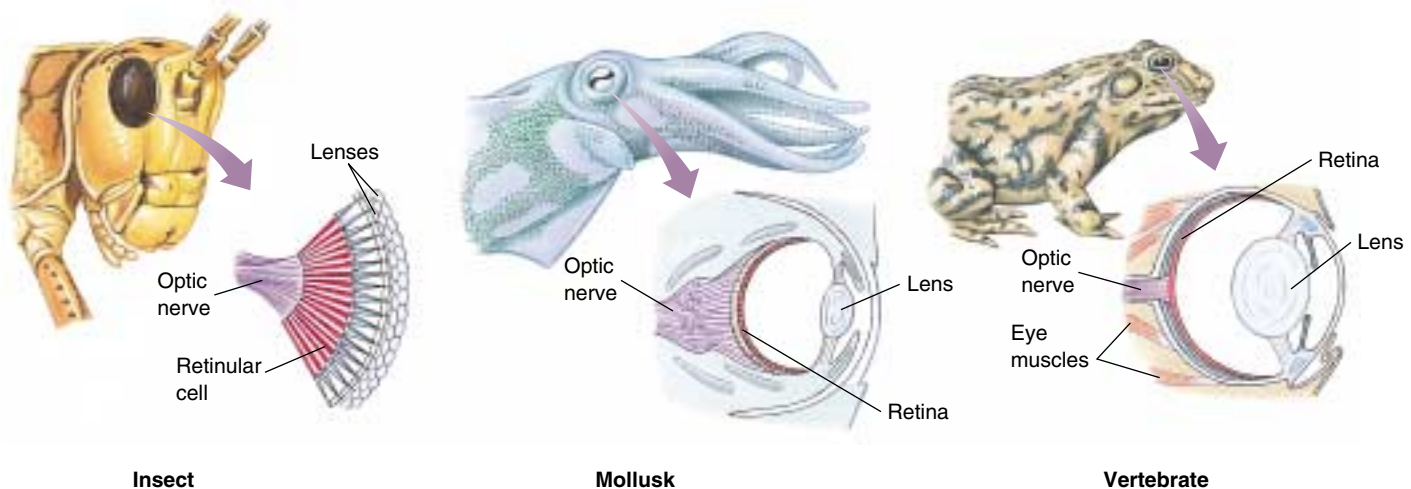


FIGURE 55.17
Eyes in three phyla of animals. Although they are superficially similar, these eyes differ greatly in structure and are not homologous. Each has evolved separately and, despite the apparent structural complexity, has done so from simpler structures.

lens and attaches it to the circular ciliary muscle. When the ciliary muscle contracts, it puts slack in the suspensory ligament and the lens becomes more rounded and powerful. This is required for close vision; in far vision, the ciliary muscles relax, moving away from the lens and tightening the suspensory ligament. The lens thus becomes more flattened and less powerful, keeping the image focused on the retina. People who are nearsighted or farsighted do not properly focus the image on the retina (figure 55.19). Interestingly, the lens of an amphibian or a fish does not change shape; these animals instead focus images by moving their lens in and out, just as you would do to focus a camera.

Annelids, mollusks, arthropods, and vertebrates have independently evolved image-forming eyes. The vertebrate eye admits light through a pupil and then focuses this light by means of an adjustable lens onto the retina at the back of the eye.

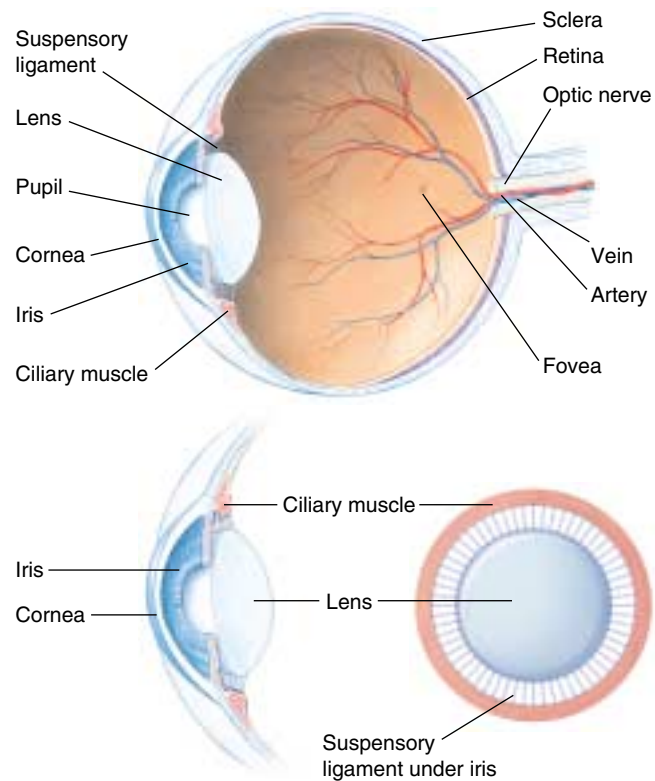


FIGURE 55.18

Structure of the human eye. The transparent cornea and lens focus light onto the retina at the back of the eye, which contains the rods and cones. The center of each eye's visual field is focused on the fovea. Focusing is accomplished by contraction and relaxation of the ciliary muscle, which adjusts the curvature of the lens.

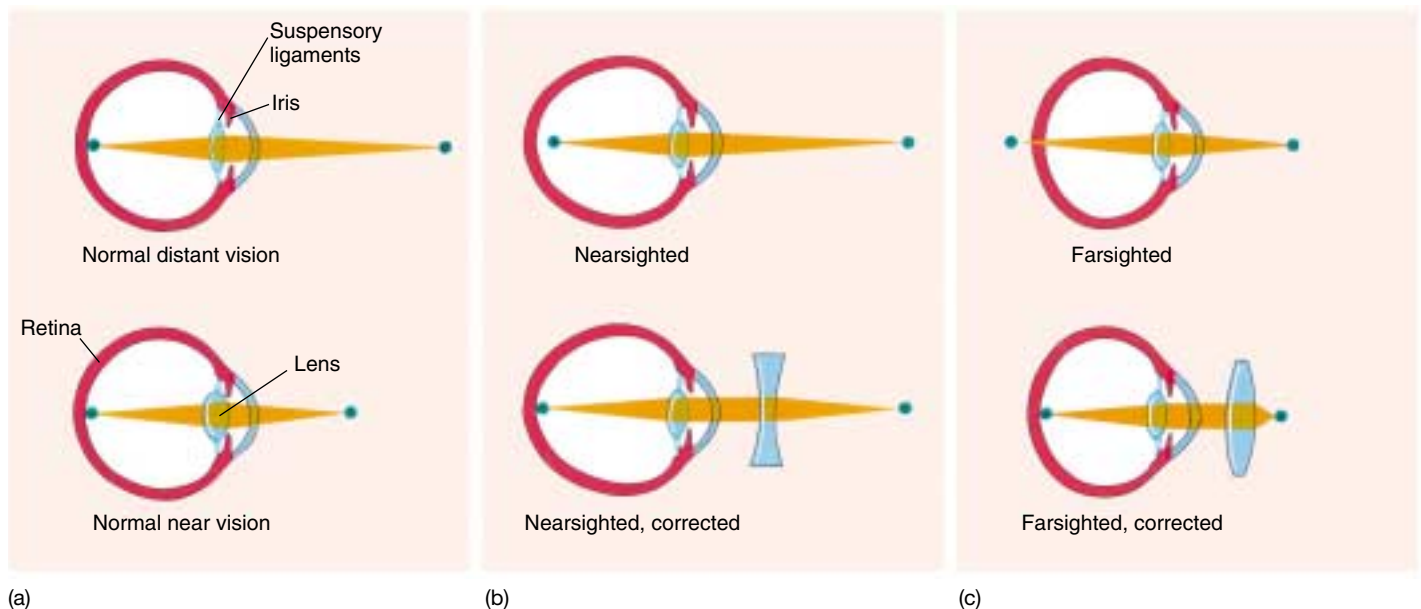


FIGURE 55.19

Focusing the human eye. (a) In people with normal vision, the image remains focused on the retina in both near and far vision because of changes produced in the curvature of the lens. When a person with normal vision stands 20 feet or more from an object, the lens is in its least convex form and the image is focused on the retina. (b) In nearsighted people, the image comes to a focus in front of the retina and the image thus appears blurred. (c) In farsighted people, the focus of the image would be behind the retina because the distance from the lens to the retina is too short.

Vertebrate Photoreceptors

The vertebrate retina contains two kinds of photoreceptors, called rods and cones (figure 55.20). Rods are responsible for black-and-white vision when the illumination is dim, while cones are responsible for high visual acuity (sharpness) and color vision. Humans have about 100 million rods and 3 million cones in each retina. Most of the cones are located in the central region of the retina known as the fovea, where the eye forms its sharpest image. Rods are almost completely absent from the fovea.

Rods and cones have the same basic cellular structure. An inner segment rich in mitochondria contains numerous vesicles filled with neurotransmitter molecules. It is connected by a narrow stalk to the outer segment, which is packed with hundreds of flattened discs stacked on top of one another. The light-capturing molecules, or photopigments, are located on the membranes of these discs.

In rods, the photopigment is called rhodopsin. It consists of the protein opsin bound to a molecule of *cis*-retinal (figure 55.21), which is derived from carotene, a photosynthetic pigment in plants. The photopigments of cones, called photopsins, are structurally very similar to rhodopsin. Humans have three kinds of cones, each of which possesses a photopsin consisting of *cis*-retinal bound to a protein with a slightly different amino acid sequence. These differences shift the *absorption maximum*—the region of the electromagnetic spectrum that is best absorbed by the pigment—(figure 55.22). The absorption maximum of the *cis*-retinal in rhodopsin is 500 nanometers (nm); the absorption maxima of the three kinds of cone photopsins, in contrast, are 455 nm (blue-absorbing), 530 nm (green-absorbing), and 625 nm (red-absorbing). These differences in the light-absorbing properties of the photopsins are responsible for the different color sensitivities of the three kinds of cones, which are often referred to as simply blue, green, and red cones.

Most vertebrates, particularly those that are diurnal (active during the day), have color vision, as do many insects. Indeed, honeybees can see light in the near-ultraviolet range, which is invisible to the human eye. Color vision requires the presence of more than one photopigment in different receptor cells, but not all animals with color vision have the three-cone system characteristic of humans and other primates. Fish, turtles, and birds, for example, have four or five kinds of cones; the “extra” cones enable these animals to see near-ultraviolet light. Many mammals (such as squirrels), on the other hand, have only two types of cones.

The retina is made up of three layers of cells (figure 55.23): the layer closest to the external surface of the eyeball consists of the rods and cones, the next layer contains bipolar cells, and the layer closest to the cavity of the eye is composed of ganglion cells. Thus, light must first pass through the ganglion cells and bipolar cells in order to reach the photoreceptors! The rods and cones synapse

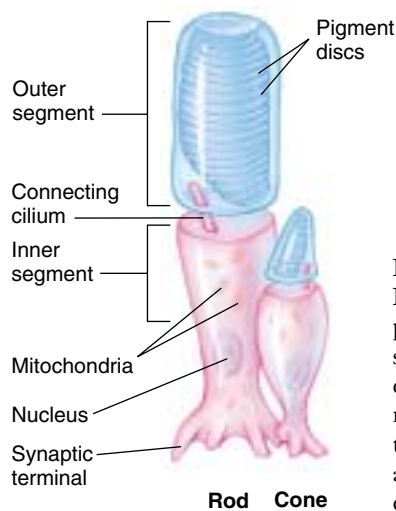


FIGURE 55.20
Rods and cones. The pigment-containing outer segment in each of these cells is separated from the rest of the cell by a partition through which there is only a narrow passage, the connective cilium.

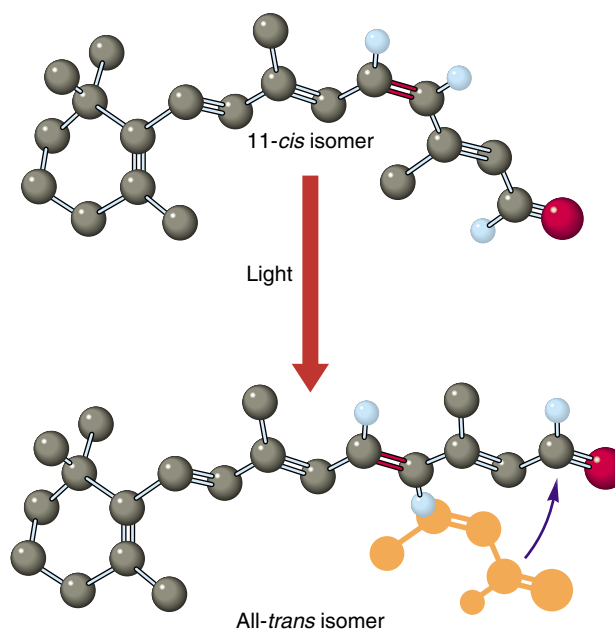


FIGURE 55.21
Absorption of light. When light is absorbed by a photopigment, the 11-*cis* isomer of retinal, the light-capturing portion of the pigment undergoes a change in shape: the linear end of the molecule (at the right in this diagram) rotates about a double bond (indicated here in red). The resulting isomer is referred to as all-*trans* retinal. This change in retinal's shape initiates a chain of events that leads to hyperpolarization of the photoreceptor.

with the bipolar cells, and the bipolar cells synapse with the ganglion cells, which transmit impulses to the brain via the optic nerve. The flow of sensory information in the retina is therefore opposite to the path of light through the retina. It should also be noted that the retina contains two additional types of neurons, horizontal cells and amacrine cells. Stimulation of horizontal cells by photoreceptors at

the center of a spot of light on the retina can inhibit the response of photoreceptors peripheral to the center. This lateral inhibition enhances contrast and sharpens the image.

Sensory Transduction in Photoreceptors

The transduction of light energy into nerve impulses follows a sequence that is the inverse of the usual way that sensory stimuli are detected. This is because, in the dark, the photoreceptors release an inhibitory neurotransmitter that hyperpolarizes the bipolar neurons. Thus inhibited, the bipolar neurons do not release excitatory neurotransmitter to the ganglion cells. Light *inhibits* the photoreceptors from releasing their inhibitory neurotransmitter, and by this means, *stimulates* the bipolar cells and thus the ganglion cells, which transmit action potentials to the brain.

A rod or cone contains many Na^+ channels in the plasma membrane of its outer segment, and in the dark, many of these channels are open. As a consequence, Na^+ continuously diffuses into the outer segment and across the narrow stalk to the inner segment. This flow of Na^+ that occurs in the absence of light is called the dark current, and it causes the membrane of a photoreceptor to be somewhat depolarized in the dark. In the light, the Na^+ channels in the outer segment rapidly close, reducing the dark current and causing the photoreceptor to hyperpolarize.

Researchers have discovered that cyclic guanosine monophosphate (cGMP) is required to keep the Na^+ channels open, and that the channels will close if the cGMP is converted into GMP. How does light cause this conversion and consequent closing of the Na^+ channels? When a photopigment absorbs light, *cis*-retinal isomerizes and dissociates from opsin in what is known as the bleaching reaction. As a result of this dissociation, the opsin protein changes shape. Each opsin is associated with over a hundred regulatory *G proteins* (see chapters 7 and 54). When the opsin changes shape, the *G proteins* dissociate, releasing subunits that activate hundreds of molecules of the enzyme *phosphodiesterase*. This enzyme converts cGMP to GMP, thus closing the Na^+ channels at a rate of about 1000 per second and inhibiting the dark current. The absorption of a single photon of light can block the entry of more than a million sodium ions, thereby causing the photoreceptor to hyperpolarize and release less inhibitory neurotransmitters. Freed from inhibition, the bipo-

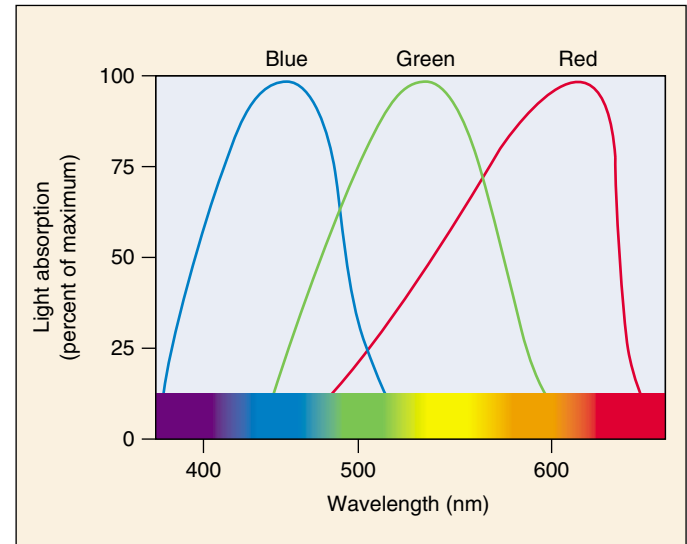


FIGURE 55.22

Color vision. The absorption maximum of *cis*-retinal in the rhodopsin of rods is 500 nanometers (nm). However, the “blue cones” have their maximum light absorption at 455 nm; the “green cones” at 530 nm, and the red cones at 625 nm. The brain perceives all other colors from the combined activities of these three cones’ systems.

lar cells activate ganglion cells, which transmit action potentials to the brain.

Photoreceptor rods and cones contain the photopigment *cis*-retinal, which dissociates in response to light and indirectly activates bipolar neurons and then ganglion cells.

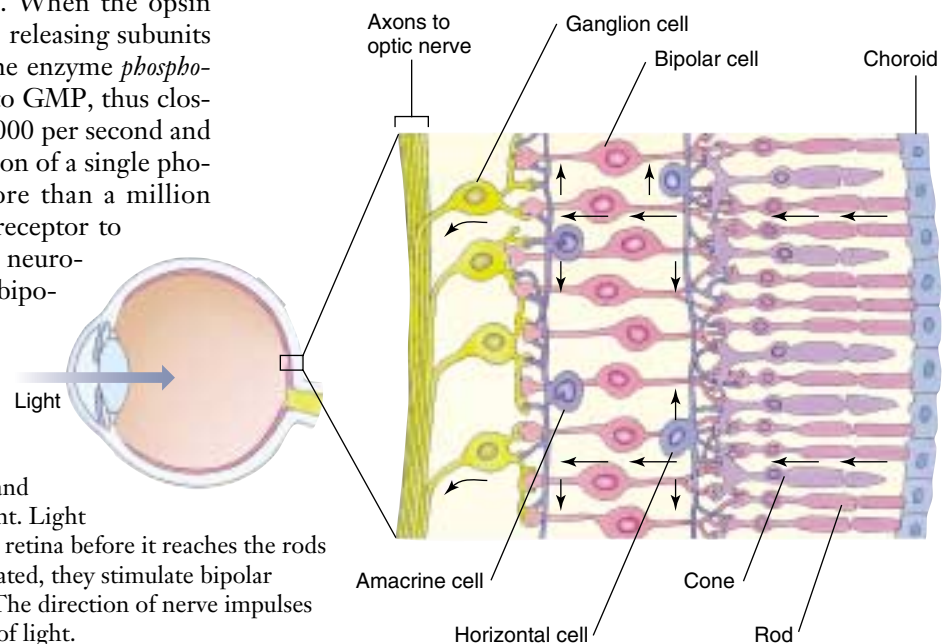


FIGURE 55.23

Structure of the retina. Note that the rods and cones are at the rear of the retina, not the front. Light passes through four other types of cells in the retina before it reaches the rods and cones. Once the photoreceptors are activated, they stimulate bipolar cells, which in turn stimulate ganglion cells. The direction of nerve impulses in the retina is thus opposite to the direction of light.

Visual Processing in the Vertebrate Retina

Action potentials propagated along the axons of ganglion cells are relayed through structures called the lateral geniculate nuclei of the thalamus and projected to the occipital lobe of the cerebral cortex (figure 55.24). There the brain interprets this information as light in a specific region of the eye's receptive field. The pattern of activity among the ganglion cells across the retina encodes a point-to-point map of the receptive field, allowing the retina and brain to image objects in visual space. In addition, the frequency of impulses in each ganglion cell provides information about the light intensity at each point, while the relative activity of ganglion cells connected (through bipolar cells) with the three types of cones provides color information.

The relationship between receptors, bipolar cells, and ganglion cells varies in different parts of the retina. In the fovea, each cone makes a one-to-one connection with a bipolar cell, and each bipolar cell synapses, in turn, with one ganglion cell. This point-to-point relationship is responsible for the high acuity of foveal vision. Outside the fovea, many rods can converge on a single bipolar cell, and many bipolar cells can converge on a single ganglion cell. This convergence permits the summation of neural activity, making the area of the retina outside of the fovea more sensitive to dim light than the fovea, but at the expense of acuity and color vision. This is why dim objects, such as faint stars at night, are best seen when you don't look directly at them. It has been said that we use the periphery of the eye as a detector and the fovea as an inspector.

Color blindness is due to an inherited lack of one or more types of cones. People with normal color vision are *trichromats*; those with only two types of cones are *dichromats*. People with this condition may lack red cones (have *protanopia*), for example, and have difficulty distinguishing red from green. Men are far more likely to be color blind than women, because the trait for color blindness is carried on the X chromosome; men have only one X chromosome per cell, whereas women have two X chromosomes and so can carry the trait in a recessive state.

Binocular Vision

Primates (including humans) and most predators have two eyes, one located on each side of the face. When both eyes are trained on the same object, the image that each sees is slightly different because each eye views the object from a different angle. This slight displacement of the images (an effect called **parallax**) permits **binocular vision**, the ability to perceive three-dimensional images and to sense depth. Having their eyes facing forward maximizes the field of overlap in which this stereoscopic vision occurs.

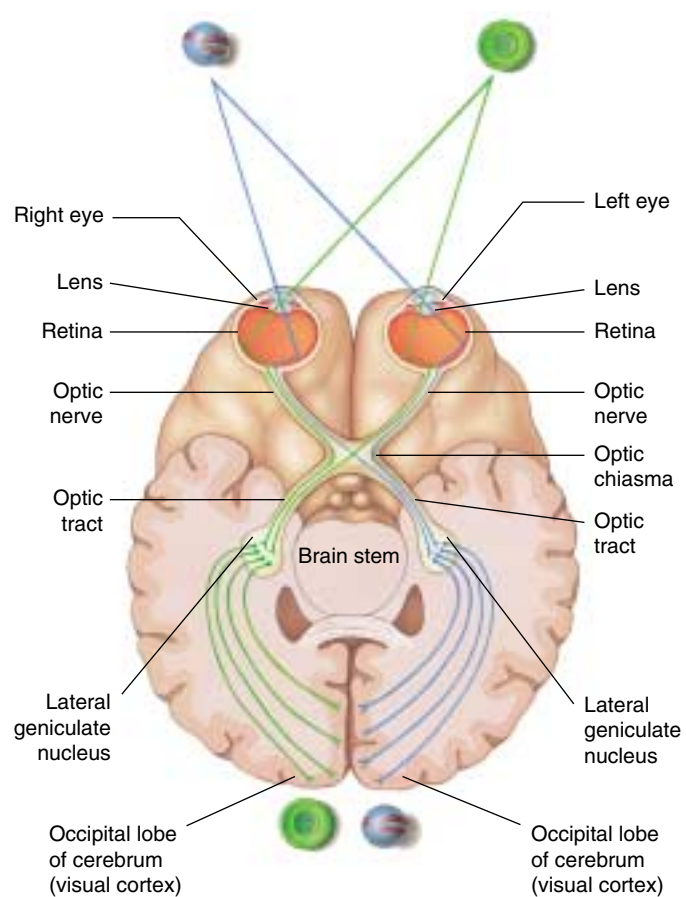


FIGURE 55.24

The pathway of visual information. Action potentials in the optic nerves are relayed from the retina to the lateral geniculate nuclei, and from there to the visual cortex of the occipital lobes. Notice that the medial fibers of the optic nerves cross to the other side at the optic chiasm, so that each hemisphere of the cerebrum receives input from both eyes.

In contrast, prey animals generally have eyes located to the sides of the head, preventing binocular vision but enlarging the overall receptive field. Depth perception is less important to prey than detection of potential enemies from any quarter. The eyes of the American Woodcock, for example, are located at exactly opposite sides of its skull so that it has a 360-degree field of view without turning its head! Most birds have laterally placed eyes and, as an adaptation, have two foveas in each retina. One fovea provides sharp frontal vision, like the single fovea in the retina of mammals, and the other fovea provides sharper lateral vision.

The axons of ganglion cells transmit action potentials to the thalamus, which in turn relays visual information to the occipital lobe of the brain. The fovea provides high visual acuity, whereas the retina outside the fovea provides high sensitivity to dim light. Binocular vision with overlapping visual fields provides depth perception.

55.5 Some vertebrates use heat, electricity, or magnetism for orientation.

Diversity of Sensory Experiences

Vision is the primary sense used by all vertebrates that live in a light-filled environment, but visible light is by no means the only part of the electromagnetic spectrum that vertebrates use to sense their environment.

Heat

Electromagnetic radiation with wavelengths longer than those of visible light is too low in energy to be detected by photoreceptors. Radiation from this *infrared* (“below red”) portion of the spectrum is what we normally think of as radiant heat. Heat is an extremely poor environmental stimulus in water because water has a high thermal capacity and readily absorbs heat. Air, in contrast, has a low thermal capacity, so heat in air is a potentially useful stimulus. However, the only vertebrates known to have the ability to sense infrared radiation are the snakes known as pit vipers.

The pit vipers possess a pair of heat-detecting pit organs located on either side of the head between the eye and the nostril (figure 55.25). The pit organs permit a blindfolded rattlesnake to accurately strike at a warm, dead rat. Each pit organ is composed of two chambers separated by a membrane. The infrared radiation falls on the membrane and warms it. Thermal receptors on the membrane are stimulated. The nature of the pit organ’s thermal receptor is not known; it probably consists of temperature-sensitive neurons innervating the two chambers. The two pit organs appear to provide stereoscopic information, in much the same way that two eyes do. Indeed, the information transmitted from the pit organs is processed by the visual center of the snake brain.

Electricity

While air does not readily conduct an electrical current, water is a good conductor. All aquatic animals generate electrical currents from contractions of their muscles. A number of different groups of fishes can detect these electrical currents. The *electrical fish* even have the ability to produce electrical discharges from specialized electrical organs. Electrical fish use these weak discharges to locate their prey and mates and to construct a three-dimensional image of their environment even in murky water.

The elasmobranchs (sharks, rays, and skates) have electroreceptors called the ampullae of Lorenzini. The receptor cells are located in sacs that open through jelly-filled canals to pores on the body surface. The jelly is a very good conductor, so a negative charge in the opening of the canal can depolarize the receptor at the base, causing the release of neurotransmitter and increased activity of sensory neurons. This allows sharks, for example, to detect the electrical fields generated by the muscle contrac-

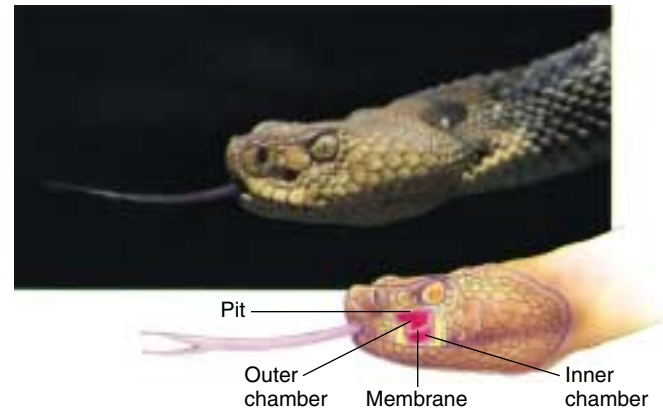


FIGURE 55.25

“Seeing” heat. The depression between the nostril and the eye of this rattlesnake opens into the pit organ. In the cutaway portion of the diagram, you can see that the organ is composed of two chambers separated by a membrane. Snakes known as pit vipers have this unique ability to sense infrared radiation (heat).

tions of their prey. Although the ampullae of Lorenzini were lost in the evolution of teleost fish (most of the bony fish), electroreception reappeared in some groups of teleost fish that use sensory structures analogous to the ampullae of Lorenzini. Electroreceptors evolved yet another time, independently, in the duck-billed platypus, an egg-laying mammal. The receptors in its bill can detect the electrical currents created by the contracting muscles of shrimp and fish, enabling the mammal to detect its prey at night and in muddy water.

Magnetism

Eels, sharks, bees, and many birds appear to navigate along the magnetic field lines of the earth. Even some bacteria use such forces to orient themselves. Birds kept in blind cages, with no visual cues to guide them, will peck and attempt to move in the direction in which they would normally migrate at the appropriate time of the year. They will not do so, however, if the cage is shielded from magnetic fields by steel. Indeed, if the magnetic field of a blind cage is deflected 120° clockwise by an artificial magnet, a bird that normally orients to the north will orient toward the east-southeast. There has been much speculation about the nature of the magnetic receptors in these vertebrates, but the mechanism is still very poorly understood.

Pit vipers can locate warm prey by infrared radiation (heat), and many aquatic vertebrates can locate prey and ascertain the contours of their environment by means of electroreceptors.



Summary

Questions

Media Resources

55.1 Animals employ a wide variety of sensory receptors.

- Mechanoreceptors, chemoreceptors, and photoreceptors are responsive to different categories of sensory stimuli; interoceptors and exteroceptors respond to stimuli that originate in the internal and external environments, respectively.

1. Can you name a sensory receptor that does not produce a membrane depolarization?



- Introduction to sense organs
- Receptors and sensations
- Somatic senses

55.2 Mechanical and chemical receptors sense the body's condition.

- Muscle spindles respond to stretching of the skeletal muscle.
- The sensory organs of taste are taste buds, scattered over the surface of a fish's body but located on the papillae of the tongue in terrestrial vertebrates.
- Chemoreceptors in the aortic and carotid bodies sense the blood pH and oxygen levels, helping to regulate breathing.
- Hair cells in the membranous labyrinth of the inner ear provide a sense of acceleration.

2. What mechanoreceptors detect muscle stretch and the tension on a tendon?



- Smell
- Taste
- Sense of balance
- Sense of rotational acceleration

3. What structures in the vertebrate ear detect changes in the body's position with respect to gravity? What structures detect angular motion?



- Sense of taste
- Sense of smell
- Equilibrium

55.3 Auditory receptors detect pressure waves in the air.

- In terrestrial vertebrates, sound waves cause vibrations of ear membranes.
- Different pitches of sounds vibrate different regions of the basilar membrane, and therefore stimulate different hair cells.
- Bats and some other vertebrates use sonar to provide a sense of "lightless vision."

4. How are sound waves transmitted and amplified through the middle ear? How is the pitch of the sound determined?



- Art Activity
Human ear anatomy



- Hearing

55.4 Optic receptors detect light over a broad range of wavelengths.

- A flexible lens focuses light onto the retina, which contains the photoreceptors.
- Light causes the photodissociation of the visual pigment, thereby blocking the dark current and hyperpolarizing the photoreceptor; this inverse effect stops the inhibitory effect of the photoreceptor and thereby activates the bipolar cells.

5. How does focusing in fishes and amphibians differ from that in other vertebrates?



- Art Activity
Human eye anatomy

6. When a photoreceptor absorbs light, what happens to the Na⁺ channels in its outer segment?



- Vision

55.5 Some vertebrates use heat, electricity, or magnetism for orientation.

- The pit organs of snakes allows them to detect the position and movements of prey. Many aquatic vertebrates can detect electrical currents produced by muscular contraction. Some vertebrates can orient themselves using the earth's magnetic field.

7. Why do rattlesnakes strike a moving lightbulb?



- Chemoreceptors

8. How do sharks detect their prey? Why don't terrestrial vertebrates have this sense?

56

The Endocrine System

Concept Outline

56.1 Regulation is often accomplished by chemical messengers.

Types of Regulatory Molecules. Regulatory molecules may function as neurotransmitters, hormones, or as organ-specific regulators.

Endocrine Glands and Hormones. Endocrine glands secrete molecules called hormones into the blood.

Paracrine Regulation. Paracrine regulators act within organs that produce them.

56.2 Lipophilic and polar hormones regulate their target cells by different means.

Hormones That Enter Cells. Steroid and thyroid hormones act by entering target cells and stimulating specific genes.

Hormones That Do Not Enter Cells. All other hormones bind to receptors on the cell surface and activate second-messenger molecules within the target cells.

56.3 The hypothalamus controls the secretions of the pituitary gland.

The Posterior Pituitary Gland. The posterior pituitary receives and releases hormones from the hypothalamus.

The Anterior Pituitary Gland. The anterior pituitary produces a variety of hormones under stimulation from hypothalamic releasing hormones.

56.4 Endocrine glands secrete hormones that regulate many body functions.

The Thyroid and Parathyroid Glands. The thyroid hormones regulate metabolism; the parathyroid glands regulate calcium balance.

The Adrenal Glands. The adrenal medulla secretes epinephrine during the fight-or-flight reaction, while the adrenal cortex secretes steroid hormones that regulate glucose and mineral balance.

The Pancreas. The islets of Langerhans in the pancreas secrete insulin, which acts to lower blood glucose, and glucagon, which acts to raise blood glucose.

Other Endocrine Glands. The gonads, pineal gland, thymus, kidneys, and other organs secrete important hormones that have a variety of functions.



FIGURE 56.1

The endocrine system controls when animals breed. These Japanese macaques live in a close-knit community whose members cooperate to ensure successful breeding and raising of offspring. Not everybody breeds at the same time because hormone levels vary among individuals.

The tissues and organs of the vertebrate body cooperate to maintain homeostasis of the body's internal environment and control other body functions such as reproduction. Homeostasis is achieved through the actions of many regulatory mechanisms that involve all the organs of the body. Two systems, however, are devoted exclusively to the regulation of the body organs: the nervous system and the endocrine system (figure 56.1). Both release regulatory molecules that control the body organs by first binding to receptor proteins in the cells of those organs. In this chapter we will examine these regulatory molecules, the cells and glands that produce them, and how they function to regulate the body's activities.

56.1 Regulation is often accomplished by chemical messengers.

Types of Regulatory Molecules

As we discussed in chapter 54, the axons of neurons secrete chemical messengers called *neurotransmitters* into the synaptic cleft. These chemicals diffuse only a short distance to the postsynaptic membrane, where they bind to their receptor proteins and stimulate the postsynaptic cell (another neuron, or a muscle or gland cell). Synaptic transmission generally affects only the one postsynaptic cell that receives the neurotransmitter.

A **hormone** is a regulatory chemical that is secreted into the blood by an **endocrine gland** or an organ of the body exhibiting an endocrine function. The blood carries the hormone to every cell in the body, but only the **target cells** for a given hormone can respond to it. Thus, the difference between a neurotransmitter and a hormone is not in the chemical nature of the regulatory molecule, but rather in the way it is transported to its target cells, and its distance from these target cells. A chemical regulator called norepinephrine, for example, is released as a neurotransmitter by sympathetic nerve endings and is also secreted by the adrenal gland as a hormone.

Some specialized neurons secrete chemical messengers into the blood rather than into a narrow synaptic cleft. In these cases, the chemical that the neurons secrete is some-

times called a **neurohormone**. The distinction between the nervous system and endocrine system blurs when it comes to such molecules. Indeed, because some neurons in the brain secrete hormones, the brain can be considered an endocrine gland!

In addition to the chemical messengers released as neurotransmitters and as hormones, other chemical regulatory molecules are released and act *within* an organ. In this way, the cells of an organ regulate one another. This type of regulation is not endocrine, because the regulatory molecules work without being transported by the blood, but is otherwise similar to the way that hormones regulate their target cells. Such regulation is called **paracrine**. Another type of chemical messenger that is released into the environment is called a *pheromone*. These messengers aid in the communication between animals, not in the regulation within an animal. A comparison of the different types of chemical messengers used for regulation is given in figure 56.2.

Regulatory molecules released by axons at a synapse are neurotransmitters, those released by endocrine glands into the blood are hormones, and those that act within the organ in which they are produced are paracrine regulators.

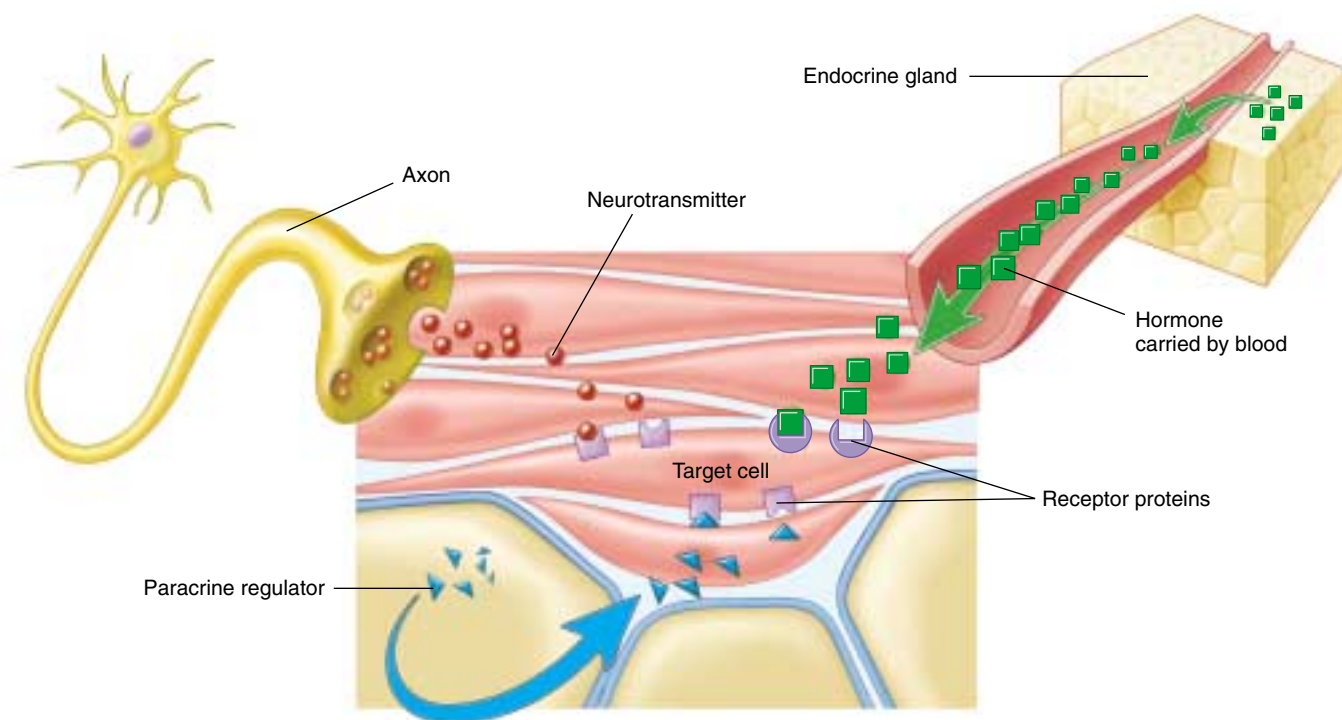


FIGURE 56.2

The functions of organs are influenced by neural, paracrine, and endocrine regulators. Each type of chemical regulator binds in a specific fashion to receptor proteins on the surface of or within the cells of target organs.

Endocrine Glands and Hormones

The endocrine system (figure 56.3) includes all of the organs that function exclusively as endocrine glands—such as the thyroid gland, pituitary gland, adrenal glands, and so on (table 56.1)—as well as organs that secrete hormones in addition to other functions. Endocrine glands lack ducts and thus must secrete into surrounding blood capillaries, unlike exocrine glands, which secrete their products into a duct.

Hormones secreted by endocrine glands belong to four different chemical categories:

1. **Polypeptides.** These hormones are composed of chains of amino acids that are shorter than about 100 amino acids. Some important examples include insulin and antidiuretic hormone (ADH).
2. **Glycoproteins.** These are composed of a polypeptide significantly longer than 100 amino acids to which is attached a carbohydrate. Examples include follicle-stimulating hormone (FSH) and luteinizing hormone (LH).
3. **Amines.** Derived from the amino acids tyrosine and tryptophan, they include hormones secreted by the adrenal medulla, thyroid, and pineal glands.
4. **Steroids.** These hormones are lipids derived from cholesterol, and include the hormones testosterone, estradiol, progesterone, and cortisol.

Steroid hormones can be subdivided into **sex steroids**, secreted by the testes, ovaries, placenta, and adrenal cortex, and **corticosteroids**, secreted only by the adrenal cortex (the outer portion of the adrenal gland). The corticosteroids include cortisol, which regulates glucose balance, and aldosterone, which regulates salt balance.

The amine hormones secreted by the adrenal medulla (the inner portion of the adrenal gland), known as **catecholamines**, include epinephrine (adrenaline) and norepinephrine (noradrenaline). These are derived from the amino acid tyrosine. Another hormone derived from tyrosine is thyroxine, secreted by the thyroid gland. The pineal gland secretes a different amine hormone, melatonin, derived from tryptophan.

All hormones may be categorized as lipophilic (fat-soluble) or hydrophilic (water-soluble). The **lipophilic hormones** include the steroid hormones and thyroxine; all other hormones are water-soluble. This distinction is important in understanding how these hormones regulate their target cells.

Neural and Endocrine Interactions

The endocrine system is an extremely important regulatory system in its own right, but it also interacts and cooperates with the nervous system to regulate the activities of the other organ systems of the body. The secretory activity of many endocrine glands is controlled by the nervous system. Among such glands are the adrenal medulla, posterior pitu-

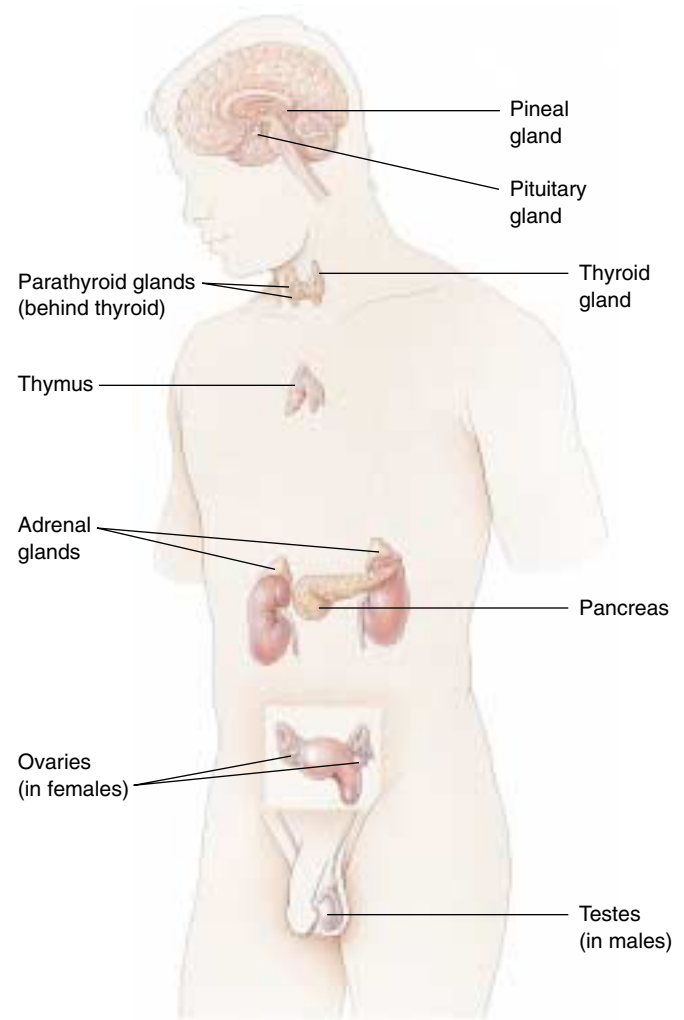


FIGURE 56.3

The human endocrine system. The major endocrine glands are shown, but many other organs secrete hormones in addition to their primary functions.

itary, and pineal gland. These three glands are derived from the neural ectoderm (to be discussed in chapter 60), the same embryonic tissue layer that forms the nervous system. The major site for neural regulation of the endocrine system, however, is the brain's regulation of the anterior pituitary gland. As we'll see, the hypothalamus controls the hormonal secretions of the anterior pituitary, which in turn regulates other endocrine glands. On the other hand, the secretion of a number of hormones is largely independent of neural control. The release of insulin by the pancreas and aldosterone by the adrenal cortex, for example, are stimulated primarily by increases in the blood concentrations of glucose and potassium (K^+), respectively.

Any organ that secretes a hormone from a ductless gland is part of the endocrine system. Hormones may be any of a variety of different chemicals.

Table 56.1 Principal Endocrine Glands and Their Hormones*

Endocrine Gland and Hormone	Target Tissue	Principal Actions	Chemical Nature
POSTERIOR LOBE OF PITUITARY			
Antidiuretic hormone (ADH)	Kidneys	Stimulates reabsorption of water; conserves water	Peptide (9 amino acids)
Oxytocin	Uterus	Stimulates contraction	Peptide
	Mammary glands	Stimulates milk ejection	(9 amino acids)
ANTERIOR LOBE OF PITUITARY			
Growth hormone (GH)	Many organs	Stimulates growth by promoting protein synthesis and fat breakdown	Protein
Adrenocorticotrophic hormone (ACTH)	Adrenal cortex	Stimulates secretion of adrenal cortical hormones such as cortisol	Peptide (39 amino acids)
Thyroid-stimulating hormone (TSH)	Thyroid gland	Stimulates thyroxine secretion	Glycoprotein
Luteinizing hormone (LH)	Gonads	Stimulates ovulation and corpus luteum formation in females; stimulates secretion of testosterone in males	Glycoprotein
Follicle-stimulating hormone (FSH)	Gonads	Stimulates spermatogenesis in males; stimulates development of ovarian follicles in females	Glycoprotein
Prolactin (PRL)	Mammary glands	Stimulates milk production	Protein
Melanocyte-stimulating hormone (MSH)	Skin	Stimulates color change in reptiles and amphibians; unknown function in mammals	Peptide (two forms; 13 and 22 amino acids)
THYROID GLAND			
Thyroxine (thyroid hormone)	Most cells	Stimulates metabolic rate; essential to normal growth and development	Iodinated amino acid
Calcitonin	Bone	Lowers blood calcium level by inhibiting loss of calcium from bone	Peptide (32 amino acids)
PARATHYROID GLANDS			
Parathyroid hormone	Bone, kidneys, digestive tract	Raises blood calcium level by stimulating bone breakdown; stimulates calcium reabsorption in kidneys; activates vitamin D	Peptide (34 amino acids)

*These are hormones released from endocrine glands. As discussed previously, many hormones are released from other body organs.

Paracrine Regulation

Paracrine regulation occurs in many organs and among the cells of the immune system. Some of these regulatory molecules are known as **cytokines**, particularly if they regulate different cells of the immune system. Other paracrine regulators are called **growth factors**, because they promote growth and cell division in specific organs. Examples include *platelet-derived growth factor*, *epidermal growth factor*, and the *insulin-like growth factors* that stimulate cell division and proliferation of their target cells. *Nerve growth factor* is a regulatory molecule that belongs to a family of paracrine regulators of the nervous system called **neurotrophins**.

Nitric oxide, which can function as a neurotransmitter (see chapter 54), is also produced by the endothelium of blood vessels. In this context, it is a paracrine regulator because it diffuses to the smooth muscle layer of the blood vessel and promotes vasodilation. The endothelium of blood vessels also

produces other paracrine regulators, including *endothelin*, which stimulates vasoconstriction, and *bradykinin*, which promotes vasodilation. This paracrine regulation supplements the regulation of blood vessels by autonomic nerves.

The most diverse group of paracrine regulators are the **prostaglandins**. A prostaglandin is a 20-carbon-long fatty acid that contains a five-member carbon ring. This molecule is derived from the precursor molecule *arachidonic acid*, released from phospholipids in the cell membrane under hormonal or other stimulation. Prostaglandins are produced in almost every organ and participate in a variety of regulatory functions, including:

- 1. Immune system.** Prostaglandins promote many aspects of inflammation, including pain and fever. Drugs that inhibit prostaglandin synthesis help to alleviate these symptoms.
- 2. Reproductive system.** Prostaglandins may play a

Table 56.1 Principal Endocrine Glands and Their Hormones

Endocrine Gland and Hormone	Target Tissue	Principal Actions	Chemical Nature
ADRENAL MEDULLA			
Epinephrine (adrenaline) and norepinephrine (noradrenaline)	Smooth muscle, cardiac muscle, blood vessels	Initiate stress responses; raise heart rate, blood pressure, metabolic rate; dilate blood vessels; mobilize fat; raise blood glucose level	Amino acid derivatives
ADRENAL CORTEX			
Aldosterone	Kidney tubules	Maintains proper balance of Na ⁺ and K ⁺ ions	Steroid
Cortisol	Many organs	Adaptation to long-term stress; raises blood glucose level; mobilizes fat	Steroid
PANCREAS			
Insulin	Liver, skeletal muscles, adipose tissue	Lowers blood glucose level; stimulates storage of glycogen in liver	Peptide (51 amino acids)
Glucagon	Liver, adipose tissue	Raises blood glucose level; stimulates breakdown of glycogen in liver	Peptide (29 amino acids)
OVARY			
Estradiol	General	Stimulates development of secondary sex characteristics in females	Steroid
Progesterone	Female reproductive structures	Stimulates growth of sex organs at puberty and monthly preparation of uterus for pregnancy	Steroid
	Uterus	Completes preparation for pregnancy	
	Mammary glands	Stimulates development	
TESTIS			
Testosterone	Many organs	Stimulates development of secondary sex characteristics in males and growth spurt at puberty	Steroid
	Male reproductive structures	Stimulates development of sex organs; stimulates spermatogenesis	
PINEAL GLAND			
Melatonin	Gonads, pigment cells	Function not well understood; influences pigmentation in some vertebrates; may control biorhythms in some animals; may influence onset of puberty in humans	Amino acid derivative

role in ovulation. Excessive prostaglandin production may be involved in premature labor, endometriosis, or dysmenorrhea (painful menstrual cramps).

- 3. Digestive system.** Prostaglandins produced by the stomach and intestines may inhibit gastric secretions and influence intestinal motility and fluid absorption.
- 4. Respiratory system.** Some prostaglandins cause constriction, whereas others cause dilation of blood vessels in the lungs and of bronchiolar smooth muscle.
- 5. Circulatory system.** Prostaglandins are needed for proper function of blood platelets in the process of blood clotting.
- 6. Urinary system.** Prostaglandins produced in the renal medulla cause vasodilation, resulting in increased renal blood flow and increased excretion of urine.

The synthesis of prostaglandins are inhibited by aspirin. Aspirin is the most widely used of the **nonsteroidal anti-**

inflammatory drugs (NSAIDs), a class of drugs that also includes indomethacin and ibuprofen. These drugs produce their effects because they specifically inhibit the enzyme cyclooxygenase-2 (cox-2), needed to produce prostaglandins from arachidonic acid. Through this action, the NSAIDs inhibit inflammation and associated pain. Unfortunately, NSAIDs also inhibit another similar enzyme, cox-1, which helps maintain the wall of the digestive tract, and in so doing can produce severe unwanted side effects, including gastric bleeding and prolonged clotting time. A new kind of pain reliever, celecoxib (Celebrex), inhibits cox-2 but not cox-1, a potentially great benefit to arthritis sufferers and others who must use pain relievers regularly.

The neural and endocrine control systems are supplemented by paracrine regulators, including the prostaglandins, which perform many diverse functions.

56.2 Lipophilic and polar hormones regulate their target cells by different means.

Hormones That Enter Cells

As we mentioned previously, hormones can be divided into those that are lipophilic (lipid-soluble) and those that are hydrophilic (water-soluble). The lipophilic hormones—all of the steroid hormones (figure 56.4) and thyroxine—as well as other lipophilic regulatory molecules (including the retinoids, or vitamin A) can easily enter cells. This is because the lipid portion of the cell membrane does not present a barrier to the entry of lipophilic regulators. Therefore, all lipophilic regulatory molecules have a similar mechanism of action. Water-soluble hormones, in contrast, cannot pass through cell membranes. They must regulate their target cells through different mechanisms.

Steroid hormones are lipids themselves and thus lipophilic; thyroxine is lipophilic because it is derived from a nonpolar amino acid. Because these hormones are not water-soluble, they don't dissolve in plasma but rather travel in the blood attached to protein carriers. When the hormones arrive at their target cells, they dissociate from their carriers and pass through the plasma membrane of

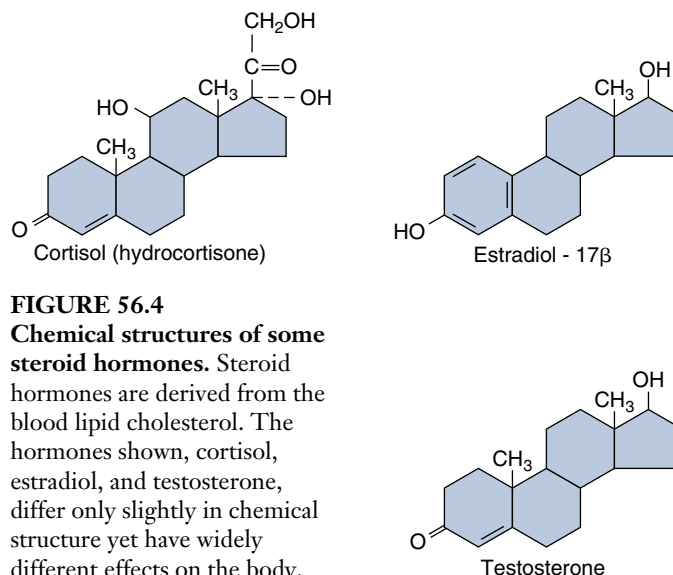


FIGURE 56.4
Chemical structures of some steroid hormones. Steroid hormones are derived from the blood lipid cholesterol. The hormones shown, cortisol, estradiol, and testosterone, differ only slightly in chemical structure yet have widely different effects on the body. Steroid hormones are secreted by the adrenal cortex, testes, ovaries, and placenta.

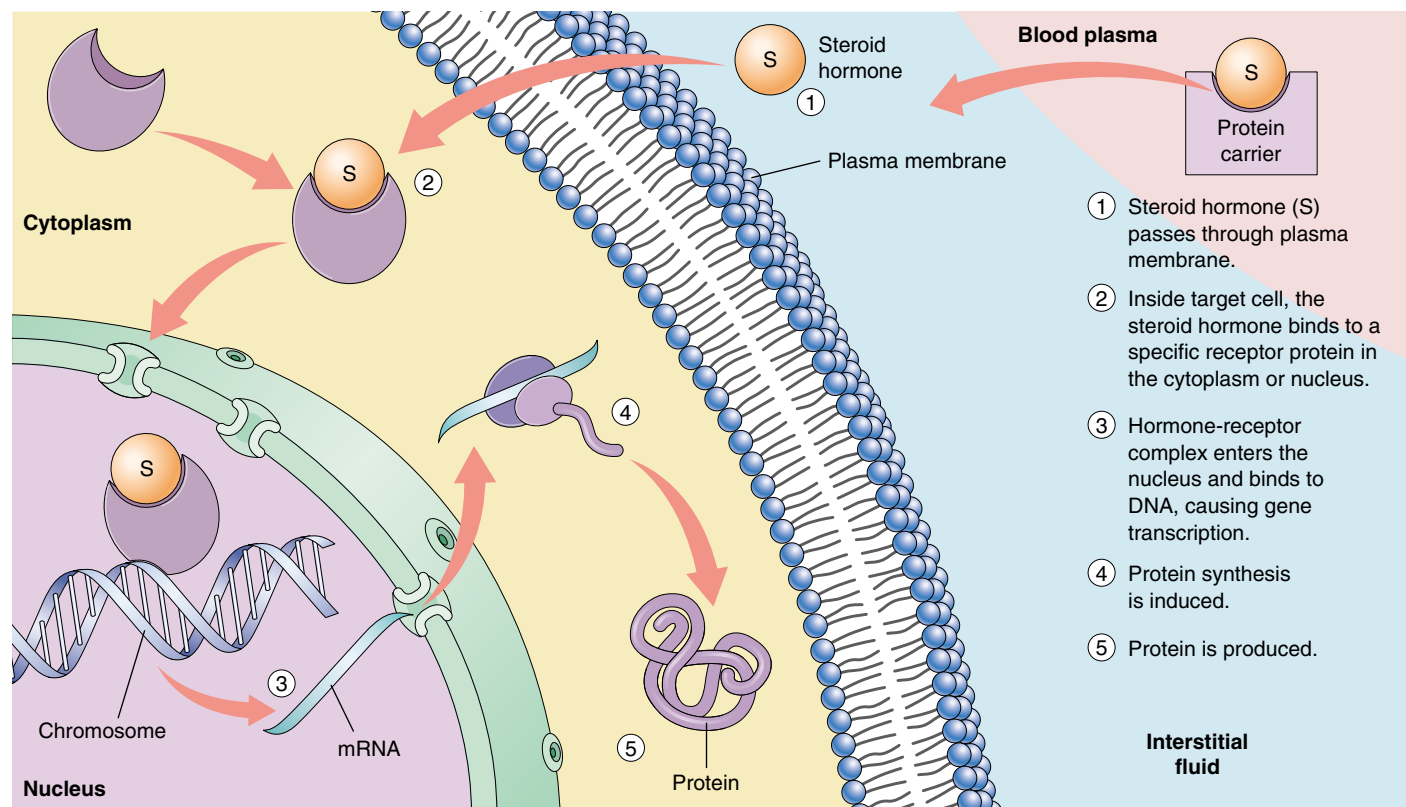


FIGURE 56.5

The mechanism of steroid hormone action. Steroid hormones are lipid-soluble and thus readily diffuse through the plasma membrane of cells. They bind to receptor proteins in either the cytoplasm or nucleus (not shown). If the steroid binds to a receptor in the cytoplasm, the hormone-receptor complex moves into the nucleus. The hormone-receptor complex then binds to specific regions of the DNA, stimulating the production of messenger RNA (mRNA).

the cell (figure 56.5). Some steroid hormones then bind to very specific receptor proteins in the cytoplasm, and then move as a hormone-receptor complex into the nucleus. Other steroids travel directly into the nucleus before encountering their receptor proteins. Whether the steroid finds its receptor in the nucleus or translocates with its receptor to the nucleus from the cytoplasm, the rest of the story is the same.

The hormone receptor, activated by binding to the lipophilic hormone, is now also able to bind to specific regions of the DNA. These DNA regions are known as the hormone response elements. The binding of the hormone-receptor complex has a direct effect on the level of transcription at that site by activating genetic transcription. This produces messenger RNA (mRNA), which then codes for the production of specific proteins. These proteins often have enzymatic activity that changes the metabolism of the target cell in a specific fashion.

The thyroid hormone's mechanism of action resembles that of the steroid hormones. Thyroxine contains four iodines and so is often abbreviated T_4 (for *tetraiodothyronine*). The thyroid gland also secretes smaller amounts of a similar molecule that has only three iodines, called *triiodothyronine* (and abbreviated T_3). Both hormones enter target cells, but all of the T_4 that enters is changed into T_3 (figure 56.6). Thus, only the T_3 form of the hormone enters the nucleus and binds to nuclear receptor proteins. The hormone-receptor complex, in turn, binds to the appropriate hormone response elements on DNA.

The lipophilic hormones pass through the target cell's plasma membrane and bind to intracellular receptor proteins. The hormone-receptor complex then binds to specific regions of DNA, thereby activating genes and regulating the target cells.

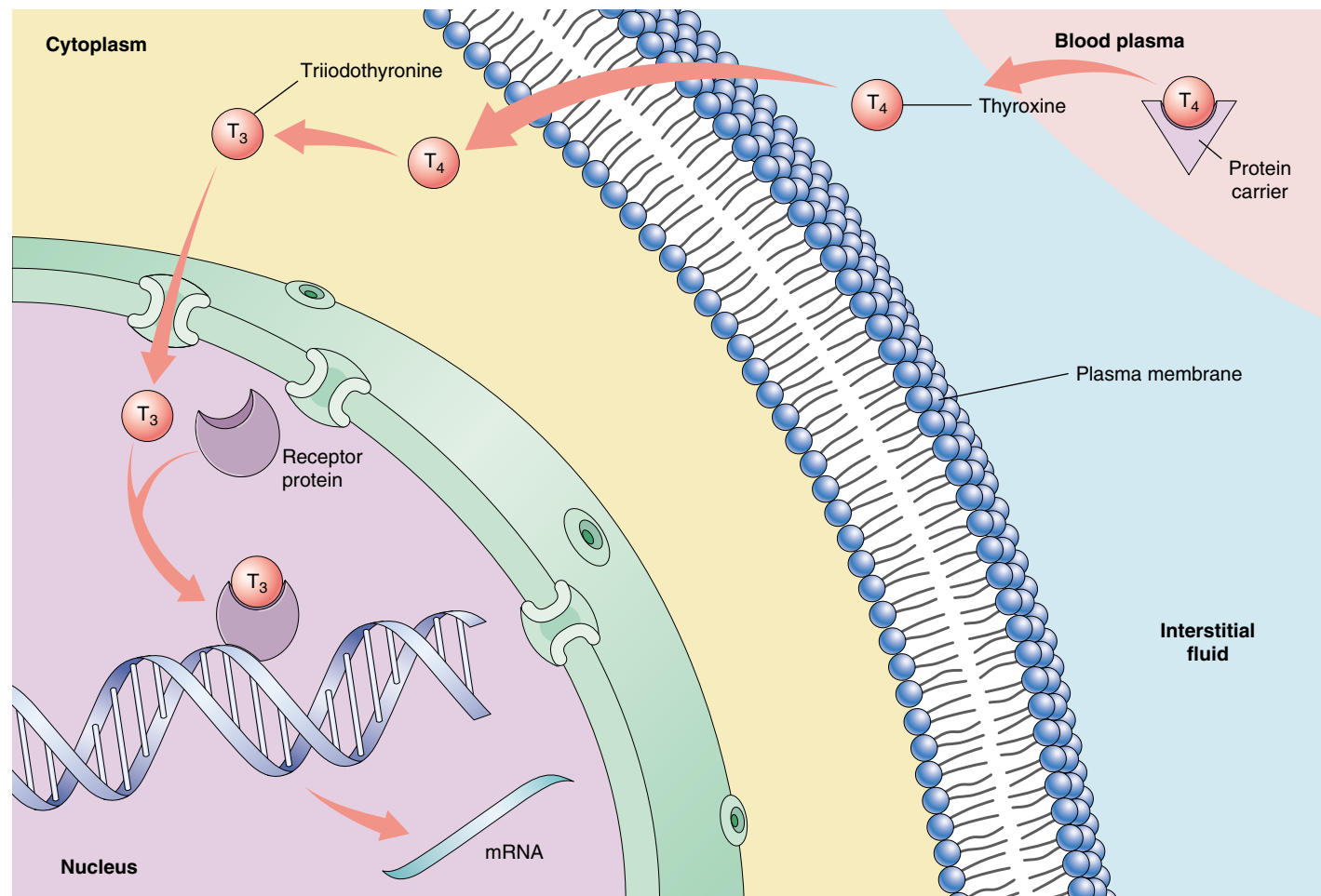


FIGURE 56.6

The mechanism of thyroxine action. Thyroxine contains four iodines. When it enters the target cell, thyroxine is changed into triiodothyronine, with three iodines. This hormone moves into the nucleus and binds to nuclear receptors. The hormone-receptor complex then binds to regions of the DNA and stimulates gene transcription.

Hormones That Do Not Enter Cells

Hormones that are too large or too polar to cross the plasma membranes of their target cells include all of the peptide and glycoprotein hormones, as well as the catecholamine hormones epinephrine and norepinephrine. These hormones bind to receptor proteins located on the outer surface of the plasma membrane—the hormones do *not* enter the cell. If you think of the hormone as a messenger sent from an endocrine gland to the target cell, it is evident that a second messenger is needed within the target cell to produce the effects of the hormone. A number of different molecules in the cell can serve as second messengers, as we saw in chapter 7. The interaction between the hormone and its receptor activates mechanisms in the plasma membrane that increase the concentration of the second messengers within the target cell cytoplasm.

The binding of a water-soluble hormone to its receptor is reversible and usually very brief. After the hormone binds to its receptor and activates a second-messenger system, it dissociates from the receptor and may travel in the blood to another target cell somewhere else in the body. Eventually, enzymes (primarily in the liver) degrade the hormone by converting it into inactive derivatives.

The Cyclic AMP Second-Messenger System

The action of the hormone epinephrine can serve as an example of a second-messenger system. Epinephrine can bind to two categories of receptors, called *alpha* (α)- and *beta* (β)-*adrenergic receptors*. The interaction of epinephrine with each type of receptor activates a different second-messenger system in the target cell.

In the early 1960s, Earl Sutherland showed that cyclic adenosine monophosphate, or **cyclic AMP (cAMP)**, serves as a second messenger when epinephrine binds to β -adrenergic receptors on the plasma membranes of liver cells (figure 56.7). The cAMP second-messenger system was the first such system to be described.

The β -adrenergic receptors are associated with membrane proteins called *G proteins* (see chapters 7 and 54). Each G protein is composed of three subunits, and the binding of epinephrine to its receptor causes one of the G protein subunits to dissociate from the other two. This subunit then diffuses within the plasma membrane until it encounters **adenylyl cyclase**, a membrane enzyme that is inactive until it binds to the G protein subunit. When activated by the G protein subunit, adenylyl cyclase catalyzes the formation of cAMP from ATP. The cAMP formed at the inner surface of the plasma membrane diffuses within the cytoplasm, where it binds to and activates **protein kinase-A**, an enzyme that adds phosphate groups to specific cellular proteins.

The identities of the proteins that are phosphorylated by protein kinase-A varies from one cell type to the next, and this variation is one of the reasons epinephrine has such di-

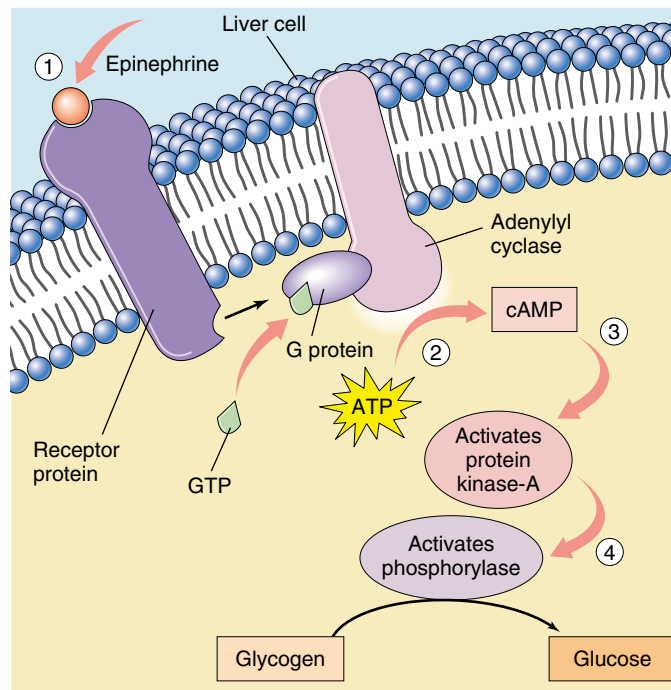


FIGURE 56.7

The action of epinephrine on a liver cell. (1) Epinephrine binds to specific receptor proteins on the cell surface. (2) Acting through intermediary G proteins, the hormone-bound receptor activates the enzyme adenylyl cyclase, which converts ATP into cyclic AMP (cAMP). (3) Cyclic AMP performs as a second messenger and activates protein kinase-A, an enzyme that was previously present in an inactive form. (4) Protein kinase-A phosphorylates and thereby activates the enzyme phosphorylase, which catalyzes the hydrolysis of glycogen into glucose.

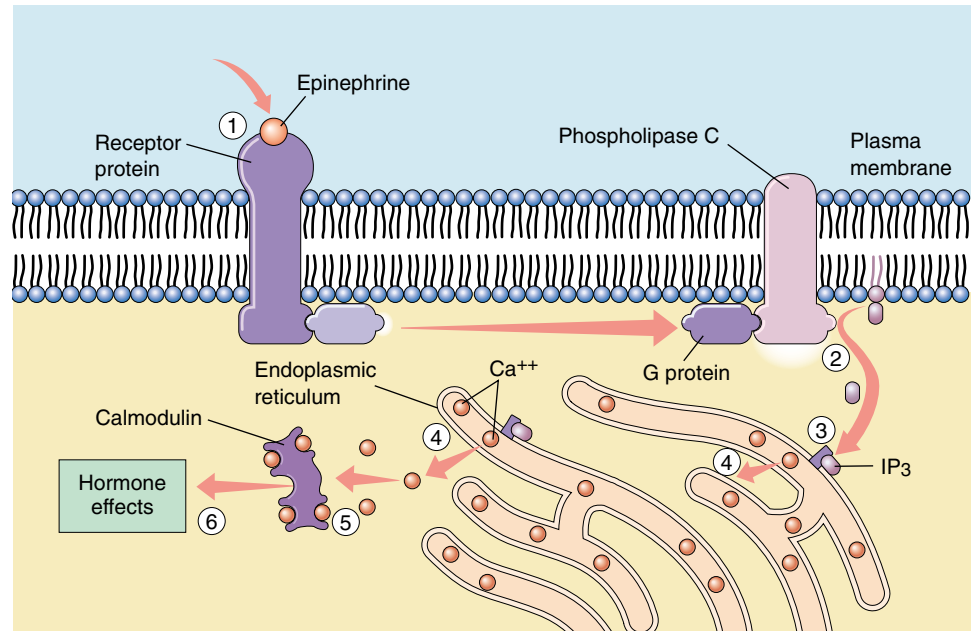
verse effects on different tissues. In liver cells, protein kinase-A phosphorylates and thereby activates another enzyme, phosphorylase, which converts glycogen into glucose. Through this multistep mechanism, epinephrine causes the liver to secrete glucose into the blood during the fight-or-flight reaction, when the adrenal medulla is stimulated by the sympathetic division of the autonomic nervous system (see chapter 54). In cardiac muscle cells, protein kinase-A phosphorylates a different set of cellular proteins, which cause the heart to beat faster and more forcefully.

The IP_3/Ca^{++} Second-Messenger System

When epinephrine binds to α -adrenergic receptors, it doesn't activate adenylyl cyclase and cause the production of cAMP. Instead, through a different type of G protein, it activates another membrane-bound enzyme, phospholipase C (figure 56.8). This enzyme cleaves certain membrane phospholipids to produce the second messenger, inositol

FIGURE 56.8

The $\text{IP}_3/\text{Ca}^{++}$ second-messenger system. (1) The hormone epinephrine binds to specific receptor proteins on the cell surface. (2) Acting through G proteins, the hormone-bound receptor activates the enzyme phospholipase C, which converts membrane phospholipids into inositol trisphosphate (IP_3). (3) IP_3 diffuses through the cytoplasm and binds to receptors on the endoplasmic reticulum. (4) The binding of IP_3 to its receptors stimulates the endoplasmic reticulum to release Ca^{++} into the cytoplasm. (5) Some of the released Ca^{++} binds to a regulatory protein called calmodulin. (6) The Ca^{++} /calmodulin complex activates other intracellular proteins, ultimately producing the effects of the hormone.



trisphosphate (IP_3). IP_3 diffuses into the cytoplasm from the plasma membrane and binds to receptors located on the surface of the endoplasmic reticulum.

Recall from chapter 5 that the endoplasmic reticulum is a system of membranous sacs and tubes that serves a variety of functions in different cells. One of its functions is to accumulate Ca^{++} by actively transporting Ca^{++} out of the cytoplasm. Other pumps transport Ca^{++} from the cytoplasm through the plasma membrane to the extracellular fluid. These two mechanisms keep the concentration of Ca^{++} in the cytoplasm very low. Consequently, there is an extremely steep concentration gradient for Ca^{++} between the cytoplasm and the inside of the endoplasmic reticulum, and between the cytoplasm and the extracellular fluid.

When IP_3 binds to its receptors on the endoplasmic reticulum, it stimulates the endoplasmic reticulum to release its stored Ca^{++} . Calcium channels in the plasma membrane may also open, allowing Ca^{++} to diffuse into the cell from the extracellular fluid. Some of the Ca^{++} that has suddenly entered the cytoplasm then binds to a protein called calmodulin, which has regulatory functions analogous to those of cyclic AMP. One of the actions of calmodulin is to activate another type of protein kinase, resulting in the phosphorylation of a different set of cellular proteins.

What is the advantage of having multiple second-messenger systems? Consider the antagonistic actions of epinephrine and insulin on liver cells. Epinephrine uses cAMP as a second messenger to promote the hydrolysis of glycogen to glucose, while insulin stimulates the synthesis of glycogen from glucose. Clearly, insulin cannot use cAMP as a second messenger. Although the exact mechanism of in-

sulin's action is still not well understood, insulin may act in part through the $\text{IP}_3/\text{Ca}^{++}$ second-messenger system.

Not all large polar hormones act by increasing the concentration of a second messenger in the cytoplasm of the target cell. Others cause a change in the shape of a membrane protein called an ion channel (see chapters 6 and 54). If these channels are normally “closed,” then a change in shape will open them allowing a particular ion to enter or leave the cell depending on its concentration gradient. If an ion channel is normally open, a chemical messenger can cause it to close. For example, some hormones open Ca^{++} channels on smooth muscle cell membranes; other hormones close them. This will increase or decrease, respectively, the amount of muscle contraction.

The molecular mechanism for changing the shape of an ion channel is similar to that for activating a second messenger. The hormone first binds to a receptor protein on the outer surface of the target cell. This receptor protein may then use a G protein to signal the ion channel to change shape.

Although G proteins play a major role in many hormone functions they don't seem to be necessary for all identified actions of hormones on target cells. In the cases where G proteins are not involved, the receptor protein is connected directly to the enzyme or ion channel.

The water-soluble hormones cannot pass through the plasma membrane; they must rely on second messengers within the target cells to mediate their actions. Such second messengers include cyclic AMP (cAMP), inositol trisphosphate (IP_3), and Ca^{++} . In many cases, the second messengers activate previously inactive enzymes.

56.3 The hypothalamus controls the secretions of the pituitary gland.

The Posterior Pituitary Gland

The **pituitary gland** hangs by a stalk from the hypothalamus of the brain (figure 56.9) posterior to the optic chiasm (see chapter 54). A microscopic view reveals that the gland consists of two parts. The anterior portion appears glandular and is called the **anterior pituitary**; the posterior portion appears fibrous and is the **posterior pituitary**. These two portions of the pituitary gland have different embryonic origins, secrete different hormones, and are regulated by different control systems.

The Posterior Pituitary Gland

The posterior pituitary appears fibrous because it contains axons that originate in cell bodies within the hypothalamus and extend along the stalk of the pituitary as a tract of

fibers. This anatomical relationship results from the way that the posterior pituitary is formed in embryonic development. As the floor of the third ventricle of the brain forms the hypothalamus, part of this neural tissue grows downward to produce the posterior pituitary. The hypothalamus and posterior pituitary thus remain interconnected by a tract of axons.

The endocrine role of the posterior pituitary gland first became evident in 1912, when a remarkable medical case was reported: a man who had been shot in the head developed the need to urinate every 30 minutes or so, 24 hours a day. The bullet had lodged in his pituitary gland. Subsequent research demonstrated that removal of this gland produces the same symptoms. Pituitary extracts were found to contain a substance that makes the kidneys conserve water, and in the early 1950s investigators isolated a peptide from the posterior pituitary, **antidiuretic hormone**

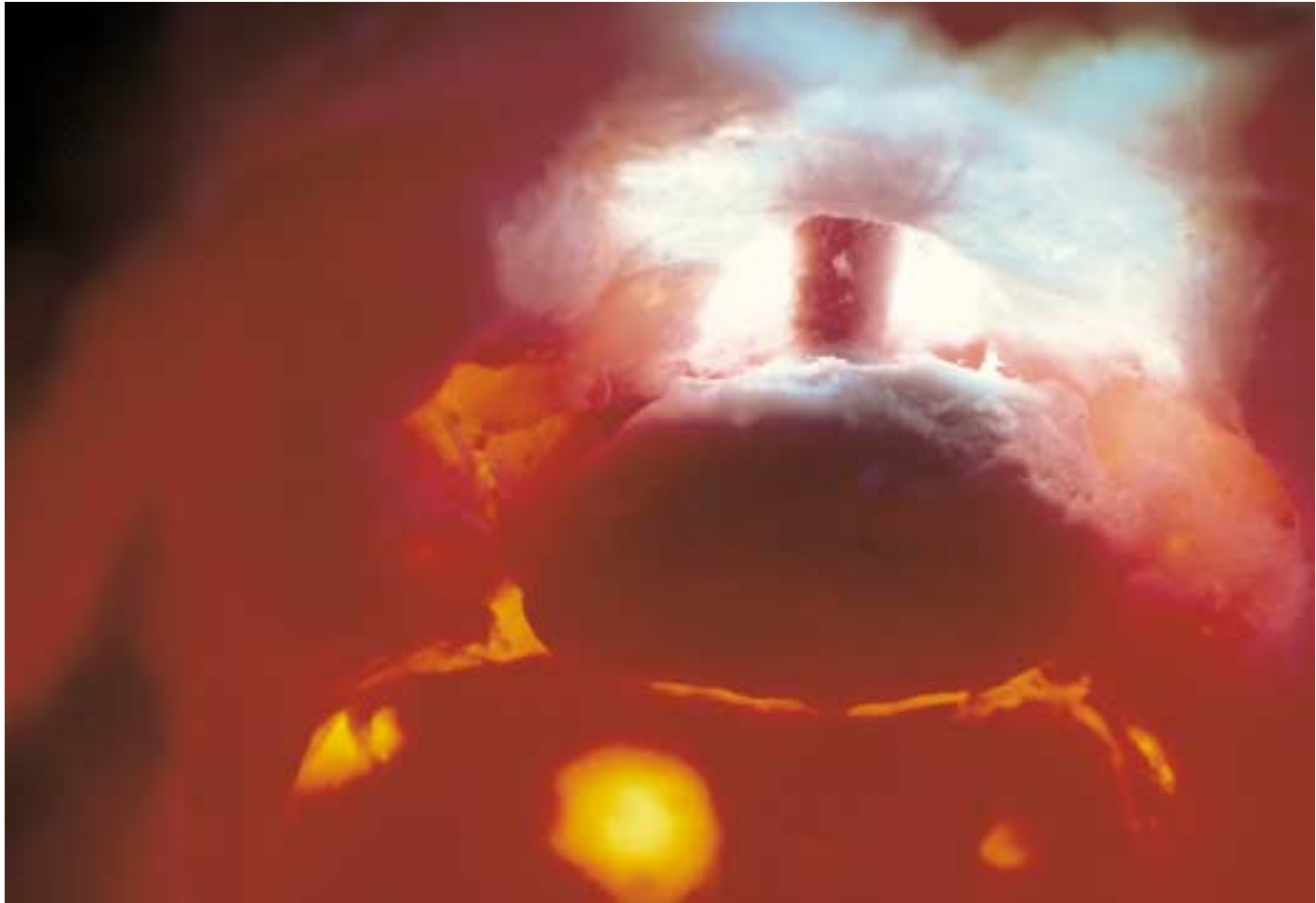
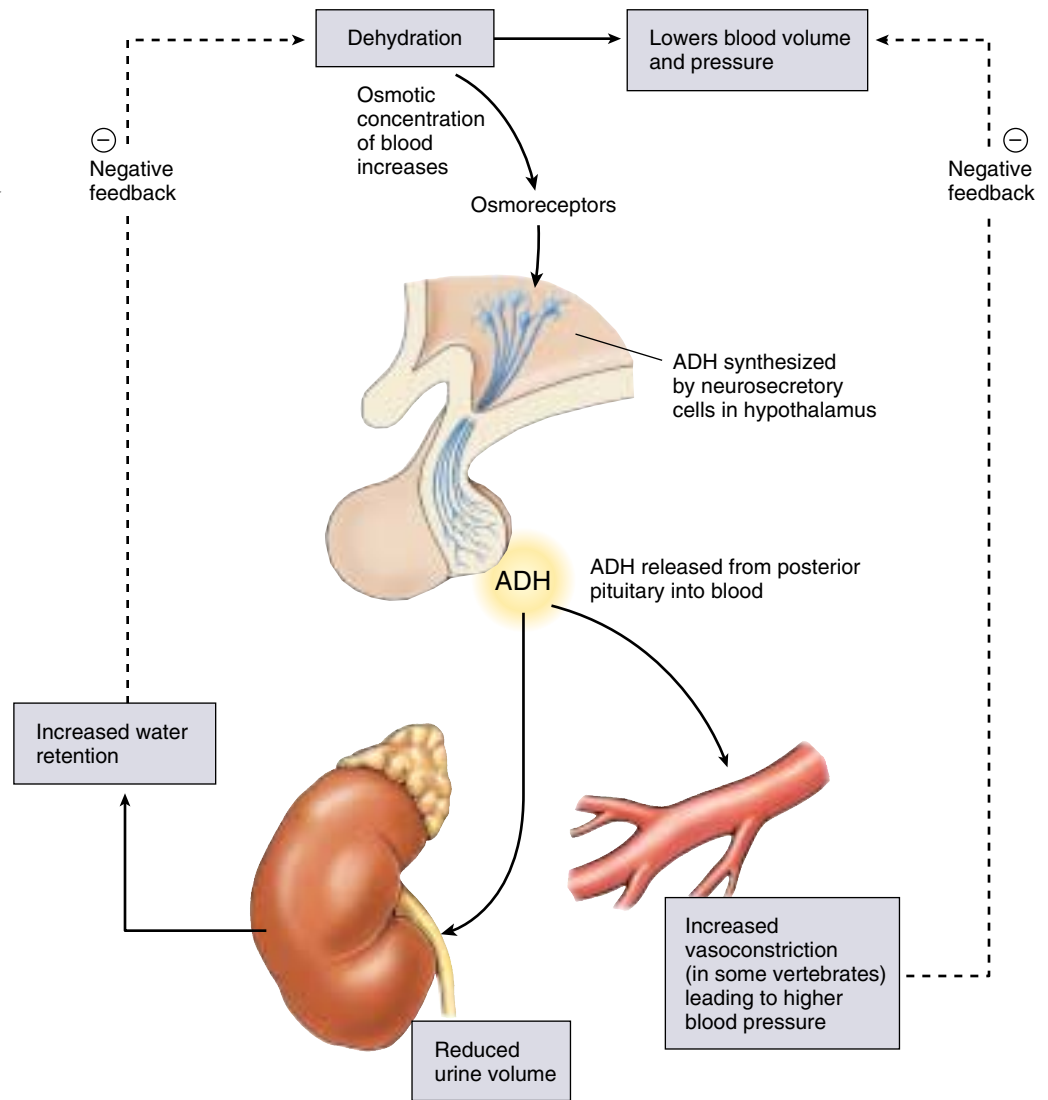


FIGURE 56.9

The pituitary gland hangs by a short stalk from the hypothalamus. The pituitary gland (the oval structure hanging from the stalk), shown here enlarged 15 times, regulates hormone production in many of the body's endocrine glands.

FIGURE 56.10

The effects of antidiuretic hormone (ADH). An increase in the osmotic concentration of the blood stimulates the posterior pituitary gland to secrete ADH, which promotes water retention by the kidneys. This works as a negative feedback loop to correct the initial disturbance of homeostasis.



(ADH, also known as **vasopressin**), that stimulates water retention by the kidneys (figure 56.10). When ADH is missing, the kidneys do not retain water and excessive quantities of urine are produced. This is why the consumption of alcohol, which inhibits ADH secretion, leads to frequent urination.

The posterior pituitary also secretes **oxytocin**, a second peptide hormone which, like ADH, is composed of nine amino acids. Oxytocin stimulates the milk-ejection reflex, so that contraction of the smooth muscles around the mammary glands and ducts causes milk to be ejected from the ducts through the nipple. During suckling, sensory receptors in the nipples send impulses to the hypothalamus, which triggers the release of oxytocin. Oxytocin is also needed to stimulate uterine contractions in women during childbirth. Oxytocin secretion continues after childbirth in a woman who is breast-feeding, which is why the uterus of a nursing mother returns to its normal size after pregnancy more quickly than does the uterus of a mother who does not breast-feed.

ADH and oxytocin are actually *produced* by neuron cell bodies located in the hypothalamus. These two hormones are transported along the axon tract that runs from the hypothalamus to the posterior pituitary and are stored in the posterior pituitary. In response to the appropriate stimulation—increased blood plasma osmolarity in the case of ADH, the suckling of a baby in the case of oxytocin—the hormones are released by the posterior pituitary into the blood. Because this reflex control involves both the nervous and endocrine systems, the secretion of ADH and oxytocin are **neuroendocrine reflexes**.

The posterior pituitary gland contains axons originating from neurons in the hypothalamus. These neurons produce ADH and oxytocin, which are stored in and released from the posterior pituitary gland in response to neural stimulation from the hypothalamus.

The Anterior Pituitary Gland

The anterior pituitary, unlike the posterior pituitary, does not develop from a downgrowth of the brain; instead, it develops from a pouch of epithelial tissue that pinches off from the roof of the embryo's mouth. Because it is epithelial tissue, the anterior pituitary is a complete gland—it produces the hormones it secretes. Many, but not all, of these hormones stimulate growth in their target organs, including other endocrine glands. Therefore, the hormones of the anterior pituitary gland are collectively termed *tropic hormones* (Greek *trophe*, “nourishment”), or *tropins*. When the target organ of a tropic hormone is another endocrine gland, that gland is stimulated by the tropic hormone to secrete its own hormones.

The hormones produced and secreted by different cell types in the anterior pituitary gland (figure 56.11) include the following:

1. **Growth hormone (GH, or *somatotropin*)** stimulates the growth of muscle, bone (indirectly), and other tissues and is also essential for proper metabolic regulation.
2. **Adrenocorticotrophic hormone (ACTH, or *corticotropin*)** stimulates the adrenal cortex to produce corticosteroid hormones, including cortisol (in humans) and corticosterone (in many other vertebrates), which regulate glucose homeostasis.
3. **Thyroid-stimulating hormone (TSH, or *thyrotropin*)** stimulates the thyroid gland to produce thyroxine, which in turn stimulates oxidative respiration.
4. **Luteinizing hormone (LH)** is needed for ovulation and the formation of a corpus luteum in the female menstrual cycle (see chapter 59). It also stimulates the testes to produce testosterone, which is needed for sperm production and for the development of male secondary sexual characteristics.

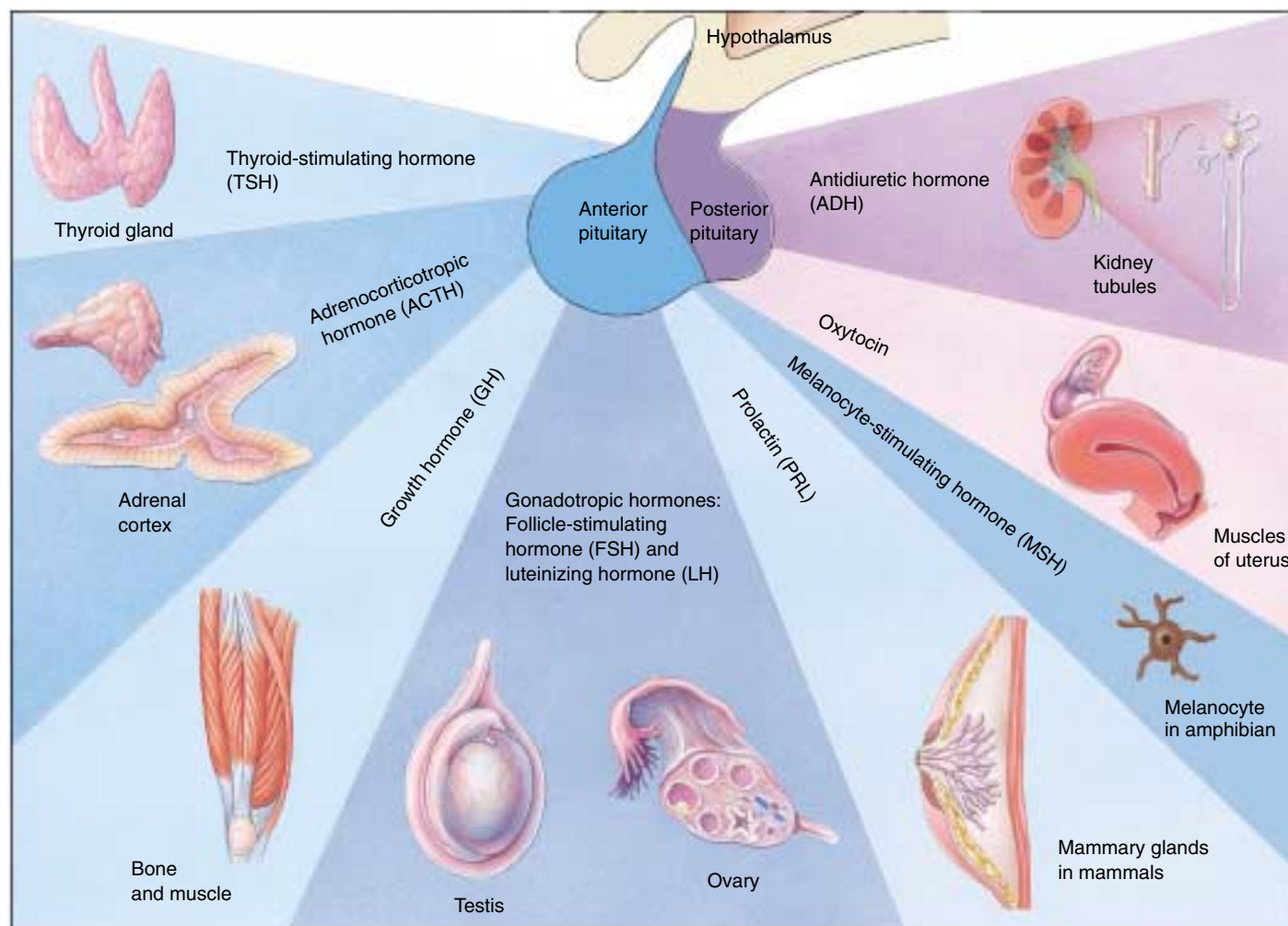


FIGURE 56.11

The major hormones of the anterior and posterior pituitary glands. Only a few of the actions of these hormones are shown.

5. **Follicle-stimulating hormone (FSH)** is required for the development of ovarian follicles in females. In males, it is required for the development of sperm. FSH and LH are both referred to as *gonadotropins*.
6. **Prolactin (PRL)** stimulates the mammary glands to produce milk in mammals. It also helps regulate kidney function in vertebrates, the production of “crop milk” (nutritional fluid fed to chicks by regurgitation) in some birds, and acts on the gills of fish that travel from salt to fresh water to promote sodium retention.
7. **Melanocyte-stimulating hormone (MSH)** stimulates the synthesis and dispersion of melanin pigment, which darken the epidermis of some fish, amphibians, and reptiles. MSH has no known specific function in mammals, but abnormally high amounts of ACTH can cause skin darkening because it contains the amino acid sequence of MSH within its structure.

Growth Hormone

The importance of the anterior pituitary gland first became understood in 1909, when a 38-year-old South Dakota farmer was cured of the growth disorder *acromegaly* by the surgical removal of a pituitary tumor. Acromegaly is a form of gigantism in which the jaw begins to protrude and other facial features thicken. It was discovered that gigantism is almost always associated with pituitary tumors. Robert Wadlow, born in 1928 in Alton, Illinois, stood 8 feet, 11 inches tall and weighed 485 pounds before he died from infection at age 22 (figure 56.12). He was the tallest human being ever recorded, and he was still growing the year he died.

We now know that gigantism is caused by the excessive secretion of growth hormone (GH) by the anterior pituitary gland in a growing child. GH stimulates protein synthesis and growth of muscles and connective tissues; it also indirectly promotes the elongation of bones by stimulating cell division in the cartilaginous epiphyseal growth plates of bones. Researchers found that this stimulation does not occur in the absence of blood plasma, suggesting that bone cells lack receptors for GH and that the stimulation by GH was indirect. We now know that GH stimulates the production of **insulin-like growth factors**, which are produced by the liver and secreted into the blood in response to stimulation by GH. The insulin-like growth factors then stimulate growth of the epiphyseal growth plates and thus elongation of the bones.

When a person's skeletal growth plates have converted from cartilage into bone, however, GH can no longer cause an increase in height. Therefore, excessive GH secretion in an adult produces bone and soft tissue deformities in the condition called acromegaly. A deficiency in GH secretion during childhood results in pituitary dwarfism, a failure to achieve normal growth.



FIGURE 56.12

The Alton giant. This photograph of Robert Wadlow of Alton, Illinois, taken on his 21st birthday, shows him at home with his father and mother and four siblings. Born normal size, he developed a growth hormone-secreting pituitary tumor as a young child and never stopped growing during his 22 years of life.

Other Anterior Pituitary Hormones

Prolactin is like growth hormone in that it acts on organs that are not endocrine glands. In addition to its stimulation of milk production in mammals and “crop milk” production in birds, prolactin has varied effects on electrolyte balance by acting on the kidneys, the gills of fish, and the salt glands of marine birds (discussed in chapter 58). Unlike growth hormone and prolactin, the other anterior pituitary hormones act on specific glands.

Some of the anterior pituitary hormones that act on specific glands have common names, such as thyroid-stimulating hormone (TSH), and alternative names that emphasize the tropic nature of the hormone, such as thyrotropin. TSH stimulates only the thyroid gland, and adrenocorticotropic hormone (ACTH) stimulates only the adrenal cortex (outer portion of the adrenal glands). Follicle-stimulating hormone (FSH) and luteinizing hormone (LH) act only on the gonads (testes and ovaries); hence, they are collectively called *gonadotropic hormones*. Although both FSH and LH act on the gonads, they each act on different target cells in the gonads of both females and males.

Hypothalamic Control of Anterior Pituitary Gland Secretion

The anterior pituitary gland, unlike the posterior pituitary, is not derived from the brain and does not receive an axon tract from the hypothalamus. Nevertheless, the hypothalamus controls production and secretion of the anterior pituitary hormones. This control is exerted hormonally rather than by means of nerve axons. Neurons in the hypothalamus secrete releasing hormones and inhibiting hormones into blood capillaries at the base of the hypothalamus (figure 56.13). These capillaries drain into small veins that run within the stalk of the pituitary to a second bed of capillaries in the anterior pituitary. This unusual system of vessels is known as the hypothalamohypophyseal portal system (another name for the pituitary is the hypophysis). It is called a portal system because it has a second capillary bed downstream from the first; the only other body location with a similar system is the liver, where capillaries receive blood drained from the gastrointestinal tract (via the hepatic portal vein—see chapter 52). Because the second bed of capillaries receives little oxygen from such vessels, the vessels must be delivering something else of importance.

Each releasing hormone delivered by the hypothalamohypophyseal system regulates the secretion of a specific anterior pituitary hormone. For example, *thyrotropin-releasing hormone (TRH)* stimulates the release of TSH, *corticotropin-releasing hormone (CRH)* stimulates the release of ACTH, and *gonadotropin-releasing hormone (GnRH)* stimulates the release of FSH and LH. A releasing hormone for growth hormone, called *growth hormone-releasing hormone (GHRH)* has also been discovered, and a releasing hormone for prolactin has been postulated but has thus far not been identified.

The hypothalamus also secretes hormones that *inhibit* the release of certain anterior pituitary hormones. To date, three such hormones have been discovered: *somatostatin* inhibits the secretion of GH; *prolactin-inhibiting factor (PIF)*, found to be the neurotransmitter dopamine, inhibits the secretion of prolactin; and *melanotropin-inhibiting hormone (MIH)* inhibits the secretion of MSH.

Negative Feedback Control of Anterior Pituitary Gland Secretion

Because hypothalamic hormones control the secretions of the anterior pituitary gland, and the anterior pituitary hormones control the secretions of some other endocrine glands, it may seem that the hypothalamus is in charge of hormonal secretion for the whole body. This idea is not valid, however, for two reasons. First, a number of endocrine organs, such as the adrenal medulla and the pancreas, are not directly regulated by this control system. Second, the hypothalamus and the anterior pituitary gland are themselves partially controlled by the very hormones whose secretion they stimulate! In most cases this is an in-

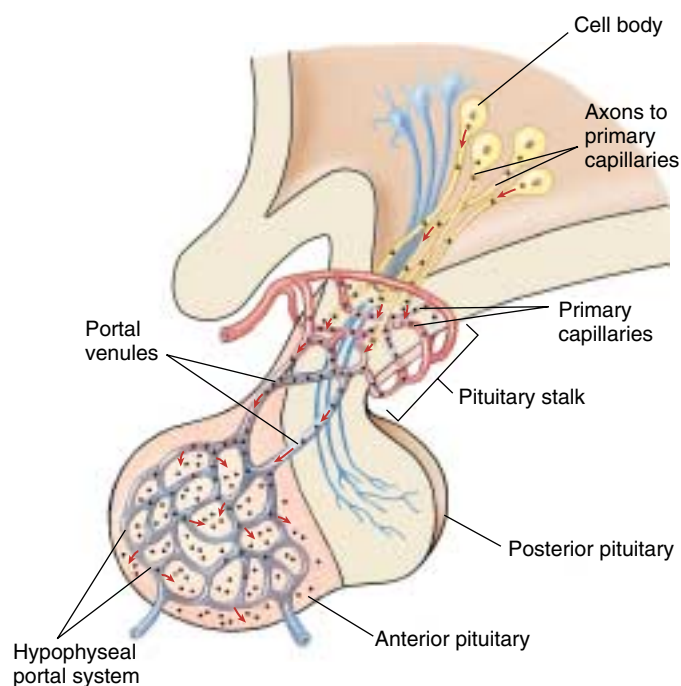


FIGURE 56.13

Hormonal control of the anterior pituitary gland by the hypothalamus. Neurons in the hypothalamus secrete hormones that are carried by short blood vessels directly to the anterior pituitary gland, where they either stimulate or inhibit the secretion of anterior pituitary hormones.

hibitory control, where the target gland hormones inhibit the secretions of the hypothalamus and anterior pituitary (figure 56.14). This type of control system is called negative feedback inhibition and acts to maintain relatively constant levels of the target cell hormone.

Let's consider the hormonal control of the thyroid gland. The hypothalamus secretes TRH into the hypothalamohypophyseal portal system, which stimulates the anterior pituitary gland to secrete TSH, which in turn stimulates the thyroid gland to release thyroxine. Among thyroxine's many target organs are the hypothalamus and the anterior pituitary gland themselves. Thyroxine acts upon these organs to inhibit their secretion of TRH and TSH, respectively (figure 56.15). This negative feedback inhibition is essential for homeostasis because it keeps the thyroxine levels fairly constant.

To illustrate the importance of negative feedback inhibition, we will examine a person who lacks sufficient iodine in the diet. Without iodine, the thyroid gland cannot produce thyroxine (which contains four iodines per molecule). As a result, thyroxine levels in the blood fall drastically, and the hypothalamus and anterior pituitary receive far less negative feedback inhibition than is normal. This reduced inhibition causes an elevated secretion of TRH and TSH. The high levels of TSH stim-

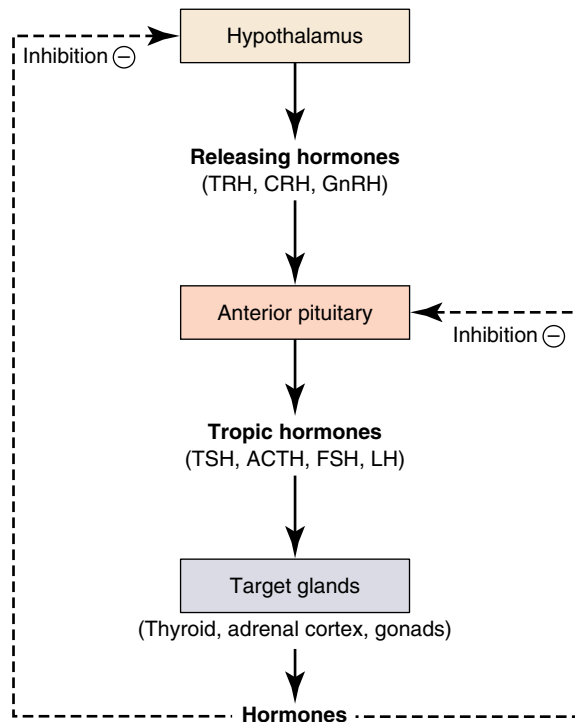


FIGURE 56.14

Negative feedback inhibition. The hormones secreted by some endocrine glands feed back to inhibit the secretion of hypothalamic releasing hormones and anterior pituitary tropic hormones.

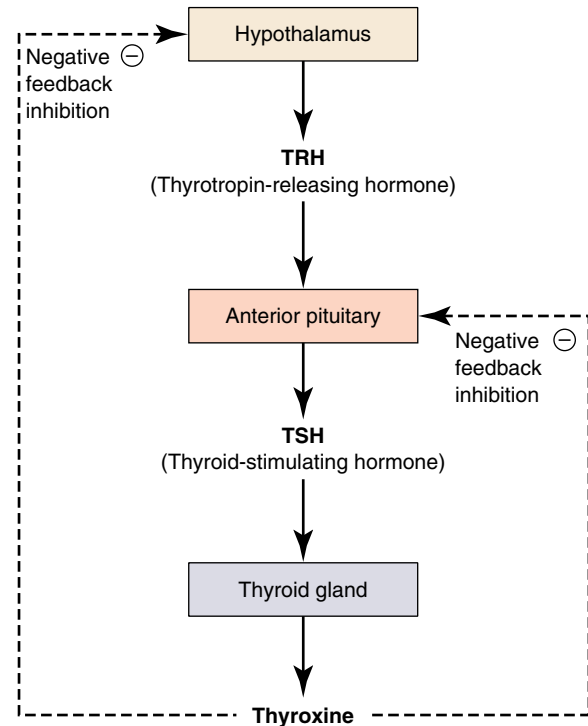


FIGURE 56.15

Regulation of thyroxine secretion. The hypothalamus secretes TRH, which stimulates the anterior pituitary to secrete TSH. The TSH then stimulates the thyroid to secrete thyroxine, which exerts negative feedback control of the hypothalamus and anterior pituitary.

ulate the thyroid gland to grow, but it still cannot produce thyroxine without iodine. The consequence of this interruption of the normal inhibition by thyroxine is an enlarged thyroid gland, a condition known as a goiter (figure 56.16).

Positive feedback in the control of the hypothalamus and anterior pituitary by the target glands is not common because positive feedback cannot maintain constancy of the internal environment (homeostasis). Positive feedback accentuates change, driving the change in the same direction. One example of positive control involves the control of ovulation, an explosive event that culminates in the expulsion of the egg cell from the ovary. In that case, an ovarian hormone, estradiol, actually stimulates the secretion of an anterior pituitary hormone, LH. This will be discussed in detail in chapter 59.

The hypothalamus controls the anterior pituitary gland by means of hormones, and the anterior pituitary gland controls some other glands through the hormones it secretes. However, both the hypothalamus and the anterior pituitary gland are controlled by other glands through negative feedback inhibition.



FIGURE 56.16

A person with a goiter. This condition is caused by a lack of iodine in the diet. As a result, thyroxine secretion is low, so there is less negative feedback inhibition of TSH. The elevated TSH secretion, in turn, stimulates the thyroid to grow and produce the goiter.

56.4 Endocrine glands secrete hormones that regulate many body functions.

The Thyroid and Parathyroid Glands

The endocrine glands that are regulated by the anterior pituitary, and those endocrine glands that are regulated by other means, help to control metabolism, electrolyte balance, and reproductive functions. Some of the major endocrine glands will be considered in this section.

The Thyroid Gland

The thyroid gland (Greek *thyros*, “shield”) is shaped like a shield and lies just below the Adam’s apple in the front of the neck. We have already mentioned that the thyroid gland secretes thyroxine and smaller amounts of triiodothyronine (T_3), which stimulate oxidative respiration in most cells in the body and, in so doing, help set the body’s basal metabolic rate (see chapter 51). In children, these thyroid hormones also promote growth and stimulate maturation of the central nervous system. Children with underactive thyroid glands are therefore stunted in their growth and suffer severe mental retardation, a condition called cretinism. This differs from pituitary dwarfism, which results from inadequate GH and is not associated with abnormal intellectual development.

People who are hypothyroid (whose secretion of thyroxine is too low) can take thyroxine orally, as pills. Only thyroxine and the steroid hormones (as in contraceptive pills), can be taken orally because they are nonpolar and can pass through the plasma membranes of intestinal epithelial cells without being digested.

There is an additional function of the thyroid gland that is unique to amphibians—thyroid hormones are needed for the metamorphosis of the larvae into adults (figure 56.17). If the thyroid gland is removed from a tadpole, it will not change into a frog. Conversely, if an immature tadpole is fed pieces of a thyroid gland, it will undergo premature metamorphosis and become a miniature frog!

The thyroid gland also secretes calcitonin, a peptide hormone that plays a role in maintaining proper levels of calcium (Ca^{++}) in the blood. When the blood Ca^{++} concentration rises too high, calcitonin stimulates the uptake of Ca^{++} into bones, thus lowering its level in the blood. Although calcitonin may be important in the physiology of some vertebrates, its significance in normal human

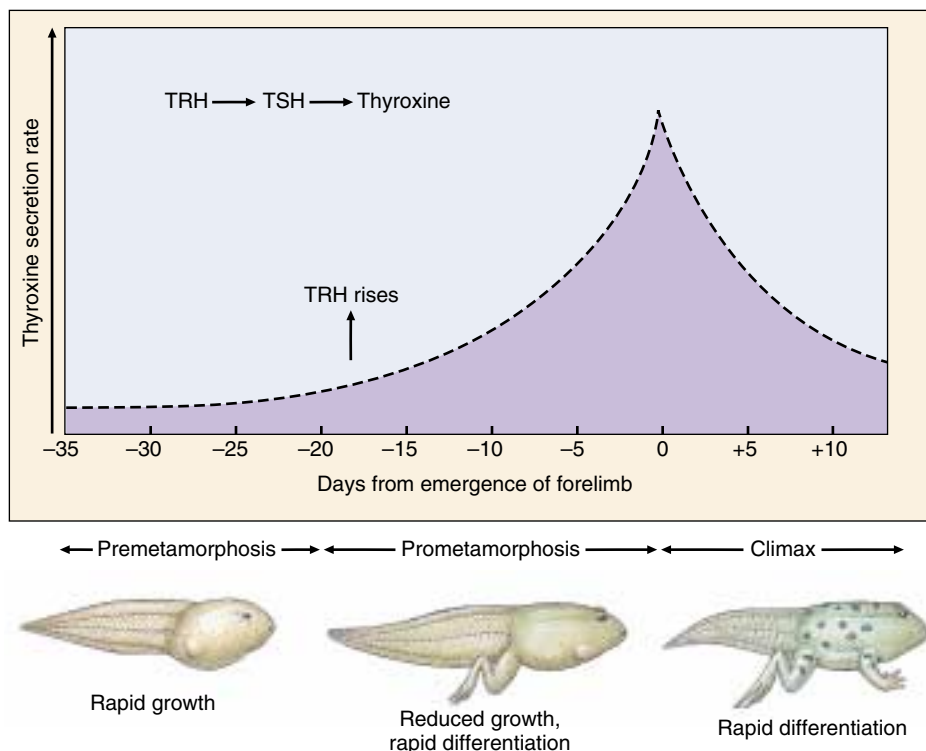


FIGURE 56.17

Thyroxine triggers metamorphosis in amphibians. In tadpoles at the premetamorphic stage, the hypothalamus releases TRH (thyrotropin-releasing hormone), which causes the anterior pituitary to secrete TSH (thyroid-stimulating hormone). TSH then acts on the thyroid gland, which secretes thyroxine. The hindlimbs then begin to form. As metamorphosis proceeds, thyroxine reaches its maximal level, after which the forelimbs begin to form.

physiology is controversial, and it appears less important in the day-to-day regulation of Ca^{++} levels. A hormone that plays a more important role in Ca^{++} homeostasis is secreted by the parathyroid glands, described in the next section.

The Parathyroid Glands and Calcium Homeostasis

The parathyroid glands are four small glands attached to the thyroid. Because of their size, researchers ignored them until well into this century. The first suggestion that these organs have an endocrine function came from experiments on dogs: if their parathyroid glands were removed, the Ca^{++} concentration in the dogs’ blood plummeted to less than half the normal value. The Ca^{++} concentration returned to normal when an extract of parathyroid gland was administered. However, if too much of the extract was administered, the dogs’ Ca^{++} levels rose far above normal as the calcium phosphate crystals in their bones was dissolved. It was clear that the parathyroid glands produce a hormone that stimulates the release of Ca^{++} from bone.

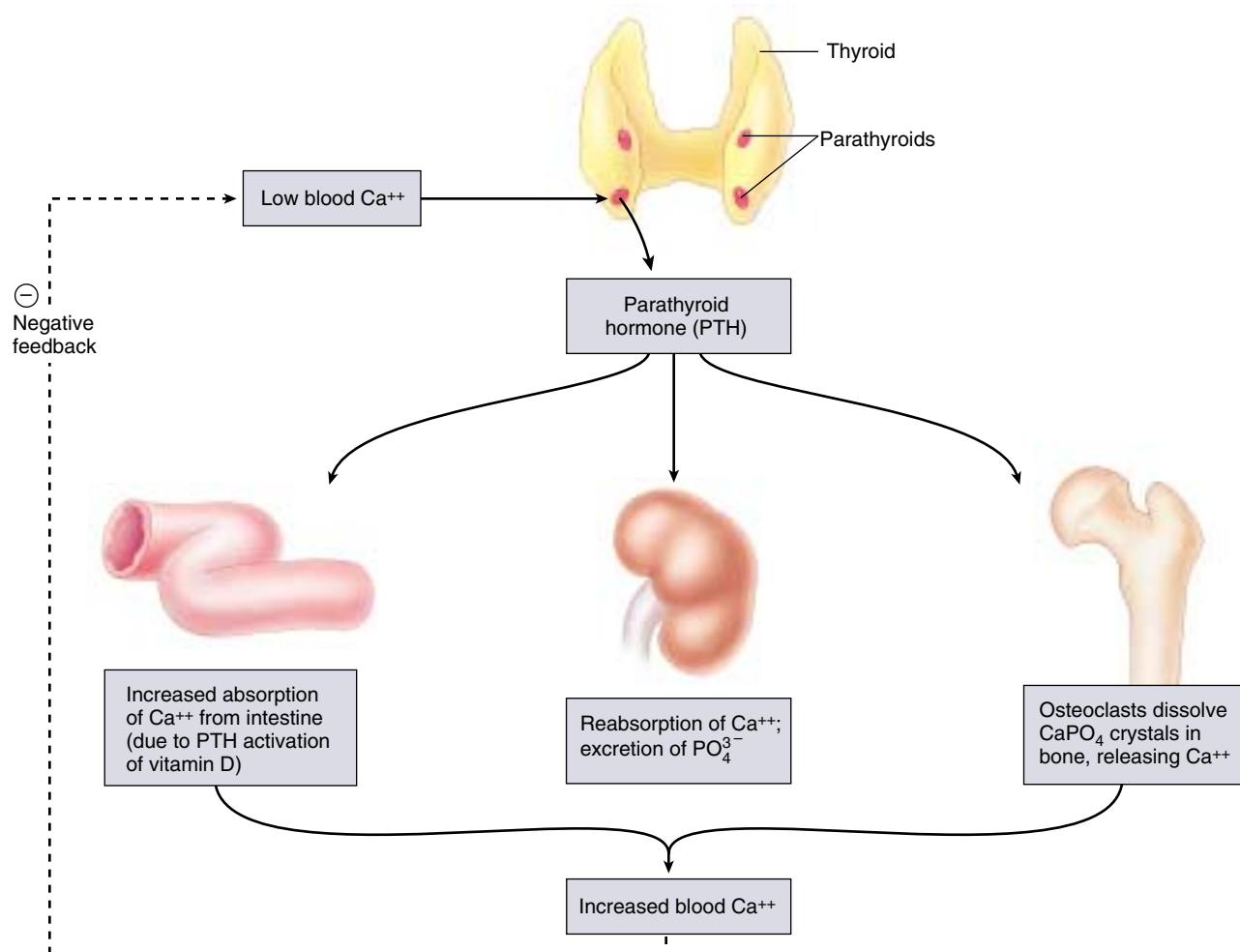


FIGURE 56.18

Regulation of blood Ca⁺⁺ levels by parathyroid hormone (PTH). When blood Ca⁺⁺ levels are low, parathyroid hormone (PTH) is released by the parathyroid glands. PTH directly stimulates the dissolution of bone and the reabsorption of Ca⁺⁺ by the kidneys. PTH indirectly promotes the intestinal absorption of Ca⁺⁺ by stimulating the production of the active form of vitamin D.

The hormone produced by the parathyroid glands is a peptide called parathyroid hormone (PTH). It is one of only two hormones in humans that are absolutely essential for survival (the other is aldosterone, which will be discussed in the next section). PTH is synthesized and released in response to falling levels of Ca⁺⁺ in the blood. This cannot be allowed to continue uncorrected, because a significant fall in the blood Ca⁺⁺ level can cause severe muscle spasms. A normal blood Ca⁺⁺ is important for the functioning of muscles, including the heart, and for proper functioning of the nervous and endocrine systems.

PTH stimulates the osteoclasts (bone cells) in bone to dissolve the calcium phosphate crystals of the bone matrix and release Ca⁺⁺ into the blood (figure 56.18). PTH also stimulates the kidneys to reabsorb Ca⁺⁺ from the urine and leads to the activation of vitamin D, needed for the absorption of Ca⁺⁺ from food in the intestine.

Vitamin D is produced in the skin from a cholesterol derivative in response to ultraviolet light. It is called a vitamin

because a dietary source is needed to supplement the amount that the skin produces. Secreted into the blood from the skin, vitamin D is actually an inactive form of a hormone. In order to become activated, the molecule must gain two hydroxyl groups (—OH); one of these is added by an enzyme in the liver, the other by an enzyme in the kidneys. The enzyme needed for this final step is stimulated by PTH, thereby producing the active form of vitamin D known as 1, 25-dihydroxyvitamin D. This hormone stimulates the intestinal absorption of Ca⁺⁺ and thereby helps to raise blood Ca⁺⁺ levels so that bone can become properly mineralized. A diet deficient in vitamin D thus leads to poor bone formation, a condition called rickets.

Thyroxine helps to set the basal metabolic rate by stimulating the rate of cell respiration throughout the body; this hormone is also needed for amphibian metamorphosis. Parathyroid hormone acts to raise the blood Ca⁺⁺ levels, in part by stimulating osteoclasts to dissolve bone.

The Adrenal Glands

The adrenal glands are located just above each kidney (figure 56.19). Each gland is composed of an inner portion, the adrenal medulla, and an outer layer, the adrenal cortex.

The Adrenal Medulla

The adrenal medulla receives neural input from axons of the sympathetic division of the autonomic nervous system, and it secretes epinephrine and norepinephrine in response to stimulation by these axons. The actions of these hormones trigger “alarm” responses similar to those elicited by the sympathetic division, helping to prepare the body for “fight or flight.” Among the effects of these hormones are an increased heart rate, increased blood pressure, dilation of the bronchioles, elevation in blood glucose, and reduced blood flow to the skin and digestive organs. The actions of epinephrine released as a hormone supplement those of norepinephrine released as a sympathetic nerve neurotransmitter.

The Adrenal Cortex: Homeostasis of Glucose and Na^+

The hormones from the adrenal cortex are all steroids and are referred to collectively as *corticosteroids*. Cortisol (also called hydrocortisone) and related steroids secreted by the adrenal cortex act on various cells in the body to maintain glucose homeostasis. In mammals, these hormones are referred to as glucocorticoids. The glucocorticoids stimulate the breakdown of muscle protein into amino acids, which are carried by the blood to the liver. They also stimulate the liver to produce the enzymes needed for gluconeogenesis, the conversion of amino acids into glucose. This creation of glucose from protein is particularly important during very long periods of fasting or exercise, when blood glucose levels might otherwise become dangerously low.

In addition to regulating glucose metabolism, the glucocorticoids modulate some aspects of the immune response. Glucocorticoids are given medically to suppress the immune system in persons with immune disorders, such as rheumatoid arthritis. Derivatives of cortisol, such as prednisone, have widespread medical use as antiinflammatory agents.

Aldosterone, the other major corticosteroid, is classified as a mineralocorticoid because it helps regulate mineral balance through two functions. One of the functions of aldosterone is to stimulate the kidneys to reabsorb Na^+ from the urine. (Urine is formed by filtration of blood plasma, so the blood levels of Na^+ will decrease if Na^+ is not reabsorbed from the urine; see chapter 58.) Sodium is the major extracellular solute and is needed for the maintenance of normal blood volume and pressure. Without aldosterone, the kidneys would lose excessive amounts of blood Na^+ in the urine, followed by Cl^- and water; this would cause the blood volume and pressure to fall. By stimulating the kidneys to reabsorb salt and water, aldosterone thus maintains the normal blood volume and pressure essential to life.

The other function of aldosterone is to stimulate the kidneys to secrete K^+ into the urine. Thus, when aldosterone levels are too low, the concentration of K^+ in the blood may rise to dangerous levels. Because of these essential functions performed by aldosterone, removal of the adrenal glands, or diseases that prevent aldosterone secretion, are invariably fatal without hormone therapy.

The adrenal medulla is stimulated by sympathetic neurons to secrete epinephrine and norepinephrine during the fight-or-flight reaction. The adrenal cortex is stimulated to secrete its steroid hormones by ACTH from the anterior pituitary. Cortisol helps to regulate blood glucose and aldosterone acts to regulate blood Na^+ and K^+ levels.

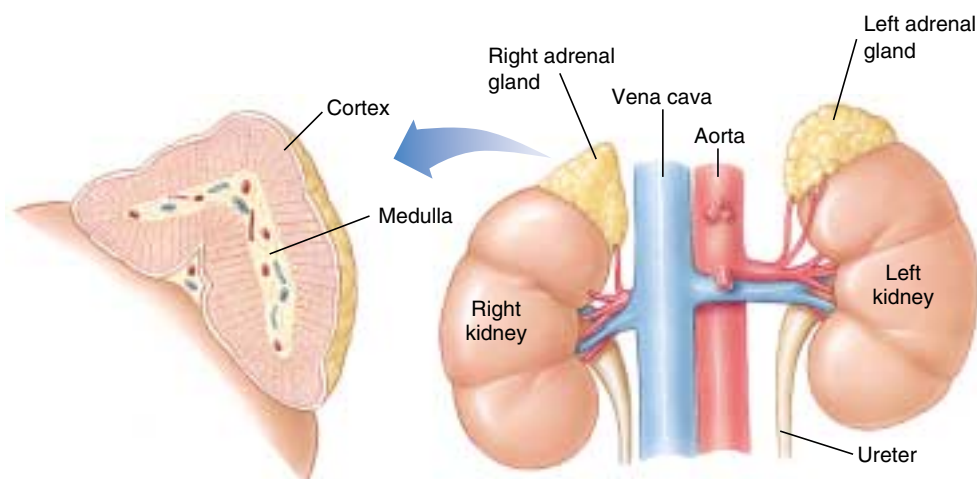


FIGURE 56.19

The adrenal glands. The inner portion of the gland, the adrenal medulla, produces epinephrine and norepinephrine, which initiate a response to stress. The outer portion of the gland, the adrenal cortex, produces steroid hormones that influence blood glucose levels.

The Pancreas

The pancreas is located adjacent to the stomach and is connected to the duodenum of the small intestine by the pancreatic duct. It secretes bicarbonate ions and a variety of digestive enzymes into the small intestine through this duct (see chapter 51), and for a long time the pancreas was thought to be solely an exocrine gland. In 1869, however, a German medical student named Paul Langerhans described some unusual clusters of cells scattered throughout the pancreas; these clusters came to be called islets of Langerhans. Laboratory workers later observed that the surgical removal of the pancreas caused glucose to appear in the urine, the hallmark of the disease diabetes mellitus. This suggested that the pancreas, specifically the islets of Langerhans, might be producing a hormone that prevents this disease.

That hormone is insulin, secreted by the beta (β) cells of the islets. Insulin was not isolated until 1922, when two young doctors working in a Toronto hospital succeeded where many others had not. On January 11, 1922, they injected an extract purified from beef pancreas into a 13-year-old diabetic boy, whose weight had fallen to 65 pounds and who was not expected to survive. With that single injection, the glucose level in the boy's blood fell 25%. A more potent extract soon brought the level down to near normal. The doctors had achieved the first instance of successful insulin therapy.

Two forms of diabetes mellitus are now recognized. People with *type I*, or insulin-dependent diabetes mellitus, lack the insulin-secreting β cells. Treatment for these patients therefore consists of insulin injections. (Because insulin is a peptide hormone, it would be digested if taken orally and must instead be injected into the blood.) In the past, only insulin extracted from the pancreas of pigs or cattle was available, but today people with insulin-dependent diabetes can inject themselves with human insulin produced by genetically engineered bacteria. Active research on the possibility of transplanting islets of Langerhans holds much promise of a lasting treatment for these patients. Most diabetic patients, however, have *type II*, or non-insulin-dependent diabetes mellitus. They generally have normal or even above-normal levels of insulin in their blood, but their cells have a reduced sensitivity to insulin. These

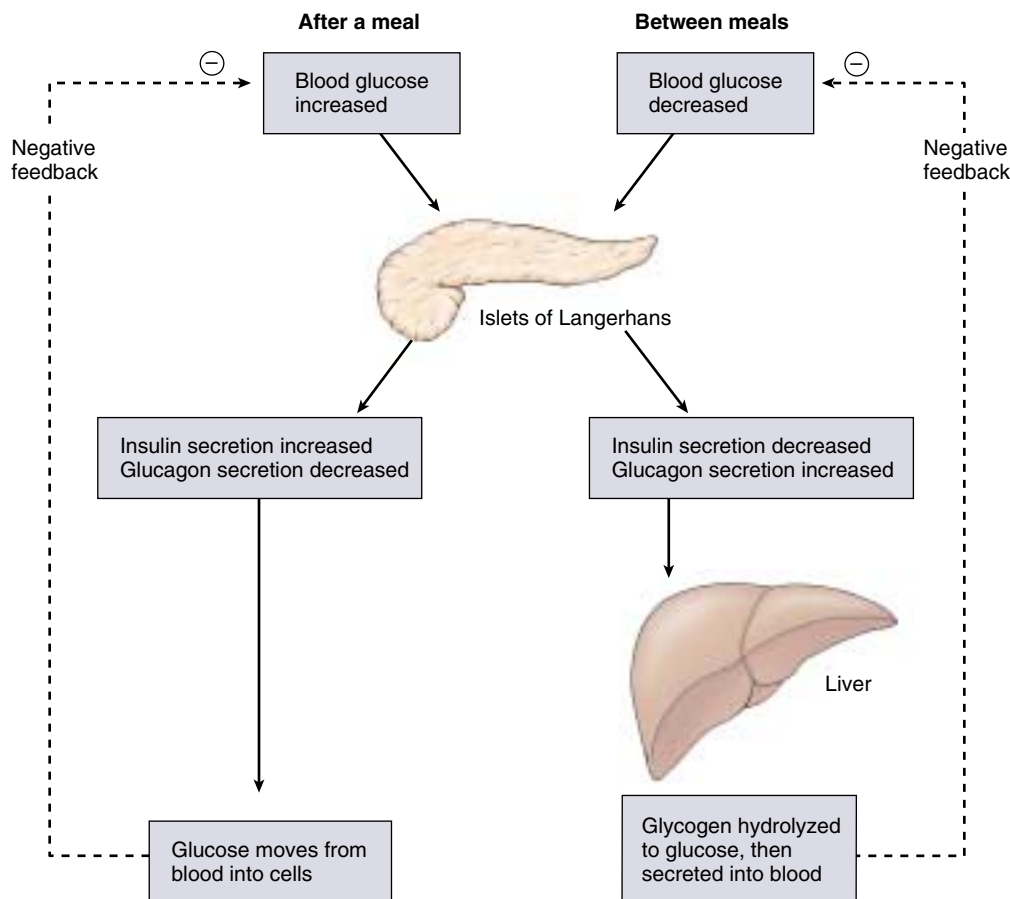


FIGURE 56.20

The antagonistic actions of insulin and glucagon on blood glucose. Insulin stimulates the cellular uptake of blood glucose into skeletal muscles and the liver after a meal. Glucagon stimulates the hydrolysis of liver glycogen between meals, so that the liver can secrete glucose into the blood. These antagonistic effects help to maintain homeostasis of the blood glucose concentration.

people do not require insulin injections and can usually control their diabetes through diet and exercise.

The islets of Langerhans produce another hormone; the alpha (α) cells of the islets secrete glucagon, which acts antagonistically to insulin (figure 56.20). When a person eats carbohydrates, the blood glucose concentration rises. This stimulates the secretion of insulin by β cells and inhibits the secretion of glucagon by the α cells. Insulin promotes the cellular uptake of glucose into liver and muscle cells, where it is stored as glycogen, and into adipose cells, where it is stored as fat. Between meals, when the concentration of blood glucose falls, insulin secretion decreases and glucagon secretion increases. Glucagon promotes the hydrolysis of stored glycogen in the liver and fat in adipose tissue. As a result, glucose and fatty acids are released into the blood and can be taken up by cells and used for energy.

The β cells of the islets of Langerhans secrete insulin, and the α cells secrete glucagon. These two hormones have antagonistic actions on the blood glucose concentration; insulin lowers and glucagon raises blood glucose.

Other Endocrine Glands

Sexual Development, Biological Clocks, and Immune Regulation in Vertebrates

The ovaries and testes are important endocrine glands, producing the steroid sex hormones called androgens (including estrogens, progesterone, and testosterone), to be described in detail in chapter 59. During embryonic development, testosterone production in the embryo is critical for the development of male sex organs. In mammals, androgens are responsible for the development of secondary sexual characteristics at puberty. These characteristics include breasts in females, body hair, and increased muscle mass in males. Because of this, some bodybuilders illegally take androgens to increase muscle mass. In addition to being illegal, this practice can cause liver disorders as well as a number of other serious side effects. In females, androgens are especially important in maintaining the sexual cycle. Estrogen and progesterone produced in the ovaries are critical regulators of the menstrual and ovarian cycles. During pregnancy, estrogen production in the placenta maintains the uterine lining, which protects and nourishes the developing embryo.

Another major endocrine gland is the pineal gland, located in the roof of the third ventricle of the brain in most vertebrates (see figure 54.27). It is about the size of a pea and is shaped like a pine-cone (hence its name). The pineal gland evolved from a median light-sensitive eye (sometimes called a “third eye,” although it could not form images) at the top of the skull in primitive vertebrates. This pineal eye is still present in primitive fish (cyclostomes) and some reptiles. In other vertebrates, however, the pineal gland is buried deep in the brain and functions as an endocrine gland by secreting the hormone melatonin. One of the actions of melatonin is to cause blanching of the skin of lower vertebrates by reducing the dispersal of melanin granules.

The secretion of melatonin is stimulated by activity of the *suprachiasmatic nucleus* (SCN) of the hypothalamus. The SCN is known to function as the major biological clock in vertebrates, entraining (synchronizing) various body processes to a circadian rhythm (one that repeats every 24 hours). Through regulation by the SCN, the secretion of melatonin by the pineal gland is entrained to cycles of light and dark, decreasing during the day and increasing at night. This daily cycling of melatonin release regulates body cycles such as sleep/wake cycles and temperature cycles. In some vertebrates, melatonin helps to regulate reproductive physiology in species with distinct breeding seasons, but the role of melatonin in human reproduction is controversial.

There are a variety of hormones secreted by non-endocrine organs. The thymus is the site of production of particular lymphocytes called T cells, and it secretes a number of hormones that function in the regulation of the immune system. The right atrium of the heart secretes atrial natriuretic hormone, which stimulates the



FIGURE 56.21

A molting cicada. This adult insect is emerging from its old cuticle.

kidneys to excrete salt and water in the urine. This hormone, therefore, acts antagonistically to aldosterone, which promotes salt and water retention. The kidneys secrete erythropoietin, a hormone that stimulates the bone marrow to produce red blood cells. Other organs such as the liver, stomach, and small intestines secrete hormones; even the skin has an endocrine function: it secretes vitamin D. The gas nitric oxide, made by many different cells, controls blood pressure by dilating arteries. The drug sildenafil (Viagra) counters impotence by causing nitric oxide to dilate the blood vessels of the penis.

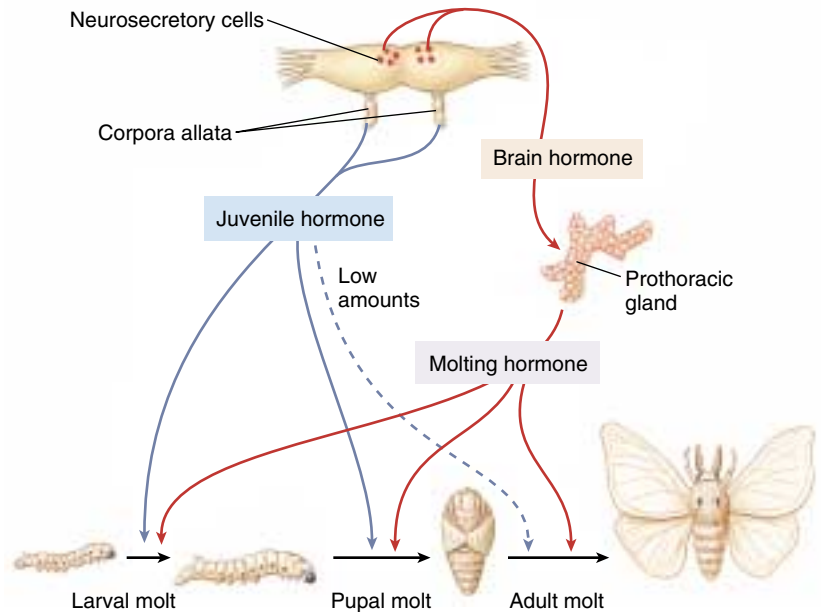
Molting and Metamorphosis in Insects

As insects grow during postembryonic development, their hardened exoskeletons do not expand. To overcome this problem, insects undergo a series of molts wherein they shed their old exoskeleton (figure 56.21) and secrete a new,

FIGURE 56.22

The hormonal control of metamorphosis in the silkworm moth, *Bombyx mori*.

While molting hormone (ecdysone), produced by the prothoracic gland, triggers when molting will occur, juvenile hormone, produced by bodies near the brain called the corpora allata, determines the result of a particular molt. High levels of juvenile hormone inhibit the formation of the pupa and adult forms. At the late stages of metamorphosis, therefore, it is important that the corpora allata not produce large amounts of juvenile hormone.



larger one. In one molt, a juvenile insect, or larvae, often undergoes a radical transformation to the adult. This process is called metamorphosis. Hormonal secretions influence both molting and metamorphosis in insects. Prior to molting, neurosecretory cells on the surface of the brain secrete a small peptide, **brain hormone**, which in turn stimulates a gland in the thorax called the prothoracic gland to produce *molting hormone*, or *ecdysone* (figure 56.22). High levels of ecdysone bring about the biochemical and behavioral changes that cause molting to occur. Another pair of endocrine glands near the brain called the corpora allata produce a hormone called **juvenile hormone**. High levels of juvenile hormone prevent the transformation to the adult and result in a larval to larval molt. If the level of juvenile hormone is low, however, the molt will result in metamorphosis.

Endocrine Disrupting Chemicals

Because target cells are very sensitive to hormones, these hormones are in very low concentrations in the blood. Therefore, small changes in concentrations can make a big difference in how the target organs function. Unfortunately, scientists are now finding that the endocrine system is not sufficiently protected from the outside world. Some man-made chemicals (and even a few plant-produced chemicals) can enter the body and interrupt normal endocrine function. These chemicals may be ones we manufacture for some other purpose and accidentally leak into the environment, ones that are industrial waste products which we “throw away” into the environment, or ones we purposefully release into the environment such as pesticides. These environmental contaminants get into the food

we eat and the air we breathe. They are everywhere on earth and cannot be avoided. Some can last for years in the environment, and just as long in an animal’s body. Those chemicals that interfere with hormone function are called *endocrine disrupting chemicals*.

Any chemical that can bind to receptor proteins and mimic the effects of the hormone is called a hormone *agonist*. Any chemical that binds to receptor proteins and has no effect but blocks the hormone from binding is called a hormone *antagonist*. Endocrine disrupting chemicals can also interfere by binding to the hormone’s protein carriers in the blood. So far, endocrine disrupting chemicals have been shown to interfere with reproductive hormones, thyroid hormones, and the immune system chemical messengers. These effects are not lethal, but may make individuals vulnerable in their environment. If they are having problems reproducing, maintaining the proper metabolic rate, or fighting off infections, then their numbers will decrease (perhaps even leading to extinction). These environmental contaminants may be harming humans in addition to other species. Laws have been passed requiring the testing of thousands of chemicals to see if they have endocrine disrupting potential, and the Environmental Protection Agency (EPA) must include this testing in their standard protocols before approving any new compounds.

Sex steroid hormones from the gonads regulate reproduction, melatonin secreted by the pineal gland helps regulate circadian rhythms, and thymus hormones help regulate the vertebrate immune system. Molting hormone, or ecdysone, and juvenile hormone regulate metamorphosis and molting in insects.



Chapter 56

Summary

Questions

Media Resources

56.1 Regulation is often accomplished by chemical messengers.

- Endocrine glands secrete hormones into the blood, which are then transported to target cells.
- Hormones may be lipophilic, such as the steroid hormones and thyroxine, or polar, such as amine, polypeptide, and glycoprotein hormones.
- Prostaglandins and other paracrine regulatory molecules are produced by one cell type and regulate different cells within the same organ.

1. What is the definition of a hormone? How do hormones reach their target cells? Why are only certain cells capable of being target cells for a particular hormone?
2. How do hormones and paracrine regulators differ from one another?



- Art Activity:
–Endocrine System



- Endocrine System Regulation

56.2 Lipophilic and polar hormones regulate their target cells by different means.

- Lipid-soluble hormones enter their target cells, bind to intracellular receptor proteins, and the complex then binds to hormone response elements on the DNA, activating specific genes.
- Polar hormones do not enter their target cells, but instead bind to receptor proteins on the cell membrane and activate second-messenger systems or control ion channels.

3. How does epinephrine result in the production of cAMP in its target cells? How does cAMP bring about specific changes inside target cells?



- Peptide Hormone Action



- Activity:
–Peptide Hormones
–Steroid Hormones
• Types of Hormones

56.3 The hypothalamus controls the secretions of the pituitary gland.

- Axons from neurons in the hypothalamus enter the posterior pituitary, carrying ADH and oxytocin; the posterior pituitary stores these hormones and secretes them in response to neural activity.
- The anterior pituitary produces and secretes a variety of hormones, many of which control other endocrine glands; the anterior pituitary, however, is itself controlled by the hypothalamus via releasing and inhibiting hormones secreted by the hypothalamus.

4. Where are hormones secreted by the posterior pituitary gland actually *produced*?

5. Why are the hormones of the anterior pituitary gland called tropic hormones?

6. How does the hypothalamus regulate the secretion of the anterior pituitary?



- Portal System



- Hypothalamus
• Pituitary Gland

56.4 Endocrine glands secrete hormones that regulate many body functions.

- The thyroid secretes thyroxine and triiodothyronine, which set the basal metabolic rate by stimulating the rate of cell respiration in most cells of the body.
- The adrenal cortex secretes cortisol, which regulates glucose balance, and aldosterone, which regulates Na⁺ and K⁺ balance.
- The β cells of the islets of Langerhans in the pancreas secrete insulin, which lowers the blood glucose; glucagon, secreted by the α cells, raises the blood glucose level.

7. What hormones are produced by the adrenal cortex? What functions do these hormones serve? What stimulates the secretion of these hormones?

8. What pancreatic hormone is produced when the body's blood glucose level becomes elevated?



- Parathyroid Hormone
• Glucose Regulation



- Thyroid Gland
• Parathyroid Glands
• Adrenal Glands
• Pancreas



57

The Immune System

Concept Outline

57.1 Many of the body's most effective defenses are nonspecific.

Skin: The First Line of Defense. The skin provides a barrier and chemical defenses against foreign bodies.

Cellular Counterattack: The Second Line of Defense. Neutrophils and macrophages kill through phagocytosis; natural killer cells kill by making pores in cells.

The Inflammatory Response. Histamines, phagocytotic cells, and fever may all play a role in local inflammations.

57.2 Specific immune defenses require the recognition of antigens.

The Immune Response: The Third Line of Defense. Lymphocytes target specific antigens for attack.

Cells of the Specific Immune System. B cells and T cells serve different functions in the immune response.

Initiating the Immune Response. T cells must be activated by an antigen-presenting cell.

57.3 T cells organize attacks against invading microbes.

T cells: The Cell-Mediated Immune Response. T cells respond to antigens when presented by MHC proteins.

57.4 B cells label specific cells for destruction.

B Cells: The Humoral Immune Response. Antibodies secreted by B cells label invading microbes for destruction.

Antibodies. Genetic recombination generates millions of B cells, each specialized to produce a particular antibody.

Antibodies in Medical Diagnosis. Antibodies react against certain blood types and pregnancy hormones.

57.5 All animals exhibit nonspecific immune response but specific ones evolved in vertebrates.

Evolution of the Immune System. Invertebrates possess immune elements analogous to those of vertebrates.

57.6 The immune system can be defeated.

T Cell Destruction: AIDS. The AIDS virus suppresses the immune system by selectively destroying helper T cells.

Antigen Shifting. Some microbes change their surface antigens and thus evade the immune system.

Autoimmunity and Allergy. The immune system sometimes causes disease by attacking its own antigens.



FIGURE 57.1

The influenza epidemic of 1918–1919 killed 22 million people in 18 months. With 25 million Americans infected, the Red Cross often worked around the clock.

When you consider how animals defend themselves, it is natural to think of turtles, armadillos, and other animals covered like tanks with heavy plates of armor. However, armor offers no protection against the greatest dangers vertebrates face—microorganisms and viruses. We live in a world awash with attackers too tiny to see with the naked eye, and no vertebrate could long withstand their onslaught unprotected. We survive because we have evolved a variety of very effective defenses against this constant attack. As we review these defenses, it is important to keep in mind that they are far from perfect. Some 22 million Americans and Europeans died from influenza over an 18-month period in 1918–1919 (figure 57.1), and more than 3 million people will die of malaria this year. Attempts to improve our defenses against infection are among the most active areas of scientific research today.

57.1 Many of the body's most effective defenses are nonspecific.

Skin: The First Line of Defense

The vertebrate is defended from infection the same way knights defended medieval cities. “Walls and moats” make entry difficult; “roaming patrols” attack strangers; and “sentries” challenge anyone wandering about and call patrols if a proper “ID” is not presented.

1. **Walls and moats.** The outermost layer of the vertebrate body, the **skin**, is the first barrier to penetration by microbes. Mucous membranes in the respiratory and digestive tracts are also important barriers that protect the body from invasion.
2. **Roaming patrols.** If the first line of defense is penetrated, the response of the body is to mount a **cellular counterattack**, using a battery of cells and chemicals that kill microbes. These defenses act very rapidly after the onset of infection.
3. **Sentries.** Lastly, the body is also guarded by mobile cells that patrol the bloodstream, scanning the surfaces of every cell they encounter. They are part of the **immune system**. One kind of immune cell aggressively attacks and kills any cell identified as foreign, whereas the other type marks the foreign cell or virus for elimination by the roaming patrols.

The Skin as a Barrier to Infection

The skin is the largest organ of the vertebrate body, accounting for 15% of an adult human's total weight. The skin not only defends the body by providing a nearly impenetrable barrier, but also reinforces this defense with chemical weapons on the surface. Oil and sweat glands give the skin's surface a pH of 3 to 5, acidic enough to inhibit the growth of many microorganisms. Sweat also contains the enzyme **lysozyme**, which digests bacterial cell walls. In addition to defending the body against invasion by viruses and microorganisms, the skin prevents excessive loss of water to the air through evaporation.

The epidermis of skin is approximately 10 to 30 cells thick, about as thick as this page. The outer layer, called the stratum corneum, contains cells that are continuously abraded, injured, and worn by friction and stress during the body's many activities. The body deals with this damage not by repairing the cells, but by replacing them. Cells are shed continuously from the stratum corneum and are replaced by new cells produced in the innermost layer of the epidermis, the stratum basale, which contains some of the most actively dividing cells in the vertebrate body. The cells formed in this layer migrate upward and enter a broad intermediate stratum spinosum layer. As they move upward they form the protein keratin, which makes skin tough and water-resistant. These new cells eventually ar-

rive at the stratum corneum, where they normally remain for about a month before they are shed and replaced by newer cells from below. Psoriasis, which afflicts some 4 million Americans, is a chronic skin disorder in which epidermal cells are replaced every 3 to 4 days, about eight times faster than normal.

The dermis of skin is 15 to 40 times thicker than the epidermis. It provides structural support for the epidermis and a matrix for the many blood vessels, nerve endings, muscles, and other structures situated within skin. The wrinkling that occurs as we grow older takes place in the dermis, and the leather used to manufacture belts and shoes is derived from very thick animal dermis.

The layer of subcutaneous tissue below the dermis contains primarily adipose cells. These cells act as shock absorbers and provide insulation, conserving body heat. Subcutaneous tissue varies greatly in thickness in different parts of the body. It is nonexistent in the eyelids, is a half-centimeter thick or more on the soles of the feet, and may be much thicker in other areas of the body, such as the buttocks and thighs.

Other External Surfaces

In addition to the skin, two other potential routes of entry by viruses and microorganisms must be guarded: the *digestive tract* and the *respiratory tract*. Recall that both the digestive and respiratory tracts open to the outside and their surfaces must also protect the body from foreign invaders. Microbes are present in food, but many are killed by saliva (which also contains lysozyme), by the very acidic environment of the stomach, and by digestive enzymes in the intestine. Microorganisms are also present in inhaled air. The cells lining the smaller bronchi and bronchioles secrete a layer of sticky mucus that traps most microorganisms before they can reach the warm, moist lungs, which would provide ideal breeding grounds for them. Other cells lining these passages have cilia that continually sweep the mucus toward the glottis. There it can be swallowed, carrying potential invaders out of the lungs and into the digestive tract. Occasionally, an infectious agent, called a pathogen, will enter the digestive and respiratory systems and the body will use defense mechanisms such as vomiting, diarrhea, coughing, and sneezing to expel the pathogens.

The surface defenses of the body consist of the skin and the mucous membranes lining the digestive and respiratory tracts, which eliminate many microorganisms before they can invade the body tissues.

Cellular Counterattack: The Second Line of Defense

The surface defenses of the vertebrate body are very effective but are occasionally breached, allowing invaders to enter the body. At this point, the body uses a host of non-specific cellular and chemical devices to defend itself. We refer to this as the second line of defense. These devices all have one property in common: they respond to *any* microbial infection without pausing to determine the invader's identity.

Although these cells and chemicals of the nonspecific immune response roam through the body, there is a central location for the collection and distribution of the cells of the immune system; it is called the lymphatic system (see chapter 52). The lymphatic system consists of a network of lymphatic capillaries, ducts, nodes and lymphatic organs (figure 57.2), and although it has other functions involved with circulation, it also stores cells and other agents used in the immune response. These cells are distributed throughout the body to fight infections, and also stored in the lymph nodes where foreign invaders can be eliminated as body fluids pass through.

Cells That Kill Invading Microbes

Perhaps the most important of the vertebrate body's non-specific defenses are white blood cells called leukocytes that circulate through the body and attack invading microbes within tissues. There are three basic kinds of these cells, and each kills invading microorganisms differently.

Macrophages ("big eaters") are large, irregularly shaped cells that kill microbes by ingesting them through *phagocytosis*, much as an amoeba ingests a food particle (figure 57.3). Within the macrophage, the membrane-bound vacuole containing the bacterium fuses with a lysosome. Fusion activates lysosomal enzymes that kill the microbe by liberating large quantities of oxygen free-radicals. Macrophages also engulf viruses, cellular debris, and dust particles in the lungs. Macrophages circulate continuously in the extracellular fluid, and their phagocytic actions supplement those of the specialized phagocytic cells that are part of the structure of the liver, spleen, and bone marrow. In response to an infection, monocytes (an undifferentiated leukocyte) found in the blood squeeze through capillaries to enter the connective tissues. There, at the site of the infection, the monocytes are transformed into additional macrophages.

Neutrophils are leukocytes that, like macrophages, ingest and kill bacteria by phagocytosis. In addition, neutrophils release chemicals (some of which are identical to household bleach) that kill other bacteria in the neighborhood as well as neutrophils themselves.

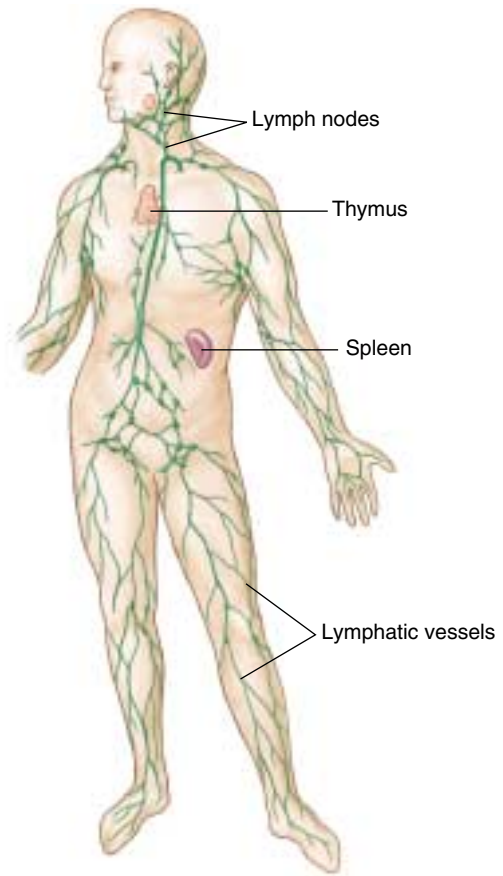


FIGURE 57.2

The lymphatic system. The lymphatic system consists of lymphatic vessels, lymph nodes, and lymphatic organs, including the spleen and thymus gland.



FIGURE 57.3

A macrophage in action (1800 $\times$). In this scanning electron micrograph, a macrophage is "fishing" with long, sticky cytoplasmic extensions. Bacterial cells that come in contact with the extensions are drawn toward the macrophage and engulfed.

Natural killer cells do not attack invading microbes directly. Instead, they kill cells of the body that have been infected with viruses. They kill not by phagocytosis, but rather by creating a hole in the plasma membrane of the target cell (figure 57.4). Proteins, called *perforins*, are released from the natural killer cells and insert into the membrane of the target cell, forming a pore. This pore allows water to rush into the target cell, which then swells and bursts. Natural killer cells also attack cancer cells, often before the cancer cells have had a chance to develop into a detectable tumor. The vigilant surveillance by natural killer cells is one of the body's most potent defenses against cancer.

Proteins That Kill Invading Microbes

The cellular defenses of vertebrates are enhanced by a very effective chemical defense called the *complement system*. This system consists of approximately 20 different proteins that circulate freely in the blood plasma. When they encounter a bacterial or fungal cell wall, these proteins aggregate to form a *membrane attack complex* that inserts itself into the foreign cell's plasma membrane, forming a pore like that produced by natural killer cells (figure 57.5). Water enters the foreign cell through this pore, causing the cell to swell and burst. Aggregation of the complement proteins is also triggered by the binding of antibodies to invading microbes, as we will see in a later section.

The proteins of the complement system can augment the effects of other body defenses. Some amplify the inflammatory response (discussed next) by stimulating histamine release; others attract phagocytes to the area of infection; and still others coat invading microbes, roughening the microbes' surfaces so that phagocytes may attach to them more readily.

Another class of proteins that play a key role in body defense are interferons. There are three major categories of interferons: *alpha*, *beta*, and *gamma*. Almost all cells in the body make alpha and beta interferons. These polypeptides act as messengers that protect normal cells in the vicinity of infected cells from becoming infected. Though viruses are still able to penetrate the neighboring cells, the alpha and beta interferons prevent viral replication and protein assembly in these cells. Gamma interferon is produced only by particular lymphocytes and natural killer cells. The secretion of gamma interferon by these cells is part of the immunological defense against infection and cancer, as we will describe later.

A patrolling army of macrophages, neutrophils, and natural killer cells attacks and destroys invading viruses and bacteria and eliminates infected cells. In addition, a system of proteins called complement may be activated to destroy foreign cells, and body cells infected with a virus secrete proteins called interferons that protect neighboring cells.

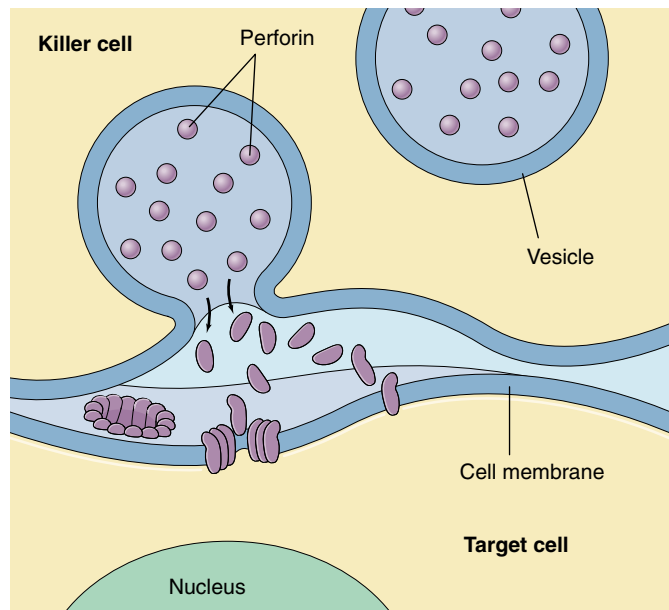


FIGURE 57.4

How natural killer cells kill target cells. The initial event, the tight binding of the killer cell to the target cell, causes vesicles loaded with *perforin* molecules within the killer cell to move to the plasma membrane and disgorge their contents into the intercellular space over the target cell. The perforin molecules insert into the plasma membrane of the target cell like staves of a barrel, forming a pore that admits water and ruptures the cell.

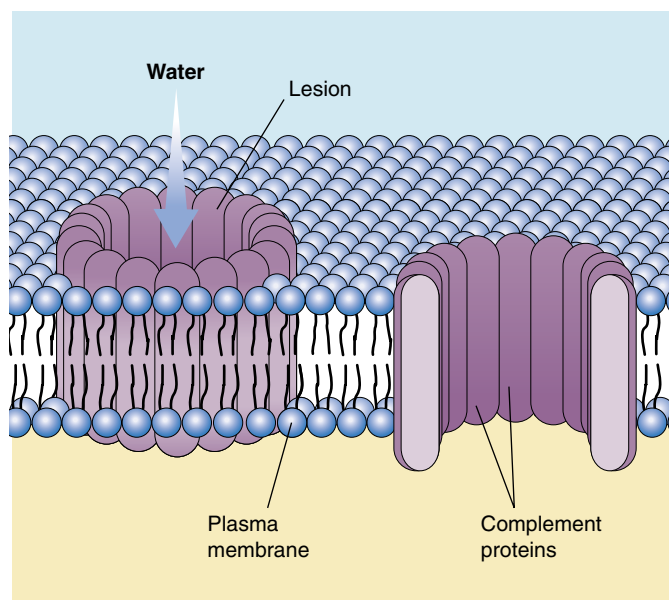


FIGURE 57.5

How complement creates a hole in a cell membrane. As the diagram shows, the complement proteins form a complex transmembrane pore resembling the perforin-lined pores formed by natural killer cells.

The Inflammatory Response

The inflammatory response is a localized, nonspecific response to infection. Infected or injured cells release chemical alarm signals, most notably histamine and prostaglandins. These chemicals promote the dilation of local blood vessels, which increases the flow of blood to the site of infection or injury and causes the area to become red and warm. They also increase the permeability of capillaries in the area, producing the edema (tissue swelling) so often associated with infection. The more permeable capillaries allow phagocytes (monocytes and neutrophils) to migrate from the blood to the extracellular fluid, where they can attack bacteria. Neutrophils arrive first, spilling out chemicals that kill the bacteria in the vicinity (as well as tissue cells and themselves); the *pus* associated with some infections is a mixture of dead or dying pathogens, tissue cells, and neutrophils. Monocytes follow, become macrophages and engulf pathogens and the remains of the dead cells (figure 57.6).

The Temperature Response

Macrophages that encounter invading microbes release a regulatory molecule called interleukin-1, which is carried

by the blood to the brain. Interleukin-1 and other pyrogens (Greek *pyr*, “fire”) such as bacterial endotoxins cause neurons in the hypothalamus to raise the body’s temperature several degrees above the normal value of 37°C (98.6°F). The elevated temperature that results is called a fever.

Experiments with lizards, which regulate their body temperature by moving to warmer or colder locations, demonstrate that infected lizards choose a warmer environment—they give themselves a fever! Further, if lizards are prevented from elevating their body temperature, they have a slower recovery from their infection. Fever contributes to the body’s defense by stimulating phagocytosis and causing the liver and spleen to store iron, reducing blood levels of iron, which bacteria need in large amounts to grow. However, very high fevers are hazardous because excessive heat may inactivate critical enzymes. In general, temperatures greater than 39.4°C (103°F) are considered dangerous for humans, and those greater than 40.6°C (105°F) are often fatal.

Inflammation aids the fight against infection by increasing blood flow to the site and raising temperature to retard bacterial growth.

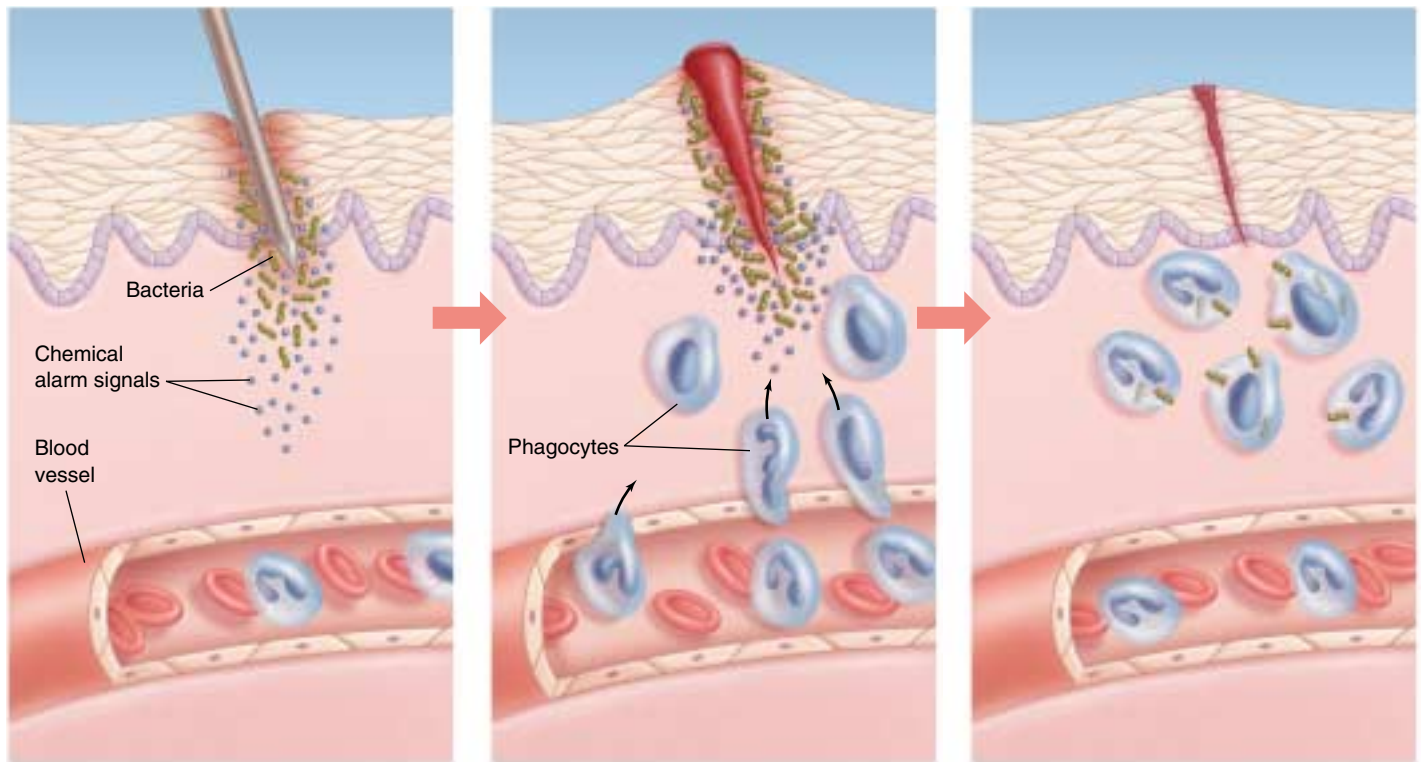


FIGURE 57.6

The events in a local inflammation. When an invading microbe has penetrated the skin, chemicals, such as histamine and prostaglandins, cause nearby blood vessels to dilate. Increased blood flow brings a wave of phagocytic cells, which attack and engulf invading bacteria.

57.2 Specific immune defenses require the recognition of antigens.

The Immune Response: The Third Line of Defense

Few of us pass through childhood without contracting some sort of infection. Chicken pox, for example, is an illness that many of us experience before we reach our teens. It is a disease of childhood, because most of us contract it as children and *never catch it again*. Once you have had the disease, you are usually immune to it. Specific immune defense mechanisms provide this immunity.

Discovery of the Immune Response

In 1796, an English country doctor named Edward Jenner carried out an experiment that marks the beginning of the study of immunology. Smallpox was a common and deadly disease in those days. Jenner observed, however, that milkmaids who had caught a much milder form of “the pox” called cowpox (presumably from cows) rarely caught smallpox. Jenner set out to test the idea that cowpox conferred protection against smallpox. He infected people with cowpox (figure 57.7), and as he had predicted, many of them became immune to smallpox.

We now know that smallpox and cowpox are caused by two different viruses with similar surfaces. Jenner’s patients who were injected with the cowpox virus mounted a defense that was also effective against a later infection of the smallpox virus. Jenner’s procedure of injecting a harmless microbe in order to confer resistance to a dangerous one is called **vaccination**. Modern attempts to develop resistance to malaria, herpes, and other diseases often involve delivering antigens via a harmless vaccinia virus related to cowpox virus.

Many years passed before anyone learned how exposure to an infectious agent can confer resistance to a disease. A key step toward answering this question was taken more than a half-century later by the famous French scientist Louis Pasteur. Pasteur was studying fowl cholera, and he isolated a culture of bacteria from diseased chickens that would produce the disease if injected into healthy birds. Before departing on a two-week vacation, he accidentally left his bacterial culture out on a shelf. When he returned, he injected this old culture into healthy birds and found that it had been weakened; the injected birds became only slightly ill and then recovered. Surprisingly, however, those



FIGURE 57.7
The birth of immunology. This famous painting shows Edward Jenner inoculating patients with cowpox in the 1790s and thus protecting them from smallpox.

birds did not get sick when subsequently infected with fresh fowl cholera. They remained healthy even if given massive doses of active fowl cholera bacteria that did produce the disease in control chickens. Clearly, something about the bacteria could elicit immunity as long as the bacteria did not kill the animals first. We now know that molecules protruding from the surfaces of the bacterial cells evoked active immunity in the chickens.

Key Concepts of Specific Immunity

An **antigen** is a molecule that provokes a specific immune response. Antigens are large, complex molecules such as proteins; they are generally foreign to the body, usually present on the surface of pathogens. A large antigen may have several parts, and each stimulates a different specific immune response. In this case, the different parts are known as **antigenic determinant sites**, and each

serves as a different antigen. Particular lymphocytes have receptor proteins on their surfaces that recognize an antigen and direct a specific immune response against either the antigen or the cell that carries the antigen.

Lymphocytes called B cells respond to antigens by producing proteins called **antibodies**. Antibody proteins are secreted into the blood and other body fluids and thus provide **humoral immunity**. (The term *humor* here is used in its ancient sense, referring to a body fluid.) Other lymphocytes called T cells do not secrete antibodies but instead directly attack the cells that carry the specific antigens. These cells are thus described as producing **cell-mediated immunity**.

The specific immune responses protect the body in two ways. First, an individual can gain immunity by being exposed to a *pathogen* (disease-causing agent) and perhaps getting the disease. This is *acquired immunity*, such as the resistance to the chicken pox that you acquire after having the disease in childhood. Another term for this process is **active immunity**. Second, an individual can gain immunity by obtaining the antibodies from another individual. This happened to you before you were born, with antibodies made by your mother being transferred to you across the placenta. Immunity gained in this way is called **passive immunity**.

Antigens are molecules, usually foreign, that provoke a specific immune attack. This immune attack may involve secreted proteins called antibodies, or it may invoke a cell-mediated attack.

Cells of the Specific Immune System

The immune defense mechanisms of the body involve the actions of white blood cells, or leukocytes. Leukocytes include neutrophils, eosinophils, basophils, and monocytes, all of which are phagocytic and are involved in the second line of defense, as well as two types of lymphocytes (*T cells* and *B cells*), which are not phagocytic but are critical to the specific immune response (table 57.1), the third line of defense. T cells direct the cell-mediated response, B cells the humoral response.

After their origin in the bone marrow, T cells migrate to the thymus (hence the designation “T”), a gland just above the heart. There they develop the ability to identify microorganisms and viruses by the antigens exposed on their surfaces. Tens of millions of different T cells are made, each specializing in the recognition of one particular antigen. No invader can escape being recognized by at least a few T cells. There are four principal kinds of T cells: inducer T cells oversee the development of T cells in the thymus; helper T cells (often symbolized T_H) initiate the immune response; cytotoxic (“cell-poisoning”) T cells (often symbolized T_C) lyse cells that have been infected by viruses; and suppressor T cells terminate the immune response.

Unlike T cells, B cells do not travel to the thymus; they complete their maturation in the bone marrow. (B cells are so named because they were originally characterized in a region of chickens called the bursa.) From the bone marrow, B cells are released to circulate in the blood and lymph. Individual B cells, like T cells, are specialized to recognize particular foreign antigens. When a B cell encounters the antigen to which it is targeted, it begins to divide rapidly, and its progeny differentiate into plasma cells and memory cells. Each plasma cell is a miniature factory producing antibodies that stick like flags to that antigen wherever it occurs in the body, marking any cell bearing the antigen for destruction. The immunity that Pasteur observed resulted from such antibodies and from the continued presence of the B cells that produced them.

The lymphocytes, T cells and B cells, are involved in the specific immune response. T cells develop in the thymus while B cells develop in the bone marrow.

Table 57.1 Cells of the Immune System

Cell Type	Function
Helper T cell	Commander of the immune response; detects infection and sounds the alarm, initiating both T cell and B cell responses
Inducer T cell	Not involved in the immediate response to infection; mediates the maturation of other T cells in the thymus
Cytotoxic T cell	Detects and kills infected body cells; recruited by helper T cells
Suppressor T cell	Dampens the activity of T and B cells, scaling back the defense after the infection has been checked
B cell	Precursor of plasma cell; specialized to recognize specific foreign antigens
Plasma cell	Biochemical factory devoted to the production of antibodies directed against specific foreign antigens
Mast cell	Initiator of the inflammatory response, which aids the arrival of leukocytes at a site of infection; secretes histamine and is important in allergic responses
Monocyte	Precursor of macrophage
Macrophage	The body's first cellular line of defense; also serves as antigen-presenting cell to B and T cells and engulfs antibody-covered cells
Natural killer cell	Recognizes and kills infected body cells; natural killer (NK) cell detects and kills cells infected by a broad range of invaders; killer (K) cell attacks only antibody-coated cells

Initiating the Immune Response

To understand how the third line of defense works, imagine you have just come down with the flu. Influenza viruses enter your body in small water droplets inhaled into your respiratory system. If they avoid becoming ensnared in the mucus lining the respiratory membranes (first line of defense), and avoid consumption by macrophages (second line of defense), the viruses infect and kill mucous membrane cells.

At this point macrophages initiate the immune defense. Macrophages inspect the surfaces of all cells they encounter. The surfaces of most vertebrate cells possess glycoproteins produced by a group of genes called the **major histocompatibility complex (MHC)**. These glycoproteins are called **MHC proteins** or, specifically in humans, **human leukocyte antigens (HLA)**. The genes encoding the MHC proteins are highly polymorphic (have many forms); for example, the human MHC proteins are specified by genes that are the most polymorphic known, with nearly 170 alleles each. Only rarely will two individuals have the same combination of alleles, and the MHC proteins are thus different for each individual, much as fingerprints are. As a result, the MHC proteins on the tissue cells serve as self markers that enable the individual's immune system to distinguish its cells from foreign cells, an ability called **self-versus-nonself**

recognition. T cells of the immune system will recognize a cell as self or nonself by the MHC proteins present on the cell surface.

When a foreign particle, such as a virus, infects the body, it is taken in by cells and partially digested. Within the cells, the viral antigens are processed and moved to the surface of the plasma membrane. The cells that perform this function are known as **antigen-presenting cells** (figure 57.8). At the membrane, the processed antigens are complexed with the MHC proteins. This enables T cells to recognize antigens presented to them associated with the MHC proteins.

There are two classes of MHC proteins. MHC-I is present on every nucleated cell of the body. MHC-II, however, is found only on macrophages, B cells, and a subtype of T cells called CD4⁺ T cells (table 57.2). These three cell types work together in one form of the immune response, and their MHC-II markers permit them to recognize one another. Cytotoxic T lymphocytes, which act to destroy infected cells as previously described, can only interact with antigens presented to them with MHC-I proteins. Helper T lymphocytes, whose functions will soon be described, can interact only with antigens presented with MHC-II proteins. These restrictions result from the presence of coreceptors, which are proteins associated with the T cell receptors. The coreceptor known as CD8 is associated with the cytotoxic T cell receptor (these cells can therefore be indicated as CD8⁺). The CD8 coreceptor can interact only with the MHC-I proteins of an infected cell. The coreceptor known as CD4 is associated with the helper T cell receptor (these cells can thus be indicated as CD4⁺) and interacts only with the MHC-II proteins of another lymphocyte (figure 57.9).

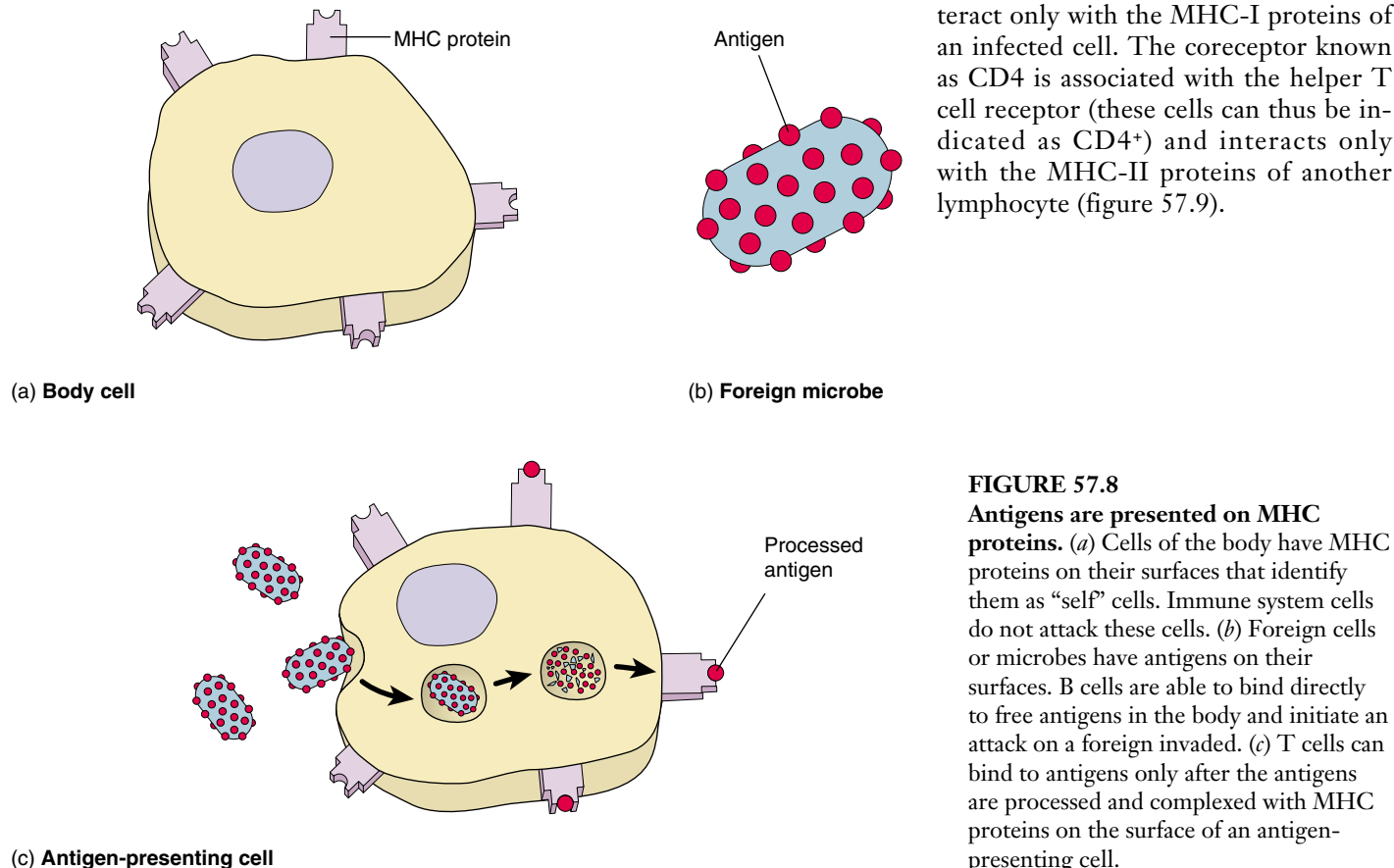


FIGURE 57.8

Antigens are presented on MHC proteins.

(a) Cells of the body have MHC proteins on their surfaces that identify them as “self” cells. Immune system cells do not attack these cells. (b) Foreign cells or microbes have antigens on their surfaces. B cells are able to bind directly to free antigens in the body and initiate an attack on a foreign invader. (c) T cells can bind to antigens only after the antigens are processed and complexed with MHC proteins on the surface of an antigen-presenting cell.

Table 57.2 Key Cell Surface Proteins of the Immune System				
Immune Receptors			MHC Proteins	
Cell Type	T Receptor	B Receptor	MHC-I	MHC-II
B cell	–	+	+	+
CD4 ⁺ T cell	+	–	+	+
CD8 ⁺ T cell	+	–	+	–
Macrophage	–	–	+	+

Note: CD4⁺ T cells include inducer T cells and helper T cells; CD8⁺ T cells include cytotoxic T cells and suppressor T cells. + means present; – means absent.

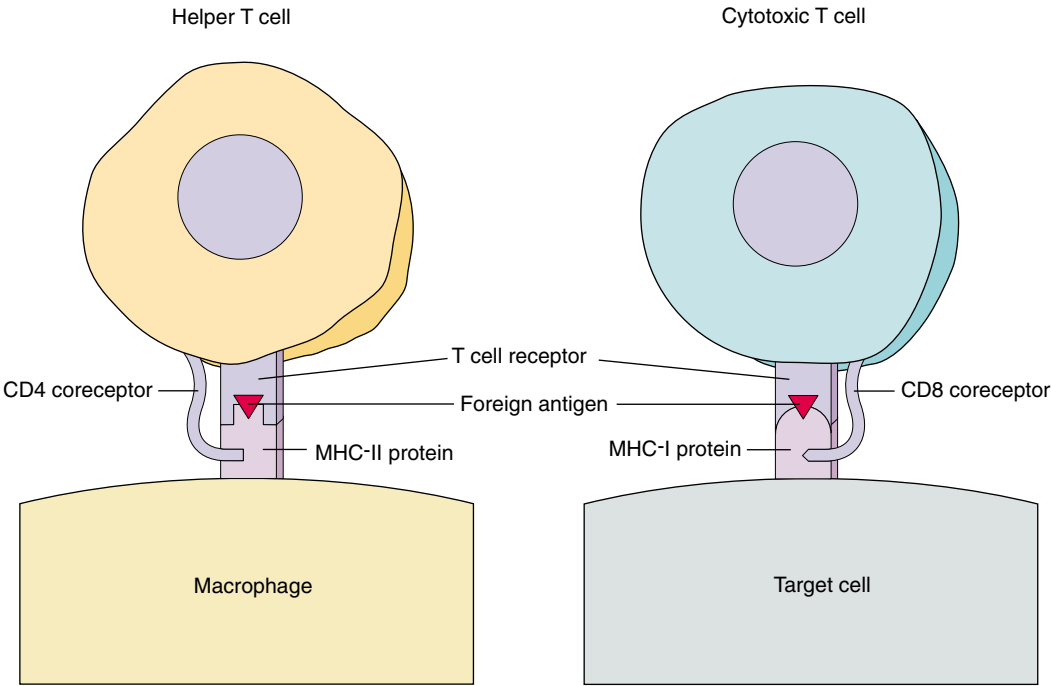


FIGURE 57.9
T cells bind to foreign antigens in conjunction with MHC proteins. The CD4 coreceptor on helper T cells requires that these cells interact with class-2 MHC (or MHC-II) proteins. The CD8 coreceptor on cytotoxic T cells requires that these cells interact only with cells bearing class-1 MHC (or MHC-I) proteins.

Macrophages encounter foreign particles in the body, partially digest the virus particles, and present the foreign antigens in a complex with the MHC-II proteins on its membrane. This combination of MHC-II proteins and foreign antigens is required for interaction with the receptors on the surface of helper T cells. At the same time, macrophages that encounter antigens or antigen-presenting cells release a protein called **interleukin-1** that acts as a chemical alarm signal (discussed in the next section). Helper T cells respond to interleukin-1 by simultaneously initiating two parallel lines of immune system defense: the

cell-mediated response carried out by T cells and the humoral response carried out by B cells.

Antigen-presenting cells must present foreign antigens together with MHC-II proteins in order to activate helper T cells, which have the CD4 coreceptor. Cytotoxic T cells use the CD8 coreceptor and must interact with foreign antigens presented on MHC-I proteins.

57.3 T cells organize attacks against invading microbes.

T cells: The Cell-Mediated Immune Response

The cell-mediated immune response, carried out by T cells, protects the body from virus infection and cancer, killing abnormal or virus-infected body cells.

Once a helper T cell that initiates this response is presented with foreign antigen together with MHC proteins by a macrophage or other antigen-presenting cell, a complex series of steps is initiated. An integral part of this process is the secretion of autocrine regulatory molecules known generally as **cytokines**, or more specifically as **lymphokines** if they are secreted by lymphocytes.

When a cytokine is first discovered, it is named according to its biological activity (such as *B cell-stimulating factor*). However, because each cytokine has many different actions, such names can be misleading. Scientists have thus agreed to

use the name **interleukin**, followed by a number, to indicate a cytokine whose amino acid sequence has been determined. *Interleukin-1*, for example, is secreted by macrophages and can activate the T cell system. B cell-stimulating factor, now called interleukin-4, is secreted by T cells and is required for the proliferation and clone development of B cells. *Interleukin-2* is released by helper T cells and, among its effects, is required for the activation of cytotoxic T lymphocytes. We will consider the actions of the cytokines as we describe the development of the T cell immune response.

Cell Interactions in the T Cell Response

When macrophages process the foreign antigens, they secrete **interleukin-1**, which stimulates cell division and proliferation of T cells (figure 57.10). Once the helper T cells have been activated by the antigens presented to them by

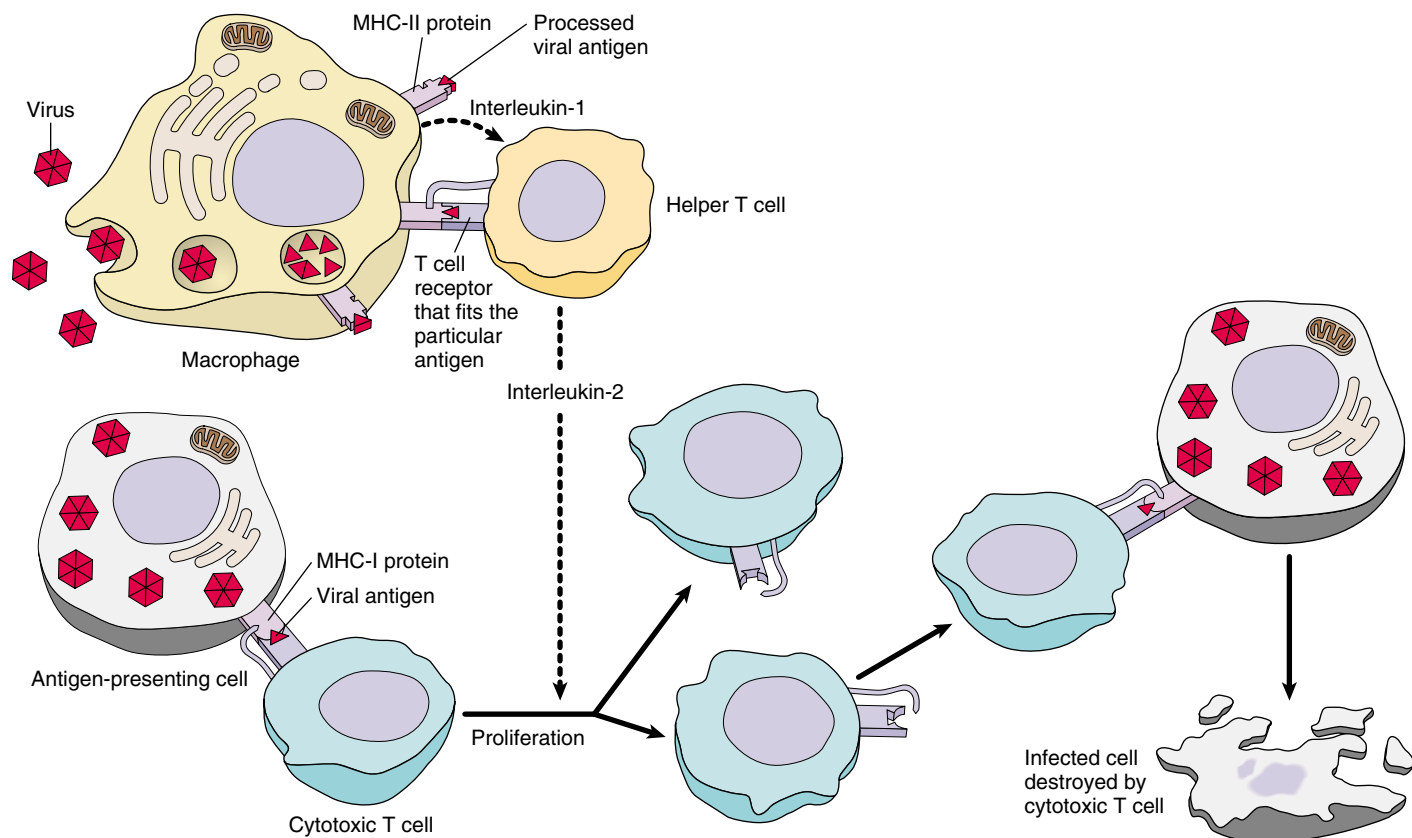
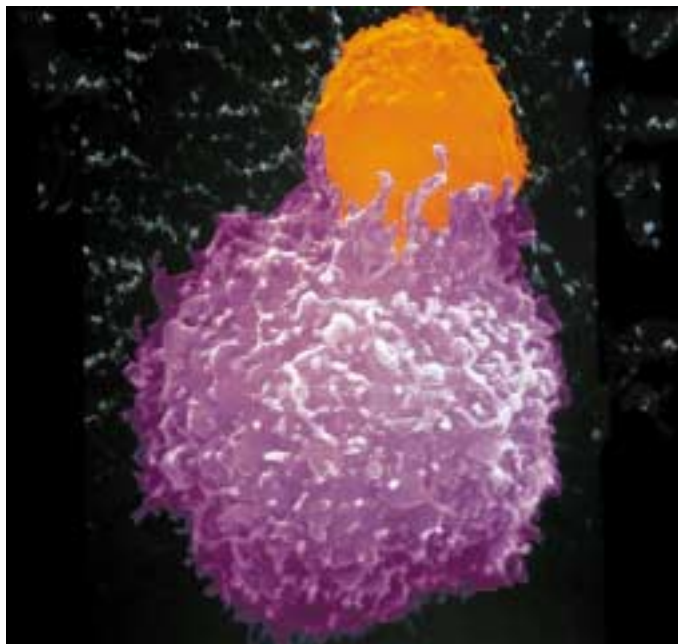
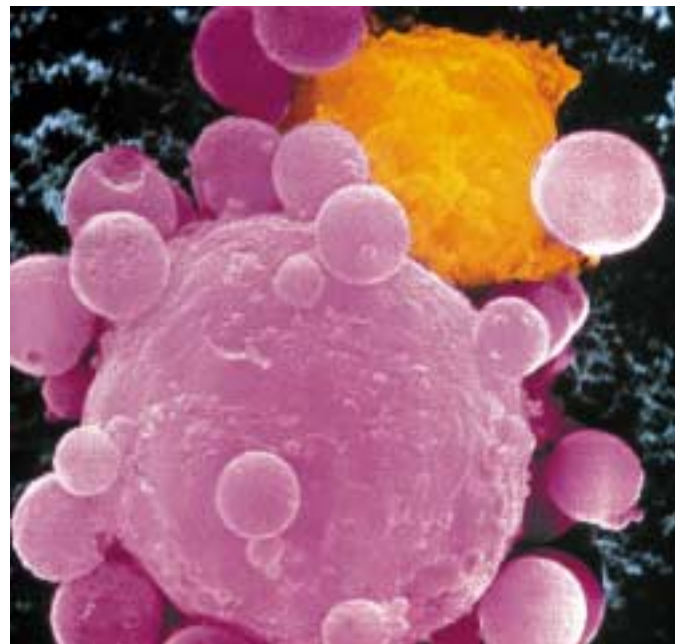


FIGURE 57.10

The T cell immune defense. After a macrophage has processed an antigen, it releases interleukin-1, signaling helper T cells to bind to the antigen-MHC protein complex. This triggers the helper T cell to release interleukin-2, which stimulates the multiplication of cytotoxic T cells. In addition, proliferation of cytotoxic T cells is stimulated when a T cell with a receptor that fits the antigen displayed by an antigen-presenting cell binds to the antigen-MHC protein complex. Body cells that have been infected by the antigen are destroyed by the cytotoxic T cells. As the infection subsides, suppressor T cells “turn off” the immune response.



(a)



(b)

FIGURE 57.11

Cytotoxic T cells destroy cancer cells. (a) The cytotoxic T cell (*orange*) comes into contact with a cancer cell (*pink*). (b) The T cell recognizes that the cancer cell is “nonself” and causes the destruction of the cancer.

the macrophages, they secrete the cytokines known as macrophage colony-stimulating factor and gamma interferon, which promote the activity of macrophages. In addition, the helper T cells secrete **interleukin-2**, which stimulates the proliferation of cytotoxic T cells that are specific for the antigen. (Interleukin-2 also stimulates B cells, as we will see in the next section.) Cytotoxic T cells can destroy infected cells only if those cells display the foreign antigen together with their MHC-I proteins (see figure 57.10).

T Cells in Transplant Rejection and Surveillance against Cancer

Cytotoxic T cells will also attack any foreign version of MHC-I as if it signaled a virus-infected cell. Therefore, even though vertebrates did not evolve the immune system as a defense against tissue transplants, their immune systems will attack transplanted tissue and cause graft rejection. Recall that the MHC proteins are polymorphic, but because of their genetic basis, the closer that two individuals are related, the less variance in their MHC proteins and the more likely they will tolerate each other's tissues—this is why relatives are often sought for kidney transplants. The drug cyclosporin inhibits graft rejection by inactivating cytotoxic T cells.

As tumors develop, they reveal surface antigens that can stimulate the immune destruction of the tumor cells. Tumor antigens activate the immune system, initiating an attack primarily by cytotoxic T cells (figure 57.11) and natural killer cells. The concept of **immunological surveillance** against

cancer was introduced in the early 1970s to describe the proposed role of the immune system in fighting cancer.

The production of human interferons by genetically engineered bacteria has made large amounts of these substances available for the experimental treatment of cancer. Thus far, interferons have proven to be a useful addition to the treatment of particular forms of cancer, including some types of lymphomas, renal carcinoma, melanoma, Kaposi's sarcoma, and breast cancer.

Interleukin-2 (IL-2), which activates both cytotoxic T cells and B cells, is now also available for therapeutic use through genetic-engineering techniques. Particular lymphocytes from cancer patients have been removed, treated with IL-2, and given back to the patients together with IL-2 and gamma interferon. Scientists are also attempting to identify specific antigens and their genes that may become uniquely expressed in cancer cells, in an effort to help the immune system to better target cancer cells for destruction.

Helper T cells are only activated when a foreign antigen is presented together with MHC antigens by a macrophage or other antigen-presenting cells. The helper T cells are also stimulated by interleukin-1 secreted by the macrophages, and, when activated, secrete a number of lymphokines. Interleukin-2, secreted by helper T cells, activates both cytotoxic T cells and B cells. Cytotoxic T cells destroy infected cells, transplanted cells, and cancer cells by cell-mediated attack.

57.4 B cells label specific cells for destruction.

B Cells: The Humoral Response

B cells also respond to helper T cells activated by interleukin-1. Like cytotoxic T cells, B cells have receptor proteins on their surface, one type of receptor for each type of B cell. B cells recognize invading microbes much as cytotoxic T cells recognize infected cells, but unlike cytotoxic T cells, they do not go on the attack themselves. Rather, they mark the pathogen for destruction by mechanisms that have no “ID check” system of their own. Early in the immune response, the markers placed by B cells alert complement proteins to attack the cells carrying them. Later in the immune response, the markers placed by B cells activate macrophages and natural killer cells.

The way B cells do their marking is simple and fool-proof. Unlike the receptors on T cells, which bind only to antigen-MHC protein complexes on antigen-presenting cells, B cell receptors can bind to free, unprocessed anti-

gens. When a B cell encounters an antigen, antigen particles will enter the B cell by endocytosis and get processed. Helper T cells that are able to recognize the specific antigen will bind to the antigen-MHC protein complex on the B cell and release interleukin-2, which stimulates the B cell to divide. In addition, free, unprocessed antigens stick to antibodies on the B cell surface. This antigen exposure triggers even more B cell proliferation. B cells divide to produce long-lived memory B cells and plasma cells that serve as short-lived antibody factories (figure 57.12). The antibodies are released into the blood plasma, lymph, and other extracellular fluids. Figure 57.13 summarizes the roles of helper T cells, which are essential in both the cell-mediated and humoral immune responses.

Antibodies are proteins in a class called **immunoglobulins** (abbreviated Ig), which is divided into subclasses based on the structures and functions of the

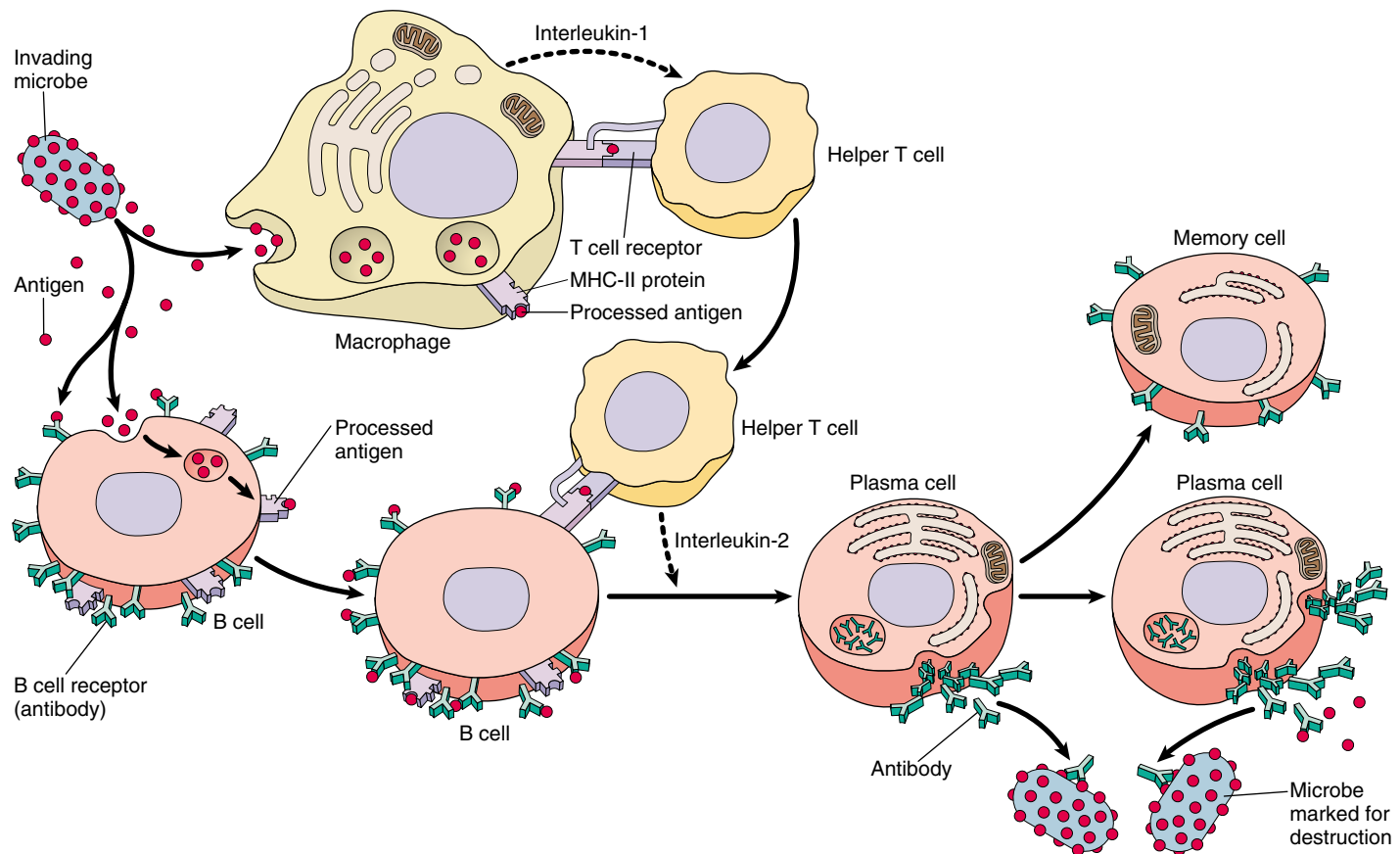


FIGURE 57.12

The B cell immune defense. Invading particles are bound by B cells, which interact with helper T cells and are activated to divide. The multiplying B cells produce either memory B cells or plasma cells that secrete antibodies which bind to invading microbes and tag them for destruction by macrophages.

antibodies. The different immunoglobulin subclasses are as follows:

1. **IgM.** This is the first type of antibody to be secreted during the primary response and they serve as receptors on the lymphocyte surface. These antibodies also promote agglutination reactions (causing antigen-containing particles to stick together, or agglutinate).
2. **IgG.** This is the major form of antibody in the blood plasma and is secreted in a secondary response.
3. **IgD.** These antibodies serve as receptors for antigens on the B cell surface. Their other functions are unknown.
4. **IgA.** This is the major form of antibody in external secretions, such as saliva and mother's milk.
5. **IgE.** This form of antibodies promotes the release of histamine and other agents that aid in attacking a pathogen. Unfortunately, they sometimes trigger a full-blown response when a harmless antigen enters the body producing allergic symptoms, such as those of hay fever.

Each B cell has on its surface about 100,000 IgM or IgD receptors. Unlike the receptors on T cells, which bind only to antigens presented by certain cells, B receptors can bind to *free* antigens. This provokes a primary response in which antibodies of the IgM class are secreted, and also stimulates cell division and clonal expansion. Upon subsequent exposure, the plasma cells secrete large amounts of antibodies that are generally of the IgG class. Although plasma cells live only a few days, they produce a vast number of antibodies. In fact, antibodies constitute about 20% by weight of the total protein in blood plasma. Production of IgG antibodies peaks after about three weeks (figure 57.14).

When IgM (and to a lesser extent IgG) antibodies bind to antigens on a cell, they cause the aggregation of complement proteins. As we mentioned earlier, these proteins form a pore that pierces the plasma membrane of the infected cell (see figure 57.5), allowing water to enter and causing the cell to burst. In contrast, when IgG antibodies bind to antigens on a cell, they serve as markers that stimulate phagocytosis by macrophages. Because certain complement proteins attract phagocytic cells, activation of complement is generally accompanied by increased phagocytosis. Notice that antibodies don't kill invading pathogens directly; rather, they cause destruction of the pathogens by activating the complement system and by targeting the pathogen for attack by phagocytic cells.

In the humoral immune response, B cells recognize antigens and divide to produce plasma cells, producing large numbers of circulating antibodies directed against those antigens. IgM antibodies are produced first, and they activate the complement system. Thereafter, IgG antibodies are produced and promote phagocytosis.

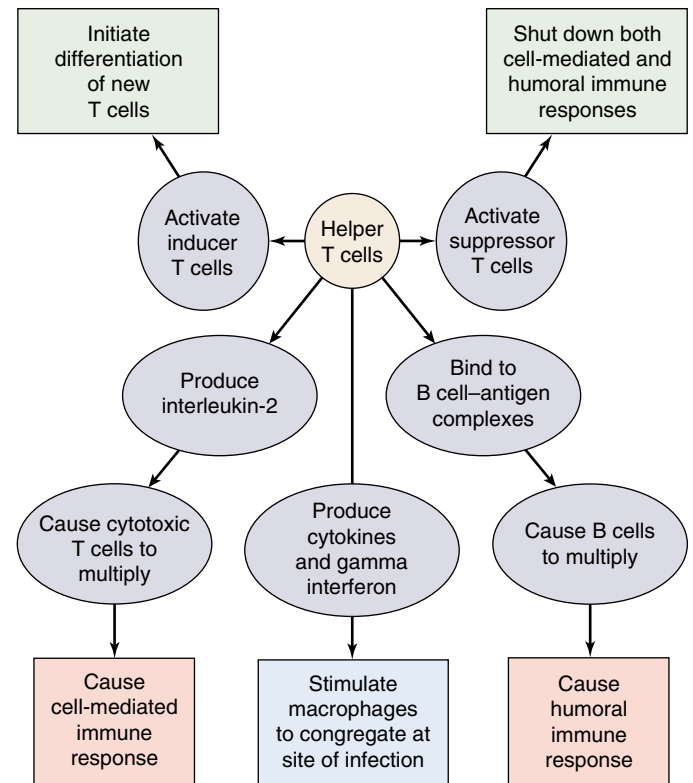


FIGURE 57.13
The many roles of helper T cells. Helper T cells, through their secretion of lymphokines and interaction with other cells of the immune system, participate in every aspect of the immune response.

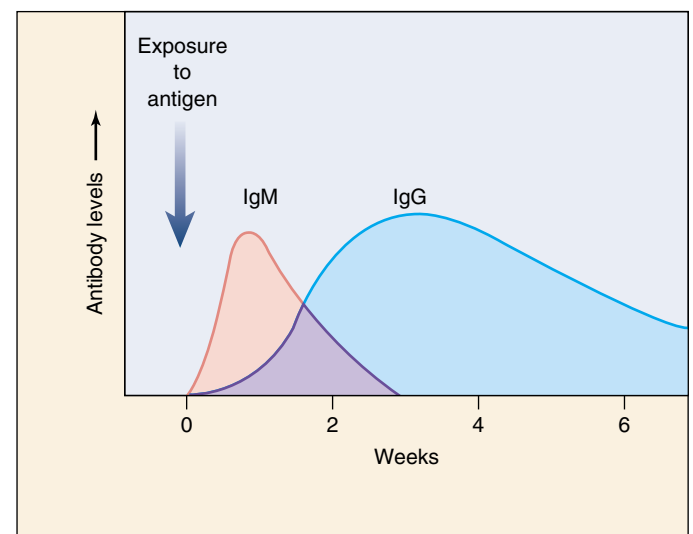


FIGURE 57.14
IgM and IgG antibodies. The first antibodies produced in the humoral immune response are IgM antibodies, which are very effective at activating the complement system. This initial wave of antibody production peaks after about one week and is followed by a far more extended production of IgG antibodies.

Antibodies

Structure of Antibodies

Each antibody molecule consists of two identical short polypeptides, called **light chains**, and two identical long polypeptides, called **heavy chains** (figure 57.15). The four chains in an antibody molecule are held together by disulfide (—S—S—) bonds, forming a Y-shaped molecule (figure 57.16).

Comparing the amino acid sequences of different antibody molecules shows that the specificity of antibodies for antigens resides in the two arms of the Y, which have a variable amino acid sequence. The amino acid sequence of the polypeptides in the stem of the Y is constant within a given class of immunoglobulins. Most of the sequence variation between antibodies of different specificity is found in the variable region of each arm. Here, a cleft forms that acts as the binding site for the antigen. Both arms always have exactly the same cleft and so bind to the same antigen.

Antibodies with the same variable segments have identical clefts and therefore recognize the same antigen, but they may differ in the stem portions of the antibody molecule. The stem is formed by the so-called “constant” regions of the heavy chains. In mammals there are five different classes of heavy chain that form five classes of immunoglobulins: IgM, IgG, IgA, IgD, and IgE. We have already discussed the roles of IgM and IgG antibodies in the humoral immune response.

IgE antibodies bind to **mast cells**. The heavy-chain stems of the IgE antibody molecules insert into receptors on the mast cell plasma membrane, in effect creating B receptors on the mast cell surface. When these cells encounter the specific antigen recognized by the arms of the antibody, they initiate the inflammatory response by releasing histamine. The resulting vasodilation and increased capillary permeability enable lymphocytes, macrophages, and complement proteins to more easily reach the site where the mast cell encountered the antigen. The IgE antibodies are involved in allergic reactions and will be discussed in more detail in a later section.

IgA antibodies are present in secretions such as milk, mucus, and saliva. In milk, these antibodies are thought to provide immune protection to nursing infants, whose own immune systems are not yet fully developed.

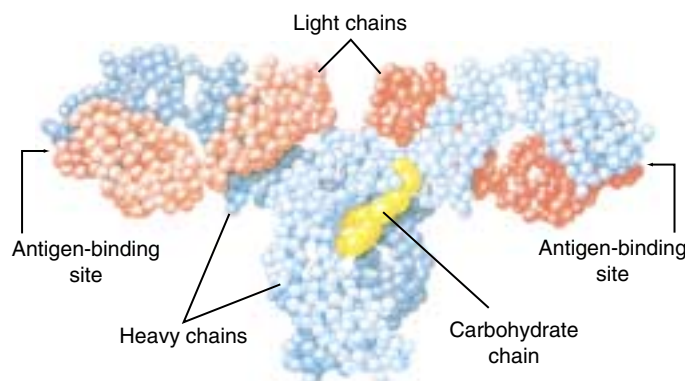


FIGURE 57.15

The structure of an antibody molecule. In this molecular model of an antibody molecule, each amino acid is represented by a small sphere. The heavy chains are colored blue; the light chains are red. The four chains wind about one another to form a Y shape, with two identical antigen-binding sites at the arms of the Y and a stem region that directs the antibody to a particular portion of the immune response.

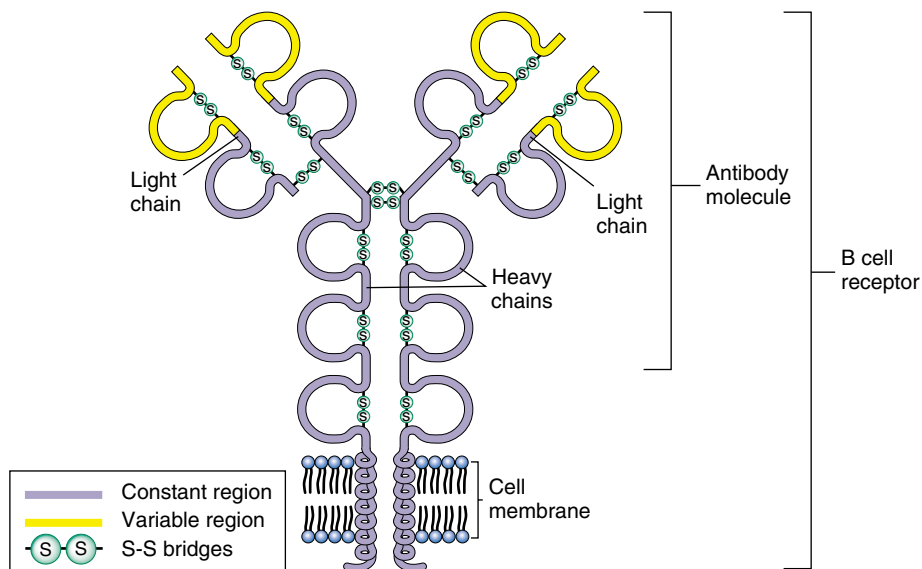


FIGURE 57.16

Structure of an antibody as a B cell receptor. The receptor molecules are characterized by domains of about 100 amino acids (represented as loops) joined by —S—S— covalent bonds. Each receptor has a constant region (purple) and a variable region (yellow). The receptor binds to antigens at the ends of its two variable regions.

Antibody Diversity

The vertebrate immune system is capable of recognizing as foreign millions of nonself molecules presented to it. Although vertebrate chromosomes contain only a few hundred receptor-encoding genes, it is estimated that human B cells can make between 10^6 and 10^9 different antibody molecules. How do vertebrates generate millions of different antigen receptors when their chromosomes con-

tain only a few hundred copies of the genes encoding those receptors?

The answer to this question is that in the B cell the millions of immune receptor genes do not have to be inherited at conception because they do not exist as single sequences of nucleotides. Rather, they are assembled by stitching together three or four DNA segments that code for different parts of the receptor molecule. When an antibody is assembled, the different sequences of DNA are brought together to form a composite gene (figure 57.17). This process is called **somatic rearrangement**. For example, combining DNA in different ways can produce 16,000 different heavy chains and about 1200 different light chains (in mouse antibodies).

Two other processes generate even more sequences. First, the DNA segments are often joined together with one or two nucleotides off-register, shifting the reading frame during gene transcription and so generating a totally different sequence of amino acids in the protein. Second, random mistakes occur during successive DNA replications as the lymphocytes divide during clonal expansion. Both mutational processes produce changes in amino acid sequences, a phenomenon known as **somatic mutation** because it takes place in a somatic cell, a B cell rather than in a gamete.

Because a B cell may end up with any heavy-chain gene and any light-chain gene during its maturation, the total number of different antibodies possible is staggering: 16,000 heavy-chain combinations $\times$ 1200 light-chain combinations = 19 million different possible antibodies. If one also takes into account the changes induced by somatic mutation, the total can exceed 200 million! It should be understood that, although this discussion has centered on B cells and their receptors, the receptors on T cells are as diverse as those on B cells because they also are subject to similar somatic rearrangements and mutations.

Immunological Tolerance

A mature animal's immune system normally does not respond to that animal's own tissue. This acceptance of self cells is known as **immunological tolerance**. The immune system of an embryo, on the other hand, is able to respond to both foreign and self molecules, but it loses the ability to respond to self molecules as its development proceeds. Indeed, if foreign tissue is introduced into an embryo before its immune system has developed, the mature animal that results will not recognize that tissue as foreign and will accept grafts of similar tissue without rejection.

There are two general mechanisms for immunological tolerance: clonal deletion and clonal suppression. During the normal maturation of hemopoietic stem cells in an embryo, fetus, or newborn, most lymphocyte clones that have receptors for self antigens are either eliminated (clonal deletion) or suppressed (clonal suppression). The cells "learn" to identify self antigens because the antigens are encountered very frequently. If a receptor is activated fre-

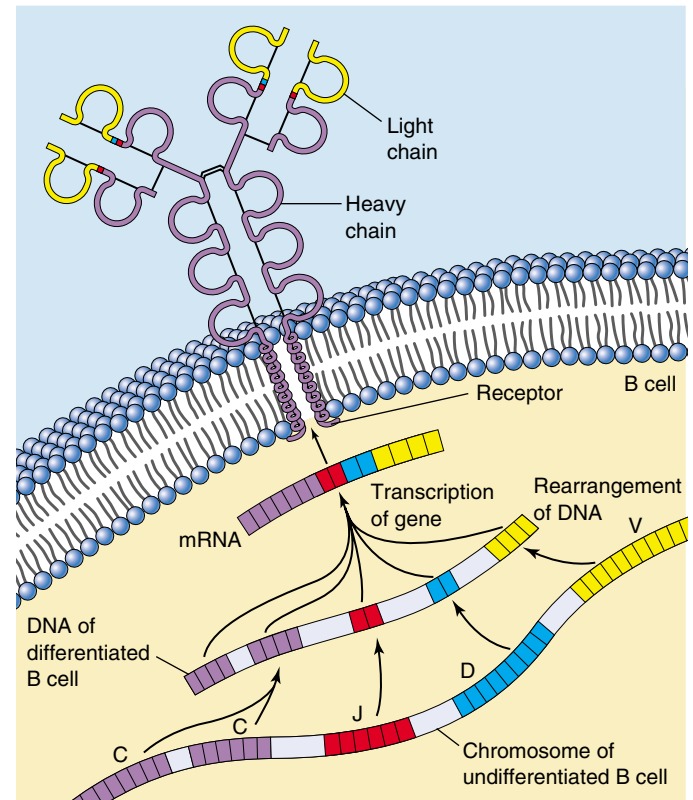


FIGURE 57.17

The lymphocyte receptor molecule is produced by a composite gene. Different regions of the DNA code for different regions of the receptor structure (C, constant regions; J, joining regions; D, diversity regions; and V, variable regions) and are brought together to make a composite gene that codes for the receptor. Through different somatic rearrangements of these DNA segments, an enormous number of different receptor molecules can be produced.

quently, it is assumed that the cell is recognizing a self antigen and the lymphocytes are eliminated or suppressed. Thus, the only clones that survive this phase of development are those that are directed against foreign rather than self molecules.

Immunological tolerance sometimes breaks down, causing either B cells or T cells (or both) to recognize their own tissue antigens. This loss of immune tolerance results in autoimmune disease. Myasthenia gravis, for example, is an autoimmune disease in which individuals produce antibodies directed against acetylcholine receptors on their own skeletal muscle cells, causing paralysis. Autoimmunity will be discussed in more detail later in this chapter.

An antibody molecule is composed of constant and variable regions. The variable regions recognize a specific antigen because they possess clefts into which the antigen can fit. Lymphocyte receptors are encoded by genes that are assembled by somatic rearrangement and mutation of the DNA.

Active Immunity through Clonal Selection

As we discussed earlier, B and T cells have receptors on their cell surfaces that recognize and bind to specific antigens. When a particular antigen enters the body, it must, by chance, encounter the specific lymphocyte with the appropriate receptor in order to provoke an immune response. The first time a pathogen invades the body, there are only a few B or T cells that may have the receptors that can recognize the invader's antigens. Binding of the antigen to its receptor on the lymphocyte surface, however, stimulates cell division and produces a *clone* (a population of genetically identical cells). This process is known as **clonal selection**. In this first encounter, there are only a few cells that can mount an immune response and the response is relatively weak. This is called a **primary immune response** (figure 57.18).

If the primary immune response involves B cells, some become plasma cells that secrete antibodies, and some become memory cells. Because a clone of memory cells specific for that antigen develops after the primary response, the immune response to a second infection by the same pathogen is swifter and stronger. The next time the body is invaded by the same pathogen, the immune system is ready. As a result of the first infection, there is now a large clone of lymphocytes that can recognize that pathogen (figure 57.19). This more effective response, elicited by subsequent exposures to an antigen, is called a **secondary immune response**.

Memory cells can survive for several decades, which is why people rarely contract chicken pox a second time after they have had it once. Memory cells are also the reason that vaccinations are effective. The vaccine triggers the primary response so that if the actual pathogen is encountered later, the large and rapid secondary response occurs and stops the infection before it can start. The viruses causing childhood diseases have surface antigens that change little from year to year, so the same antibody is effective for decades.

Figure 57.20 summarizes how the cellular and humoral lines of defense work together to produce the body's specific immune response.

Active immunity is produced by clonal selection and expansion. This occurs because interaction of an antigen with its receptor on the lymphocyte surface stimulates cell division, so that more lymphocytes are available to combat subsequent exposures to the same antigen.

FIGURE 57.19

The clonal selection theory of active immunity. In response to interaction with an antigen that binds specifically to its surface receptors, a B cell divides many times to produce a clone of B cells. Some of these become plasma cells that secrete antibodies for the primary response, while others become memory cells that await subsequent exposures to the antigen for the mounting of a secondary immune response.

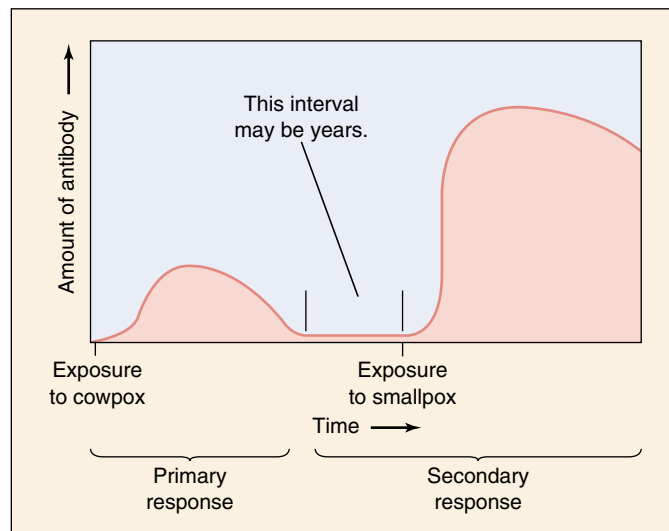
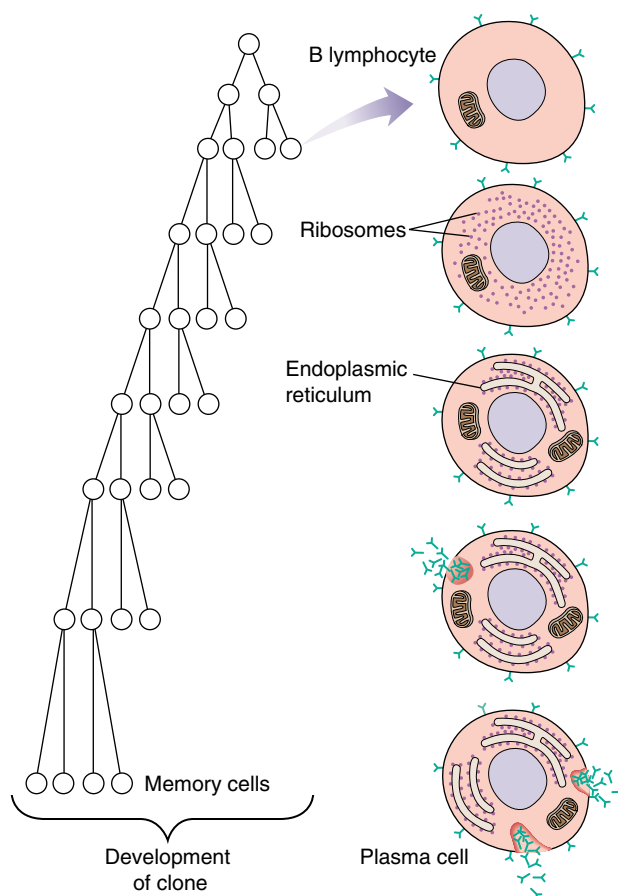


FIGURE 57.18

The development of active immunity. Immunity to smallpox in Jenner's patients occurred because their inoculation with cowpox stimulated the development of lymphocyte clones with receptors that could bind not only to cowpox but also to smallpox antigens. As a result of clonal selection, a second exposure, this time to smallpox, stimulates the immune system to produce large amounts of the antibody more rapidly than before.



THE IMMUNE RESPONSE

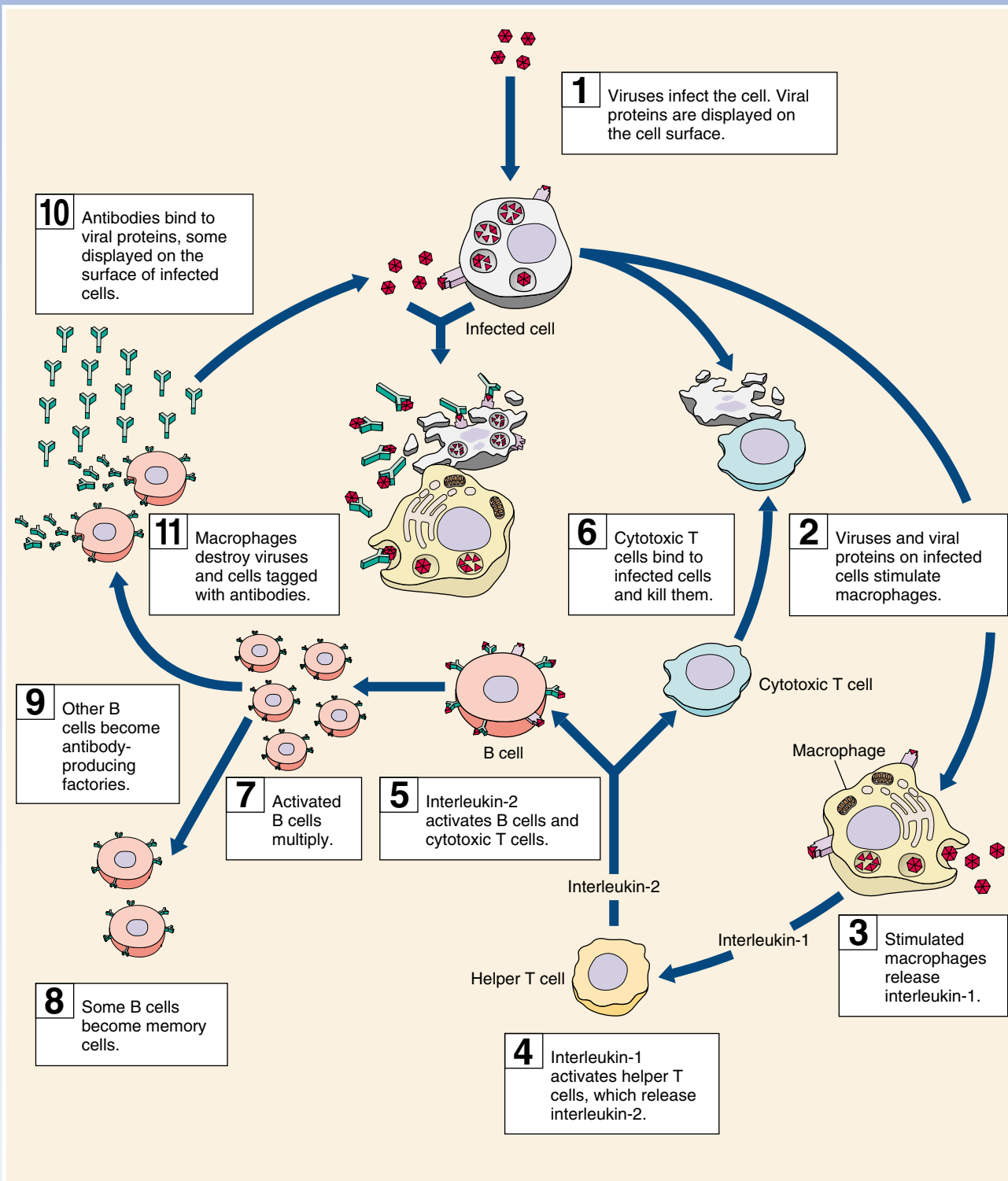


FIGURE 57.20
Overview of the specific immune response.

Antibodies in Medical Diagnosis

Blood Typing

The blood type denotes the class of antigens found on the red blood cell surface. Red blood cell antigens are clinically important because their types must be matched between donors and recipients for blood transfusions. There are several groups of red blood cell antigens, but the major group is known as the **ABO system**. In terms of the antigens present on the red blood cell surface, a person may be *type A* (with only A antigens), *type B* (with only B antigens), *type AB* (with both A and B antigens), or *type O* (with neither A nor B antigens).

The immune system is tolerant to its own red blood cell antigens. A person who is type A, for example, does not produce anti-A antibodies. Surprisingly, however, people with type A blood do make antibodies against the B antigen, and conversely, people with blood type B make antibodies against the A antigen. This is believed to result from the fact that antibodies made in response to some common bacteria cross-react with the A or B antigens. A person who is type A, therefore, acquires antibodies that can react with B antigens by exposure to these bacteria but does not develop antibodies that can react with A antigens. People who are type AB develop tolerance to both antigens and thus do not produce either anti-A or anti-B antibodies. Those who are type O, in contrast, do not develop tolerance to either antigen and, therefore, have both anti-A and anti-B antibodies in their plasma.

If type A blood is mixed on a glass slide with serum from a person with type B blood, the anti-A antibodies in the serum will cause the type A red blood cells to clump together, or **agglutinate** (figure 57.21). These tests allow the blood types to be matched prior to transfusions, so that agglutination will not occur in the blood vessels, where it could lead to inflammation and organ damage.

Rh Factor. Another group of antigens found in most red blood cells is the *Rh factor* (Rh stands for rhesus monkey, in which these antigens were first discovered). People who have these antigens are said to be **Rh-positive**, whereas those who do not are **Rh-negative**. There are fewer Rh-negative people because this condition is recessive to Rh-positive. The Rh factor is of particular signifi-

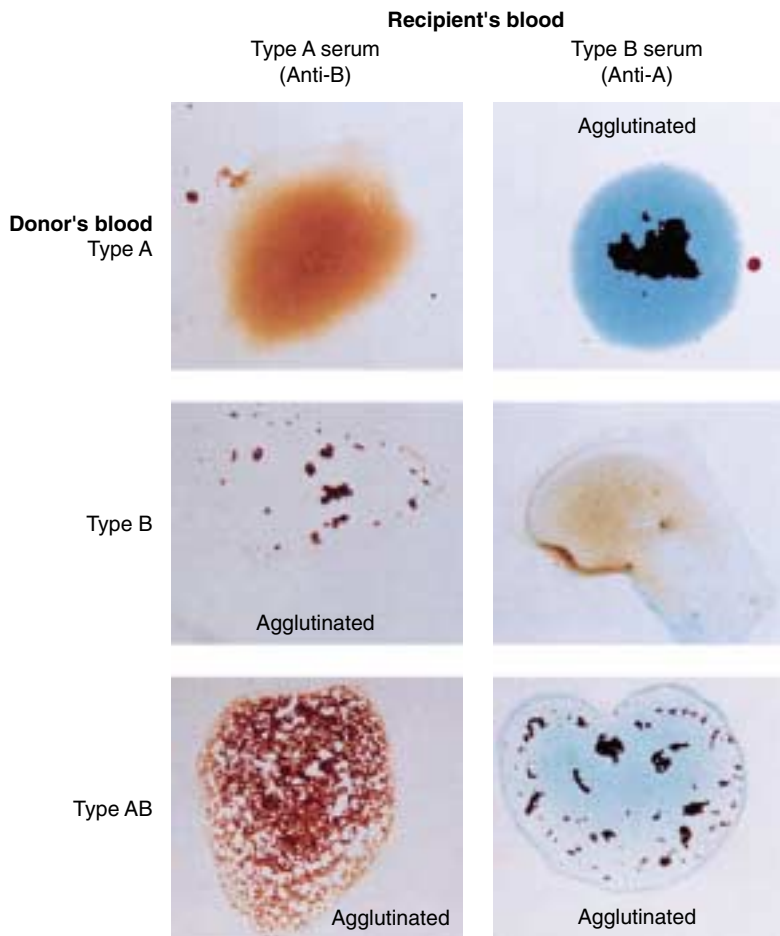


FIGURE 57.21

Blood typing: Agglutination of the red blood cells is seen when blood types are mixed with sera containing antibodies against the ABO antigens. Note that no agglutination would be seen if type O blood (not shown) were used.

cance when Rh-negative mothers give birth to Rh-positive babies.

Because the fetal and maternal blood are normally kept separate across the placenta (see chapter 60), the Rh-negative mother is not usually exposed to the Rh antigen of the fetus during the pregnancy. At the time of birth, however, a variable degree of exposure may occur, and the mother's immune system may become sensitized and produce antibodies against the Rh antigen. If the woman does produce antibodies against the Rh factor, these antibodies can cross the placenta in subsequent pregnancies and cause hemolysis of the Rh-positive red blood cells of the fetus. The baby is therefore born anemic, with a condition called *erythroblastosis fetalis*, or *hemolytic disease of the newborn*.

Erythroblastosis fetalis can be prevented by injecting the Rh-negative mother with an antibody preparation against the Rh factor within 72 hours after the birth of each Rh-positive baby. This is a type of passive immunization in which the injected antibodies inactivate the Rh antigens and thus prevent the mother from becoming actively immunized to them.

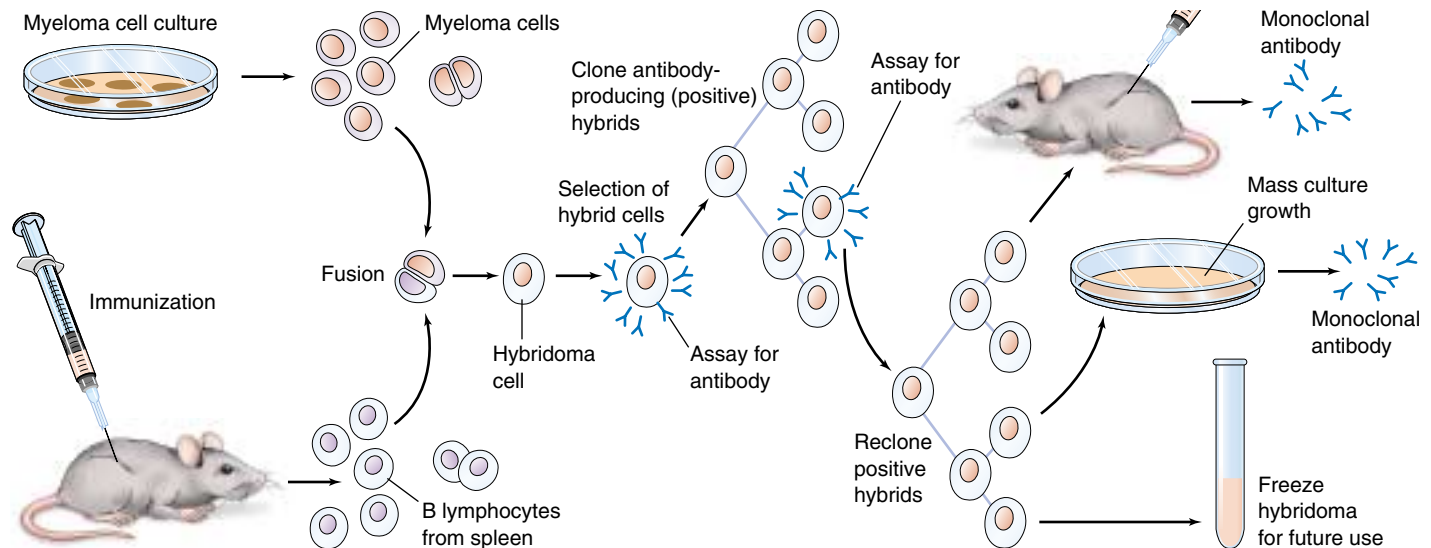


FIGURE 57.22

The production of monoclonal antibodies. These antibodies are produced by cells that arise from successive divisions of a single B cell, and hence all of the antibodies target a single antigenic determinant site. Such antibodies are used for a variety of medical applications, including pregnancy testing.

Monoclonal Antibodies

Antibodies are commercially prepared for use in medical diagnosis and research. In the past, antibodies were obtained by chemically purifying a specific antigen and then injecting this antigen into animals. However, because an antigen typically has many different antigenic determinant sites, the antibodies obtained by this method were *polyclonal*; they stimulated the development of different B-cell clones with different specificities. This decreased their sensitivity to a particular antigenic site and resulted in some degree of cross-reaction with closely related antigen molecules.

Monoclonal antibodies, by contrast, exhibit specificity for one antigenic determinant only. In the preparation of monoclonal antibodies, an animal (frequently, a mouse) is injected with an antigen and subsequently killed. B lymphocytes are then obtained from the animal's spleen and placed in thousands of different *in vitro* incubation vessels. These cells soon die, however, unless they are hybridized with cancerous multiple myeloma cells. The fusion of a B lymphocyte with a cancerous cell produces a hybrid that undergoes cell division and produces a clone called a *hybridoma*. Each hybridoma secretes large amounts of identical, monoclonal antibodies. From among the thousands of hybridomas produced in this way, the one that produces the desired antibody is cultured for large-scale production, and the rest are discarded (figure 57.22).

The availability of large quantities of pure monoclonal antibodies has resulted in the development of much more sensitive clinical laboratory tests. Modern pregnancy tests, for example, use particles (latex rubber or red blood cells) that are covered with monoclonal antibodies produced against a pregnancy hormone (abbreviated hCG—see

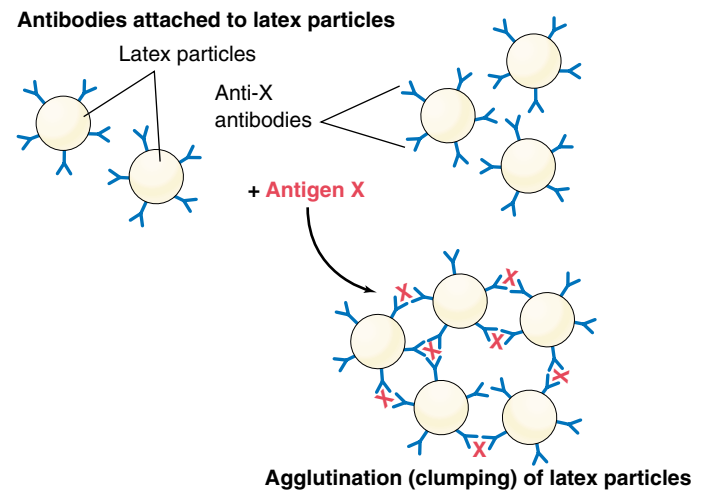


FIGURE 57.23

Using monoclonal antibodies to detect an antigen. In many clinical tests (such as pregnancy testing), the monoclonal antibodies are bound to particles of latex, which agglutinate in the presence of the antigen.

chapter 59) as the antigen. When these particles are mixed with a sample that contains this hormone antigen from a pregnant woman, the antigen-antibody reaction causes a visible agglutination of the particles (figure 57.23).

Agglutination occurs because different antibodies exist for the ABO and Rh factor antigens on the surface of red blood cells. Monoclonal antibodies are commercially produced antibodies that react against one specific antigen.

57.5 All animals exhibit nonspecific immune response but specific ones evolved in vertebrates.

Evolution of the Immune System

All organisms possess mechanisms to protect themselves from the onslaught of smaller organisms and viruses. Bacteria defend against viral invasion by means of *restriction endonucleases*, enzymes that degrade any foreign DNA lacking the specific pattern of DNA methylation characteristic of that bacterium. Multicellular organisms face a more difficult problem in defense because their bodies often take up whole viruses, bacteria, or fungi instead of naked DNA.

Invertebrates

Invertebrate animals solve this problem by marking the surfaces of their cells with proteins that serve as “self” labels. Special amoeboid cells in the invertebrate attack and engulf any invading cells that lack such labels. By looking for the absence of specific markers, invertebrates employ a *negative* test to recognize foreign cells and viruses. This method provides invertebrates with a very effective surveillance system, although it has one great weakness: any microorganism or virus with a surface protein resembling the invertebrate self marker will not be recognized as foreign. An invertebrate has no defense against such a “copycat” invader.

In 1882, Russian zoologist Elie Metchnikoff became the first to recognize that invertebrate animals possess immune defenses. On a beach in Sicily, he collected the tiny transparent larva of a common starfish. Carefully he pierced it with a rose thorn. When he looked at the larva the next morning, he saw a host of tiny cells covering the surface of the thorn as if trying to engulf it (figure 57.24). The cells were attempting to defend the larva by ingesting the invader by phagocytosis (described in chapter 6). For this discovery of what came to be known as the **cellular immune response**, Metchnikoff was awarded the 1908 Nobel Prize in Physiology or Medicine, along with Paul Ehrlich for his work on the other major part of the immune defense, the antibody or **humoral immune response**. The invertebrate immune response shares several elements with the vertebrate one.

Phagocytes. All animals possess phagocytic cells that attack invading microbes. These phagocytic cells travel through the animal’s circulatory system or circulate within the fluid-filled body cavity. In simple animals like sponges that lack either a circulatory system or a body cavity, the phagocytic cells circulate among the spaces between cells.

Distinguishing Self from Nonself. The ability to recognize the difference between cells of one’s own body and those of another individual appears to have evolved early in the history of life. Sponges, thought to be the oldest animals, attack grafts from other sponges, as do insects and starfish. None of these invertebrates, however, exhibit



FIGURE 57.24
Discovering the cellular immune response in invertebrates.
In a Nobel-Prize-winning experiment, the Russian zoologist Metchnikoff pierced the larva of a starfish with a rose thorn and the next day found tiny phagocytic cells covering the thorn.

any evidence of immunological memory; apparently, the antibody-based humoral immune defense did not evolve until the vertebrates.

Complement. While invertebrates lack complement, many arthropods (including crabs and a variety of insects) possess an analogous nonspecific defense called the prophenyloxidase (proPO) system. Like the vertebrate complement defense, the proPO defense is activated as a cascade of enzyme reactions, the last of which converts the inactive protein prophenyloxidase into the active enzyme phenyloxidase. Phenyloxidase both kills microbes and aids in encapsulating foreign objects.

Lymphocytes. Invertebrates also lack lymphocytes, but annelid earthworms and other invertebrates do possess lymphocyte-like cells that may be evolutionary precursors of lymphocytes.

Antibodies. All invertebrates possess proteins called lectins that may be the evolutionary forerunners of antibodies. Lectins bind to sugar molecules on cells, making the cells stick to one another. Lectins isolated from sea urchins, mollusks, annelids, and insects appear to tag invading microorganisms, enhancing phagocytosis. The genes encoding vertebrate antibodies are part of a very ancient gene family, the immunoglobulin superfamily. Proteins in

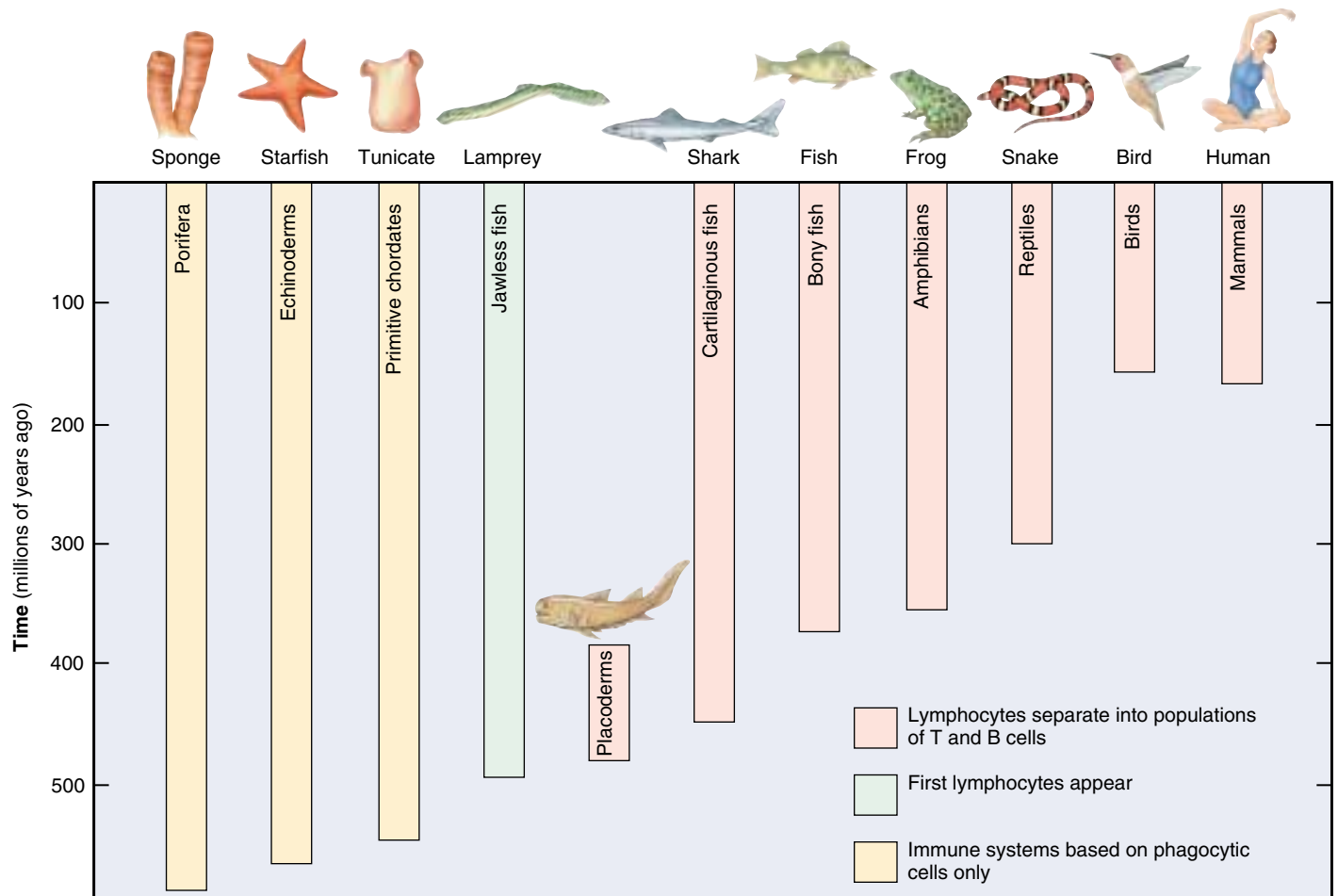


FIGURE 57.25

How immune systems evolved. Lampreys were the first vertebrates to possess an immune system based on lymphocytes, although distinct B and T cells did not appear until the jawed fishes evolved. By the time sharks and other cartilaginous fish appeared, the vertebrate immune response was fully formed.

this group all have a characteristic recognition structure called the Ig fold. The fold probably evolved as a self-recognition molecule in early metazoans. Insect immunoglobulins have been described in moths, grasshoppers, and flies that bind to microbial surfaces and promote their destruction by phagocytes. The antibody immune response appears to have evolved from these earlier, less complex systems.

Vertebrates

The earliest vertebrates of which we have any clear information, the jawless lampreys that first evolved some 500 million years ago, possess an immune system based on lymphocytes. At this early stage of vertebrate evolution, however, lampreys lack distinct populations of B and T cells such as found in all higher vertebrates (figure 57.25).

With the evolution of fish with jaws, the modern vertebrate immune system first arose. The oldest surviving group of jawed fishes are the sharks, which evolved some 450 mil-

lion years ago. By then, the vertebrate immune defense had fully evolved. Sharks have an immune response much like that seen in mammals, with a cellular response carried out by T-cell lymphocytes and an antibody-mediated humoral response carried out by B cells. The similarities of the cellular and humoral immune defenses are far more striking than the differences. Both sharks and mammals possess a thymus that produces T cells and a spleen that is a rich source of B cells. Four hundred fifty million years of evolution did little to change the antibody molecule—the amino acid sequences of shark and human antibody molecules are very similar. The most notable difference between sharks and mammals is that their antibody-encoding genes are arrayed somewhat differently.

The sophisticated two-part immune defense of mammals evolved about the time jawed fishes appeared. Before then, animals utilized a simpler immune defense based on mobile phagocytic cells.

57.6 The immune system can be defeated.

T Cell Destruction: AIDS

One mechanism for defeating the vertebrate immune system is to attack the immune mechanism itself. Helper T cells and inducer T cells are $CD4^+$ T cells. Therefore, any pathogen that inactivates $CD4^+$ T cells leaves the immune system unable to mount a response to *any* foreign antigen. Acquired immune deficiency syndrome (AIDS) is a deadly disease for just this reason. The AIDS retrovirus, called human immunodeficiency virus (HIV), mounts a direct attack on $CD4^+$ T cells because it recognizes the $CD4$ coreceptors associated with these cells.

HIV's attack on $CD4^+$ T cells cripples the immune system in at least three ways. First, HIV-infected cells die only after releasing replicated viruses that infect other $CD4^+$ T cells, until the entire population of $CD4^+$ T cells is destroyed (figure 57.26). In a normal individual, $CD4^+$ T cells make up 60 to 80% of circulating T cells; in AIDS patients, $CD4^+$ T cells often become too rare to detect (figure 57.27). Second, HIV causes infected $CD4^+$ T cells to secrete a soluble suppressing factor that blocks other T cells from responding to the HIV antigen. Finally, HIV may block transcription of MHC genes, hindering the recognition and destruction of infected $CD4^+$ T cells and thus protecting those cells from any remaining vestiges of the immune system.

The combined effect of these responses to HIV infection is to wipe out the human immune defense. With no defense against infection, any of a variety of otherwise commonplace infections proves fatal. With no ability to recognize and destroy cancer cells when they arise, death by cancer becomes far more likely. Indeed, AIDS was first recognized as a disease because of a cluster of cases of an unusually rare form of cancer. More AIDS victims die of cancer than from any other cause.

Although HIV became a human disease vector only recently, possibly through transmission to humans from chimpanzees in Central Africa, it is already clear that AIDS is one of the most serious diseases in human history (figure 57.28). The fatality rate of AIDS is 100%; no patient exhibiting the symptoms of AIDS has ever been known to survive more than a few years without treatment. Aggressive treatments can prolong life but how much longer has not been determined. However, the disease is *not* highly contagious, as it is transmitted from one individual to another through the transfer of internal body fluids, typically in semen and in blood during transfusions. Not all individuals exposed to HIV (as judged by anti-HIV antibodies in their blood) have yet acquired the disease.

Until recently, the only effective treatment for slowing the progression of the disease involved treatment with drugs such as AZT that inhibit the activity of reverse transcriptase, the enzyme needed by the virus to produce DNA from RNA. Recently, however, a new type of drug has be-

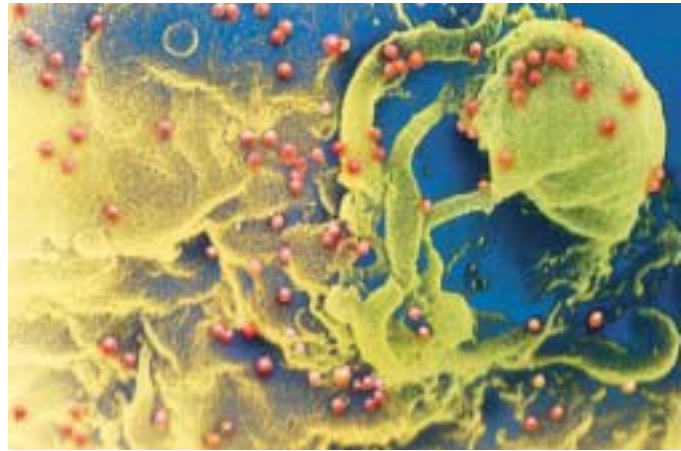


FIGURE 57.26

HIV, the virus that causes AIDS. Viruses released from infected $CD4^+$ T cells soon spread to neighboring $CD4^+$ T cells, infecting them in turn. The individual viruses, colored blue in this scanning electron micrograph, are extremely small; over 200 million would fit on the period at the end of this sentence.

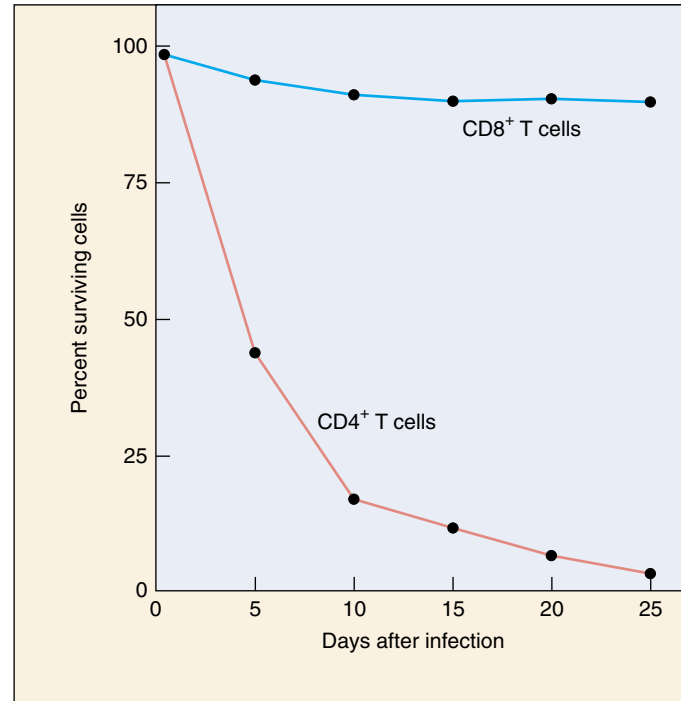


FIGURE 57.27

Survival of T cells in culture after exposure to HIV. The virus has little effect on the number of $CD8^+$ T cells, but it causes the number of $CD4^+$ T cells (this group includes helper T cells) to decline dramatically.

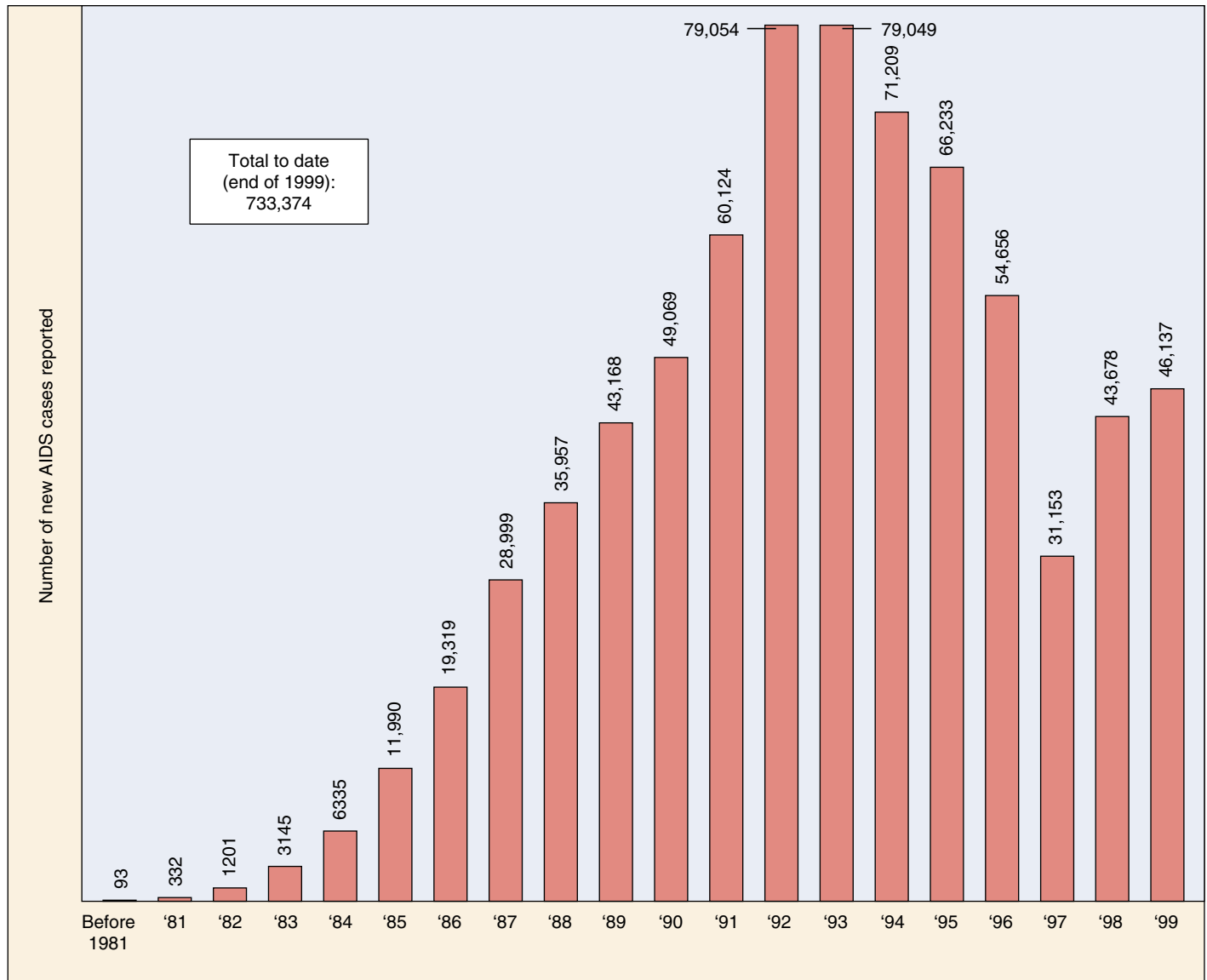


FIGURE 57.28

The AIDS epidemic in the United States: new cases. The U.S. Centers for Disease Control and Prevention (CDC) reports that 43,678 new AIDS cases were reported in 1998 and 46,137 new cases in 1999, with a total of 733,374 cases and 390,692 deaths in the United States. Over 1.5 million other individuals are thought to be infected with the HIV virus in the United States, and 14 million worldwide. The 100,000th AIDS case was reported in August 1989, eight years into the epidemic; the next 100,000 cases took just 26 months; the third 100,000 cases took barely 19 months (May 1993), and the fourth 100,000 took only 13 months (June 1994). The extraordinarily high numbers seen in 1992 reflect an expansion of the definition of what constitutes an AIDS case.

Source: Data from U.S. Centers for Disease Control and Prevention, Atlanta, GA.

come available that acts to inhibit protease, an enzyme needed for viral assembly. Treatments that include a combination of reverse transcriptase inhibitors and protease inhibitors (p. 672) appear to lower levels of HIV, though they are very costly. Efforts to develop a vaccine against AIDS continue, both by splicing portions of the HIV surface protein gene into vaccinia virus and by attempting to develop a harmless strain of HIV. These approaches, while promising, have not yet proved successful and are limited by the

fact that different strains of HIV seem to possess different surface antigens. Like the influenza virus, HIV engages in some form of antigen shifting, making it difficult to develop an effective vaccine.

AIDS destroys the ability of the immune system to mount a defense against any infection. HIV, the virus that causes AIDS, induces a state of immune deficiency by attacking and destroying CD4⁺ T cells.

Antigen Shifting

A second way that a pathogen may defeat the immune system is to mutate frequently so that it varies the nature of its surface antigens. The virus which causes influenza uses this mechanism, and so we have to be immunized against a different strain of this virus periodically. This way of escaping immune attack is known as antigen shifting, and is practiced very effectively by trypanosomes, the protists responsible for sleeping sickness (see chapter 35). Trypanosomes possess several thousand different versions of the genes encoding their surface protein, but the cluster containing these genes has no promoter and so is not transcribed as a unit. The necessary promoter is located within a transposable element that jumps at random from one position to another within the cluster, transcribing a different surface protein gene with every move. Because such moves occur in at least one cell of an infective trypanosome population every few weeks, the vertebrate immune system is unable to mount an effective defense against trypanosome infection. By the time a significant number of antibodies have been generated against one form of trypanosome surface protein, another form is already present in the trypanosome population that survives immunological attack, and the infection cycle is renewed. People with sleeping sickness rarely rid themselves of the infection.

Although this mechanism of mutation to alter surface proteins seems very “directed” or intentional on the part of the pathogen, it is actually the process of evolution by natural selection at work. We usually think of evolution as requiring thousands of years to occur, and not in the time frame of weeks. However, evolution can occur whenever mutations are passed on to offspring that provide an organism with a competitive advantage. In the case of viruses, bacteria, and other pathogenic agents, their generation times are on the order of hours. Thus, in the time frame of a week, the population has gone through millions of cell divisions. Looking at it from this perspective, it is easy to see how random mutations in the genes for the surface antigens could occur and change the surface of the pathogen in as little as a week’s time.

How Malaria Hides from the Immune System

Every year, about a half-million people become infected with the protozoan parasite *Plasmodium falciparum*, which multiplies in their bodies to cause the disease malaria. The plasmodium parasites enter the red blood cells and consume the hemoglobin of their hosts. Normally this sort of damage to a red blood cell would cause the damaged cell to be transported to the spleen for disassembly, destroying the plasmodium as well. The plasmodium avoids this fate, however, by secreting knoblike proteins that extend through the surface of the red blood cell and anchor the cell to the inner surface of the blood vessel.

Over the course of several days, the immune system of the infected person slowly brings the infection under control. During this time, however, a small proportion of the plasmodium parasites change their knob proteins to a form different from those that sensitized the immune system. Cells infected with these individuals survive the immune response, only to start a new wave of infection.

Scientists have recently discovered how the malarial parasite carries out this antigen-shifting defense. About 6% of the total DNA of the plasmodium is devoted to encoding a block of some 150 *var* genes, which are shifted on and off in multiple combinations. Each time a plasmodium divides, it alters the pattern of *var* gene expression about 2%, an incredibly rapid rate of antigen shifting. The exact means by which this is done is not yet completely understood.

DNA Vaccines May Get around Antigen Shifting

Vaccination against diseases such as smallpox, measles, and polio involves introducing into your body a dead or disabled pathogen, or a harmless microbe with pathogen proteins displayed on its surface. The vaccination triggers an immune response against the pathogen, and the bloodstream of the vaccinated person contains B cells which will remember and quickly destroy the pathogen in future infections. However, for some diseases, vaccination is nearly impossible because of antigen shifting; the pathogens change over time, and the B cells no longer recognize them. Influenza, as we have discussed, presents different surface proteins yearly. The trypanosomes responsible for sleeping sickness change their surface proteins every few weeks.

A new type of vaccine, based on DNA, may prove to be effective against almost any disease. The vaccine makes use of the killer T cells instead of the B cells of the immune system. DNA vaccines consist of a plasmid, a harmless circle of bacterial DNA, that contains a gene from the pathogen that encodes an internal protein, one which is critical to the function of the pathogen and does not change. When this plasmid is injected into cells, the gene they carry is transcribed into protein but is not incorporated into the DNA of the cell’s nucleus. Fragments of the pathogen protein are then stuck on the cell’s membrane, marking it for destruction by T cells. In actual infections later, the immune system will be able to respond immediately. Studies are now underway to isolate the critical, unchanging proteins of pathogens and to investigate fully the use of the vaccines in humans.

Antigen shifting refers to the way a pathogen may defeat the immune system by changing its surface antigens and thereby escaping immune recognition. Pathogens that employ this mechanism include flu viruses, trypanosomes, and the protozoans that cause malaria.

Autoimmunity and Allergy

The previous section described ways that pathogens can elude the immune system to cause diseases. There is another way the immune system can fail; it can itself be the agent of disease. Such is the case with autoimmune diseases and allergies—the immune system is the cause of the problem, not the cure.

Autoimmune Diseases

Autoimmune diseases are produced by failure of the immune system to recognize and tolerate self antigens. This failure results in the activation of autoreactive T cells and the production of autoantibodies by B cells, causing inflammation and organ damage. There are over 40 known or suspected autoimmune diseases that affect 5 to 7% of the population. For reasons that are not understood, two-thirds of the people with autoimmune diseases are women.

Autoimmune diseases can result from a variety of mechanisms. The self antigen may normally be hidden from the immune system, for example, so that the immune system treats it as foreign if exposure later occurs. This occurs when a protein normally trapped in the thyroid follicles triggers autoimmune destruction of the thyroid (Hashimoto's thyroiditis). It also occurs in systemic lupus erythematosus, in which antibodies are made to nucleoproteins. Because the immune attack triggers inflammation, and inflammation causes organ damage, the immune system must be suppressed to alleviate the symptoms of autoimmune diseases. Immune suppression is generally accomplished with corticosteroids (including hydrocortisone) and by nonsteroidal antiinflammatory drugs, including aspirin.

Allergy

The term *allergy*, often used interchangeably with *hypersensitivity*, refers to particular types of abnormal immune responses to antigens, which are called *allergens* in these cases. There are two major forms of allergy: (1) **immediate hypersensitivity**, which is due to an abnormal B-cell response to an allergen that produces symptoms within seconds or minutes, and (2) **delayed hypersensitivity**, which is an abnormal T cell response that produces symptoms within about 48 hours after exposure to an allergen.

Immediate hypersensitivity results from the production of antibodies of the IgE subclass instead of the normal IgG antibodies. Unlike IgG antibodies, IgE antibodies do not circulate in the blood. Instead, they attach to tissue mast

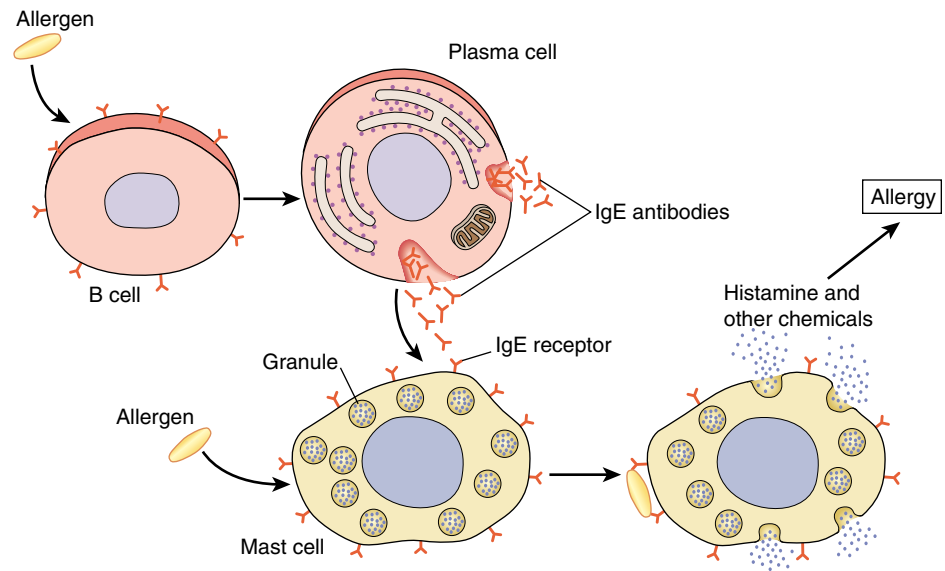


FIGURE 57.29

An allergic reaction. This is an immediate hypersensitivity response, in which B cells secrete antibodies of the IgE class. These antibodies attach to the plasma membranes of mast cells, which secrete histamine in response to antigen-antibody binding.

cells and basophils, which have membrane receptors for these antibodies. When the person is again exposed to the same allergen, the allergen binds to the antibodies attached to the mast cells and basophils. This stimulates these cells to secrete various chemicals, including histamine, which produce the symptom of the allergy (figure 57.29).

Allergens that provoke immediate hypersensitivity include various foods, bee stings, and pollen grains. The most common allergy of this type is seasonal hay fever, which may be provoked by ragweed (*Ambrosia*) pollen grains. These allergic reactions are generally mild, but in some allergies (as to penicillin or peanuts in susceptible people) the widespread and excessive release of histamine may cause **anaphylactic shock**, an uncontrolled fall in blood pressure.

In delayed hypersensitivity, symptoms take a longer time (hours to days) to develop than in immediate hypersensitivity. This may be due to the fact that immediate hypersensitivity is mediated by antibodies, whereas delayed hypersensitivity is a T cell response. One of the best-known examples of delayed hypersensitivity is **contact dermatitis**, caused by poison ivy, poison oak, and poison sumac. Because the symptoms are caused by the secretion of lymphokines rather than by the secretion of histamine, treatment with antihistamines provides little benefit. At present, corticosteroids are the only drugs that can effectively treat delayed hypersensitivity.

Autoimmune diseases are produced when the immune system fails to tolerate self antigens.



Summary

Questions

Media Resources

57.1 Many of the body's most effective defenses are nonspecific.

- Nonspecific defenses include physical barriers such as the skin, phagocytic cells, killer cells, and complement proteins.
- The inflammatory response aids the mobilization of defensive cells at infected sites.

1. How do macrophages destroy foreign cells?
2. How does the complement system participate in defense against infection?



- Art Activity:
Human skin anatomy

57.2 Specific immune defenses require the recognition of antigens.

- Lymphocytes called B cells secrete antibodies and produce the humoral response; lymphocytes called T cells are responsible for cell-mediated immunity.

3. On what types of cells are the two classes of MHC proteins found?



- Specific immunity
- Lymphocytes
- Cell mediated immunity

57.3 T cells organize attacks against invading microbes.

- T cells only respond to antigens presented to them by macrophages or other antigen-presenting cells together with MHC proteins.
- Cytotoxic T cells kill cells that have foreign antigens presented together with MHC-I proteins.

4. In what two ways do macrophages activate helper T cells? How do helper T cells stimulate the proliferation of cytotoxic T cells?



- T-cell function

57.4 B cells label specific cells for destruction.

- The antibody molecules consist of two heavy and two light polypeptide regions arranged like a "Y"; the ends of the two arms bind to antigens.
- An individual can produce a tremendous variety of different antibodies because the genes which produce those antibodies recombine extensively.
- Active immunity occurs when an individual gains immunity by prior exposure to a pathogen; passive immunity is produced by the transfer of antibodies from one individual to another.

5. How do IgM and IgG antibodies differ in triggering destruction of infected cells?
6. How does the clonal selection model help to explain active immunity?
7. How are lymphocytes able to produce millions of different types of immune receptors?



- Clonal selection



- Activity:
Plasma cell production

57.5 All animals exhibit nonspecific immune response but specific ones evolved in vertebrates.

- The immune system evolved in animals from a strictly nonspecific immune response in invertebrates to the two-part immune defense found in mammals.

8. Compare insect and mammalian immune defenses.



- Phagocytic cells

57.6 The immune system can be defeated.

- Flu viruses, trypanosomes, and the protozoan that causes malaria are able to evade the immune system by mutating the genes that produce their surface antigens. In autoimmune diseases, the immune system targets the body's own antigens.

9. What might cause an immune attack of self antigens?
10. How does HIV defeat human immune defenses?



- Abnormalities

58

Maintaining the Internal Environment

Concept Outline

58.1 The regulatory systems of the body maintain homeostasis.

The Need to Maintain Homeostasis. Regulatory mechanisms maintain homeostasis through negative feedback loops.

Antagonistic Effectors and Positive Feedback. Antagonistic effectors cause opposite changes, while positive feedback pushes changes further in the same way.

58.2 The extracellular fluid concentration is constant in most vertebrates.

Osmolality and Osmotic Balance. Vertebrates have to cope with the osmotic gain or loss of body water.

Osmoregulatory Organs. Invertebrates have a variety of organs to regulate water balance; kidneys are the osmoregulatory organs of most vertebrates.

Evolution of the Vertebrate Kidney. Freshwater bony fish produce a dilute urine and marine bony fish produce an isotonic urine. Only birds and mammals can retain so much water that they produce a concentrated urine.

58.3 The functions of the vertebrate kidney are performed by nephrons.

The Mammalian Kidney. Each kidney contains nephrons that produce a filtrate which is modified by reabsorption and secretion to produce urine.

Transport Processes in the Mammalian Nephron. The nephron tubules of birds and mammals have loops of Henle, which function to draw water out of the tubule and back into the blood.

Ammonia, Urea, and Uric Acid. The breakdown of protein and nucleic acids yields nitrogen, which is excreted as ammonia in bony fish, as urea in mammals, and as uric acid in reptiles and birds.

58.4 The kidney is regulated by hormones.

Hormones Control Homeostatic Functions.

Antidiuretic hormone promotes water retention and the excretion of a highly concentrated urine. Aldosterone stimulates the retention of salt and water, whereas atrial natriuretic hormone promotes the excretion of salt and water.



FIGURE 58.1

Regulating body temperature with water. One of the ways an elephant can regulate its temperature is to spray water on its body. Water also cycles through the elephant's body in enormous quantities each day and helps to regulate its internal environment.

The first vertebrates evolved in seawater, and the physiology of all vertebrates reflects this origin. Approximately two-thirds of every vertebrate's body is water. If the amount of water in the body of a vertebrate falls much lower than this, the animal will die. In this chapter, we discuss the various mechanisms by which animals avoid gaining or losing too much water. As we shall see, these mechanisms are closely tied to the way animals exploit the varied environments in which they live and to the regulatory systems of the body (figure 58.1).

58.1 The regulatory systems of the body maintain homeostasis.

The Need to Maintain Homeostasis

As the animal body has evolved, specialization has increased. Each cell is a sophisticated machine, finely tuned to carry out a precise role within the body. Such specialization of cell function is possible only when extracellular conditions are kept within narrow limits. Temperature, pH, the concentrations of glucose and oxygen, and many other factors must be held fairly constant for cells to function efficiently and interact properly with one another.

Homeostasis may be defined as the dynamic constancy of the internal environment. The term *dynamic* is used because conditions are never absolutely constant, but fluctuate continuously within narrow limits. Homeostasis is essential for life, and most of the regulatory mechanisms of the vertebrate body that are not devoted to reproduction are concerned with maintaining homeostasis.

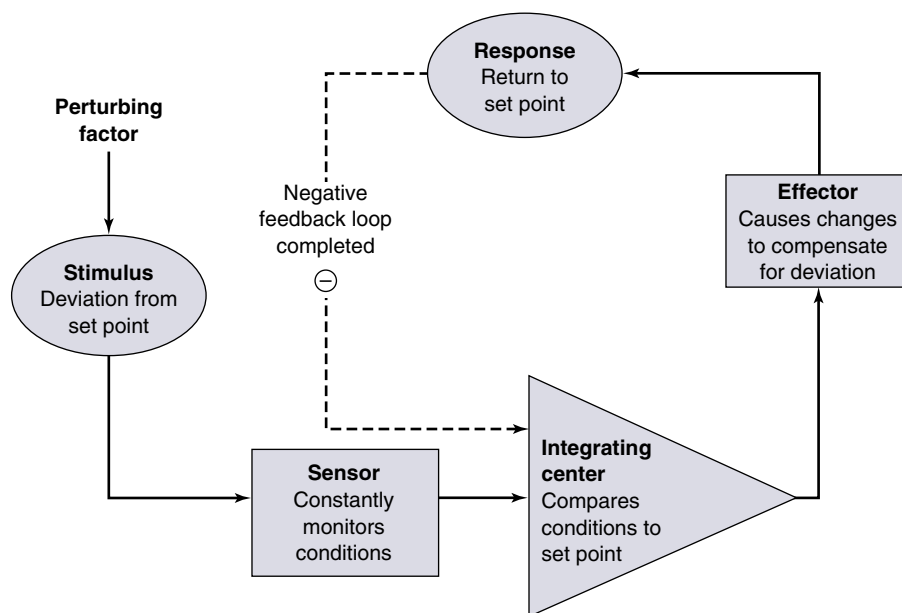


FIGURE 58.2

A generalized diagram of a negative feedback loop. Negative feedback loops maintain a state of homeostasis, or dynamic constancy of the internal environment, by correcting deviations from a set point.

Negative Feedback Loops

To maintain internal constancy, the vertebrate body must have **sensors** that are able to measure each condition of the internal environment (figure 58.2). These constantly monitor the extracellular conditions and relay this information (usually via nerve signals) to an **integrating center**, which contains the “*set point*” (the proper value for that condition). This set point is analogous to the temperature setting on a house thermostat. In a similar manner, there are set points for body temperature, blood glucose concentration, the tension on a tendon, and so on. The integrating center is often a particular region of the brain or spinal cord, but in some cases it can also be cells of endocrine glands. It receives messages from several sensors, weighing the relative strengths of each sensor input, and then determines whether the value of the condition is deviating from the set point. When a deviation in a condition occurs (the “stimulus”), the integrating center sends a message to increase or decrease the activity of particular **effectors**. Effectors are generally muscles or glands, and can change the value of the condition in question back toward the set point value (the “response”).

To return to the idea of a home thermostat, suppose you set the thermostat at a set point of 70°F. If the temperature of the house rises sufficiently above the set point, the thermostat (equivalent to an integrating center) receives this

input from a temperature sensor, like a thermometer (a sensor) within the wall unit. It compares the actual temperature to its set point. When these are different, it sends a signal to an effector. The effector in this case may be an air conditioner, which acts to reverse the deviation from the set point.

In a human, if the body temperature exceeds the set point of 37°C, sensors in a part of the brain detect this deviation. Acting via an integrating center (also in the brain), these sensors stimulate effectors (including sweat glands) that lower the temperature (figure 58.3). One can think of the effectors as “defending” the set points of the body against deviations. Because the activity of the effectors is influenced by the effects they produce, and because this regulation is in a negative, or reverse, direction, this type of control system is known as a *negative feedback loop*.

The nature of the negative feedback loop becomes clear when we again refer to the analogy of the thermostat and air conditioner. After the air conditioner has been on for some time, the room temperature may fall significantly below the set point of the thermostat. When this occurs, the air conditioner will be turned off. The effector (air conditioner) is turned on by a high temperature; and, when activated, it produces a negative change (lowering of the temperature) that ultimately causes the effector to be turned off. In this way, constancy is maintained.

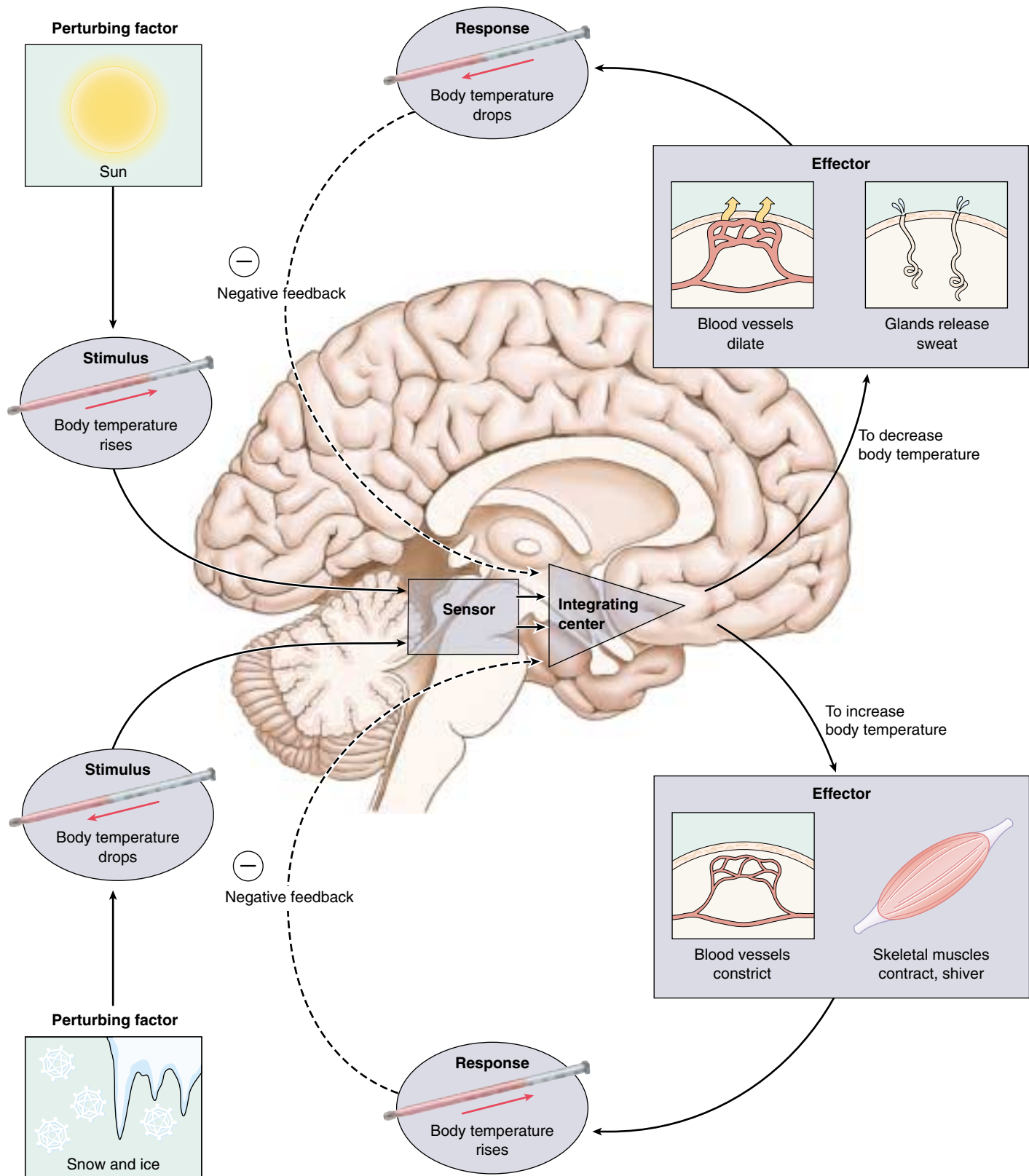


FIGURE 58.3

Negative feedback loops keep the body temperature within a normal range. An increase (*top*) or decrease (*bottom*) in body temperature is sensed by the brain. The integrating center in the brain then processes the information and activates effectors, such as surface blood vessels, sweat glands, and skeletal muscles. When the body temperature returns to normal, negative feedback prevents further stimulation of the effectors by the integrating center.

Regulating Body Temperature

Humans, together with other mammals and with birds, are *endothermic*; they can maintain relatively constant body temperatures independent of the environmental temperature. When the temperature of your blood exceeds 37°C (98.6°F), neurons in a part of the brain called the hypothalamus detect the temperature change. Acting through the control of motor neurons, the hypothalamus responds by promoting the dissipation of heat through sweating, dilation of blood vessels in the skin, and other mechanisms. These responses tend to counteract the rise in body temperature. When body temperature falls, the hypothalamus coordinates a different set of responses, such as shivering and the constriction of blood vessels in the skin, which help to raise body temperature and correct the initial challenge to homeostasis.

Vertebrates other than mammals and birds are *ectothermic*; their body temperatures are more or less dependent on the environmental temperature. However, to the extent that it is possible, many ectothermic vertebrates attempt to maintain some degree of temperature homeostasis. Certain large fish, including tuna, swordfish, and some sharks, for example, can maintain parts of their body at a significantly higher temperature than that of the water. Reptiles attempt to maintain a constant body temperature through behavioral means—by placing themselves in varying locations of sun and shade (see chapter 29). That’s why you frequently see lizards basking in the sun. Sick lizards even give themselves a “fever” by seeking warmer locations!

Most invertebrates do not employ feedback regulation to physiologically control their body temperature. Instead, they use behavior to adjust their temperature. Many butterflies, for example, must reach a certain body temperature before they can fly. In the cool of the morning they orient so as to maximize their absorption of sunlight. Moths and many other insects use a shivering reflex to warm their thoracic flight muscles (figure 58.4).

Regulating Blood Glucose

When you digest a carbohydrate-containing meal, you absorb glucose into your blood. This causes a temporary rise in the blood glucose concentration, which is brought back down in a few hours. What counteracts the rise in blood glucose following a meal?

Glucose levels within the blood are constantly monitored by a sensor, the islets of Langerhans in the pancreas. When levels increase, the islets secrete the hormone *insulin*, which stimulates the uptake of blood glucose into muscles, liver, and adipose tissue. The islets are, in this case, the sensor and the integrating center. The muscles, liver, and adipose cells are the effectors, taking up glucose to control the levels. The muscles and liver can convert the glucose into the polysaccharide glycogen; adipose cells can convert glucose into fat. These actions lower the blood glucose (figure 58.5) and help to store energy in forms that the body can use later.

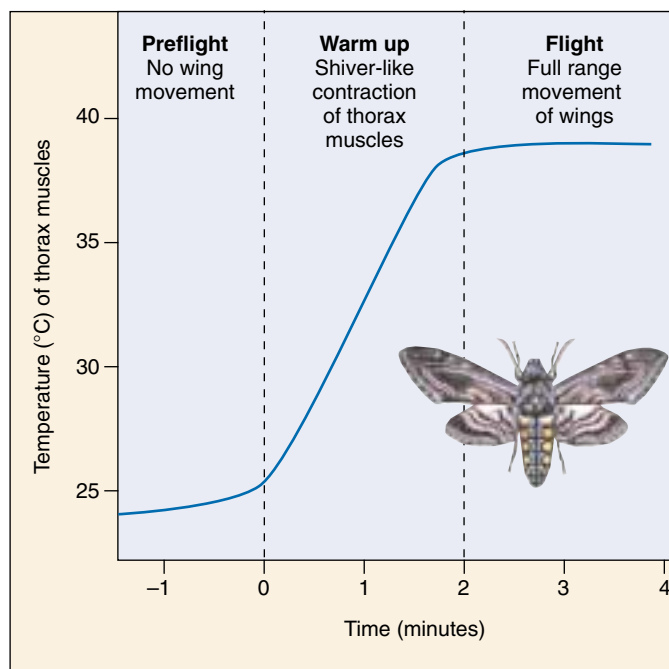


FIGURE 58.4

Thermoregulation in insects. Some insects, such as the sphinx moth, contract their thoracic muscles to warm up for flight.

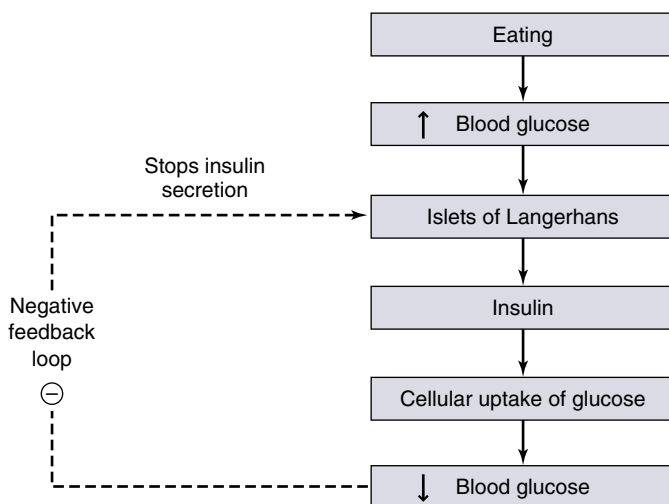


FIGURE 58.5

The negative feedback control of blood glucose. The rise in blood glucose concentration following a meal stimulates the secretion of insulin from the islets of Langerhans in the pancreas. Insulin is a hormone that promotes the entry of glucose in skeletal muscle and other tissue, thereby lowering the blood glucose and compensating for the initial rise.

Negative feedback mechanisms correct deviations from a set point for different internal variables. In this way, body temperature and blood glucose, for example, are kept within normal limits.

Antagonistic Effectors and Positive Feedback

The negative feedback mechanisms that maintain homeostasis often oppose each other to produce a finer degree of control. In a few cases positive feedback mechanisms, which push a change further in the same direction, are used by the body.

Antagonistic Effectors

Most factors in the internal environment are controlled by several effectors, which often have antagonistic actions. Control by antagonistic effectors is sometimes described as “push-pull,” in which the increasing activity of one effector is accompanied by decreasing activity of an antagonistic effector. This affords a finer degree of control than could be achieved by simply switching one effector on and off.

Room temperature can be maintained, for example, by simply turning an air conditioner on and off, or by just turning a heater on and off. A much more stable temperature, however, can be achieved if the air conditioner and heater are both controlled by a thermostat (figure 58.6). Then the heater is turned on when the air conditioner shuts off, and vice versa. Antagonistic effectors are similarly involved in the control of body temperature and blood glucose. Whereas insulin, for example, lowers blood glucose following a meal, other hormones act to raise the blood glucose concentration between meals, especially when a person is exercising. The heart rate is similarly controlled by antagonistic effectors. Stimulation of one group of nerve fibers increases the heart rate, while stimulation of another group slows the heart rate.

Positive Feedback Loops

Feedback loops that accentuate a disturbance are called positive feedback loops. In a positive feedback loop, perturbations cause the effector to drive the value of the controlled variable even *farther* from the set point. Hence, systems in which there is positive feedback are highly unstable, analogous to a spark that ignites an explosion. They do not help to maintain homeostasis. Nevertheless, such systems are important components of some physiological mechanisms. For example, positive feedback occurs in blood clotting, where one clotting factor activates another in a cascade that leads quickly to the formation of a clot. Positive feedback also plays a role in the contractions of the uterus during childbirth (figure 58.7). In this case, stretching of the uterus by the fetus stimulates contraction, and contraction causes further stretching; the cycle continues until the fetus is expelled from the uterus. In the body, most positive feedback systems act as part of some larger mechanism that maintains homeostasis. In the examples we’ve described, formation of a blood clot stops bleeding and hence tends to keep blood volume constant, and expulsion of the fetus reduces the contractions of the uterus.

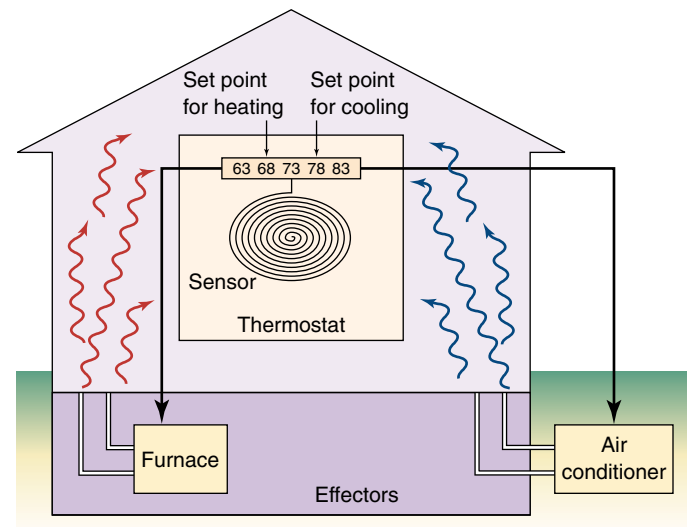


FIGURE 58.6

Room temperature is maintained by antagonistic effectors. If a thermostat senses a low temperature, the heater is turned on and the air conditioner is turned off. If the temperature is too high, the air conditioner is activated, and the heater is turned off.

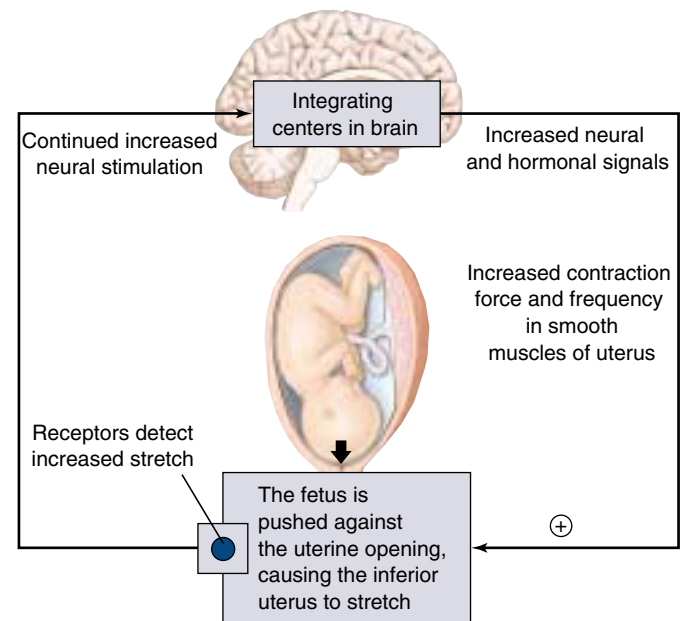


FIGURE 58.7

An example of positive feedback during childbirth. This is one of the few examples of positive feedback that operate in the vertebrate body.

Antagonistic effectors that act antagonistically to each other are more effective than effectors that act alone. Positive feedback mechanisms accentuate changes and have limited functions in the body.

58.2 The extracellular fluid concentration is constant in most vertebrates.

Osmolality and Osmotic Balance

Water in an animal's body is distributed between the intracellular and extracellular compartments (figure 58.8). In order to maintain osmotic balance, the extracellular compartment of an animal's body (including its blood plasma) must be able to take water from its environment or to excrete excess water into its environment. Inorganic ions must also be exchanged between the extracellular body fluids and the external environment to maintain homeostasis. Such exchanges of water and electrolytes between the body and the external environment occur across specialized epithelial cells and, in most vertebrates, through a filtration process in the kidneys.

Most vertebrates maintain homeostasis in regard to the total solute concentration of their extracellular fluids and in regard to the concentration of specific inorganic ions. Sodium (Na^+) is the major cation in extracellular fluids, and chloride (Cl^-) is the major anion. The divalent cations, calcium (Ca^{++}) and magnesium (Mg^{++}), as well as other ions, also have important functions and must be maintained at their proper concentrations.

Osmolality and Osmotic Pressure

Osmosis is the diffusion of water across a membrane, and it always occurs from a more dilute solution (with a lower solute concentration) to a less dilute solution (with a higher solute concentration). Because the total solute concentration of a solution determines its osmotic behavior, the total moles of solute per kilogram of water is expressed as the **osmolality** of the solution. Solutions that have the same osmolality are *isosmotic*. A solution with a lower or higher osmolality than another is called *hypoosmotic* or *hyperosmotic*, respectively.

If one solution is hyperosmotic compared with another, and if the two solutions are separated by a semipermeable membrane, water may move by osmosis from the more dilute solution to the hyperosmotic one. In this case, the hyperosmotic solution is also **hypertonic** ("higher strength") compared with the other solution, and it has a higher osmotic pressure. The **osmotic pressure** of a solution is a measure of its tendency to take in water by osmosis. A cell placed in a hypertonic solution, which has a higher osmotic pressure than the cell cytoplasm, will lose water to the surrounding solution and shrink. A cell placed in a **hypotonic** solution, in contrast, will gain water and expand.

If a cell is placed in an isosmotic solution, there may be no net water movement. In this case, the isosmotic solution can also be said to be **isotonic**. Isotonic solutions such as normal saline and 5% dextrose are used in medical care to bathe exposed tissues and to be given as intravenous fluids.

Osmoconformers and Osmoregulators

Most marine invertebrates are **osmoconformers**; the osmolality of their body fluids is the same as that of seawater (although the concentrations of particular solutes, such as magnesium ion, are not equal). Because the extracellular fluids are isotonic to seawater, there is no osmotic gradient and no tendency for water to leave or enter the body. Therefore, osmoconformers are in osmotic equilibrium with their environment. Among the vertebrates, only the primitive hagfish are strict osmoconformers. The sharks and their relatives in the class Chondrichthyes (cartilaginous fish) are also isotonic to seawater, even though their blood level of NaCl is lower than that of seawater; the difference in total osmolality is made up by retaining urea at a high concentration in their blood plasma.

All other vertebrates are osmoregulators—that is, animals that maintain a relatively constant blood osmolality despite the different concentration in the surrounding environment. The maintenance of a relatively constant body fluid osmolality has permitted vertebrates to exploit a wide variety of ecological niches. Achieving this constancy, however, requires continuous regulation.

Freshwater vertebrates have a much higher solute concentration in their body fluids than that of the surrounding water. In other words, they are hypertonic to their environment. Because of their higher osmotic pressure, water tends to enter their bodies. Consequently, they must prevent water from entering their bodies as much as possible and eliminate the excess water that does enter. In addition, they tend to lose inorganic ions to their environment and so must actively transport these ions back into their bodies.

In contrast, most marine vertebrates are hypotonic to their environment; their body fluids have only about one-third the osmolality of the surrounding seawater. These animals are therefore in danger of losing water by osmosis and must retain water to prevent dehydration. They do this by drinking seawater and eliminating the excess ions through their kidneys and gills.

The body fluids of terrestrial vertebrates have a higher concentration of water than does the air surrounding them. Therefore, they tend to lose water to the air by evaporation from the skin and lungs. All reptiles, birds, and mammals, as well as amphibians during the time when they live on land, face this problem. These vertebrates have evolved excretory systems that help them retain water.

Marine invertebrates are isotonic with their environment, but most vertebrates are either hypertonic or hypotonic to their environment and thus tend to gain or lose water. Physiological mechanisms help most vertebrates to maintain a constant blood osmolality and constant concentrations of individual ions.

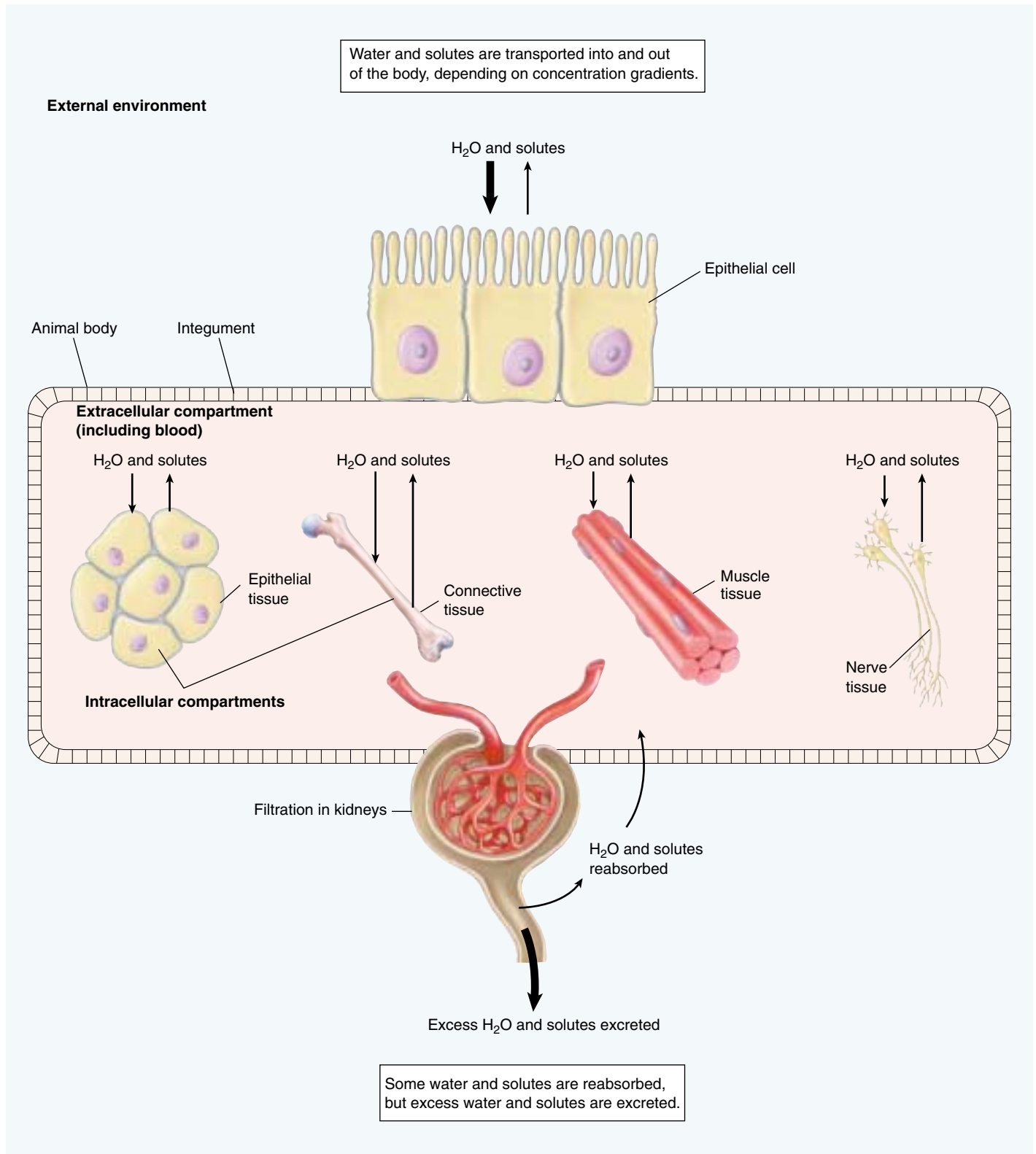


FIGURE 58.8

The interaction between intracellular and extracellular compartments of the body and the external environment. Water can be taken in from the environment or lost to the environment. Exchanges of water and solutes between the extracellular fluids of the body and the environment occur across transport epithelia, and water and solutes can be filtered out of the blood by the kidneys. Overall, the amount of water and solutes that enters and leaves the body must be balanced in order to maintain homeostasis.

Osmoregulatory Organs

Animals have evolved a variety of mechanisms to cope with problems of water balance. In many animals, the removal of water or salts from the body is coupled with the removal of metabolic wastes through the excretory system. Protists employ contractile vacuoles for this purpose, as do sponges. Other multicellular animals have a system of excretory tubules (little tubes) that expel fluid and wastes from the body.

In flatworms, these tubules are called *protonephridia*, and they branch throughout the body into bulblike **flame cells** (figure 58.9). While these simple excretory structures open to the outside of the body, they do not open to the inside of the body. Rather, cilia within the flame cells must draw in fluid from the body. Water and metabolites are then reabsorbed, and the substances to be excreted are expelled through excretory pores.

Other invertebrates have a system of tubules that open both to the inside and to the outside of the body. In the earthworm, these tubules are known as *metanephridia* (figure 58.10). The metanephridia obtain fluid from the body cavity through a process of filtration into funnel-shaped structures called *nephrostomes*. The term *filtration* is used because the fluid is formed under pressure and passes through small openings, so that molecules larger than a certain size are excluded. This filtered fluid is isotonic to the fluid in the coelom, but as it passes through the tubules of the metanephridia, NaCl is removed by active transport processes. A general term for transport out of the tubule and into the surrounding body fluids is *reabsorption*. Because salt is reabsorbed from the filtrate, the urine excreted is more dilute than the body fluids (is hypotonic). The kidneys of mollusks and the excretory organs of crustaceans (called *antennal glands*) also produce urine by filtration and reclaim certain ions by reabsorption.

The excretory organs in insects are the Malpighian tubules (figure 58.11), extensions of the digestive tract that branch off anterior to the hindgut. Urine is not formed by filtration in these tubules, because there is no pressure difference between the blood in the body cavity and the tubule. Instead, waste molecules and potassium (K^+) ions are secreted into the tubules by active transport. Secretion is the opposite of reabsorption—ions or molecules are transported from the body fluid into the tubule. The secretion of K^+ creates an osmotic gradient that causes water to enter the tubules by osmosis from the body's

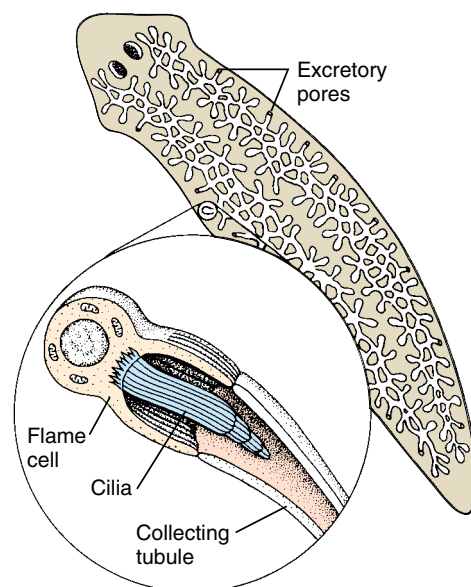


FIGURE 58.9

The protonephridia of flatworms. A branching system of tubules, bulblike flame cells, and excretory pores make up the protonephridia of flatworms. Cilia inside the flame cells draw in fluids from the body by their beating action. Substances are then expelled through pores which open to the outside of the body.

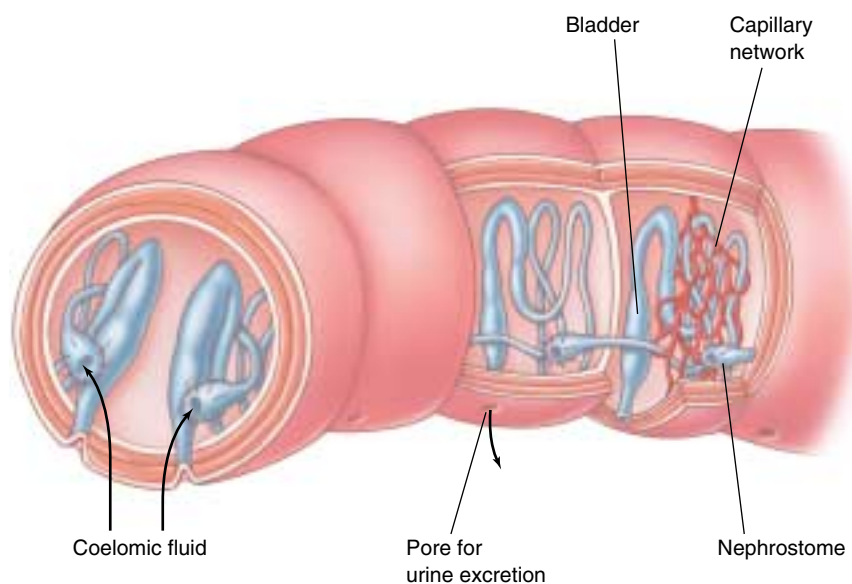


FIGURE 58.10

The metanephridia of annelids. Most invertebrates, such as the annelid shown here, have metanephridia. These consist of tubules that receive a filtrate of coelomic fluid, which enters the funnel-like nephrostomes. Salt can be reabsorbed from these tubules, and the fluid that remains, urine, is released from pores into the external environment.

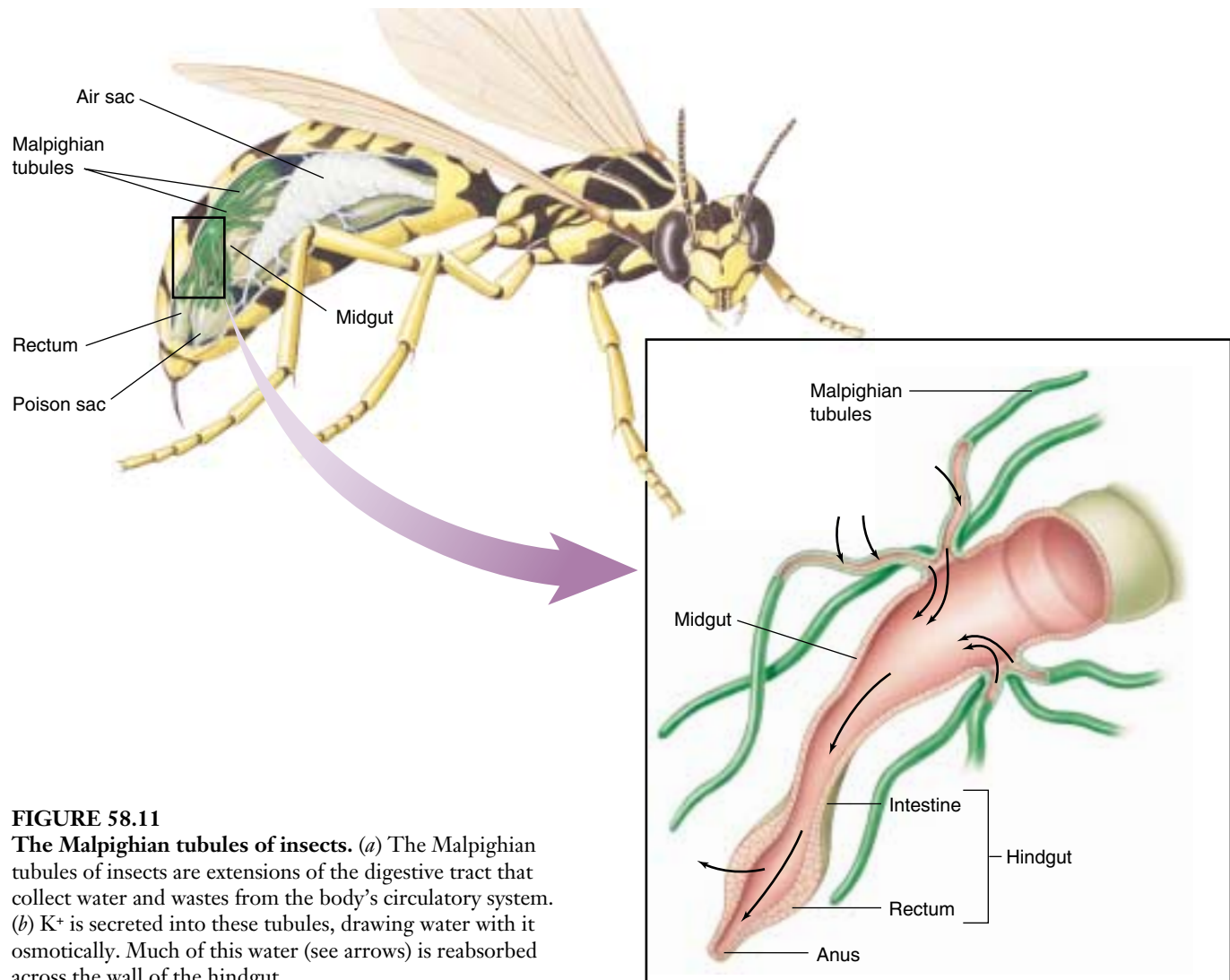


FIGURE 58.11

The Malpighian tubules of insects. (a) The Malpighian tubules of insects are extensions of the digestive tract that collect water and wastes from the body's circulatory system. (b) K^+ is secreted into these tubules, drawing water with it osmotically. Much of this water (see arrows) is reabsorbed across the wall of the hindgut.

open circulatory system. Most of the water and K^+ is then reabsorbed into the circulatory system through the epithelium of the hindgut, leaving only small molecules and waste products to be excreted from the rectum along with feces. Malpighian tubules thus provide a very efficient means of water conservation.

The kidneys of vertebrates, unlike the Malpighian tubules of insects, create a tubular fluid by filtration of the blood under pressure. In addition to containing waste products and water, the filtrate contains many small molecules that are of value to the animal, including glucose, amino acids, and vitamins. These molecules and most of the water are reabsorbed from the tubules into the blood, while wastes remain in the filtrate. Additional wastes may be secreted by the tubules and added to the filtrate, and the final waste product, urine, is eliminated from the body.

It may seem odd that the vertebrate kidney should filter out almost everything from blood plasma (except proteins, which are too large to be filtered) and then spend energy to take back or reabsorb what the body needs. But selective reabsorption provides great flexibility, because various vertebrate groups have evolved the ability to reabsorb different molecules that are especially valuable in particular habitats. This flexibility is a key factor underlying the successful colonization of many diverse environments by the vertebrates.

Many invertebrates filter fluid into a system of tubules and then reabsorb ions and water, leaving waste products for excretion. Insects create an excretory fluid by secreting K^+ into tubules, which draws water osmotically. The vertebrate kidney produces a filtrate that enters tubules and is modified to become urine.

Evolution of the Vertebrate Kidney

The kidney is a complex organ made up of thousands of repeating units called **nephrons**, each with the structure of a bent tube (figure 58.12). Blood pressure forces the fluid in blood past a filter, called the *glomerulus*, at the top of each nephron. The glomerulus retains blood cells, proteins, and other useful large molecules in the blood but allows the water, and the small molecules and wastes dissolved in it, to pass through and into the bent tube part of the nephron. As the filtered fluid passes through the nephron tube, useful sugars and ions are recovered from it by active transport, leaving the water and metabolic wastes behind in a fluid urine.

Although the same basic design has been retained in all vertebrate kidneys, there have been a few modifications. Because the original glomerular filtrate is isotonic to blood, all vertebrates can produce a urine that is isotonic to blood by reabsorbing ions and water in equal proportions or hypotonic to blood—that is, more dilute than the blood, by reabsorbing relatively less water blood. Only birds and mammals can reabsorb enough water from their glomerular filtrate to produce a urine that is hypertonic to blood—that is, more concentrated than the blood, by reabsorbing relatively more water.

Freshwater Fish

Kidneys are thought to have evolved first among the freshwater teleosts, or bony fish. Because the body fluids of a freshwater fish have a greater osmotic concentration than the surrounding water, these animals face two serious problems: (1) water tends to enter the body from the environment; and (2) solutes tend to leave the body and

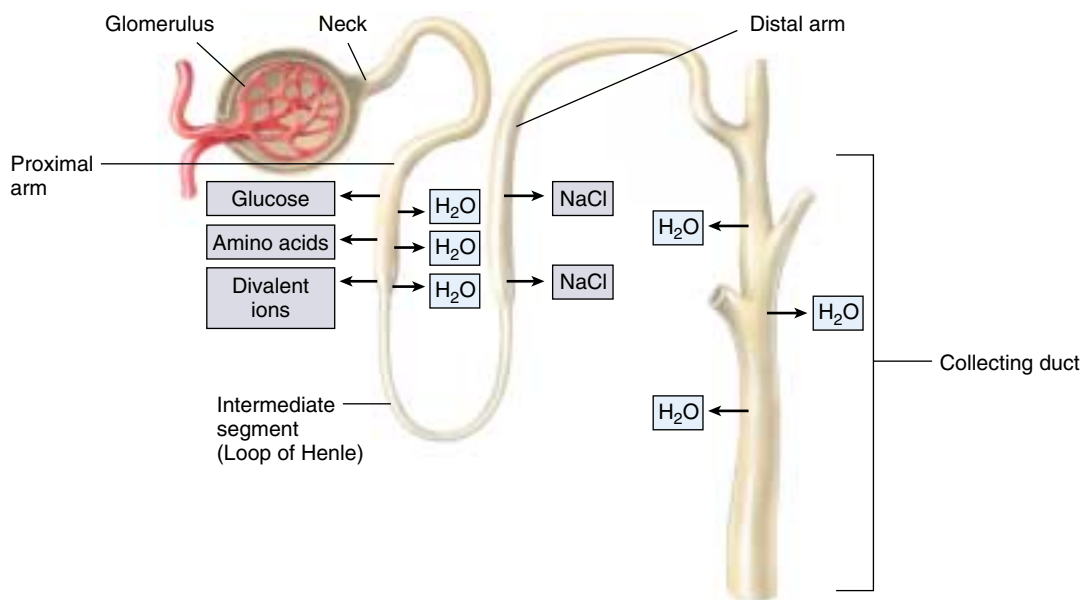
enter the environment. Freshwater fish address the first problem by *not* drinking water and by excreting a large volume of dilute urine, which is hypotonic to their body fluids. They address the second problem by reabsorbing ions across the nephron tubules, from the glomerular filtrate back into the blood. In addition, they actively transport ions across their gill surfaces from the surrounding water into the blood.

Marine Bony Fish

Although most groups of animals seem to have evolved first in the sea, marine bony fish (teleosts) probably evolved from freshwater ancestors, as was mentioned in chapter 48. They faced significant new problems in making the transition to the sea because their body fluids are hypotonic to the surrounding seawater. Consequently, water tends to leave their bodies by osmosis across their gills, and they also lose water in their urine. To compensate for this continuous water loss, marine fish drink large amounts of seawater (figure 58.13).

Many of the divalent cations (principally Ca^{++} and Mg^{++}) in the seawater that a marine fish drinks remain in the digestive tract and are eliminated through the anus. Some, however, are absorbed into the blood, as are the monovalent ions K^+ , Na^+ , and Cl^- . Most of the monovalent ions are actively transported out of the blood across the gill surfaces, while the divalent ions that enter the blood are secreted into the nephron tubules and excreted in the urine. In these two ways, marine bony fish eliminate the ions they get from the seawater they drink. The urine they excrete is isotonic to their body fluids. It is more concentrated than the urine of freshwater fish, but not as concentrated as that of birds and mammals.

FIGURE 58.12
The basic organization of the vertebrate nephron. The nephron tubule of the freshwater fish is a basic design that has been retained in the kidneys of marine fish and terrestrial vertebrates that evolved later. Sugars, amino acids, and divalent ions such as Ca^{++} are recovered in the proximal arm; monovalent ions such as Na^+ and Cl^- are recovered in the distal arm; and water is recovered in the collecting duct.



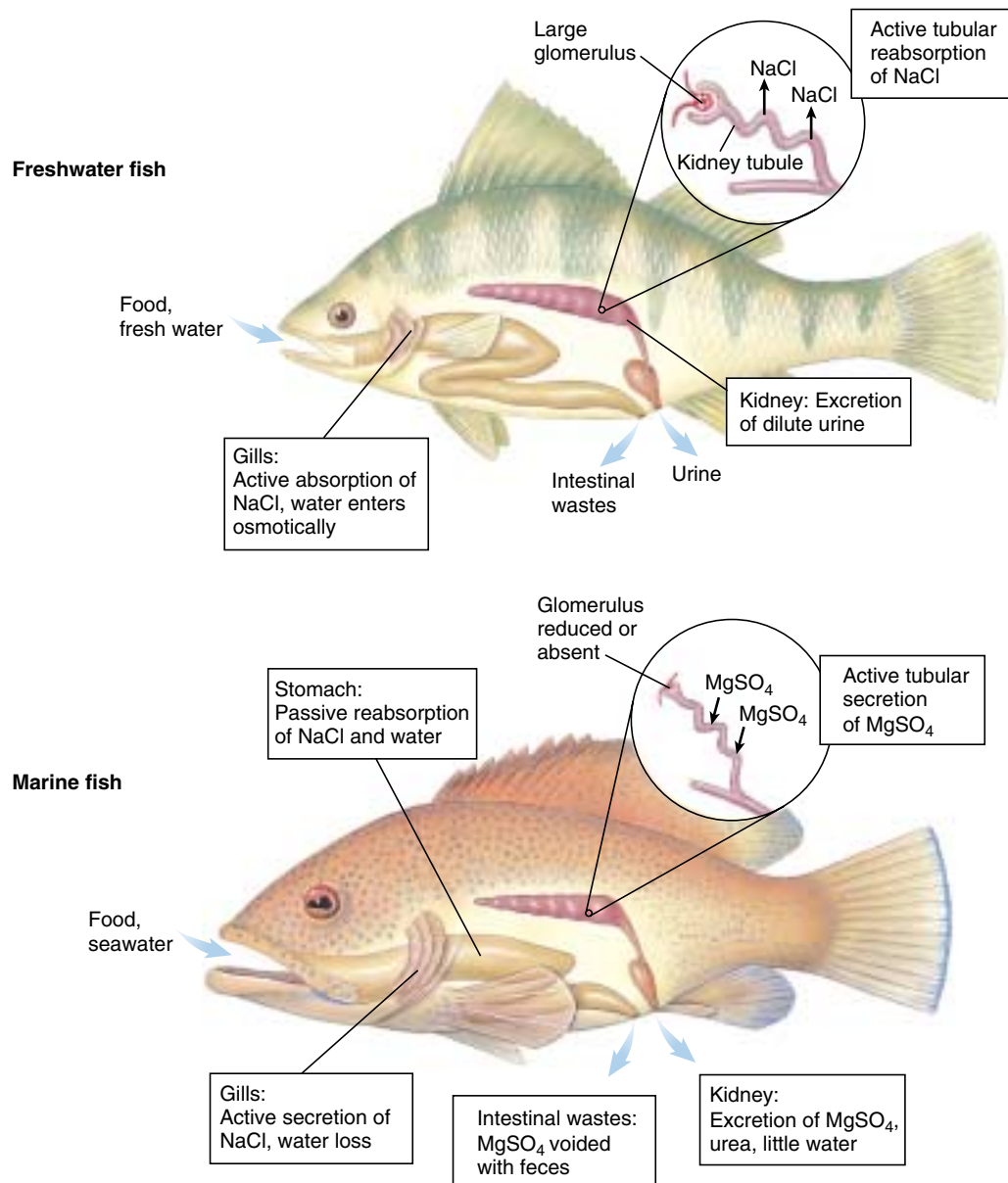


FIGURE 58.13

Freshwater and marine teleosts (bony fish) face different osmotic problems. Whereas the freshwater teleost is hypertonic to its environment, the marine teleost is hypotonic to seawater. To compensate for its tendency to take in water and lose ions, a freshwater fish excretes dilute urine, avoids drinking water, and reabsorbs ions across the nephron tubules. To compensate for its osmotic loss of water, the marine teleost drinks seawater and eliminates the excess ions through active transport across epithelia in the gills and kidneys.

Cartilaginous Fish

The elasmobranchs, including sharks and rays, are by far the most common subclass in the class Chondrichthyes (cartilaginous fish). Elasmobranchs have solved the osmotic problem posed by their seawater environment in a different way than have the bony fish. Instead of having body fluids that are hypotonic to seawater, so that they have to continuously drink seawater and actively pump out ions, the elasmobranchs reabsorb urea from the nephron tubules and main-

tain a blood urea concentration that is 100 times higher than that of mammals. This added urea makes their blood approximately isotonic to the surrounding sea. Because there is no net water movement between isotonic solutions, water loss is prevented. Hence, these fishes do not need to drink seawater for osmotic balance, and their kidneys and gills do not have to remove large amounts of ions from their bodies. The enzymes and tissues of the cartilaginous fish have evolved to tolerate the high urea concentrations.

Amphibians and Reptiles

The first terrestrial vertebrates were the amphibians, and the amphibian kidney is identical to that of freshwater fish. This is not surprising, because amphibians spend a significant portion of their time in fresh water, and when on land, they generally stay in wet places. Amphibians produce a very dilute urine and compensate for their loss of Na⁺ by actively transporting Na⁺ across their skin from the surrounding water.

Reptiles, on the other hand, live in diverse habitats. Those living mainly in fresh water occupy a habitat simi-

lar to that of the freshwater fish and amphibians and thus have similar kidneys. Marine reptiles, including some crocodilians, sea turtles, sea snakes, and one lizard, possess kidneys similar to those of their freshwater relatives but face opposite problems; they tend to lose water and take in salts. Like marine teleosts (bony fish), they drink the seawater and excrete an isotonic urine. Marine teleosts eliminate the excess salt by transport across their gills, while marine reptiles eliminate excess salt through salt glands located near the nose or the eye (figure 58.14).

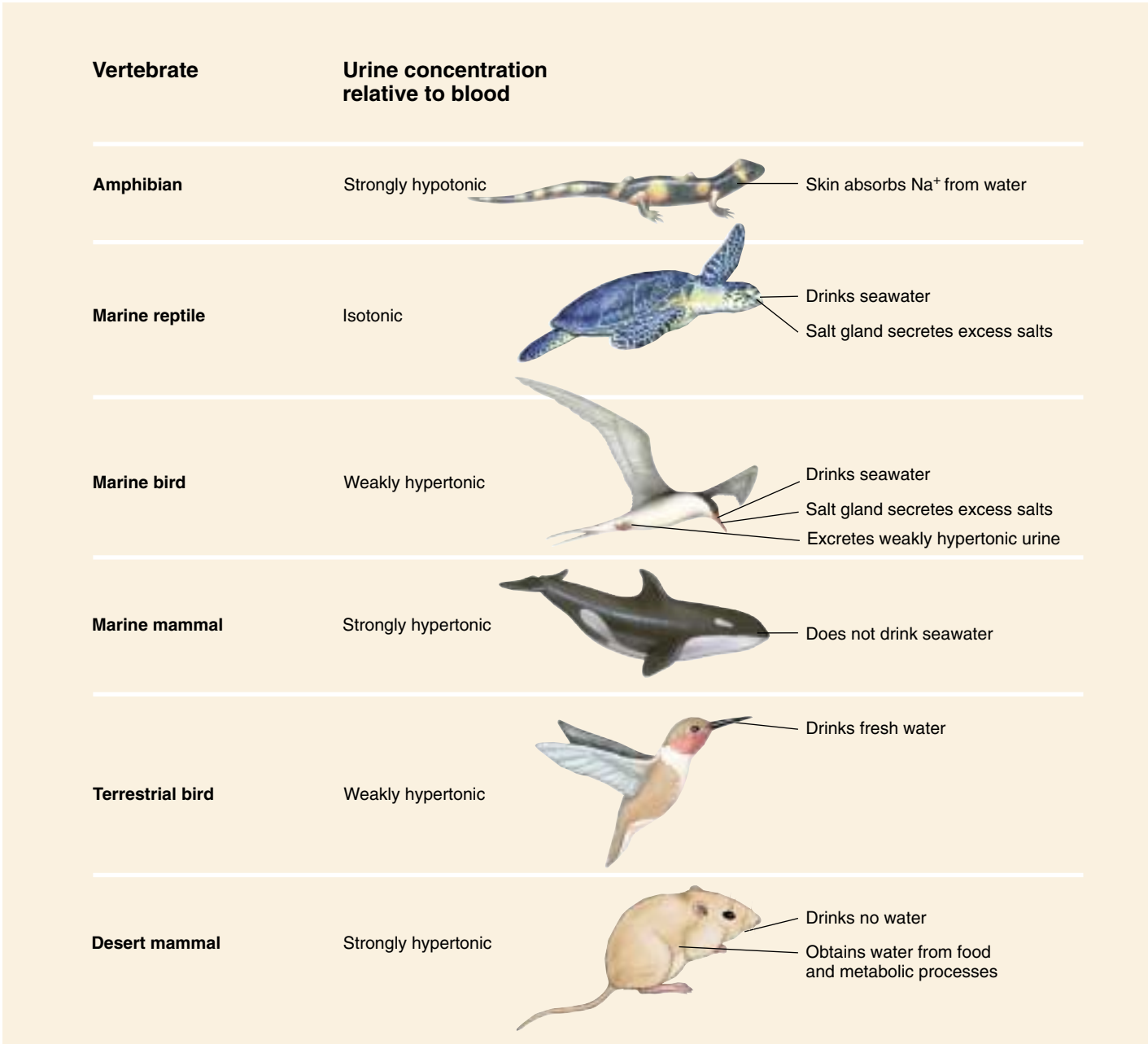


FIGURE 58.14
Osmoregulation by some vertebrates. Only birds and mammals can produce a hypertonic urine and thereby retain water efficiently, but marine reptiles and birds can drink seawater and excrete the excess salt through salt glands.

The kidneys of terrestrial reptiles also reabsorb much of the salt and water in their nephron tubules, helping somewhat to conserve blood volume in dry environments. Like fish and amphibians, they cannot produce urine that is more concentrated than the blood plasma. However, when their urine enters their cloaca (the common exit of the digestive and urinary tracts), additional water can be reabsorbed.

Mammals and Birds

Mammals and birds are the only vertebrates able to produce urine with a higher osmotic concentration than their body fluids. This allows these vertebrates to excrete their waste products in a small volume of water, so that more water can be retained in the body. Human kidneys can produce urine that is as much as 4.2 times as concentrated as blood plasma, but the kidneys of some other mammals are even more efficient at conserving water. For example, camels, gerbils, and pocket mice of the genus *Perognathus* can excrete urine 8, 14, and 22 times as concentrated as their blood plasma, respectively. The kidneys of the kangaroo rat (figure 58.15) are so efficient it never has to drink water; it can obtain all the water it needs from its food and from water produced in aerobic cell respiration!

The production of hypertonic urine is accomplished by the *loop of Henle* portion of the nephron (see figure 58.18), found only in mammals and birds. A nephron with a long loop of Henle extends deeper into the renal medulla, where the hypertonic osmotic environment draws out more water, and so can produce more concentrated urine. Most mammals have some nephrons with short loops and other nephrons with loops that are much longer (see figure 58.17). Birds, however, have relatively few or no nephrons with long loops, so they cannot produce urine that is as concentrated as that of mammals. At most, they can only reabsorb enough water to produce a urine that is about twice the concentration of their blood. Marine birds solve the problem of water loss by drinking salt water and then excreting the excess salt from salt glands near the eyes (figure 58.16).

The moderately hypertonic urine of a bird is delivered to its cloaca, along with the fecal material from its digestive tract. If needed, additional water can be absorbed across the wall of the cloaca to produce a semisolid white paste or pellet, which is excreted.

The kidneys of freshwater fish must excrete copious amounts of very dilute urine, while marine teleosts drink seawater and excrete an isotonic urine. The basic design and function of the nephron of freshwater fishes have been retained in the terrestrial vertebrates. Modifications, particularly the presence of a loop of Henle, allow mammals and birds to reabsorb more water and produce a hypertonic urine.



FIGURE 58.15

The kangaroo rat, *Dipodomys panamintensis*. This mammal has very efficient kidneys that can concentrate urine to a high degree by reabsorbing water, thereby minimizing water loss from the body. This feature is extremely important to the kangaroo rat's survival in dry or desert habitats.

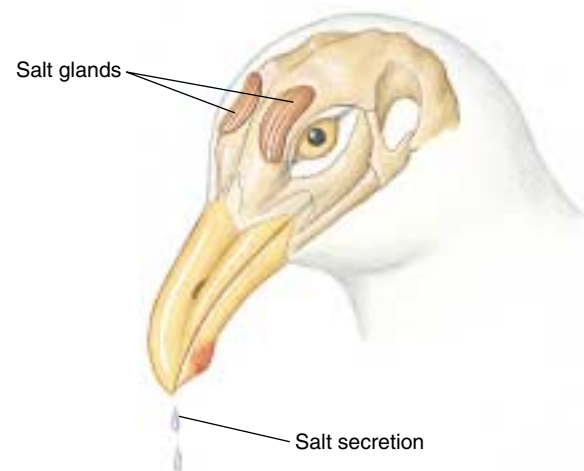


FIGURE 58.16

Marine birds drink seawater and then excrete the salt through salt glands. The extremely salty fluid excreted by these glands can then dribble down the beak.

58.3 The functions of the vertebrate kidney are performed by nephrons.

The Mammalian Kidney

In humans, the kidneys are fist-sized organs located in the region of the lower back. Each kidney receives blood from a renal artery, and it is from this blood that urine is produced. Urine drains from each kidney through a **ureter**, which carries the urine to a **urinary bladder**. Within the kidney, the mouth of the ureter flares open to form a funnel-like structure, the *renal pelvis*. The renal pelvis, in turn, has cup-shaped extensions that receive urine from the renal tissue. This tissue is divided into an outer **renal cortex** and an inner **renal medulla** (figure 58.17). Together, these structures perform filtration, reabsorption, secretion, and excretion.

Nephron Structure and Filtration

On a microscopic level, each kidney contains about one million functioning *nephrons*. Mammalian kidneys contain a mixture of juxtamedullary nephrons, which have long loops which dip deeply into the medulla, and cortical nephrons with shorter loops (see figure 58.17). The significance of the length of the loops will be explained a little later.

Each nephron consists of a long tubule and associated small blood vessels. First, blood is carried by an *afferent arteriole* to a tuft of capillaries in the renal cortex, the **glomerulus** (figure 58.18). Here the blood is filtered as the blood pressure forces fluid through the porous capillary walls. Blood cells and plasma proteins are too large to enter

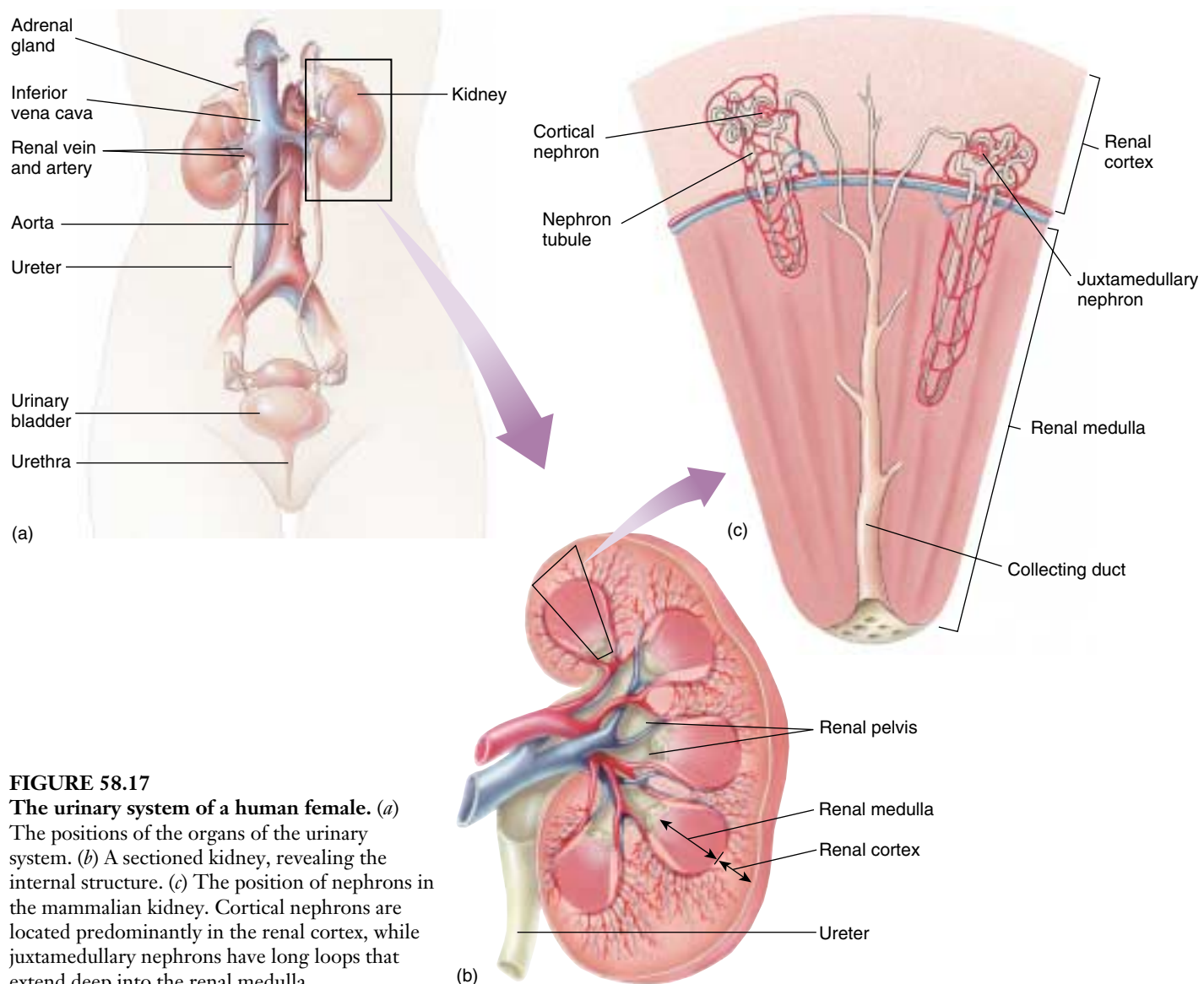


FIGURE 58.17

The urinary system of a human female. (a) The positions of the organs of the urinary system. (b) A sectioned kidney, revealing the internal structure. (c) The position of nephrons in the mammalian kidney. Cortical nephrons are located predominantly in the renal cortex, while juxtamedullary nephrons have long loops that extend deep into the renal medulla.

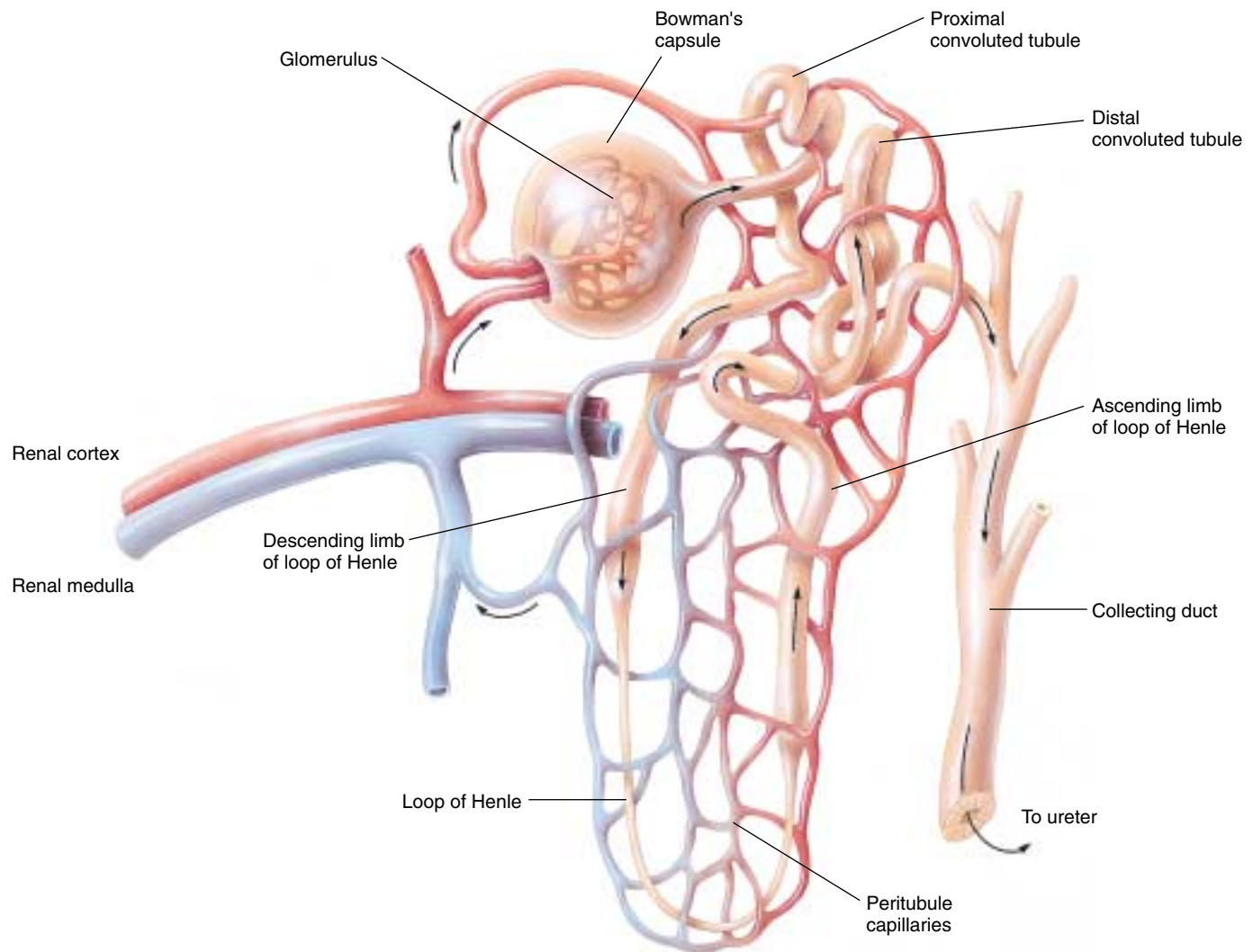


FIGURE 58.18

A nephron in a mammalian kidney. The nephron tubule is surrounded by peritubular capillaries, which carry away molecules and ions that are reabsorbed from the filtrate.

this **glomerular filtrate**, but large amounts of water and dissolved molecules leave the vascular system at this step. The filtrate immediately enters the first region of the nephron tubules. This region, *Bowman's capsule*, envelops the glomerulus much as a large, soft balloon surrounds your fist if you press your fist into it. The capsule has slit openings so that the glomerular filtrate can enter the system of nephron tubules.

After the filtrate enters Bowman's capsule it goes into a portion of the nephron called the **proximal convoluted tubule**, located in the cortex. The fluid then moves down into the medulla and back up again into the cortex in a **loop of Henle**. Only the kidneys of mammals and birds have loops of Henle, and this is why only birds and mammals have the ability to concentrate their urine. After leaving the loop, the fluid is delivered to a **distal convo-**

luted tubule in the cortex that next drains into a **collecting duct**. The collecting duct again descends into the medulla, where it merges with other collecting ducts to empty its contents, now called urine, into the renal pelvis.

Blood components that were not filtered out of the glomerulus drain into an *efferent arteriole*, which then empties into a second bed of capillaries called *peritubular capillaries* that surround the tubules. This is the only location in the body where two capillary beds occur in series. The glomerulus is drained by an arteriole and this second arteriole delivers blood to a second capillary bed, the peritubular capillaries. As described later, the peritubular capillaries are needed for the processes of reabsorption and secretion.

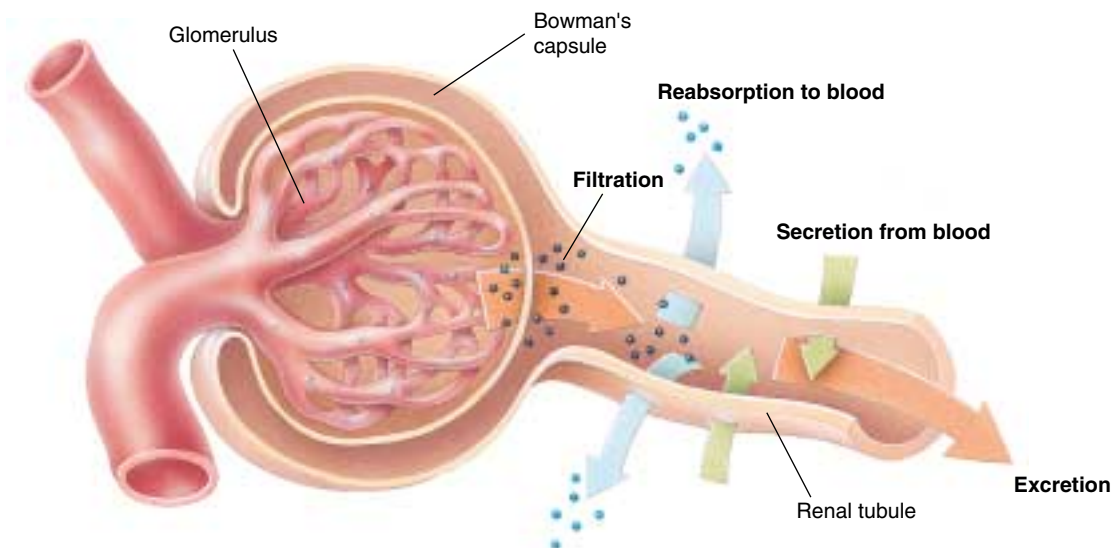


FIGURE 58.19
Four functions of the kidney. Molecules enter the urine by *filtration* out of the glomerulus and by *secretion* into the tubules from surrounding peritubular capillaries. Molecules that entered the filtrate can be returned to the blood by *reabsorption* from the tubules into surrounding peritubular capillaries, or they may be eliminated from the body by *excretion* through the tubule to a ureter, then to the bladder.

Reabsorption and Secretion

Most of the water and dissolved solutes that enter the glomerular filtrate must be returned to the blood (figure 58.19), or the animal would literally urinate to death. In a human, for example, approximately 2000 liters of blood passes through the kidneys each day, and 180 liters of water leaves the blood and enters the glomerular filtrate. Because we only have a total blood volume of about 5 liters and only produce 1 to 2 liters of urine per day, it is obvious that each liter of blood is filtered many times per day and most of the filtered water is reabsorbed. The reabsorption of water occurs as a consequence of salt (NaCl) reabsorption through mechanisms that will be described shortly.

The reabsorption of glucose, amino acids, and many other molecules needed by the body is driven by active transport carriers. As in all carrier-mediated transport, a maximum rate of transport is reached whenever the carriers are saturated (see chapter 6). For the renal glucose carriers, saturation occurs when the concentration of glucose in the blood (and thus in the glomerular filtrate) is about 180 milligrams per 100 milliliters of blood. If a person has a blood glucose concentration in excess of this amount, as happens in untreated diabetes mellitus, the glucose left untransported in the filtrate is expelled in the urine. Indeed, the presence of glucose in the urine is diagnostic of diabetes mellitus.

The secretion of foreign molecules and particular waste products of the body involves the transport of these molecules across the membranes of the blood capillaries and kidney tubules into the filtrate. This process is similar to reabsorption, but it proceeds in the opposite direction. Some secreted molecules are eliminated in the urine so rapidly that they may be cleared from the blood in a single pass through the kidneys. This rapid elimination ex-

plains why penicillin, which is secreted by the nephrons, must be administered in very high doses and several times per day.

Excretion

A major function of the kidney is the elimination of a variety of potentially harmful substances that animals eat and drink. In addition, urine contains nitrogenous wastes, such as urea and uric acid, that are products of the catabolism of amino acids and nucleic acids. Urine may also contain excess K^+ , H^+ , and other ions that are removed from the blood. Urine's generally high H^+ concentration (pH 5 to 7) helps maintain the acid-base balance of the blood within a narrow range (pH 7.35 to 7.45). Moreover, the excretion of water in urine contributes to the maintenance of blood volume and pressure; the larger the volume of urine excreted, the lower the blood volume.

The purpose of kidney function is therefore homeostasis—the kidneys are critically involved in maintaining the constancy of the internal environment. When disease interferes with kidney function, it causes a rise in the blood concentration of nitrogenous waste products, disturbances in electrolyte and acid-base balance, and a failure in blood pressure regulation. Such potentially fatal changes highlight the central importance of the kidneys in normal body physiology.

The mammalian kidney is divided into a cortex and medulla and contains microscopic functioning units called nephrons. The nephron tubules receive a blood filtrate from the glomeruli and modify this filtrate to produce urine, which empties into the renal pelvis and is expelled from the kidney through the ureter.

Transport Processes in the Mammalian Nephron

As previously described, approximately 180 liters (in a human) of isotonic glomerular filtrate enters the Bowman's capsules each day. After passing through the remainder of the nephron tubules, this volume of fluid would be lost as urine if it were not reabsorbed back into the blood. It is clearly impossible to produce this much urine, yet water is only able to pass through a cell membrane by osmosis, and osmosis is not possible between two isotonic solutions. Therefore, some mechanism is needed to create an osmotic gradient between the glomerular filtrate and the blood, allowing reabsorption.

Proximal Tubule

Approximately two-thirds of the NaCl and water filtered into Bowman's capsule is immediately reabsorbed across the walls of the proximal convoluted tubule. This reabsorption is driven by the active transport of Na^+ out of the filtrate and into surrounding peritubular capillaries. Cl^- follows Na^+ passively because of electrical attraction, and water follows them both because of osmosis. Because NaCl and water are removed from the filtrate in proportionate amounts, the filtrate that remains in the tubule is still isotonic to the blood plasma.

Although only one-third of the initial volume of filtrate remains in the nephron tubule after the initial reabsorption of NaCl and water, it still represents a large volume (60 L out of the original 180 L of filtrate produced per day by both human kidneys). Obviously, no animal can excrete that much urine, so most of this water must also be reabsorbed. It is reabsorbed primarily across the wall of the collecting duct because the interstitial fluid of the renal medulla surrounding the collecting ducts is hypertonic. The hypertonic renal medulla draws water out of the collecting duct by osmosis, leaving behind a hypertonic urine for excretion.

Loop of Henle

The reabsorption of much of the water in the tubular filtrate thus depends on the creation of a hypertonic renal medulla; the more hypertonic the medulla is, the steeper the osmotic gradient will be and the more water will leave the collecting ducts. It is the loops of Henle that create the hypertonic renal medulla in the following manner (figure 58.20):

1. The ascending limb of the loop actively extrudes Na^+ , and Cl^- follows. The mechanism that extrudes NaCl from the ascending limb of the loop differs from that which extrudes NaCl from the proximal tubule, but the most important difference is that the ascending limb is *not permeable to water*. As Na^+ exits, the fluid within the ascending limb becomes increasingly dilute (hypotonic) as it enters the cortex, while the surrounding tissue becomes increasingly concentrated (hypertonic).

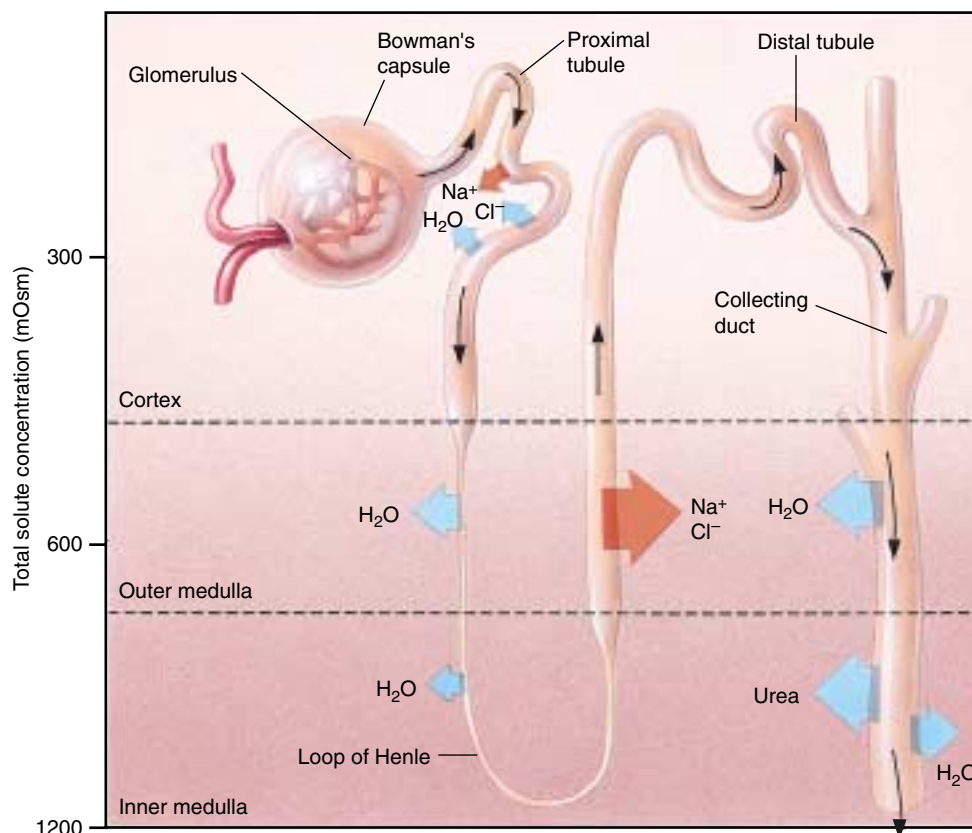


FIGURE 58.20

The reabsorption of salt and water in the mammalian kidney.

Active transport of Na^+ out of the proximal tubules is followed by the passive movement of Cl^- and water. Active extrusion of NaCl from the ascending limb of the loop of Henle creates the osmotic gradient required for the reabsorption of water from the collecting duct. The changes in osmolality from the cortex to the medulla is indicated to the left of the figure.

2. The NaCl pumped out of the ascending limb of the loop is trapped within the surrounding interstitial fluid. This is because the peritubular capillaries in the medulla also have loops, called *vasa recta*, so that NaCl can diffuse from the blood leaving the medulla to the blood entering the medulla. Thus, the vasa recta functions in a countercurrent exchange, similar to that described for oxygen in the countercurrent flow of water and blood in the gills of fish (see chapter 53). In the case of the renal medulla, the diffusion of NaCl between the blood vessels keeps much of the NaCl within the interstitial fluid, making it hypertonic.
3. The descending limb is permeable to water, so water leaves by osmosis as the fluid descends into the hypertonic renal medulla. This water enters the blood vessels of the vasa recta and is carried away in the general circulation.
4. The loss of water from the descending limb multiplies the concentration that can be achieved at each level of the loop through the active extrusion of NaCl by the ascending limb. The longer the loop of Henle, the longer the region of interaction between the descending and ascending limbs, and the greater the total concentration that can be achieved. In a human kidney, the concentration of filtrate entering the loop is 300 milliosmolal, and this concentration is multiplied to more than 1200 milliosmolal at the bottom of the longest loops of Henle in the renal medulla.

Because fluid flows in opposite directions in the two limbs of the loop, the action of the loop of Henle in creating a hypertonic renal medulla is known as the *countercurrent multiplier system*. The high solute concentration of the renal medulla is primarily the result of NaCl accumulation by the countercurrent multiplier system, but urea also contributes to the total osmolality of the medulla. This is because the descending limb of the loop of Henle and the collecting duct are permeable to urea, which leaves these regions of the nephron by diffusion.

Distal Tubule and Collecting Duct

Because NaCl was pumped out of the ascending limb, the filtrate that arrives at the distal convoluted tubule and enters the collecting duct in the renal cortex is hypotonic (with a concentration of only 100 mOsm). The collecting duct carrying this dilute fluid now plunges into the medulla. As a result of the hypertonic interstitial fluid of the renal medulla, there is a strong osmotic gradient that pulls water out of the collecting duct and into surrounding blood vessels.

The osmotic gradient is normally constant, but the permeability of the collecting duct to water is adjusted by a hormone, **antidiuretic hormone (ADH)**, also called **vasopressin**, discussed in chapters 52 and 56. When an animal

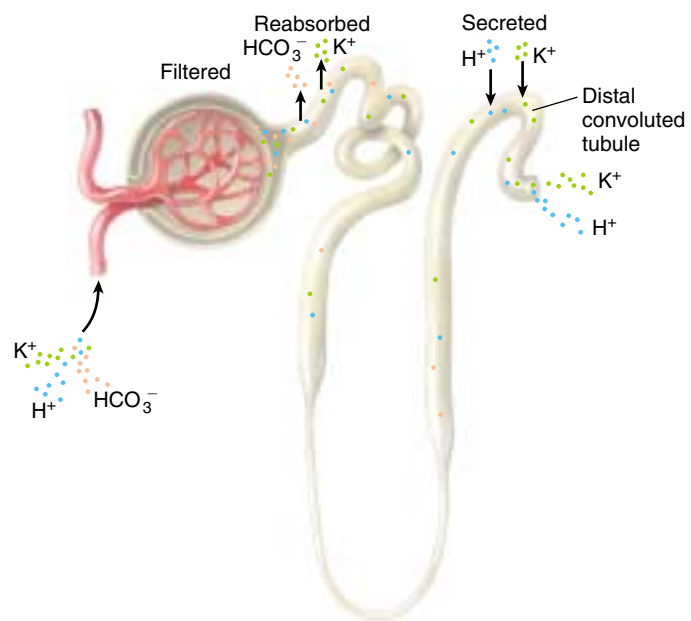


FIGURE 58.21

The nephron controls the amounts of K^+ , H^+ , and HCO_3^- excreted in the urine. K^+ is completely reabsorbed in the proximal tubule and then secreted in varying amounts into the distal tubule. HCO_3^- is filtered but normally completely reabsorbed. H^+ is filtered and also secreted into the distal tubule, so that the final urine has an acidic pH.

needs to conserve water, the posterior pituitary gland secretes more ADH, and this hormone increases the number of water channels in the plasma membranes of the collecting duct cells. This increases the permeability of the collecting ducts to water so that more water is reabsorbed and less is excreted in the urine. The animal thus excretes a hypertonic urine.

In addition to the regulation of water balance, the kidneys regulate the balance of electrolytes in the blood by reabsorption and secretion. For example, the kidneys reabsorb K^+ in the proximal tubule and then secrete an amount of K^+ needed to maintain homeostasis into the distal convoluted tubule (figure 58.21). The kidneys also maintain acid-base balance by excreting H^+ into the urine and reabsorbing bicarbonate (HCO_3^-), as previously described.

The loop of Henle creates a hypertonic renal medulla as a result of the active extrusion of NaCl from the ascending limb and the interaction with the descending limb. The hypertonic medulla then draws water osmotically from the collecting duct, which is permeable to water under the influence of antidiuretic hormone.

Ammonia, Urea, and Uric Acid

Amino acids and nucleic acids are nitrogen-containing molecules. When animals catabolize these molecules for energy or convert them into carbohydrates or lipids, they produce nitrogen-containing by-products called **nitrogenous wastes** (figure 58.22) that must be eliminated from the body.

The first step in the metabolism of amino acids and nucleic acids is the removal of the amino ($-\text{NH}_2$) group and its combination with H^+ to form **ammonia** (NH_3) in the liver. Ammonia is quite toxic to cells and therefore is safe only in very dilute concentrations. The excretion of ammonia is not a problem for the bony fish and tadpoles, which eliminate most of it by diffusion through the gills and less by excretion in very dilute urine. In elasmobranchs, adult amphibians, and mammals, the nitrogenous wastes are eliminated in the far less toxic form of **urea**. Urea is water-soluble and so can be excreted in large amounts in the urine. It is carried in the bloodstream from its place of synthesis in the liver to the kidneys where it is excreted in the urine.

Reptiles, birds, and insects excrete nitrogenous wastes in the form of **uric acid**, which is only slightly soluble in water. As a result of its low solubility, uric acid precipitates

and thus can be excreted using very little water. Uric acid forms the pasty white material in bird droppings called *guano*. The ability to synthesize uric acid in these groups of animals is also important because their eggs are encased within shells, and nitrogenous wastes build up as the embryo grows within the egg. The formation of uric acid, while a lengthy process that requires considerable energy, produces a compound that crystallizes and precipitates. As a precipitate, it is unable to affect the embryo's development even though it is still inside the egg.

Mammals also produce some uric acid, but it is a waste product of the degradation of purine nucleotides (see chapter 3), not of amino acids. Most mammals have an enzyme called *uricase*, which converts uric acid into a more soluble derivative, **allantoin**. Only humans, apes, and the dalmatian dog lack this enzyme and so must excrete the uric acid. In humans, excessive accumulation of uric acid in the joints produces a condition known as *gout*.

The metabolic breakdown of amino acids and nucleic acids produces ammonia as a by-product. Ammonia is excreted by bony fish, but other vertebrates convert nitrogenous wastes into urea and uric acid, which are less toxic nitrogenous wastes.

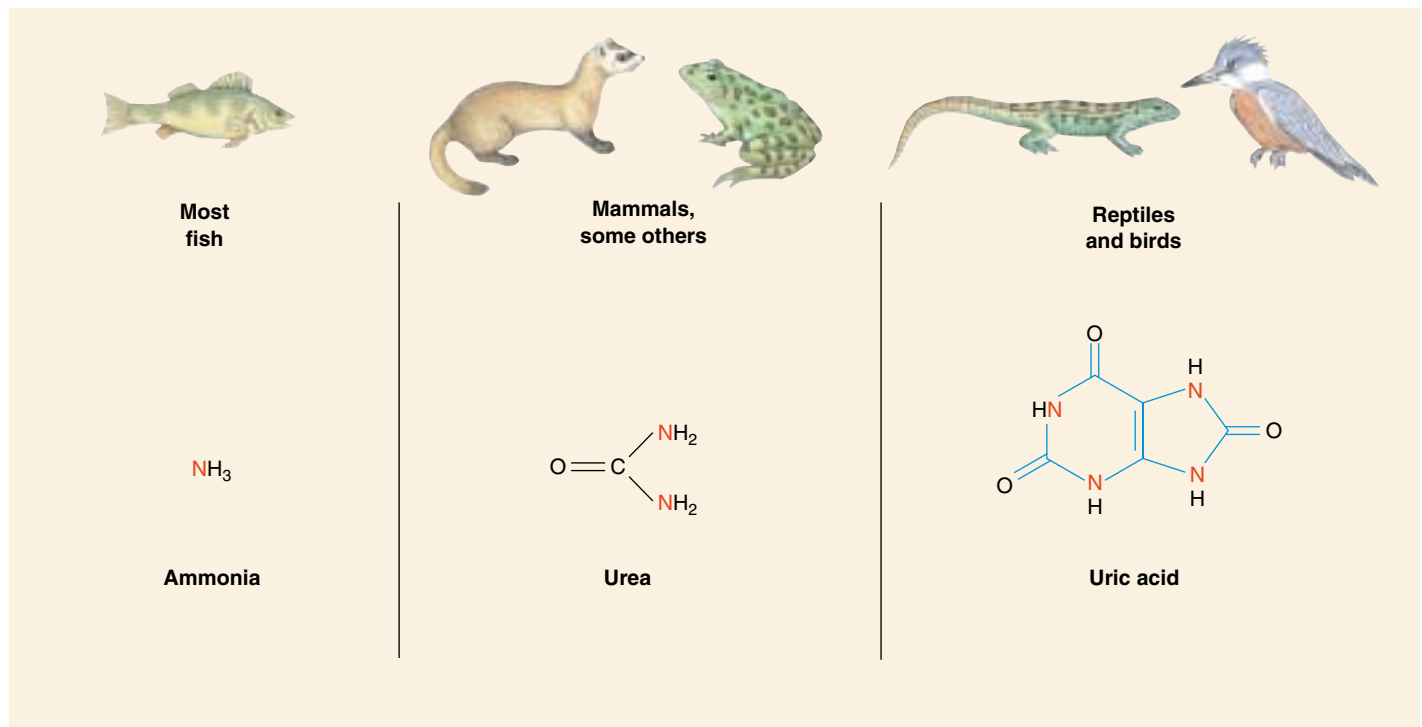


FIGURE 58.22

Nitrogenous wastes. When amino acids and nucleic acids are metabolized, the immediate by-product is ammonia, which is quite toxic but which can be eliminated through the gills of teleost fish. Mammals convert ammonia into urea, which is less toxic. Birds and terrestrial reptiles convert it instead into uric acid, which is insoluble in water.

58.4 The kidney is regulated by hormones.

Hormones Control Homeostatic Functions

In mammals and birds, the amount of water excreted in the urine, and thus the concentration of the urine, varies according to the changing needs of the body. Acting through the mechanisms described next, the kidneys will excrete a hypertonic urine when the body needs to conserve water. If an animal drinks too much water, the kidneys will excrete a hypotonic urine. As a result, the volume of blood, the blood pressure, and the osmolality of blood plasma are maintained relatively constant by the kidneys, no matter how much water you drink. The kidneys also regulate the plasma K^+ and Na^+ concentrations and blood pH within very narrow limits. These homeostatic functions of the kidneys are coordinated primarily by hormones (see chapter 56).

Antidiuretic Hormone

Antidiuretic hormone (ADH) is produced by the hypothalamus and secreted by the posterior pituitary gland. The primary stimulus for ADH secretion is an increase in the osmolality of the blood plasma. The osmolality of plasma increases when a person is dehydrated or when a person eats salty food. Osmoreceptors in the hypothalamus respond to the elevated blood osmolality by sending more nerve signals to the integration center (also in the hypothalamus). This, in turn, triggers a sensation of thirst and an increase in the secretion of ADH (figure 58.23).

ADH causes the walls of the collecting ducts in the kidney to become more permeable to water. This occurs because water channels are contained within the membranes of intracellular vesicles in the epithelium of the collecting ducts, and ADH stimulates the fusion of the vesicle membrane with the plasma membrane, similar to the process of exocytosis. When the secretion of ADH is reduced, the plasma membrane pinches in to form new vesicles that contain the water channels, so that the plasma membrane becomes less permeable to water.

Because the extracellular fluid in the renal medulla is hypertonic to the filtrate in the collecting ducts, water leaves the filtrate by osmosis and is reabsorbed into the blood. Under conditions of maximal ADH secretion, a person excretes only 600 milliliters of highly concentrated urine per day. A person who lacks ADH due to pituitary damage has the disorder known as *diabetes insipidus* and constantly ex-

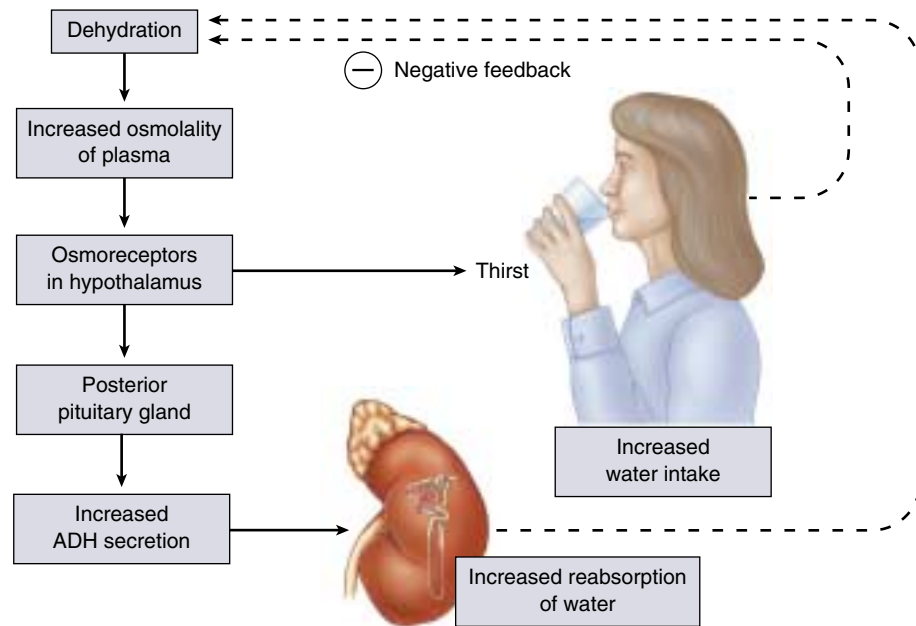


FIGURE 58.23

Antidiuretic hormone stimulates the reabsorption of water by the kidneys. This action completes a negative feedback loop and helps to maintain homeostasis of blood volume and osmolality.

cretes a large volume of dilute urine. Such a person is in danger of becoming severely dehydrated and succumbing to dangerously low blood pressure.

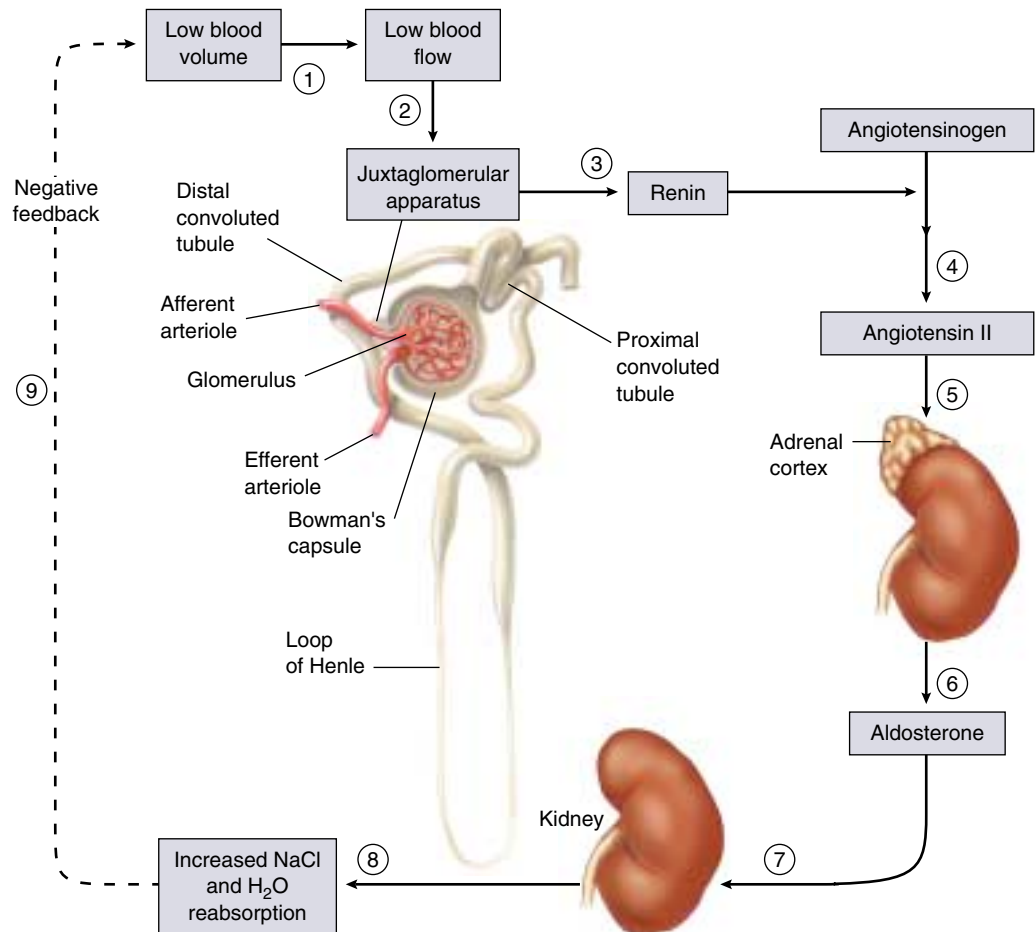
Aldosterone and Atrial Natriuretic Hormone

Sodium ion is the major solute in the blood plasma. When the blood concentration of Na^+ falls, therefore, the blood osmolality also falls. This drop in osmolality inhibits ADH secretion, causing more water to remain in the collecting duct for excretion in the urine. As a result, the blood volume and blood pressure decrease. A decrease in extracellular Na^+ also causes more water to be drawn into cells by osmosis, partially offsetting the drop in plasma osmolality but further decreasing blood volume and blood pressure. If Na^+ deprivation is severe, the blood volume may fall so low that there is insufficient blood pressure to sustain life. For this reason, salt is necessary for life. Many animals have a “salt hunger” and actively seek salt, such as the deer at “salt licks.”

A drop in blood Na^+ concentration is normally compensated by the kidneys under the influence of the hormone aldosterone, which is secreted by the adrenal cortex. Aldosterone stimulates the distal convoluted tubules to reabsorb Na^+ , decreasing the excretion of Na^+ in the urine. Indeed, under conditions of maximal aldosterone secretion, Na^+ may be completely absent from the urine. The reabsorp-

FIGURE 58.24**A lowering of blood volume activates the renin-angiotensin-aldosterone system.**

(1) Low blood volume accompanies a decrease in blood Na^+ levels. (2) Reduced blood flow past the juxtaglomerular apparatus triggers (3) the release of renin into the blood, which catalyzes the production of angiotensin I from angiotensinogen. (4) Angiotensin I converts into an active form, angiotensin II. (5) Angiotensin II stimulates blood vessel constriction and (6) the release of aldosterone from the adrenal cortex. (7) Aldosterone stimulates the reabsorption of Na^+ in the distal convoluted tubules. (8) Increased Na^+ reabsorption is followed by the reabsorption of Cl^- and water. (9) This increases blood volume. An increase in blood volume may also trigger the release of atrial natriuretic hormone that inhibits the release of aldosterone. These two systems work together to maintain homeostasis.



tion of Na^+ is followed by Cl^- and by water, so aldosterone has the net effect of promoting the retention of both salt and water. It thereby helps to maintain blood volume and pressure.

The secretion of aldosterone in response to a decreased blood level of Na^+ is indirect. Because a fall in blood Na^+ is accompanied by a decreased blood volume, there is a reduced flow of blood past a group of cells called the juxtaglomerular apparatus, located in the region of the kidney between the distal convoluted tubule and the afferent arteriole (figure 58.24). The juxtaglomerular apparatus responds by secreting the enzyme *renin* into the blood, which catalyzes the production of the polypeptide angiotensin I from the protein angiotensinogen (see chapter 52). Angiotensin I is then converted by another enzyme into angiotensin II, which stimulates blood vessels to constrict and the adrenal cortex to secrete aldosterone. Thus, homeostasis of blood volume and pressure can be maintained by the activation of this renin-angiotensin-aldosterone system.

In addition to stimulating Na^+ reabsorption, aldosterone also promotes the secretion of K^+ into the distal convoluted tubules. Consequently, aldosterone lowers the blood K^+

concentration, helping to maintain constant blood K^+ levels in the face of changing amounts of K^+ in the diet. People who lack the ability to produce aldosterone will die if untreated because of the excessive loss of salt and water in the urine and the buildup of K^+ in the blood.

The action of aldosterone in promoting salt and water retention is opposed by another hormone, atrial natriuretic hormone (ANH, see chapter 52). This hormone is secreted by the right atrium of the heart in response to an increased blood volume, which stretches the atrium. Under these conditions, aldosterone secretion from the adrenal cortex will decrease and atrial natriuretic hormone secretion will increase, thus promoting the excretion of salt and water in the urine and lowering the blood volume.

ADH stimulates the insertion of water channels into the cells of the collecting duct, making the collecting duct more permeable to water. Thus, ADH stimulates the reabsorption of water and the excretion of a hypertonic urine. Aldosterone promotes the reabsorption of NaCl and water across the distal convoluted tubule, as well as the secretion of K^+ into the tubule. ANH decreases NaCl reabsorption.



Summary

Questions

Media Resources

58.1 The regulatory systems of the body maintain homeostasis.

- Negative feedback loops maintain nearly constant extracellular conditions in the internal environment of the body, a condition called homeostasis.
- Antagonistic effectors afford an even finer degree of control.

1. What is homeostasis? What is a negative feedback loop? Give an example of how homeostasis is maintained by a negative feedback loop.



- Osmoregulation

58.2 The extracellular fluid concentration is constant in most vertebrates.

- Osmoconformers maintain a tissue fluid osmolality equal to that of their environment, whereas osmoregulators maintain a constant blood osmolality that is different from that of their environment.
- Insects eliminate water by secreting K^+ into Malpighian tubules and the water follows the K^+ by osmosis.
- The kidneys of most vertebrates eliminate water by filtering blood into nephron tubules.
- Freshwater bony fish are hypertonic to their environment, and saltwater bony fish are hypotonic to their environment; these conditions place different demands upon their kidneys and other regulatory systems.
- Birds and mammals are the only vertebrates that have loops of Henle and thus are capable of producing a hypertonic urine.

2. What is the difference between an osmoconformer and an osmoregulator? What are examples of each?



- Body fluid distribution
- Water balance

3. How does the body fluid osmolality of a freshwater vertebrate compare with that of its environment? Does water tend to enter or exit its body? What must it do to maintain proper body water levels?

4. In what type of animal are Malpighian tubules found? By what mechanism is fluid caused to flow into these tubules? How is this fluid further modified before it is excreted?

58.3 The functions of the vertebrate kidney are performed by nephrons.

- The primary function of the kidneys is homeostasis of blood volume, pressure, and composition, including the concentration of particular solutes in the blood and the blood pH.
- Bony fish remove the amine portions of amino acids and excrete them as ammonia across the gills.
- Elasmobranchs, adult amphibians, and mammals produce and excrete urea, which is quite soluble but much less toxic than ammonia.
- Insects, reptiles, and birds produce uric acid from the amino groups in amino acids; this precipitates, so that little water is required for its excretion.

5. What drives the movement of fluid from the blood to the inside of the nephron tubule at Bowman's capsule?



- Bioethics case study: Kidney transplant

6. In what portion of the nephron is most of the $NaCl$ and water reabsorbed from the filtrate?



- Art activities
Urinary system
Anatomy of kidney and lobe
Nephron anatomy

7. What causes water reabsorption from the collecting duct? How is this influenced by antidiuretic hormone?



- Kidney function

58.4 The kidney is regulated by hormones.

- Antidiuretic hormone is secreted by the posterior pituitary gland in response to an increase in blood osmolality, and acts to increase the number of water channels in the walls of the collecting ducts.

8. What effects does aldosterone have on kidney function? How is the secretion of aldosterone stimulated?



- Kidney function

59

Sex and Reproduction

Concept Outline

59.1 Animals employ both sexual and asexual reproductive strategies.

Asexual and Sexual Reproduction. Some animals reproduce asexually, but most reproduce sexually; male and female are usually different individuals, but not always.

59.2 The evolution of reproduction among the vertebrates has led to internalization of fertilization and development.

Fertilization and Development. Among vertebrates that have internal fertilization, the young are nourished by egg yolk or from their mother's blood.

Fish and Amphibians. Most bony fish and amphibians have external fertilization, while most cartilaginous fish have internal fertilization.

Reptiles and Birds. Most reptiles and all birds lay eggs externally, and the young develop inside the egg.

Mammals. Monotremes lay eggs, marsupials have pouches where their young develop, and placental mammals have placentas that nourish the young within the uterus.

59.3 Male and female reproductive systems are specialized for different functions.

Structure and Function of the Male Reproductive System. The testes produce sperm and secrete the male sex hormone, testosterone.

Structure and Function of the Female Reproductive System. An egg cell within an ovarian follicle develops and is released from the ovary; the egg cell travels into the female reproductive tract, which undergoes cyclic changes due to hormone secretion.

59.4 The physiology of human sexual intercourse is becoming better known.

Physiology of Human Sexual Intercourse. The human sexual response can be divided into four phases: excitement, plateau, orgasm, and resolution.

Birth Control. Various methods of birth control are employed, including barriers to fertilization, prevention of ovulation, and prevention of the implantation.



FIGURE 59.1

The bright color of male golden toads serves to attract mates. The rare golden toads of the Monteverde Cloud Forest Reserve of Costa Rica are nearly voiceless and so use bright colors to attract mates. Always rare, they may now be extinct.

The cry of a cat in heat, insects chirping outside the window, frogs croaking in swamps, and wolves howling in a frozen northern forest are all sounds of evolution's essential act, reproduction. These distinct vocalizations, as well as the bright coloration characteristic of some animals like the tropical golden toads of figure 59.1, function to attract mates. Few subjects pervade our everyday thinking more than sex, and few urges are more insistent. This chapter deals with sex and reproduction among the vertebrates, including humans.

59.1 Animals employ both sexual and asexual reproductive strategies.

Asexual and Sexual Reproduction

Asexual reproduction is the primary means of reproduction among the protists, cnidaria, and tunicates, but it may also occur in some of the more complex animals. Indeed, the formation of identical twins (by the separation of two identical cells of a very early embryo) is a form of asexual reproduction.

Through mitosis, genetically identical cells are produced from a single parent cell. This permits asexual reproduction to occur in protists by division of the organism, or **fission**. Cnidaria commonly reproduce by **budding**, where a part of the parent's body becomes separated from the rest and differentiates into a new individual. The new individual may become an independent animal or may remain attached to the parent, forming a colony.

Sexual reproduction occurs when a new individual is formed by the union of two sex cells, or **gametes**, a term that includes **sperm** and **eggs** (or ova). The union of sperm and egg cells produces a fertilized egg, or **zygote**, that develops by mitotic division into a new multicellular organism. The zygote and the cells it forms by mitosis are diploid; they contain both members of each homologous pair of chromosomes. The gametes, formed by meiosis in the sex organs, or **gonads**—the **testes** and **ovaries**—are haploid (see chapter 12). The process of spermatogenesis (sperm formation) and oogenesis (egg formation) will be described in later sections. For a more detailed discussion of asexual and sexual reproduction, see chapter 12.

Different Approaches to Sex

Parthenogenesis (virgin birth) is common in many species of arthropods; some species are exclusively parthenogenic (and all female), while others switch between sexual reproduction and parthenogenesis in different generations. In honeybees, for example, a queen bee mates only once and stores the sperm. She then can control the release of sperm. If no sperm are released, the eggs develop parthenogenetically into drones, which are males; if sperm are allowed to fertilize the eggs, the fertilized eggs develop into other queens or worker bees, which are female.



(a)



(b)

FIGURE 59.2

Hermaphroditism and protogyny. (a) The hamlet bass (genus *Hypoplectrus*) is a deep-sea fish that is a hermaphrodite—both male and female at the same time. In the course of a single pair-mating, one fish may switch sexual roles as many as four times, alternately offering eggs to be fertilized and fertilizing its partner's eggs. Here the fish acting as a male curves around its motionless partner, fertilizing the upward-floating eggs. (b) The bluehead wrasse, *Thalassoma bifasciatum*, is protogynous—females sometimes turn into males. Here a large male, or sex-changed female, is seen among females, typically much smaller.

The Russian biologist Ilya Darevsky reported in 1958 one of the first cases of unusual modes of reproduction among vertebrates. He observed that some populations of small lizards of the genus *Lacerta* were exclusively female, and suggested that these lizards could lay eggs that were viable even if they were not fertilized. In other words, they were capable of asexual reproduction in the absence of sperm, a type of parthenogenesis. Further work has shown that parthenogenesis also occurs among populations of other lizard genera.

Another variation in reproductive strategies is **hermaphroditism**, when one individual has both testes and ovaries, and so can produce both sperm and eggs (figure 59.2a). A tapeworm is hermaphroditic and can fertilize itself, a useful strategy because it is unlikely to encounter another tapeworm. Most hermaphroditic animals, however, require another individual to reproduce. Two earthworms, for example, are required for reproduction—each functions as both male and female, and each leaves the encounter with fertilized eggs.

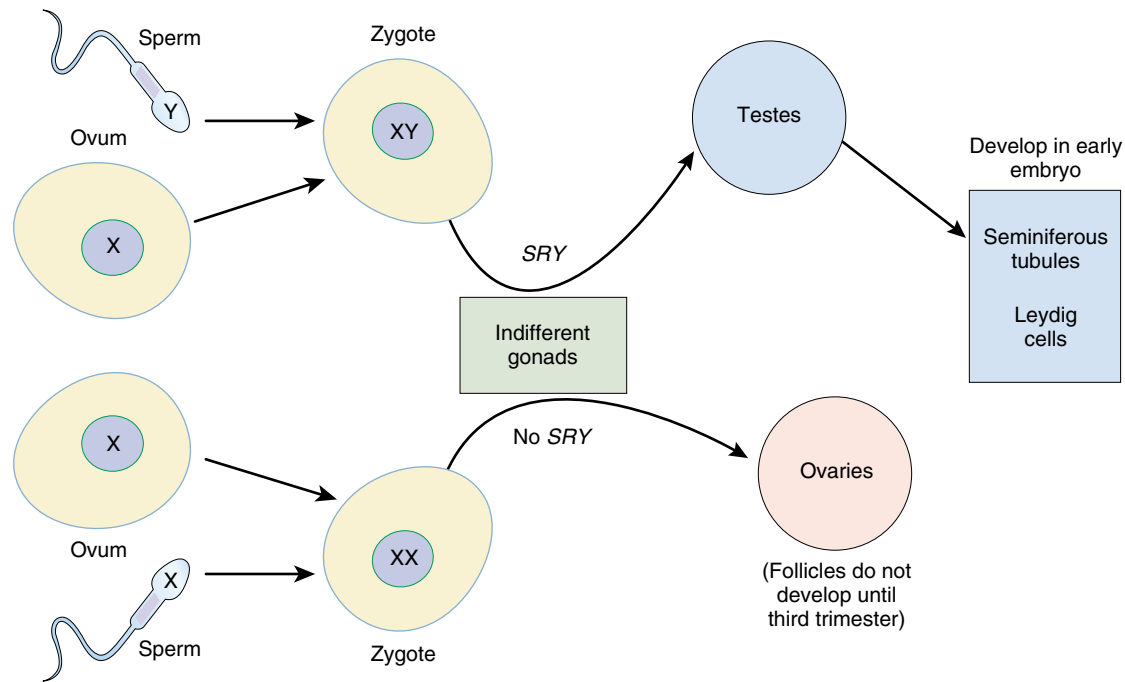


FIGURE 59.3

Sex determination in mammals is made by a region of the Y chromosome designated *SRY*. Testes are formed when the Y chromosome and *SRY* are present; ovaries are formed when they are absent.

There are some deep-sea fish that are hermaphrodites—both male and female at the same time. Numerous fish genera include species in which individuals can change their sex, a process called *sequential hermaphroditism*. Among coral reef fish, for example, both **protogyny** (“first female,” a change from female to male) and **protandry** (“first male,” a change from male to female) occur. In fish that practice protogyny (figure 59.2*b*), the sex change appears to be under social control. These fish commonly live in large groups, or schools, where successful reproduction is typically limited to one or a few large, dominant males. If those males are removed, the largest female rapidly changes sex and becomes a dominant male.

Sex Determination

Among the fish just described, and in some species of reptiles, environmental changes can cause changes in the sex of the animal. In mammals, the sex is determined early in embryonic development. The reproductive systems of human males and females appear similar for the first 40 days after conception. During this time, the cells that will give rise to ova or sperm migrate from the yolk sac to the embryonic gonads, which have the potential to become either ovaries in females or testes in males. For this reason, the embry-

onic gonads are said to be “indifferent.” If the embryo is a male, it will have a Y chromosome with a gene whose product converts the indifferent gonads into testes. In females, which lack a Y chromosome, this gene and the protein it encodes are absent, and the gonads become ovaries. Recent evidence suggests that the sex-determining gene may be one known as *SRY* (for “sex-determining region of the Y chromosome”) (figure 59.3). The *SRY* gene appears to have been highly conserved during the evolution of different vertebrate groups.

Once testes form in the embryo, the testes secrete testosterone and other hormones that promote the development of the male external genitalia and accessory reproductive organs. If the embryo lacks testes (the ovaries are nonfunctional at this stage), the embryo develops female external genitalia and sex accessory organs. In other words, all mammalian embryos will develop female sex accessory organs and external genitalia unless they are masculinized by the secretions of the testes.

Sexual reproduction is most common among animals, but many reproduce asexually by fission, budding, or parthenogenesis. Sexual reproduction generally involves the fusion of gametes derived from different individuals of a species, but some species are hermaphroditic.

59.2 The evolution of reproduction among the vertebrates has led to internalization of fertilization and development.

Fertilization and Development

Vertebrate sexual reproduction evolved in the ocean before vertebrates colonized the land. The females of most species of marine bony fish produce eggs or ova in batches and release them into the water. The males generally release their sperm into the water containing the eggs, where the union of the free gametes occurs. This process is known as **external fertilization**.

Although seawater is not a hostile environment for gametes, it does cause the gametes to disperse rapidly, so their release by females and males must be almost simultaneous. Thus, most marine fish restrict the release of their eggs and sperm to a few brief and well-defined periods. Some reproduce just once a year, while others do so more frequently. There are few seasonal cues in the ocean that organisms can use as signals for synchronizing reproduction, but one all-pervasive signal is the cycle of the moon. Once each month, the moon approaches closer to the earth than usual, and when it does, its increased gravitational attraction causes somewhat higher tides. Many marine organisms sense the tidal changes and entrain the production and release of their gametes to the lunar cycle.

The invasion of land posed the new danger of desiccation, a problem that was especially severe for the small and vulnerable gametes. On land, the gametes could not simply be released near each other, as they would soon dry up and perish. Consequently, there was intense selective pressure for terrestrial vertebrates (as well as some groups of fish) to evolve **internal fertilization**, that is, the introduction of male gametes into the female repro-

ductive tract. By this means, fertilization still occurs in a nondesiccating environment, even when the adult animals are fully terrestrial. The vertebrates that practice internal fertilization have three strategies for embryonic and fetal development:

1. **Oviparity.** This is found in some bony fish, most reptiles, some cartilaginous fish, some amphibians, a few mammals, and all birds. The eggs, after being fertilized internally, are deposited outside the mother's body to complete their development.
2. **Ovoviviparity.** This is found in some bony fish (including mollies, guppies, and mosquito fish), some cartilaginous fish, and many reptiles. The fertilized eggs are retained within the mother to complete their development, but the embryos still obtain all of their nourishment from the egg yolk. The young are fully developed when they are hatched and released from the mother.
3. **Viviparity.** This is found in most cartilaginous fish, some amphibians, a few reptiles, and almost all mammals. The young develop within the mother and obtain nourishment directly from their mother's blood, rather than from the egg yolk (figure 59.4).

Fertilization is external in most fish but internal in most other vertebrates. Depending upon the relationship of the developing embryo to the mother and egg, those vertebrates with internal fertilization may be classified as oviparous, ovoviviparous, or viviparous.

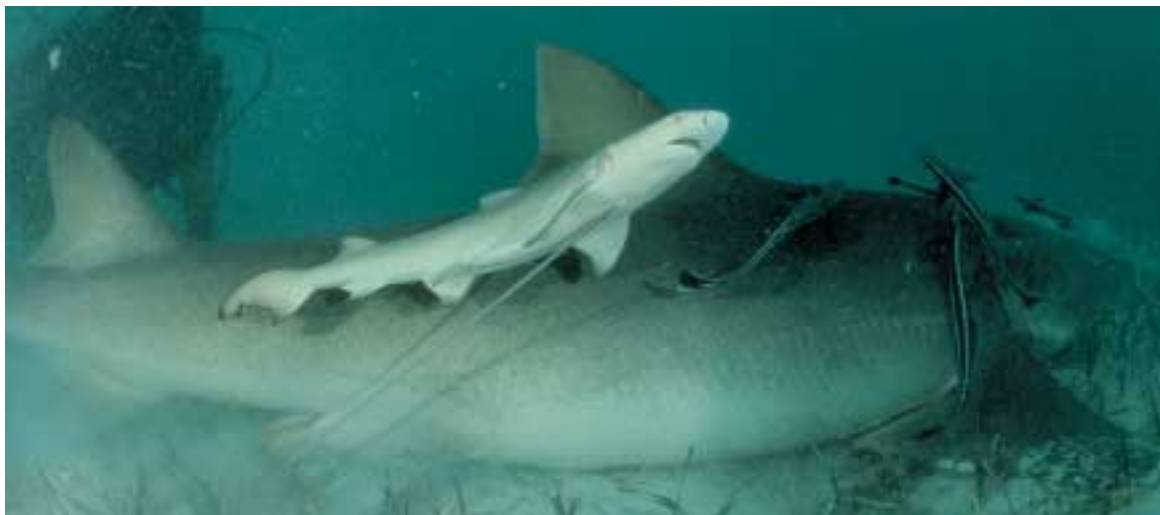


FIGURE 59.4
Viviparous fish
carry live, mobile
young within their
bodies. The young
complete their
development within
the body of the
mother and are then
released as small but
competent adults.
Here a lemon shark
has just given birth
to a young shark,
which is still
attached by the
umbilical cord.

Fish and Amphibians

Most fish and amphibians, unlike other vertebrates, reproduce by means of external fertilization.

Fish

Fertilization in most species of bony fish (teleosts) is external, and the eggs contain only enough yolk to sustain the developing embryo for a short time. After the initial supply of yolk has been exhausted, the young fish must seek its food from the waters around it. Development is speedy, and the young that survive mature rapidly. Although thousands of eggs are fertilized in a single mating, many of the resulting individuals succumb to microbial infection or predation, and few grow to maturity.

In marked contrast to the bony fish, fertilization in most cartilaginous fish is internal. The male introduces sperm into the female through a modified pelvic fin. Development of the young in these vertebrates is generally viviparous.

Amphibians

The amphibians invaded the land without fully adapting to the terrestrial environment, and their life cycle is still tied to the water. Fertilization is external in most amphibians, just as it is in most species of bony fish. Gametes from both males and females are released through the cloaca. Among the frogs and toads, the male grasps the female and discharges fluid containing the sperm onto the eggs as they are released into the water (figure 59.5). Although the eggs of most amphibians develop in the water, there are some interesting exceptions. In two species of frogs, for example, the eggs develop in the vocal sacs and stomach, and the young frogs leave through their mother's mouth (figure 59.6)!

The time required for development of amphibians is much longer than that for fish, but amphibian eggs do not include a significantly greater amount of yolk. Instead, the process of development in most amphibians is divided into embryonic, larval, and adult stages, in a way reminiscent of the life cycles found in some insects. The embryo develops within the egg, obtaining nutrients from the yolk. After hatching from the egg, the aquatic larva then functions as a free-swimming, food-gathering machine, often for a considerable period of time. The larvae may increase in size rapidly; some tadpoles, which are the larvae of frogs and toads, grow in a matter of weeks from creatures no bigger than the tip of a pencil into individuals as big as a goldfish. When the larva has grown to a sufficient size, it undergoes a developmental transition, or metamorphosis, into the terrestrial adult form.

The eggs of most bony fish and amphibians are fertilized externally. In amphibians the eggs develop into a larval stage that undergoes metamorphosis.



FIGURE 59.5

The eggs of frogs are fertilized externally. When frogs mate, as these two are doing, the clasp of the male induces the female to release a large mass of mature eggs, over which the male discharges his sperm.

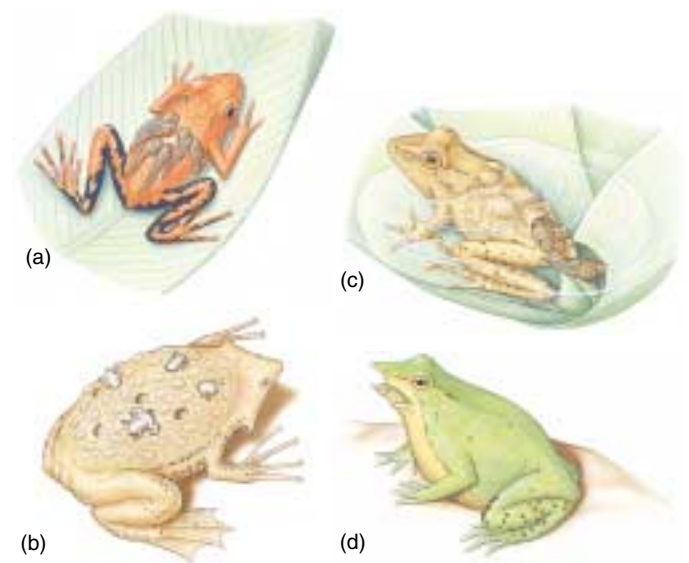


FIGURE 59.6

Different ways young develop in frogs. (a) In the poison arrow frog, the male carries the tadpoles on his back. (b) In the female Surinam frog, froglets develop from eggs in special brooding pouches on the back. (c) In the South American pygmy marsupial frog, the female carries the developing larvae in a pouch on her back. (d) Tadpoles of the Darwin's frog develop into froglets in the vocal pouch of the male and emerge from the mouth.

Reptiles and Birds

Most reptiles and all birds are oviparous—after the eggs are fertilized internally, they are deposited outside of the mother's body to complete their development. Like most vertebrates that fertilize internally, male reptiles utilize a tubular organ, the penis, to inject sperm into the female (figure 59.7). The penis, containing erectile tissue, can become quite rigid and penetrate far into the female reproductive tract. Most reptiles are oviparous, laying eggs and then abandoning them. These eggs are surrounded by a leathery shell that is deposited as the egg passes through the oviduct, the part of the female reproductive tract leading from the ovary. A few species of reptiles are ovoviviparous or viviparous, forming eggs that develop into embryos within the body of the mother.

All birds practice internal fertilization, though most male birds lack a penis. In some of the larger birds (including swans, geese, and ostriches), however, the male cloaca extends to form a false penis. As the egg passes along the oviduct, glands secrete albumin proteins (the egg white) and the hard, calcareous shell that distinguishes bird eggs from reptilian eggs. While modern reptiles are poikilotherms (animals whose body temperature varies with the temperature of their environment), birds are homeotherms (animals that maintain a relatively constant body temperature independent of environmental temperatures). Hence, most birds incubate their eggs after laying them to keep them warm (figure 59.8). The young that hatch from the eggs of most bird species are unable to survive unaided, as their development is still incomplete. These young birds are fed and nurtured by their parents, and they grow to maturity gradually.

The shelled eggs of reptiles and birds constitute one of the most important adaptations of these vertebrates to life on land, because shelled eggs can be laid in dry places. Such eggs are known as amniotic eggs because the embryo develops within a fluid-filled cavity surrounded by a membrane called the amnion. The amnion is an extraembryonic membrane—that is, a membrane formed from embryonic cells but located outside the body of the embryo. Other extraembryonic membranes in amniotic eggs include the chorion, which lines the inside of the eggshell, the yolk sac, and the allantois. In contrast, the eggs of fish and amphibians contain only one extraembryonic membrane, the yolk sac. The viviparous mammals, including humans, also have extraembryonic membranes that will be described in chapter 60.



FIGURE 59.7

The introduction of sperm by the male into the female's body is called copulation. Reptiles such as these turtles were the first terrestrial vertebrates to develop this form of reproduction, which is particularly suited to a terrestrial environment.



FIGURE 59.8

Crested penguins incubating their egg. This nesting pair is changing the parental guard in a stylized ritual.

Most reptiles and all birds are oviparous, laying amniotic eggs that are protected by watertight membranes from desiccation. Birds, being homeotherms, must keep the eggs warm by incubation.

Mammals

Some mammals are seasonal breeders, reproducing only once a year, while others have shorter reproductive cycles. Among the latter, the females generally undergo the reproductive cycles, while the males are more constant in their reproductive activity. Cycling in females involves the periodic release of a mature ovum from the ovary in a process known as ovulation. Most female mammals are “in heat,” or sexually receptive to males, only around the time of ovulation. This period of sexual receptivity is called **estrus**, and the reproductive cycle is therefore called an **estrous cycle**. The females continue to cycle until they become pregnant.

In the estrous cycle of most mammals, changes in the secretion of follicle-stimulating hormone (FSH) and luteinizing hormone (LH) by the anterior pituitary gland cause changes in egg cell development and hormone secretion in the ovaries. Humans and apes have menstrual cycles that are similar to the estrous cycles of other mammals in their cyclic pattern of hormone secretion and ovulation. Unlike mammals with estrous cycles, however, human and ape females bleed when they shed the inner lining of their uterus, a process called menstruation, and may engage in copulation at any time during the cycle.

Rabbits and cats differ from most other mammals in that they are induced ovulators. Instead of ovulating in a cyclic fashion regardless of sexual activity, the females ovulate only after copulation as a result of a reflex stimulation of LH secretion (described later). This makes these animals extremely fertile.

The most primitive mammals, the **monotremes** (consisting solely of the duck-billed platypus and the echidna), are oviparous, like the reptiles from which they evolved. They incubate their eggs in a nest (figure 59.9a) or specialized pouch, and the young hatchlings obtain milk from their mother’s mammary glands by licking her skin, as monotremes lack nipples. All other mammals are viviparous, and are divided into two subcategories based on how they nourish their young. The **marsupials**, a group that includes opossums and kangaroos, give birth to fetuses that are incompletely developed. The fetuses complete their development in a pouch of their mother’s skin, where they can obtain nourishment from nipples of the mammary glands (figure 59.9b). The **placental mammals** (figure 59.9c) retain their young for a much longer period of development within the mother’s uterus. The fetuses are nourished by a structure known as the placenta, which is derived from both an extraembryonic membrane (the chorion) and the mother’s uterine lining. Because the fetal and maternal blood vessels are in very close proximity in the placenta, the fetus can obtain nutrients by diffusion from the mother’s blood. The functioning of the placenta is discussed in more detail in chapter 60.

Among mammals that are not seasonal breeders, the females undergo shorter cyclic variations in ovarian function. These are estrous cycles in most mammals and menstrual cycles in humans and apes. Some mammals are induced ovulators, ovulating in response to copulation.



(a)



(b)



(c)

FIGURE 59.9

Reproduction in mammals. (a) Monotremes, like the duck-billed platypus shown here, lay eggs in a nest. (b) Marsupials, such as this kangaroo, give birth to small fetuses which complete their development in a pouch. (c) In placental mammals, like this domestic cat, the young remain inside the mother’s uterus for a longer period of time and are born relatively more developed.

59.3 Male and female reproductive systems are specialized for different functions.

Structure and Function of the Male Reproductive System

The structures of the human male reproductive system, typical of mammals, are illustrated in figure 59.10. If testes form in the human embryo, they develop *seminiferous tubules* beginning at around 43 to 50 days after conception. The seminiferous tubules are the sites of sperm production. At about 9 to 10 weeks, the Leydig cells, located in the interstitial tissue between the seminiferous tubules, begin to secrete testosterone (the major male sex hormone, or androgen). Testosterone secretion during embryonic development converts indifferent structures into the male external genitalia, the *penis* and the *scrotum*, a sac that contains the testes. In the absence of testosterone, these structures develop into the female external genitalia.

In an adult, each testis is composed primarily of the highly convoluted seminiferous tubules (figure 59.11). Although the testes are actually formed within the abdominal cavity, shortly before birth they descend through an opening called the inguinal canal into the scrotum, which suspends them outside the abdominal cavity. The scrotum maintains the testes at around 34°C, slightly lower than the core body temperature (37°C). This lower temperature is required for normal sperm development in humans.

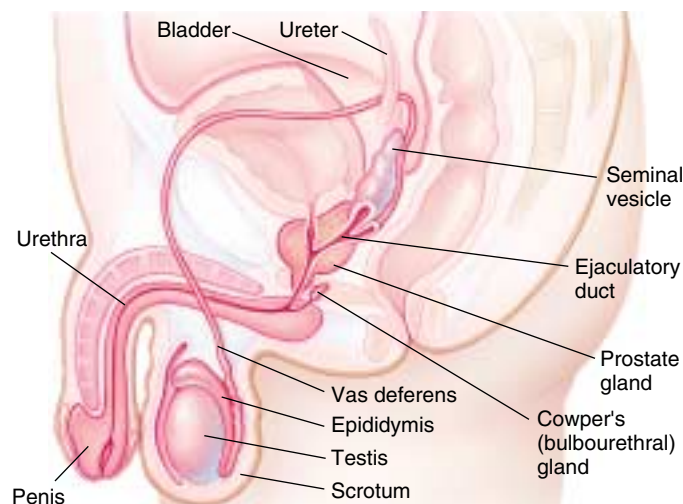


FIGURE 59.10

Organization of the human male reproductive system. The penis and scrotum are the external genitalia, the testes are the gonads, and the other organs are sex accessory organs, aiding the production and ejaculation of semen.

Production of Sperm

The wall of the seminiferous tubule consists of *germinal cells*, which become sperm by meiosis, and supporting *Sertoli cells*. The germinal cells near the outer surface of the seminiferous tubule are diploid (with 46 chromosomes in humans), while those located closer to the lumen of the tubule are haploid (with 23 chromosomes each). Each parent cell duplicates by mitosis, and one of the two daughter cells then undergoes meiosis to form sperm; the other remains as a parent cell. In that way, the male never runs out of parent cells to produce sperm. Adult males produce an average of 100 to 200 million sperm each day and can continue to do so throughout most of the rest of their lives.

The diploid daughter cell that begins meiosis is called a *primary spermatocyte*. It has 23 pairs of homologous chromosomes (in humans) and each chromosome is duplicated, with two chromatids. The first meiotic division separates the homologous chromosomes, producing two haploid *secondary spermatocytes*. However, each chromosome still consists of two duplicate chromatids. Each of these cells then undergoes the second meiotic division to separate the chromatids and produce two haploid cells, the *spermatids*. Therefore, a total of four haploid spermatids are produced by each primary spermatocyte (figure 59.11). All of these cells constitute the germinal epithelium of the seminiferous tubules because they “germinate” the gametes.

In addition to the germinal epithelium, the walls of the seminiferous tubules contain nongerminal cells known as Sertoli cells. The Sertoli cells nurse the developing sperm and secrete products required for spermatogenesis (sperm production). They also help convert the spermatids into *spermatozoa* by engulfing their extra cytoplasm.

Spermatozoa, or sperm, are relatively simple cells, consisting of a head, body, and tail (figure 59.12). The head encloses a compact nucleus and is capped by a vesicle called an acrosome, which is derived from the Golgi complex. The acrosome contains enzymes that aid in the penetration of the protective layers surrounding the egg. The body and tail provide a propulsive mechanism: within the tail is a flagellum, while inside the body are a centriole, which acts as a basal body for the flagellum, and mitochondria, which generate the energy needed for flagellar movement.

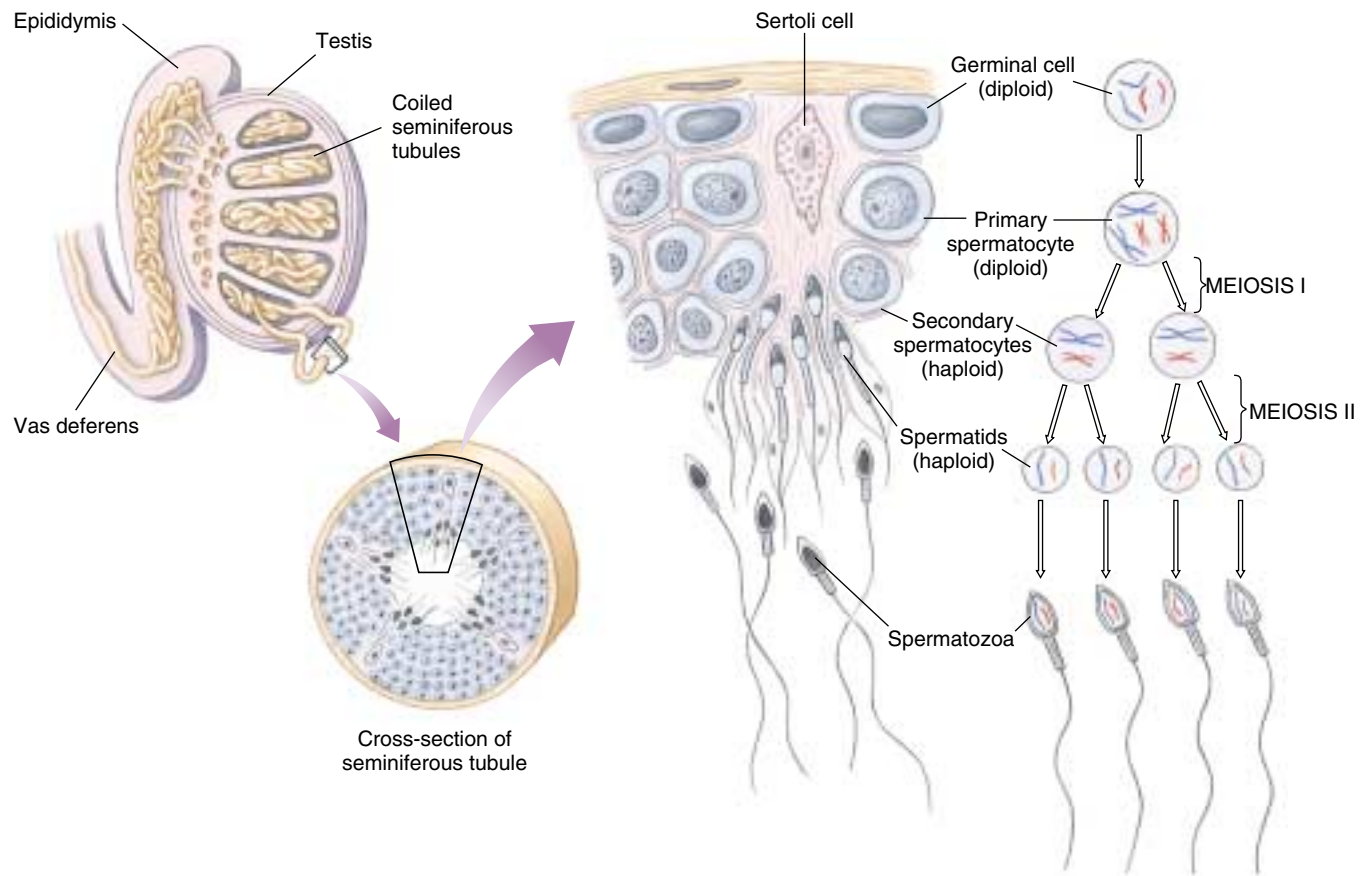
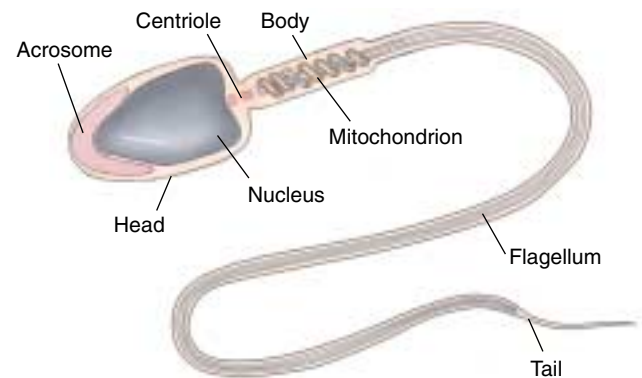


FIGURE 59.11

The testis and spermatogenesis. Inside the testis, the seminiferous tubules are the sites of spermatogenesis. Germinal cells in the seminiferous tubules give rise to spermatozoa by meiosis. Sertoli cells are nongerminal cells within the walls of the seminiferous tubules. They assist spermatogenesis in several ways, such as helping to convert spermatids into spermatozoa. A primary spermatocyte is diploid. At the end of the first meiotic division, homologous chromosomes have separated, and two haploid secondary spermatocytes form. The second meiotic division separates the sister chromatids and results in the formation of four haploid spermatids.



(a)



(b)

FIGURE 59.12

Human sperm. (a) A scanning electron micrograph. (b) A diagram of the main components of a sperm cell.

Male Accessory Sex Organs

After the sperm are produced within the seminiferous tubules, they are delivered into a long, coiled tube called the epididymis (figure 59.13). The sperm are not motile when they arrive in the epididymis, and they must remain there for at least 18 hours before their motility develops. From the epididymis, the sperm enter another long tube, the vas deferens, which passes into the abdominal cavity via the inguinal canal.

The vas deferens from each testis joins with one of the ducts from a pair of glands called the seminal vesicles (see figure 59.10), which produce a fructose-rich fluid. From this point, the vas deferens continues as the ejaculatory duct and enters the prostate gland at the base of the urinary bladder. In humans, the prostate gland is about the size of a golf ball and is spongy in texture. It contributes about 60% of the bulk of the semen, the fluid that contains the products of the testes, fluid from the seminal vesicles, and the products of the prostate gland. Within the prostate gland, the ejaculatory duct merges with the urethra from the urinary bladder. The urethra carries the semen out of the body through the tip of the penis. A pair of pea-sized bulbourethral glands secrete a fluid that lines the urethra and lubricates the tip of the penis prior to coitus (sexual intercourse).

In addition to the urethra, there are two columns of erectile tissue, the corpora cavernosa, along the dorsal side of the penis and one column, the corpus spongiosum, along the ventral side (figure 59.14). Penile erection is produced by neurons in the parasympathetic division of the autonomic nervous system. As a result of the release of nitric oxide by these neurons, arterioles in the penis dilate, causing the erectile tissue to become engorged with blood and turgid. This increased pressure in the erectile tissue compresses the veins, so blood flows into the penis but cannot flow out. The drug sildenafil (**Viagra**) prolongs erection by stimulating release of nitric oxide in the penis. Some mammals, such as the walrus, have a bone in the penis that contributes to its stiffness during erection, but humans do not.

The result of erection and continued sexual stimulation is ejaculation, the ejection from the penis of about 5 milliliters of semen containing an average of 300 million sperm. Successful fertilization requires such a high sperm count because the odds against any one sperm cell successfully completing the journey to the egg and fertilizing it are extraordinarily high, and the acrosomes of several sperm need to interact with the egg before a single sperm can penetrate the egg. Males with fewer than 20 million sperm per milliliter are generally considered sterile. Despite their large numbers, sperm constitute only about 1% of the volume of the semen ejaculated.

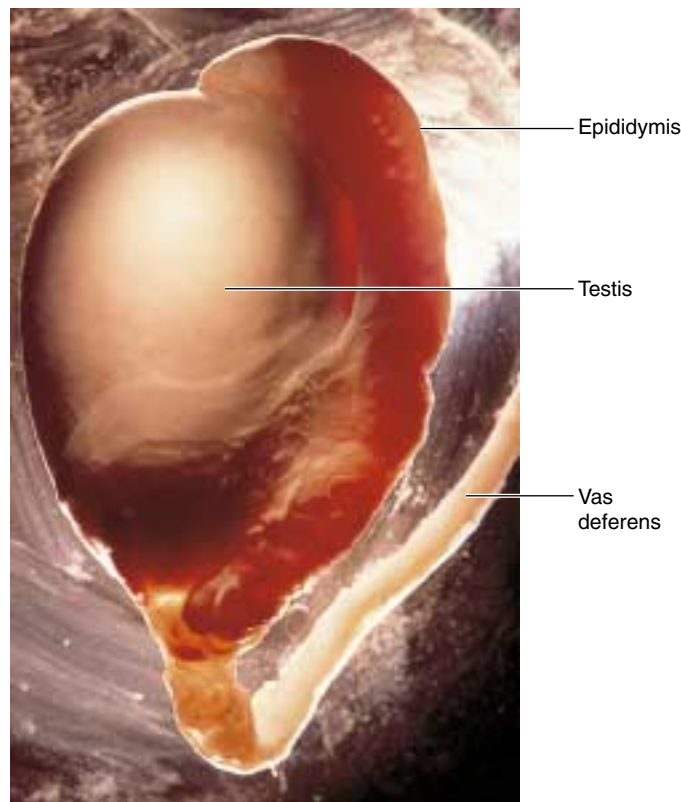


FIGURE 59.13

Photograph of the human testis. The dark, round object in the center of the photograph is a testis, within which sperm are formed. Cupped around it is the epididymis, a highly coiled passageway in which sperm complete their maturation. Mature sperm are stored in the vas deferens, a long tube that extends from the epididymis.

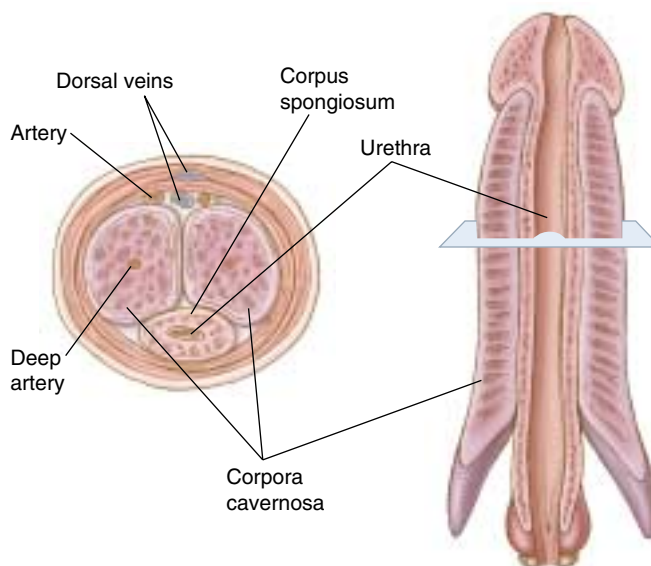


FIGURE 59.14

A penis in cross-section (left) and longitudinal section (right). Note that the urethra runs through the corpus spongiosum.

Table 59.1 Mammalian Reproductive Hormones**MALE**

Follicle-stimulating hormone (FSH)	Stimulates spermatogenesis
Luteinizing hormone (LH)	Stimulates secretion of testosterone by Leydig cells
Testosterone	Stimulates development and maintenance of male secondary sexual characteristics and accessory sex organs

FEMALE

Follicle-stimulating hormone (FSH)	Stimulates growth of ovarian follicles and secretion of estradiol
Luteinizing hormone (LH)	Stimulates ovulation, conversion of ovarian follicles into corpus luteum, and secretion of estradiol and progesterone by corpus luteum
Estradiol	Stimulates development and maintenance of female secondary sexual characteristics; prompts monthly preparation of uterus for pregnancy
Progesterone	Completes preparation of uterus for pregnancy; helps maintain female secondary sexual characteristics
Oxytocin	Stimulates contraction of uterus and milk-ejection reflex
Prolactin	Stimulates milk production

Hormonal Control of Male Reproduction

As we saw in chapter 56, the anterior pituitary gland secretes two gonadotropic hormones: FSH and LH. Although these hormones are named for their actions in the female, they are also involved in regulating male reproductive function (table 59.1). In males, FSH stimulates the Sertoli cells to facilitate sperm development, and LH stimulates the Leydig cells to secrete testosterone.

The principle of negative feedback inhibition discussed in chapter 56 applies to the control of FSH and LH secretion (figure 59.15). The hypothalamic hormone, gonadotropin-releasing hormone (GnRH), stimulates the anterior pituitary gland to secrete both FSH and LH. FSH causes the Sertoli cells to release a peptide hormone called inhibin that specifically inhibits FSH secretion. Similarly, LH stimulates testosterone secretion, and testosterone feeds back to inhibit the release of LH, both directly at the anterior pituitary gland and indirectly by reducing GnRH release. The importance of negative feedback inhibition can be demonstrated by removing the testes; in the absence of testosterone and inhibin, the secretion of FSH and LH from the anterior pituitary is greatly increased.

An adult male produces sperm continuously by meiotic division of the germinal cells lining the seminiferous tubules. Semen consists of sperm from the testes and fluid contributed by the seminal vesicles and prostate gland. Production of sperm and secretion of testosterone from the testes are controlled by FSH and LH from the anterior pituitary.

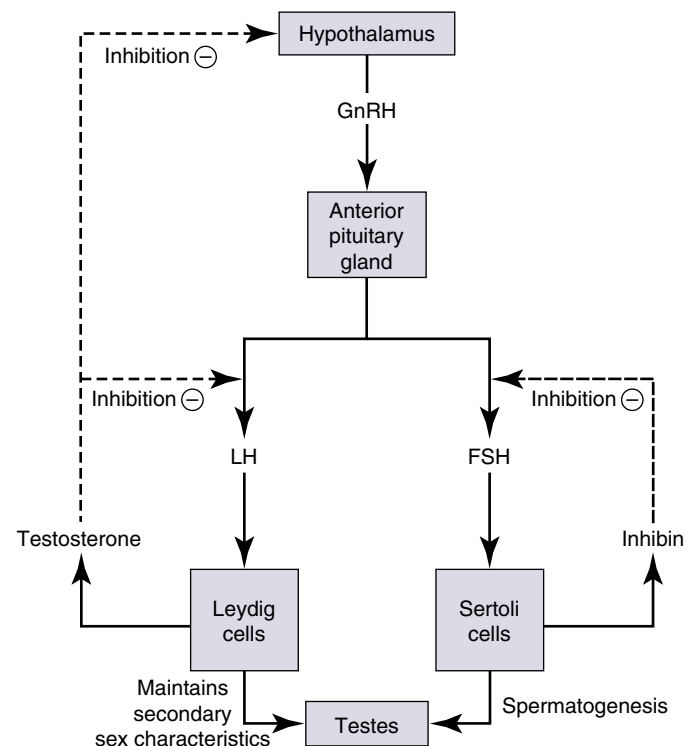


FIGURE 59.15
Hormonal interactions between the testes and anterior pituitary. LH stimulates the Leydig cells to secrete testosterone, and FSH stimulates the Sertoli cells of the seminiferous tubules to secrete inhibin. Testosterone and inhibin, in turn, exert negative feedback inhibition on the secretion of LH and FSH, respectively.

Structure and Function of the Female Reproductive System

The structures of the reproductive system in a human female are shown in figure 59.16. In contrast to the testes, the ovaries develop much more slowly. In the absence of testosterone, the female embryo develops a **clitoris** and **labia majora** from the same embryonic structures that produce a penis and scrotum in males. Thus clitoris and penis, and the labia majora and scrotum, are said to be *homologous structures*. The clitoris, like the penis, contains corpora cavernosa and is therefore erectile. The ovaries contain microscopic structures called **ovarian follicles**, which each contain an egg cell and smaller **granulosa cells**. The ovarian follicles are the functional units of the ovary.

At puberty, the granulosa cells begin to secrete the major female sex hormone estradiol (also called estrogen), triggering **menarche**, the onset of menstrual cycling. Estradiol also stimulates the formation of the **female secondary sexual characteristics**, including breast development and the production of pubic hair. In addition, estradiol and another steroid hormone, progesterone, help to

maintain the female accessory sex organs: the fallopian tubes, uterus, and vagina.

Female Accessory Sex Organs

The fallopian tubes (also called uterine tubes or oviducts) transport ova from the ovaries to the uterus. In humans, the uterus is a muscular, pear-shaped organ that narrows to form a neck, the cervix, which leads to the vagina (figure 59.17*a*). The uterus is lined with a simple columnar epithelial membrane called the endometrium. The surface of the endometrium is shed during menstruation, while the underlying portion remains to generate a new surface during the next cycle.

Mammals other than primates have more complex female reproductive tracts, where part of the uterus divides to form uterine “horns,” each of which leads to an oviduct (figure 59.17*b, c*). In cats, dogs, and cows, for example, there is one cervix but two uterine horns separated by a septum, or wall. Marsupials, such as opossums, carry the split even further, with two unconnected uterine horns, two cervices, and two vaginas. A male marsupial has a forked penis that can enter both vaginas simultaneously.

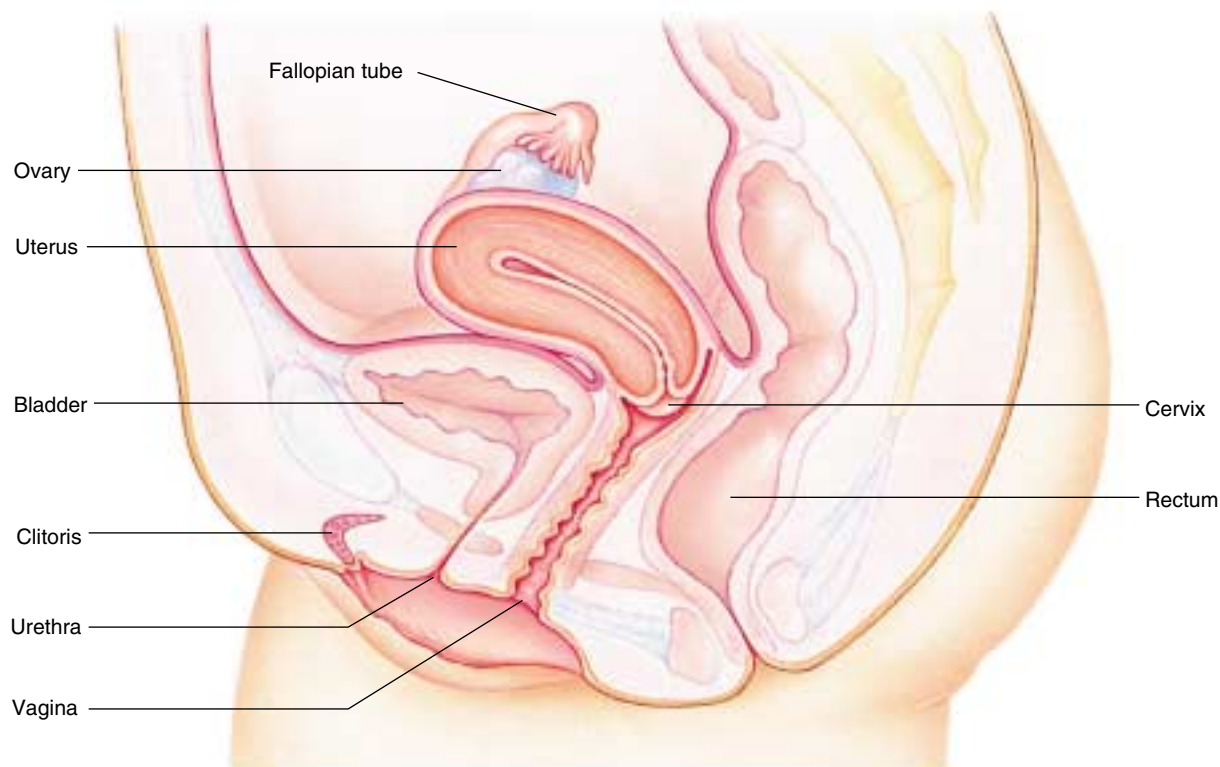


FIGURE 59.16

Organization of the human female reproductive system. The ovaries are the gonads, the fallopian tubes receive the ovulated ova, and the uterus is the womb, the site of development of an embryo if the egg cell becomes fertilized.

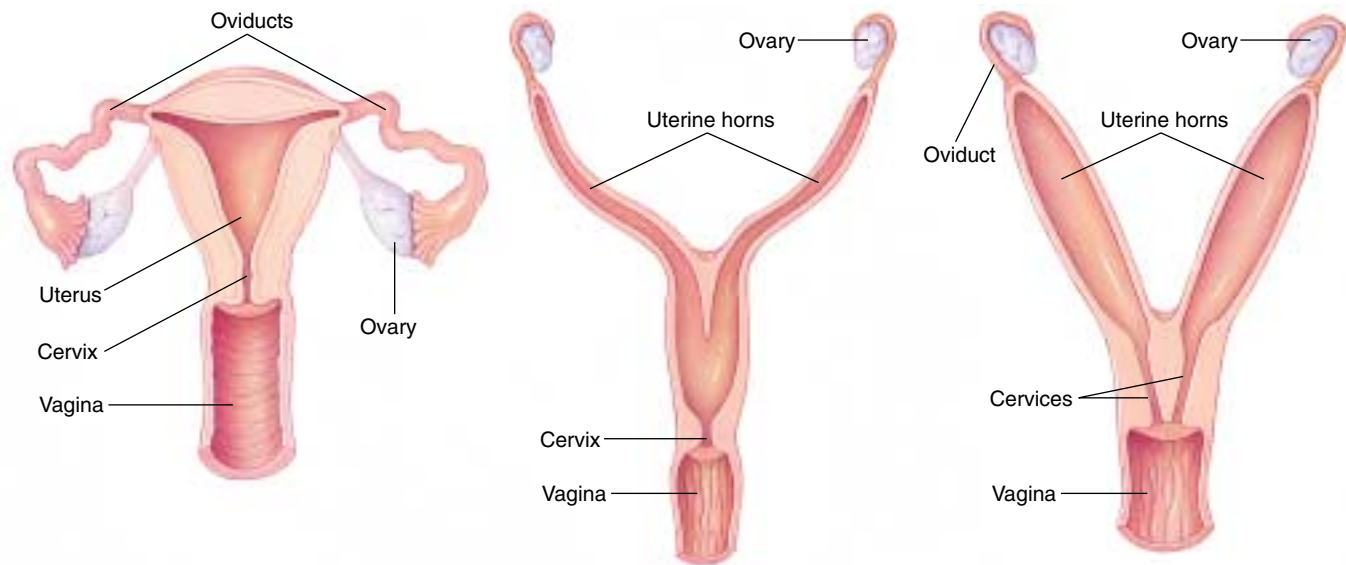


FIGURE 59.17

A comparison of mammalian uteruses. (a) Humans and other primates; (b) cats, dogs, and cows; and (c) rats, mice, and rabbits.

Menstrual and Estrous Cycles

At birth, a female's ovaries contain some 2 million follicles, each with an ovum that has begun meiosis but which is arrested in prophase of the first meiotic division. At this stage, the ova are called primary oocytes. Some of these primary-oocyte-containing follicles are stimulated to develop during each cycle. The human menstrual (Latin *mens*, "month") cycle lasts approximately one month (28 days on the average) and can be divided in terms of ovarian activity into a follicular phase and luteal phase, with the two phases separated by the event of ovulation.

Follicular Phase

During the follicular phase, a few follicles are stimulated to grow under FSH stimulation, but only one achieves full maturity as a tertiary, or Graafian, follicle (figure 59.18). This follicle forms a thin-walled blister on the surface of the ovary. The primary oocyte within the Graafian follicle completes the first meiotic division during the follicular phase. Instead of forming two equally large daughter cells, however, it produces one large daughter cell, the secondary oocyte, and one tiny daughter cell, called a polar body. Thus, the secondary oocyte acquires almost all of the cytoplasm from the primary oocyte, increasing its chances of sustaining the early embryo should the oocyte be fertilized. The polar body, on the other hand, often disintegrates. The secondary oocyte then begins the second meiotic division, but its progress is arrested at metaphase II. It is in this form that the egg cell is discharged from the ovary at ovulation, and it does not complete the second meiotic division unless it becomes fertilized in the fallopian tube.

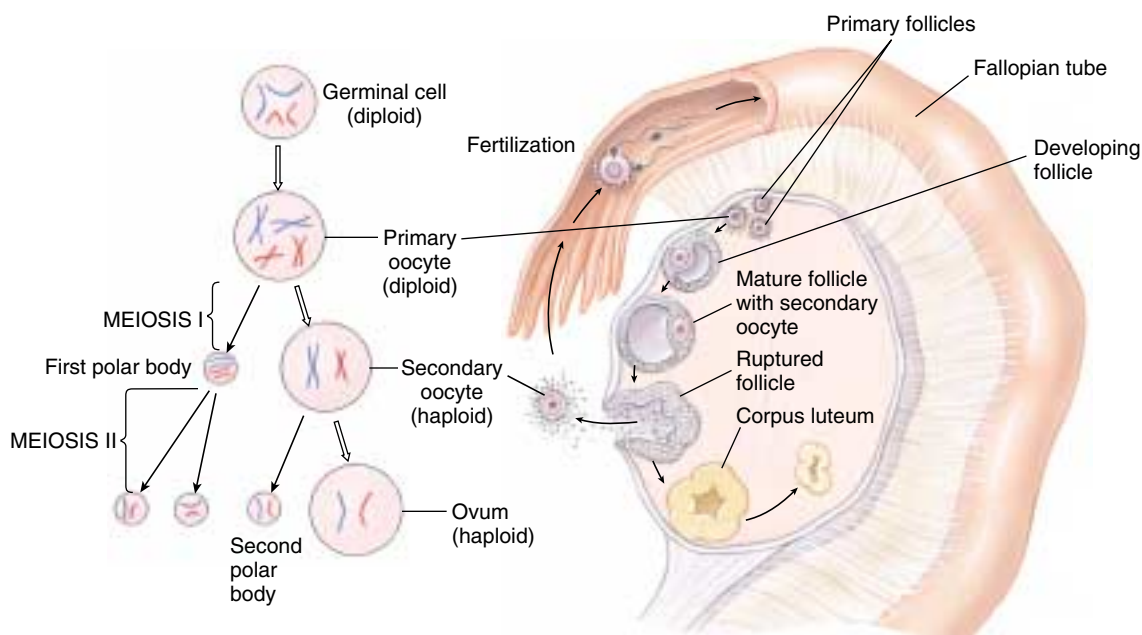


FIGURE 59.18

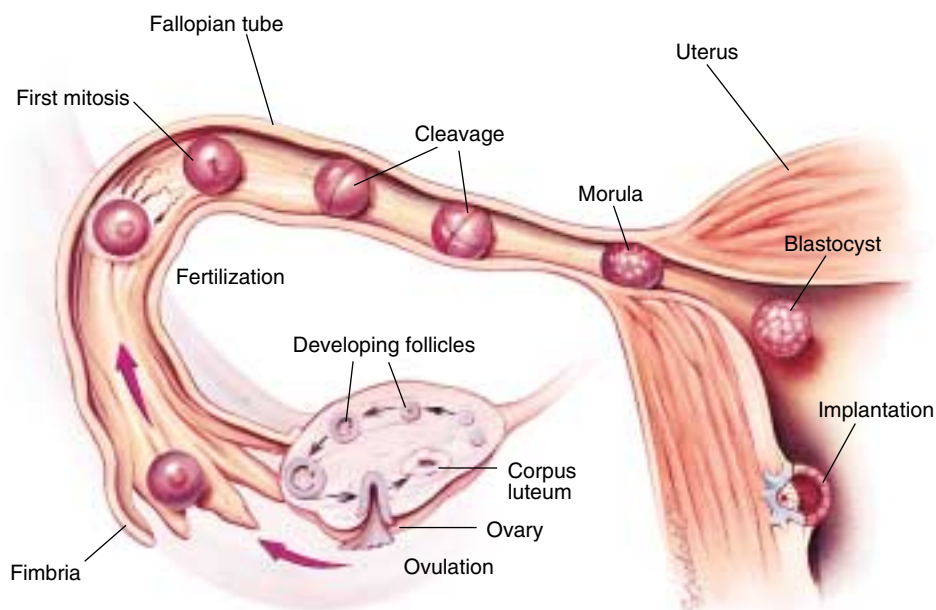
A mature Graafian follicle in a cat ovary (50 $\times$). Note the ring of granulosa cells that surrounds the secondary oocyte. This ring will remain around the egg cell when it is ovulated, and sperm must tunnel through the ring in order to reach the plasma membrane of the egg cell.

FIGURE 59.19**The meiotic events of oogenesis in humans.**

A primary oocyte is diploid. At the completion of the first meiotic division, one division product is eliminated as a polar body, while the other, the secondary oocyte, is released during ovulation. The secondary oocyte does not complete the second meiotic division until after fertilization; that division yields a second polar body and a single haploid egg, or ovum. Fusion of the haploid egg with a haploid sperm during fertilization produces a diploid zygote.

**Ovulation**

The increasing level of estradiol in the blood during the follicular phase stimulates the anterior pituitary gland to secrete LH about midcycle. This sudden secretion of LH causes the fully developed Graafian follicle to burst in the process of ovulation, releasing its secondary oocyte. The released oocyte enters the abdominal cavity near the fimbriae, the feathery projections surrounding the opening to the fallopian tube. The ciliated epithelial cells lining the fallopian tube propel the oocyte through the fallopian tube toward the uterus. If it is not fertilized, the oocyte will disintegrate within a day following ovulation. If it is fertilized, the stimulus of fertilization allows it to complete the second meiotic division, forming a fully mature ovum and a second polar body. Fusion of the two nuclei from the ovum and the sperm produces a diploid zygote (figure 59.19). Fertilization normally occurs in the upper one-third of the fallopian tube, and in a human the zygote takes approximately three days to reach the uterus, then another two to three days to implant in the endometrium (figure 59.20).

**FIGURE 59.20**

The journey of an egg. Produced within a follicle and released at ovulation, an egg is swept into a fallopian tube and carried along by waves of ciliary motion in the tube walls. Sperm journeying upward from the vagina fertilize the egg within the fallopian tube. The resulting zygote undergoes several mitotic divisions while still in the tube, so that by the time it enters the uterus, it is a hollow sphere of cells called a blastocyst. The blastocyst implants within the wall of the uterus, where it continues its development. (The egg and its subsequent stages have been enlarged for clarification.)

Luteal Phase

After ovulation, LH stimulates the empty Graafian follicle to develop into a structure called the corpus luteum (Latin, “yellow body”). For this reason, the second half of the menstrual cycle is referred to as the **luteal phase** of the cycle. The corpus luteum secretes both estradiol and another steroid hormone, progesterone. The high blood levels of estradiol and progesterone during the luteal phase now exert negative feedback inhibition of FSH and LH secretion by the anterior pituitary gland. This inhibition during the luteal phase is in contrast to the stimulation exerted by estradiol on LH secretion at midcycle, which caused ovulation. The inhibitory effect of estradiol and progesterone on FSH and LH secretion after ovulation acts as a natural contraceptive mechanism, preventing both the development of additional follicles and continued ovulation.

During the follicular phase the granulosa cells secrete increasing amounts of estradiol, which stimulates the growth of the endometrium. Hence, this portion of the cycle is also referred to as the **proliferative phase** of the endometrium. During the luteal phase of the cycle, the combination of estradiol and progesterone cause the endometrium to become more vascular, glandular, and enriched with glycogen deposits. Because of the endometrium’s glandular appearance, this portion of the cycle is known as the **secretory phase** of the endometrium (figure 59.21).

In the absence of fertilization, the corpus luteum triggers its own atrophy, or regression, toward the end of the luteal phase. It does this by secreting hormones (estradiol and progesterone) that inhibit the secretion of LH, the hormone needed for its survival. In many mammals, atrophy of the corpus luteum is assisted by luteolysin, a paracrine regulator believed to be a prostaglandin. The disappearance of the corpus luteum results in an abrupt decline in the blood concentration of estradiol and progesterone at the end of the luteal phase, causing the built-up endometrium to be sloughed off with accompanying bleeding. This process is called menstruation, and the portion of the cycle in which it occurs is known as the **menstrual phase** of the endometrium.

If the ovulated oocyte is fertilized, however, regression of the corpus luteum and subsequent menstruation is averted by the tiny embryo! It does this by secreting human chorionic gonadotropin (hCG), an LH-like hormone produced by the chorionic membrane of the embryo. By maintaining the corpus luteum, hCG keeps the levels of estradiol and progesterone high and thereby prevents menstruation, which would terminate the pregnancy. Because hCG comes from the embryonic chorion and not the mother, it is the hormone that is tested for in all pregnancy tests.

Menstruation is absent in mammals with an estrous cycle. Although such mammals do cyclically shed cells from the endometrium, they don’t bleed in the process. The estrous cycle is divided into four phases: proestrus, estrus, metestrus, and diestrus, which correspond to the proliferative, mid-cycle, secretory, and menstrual phases of the endometrium in the menstrual cycle.

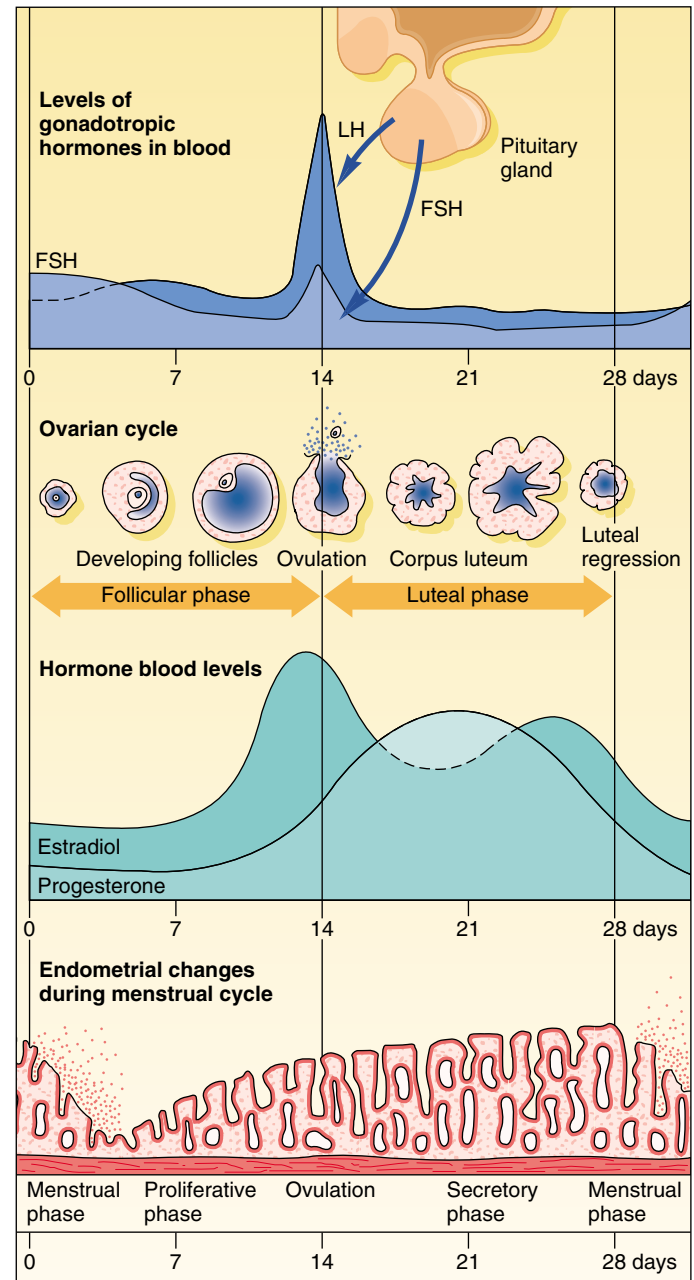


FIGURE 59.21

The human menstrual cycle. The growth and thickening of the endometrial (uterine) lining is stimulated by estradiol and progesterone. The decline in the levels of these two hormones triggers menstruation, the sloughing off of built-up endometrial tissue.

The ovarian follicles develop under FSH stimulation, and one follicle ovulates under LH stimulation. During the follicular and luteal phases, the hormones secreted by the ovaries stimulate the development of the endometrium, so an embryo can implant there if fertilization has occurred. A secondary oocyte is released from an ovary at ovulation, and it only completes meiosis if it is fertilized.

59.4 The physiology of human sexual intercourse is becoming better known.

Physiology of Human Sexual Intercourse

Few physical activities are more pleasurable to humans than sexual intercourse. The sex drive is one of the strongest drives directing human behavior, and as such, it is circumscribed by many rules and customs. Sexual intercourse acts as a channel for the strongest of human emotions such as love, tenderness, and personal commitment. Few subjects are at the same time more private and of more general interest. Here we will limit ourselves to a very narrow aspect of sexual behavior, its immediate physiological effects. The emotional consequences are no less real, but they are beyond the scope of this book.

Until relatively recently, the physiology of human sexual activity was largely unknown. Perhaps because of the prevalence of strong social taboos against the open discussion of sexual matters, no research was carried out on the subject, and detailed information was lacking. Over the past 40 years, however, investigations by William Masters and Virginia Johnson, as well as an army of researchers who followed them, have revealed much about the biological nature of human sexual activity.

The sexual act is referred to by a variety of names, including sexual intercourse, copulation, and coitus, as well as a host of informal terms. It is common to partition the physiological events that accompany intercourse into four phases—**excitement**, **plateau**, **orgasm**, and **resolution**—although there are no clear divisions between these phases.

Excitement

The sexual response is initiated by the nervous system. In both males and females, commands from the brain increase the respiratory rate, heart rate, and blood pressure. The nipples commonly harden and become more sensitive. Other changes increase the diameter of blood vessels, leading to increased circulation. In some people, these changes may produce a reddening of the skin around the face, breasts, and genitals (the sex flush). Increased circulation also leads to vasocongestion, producing erection of the male's penis and similar swelling of the female's clitoris. The female experiences changes that prepare the vagina for sexual intercourse: the labia majora and labia minora, lips of tissue that cover the opening to the vagina, swell and separate due to the increased circulation; the vaginal walls become moist; and the muscles encasing the vagina relax.

Plateau

The penetration of the vagina by the thrusting penis continuously stimulates nerve endings both in the tip of the penis and in the clitoris. The clitoris, which is now swollen,

becomes very sensitive and withdraws up into a sheath or "hood." Once it has withdrawn, the clitoris is stimulated indirectly when the thrusting movements of the penis rub the clitoral hood against the clitoris. The nervous stimulation produced by the repeated movements of the penis within the vagina elicits a continuous response in the autonomic nervous system, greatly intensifying the physiological changes initiated during the excitement phase. In the female, pelvic thrusts may begin, while in the male the penis reaches its greatest length and rigidity.

Orgasm

The climax of intercourse is reached when the stimulation is sufficient to initiate a series of reflexive muscular contractions. The nerve impulses producing these contractions are associated with other activity within the central nervous system, activity that we experience as intense pleasure. In females, the contractions are initiated by impulses in the hypothalamus, which causes the posterior pituitary gland to release large amounts of oxytocin. This hormone, in turn, causes the muscles in the uterus and around the vaginal opening to contract and the cervix to be pulled upward. Contractions occur at intervals of about one per second. There may be one to several intense peaks of contractions (orgasms), or the peaks may be more numerous but less intense.

Analogous contractions take place in the male. The first contractions, which occur in the vas deferens and prostate gland, cause *emission*, the peristaltic movement of sperm and seminal fluid into a collecting zone of the urethra located at the base of the penis. Shortly thereafter, violent contractions of the muscles at the base of the penis result in *ejaculation* of the collected semen through the penis. As in the female, the contractions are spaced about one second apart, although in the male they continue for only a few seconds and are almost invariably restricted to a single intense wave.

Resolution

After ejaculation, males rapidly lose their erection and enter a refractory period lasting 20 minutes or longer, in which sexual arousal is difficult to achieve and ejaculation is almost impossible. By contrast, many women can be aroused again almost immediately. After intercourse, the bodies of both men and women return over a period of several minutes to their normal physiological state.

Sexual intercourse is a physiological series of events leading to the ultimate deposition of sperm within the female reproductive tract. The phases are similar in males and females.

Birth Control

In most vertebrates, copulation is associated solely with reproduction. Reflexive behavior that is deeply ingrained in the female limits sexual receptivity to those periods of the sexual cycle when she is fertile. In humans and a few species of apes, the female can be sexually receptive throughout her reproductive cycle, and this extended receptivity to sexual intercourse serves a second important function—it reinforces pair-bonding, the emotional relationship between two individuals living together.

Not all human couples want to initiate a pregnancy every time they have sexual intercourse, yet sexual intercourse may be a necessary and important part of their emotional lives together. The solution to this dilemma is to find a way to avoid reproduction without avoiding sexual intercourse; this approach is commonly called birth control or contraception. A variety of approaches differing in effectiveness and in their acceptability to different couples are commonly taken to achieve birth control (figure 59.22 and table 59.2).

Abstinence

The simplest and most reliable way to avoid pregnancy is not to have sexual intercourse at all. Of all methods of birth control, this is the most certain. It is also the most limiting, because it denies a couple the emotional support of a sexual relationship.

Sperm Blockage

If sperm cannot reach the uterus, fertilization cannot occur. One way to prevent the delivery of sperm is to encase the penis within a thin sheath, or condom. Many males do not favor the use of condoms, which tend to decrease their sensory pleasure during intercourse. In principle, this method is easy to apply and foolproof, but in practice it has a failure rate of 3 to 15% because of incorrect use or inconsistent use. Nevertheless, it is the most commonly employed form of birth control in the United States. Condoms are also widely used to prevent the transmission of AIDS and other sexually transmitted diseases (STDs). Over a billion condoms were sold in the United States last year.

A second way to prevent the entry of sperm into the uterus is to place a cover over the cervix. The cover may be a relatively tight-fitting cervical cap, which is worn for days at a time, or a rubber dome called a diaphragm, which is inserted immediately before intercourse. Because



FIGURE 59.22
Four common methods of birth control. (a) Condom; (b) diaphragm and spermicidal jelly; (c) oral contraceptives; (d) Depo-Provera.

the dimensions of individual cervixes vary, a cervical cap or diaphragm must be fitted by a physician. Failure rates average 4 to 25% for diaphragms, perhaps because of the propensity to insert them carelessly when in a hurry. Failure rates for cervical caps are somewhat lower.

Sperm Destruction

A third general approach to birth control is to eliminate the sperm after ejaculation. This can be achieved in principle by washing out the vagina immediately after intercourse, before the sperm have a chance to enter the uterus. Such a procedure is called a douche (French, “wash”). The douche method is difficult to apply well, because it involves a rapid dash to the bathroom immediately after ejaculation and a very thorough washing. Its failure rate is as high as 40%. Alternatively, sperm delivered to the vagina can be destroyed there with spermicidal jellies or foams. These treatments generally require application immediately before intercourse. Their failure rates vary from 10 to 25%. The use of a spermicide with a condom increases the effectiveness over each method used independently.

Prevention of Ovulation

Since about 1960, a widespread form of birth control in the United States has been the daily ingestion of birth control pills, or oral contraceptives, by women. These pills contain analogues of progesterone, sometimes in combination with

Table 59.2 Methods of Birth Control

Device	Action	Failure Rate*	Advantages	Disadvantages
Oral contraceptive	Hormones (progesterone analogue alone or in combination with other hormones) primarily prevent ovulation	1–5, depending on type	Convenient; highly effective; provides significant noncontraceptive health benefits, such as protection against ovarian and endometrial cancers	Must be taken regularly; possible minor side effects which new formulations have reduced; not for women with cardiovascular risks (mostly smokers over age 35)
Condom	Thin sheath for penis that collects semen; “female condoms” sheath vaginal walls	3–15	Easy to use; effective; inexpensive; protects against some sexually transmitted diseases	Requires male cooperation; may diminish spontaneity; may deteriorate on the shelf
Diaphragm	Soft rubber cup covers entrance to uterus, prevents sperm from reaching egg, holds spermicide	4–25	No dangerous side effects; reliable if used properly; provides some protection against sexually transmitted diseases and cervical cancer	Requires careful fitting; some inconvenience associated with insertion and removal; may be dislodged during intercourse
Intrauterine device (IUD)	Small plastic or metal device placed in the uterus; prevents implantation; some contain copper, others release hormones	1–5	Convenient; highly effective; infrequent replacement	Can cause excess menstrual bleeding and pain; risk of perforation, infection, expulsion, pelvic inflammatory disease, and infertility; not recommended for those who eventually intend to conceive or are not monogamous; dangerous in pregnancy
Cervical cap	Miniature diaphragm covers cervix closely, prevents sperm from reaching egg, holds spermicide	Probably similar to that of diaphragm	No dangerous side effects; fairly effective; can remain in place longer than diaphragm	Problems with fitting and insertion; comes in limited number of sizes
Foams, creams, jellies, vaginal suppositories	Chemical spermicides inserted in vagina before intercourse that prevent sperm from entering uterus	10–25	Can be used by anyone who is not allergic; protect against some sexually transmitted diseases; no known side effects	Relatively unreliable; sometimes messy; must be used 5–10 minutes before each act of intercourse
Implant (levonorgestrel; Norplant)	Capsules surgically implanted under skin slowly release hormone that blocks ovulation	.03	Very safe, convenient, and effective; very long-lasting (5 years); may have nonreproductive health benefits like those of oral contraceptives	Irregular or absent periods; minor surgical procedure needed for insertion and removal; some scarring may occur
Injectable contraceptive (medroxy-progesterone; Depo-Provera)	Injection every 3 months of a hormone that is slowly released and prevents ovulation	1	Convenient and highly effective; no serious side effects other than occasional heavy menstrual bleeding	Animal studies suggest it may cause cancer, though new studies in humans are mostly encouraging; occasional heavy menstrual bleeding

*Failure rate is expressed as pregnancies per 100 actual users per year.

Source: Data from American College of Obstetricians and Gynecologists: Contraception, Patient Education Pamphlet No. AP005.ACOG, Washington, D.C., 1990.

estrogens. As described earlier, progesterone and estradiol act by negative feedback to inhibit the secretion of FSH and LH during the luteal phase of the menstrual cycle, thereby preventing follicle development and ovulation. They also cause a buildup of the endometrium. The hormones in birth control pills have the same effects. Because the pills block ovulation, no ovum is available to be fertilized. A woman generally takes the hormone-containing

pills for three weeks; during the fourth week, she takes pills without hormones (placebos), allowing the levels of those hormones in her blood to fall, which causes menstruation. Oral contraceptives provide a very effective means of birth control, with a failure rate of only 1 to 5%. In a variation of the oral contraceptive, hormone-containing capsules are implanted beneath the skin. These implanted capsules have failure rates below 1%.

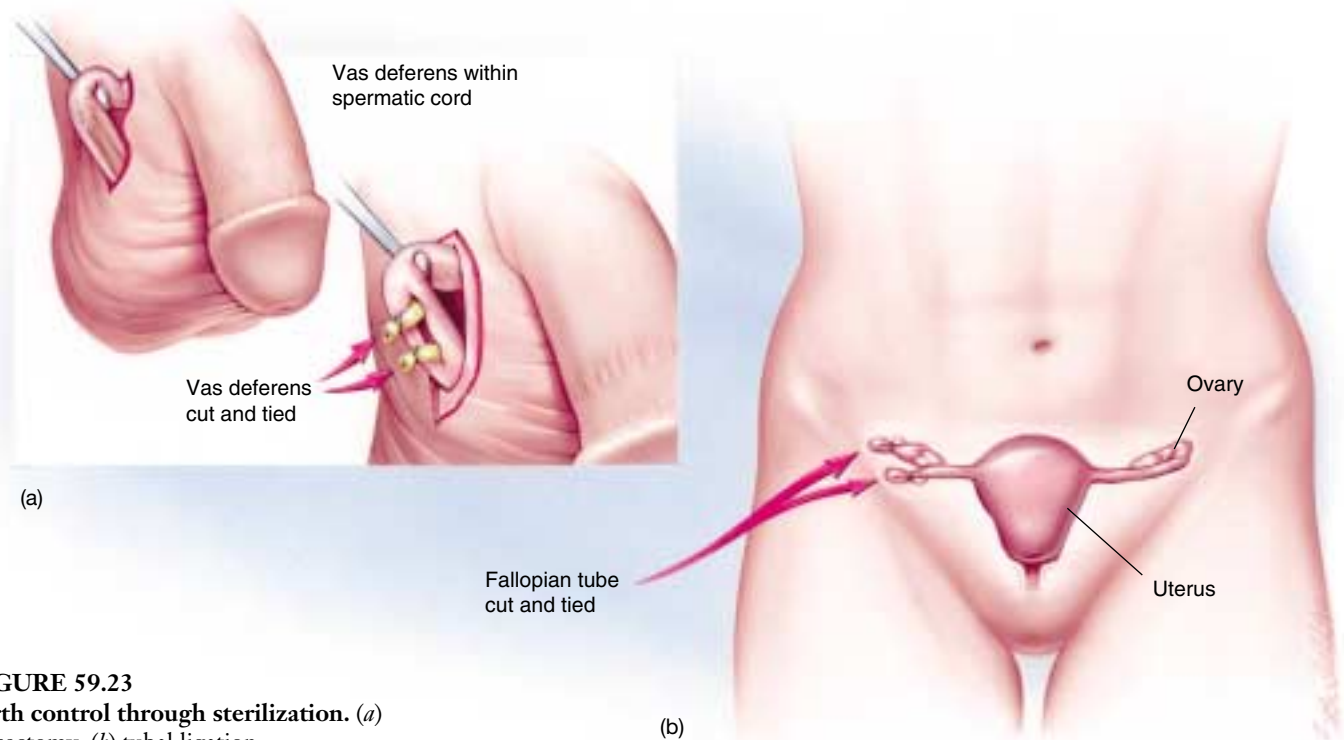


FIGURE 59.23

Birth control through sterilization. (a) Vasectomy; (b) tubal ligation.

A small number of women using birth control pills or implants experience undesirable side effects, such as blood clotting and nausea. These side effects have been reduced in newer generations of birth control pills, which contain less estrogen and different analogues of progesterone. Moreover, these new oral contraceptives provide a number of benefits, including reduced risks of endometrial and ovarian cancer, cardiovascular disease, and osteoporosis (for older women). However, they may increase the risk of contracting breast cancer and cervical cancer. The risks involved with birth control pills increase in women who smoke and increase greatly in women over 35 who smoke. The current consensus is that, for many women, the health benefits of oral contraceptives outweigh their risks, although a physician must help each woman determine the relative risks and benefits.

Prevention of Embryo Implantation

The insertion of a coil or other irregularly shaped object into the uterus is an effective means of birth control, because the irritation it produces in the uterus prevents the implantation of an embryo within the uterine wall. Such intrauterine devices (IUDs) have a failure rate of only 1 to 5%. Their high degree of effectiveness probably reflects their convenience; once they are inserted, they can be forgotten. The great disadvantage of this method is that almost a third of the women who attempt to use IUDs experience cramps, pain, and sometimes bleeding and therefore must discontinue using them.

Another method of preventing embryo implantation is the “morning after pill,” which contains 50 times the dose of estrogen present in birth control pills. The pill works by temporarily stopping ovum development, by preventing fertilization, or by stopping the implantation of a fertilized ovum. Its failure rate is 1 to 10%, but many women are uneasy about taking such high hormone doses, as side effects can be severe. This is not recommended as a regular method of birth control but rather as a method of emergency contraception.

Sterilization

A completely effective means of birth control is sterilization, the surgical removal of portions of the tubes that transport the gametes from the gonads (figure 59.23). Sterilization may be performed on either males or females, preventing sperm from entering the semen in males and preventing an ovulated oocyte from reaching the uterus in females. In males, sterilization involves a vasectomy, the removal of a portion of the vas deferens from each testis. In females, the comparable operation involves the removal of a section of each fallopian tube.

Fertilization can be prevented by a variety of birth control methods, including barrier contraceptives, hormonal inhibition, surgery, and abstinence. Efficacy rates vary from method to method.



Summary

Questions

Media Resources

59.1 Animals employ both sexual and asexual reproductive strategies.

- Parthenogenesis is a form of asexual reproduction that is practiced by many insects and some lizards.
- Among mammals, the sex is determined by the presence of a Y chromosome in males and its absence in females.

1. How are oviparity, ovoviviparity, and viviparity different?



- Introduction to reproduction

59.2 The evolution of reproduction among the vertebrates has led to internalization of fertilization and development.

- Most bony fish practice external fertilization, releasing eggs and sperm into the water where fertilization occurs. Amphibians have external fertilization and the young go through a larval stage before metamorphosis.
- Reptiles and birds are oviparous, the young developing in eggs that are deposited externally. Most mammals are viviparous, the young developing within the mother.

2. How does fetal development differ in the monotremes, marsupials, and placental mammals?



- *On Science* articles: Interactions
- Student Research: Reproductive biology of house mice
Evolution of uterine function

59.3 Male and female reproductive systems are specialized for different functions.

- Sperm leave the testes and pass through the epididymis and vas deferens; the ejaculatory duct merges with the urethra, which empties at the tip of the penis.
- An egg cell released from the ovary in ovulation is drawn by fimbria into the fallopian tube, which conducts the egg cell to the lining of the uterus, or endometrium, where it implants if fertilized.
- If fertilization does not occur, the corpus luteum regresses at the end of the cycle and the resulting fall in estradiol and progesterone secretion cause menstruation to occur in humans and apes.

3. Briefly describe the function of seminal vesicles, prostate gland, and bulbourethral glands.

4. When do the ova in a female mammal begin meiosis? When do they complete the first meiotic division?

5. What hormone is secreted by the granulosa cells in a Graafian follicle? What effect does this hormone have on the endometrium?



- Art Activities: Sperm and egg anatomy
Male reproductive system
Penis anatomy
Female reproductive system
Breast anatomy



- Spermatogenesis
- Menstruation
- Female reproductive cycle
- Oogenesis

59.4 The physiology of human sexual intercourse is becoming better known.

- The physiological events that occur in the human sexual response are grouped into four phases: excitement, plateau, orgasm, and resolution.
- Males and females have similar phases, but males enter a refractory period following orgasm that is absent in many women.
- There are a variety of methods of birth control available that range in ease of use, effectiveness, and permanence.

6. What are the four phases in the physiological events of sexual intercourse in humans? During the first phase, what events occur specifically in males, and what events occur specifically in females?

7. How do birth control pills prevent pregnancy?



- Penile erection
- Vasectomy
- Tubal ligation

60

Vertebrate Development

Concept Outline

60.1 Fertilization is the initial event in development.

Stages of Development. Fertilization of an egg cell by a sperm occurs in three stages: penetration, activation of the egg cell, and fusion of the two haploid nuclei.

60.2 Cell cleavage and the formation of a blastula set the stage for later development.

Cell Cleavage Patterns. The cytoplasm of the zygote is divided into smaller cells by a mitotic cell division in a process called cleavage.

60.3 Gastrulation forms the three germ layers of the embryo.

The Process of Gastrulation. Cells of the blastula invaginate and involute to produce an outer ectoderm, an inner endoderm layer, and a third layer, the mesoderm.

60.4 Body architecture is determined during the next stages of embryonic development.

Developmental Processes during Neurulation. The mesoderm of chordates forms a notochord, and the overlying ectoderm rolls to produce a neural tube.

How Cells Communicate during Development. Cell-to-cell contact plays a major role in selecting the paths along which cells develop.

Embryonic Development and Vertebrate Evolution. The embryonic development of a mammal includes stages that are characteristic of more primitive vertebrates.

Extraembryonic Membranes. Embryonic cells form several membranes outside of the embryo that provide protection, nourishment, and gas exchange for the embryo.

60.5 Human development is divided into trimesters.

First Trimester. A blastocyst implants into the mother's endometrium, and the formation of body organs begins during the third and fourth week.

Second and Third Trimesters. All of the major organs of the body form during the first trimester and so further growth and development take place during this time.

Birth and Postnatal Development. Birth occurs as a result of uterine contractions; the human brain continues to grow significantly after birth.



FIGURE 60.1

Development is the process that determines an organism's form and function. A human fetus at 18 weeks is not yet halfway through the 38 weeks—about 9 months—it will spend within its mother, but it has already developed many distinct behaviors, such as the sucking reflex that is so important to survival after birth.

Reproduction in all but a few vertebrates unites two haploid gametes to form a single diploid cell called a zygote. The zygote develops by a process of cell division and differentiation into a complex multicellular organism, composed of many different tissues and organs (figure 60.1). Although the process of development is a continuous series of events with some of the details varying among different vertebrate groups, we will examine vertebrate development in six stages. In this chapter, we will consider the stages of development and conclude with a description of the events that occur during human development.

60.1 Fertilization is the initial event in development.

Stages of Development

In vertebrates, as in all sexual animals, the first step in development is the union of male and female gametes, a process called **fertilization**. Fertilization is typically external in fish and amphibians, which reproduce in water, and internal in all other vertebrates. In internal fertilization, small, motile sperm are introduced into the female reproductive tract during mating. The sperm swim up the reproductive tract until they encounter a mature egg or oocyte in an oviduct, where fertilization occurs. Fertilization consists of three stages: penetration, activation, and fusion.

Penetration

As described in chapter 59, the secondary oocyte is released from a fully developed Graafian follicle at ovulation. It is surrounded by the same layer of small granulosa cells that surrounded it within the follicle (figure 60.2). Between the granulosa cells and the egg's plasma membrane is a glyco-

protein layer called the zona pellucida. The head of each sperm is capped by an organelle called the acrosome, which contains glycoprotein-digesting enzymes. These enzymes become exposed as the sperm begin to work their way into the layer of granulosa cells, and the activity of the enzymes enables the sperm to tunnel their way through the zona pellucida to the egg's plasma membrane. In sea urchins, egg cytoplasm bulges out at this point, engulfing the head of the sperm and permitting the sperm nucleus to enter the cytoplasm of the egg (figure 60.3).

Activation

The series of events initiated by sperm penetration are collectively called egg activation. In some frogs, reptiles, and birds, more than one sperm may penetrate the egg, but only one is successful in fertilizing it. In mammals, by contrast, the penetration of the first sperm initiates changes in the egg membrane that prevent the entry of other sperm. As the sperm makes contact with the oocyte membrane, there is a change in the membrane potential (see chapter 54

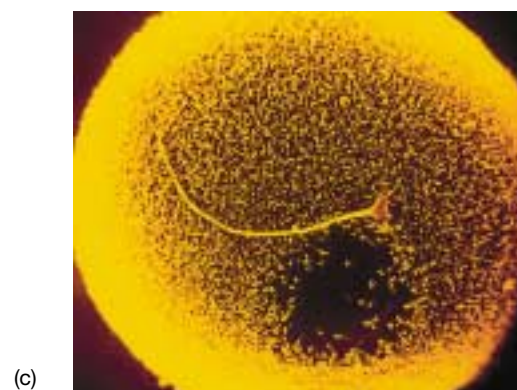
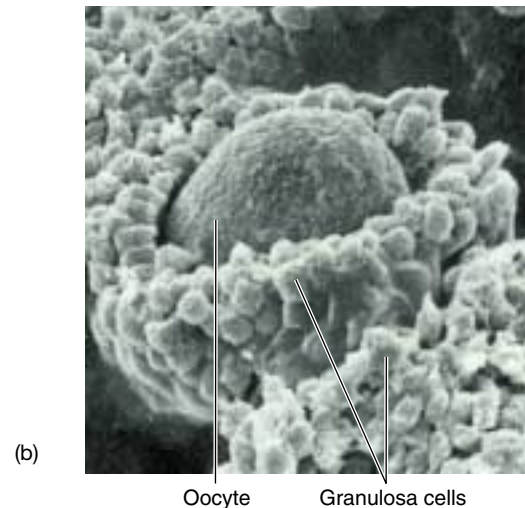
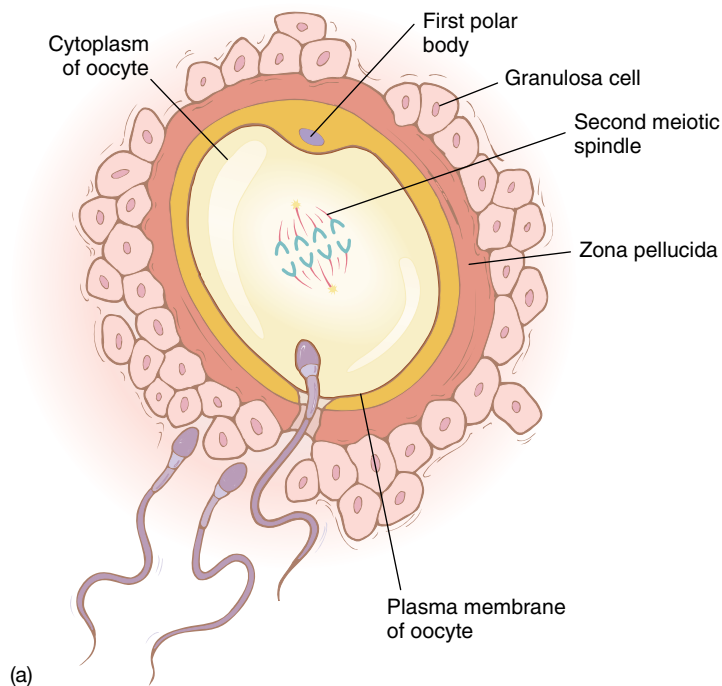
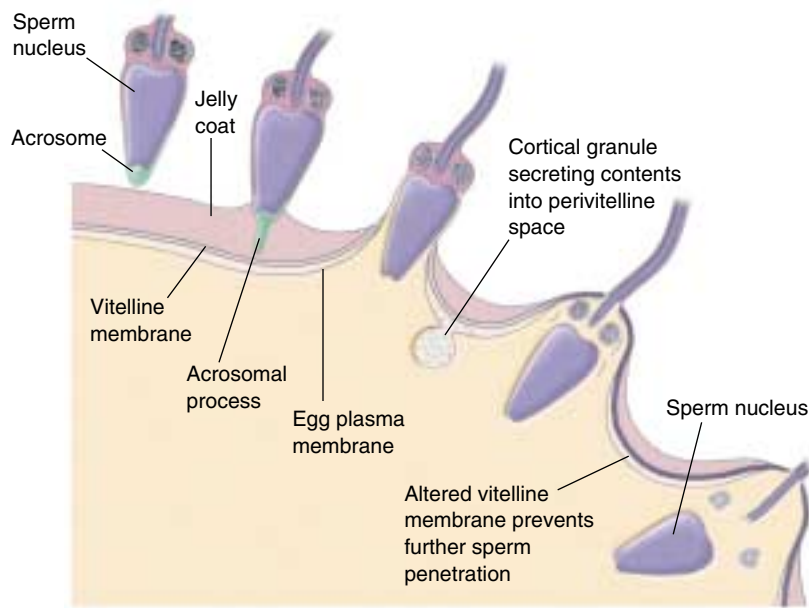
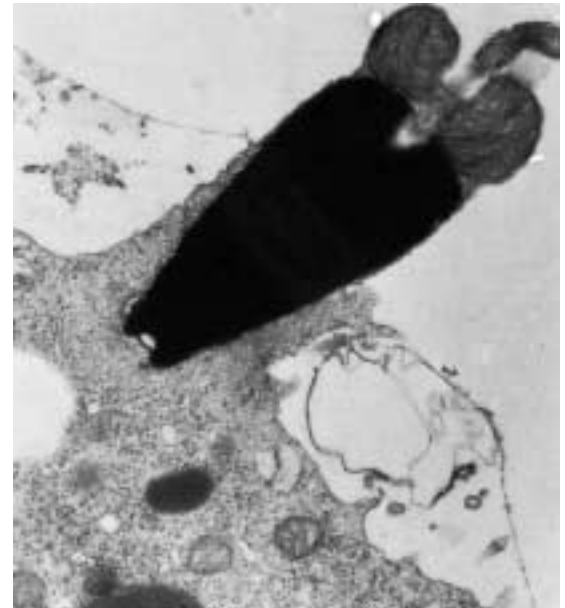


FIGURE 60.2

Mammalian reproductive cells. (a) A sperm must penetrate a layer of granulosa cells and then a layer of glycoprotein called the zona pellucida, before it reaches the oocyte membrane. This penetration is aided by digestive enzymes in the acrosome of the sperm. These scanning electron micrographs show (b) a human oocyte (90 $\times$) surrounded by numerous granulosa cells, and (c) a human sperm on an egg (3000 $\times$).



(a)



(b)

FIGURE 60.3

Sperm penetration of a sea urchin egg. (a) The stages of penetration. (b) An electron micrograph (50,000 $\times$) of penetration. Penetration in both invertebrate and vertebrate eggs is similar.

for discussion of membrane potential) that prevents other sperm from fusing with the oocyte membrane. In addition to these changes, sperm penetration can have three other effects on the egg. First, in mammals it stimulates the chromosomes in the egg nucleus to complete the second meiotic division, producing two egg nuclei. One of these nuclei is extruded from the egg as a second polar body (see chapter 59), leaving a single haploid egg nucleus within the egg.

Second, sperm penetration in some animals triggers movements of the egg cytoplasm around the point of sperm entry. These movements ultimately establish the bilateral symmetry of the developing animal. In frogs, for example, sperm penetration causes an outer pigmented cap of egg cytoplasm to rotate toward the point of entry, uncovering a gray crescent of interior cytoplasm opposite the point of penetration (figure 60.4). The position of the gray crescent determines the orientation of the first cell division. A line drawn between the point of sperm entry and the gray crescent would bisect the right and left halves of the future adult. Third, activation is characterized by a sharp increase in protein synthesis and an increase in metabolic activity in general. Experiments demonstrate that the protein synthesis in the activated oocyte is coded by mRNA that was previously produced and already present in the cytoplasm of the unfertilized egg cell.

In some vertebrates, it is possible to activate an egg without the entry of a sperm, simply by pricking the egg membrane. An egg that is activated in this way may go on to develop parthenogenetically. A few kinds of amphibians, fish, and reptiles rely entirely on parthenogenetic reproduction in nature, as we mentioned in chapter 59.

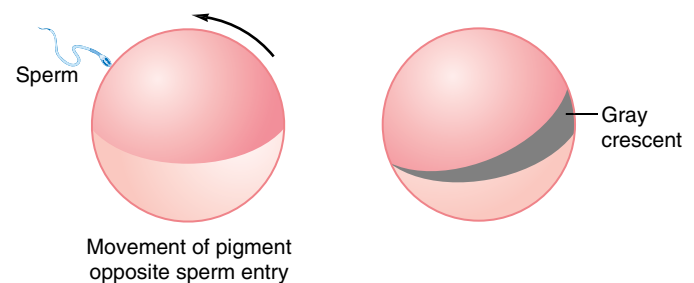


FIGURE 60.4

Gray crescent formation in frog eggs. The gray crescent appears opposite the point of penetration by the sperm.

Nuclei Fusion

The third stage of fertilization is the fusion of the entering sperm nucleus with the haploid egg nucleus to form the diploid nucleus of the zygote. This fusion is triggered by the activation of the egg. If a sperm nucleus is microinjected into an egg without activating the egg, the two nuclei will not fuse. The nature of the signals that are exchanged between the two nuclei, or sent from one to the other, is not known.

The three stages of fertilization are penetration, activation, and nuclei fusion. Penetration initiates a complex series of developmental events, including major movements of cytoplasm, which eventually lead to the fusion of the egg and sperm nuclei.

60.2 Cell cleavage and the formation of a blastula set the stage for later development.

Cell Cleavage Patterns

Following fertilization, the second major event in vertebrate reproduction is the rapid division of the zygote into a larger and larger number of smaller and smaller cells (table 60.1). This period of division, called cleavage, is not accompanied by an increase in the overall size of the embryo. The resulting tightly packed mass of about 32 cells is called a **morula**, and each individual cell in the morula is referred to as a **blastomere**. As the blastomeres continue to divide, they secrete a fluid into the center of the morula. Eventually, a hollow ball of 500 to 2000 cells, the **blastula**, is formed. The fluid-filled cavity within the blastula is known as the *blastocoel*.

The pattern of cleavage division is influenced by the presence and location of yolk, which is abundant in the eggs of many vertebrates (figure 60.5). As we discussed in the previous chapter, vertebrates have embraced a variety of reproductive strategies involving different patterns of yolk utilization.

Primitive Chordates

When eggs contain little or no yolk, cleavage occurs throughout the whole egg (figure 60.6). This pattern of cleavage, called holoblastic cleavage, was characteristic of the ancestors of the vertebrates and is still seen in groups such as the lancelets and agnathans. In these animals, holoblastic cleavage results in the formation of a symmetrical blastula composed of cells of approximately equal size.

Amphibians and Advanced Fish

The eggs of bony fish and frogs contain much more cytoplasmic yolk in one hemisphere than the other. Because yolk-rich cells divide much more slowly than those that have little yolk, holoblastic cleavage in these eggs results in a very asymmetrical blastula (figure 60.7), with large cells containing a lot of yolk at one pole and a concentrated mass of small cells containing very little yolk at the other. In these blastulas, the pole that is rich in yolk is called the vegetal pole, while the pole that is relatively poor in yolk is called the animal pole.

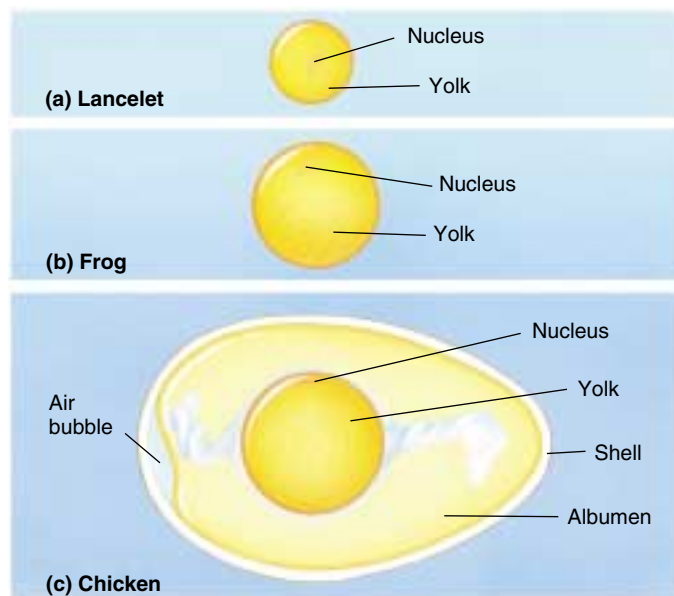


FIGURE 60.5

Yolk distribution in three kinds of eggs. (a) In the lancelet, a primitive chordate, the egg consists of a central nucleus surrounded by a small amount of yolk. (b) In a frog egg there is much more yolk, and the nucleus is displaced toward one pole. (c) Bird eggs are complexly organized, with the nucleus just under the surface of a large, central yolk.



FIGURE 60.6

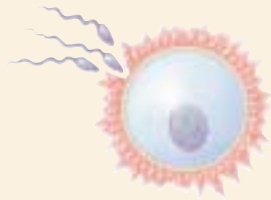
Holoblastic cleavage (3000 $\times$). In this type of cleavage, cell division occurs throughout the entire egg.



FIGURE 60.7

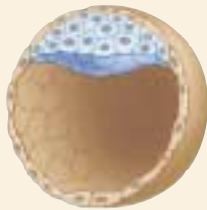
Dividing frog eggs. The closest cells in this photo (those near the animal pole) divide faster and are smaller than those near the vegetal pole.

Table 60.1 Stages of Vertebrate Development (Mammal)



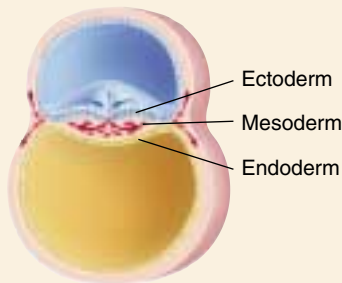
Fertilization

The haploid male and female gametes fuse to form a diploid zygote.



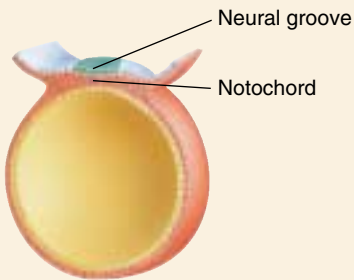
Cleavage

The zygote rapidly divides into many cells, with no overall increase in size. These divisions affect future development, since different cells receive different portions of the egg cytoplasm and, hence, different regulatory signals.



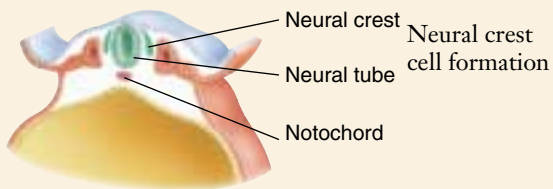
Gastrulation

The cells of the embryo move, forming three primary cell layers: ectoderm, mesoderm, and endoderm.



Neurulation

In all chordates, the first organ to form is the notochord; second is the dorsal nerve cord.



Neural crest cell formation

During neurulation, the neural crest is produced as the neural tube is formed. The neural crest gives rise to several uniquely vertebrate structures.



Organogenesis

Cells from the three primary layers combine in various ways to produce the organs of the body.

Reptiles and Birds

The eggs produced by reptiles, birds, and some fish are composed almost entirely of yolk, with a small amount of cytoplasm concentrated at one pole. Cleavage in these eggs occurs only in the tiny disc of polar cytoplasm, called the blastodisc, that lies astride the large ball of yolk material. This type of cleavage pattern is called meroblastic cleavage (figure 60.8). The resulting embryo is not spherical, but rather has the form of a thin cap perched on the yolk.

Mammals

Mammalian eggs are in many ways similar to the reptilian eggs from which they evolved, except that they contain very little yolk. Because cleavage is not impeded by yolk in mammalian eggs, it is holoblastic, forming a ball of cells surrounding a blastocoel. However, an inner cell mass is concentrated at one pole (figure 60.9). This inner cell mass is analogous to the blastodisc of reptiles and birds, and it goes on to form the developing embryo. The outer sphere of cells, called a trophoblast, is analogous to the cells that form the membranes underlying the tough outer shell of the reptilian egg. These cells have changed during the course of mammalian evolution to carry out a very different function: part of the trophoblast enters the endometrium (the epithelial lining of the uterus) and contributes to the placenta, the organ that permits exchanges between the fetal and maternal bloods. While part of the placenta is composed of fetal tissue (the trophoblast), part is composed of the modified endometrial tissue (called the *decidua basalis*) of the mother's uterus. The placenta will be discussed in more detail in a later section.

The Blastula

Viewed from the outside, the blastula looks like a simple ball of cells all resembling each other. In many animals, this appearance is misleading, as unequal distribution of developmental signals from the egg produces some degree of mosaic development, so that the cells are already committed to different developmental paths. In mammals, however, it appears that all of the blastomeres receive equivalent sets of signals, and body form is determined by cell-cell interactions. In a mammalian blastula, called a *blastocyst*, each cell is in contact with a different set of neighboring cells, and these interactions with neighboring cells are a major factor influencing the developmental fate of each cell. This positional information is of particular importance in the orientation of mammalian embryos, setting up different patterns of development along three embryonic axes: anterior-posterior, dorsal-ventral, and proximal-distal.

For a short period of time, just before they implant in the uterus, the cells of the mammalian blastocyst have the power to develop into many of the 210 different types of

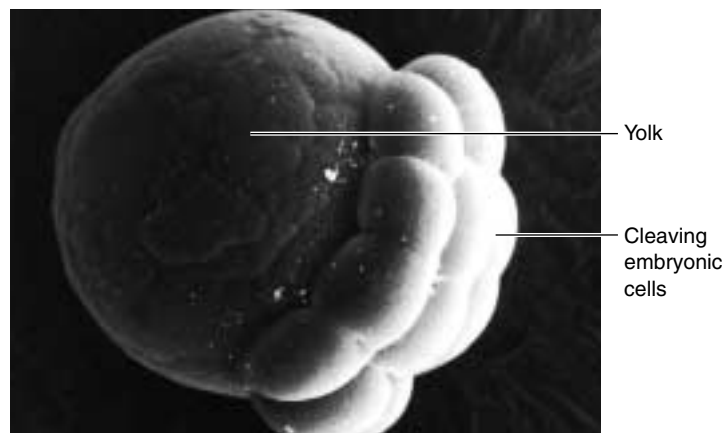


FIGURE 60.8
Meroblastic cleavage (400×). In this type of cleavage, only a portion of the egg actively divides to form a mass of cells.

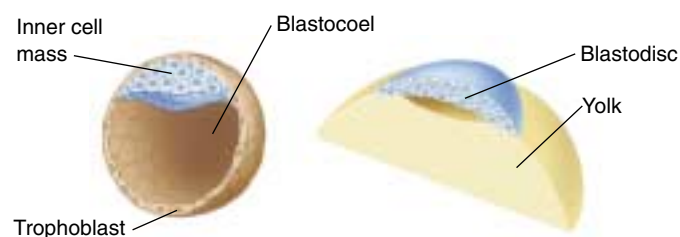


FIGURE 60.9
The embryos of mammals and birds are more similar than they seem. (a) A mammalian blastula is composed of a sphere of cells, the trophoblast, surrounding a cavity, the blastocoel, and an inner cell mass. (b) An avian (bird) blastula consists of a cap of cells, the blastodisc, resting atop a large yolk mass. The blastodisc will form an upper and a lower layer with a compressed blastocoel in between.

cells in the body—and probably all of them. Biologists have long sought to grow these cells, called *embryonic stem cells*, in tissue culture, as such stem cells might in principle be used to produce tissues for human transplant operations. Injected into a patient, for example, they might be able to respond to local signals and produce new tissue (see chapter 19). The first success in growing stem cells in culture was reported in 1998, when researchers isolated cells from the inner cell mass of human blastocysts and successfully grew them in tissue culture. These stem cells continue to grow and divide in culture indefinitely, unlike ordinary body cells, which divide only 50 or so times and then die.

A series of rapid cell divisions called cleavage transforms the zygote into a hollow ball of cells, the blastula. The cleavage pattern is influenced by the amount of yolk and its distribution in the egg.

60.3 Gastrulation forms the three germ layers of the embryo.

The Process of Gastrulation

The first visible results of cytoplasmic distribution and cell position within the blastula can be seen immediately after the completion of cleavage. Certain groups of cells **invaginate** (dent inward) and **involute** (roll inward) from the surface of the blastula in a carefully orchestrated activity called **gastrulation**. The events of gastrulation determine the basic developmental pattern of the vertebrate embryo. By the end of gastrulation, the cells of the embryo have rearranged into three primary **germ layers**: **ectoderm**, **mesoderm**, and **endoderm**. The cells in each layer have very different developmental fates. In general, the ectoderm is destined to form the epidermis and neural tissue; the mesoderm gives rise to connective tissue, skeleton, muscle, and vascular elements; and the endoderm forms the lining of the gut and its derivatives (table 60.2).

How is cell movement during gastrulation brought about? Apparently, migrating cells creep over stationary cells by means of actin filament contractions that change the shapes of the migrating cells affecting an invagination of blastula tissue. Each cell that moves possesses particular cell surface polysaccharides, which adhere to similar polysaccharides on the surfaces of the other moving cells. This interaction between cell surface molecules enables the migrating cells to adhere to one another and move as a single mass (see chapter 7).

Just as the pattern of cleavage divisions in different groups of vertebrates depends heavily on the amount and distribution of yolk in the egg, so the pattern of gastrulation among vertebrate groups depends on the shape of the blastulas produced during cleavage.

Table 60.2 Developmental Fates of the Primary Cell Layers

Ectoderm	Epidermis, central nervous system, sense organs, neural crest
Mesoderm	Skeleton, muscles, blood vessels, heart, gonads
Endoderm	Lining of digestive and respiratory tracts; liver, pancreas

Gastrulation in Primitive Chordates

In primitive chordates such as lancelets, which develop from symmetrical blastulas, gastrulation begins as the surface of the blastula invaginates into the blastocoel. About half of the blastula's cells move into the interior of the blastula, forming a structure that looks something like an indented tennis ball. Eventually, the inward-moving wall of cells pushes up against the opposite side of the blastula and then stops moving. The resulting two-layered, cup-shaped embryo is the gastrula (figure 60.10). The hollow structure resulting from the invagination is called the archenteron, and it is the progenitor of the gut. The opening of the archenteron, the future anus, is known as the blastopore.

This process produces an embryo with two cell layers: an outer ectoderm and an inner endoderm. Soon afterward, a third cell layer, the mesoderm, forms between the ectoderm and endoderm. In lancelets, the mesoderm forms from pouches that pinch off the endoderm. The appearance of these three primary cell layers sets the stage for all subsequent tissue and organ differentiation.

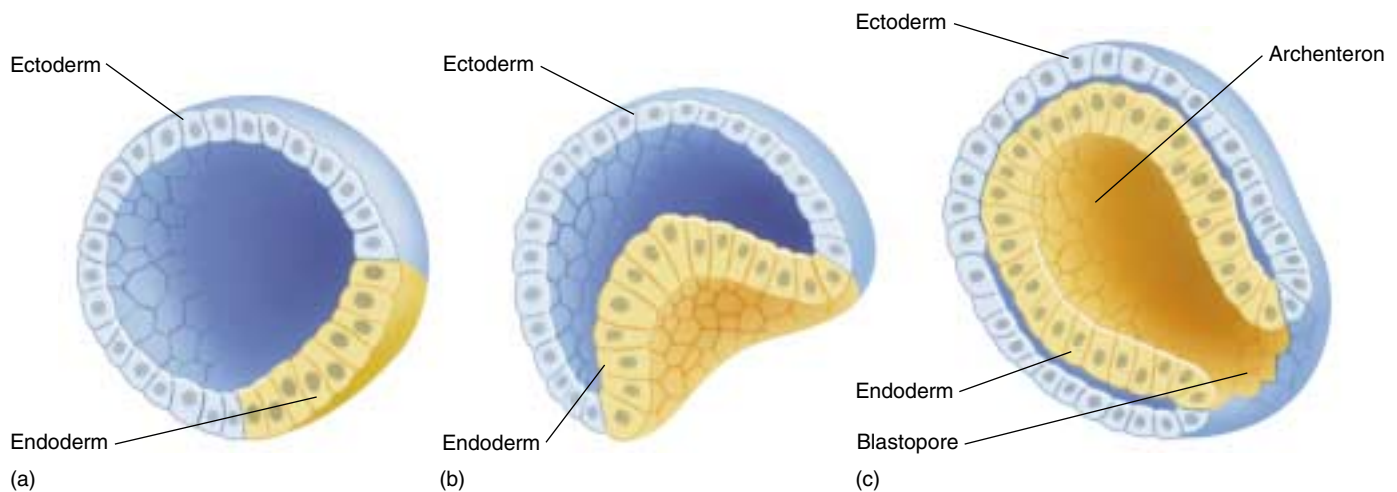


FIGURE 60.10

Gastrulation in a lancelet. In these chordates, the endoderm is formed by invagination of surface cells (*a*, *b*). This produces the primitive gut, or archenteron (*c*). Mesoderm will later be formed from pouches off the endoderm.

Gastrulation in Most Aquatic Vertebrates

In the blastulas of amphibians and those aquatic vertebrates with asymmetrical yolk distribution, the yolk-laden cells of the vegetal pole are fewer and much larger than the yolk-free cells of the animal pole. Consequently, gastrulation is more complex than it is in the lancelets. First, a layer of surface cells invaginates to form a small, crescent-shaped slit where the blastopore will soon be located. Next, cells from the animal pole involute over the dorsal lip of the blastopore (figure 60.11), at the same location as the gray crescent of the fertilized egg (see figure 60.4). As in the

lancelets, the involuting cell layer eventually presses against the inner surface of the opposite side of the embryo, eliminating the blastocoel and producing an archenteron with a blastopore. In this case, however, the blastopore is filled with yolk-rich cells, forming the yolk plug. The outer layer of cells resulting from these movements is the ectoderm, and the inner layer is the endoderm. Other cells that involute over the dorsal lip and ventral lip (the two lips of the blastopore that are separated by the yolk plug) migrate between the ectoderm and endoderm to form the third germ layer, the mesoderm (figure 60.11).

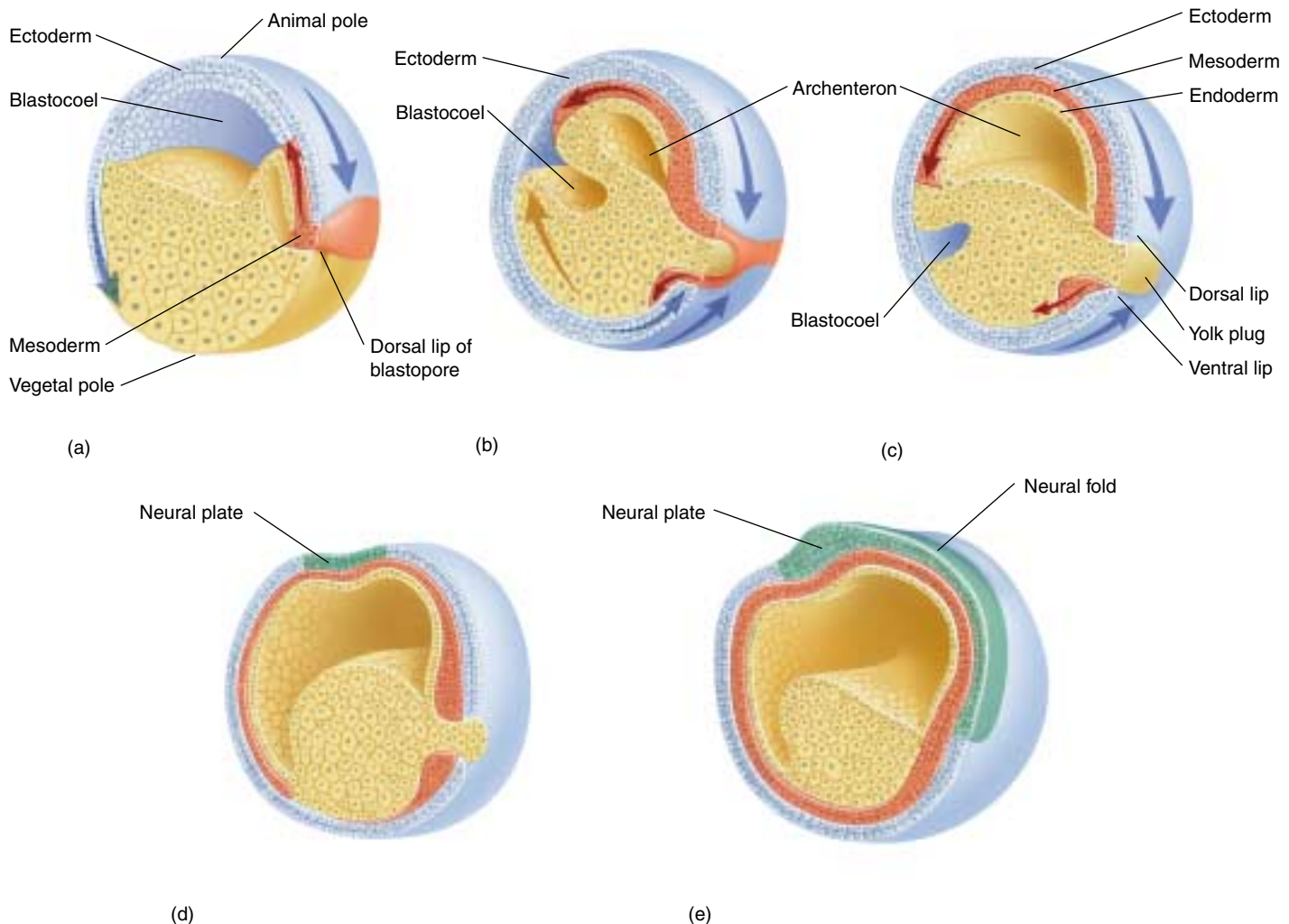


FIGURE 60.11

Frog gastrulation. (a) A layer of cells from the animal pole moves toward the yolk cells ultimately involuting through the dorsal lip of the blastopore. (b) Cells in the dorsal lip zone then involute into the hollow interior, or blastocoel, eventually pressing against the far wall. The three primary tissues (ectoderm, mesoderm, and endoderm) become distinguished. Ectoderm is shown in blue, mesoderm in red, and endoderm in yellow. (c) The movement of cells in the dorsal lip creates a new internal cavity, the archenteron, which opens to the outside through the plug of yolk remaining at the point of invagination. (d) The neural plate later forms from ectoderm. (e) This will next form a neural groove and then a neural tube as the embryo begins the process of neurulation. The cells of the neural ectoderm are shown in green.

Gastrulation in Reptiles, Birds, and Mammals

In the blastodisc of a bird or reptile and the inner cell mass of a mammal, the developing embryo is a small cap of cells rather than a sphere. No yolk separates the two sides of the embryo, and, as a result, the lower cell layer is able to differentiate into endoderm and the upper layer into ectoderm without cell movement. Just after these two primary cell layers form, the mesoderm arises by invagination and involution of cells from the upper layer. The surface cells begin moving to the midline where they involute and migrate laterally to form a mesodermal layer between the ectoderm and endoderm. A furrow along the longitudinal midline marks the site of this involution (figures 60.12 and 60.13). This furrow, analogous to an elongated blastopore, is called the **primitive streak**.

Gastrulation in lancelets involves the formation of ectoderm and endoderm by the invagination of the blastula, and the mesoderm layer forms from pouches pinched from the endoderm. In those vertebrates with extensive amounts of yolk, gastrulation requires the involution of surface cells into a blastopore or primitive streak, and the mesoderm is derived from some of these involuted cells.

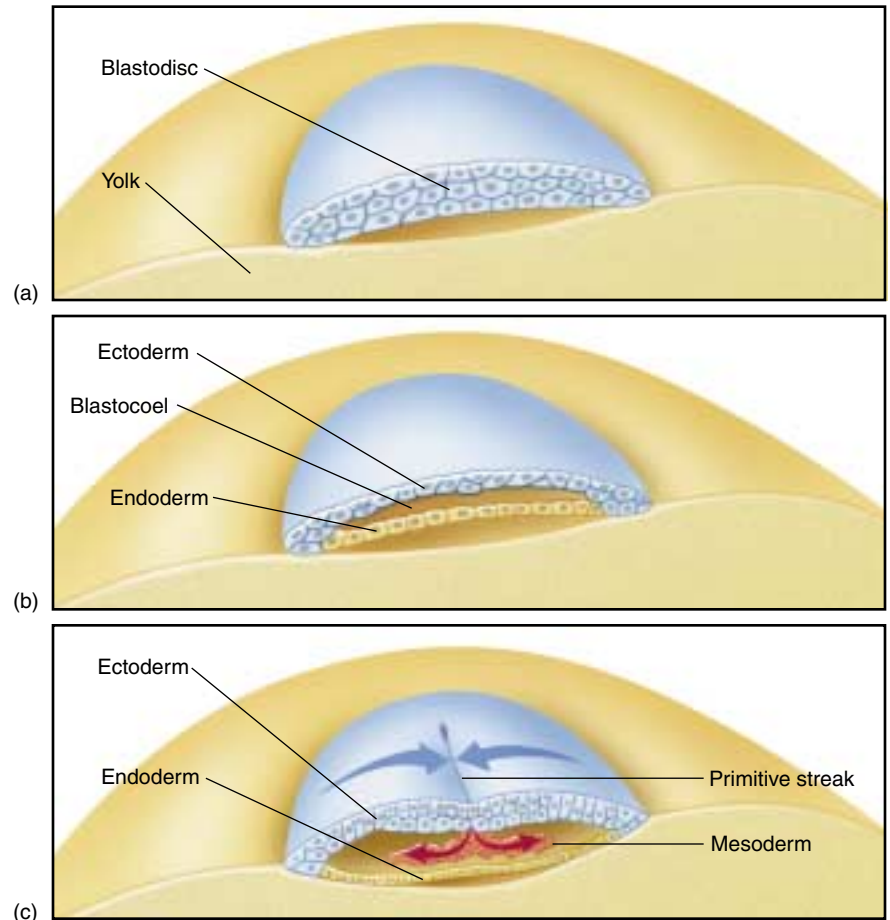


FIGURE 60.12

Gastrulation in birds. The upper layer of the blastodisc (a) differentiates into ectoderm, the lower layer into endoderm (b). Among the cells that migrate into the interior through the dorsal primitive streak are future mesodermal cells (c).

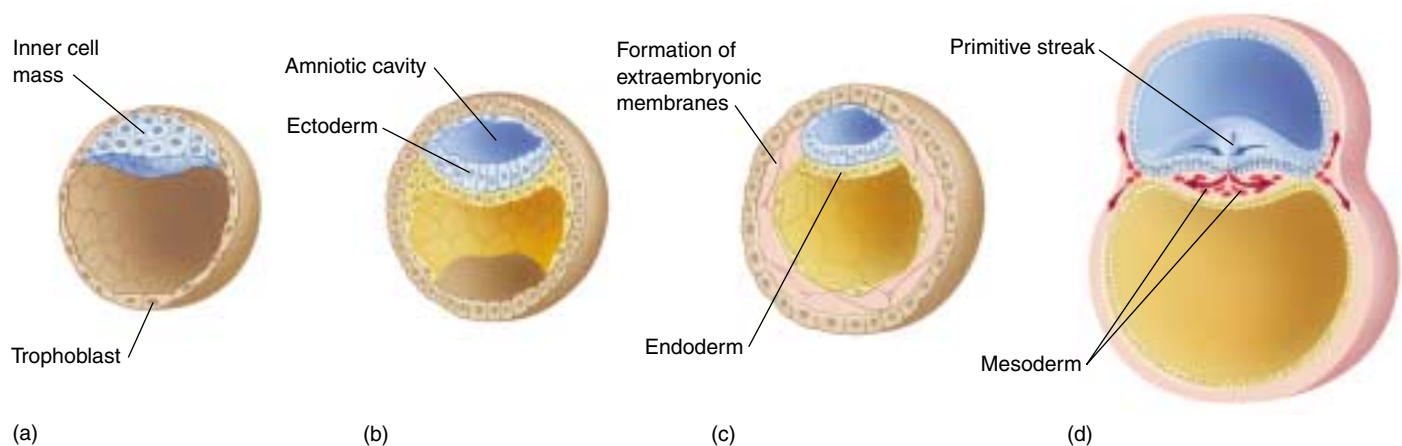


FIGURE 60.13

Mammalian gastrulation. (a) The amniotic cavity forms within the inner cell mass and its base. Layers of ectoderm and endoderm differentiate (b and c) as in the avian blastodisc. (d) A primitive streak develops, through which cells destined to become mesoderm migrate into the interior, again reminiscent of gastrulation in birds. The trophoblast has now moved further away from the embryo and begins to play a role in forming the placenta.

60.4 Body architecture is determined during the next stages of embryonic development.

Developmental Processes during Neurulation

During the next step in vertebrate development, the three primary cell layers begin their transformation into the body's tissues and organs. The process of tissue differentiation begins with the formation of two morphological features found only in chordates, the **notochord** and the hollow dorsal nerve cord. This development of the dorsal nerve cord is called **neurulation**.

The notochord is first visible soon after gastrulation is complete, forming from mesoderm along the dorsal midline of the embryo. It is a flexible rod located along the dorsal midline in the embryos of all chordates, although its function is replaced by the vertebral column when it develops from mesoderm in the vertebrates. After the notochord has been laid down, a layer of ectodermal cells situated above the notochord invaginates, forming a long crease, the **neural groove**, down the long axis of the embryo. The edges of the neural groove then move toward each other and fuse, creating a long hollow cylinder, the **neural tube** (figure 60.14), which runs beneath the surface of the embryo's back. The neural tube later differentiates into the spinal cord and brain.

The dorsal lip of the blastopore induces the formation of a notochord, and the presence of the notochord induces the overlying ectoderm to differentiate into the neural tube. The process of induction, when one embryonic region of cells influences the development of an adjacent region by changing its developmental pathway, was discussed in chapter 17 and is further examined in the next section.

While the neural tube is forming from ectoderm, the rest of the basic architecture of the body is being determined rapidly by changes in the mesoderm. On either side

of the developing notochord, segmented blocks of mesoderm tissue called **somites** form; more somites are added as development continues. Ultimately, the somites give rise to the muscles, vertebrae, and connective tissues. The mesoderm in the head region does not separate into discrete somites but remains connected as **somitomeres** and form the striated muscles of the face, jaws, and throat. Some body organs, including the kidneys, adrenal glands, and gonads, develop within another strip of mesoderm that runs alongside the somites. The remainder of the mesoderm moves out and around the endoderm and eventually surrounds it completely. As a result of this movement, the mesoderm becomes separated into two layers. The outer layer is associated with the body wall and the inner layer is associated with the gut. Between these two layers of mesoderm is the coelom (see chapter 45), which becomes the body cavity of the adult.

The Neural Crest

Neurulation occurs in all chordates, and the process in a lancelet is much the same as it is in a human. However, in vertebrates, just before the neural groove closes to form the neural tube, its edges pinch off, forming a small strip of cells, the **neural crest**, which becomes incorporated into the roof of the neural tube (figure 60.14). The cells of the neural crest later move to the sides of the developing embryo. The appearance of the neural crest was a key event in the evolution of the vertebrates because neural crest cells, after migrating to different parts of the embryo, ultimately develop into the structures characteristic of (though not necessarily unique to) the vertebrate body.

The differentiation of neural crest cells depends on their location. At the anterior end of the embryo, they merge

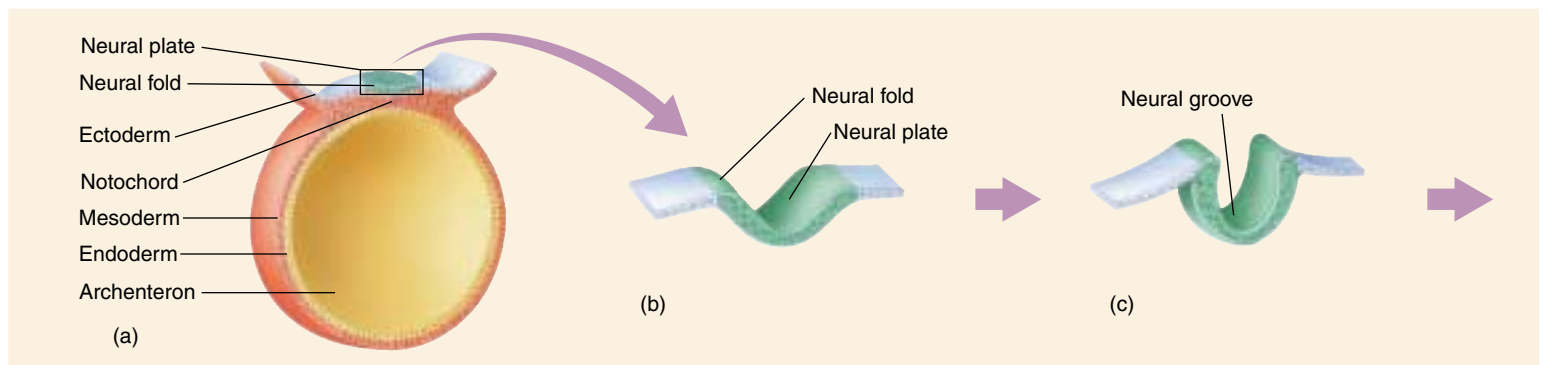


FIGURE 60.14

Mammalian neural tube formation. (a) The neural tube forms above the notochord when (b) cells of the neural plate fold together to form the (c) neural groove.

with the anterior portion of the brain, the forebrain. Nearby clusters of ectodermal cells associated with the neural crest cells thicken into **placodes**, which are distinct from neural crest cells although they arise from similar cellular interactions. Placodes subsequently develop into parts of the sense organs in the head. The neural crest and associated placodes exist in two lateral strips, which is why the vertebrate sense organs that develop from them are paired.

Neural crest cells located in more posterior positions have very different developmental fates. These cells migrate away from the neural tube to other locations in the head and trunk, where they form connections between the neural tube and the surrounding tissues. At these new locations, they contribute to the development of a variety of structures that are particularly characteristic of the vertebrates, several of which are discussed below. The migration of neural crest cells is unique in that it is not simply a change in the relative positions of cells, such as that seen in gastrulation. Instead, neural crest cells actually pass through other tissues.

The Gill Chamber

Primitive chordates such as lancelets are filter-feeders, using the rapid beating of cilia to draw water into their bodies through slits in their pharynx. These pharyngeal slits evolved into the vertebrate gill chamber, a structure that provides a greatly improved means of respiration. The evolution of the gill chamber was certainly a key event in the transition from filter-feeding to active predation.

In the development of the gill chamber, some of the neural crest cells form cartilaginous bars between the embryonic pharyngeal slits. Other neural crest cells induce portions of the mesoderm to form muscles along the cartilage, while still others form neurons that carry impulses between the nerve cord and these muscles. A major blood vessel called the aortic arch passes through each of the embryonic bars. Lined by still more neural crest cells, these

bars, with their internal blood supply, become highly branched and form the gills of the adult.

Because the stiff bars of the gill chamber can be bent inward by powerful muscles controlled by nerves, the whole structure is a very efficient pump that drives water past the gills. The gills themselves act as highly efficient oxygen exchangers, greatly increasing the respiratory capacity of the animals that possess them.

Elaboration of the Nervous System

Some neural crest cells migrate ventrally toward the notochord and form sensory neurons in the dorsal root ganglia (see chapter 54). Others become specialized as Schwann cells, which insulate nerve fibers and permit the rapid conduction of nerve impulses. Still others form the autonomic ganglia and the adrenal medulla. Cells in the adrenal medulla secrete epinephrine when stimulated by the sympathetic division of the autonomic nervous system during the fight-or-flight reaction. The similarity in the chemical nature of the hormone epinephrine and the neurotransmitter norepinephrine, released by sympathetic neurons, is understandable—both adrenal medullary cells and sympathetic neurons derive from the neural crest.

Sensory Organs and Skull

A variety of sense organs develop from the placodes. Included among them are the olfactory (smell) and lateral line (primitive hearing) organs discussed in chapter 55. Neural crest cells contribute to tooth development and to some of the facial and cranial bones of the skull.

The appearance of the neural crest in the developing embryo marked the beginning of the first truly vertebrate phase of development, as many of the structures characteristic of vertebrates derive directly or indirectly from neural crest cells.

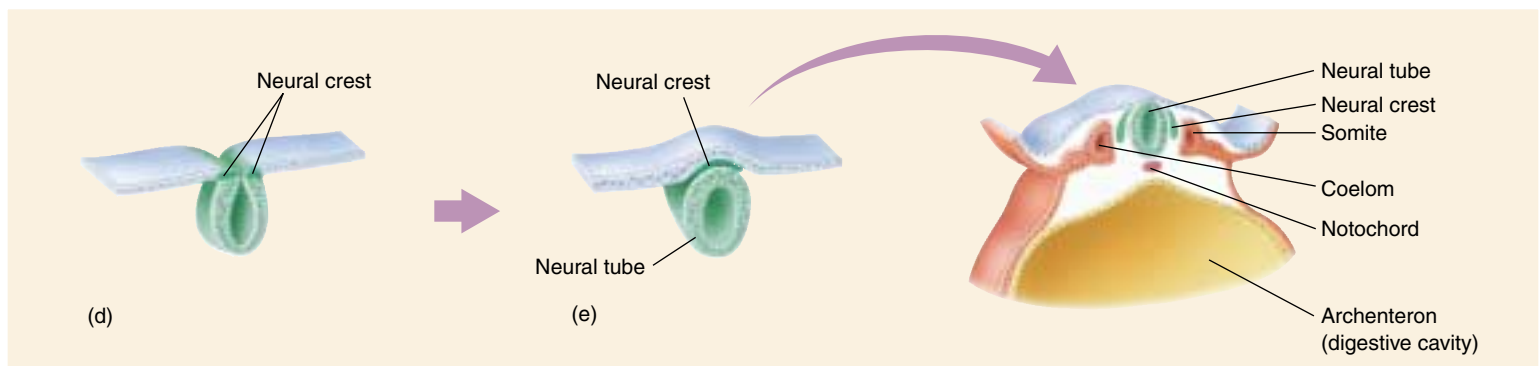


FIGURE 60.14 (continued)

(d) The neural groove eventually closes to form a hollow tube. (e) As the tube closes, some of the cells from the dorsal margin of the neural tube differentiate into the neural crest, which is characteristic of vertebrates.

How Cells Communicate during Development

In the process of vertebrate development, the relative position of particular cell layers determines, to a large extent, the organs that develop from them. By now, you may have wondered how these cell layers know where they are. For example, when cells of the ectoderm situated above the developing notochord give rise to the neural groove, how do these cells know they are above the notochord?

The solution to this puzzle is one of the outstanding accomplishments of experimental embryology, the study of how embryos form. The great German biologist Hans Spemann and his student Hilde Mangold solved it early in this century. In their investigation they removed cells from the dorsal lip of an amphibian blastula and transplanted them to a different location on another blastula (figure 60.15). (The dorsal lip region of amphibian blastu-

las develops from the grey crescent zone and is the site of origin of those mesoderm cells that later produce the notochord.) The new location corresponded to that of the animal's belly. What happened? The embryo developed *two* notochords, a normal dorsal one and a second one along its belly!

By using genetically different donor and host blastulas, Spemann and Mangold were able to show that the notochord produced by transplanting dorsal lip cells contained host cells as well as transplanted ones. The transplanted dorsal lip cells had acted as **organizers** (see also chapter 17) of notochord development. As such, these cells stimulated a developmental program in the belly cells of the embryos in which they were transplanted: the development of the notochord. The belly cells clearly contained this developmental program but would not have expressed it in the normal course of their development. The transplantation of the dorsal lip cells caused them to do so.

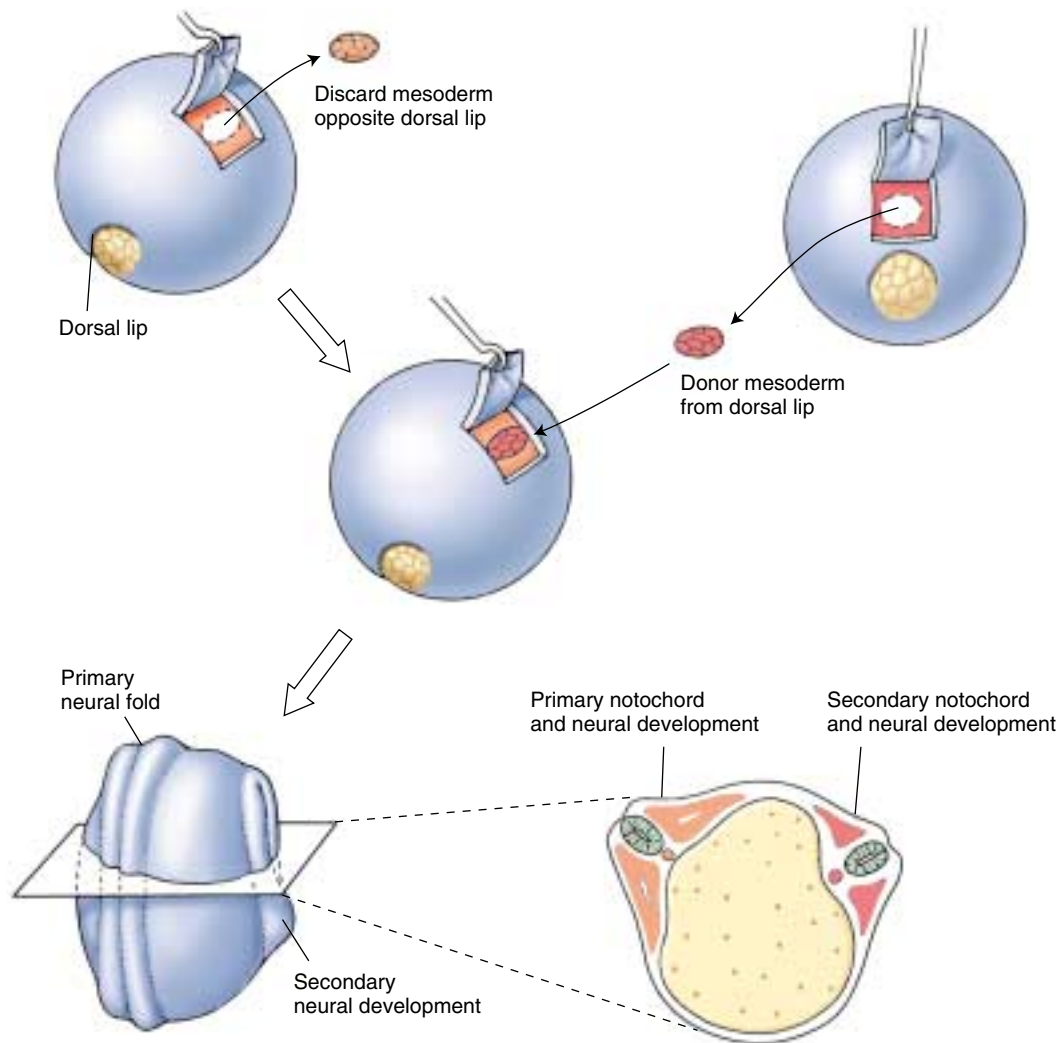


FIGURE 60.15
Spemann and Mangold's dorsal lip transplant experiment.

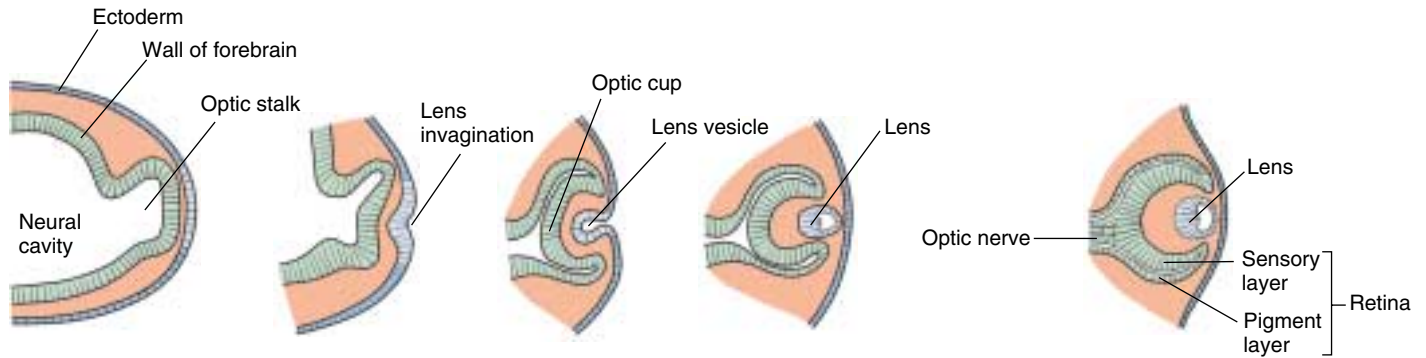


FIGURE 60.16

Development of the eye by induction. An extension of the optic stalk grows until it contacts ectoderm, which induces a section of the ectoderm to pinch off and form the lens. Other structures of the eye develop from the optic stalk.

These cells had indeed induced the ectoderm cells of the belly to form a notochord. This phenomenon as a whole is known as **induction**.

The process of induction that Spemann discovered appears to be the basic mode of development in vertebrates. Inductions between the three primary tissue types—ectoderm, mesoderm, and endoderm—are referred to as **primary inductions**. Inductions between tissues that have already been differentiated are called **secondary inductions**. The differentiation of the central nervous system during neurulation by the interaction of dorsal ectoderm and dorsal mesoderm to form the neural tube is an example of primary induction. In contrast, the differentiation of the lens of the vertebrate eye from ectoderm by interaction with tissue from the central nervous system is an example of secondary induction.

The eye develops as an extension of the forebrain, a stalk that grows outward until it comes into contact with the epidermis (figure 60.16). At a point directly above the growing stalk, a layer of the epidermis pinches off, forming a transparent lens. When the optic stalks of the two eyes have just started to project from the brain and the lenses have not yet formed, one of the budding stalks can be removed and transplanted to a region underneath a different epidermis, such as that of the belly. When Spemann performed this critical experiment, a lens still formed, this time from belly epidermis cells in the region above where the budding stalk had been transplanted.

What is the nature of the inducing signal that passes from one tissue to the other? If one imposes a nonporous barrier, such as a layer of cellophane, between the inducer and the target tissue, no induction takes place. In contrast, a porous filter, through which proteins can pass, does permit induction to occur. The induction process was discussed in detail in chapter 17. In brief, the inducer cells produce a protein factor that binds to the cells of the target tissue, initiating changes in gene expression.

The Nature of Developmental Decisions

All of the cells of the body, with the exception of a few specialized ones that have lost their nuclei, have an entire complement of genetic information. Despite the fact that all of its cells are genetically identical, an adult vertebrate contains hundreds of cell types, each expressing various aspects of the total genetic information for that individual. What factors determine which genes are to be expressed in a particular cell and which are not to be? In a liver cell, what mechanism keeps the genetic information that specifies nerve cell characteristics turned off? Does the differentiation of that particular cell into a liver cell entail the physical loss of the information specifying other cell types? No, it does not—but cells progressively lose the capacity to *express* ever-larger portions of their genomes. *Development is a process of progressive restriction of gene expression.*

Some cells become **determined** quite early in development. For example, all of the egg cells of the human female are set aside very early in the life of the embryo, yet some of these cells will not achieve differentiation into functional oocytes for more than 40 years. To a large degree, a cell's location in the developing embryo determines its fate. By changing a cell's location, an experimenter can alter its developmental destiny. However, this is only true up to a certain point in the cell's development. At some stage, every cell's ultimate fate becomes fixed, a process referred to as **commitment**. Commitment is not irreversible (entire individuals can be cloned from an individual specialized cell, as recounted in chapter 17), but rarely if ever reverses under ordinary circumstances.

When a cell is “determined,” it is possible to predict its developmental fate; when a cell is “committed,” that developmental fate cannot be altered. Determination often occurs very early in development, commitment somewhat later.

Embryonic Development and Vertebrate Evolution

The primitive chordates that gave rise to vertebrates were initially slow-moving, filter-feeding animals with relatively low metabolic rates. Many of the unique vertebrate adaptations that contribute to their varied ecological roles involve structures that arise from neural crest cells. The vertebrates became fast-swimming predators with much higher metabolic rates. This accelerated metabolism permitted a greater level of activity than was possible among the more primitive chordates. Other evolutionary changes associated with the derivatives of the neural crest provided better detection of prey, a greatly improved ability to orient spatially during prey capture, and the means to respond quickly to sensory information. The evolution of the neural crest and of the structures derived from it were thus crucial steps in the evolution of the vertebrates (figure 60.17).

Ontogeny Recapitulates Phylogeny

The patterns of development in the vertebrate groups that evolved most recently reflect in many ways the simpler patterns occurring among earlier forms. Thus, mammalian development and bird development are elaborations of reptile development, which is an elaboration of amphibian development, and so forth (figure 60.18). During the development of a mammalian embryo, traces can be seen of more primitive chordates. For example, at certain stages a human embryo possesses pharyngeal slits, which occur in all chordates and are homologous to the gill slits of fish. At later stages, a human embryo also has a tail.

In a sense, the patterns of development in chordate groups has built up in incremental steps over the evolutionary history of those groups. The developmental instructions for each new form seem to have been layered on top of the previous instructions, contributing addi-

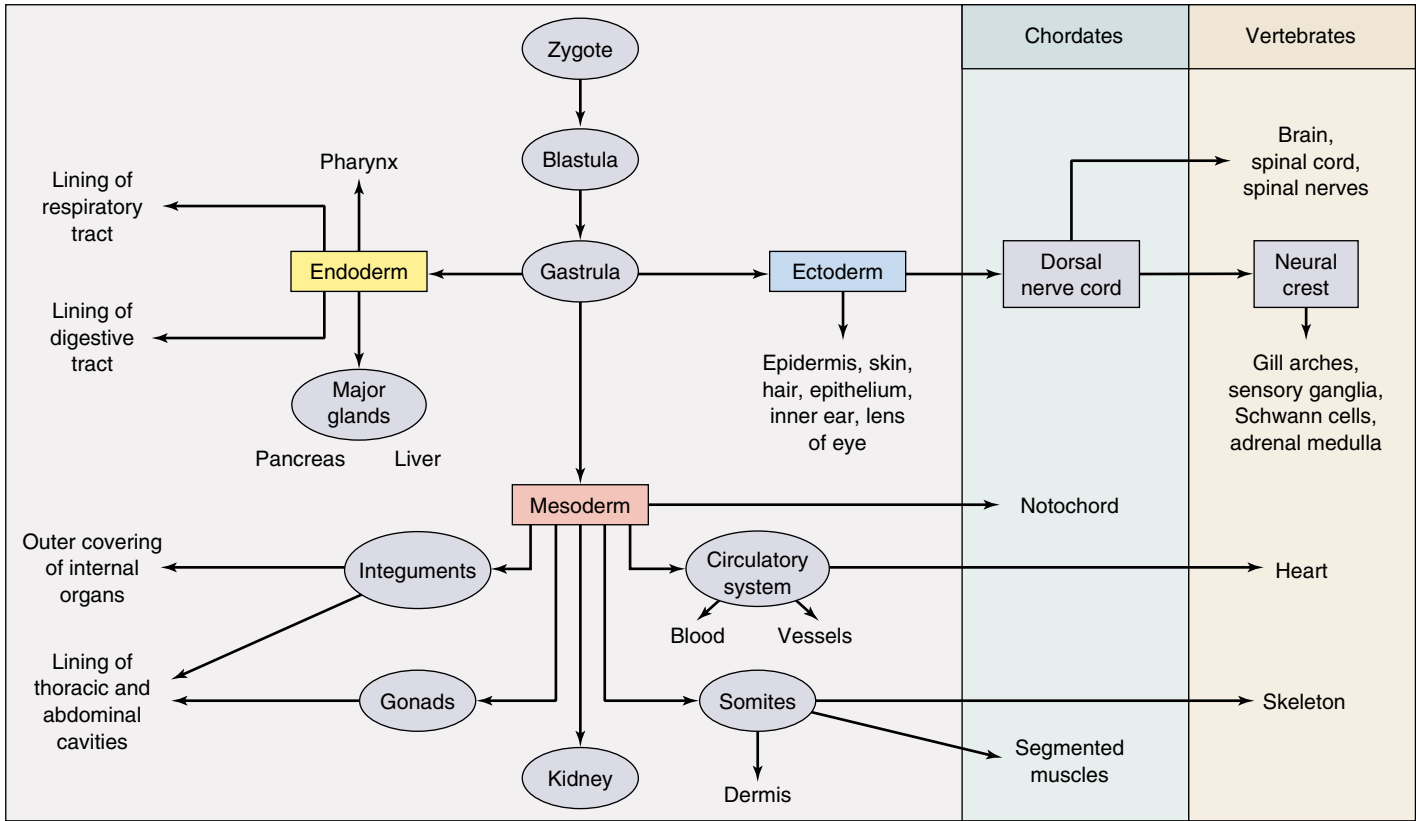


FIGURE 60.17
Derivation of the major tissue types. The three germ layers that form during gastrulation give rise to all organs and tissues in the body, but the neural crest cells that form from ectodermal tissue give rise to structures that are prevalent in the vertebrate animal such as gill arches and Schwann cells.

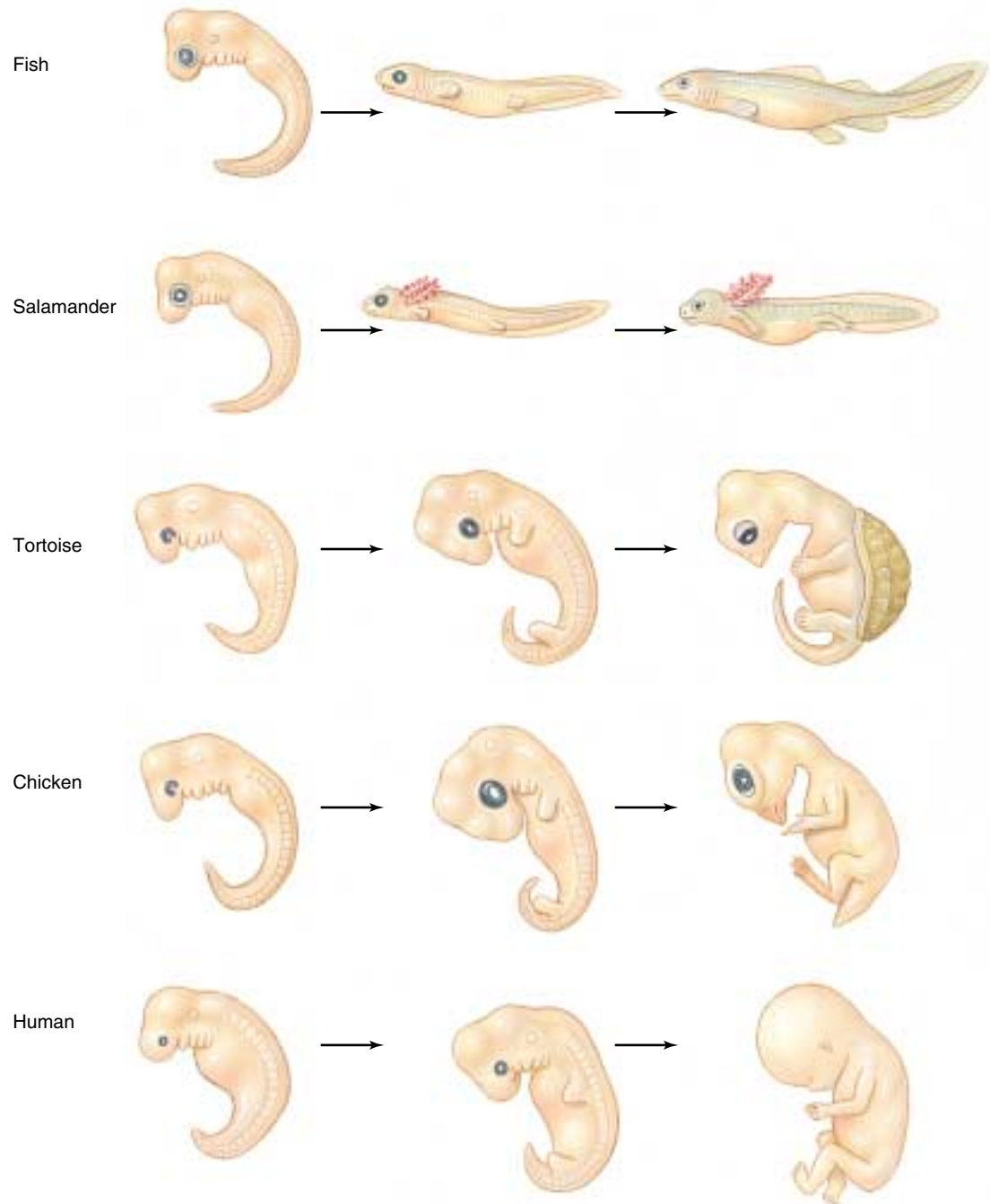


FIGURE 60.18
Embryonic development of vertebrates. Notice that the early embryonic stages of these vertebrates bear a striking resemblance to each other, even though the individuals are from different classes (fish, amphibians, reptiles, birds, and mammals). All vertebrates start out with an enlarged head region, gill slits, and a tail regardless of whether these characteristics are retained in the adults.

tional steps in the developmental journey. This hypothesis, proposed in the nineteenth century by Ernst Haeckel, is referred to as the “biogenic law.” It is usually stated as an aphorism: *ontogeny recapitulates phylogeny*; that is, embryological development (ontogeny) involves the same progression of changes that have occurred during evolution (phylogeny). However, the biogenic law is not literally true when stated in this way because embryonic stages are not reflections of *adult* ancestors. Instead, the embryonic stages of a particular vertebrate often reflect the *embryonic* stages of that vertebrate’s ancestors. Thus, the

pharyngeal slits of a mammalian embryo are *not* like the gill slits its ancestors had *when they were adults*. Rather, they are like the pharyngeal slits its ancestors had *when they were embryos*.

Vertebrates seem to have evolved largely by the addition of new instructions to the developmental program. Development of a mammal thus proceeds through a series of stages, and the earlier stages are unchanged from those that occur in the development of more primitive vertebrates.

Extraembryonic Membranes

As an adaptation to terrestrial life, the embryos of reptiles, birds, and mammals develop within a fluid-filled **amniotic membrane** (see chapter 48). The amniotic membrane and several other membranes form from embryonic cells but are located outside of the body of the embryo. For this reason, they are known as **extraembryonic membranes**. The extraembryonic membranes, later to become the **fetal membranes**, include the amnion, chorion, yolk sac, and allantois.

In birds, the **amnion** and **chorion** arise from two folds that grow to completely surround the embryo (figure 60.19). The amnion is the inner membrane that surrounds the embryo and suspends it in *amniotic fluid*, thereby mimicking the aquatic environments of fish and amphibian embryos. The chorion is located next to the eggshell and is separated from the other membranes by a cavity, the *extraembryonic coelom*. The **yolk sac** plays a critical role in the nutrition of bird and reptile embryos; it is also present in mammals, though it does not nourish the embryo. The **allantois** is derived as an outpouching of the gut and serves to store the uric acid excreted in the urine of birds. During

development, the allantois of a bird embryo expands to form a sac that eventually fuses with the overlying chorion, just under the eggshell. The fusion of the allantois and chorion form a functioning unit in which embryonic blood vessels, carried in the allantois, are brought close to the porous eggshell for gas exchange. The allantois is thus the functioning “lung” of a bird embryo.

In mammals, the embryonic cells form an inner cell mass that will become the body of the embryo and a layer of surrounding cells called the trophoblast (see figure 60.9). The trophoblast implants into the endometrial lining of its mother’s uterus and becomes the chorionic membrane (figure 60.20). The part of the chorion in contact with endometrial tissue contributes to the placenta, as is described in more detail in the next section. The allantois in mammals contributes blood vessels to the structure that will become the umbilical cord, so that fetal blood can be delivered to the placenta for gas exchange.

The extraembryonic membranes include the yolk sac, amnion, chorion, and allantois. These are derived from embryonic cells and function in a variety of ways to support embryonic development.

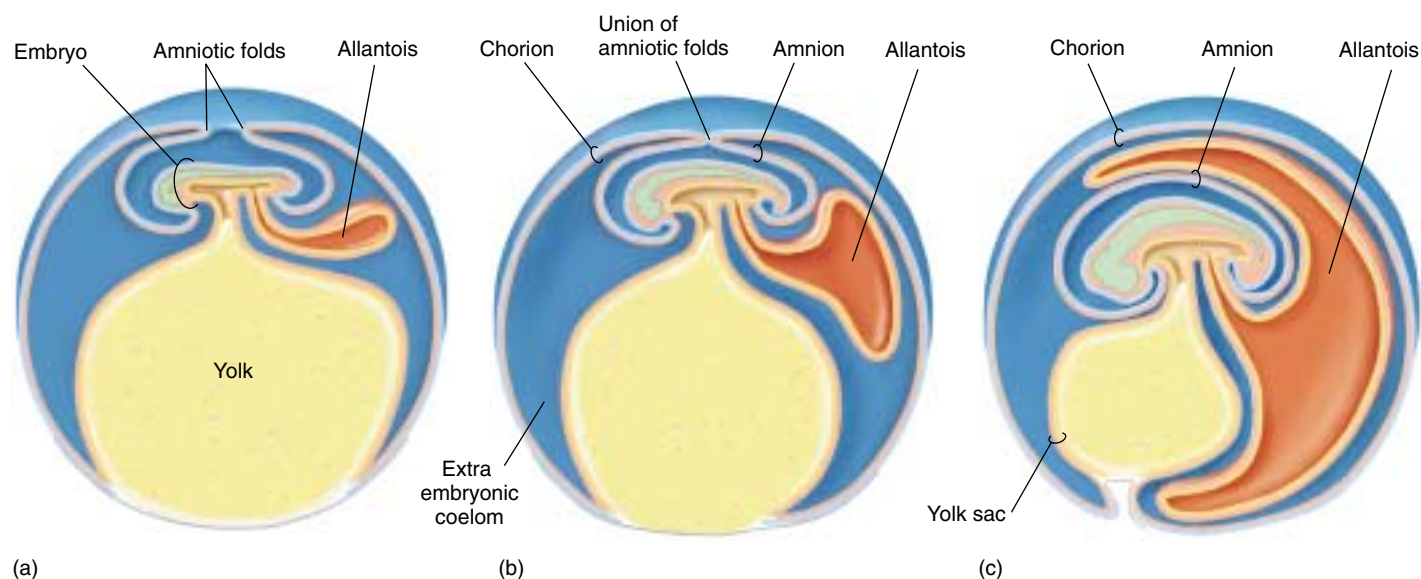


FIGURE 60.19

The extraembryonic membranes of a chick embryo. The membranes begin as amniotic folds from the embryo (a) that unite (b) to form a separate amnion and chorion (c). The allantois continues to grow until it will eventually unite with the chorion just under the eggshell.

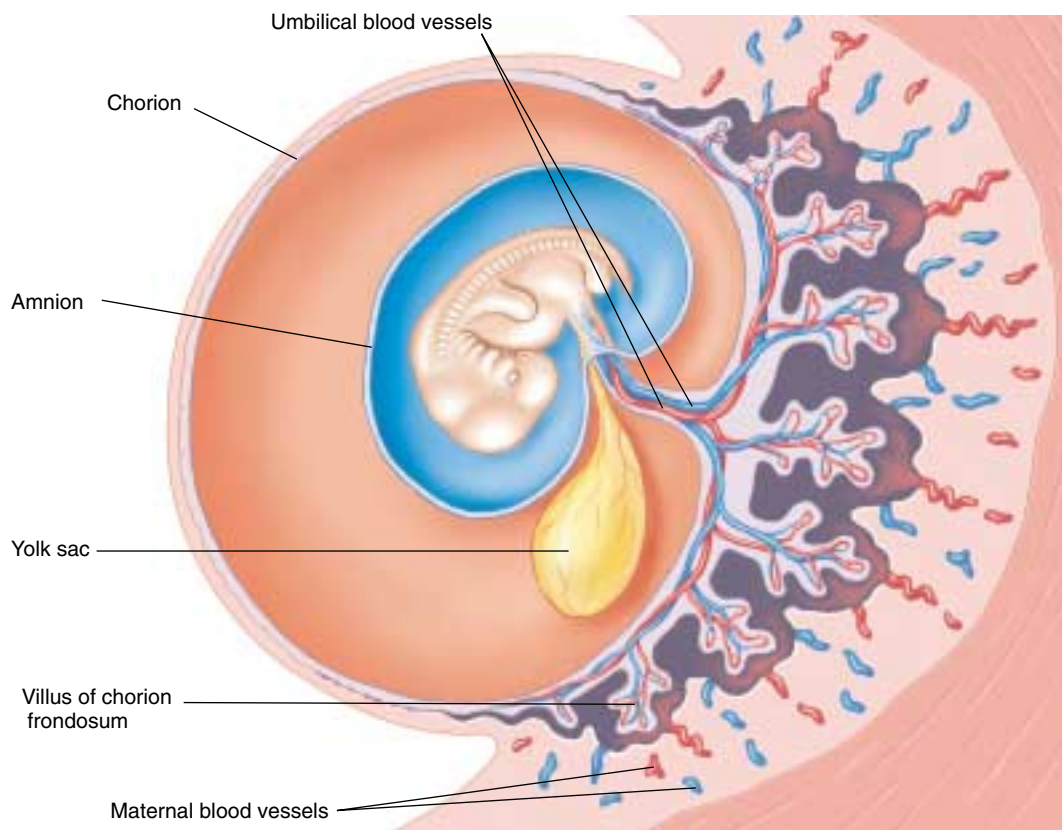
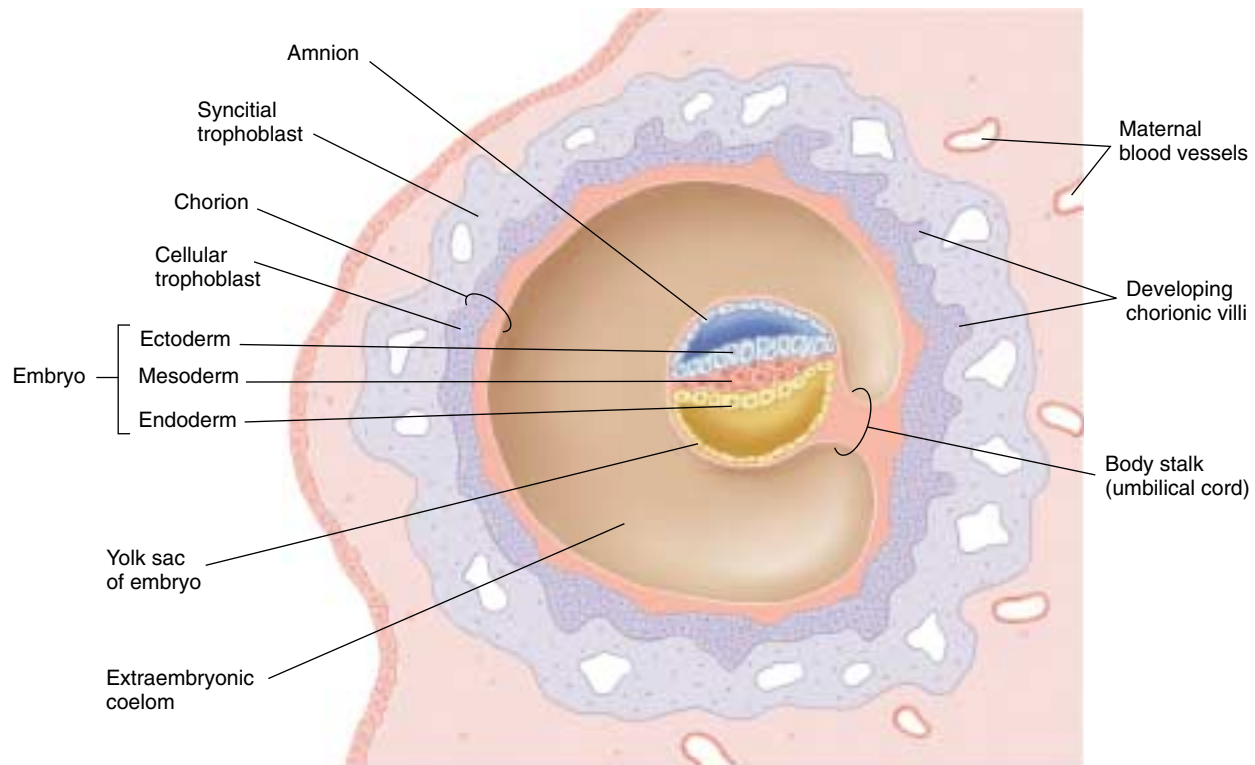


FIGURE 60.20

The extraembryonic membranes of a mammalian embryo. (a) After the embryo implants into the mother's endometrium (6–7 days after fertilization), the trophoblast becomes the chorion, and the yolk sac and amnion are produced. (b) The chorion develops extensions, called villi, that interdigitate with surrounding endometrial tissue. The embryo is encased within an amniotic sac.

60.5 Human development is divided into trimesters.

First Trimester

The development of the human embryo shows its evolutionary origins. Without an evolutionary perspective, we would be unable to account for the fact that human development proceeds in much the same way as development in a bird. In both animals, the embryo develops from a flattened collection of cells—the blastodisc in birds or the inner cell mass in humans. While the blastodisc of a bird is flattened because it is pressed against a mass of yolk, the inner cell mass of a human is flat despite the absence of a yolk mass. In humans as well as in birds, a primitive streak

forms and gives rise to the three primary germ layers. Human development, from fertilization to birth, takes an average of 266 days. This time is commonly divided into three periods, called trimesters.

The First Month

About 30 hours after fertilization, the zygote undergoes its first cleavage; the second cleavage occurs about 30 hours after that. By the time the embryo reaches the uterus (6–7 days after fertilization), it is a blastula, which in mammals is referred to as a blastocyst. As we mentioned earlier, the

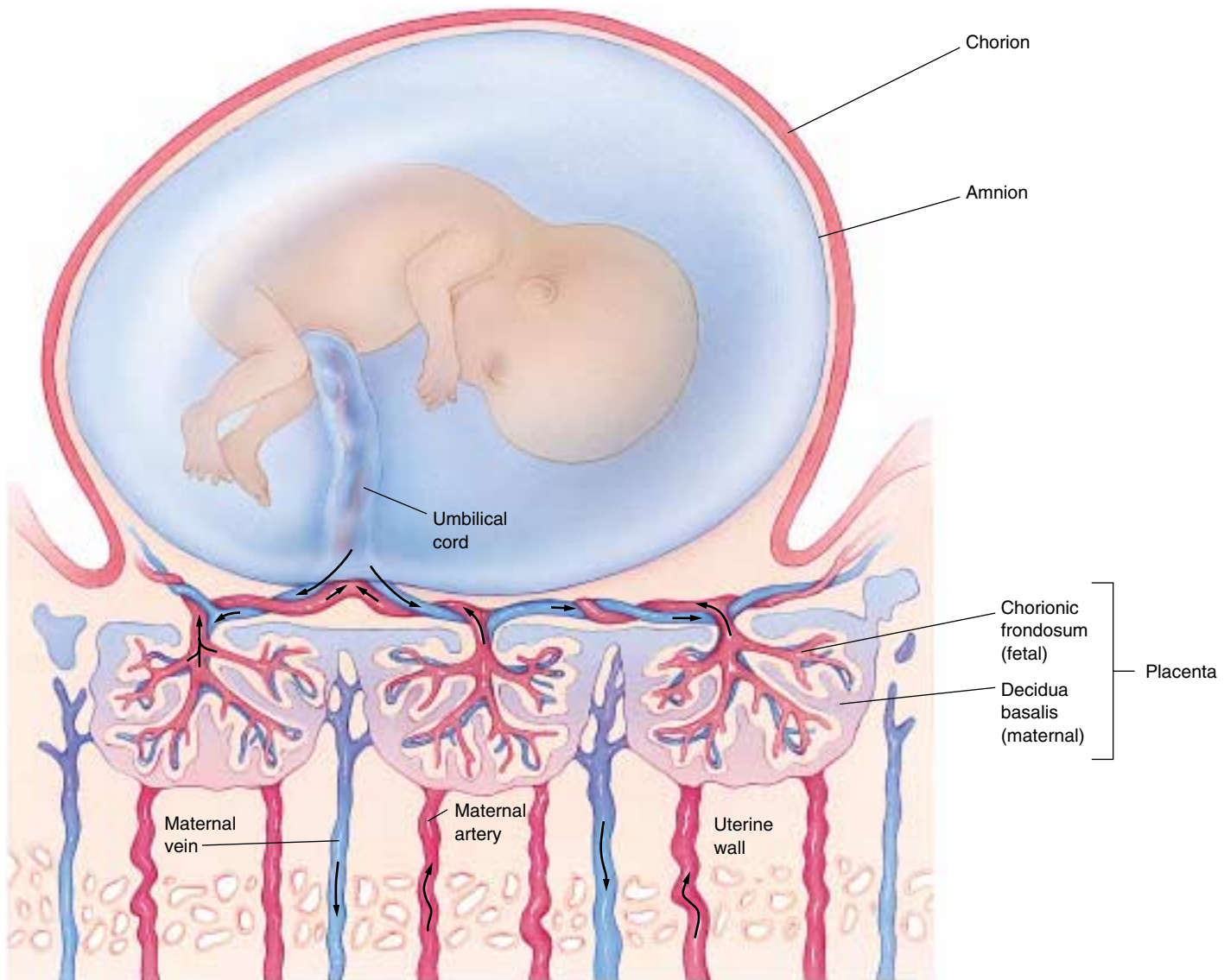


FIGURE 60.21

Structure of the placenta. The placenta contains a fetal component, the chorionic frondosum, and a maternal component, the decidua basalis. Deoxygenated fetal blood from the umbilical arteries (shown in blue) enters the placenta, where it picks up oxygen and nutrients from the mother's blood. Oxygenated fetal blood returns in the umbilical vein (shown in red) to the fetus, where it picks up oxygen and nutrients from the mother's blood.

blastocyst consists of an inner cell mass, which will become the body of the embryo, and a surrounding layer of trophoblast cells (see figure 60.9). The blastocyst begins to grow rapidly and initiates the formation of the amnion and the chorion. The blastocyst digests its way into the endometrial lining of the uterus in the process known as implantation.

During the second week after fertilization, the developing chorion forms branched extensions, the *chorionic frondosum* (fetal placenta) that protrude into the endometrium (figure 60.21). These extensions induce the surrounding endometrial tissue to undergo changes and become the *decidua basalis* (maternal placenta). Together, the chorionic frondosum and decidua basalis form a single functioning unit, the placenta (figure 60.22). Within the placenta, the mother's blood and the blood of the embryo come into close proximity but do not mix (see figure 60.21). Oxygen can thus diffuse from the mother to the embryo, and carbon dioxide can diffuse in the opposite direction. In addition to exchanging gases, the placenta provides nourishment for the embryo, detoxifies certain molecules that may pass into the embryonic circulation, and secretes hormones. Certain substances such as alcohol, drugs, and antibiotics are not stopped by the placenta and pass from the mother's bloodstream to the fetus.

One of the hormones released by the placenta is human chorionic gonadotropin (hCG), which was discussed in chapter 59. This hormone is secreted by the trophoblast cells even before they become the chorion, and is the hormone assayed in pregnancy tests. Because its action is almost identical to that of luteinizing hormone (LH), hCG maintains the mother's corpus luteum. The corpus luteum, in turn, continues to secrete estradiol and progesterone, thereby preventing menstruation and further ovulations.

Gastrulation also takes place in the second week after fertilization. The primitive streak can be seen on the surface of the embryo, and the three germ layers (ectoderm, mesoderm, and endoderm) are differentiated.

In the third week, neurulation occurs. This stage is marked by the formation of the neural tube along the dorsal axis of the embryo, as well as by the appearance of the first somites, which give rise to the muscles, vertebrae, and connective tissues. By the end of the week, over a dozen somites are evident, and the blood vessels and gut have begun to develop. At this point, the embryo is about 2 millimeters long.

Organogenesis (the formation of body organs) begins during the fourth week (figure 60.23*a*). The eyes form. The tubular heart develops its four chambers and starts to pulsate rhythmically, as it will for the rest of the individual's life. At 70 beats per minute, the heart is destined to beat more than 2.5 billion times during a lifetime of 70 years. Over 30 pairs of somites are visible by the end of the fourth week, and the arm and leg buds have begun to form. The embryo has increased in length to about 5 millimeters. Although the developmental scenario is now far advanced,



FIGURE 60.22
Placenta and fetus at seven weeks.

many women are unaware they are pregnant at this stage.

Early pregnancy is a very critical time in development because the proper course of events can be interrupted easily. In the 1960s, for example, many pregnant women took the tranquilizer thalidomide to minimize the discomforts associated with early pregnancy. Unfortunately, this drug had not been adequately tested. It interferes with limb bud development, and its widespread use resulted in many deformed babies. Organogenesis may also be disrupted during the first and second months of pregnancy if the mother contracts rubella (German measles). Most spontaneous abortions occur during this period.

The Second Month

Morphogenesis (the formation of shape) takes place during the second month (figure 60.23*b*). The miniature limbs of the embryo assume their adult shapes. The arms, legs, knees, elbows, fingers, and toes can all be seen—as well as a short bony tail! The bones of the embryonic tail, an evolutionary reminder of our past, later fuse to form the coccyx. Within the abdominal cavity, the major organs, including the liver, pancreas, and gallbladder, become evident. By the end of the second month, the embryo has grown to about 25 millimeters in length, weighs about one gram, and begins to look distinctly human.



(a)



(b)

FIGURE 60.23

The developing human. (a) Four weeks, (b) seven weeks, (c) three months, and (d) four months.

The Third Month

The nervous system and sense organs develop during the third month, and the arms and legs start to move (figure 60.23c). The embryo begins to show facial expressions and carries out primitive reflexes such as the startle reflex and sucking. The eighth week marks the transition from embryo to fetus. At this time, all of the major organs of the body have been established. What remains of development is essentially growth.

At around 10 weeks, the secretion of human chorionic gonadotropin (hCG) by the placenta declines, and the corpus luteum regresses as a result. However, menstruation does not occur because the placenta itself secretes estradiol and progesterone (figure 60.24). In fact, the amounts of these two hormones secreted by the placenta far exceed the amounts that are ever secreted by the ovaries. The high levels of estradiol and progesterone in the blood during pregnancy continue to inhibit the release of FSH and LH, thereby preventing ovulation. They also help maintain the uterus and eventually prepare it for labor and delivery, and they stimulate the development of the mammary glands in preparation for lactation after delivery.

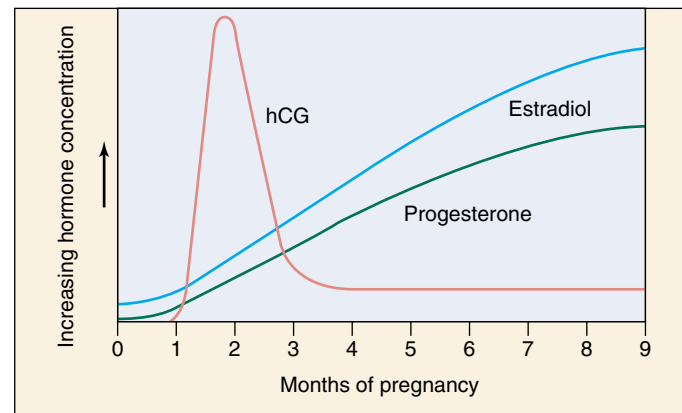


FIGURE 60.24

Hormonal secretion by the placenta. The placenta secretes chorionic gonadotropin (hCG) for 10 weeks. Thereafter, it secretes increasing amounts of estradiol and progesterone.

The embryo implants into the endometrium, differentiates the germ layers, forms the extraembryonic membranes, and undergoes organogenesis during the first month and morphogenesis during the second month.



(c)



(d)

Second and Third Trimesters

The second and third trimesters are characterized by the tremendous growth and development required for the viability of the baby after its birth.

Second Trimester

Bones actively enlarge during the fourth month (figure 60.23*d*), and by the end of the month, the mother can feel the baby kicking. During the fifth month, the head and body grow a covering of fine hair. This hair, called **lanugo**, is another evolutionary relict but is lost later in development. By the end of the fifth month, the rapid heartbeat of the fetus can be heard with a stethoscope, although it can also be detected as early as 10 weeks with a fetal monitor. The fetus has grown to about 175 millimeters in length and attained a weight of about 225 grams. Growth begins in earnest in the sixth month; by the end of that month, the baby weighs 600 grams (1.3 lbs) and is over 300 millimeters (1 ft) long. However, most of its prebirth growth is still to come. The baby cannot yet survive outside the uterus without special medical intervention.

Third Trimester

The third trimester is predominantly a period of growth rather than development. The weight of the fetus doubles several times, but this increase in bulk is not the only kind of growth that occurs. Most of the major nerve tracts in the brain, as well as many new neurons (nerve cells), are formed during this period. The developing brain produces neurons at an average rate estimated at more than 250,000 per minute! Neurological growth is far from complete at the end of the third trimester, when birth takes place. If the fetus remained in the uterus until its neurological development was complete, it would grow too large for safe delivery through the pelvis. Instead, the infant is born as soon as the probability of its survival is high, and its brain continues to develop and produce new neurons for months after birth.

The critical stages of human development take place quite early, and the following six months are essentially a period of growth. The growth of the brain is not yet complete, however, by the end of the third trimester, and must be completed postnatally.

Birth and Postnatal Development

In some mammals, changing hormone levels in the developing fetus initiate the process of birth. The fetuses of these mammals have an extra layer of cells in their adrenal cortex, called a fetal zone. Before birth, the fetal pituitary gland secretes corticotropin, which stimulates the fetal zone to secrete steroid hormones. These corticosteroids then induce the uterus of the mother to manufacture prostaglandins, which trigger powerful contractions of the uterine smooth muscles.

The adrenal glands of human fetuses lack a fetal zone, and human birth does not seem to be initiated by this mechanism. In a human, the uterus releases prostaglandins, possibly as a result of the high levels of estradiol secreted by the placenta. Estradiol also stimulates the uterus to produce more oxytocin receptors, and as a result, the uterus becomes increasingly sensitive to oxytocin. Prostaglandins begin the uterine contractions, but then sensory feedback from the uterus stimulates the release of oxytocin from the mother's posterior pituitary gland. Working together, oxytocin and prostaglandins further stimulate uterine contractions, forcing the fetus downward (figure 60.25). Initially, only a few contractions occur each hour, but the rate eventually increases to one contraction every two to three minutes. Finally, strong contractions, aided by the mother's pushing, expel the fetus, which is now a newborn baby.

After birth, continuing uterine contractions expel the placenta and associated membranes, collectively called the afterbirth. The umbilical cord is still attached to the baby, and to free the newborn, a doctor or midwife clamps and cuts the cord. Blood clotting and contraction of muscles in the cord prevent excessive bleeding.

Nursing

Milk production, or lactation, occurs in the alveoli of mammary glands when they are stimulated by the anterior pituitary hormone, prolactin. Milk from the alveoli is secreted into a series of alveolar ducts, which are surrounded by smooth muscle and lead to the nipple (figure 60.26). During pregnancy, high levels of progesterone stimulate the development of the mammary alveoli, and high levels of estradiol stimulate the development of the alveolar ducts. However, estradiol blocks the actions of prolactin on the

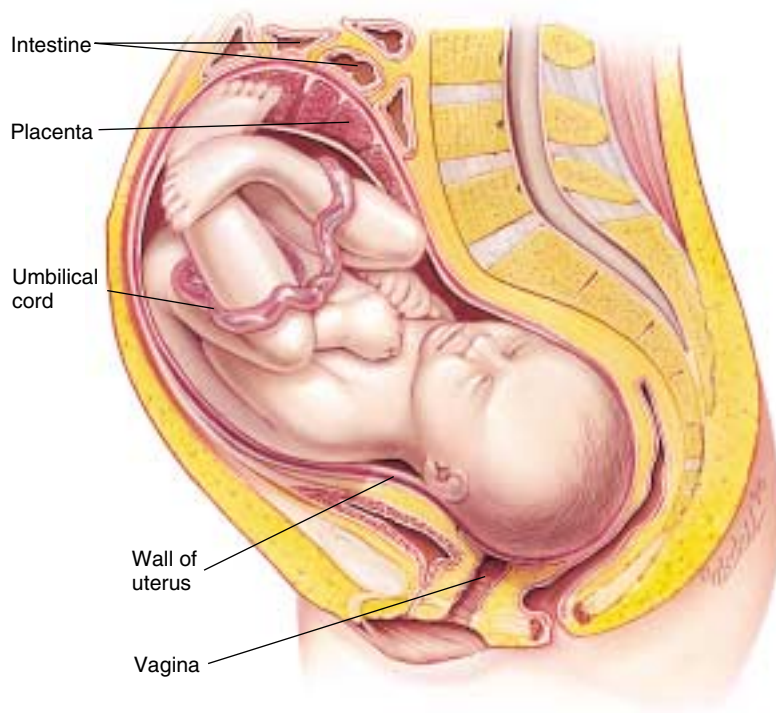


FIGURE 60.25

Position of the fetus just before birth. A developing fetus is a major addition to a woman's anatomy. The stomach and intestines are pushed far up, and there is often considerable discomfort from pressure on the lower back. In a natural delivery, the fetus exits through the vagina, which must dilate (expand) considerably to permit passage.

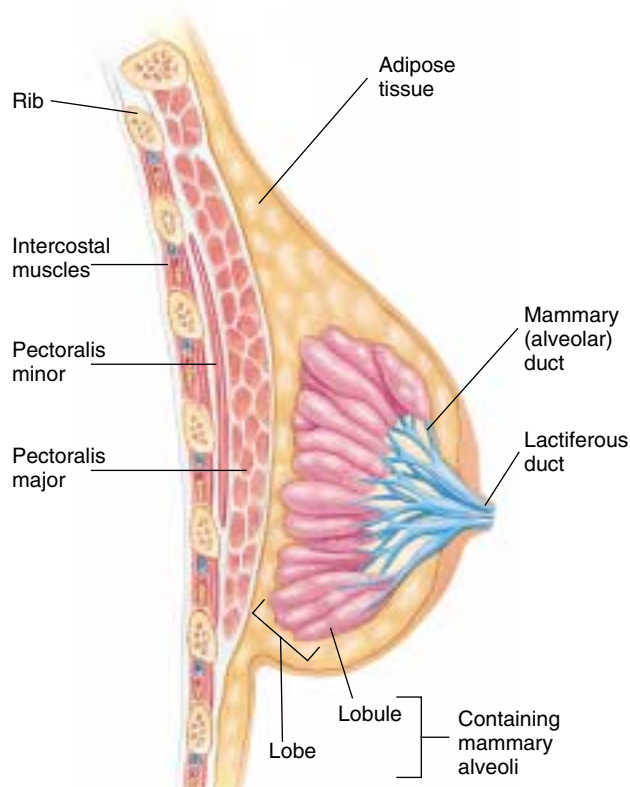


FIGURE 60.26

A sagittal section of a mammary gland. The mammary alveoli produce milk in response to stimulation by prolactin, and milk is ejected through the lactiferous duct in response to stimulation by oxytocin.

mammary glands, and it inhibits prolactin secretion by promoting the release of prolactin-inhibiting hormone from the hypothalamus. During pregnancy, therefore, the mammary glands are prepared for lactation but prevented from lactating.

When the placenta is discharged after birth, the concentrations of estradiol and progesterone in the mother's blood decline rapidly. This decline allows the anterior pituitary gland to secrete prolactin, which stimulates the mammary alveoli to produce milk. Sensory impulses associated with the baby's suckling trigger the posterior pituitary gland to release oxytocin. Oxytocin stimulates contraction of the smooth muscle surrounding the alveolar ducts, thus causing milk to be ejected by the breast. This pathway is known as the milk-ejection reflex. The secretion of oxytocin during lactation also causes some uterine contractions, as it did during labor. These contractions help to restore the tone of uterine muscles in mothers who are breastfeeding.

The first milk produced after birth is a yellowish fluid called colostrum, which is both nutritious and rich in maternal antibodies. Milk synthesis begins about three days following the birth and is referred to as when milk "comes in." Many mothers nurse for a year or longer. During this period, important pair-bonding occurs between the mother and child. When nursing stops, the accumulation of milk in the breasts signals the brain to stop prolactin secretion, and milk production ceases.

Postnatal Development

Growth of the infant continues rapidly after birth. Babies typically double their birth weight within two months. Because different organs grow at different rates and cease growing at different times, the body proportions of infants are different from those of adults. The head, for example, is disproportionately large in newborns, but after birth it grows more slowly than the rest of the body. Such a pattern of growth, in which different components grow at different rates, is referred to as allometric growth.

In most mammals, brain growth is mainly a fetal phenomenon. In chimpanzees, for instance, the brain and the cerebral portion of the skull grow very little after birth, while the bones of the jaw continue to grow. As a result, the head of an adult chimpanzee looks very different from that of a fetal chimpanzee (figure 60.27). In human infants, on the other hand, the brain and cerebral skull grow at the same rate as the jaw. Therefore, the jaw-skull proportions do not change after birth, and the head of a human adult looks very similar to that of a human fetus. It is primarily for this reason that an early human fetus seems so remarkably adultlike. The fact that the human brain continues to grow significantly for the first few years of postnatal life means that adequate nutrition and a rich, safe environment are particularly crucial during this period for the full development of the person's intellectual potential.

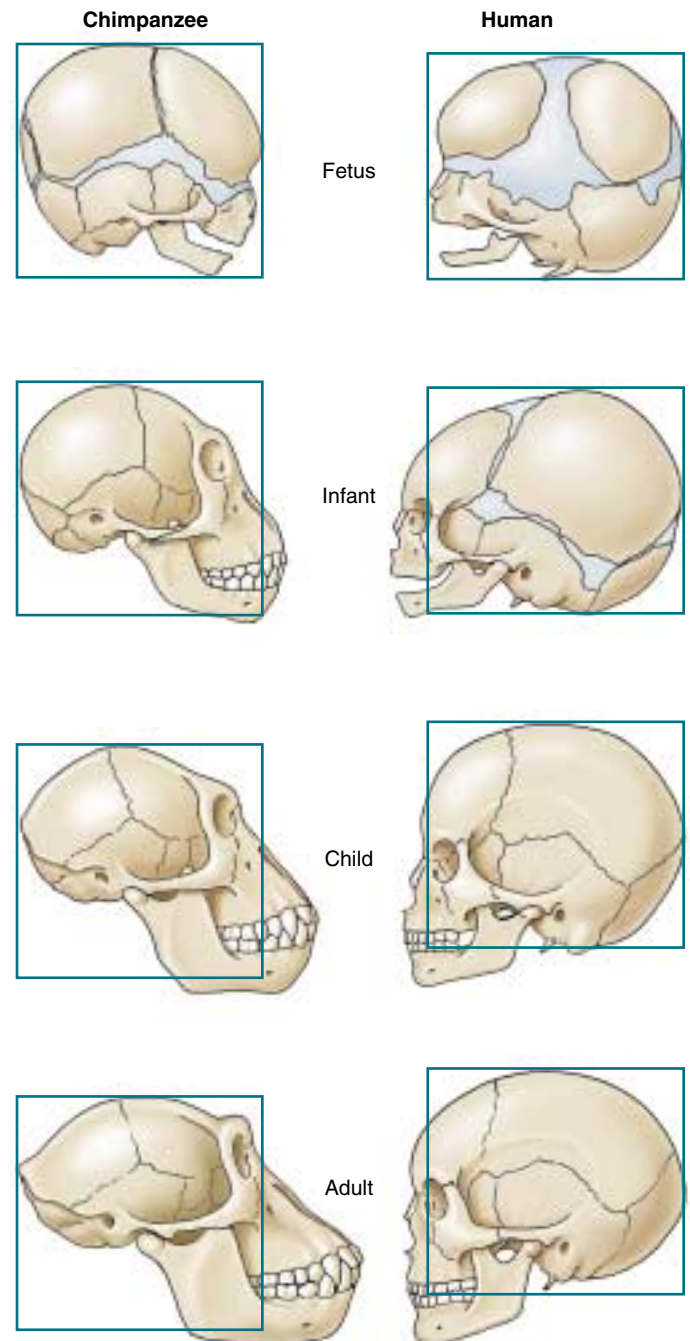


FIGURE 60.27

Allometric growth. In young chimpanzees, the jaw grows at a faster rate than the rest of the head. As a result, the adult chimpanzee head shape differs greatly from its head shape as a newborn. In humans, the difference in growth between the jaw and the rest of the head is much smaller, and the adult head shape is similar to that of the newborn.

Birth occurs in response to uterine contractions stimulated by oxytocin and prostaglandins. Lactation is stimulated by prostaglandin, but the milk-ejection reflex requires the action of oxytocin.



Summary

Questions

Media Resources

60.1 Fertilization is the initial event in development.

- Fertilization is the union of an egg and a sperm to form a zygote. It is accomplished externally in most fish and amphibians, and internally in all other vertebrates.
- The three stages of fertilization are (1) penetration, (2) activation, and (3) fusion.

1. What are the three stages of fertilization, and what happens during each stage?



• Fertilization

60.2 Cell cleavage and the formation of a blastula set the stage for later development.

- Cleavage is the rapid division of the newly formed zygote into a mass of cells, without any increase in overall size.
- The cleavages produce a hollow ball of cells, called the blastula.

2. What is the difference between holoblastic cleavage and meroblastic cleavage? What is responsible for an embryo undergoing one or the other type of cleavage?



• Early development
• Preembryonic development

60.3 Gastrulation forms the three germ layers of the embryo.

- During gastrulation, cells in the blastula change their relative positions, forming the three primary cell layers: ectoderm, mesoderm, and endoderm.
- In eggs with moderate or large amounts of yolk, cells involute down and around the yolk, through a blastopore or primitive streak to form the three germ layers.

3. What is an archenteron, and during what developmental stage does it form? What is the future fate of this opening in vertebrates?



• Cell differentiation

4. How is gastrulation in amphibians different from that in lancelets?

60.4 Body architecture is determined during the next stages of embryonic development.

- Neurulation involves the formation of a structure found only in chordates, the notochord and dorsal hollow nerve cord.
- The formation of the neural crest is the first developmental event unique to vertebrates. Most of the distinctive structures associated with vertebrates are derived from the neural crest.

5. What structure unique to chordates forms during neurulation?



• Art Activity:
Human extra-embryonic membranes

6. What are the functions of the amnion, chorion, and allantois in birds and mammals?



• Embryonic and fetal development

60.5 Human development is divided into trimesters.

- Most of the critical events in human development occur in the first month of pregnancy. Cleavage occurs during the first week, gastrulation during the second week, neurulation during the third week, and organogenesis during the fourth week.
- The second and third months of human development are devoted to morphogenesis and to the elaboration of the nervous system and sensory organs.
- During the last six months before birth, the human fetus grows considerably, and the brain produces large numbers of neurons and establishes major nerve tracts.

7. How does the placenta prevent menstruation during the first two months of pregnancy?



• Bioethics Case Study:
Critically ill newborns

8. At what time during human pregnancy does organogenesis occur?



• Human development
• Pregnancy
• Postnatal period

9. Is neurological growth complete at birth?

10. What hormone stimulates lactation (milk production)? What hormone stimulates milk ejection from the breast?